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Transaction Fraud for Banking

Monitor all the transactions throughout your entire business in real time. Prevent the most complex fraud occurrences with accuracy and scalability.

Get started

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FAQ

Was this page helpful?Â

`device_id` in Transfers or Digital Activity eventsâ

If you have the Digital Trust (DT) solution enabled in your product, the `revelock_device_id` generated by DT will be used as the default `device_id`. On the other hand, if you don't have the DT solution enabled in your product, you're recommended to send the `device_id` field from the API payload, so that you can use rules based on this field.

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File SAR

Was this page helpful?Â

In some geographies, financial institutions are required to report fraudulent activity using Suspicious Activity Reports (SAR). The Feedzai's SAR Manager component allows fraud analysts to create SARs by providing a guided experience and pre-populating several fields to make this process easier.

info

Access to the SAR functionality: This functionality applies only to specific geographies and under certain use cases. If SAR is not within the scope of your license, you can skip this page entirely.

SAR scopeâ

In some jurisdictions, reporting suspicious activity needs to be performed via a formal document named Suspicious Activity Report (SAR). For example, in the US, the scope of activity that requires filing a SAR within a fraud context is well defined, and includes:

- ACH
- Advance fee
- Business loan
- Check
- Consumer loan
- Credit/Debit card

- Healthcare/Public or private health insurance
- Mail
- Mass-marketing
- Ponzi scheme
- Pyramid scheme
- Securities fraud
- Wire
- Other (specify type of suspicious activity in space provided)

Check the FinCEN Suspicious Activity Report (FinCEN SAR) XML Schema 2.0 User Guide to learn more about US e-filing requirements.

Fraud analyst journeyâ

After finding evidence of suspicious activity, the fraud analyst journey to start a SAR is the following:

1. The analyst groups all the relevant events - both alerts and non-alerted events - into a case. The case is the starting point to create a SAR.
2. The analyst uses the cases' panel to access the created case and investigates the events linked to it:
3. Once the case is fully investigated and the analyst confirms that the events within the case pertain to illegal activity, the analyst marks the case as closed:
4. The previous action opens a pop-up window with the option to "Close with SAR":
5. After submitting the case as *Closed with SAR*, the relevant case is updated and displays the SAR Information widget:
6. Fraud analysts can now use the **Open SAR** button to start filling the report.

To learn more about how to submit and access created SARs, check the official SAR documentation.

About the Solution

Transaction Fraud for Banking (TFB) offers real-time fraud prevention to detect and stop risky transactions. This solution uses hypergranular profiles that encode the dynamic behavior of several dimensions and entities involved in each transaction. Such information will then become available to rules and machine learning models to score each transaction and make accept/decline decisions in the critical path.

TFB takes an all-round approach to fraud transaction scoring by evaluating transactions from the perspective of each individual leg (e.g. card authorizations, clearings, transfers initiations, settlements), as well as from the perspective of the end-to-end transaction. TFB provides a RESTful interface with real-time endpoints to ingest each event throughout the transaction life cycle, scores each event individually, and builds a cumulative view of the transaction as a whole.

Check the Life Cycle pages for more details about the role of each endpoint within the transaction life cycle.

TFB packages three use cases, namely Transaction Fraud for Cards (issuing), Transaction Fraud for Transfers, and Transaction Fraud for Instant Transfers, into a single risk management solution.

The following sections detail the specifics of each of these use cases.

Cards (issuing) use caseâ

The integration with the Cards API allows issuing banks to score both card present and card not-present transactions. This API is compliant with the international standard for financial transactions (**ISO 8583**) and contains additional fields that are critical for managing fraud efficiently (e.g. terminal latitude/longitude, device ID, and acceptor risk).

The Cards API supports any type of card payment including debit, credit or pre-paid cards. TFB covers the following card schemes:

- Mastercard
- VISA
- AMEX
- Discover
- UnionPay

Regarding fraud prevention, TFB covers the follow schemes associated with the Cards (issuing) use case:

- Carding (e.g. access to cloned cards)
- Card testing (e.g. bot attacks)
- Account takeover (ATO)
- Lost and stolen
- Card phishing
- Skimming
- Card-not-Present fraud (i.e. perpetrated by fraudsters or by bot attacks)

The solution pre-packages a set of rules that generate real-time recommendations to approve or decline transactions. Learn more about Cards' fraud scenarios and endpoint life cycles.

Inbound Payments use caseâ

The inbound payments use case enables the scoring of inbound events and the development of a dedicated risk strategy tailored specifically for inbound transactions.

This channel allows you to make decisions at an inbound event level, as well as investigate other inbound payment events within the transaction history widgets across all payment types.

Moreover, the Inbound Payments API follows the same definition as the Outbound Transfers API, thus ensuring a seamless integration process for clients. This API allows to segregate the strategy for inbound payments and leverage inbound payment data to make risk decisions not only on the legitimacy of received funds but also on any associated risks if these funds are subsequently used for outbound transactions.

Learn more about Inbound Payments' fraud scenarios and endpoint life cycles.

Outbound Transfers & Instant Transfers use casesâ

The main driver for the existence of two separate use cases - Outbound Transfers and Instant Transfers - is the structural difference regarding the timing for funds to settle (transfer cut-off time). In more traditional transfers like SEPA or WIRE, settlement windows are measured in business days. In modern, real-time payment types (like Faster Payments, SEPA Instant Credit, RTP, etc.) funds change hands instantaneously. This is a decisive difference because the operational workflow is different. As an example, it is possible to have a manual review of alerts in the critical path of allowing or stopping a transfer because there is enough time to make a final decision. That option is not available for instant payment types, and thus the manual revision is a post-mortem-like analysis.

On the other hand, Outbound Transfers and Instant Transfers also share transactional aspects. For example, both require information such as:

- The name of the bank to which an individual transfers funds.
- The account number of the transfer's receiver.
- The American Bankers Association routing number of the transfer's receiver bank.

Therefore, TFB tackles the two use cases with a single Outbound Transfers API. This API allows issuing banks to send information regarding traditional transfer rails (asynchronous settlement model) and real-time payments (instantaneous movement of funds). The Transfers API is compliant with the **ISO 20022** standard and provides out-of-the-box support for the following payment types:

Additionally, the TFB architecture is designed to promote gradual coverage (if needed) to more payment instruments as the business grows and the landscape of offered services and payment expands. Therefore, TFB can add support for new payments because the API fields are generic and the risk strategy is oriented to fraud scenarios (i.e. not specific to the above payment methods).

Regarding fraud prevention, TFB covers the following schemes associated with the Transfers & Instant Transfers use cases:

- Authorized push payment fraud
- ATO
- Phishing
- Money mule scams
- Fake invoice fraud
- Payroll scheme (ghost employee)
- Mandate fraud
- CEO fraud

These schemes are usually perpetrated through social engineering techniques, such as the Business Email Compromised fraud, which comprises CEO Fraud, Fake Invoice, or Payroll scheme. To effectively fight these fraud scenarios, the solution uses the transaction information and evaluates the origin of the transfer using the IP address, device, and browser information. TFB pre-packages a set of rules that generate real-time recommendations to approve or decline transactions.

Learn more about Transfers' fraud scenarios and endpoint life cycles.

Digital Activity use caseâ

The integration with the Digital Activity API allows financial institutions to detect abnormal digital activity behavior, thus building trust and reputation with their clients.

The Digital Activity connector is a plug-in component available out of the box in TFB. It provides a RESTful API that allows customers to send information at different stages of the digital activity's life cycle, including:

- Authentication
- Credential
- Device
- Profile updates
- New alias

The Digital Activity connector's main goal is to enable the injection of digital events happening in a banking session, such as login attempts, changes of profile data (e.g. email, phone number, address, password reset requests, etc). All these events allow Feedzai to create high-accurate profiles on the usual patterns of the customer (e.g. normal hours, locations, devices, etc) and use them to detect and prevent fraud scenarios such as:

- ATO
- Authorized push payment

Learn more about Digital Activity's fraud scenarios and endpoint life cycles.

Ad-hoc customer investigationsâ

The Ad-Hoc Customer Investigations API provides a mechanism for creating ad-hoc investigation points for customers initiated from the client's side, independent of Cards, Inbound Payments, or Outbound Transfers alerts. This functionality enables external systems to trigger investigations, facilitating proactive analysis and assessments for specific customers.

Learn more about Ad-hoc customer investigations' life cycles.

Device and behaviorâ

Digital Trust collects and analyzes thousands of parameters such as biometric, behavioral, device, IP, network data to create a customer profile in banking sessions. The Session Risk Score is a unique digital identifier that is created for each user. Digital Trust builds a Session Risk Score by collecting the userâs full context of biometric, behavioral, device, and network data points and continuously analyzing user behavior from login to logout across every interaction.

TFB offers a standard integration with Digital Trust's behavioral analysis and risk indicators, which enrich the solution's standard capabilities to assess risk on bank transfers and digital activity events. This solution combines the long-term and real-time metrics that TFB calculates (e.g. number of successful/failed logins in the last hour/days, number of distinct devices, usual locations, login hours, etc), and the profile augmentation that Digital Trust provides (e.g. Session Risk Score, Malware, Adware, RAT, Proxy, VPN indicators, etc).

End-to-end packageâ

The out-of-the-box capabilities of this solution for both card transactions and transfers are:

- **REST APIs:** Send information via REST API at different stages of the transaction life cycle.
- **Reference data:** Send account and cardholder information upfront via a standard API. This information is then used in the Feedzai scoring process. It also eliminates the need to bundle the information in the real-time authorization call itself.
- **Pre-configured risk engine:** Processes transactions, generates real-time accept/decline decision in the critical path, and generates review recommendation in use cases in which the settlement is non-instantaneous (e.g. Transfers). TFB pre-packages resources (e.g. plan, rules, machine learning, etc) to help you fight fraud from day 1.
- **Pre-configured case and alert management interface:** Review suspicious transactions that require manual decision. This interface includes a dashboard component to give you at-a-glance information about business and rules performance.

Next stepâ

Check How the Solution Works to get started with the product. For tailored guidance, you can check the Learning Paths page, which contains recommendations on how to access the documentation according to the user's role to get you up and running with the solution quickly.

Roles and Permissions

Was this page helpful?Â

Learn about the default roles and their associated permissions to understand how TFB manages the access within Pulse, Case Manager, RiskOps Studio, and dashboards.

Overviewâ

The following list summarizes the out-of-the-box roles in TFB with a high level information about the allowed actions for each role:

Fraud Prevention Agents - L1:

- Assign alerts to self to act on them (learn how to assign alerts to yourself in any of the Analyze Events pages)

- Mark up transactions
- Add and remove files to Alerts

Fraud Prevention Agents - L2:

- Reassign to self any alerts assigned to other agents (learn how to assign alerts to yourself in any of the Analyze Events pages)
- Mark up transactions
- Create cases in Case Manager
- Manage cases
- Add and remove files to Alerts and Cases

Fraud Prevention Manager:

- Reassign to self any alerts assigned to other agents (learn how to assign alerts to yourself in any of the Analyze Events pages)
- Act on alerts without assigning alerts to self
- The same as fraud prevention agents plus being able to override the agents' actions
- Access to dashboards
- Edit/Customize Case Manager's UI
- Manually close alerts
- Reveal detokenized card PANs
- Access and edit translation settings

Fraud Strategist:

- Maintain rules, lists, thresholds in Pulse, Case Manager and RiskOps Studio

Fraud Strategist Manager:

- The same as fraud strategists, plus being able to override the fraud strategists' actions
- Access to dashboards

Ad-Hoc Customer Investigations:

- Access to the Ad-Hoc Customer Investigations event page in Case Manager

Tenant Fraud Prevention Agents - L1:

- Mark up transactions
- Add and remove files to Alerts

Tenant Fraud Prevention Agents - L2:

- Mark up transactions
- Add and remove files to Alerts and Cases
- Create cases in Case Manager
- Manage cases

Tenant Fraud Prevention Manager:

- Mark up transactions
- Add and remove files to Alerts and Cases
- Create cases in Case Manager
- Manage cases
- Override L1 and L2 agents' actions
- Manually close alerts
- Reveal detokenized card PANs

multi-tenancy

The aforementioned **tenant** roles are only relevant if you're using the multi-tenancy add-on.

Monitoring and Auditing:

- Quality Monitoring (i.e. QA) - users with the responsibility for selecting a sample of alerts per agent on a given month and verify the accuracy of the investigations (i.e. typically an internal auditor)
- Read-only access to alerts, cases, and rules for users with a reporting responsibility
- Access to dashboards

Integrator:

- Manage API JWKs

caution

If the multi-tenancy add-on is activated, users with any of the following roles must be added to the `tfrb_landlord_resources` group and be granted the landlord role, when creating a user:

- Fraud Prevention Agents - L1
- Fraud Prevention Agents - L2
- Fraud Prevention Manager
- Fraud Strategist
- Fraud Strategist Manager
- Ad-Hoc Customer Investigations
- Monitoring and Auditing
- Integrator

Detailed viewâ

The following table shows a detailed view of each role's permissions.

Pulse permissionsâ

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	Int
4-Eye Check - Approve / Review					X						
4-Eye Check - Submit / Review				X							
Manage Application			X	X	X						
Manage Custom Services			X	X	X						
Manage Data Science			X	X	X				X		
Manage History			X	X	X					X	
Manage Jobs			X	X	X						
Manage JWK											X
Manage Lists	X	X	X	X	X		X	X	X		
Manage Outcome Combinators			X	X	X						
Manage Sandbox Evaluations			X	X	X						
Manage Workflows			X	X	X						
Read Application			X	X	X					X	X
Read Custom Services			X	X	X						X
Read Data Science			X	X	X			X	X	X	X
Read History			X	X	X					X	X
Read Jobs			X	X	X						
Read Lists	X	X	X	X	X		X	X	X	X	
Read Outcome Combinators			X	X	X						X
Read Sandbox Evaluations			X	X	X						

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	Int
Read Users											X
Read Workflows			X	X	X			X	X	X	X
Manage Users			X		X						

Case Manager permissionsâ

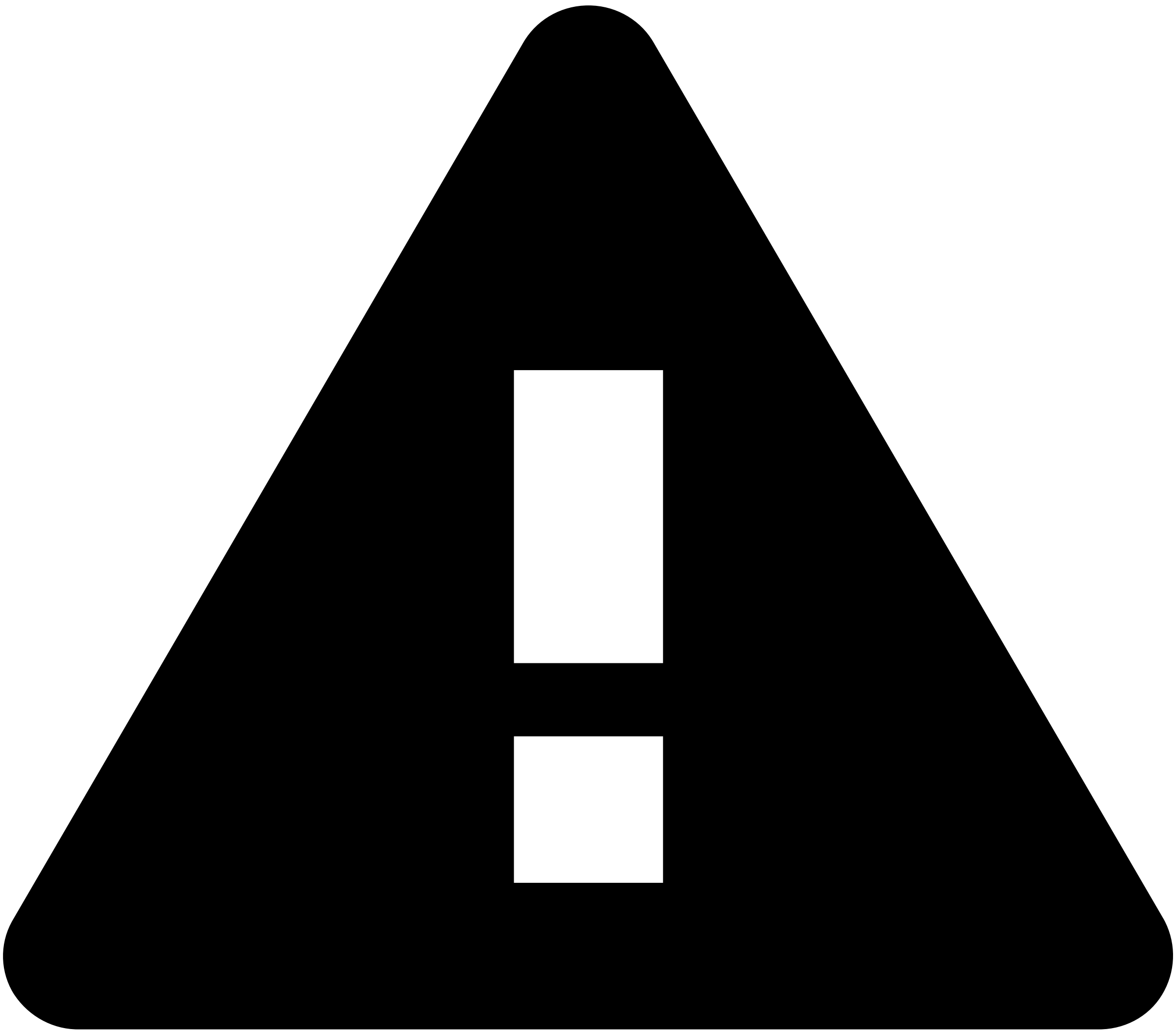
Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	In
Access to Ad-Hoc Customer Investigations event page						X					
Assign and Lock		X						X			
Manage Assign and Lock			X						X		
Assign and Lock No Steal	X						X				
Close Alert			X						X		
Configure GUI			X								
Entity Files Manage		X			X						
Manage Event Files	X	X	X				X	X	X		
Event Files Read	X	X	X	X	X		X	X	X	X	
Manage Automation Rules			X								
Manage Case Sources			X								
Manage Cases		X	X					X	X		
Manage Entity Notes	X	X	X				X	X	X		
	X	X	X				X	X	X		

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	In
Manage Events											
Manage List Items			X	X	X				X		
Manage Queues			X								
Manage Reports			X						X	X	
Manage SLAs			X								
Manage SSRs			X	X	X				X		
Manage Status Reasons			X								
Manage Translations			X								
Manage User Queues	X	X	X				X	X	X		
Manage Reasons			X								
Reveal Encrypted Values		X	X					X	X		
Test SSRs		X	X	X				X	X		
Update Lists	X	X	X	X	X		X	X	X		
Read Automation Rules			X								
Read Case Sources	X	X	X		X		X	X	X	X	
Read Cases		X	X		X			X	X	X	
Read Customers	X	X	X	X	X		X	X	X	X	
Read Dashboards			X		X					X	
Read Entity Notes				X	X					X	X
Read Event Notes				X	X					X	
Read Events	X	X	X	X	X		X	X	X	X	
Read Queues			X							X	
Read Reports			X						X	X	

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	In
Read SLAs			X							X	
Read SSRs		X	X	X	X			X	X	X	

RiskOps Studio permissionsâ

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	Int
Monitor rules			X	X	X					X	



pulse permissions

The following permissions are accessible via the Pulse base URL `/pulseviews/management/#admin/security`:

- Read users
- Manage users

Custom Roles

If you need a set of roles and permissions different from the out-of-the-box roles listed in this page, reach out to your Feedzai contact point to have custom roles created.

roles-and-permissions-table

Pulse permissionsâ

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	Int
4-Eye Check - Approve / Review					X						
4-Eye Check - Submit / Review				X							
Manage Application			X	X	X						
Manage Custom Services			X	X	X						
Manage Data Science			X	X	X				X		
Manage History			X	X	X					X	
Manage Jobs			X	X	X						
Manage JWK											X
Manage Lists	X	X	X	X	X		X	X	X		
Manage Outcome Combinators			X	X	X						
Manage Sandbox Evaluations			X	X	X						
Manage Workflows			X	X	X						
Read Application			X	X	X					X	X
			X	X	X						X

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	Int
Read Custom Services											
Read Data Science			X	X	X			X	X	X	X
Read History			X	X	X					X	X
Read Jobs			X	X	X						
Read Lists	X	X	X	X	X		X	X	X	X	
Read Outcome Combinators			X	X	X						X
Read Sandbox Evaluations			X	X	X						
Read Users											X
Read Workflows			X	X	X			X	X	X	X
Manage Users			X		X						

Case Manager permissionsâ

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	In
Access to Ad-Hoc Customer Investigations event page						X					
Assign and Lock		X						X			
Manage Assign and Lock			X						X		
Assign and Lock No Steal	X						X				
Close Alert			X						X		
Configure GUI			X								
Entity Files Manage		X			X						
	X	X	X				X	X	X		

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	In
Manage Event Files											
Event Files Read	X	X	X	X	X		X	X	X	X	
Manage Automation Rules			X								
Manage Case Sources			X								
Manage Cases		X	X					X	X		
Manage Entity Notes	X	X	X				X	X	X		
Manage Events	X	X	X				X	X	X		
Manage List Items			X	X	X				X		
Manage Queues			X								
Manage Reports			X						X	X	
Manage SLAs			X								
Manage SSRs			X	X	X				X		
Manage Status Reasons			X								
Manage Translations			X								
Manage User Queues	X	X	X				X	X	X		
Manage Reasons			X								
Reveal Encrypted Values		X	X					X	X		
Test SSRs		X	X	X				X	X		
Update Lists	X	X	X	X	X		X	X	X		
Read Automation Rules			X								
Read Case Sources	X	X	X		X		X	X	X	X	
Read Cases		X	X		X			X	X	X	

Permissions	Fraud Prevention Agents - L1	Fraud Prevention Agents - L2	Fraud Prevention Manager	Fraud Strategist	Fraud Strategist Manager	Ad-Hoc Customer Investigations	Tenant Fraud Prevention Agents - L1	Tenant Fraud Prevention Agents - L2	Tenant Fraud Prevention Manager	Monitoring and Auditing	In
Read Customers	X	X	X	X	X		X	X	X	X	
Read Dashboards			X		X					X	
Read Entity Notes				X	X					X	X
Read Event Notes				X	X					X	
Read Events	X	X	X	X	X		X	X	X	X	
Read Queues			X							X	
Read Reports			X						X	X	
Read SLAs			X							X	
Read SSRs		X	X	X	X			X	X	X	

- Operational Work
- Data Scientist & Fraud Strategist
- Understand the Data Science Framework (DSF)

On this page

Understand the Data Science Framework (DSF)

Was this page helpful?Â

The Data Science Framework (DSF), also known as the Data Science Studio, is a dedicated environment used in risk assessment strategies for building, testing, and evaluating rules and models according to the Data Science Loop:

Its main capabilities include:

- **Data preparation**: Read production data or ingest historical data, clean it, and analyze its quality and statistics to ensure it fits your schema. This is crucial to work with Rules, Models and/or ad-hoc analysis.
- **Feature engineering and plan execution**: Create and test new features using Plans, which can then be deployed to production. Plans allow the application of generic data computations, typically to transform it or extend it to a new file output.

- **Model and rule development**: Train Machine Learning (ML) models and create rule sets, and evaluate their performance against historical data to identify if further improvements are needed.
- **Ad-hoc analysis**: Study and request ad-hoc reports or analysis via the DSF.
- **Sandbox Evaluations**: Test a sequence of a Model and Rule project in the Data Science application.

What are Data Science jobs?â

Data Science (DS) jobs are jobs initiated in the DSF environment, either via Pulse or JupyterLab, and processed through the Pulse application. While Pulse manages the job lifecycle, the heavy computational workload is shifted to the cluster, which allocates **executors** (i.e. processing units) throughout its **task nodes** (i.e. the actual machines that compose the cluster) to perform the computations as a Spark job.

Refer to Spark's documentation to further understand its core functionality.

The resources allocated to each job are guided by the job configuration and workload, and bounded by the cluster configuration (i.e. size and number of task nodes, as well as the executors per node).

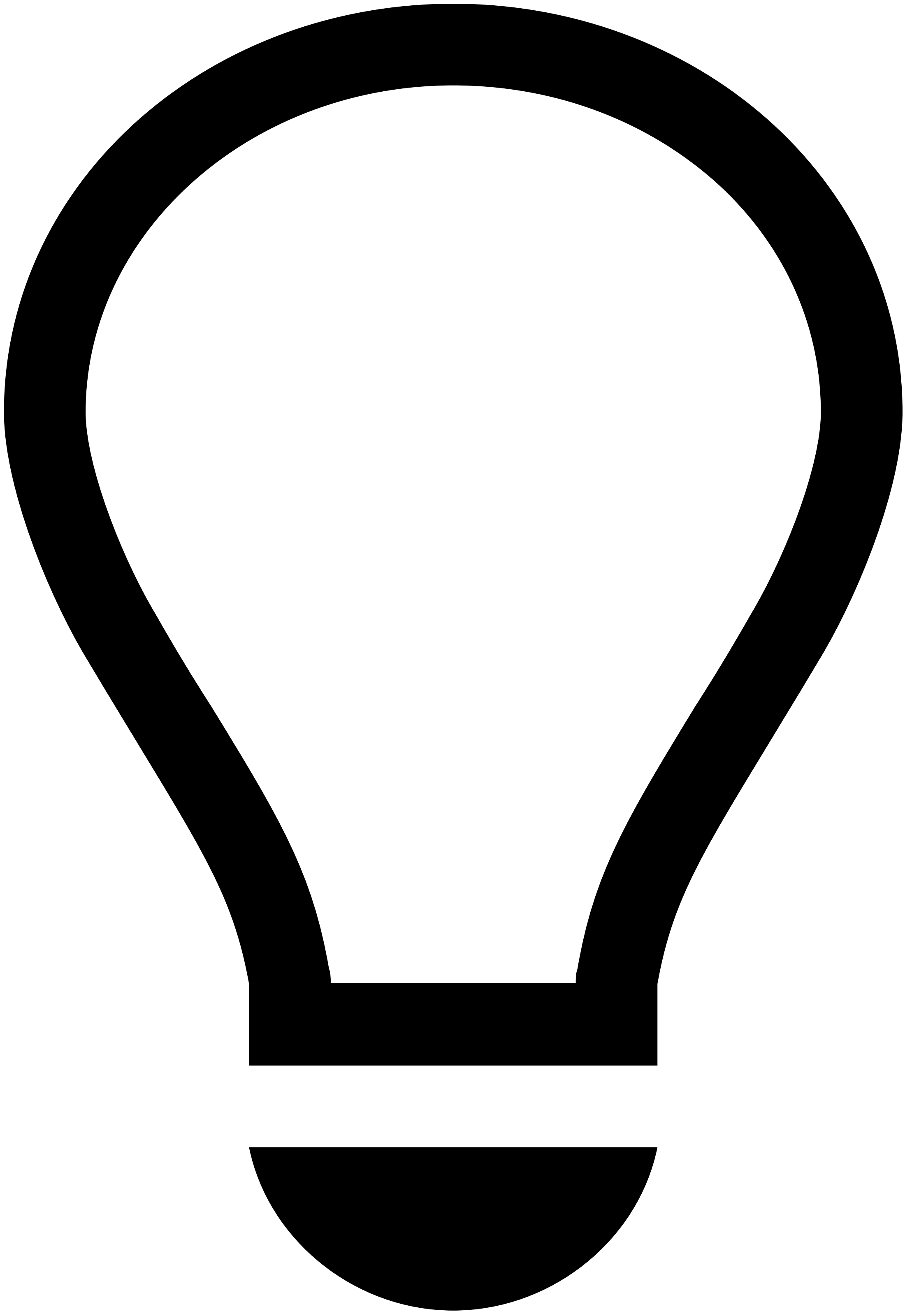
Pulse provides one default job profile that applies to all DS jobs, but it is recommended that you adapt this default profile to better fit the jobs you'll be executing. The default job profile is, essentially, a starting point to create other profiles.

What are executors?â

Executors represent the **main processing unit** that do the actual work in your job (i.e. reading data, transforming it, caching it, and writing results).

When a job is launched/scheduled, the cluster allocates executors on the task nodes (i.e. machines). These executors borrow resources (i.e. cores and memory) from the task nodes to perform the computations and execute DS jobs.

When sizing the executors, be mindful of this to avoid setting unrealistic bounds on the amount and/or size of your executors.



Practical example

If you consider a task node with 16 cores and 128GB of memory, configuring executors to have nine cores and 50GB of memory would represent poor cluster resource usage, which would, ultimately, lead to poor performance. Here's why:

After allocating one executor (with nine cores), the cluster is left with seven available. As each executor borrows nine cores, the cluster is not able to allocate any more executors for that job.
This way, it would only be possible to fit one executor, leaving seven cores unutilized.

Note that the remaining seven cores could still be used by other jobs, as long as the executors fit in the task nodes.

Dynamic executor allocation

Note that while you can explicitly define how many executors to run, the **general rule-of-thumb favors dynamic allocation of executors**. This lets the cluster decide how many executors to run depending on the workload.

Why tune DS jobs?â

Tuning DS jobs is the process of optimizing Spark configurations used by the cluster to handle specific workloads efficiently. This helps prevent:

- Executions from hanging due to the lack of requested resources.
- Executions crashing due to cheap job configurations.
- Resource starvation caused by greedy job configurations.

tip

Choosing the appropriate configuration is **trade-off between whatâ□□s available and whatâ□□s needed**.

Note that there isnâ□□t a "one-size-fits-all" configuration. Heavier jobs will take advantage of more generous configurations, while lighter jobs will perform well with more sensible ones.

Additionally, clusters are shared by all team members, so guaranteeing that everyone is mindful of their job configurations is key to avoid concurrency issues and leverage the resources appropriately.

Jobs can be tuned either within Pulse (using Job Profiles) or in JupyterLab (by providing the Spark configuration when opening a session).

Impact of tuning in the DS job life cycleâ

To better understand how job configurations affect the life cycle of a DS job, it is important to consider the scaling capacity and how resources are allocated and shared.

Scaling capacityâ

Scaling refers to the cluster's capacity to add and remove task nodes from the pool of resources at any time depending on workload. It occurs automatically based on demand given by the number of executors that are waiting to be allocated.

The scale-up strategy follows a tiered approach: the higher the demand (i.e. jobs pending/waiting for allocation), the more capacity is allocated. These tiers can be classified based on how fast more capacity is added (e.g., light, moderate, high, very high).

Capacity limits

There is an upper bound to the maximum capacity attributed to you based on your contract with Feedzai.

Resource allocation and sharingâ

Choosing adequate capacity for a job is important to allow for optimal execution times and sharing that capacity with other people in the team. To better understand how resources are allocated and shared within the cluster, it's important to know how jobs allocate capacity and what it means to have running Spark sessions.

How do jobs allocate capacity?â

When you submit a Pulse job or open a Spark session, the EMR (Elastic MapReduce) cluster registers the request and allocates resources dynamically depending on the workload. This means that if you request a job with a minimum of 1 and a maximum of 8 executors, the job begins executing as soon as the cluster is able to allocate 1 executor. If during the job execution more cluster resources become available (e.g., because other jobs have finished or the cluster has scaled up), the cluster allocates additional executors, up to a maximum of 8.

info

Note that executors are main processing units for jobs and are indivisible. This means that while executors for the same job can potentially â□□liveâ□□in a different cluster node, each executor alone must fit entirely in a given cluster node.

Refer to the What are executors? section for more details on these processing units.

On the other hand, if dynamic allocation is disabled, the job only begins executing when the cluster is able to allocate all the executors requested in the job configurations. For example, if you request a job with 8 executors (each with 12GB and 4 cores), the job will start once the cluster is able to fit those 8 executors within the pool of cluster resources available at that moment.

Refer to Applying configurations in JupyterLab and Applying configurations in Pulse for details on how to configure jobs/sessions, respectively.

warning

Disabling dynamic allocation is only recommended when debugging issues with the cluster.

Depending on the number of executors being requested by all the jobs registered in the cluster at a given moment (including jobs waiting for resources to begin execution, and dynamic allocation jobs already executing and requesting more executors), the cluster can scale up and add more nodes to its pool of resources.

In contrast, if the cluster resources are idle for a certain amount of time (e.g., because there are few jobs running) it can scale down and remove cluster nodes from the pool of resources.

What does this mean?

Both the cluster resource usage (i.e. all jobs from all users using the cluster at the same time) and the cluster total pool of resources (i.e. if the cluster is scaled up or scaled down) have an impact on the duration of your job. Therefore, the exact same job might take a different amount of time to complete on separate occasions.

When you submit your job, you can check its status (i.e. if it's running or waiting for resources) in the **YARN Web UI**. There, you can also find the total and available pool of cluster resources. These metrics can help you understand why your job hasnâ□□t started yet (e.g., if there isn't enough memory or cores to satisfy your job configurations).

You can also use the **YARN Web UI** to check the resource usage of your running jobs to make sure they aren't exhausting most of the cluster resources and adhere to fair cluster usage.

What does it mean to have running Spark sessions?â

Spark sessions are created in JupyterLab and represent the connection between your notebook and the cluster. Refer to Pulse's documentation to learn about Interacting with Pulse via JupyterLab, including a list of available notebooks.

Note that once created, Spark sessions remain active until you either:

- Close them using `spark_session.stop()`.
  OR
- Kill the notebook kernel.

warning

Be mindful of your active Spark sessions, and make sure you manually stop them if they're no longer needed. Hanging (idle) sessions can occupy cluster resources and impact the other cluster users' experience.

In some cases, Spark sessions that are idle for a long period are automatically closed, but this depends on the cluster's configuration.

Common scenariosâ

Here are some scenarios where tuning Spark configurations is most commonly needed and what changes they require:

info

The following table provides guidance in terms of how different parameters could be configured, but choosing the appropriate configuration is **trade-off between whatâs available and whatâs needed**. Be mindful of job configurations to avoid concurrency issues and leverage the resources appropriately.

There isnât a "one-size-fits-all" configuration. Heavier jobs will take advantage of more generous configurations, while lighter jobs can be executed in an acceptable timeframe with less demanding configurations, and without compromising other user's jobs.

Scenario	Considerations
Dataset analysis and rule evaluations	This scenario doesn't require you to limit the number of executors.

Scenario	Considerations
Model training	This is a single-task/executor job, so: • The number of cores per executor equals the number of cores per task. • The rule-of-thumb for job memory and cores per executor is 1/4 of the task node size for both. This allows for running four models in parallel in just one task node, which complies with the default HPT notebooks (that launch four model training jobs simultaneously). note Even though model training jobs are single-task/executor jobs, you must **allocate more than one CPU per task using `spark.task.cpus`**. This is set by default to `1`. However, if you don't override this default, you're essentially running a job with X executors but only `1` will be used (which is a waste of resources). Since model training is a single-task/executor job, **make sure the task gets all the executor cores** by matching the `spark.task.cpus` to the `spark.executors.cores`. Essentially, this allows you to increase the number of cores per executor in order to leverage model training multi-threading (as long as that property remains aligned with the number of CPUs per task).
Plan executions	In this scenario: • Memory-heavy jobs benefit from parallelization. • Both the cores and memory per executor should be maximized.
Data exploration/ manipulation	In this scenario: • The necessary configurations vary depending on the data and type of exploration/manipulation that will be performed.

Scenario	Considerations
	• A mixture of approaches is recommended.

Applying configurations in JupyterLabâ

When creating Spark sessions via JupyterLab (e.g., when using notebooks), you need to provide the Spark configuration at the beginning:

```
# Spark Session Parameters
SPARK_PARAMETERS = {
    "spark.executor.cores": "7",
    "spark.executor.memory": "8g",
    "spark.executor.memoryOverhead": "5g",
    "spark.dynamicAllocation.enabled": True,
    "spark.dynamicAllocation.initialExecutors": "2",
    "spark.dynamicAllocation.minExecutors": "1",
    "spark.dynamicAllocation.maxExecutors": "6",
    "spark.dynamicAllocation.executorIdleTimeout": "300s",
}

# Get SparkSession with default Pulse's configuration
spark = feedzai.get_spark_session(SPARK_PARAMETERS)
```

The configuration examples below can be used to configure your Spark jobs:

note

Ensure you replace the values for each as needed.

Configuration aspect	Example

Cores per executor	`SPARK_PARAMETERS = {`
	`    "spark.executor.cores": "7",`
	`}`

Configuration aspect	Example

Memory per executor	```SPARK_PARAMETERS = {
	    "spark.executor.memory": "8g",
	    "spark.executor.memoryOverhead": "5g",
	}```

Dynamic allocation	```SPARK_PARAMETERS = {
	    "spark.dynamicAllocation.enabled": True,
	    "spark.dynamicAllocation.initialExecutors": "2",
	    "spark.dynamicAllocation.minExecutors": "1",
	    "spark.dynamicAllocation.maxExecutors": "6",
	    "spark.dynamicAllocation.executorIdleTimeout": "300s",
	}```

Validating Spark session configurationsâ

Once you've completed the configuration and before running a full job, verify jobs are behaving as expected. This helps spot issues early on, before committing to executing real jobs.

To do this, run the following test job:

```
spark.range(100_000_000).count()
```

Then, via the **YARN Web UI**:

- Confirm the Spark session is in the list of **Applications**.
- Monitor the Spark session **State** (running sessions should have `State = RUNNING`, while sessions that are waiting for resources should have `State = ACCEPTED`).

Additionally, check the **Spark History Server** to confirm that your Spark session configurations were correctly set. This can be accessed in the full list of configurations via the **Environment** tab.

Applying configurations in Pulseâ

For DS jobs triggered in Pulse (e.g., plan executions, model training, rule evaluations, etc.), you can use Job Profiles to provide a Spark configuration.

Refer to Pulse's documentation to read all about Tuning Job executions with Job Profiles and learn how to:

- Create a Job Profile to apply a set of property configurations when running jobs.
- Select the profile that is more suitable for the Spark job that you want to run.
- Edit Spark properties on demand to control the configuration of the Spark job that performs the calculations for a specific Data Science resource.

Jobs triggered via the DS-API

You can also trigger Pulse jobs from JupyterLab using the **DS-API**. In these cases, the job spark configurations are managed through a combination of:

- Parameters sent through the DS-API method calls.
- Pulse Job Profiles (**not** through `SPARK_PARAMETERS`).

As an example, refer to the plan execution DS-API method documentation (see the `job_profile` and `job_properties` arguments).

Fraud Prevention Agent

Investigate alerts

File SAR

- Operational Work
- Fraud Prevention Manager

Fraud Prevention Manager

Was this page helpful?Â

Manage workload and rules

View analytics dashboards

Data Scientist & Fraud Strategist

Was this page helpful?Â

Understand the Data Science Framework (DSF)

Work with rules

Use models

Migrate resources between environments

How the Solution Works

Imagine that a cardholder performs a payment within a given merchant. The issuer bank sends this transaction to Feedzai which will respond with a real-time recommendation on whether the transaction is legitimate or fraudulent:

Read below how this recommendation is generated.

1. Feedzai's APIâ

Using the example above, the life cycle initiates when an issuer bank sends data to Feedzai via RESTful API. In this case, the transaction is sent via the cards API, typically using the Authorization endpoint. You can also check the Life Cycles to learn more about how the API works, the available endpoints, and how they play a role within the transaction life cycle.

The transaction is then processed by the Feedzai's risk engine - Pulse - which will generate a recommendation. This processing is performed in real-time by a runtime workflow, but all the resources are pre-formed beforehand. If the event is identified as suspicious, Pulse generates an alert, which is then analyzed manually by fraud analysts.

2. Enrichmentâ

TFB allows the enrichment of events in real time with extra data points that can be used for improved fraud detection. The enrichment of the events can be done through the Reference Data connector or by API integrations.

The API enrichment can be done by integrating external providers. TFB includes a standard integration with Feedzai's Digital Trust. Digital Trust offers a session analysis that extracts thousands of parameters such as biometric, behavioral, device, IP, network data, etc, throughout the session. These are data points that are incorporated into the events in session stages (e.g. login, profile update, transfer initiation) to monitor and score different transfer initiations and/or digital events.

The following diagram shows an example of a session life cycle and the different data points that can be identified on each stage.

1. Login from an unknown device and VPN connection detected.
2. Customer phone number changed with network anomalies detected.
3. RAT detected when a monetary transfer initiated.

For more information on how to detect fraud with these data points, check the Digital Trust Integration page.

3. Rules and data science work in Pulseâ

The user interaction with Pulse is based on two fundamental concepts:

- **Resource configuration:** The resources (i.e. schemas, plans, models, rules, TrustScore, and lists) are developed beforehand and only then defined in a runtime workflow.
- **Runtime workflow:** Represents what Pulse executes in runtime.

The configuration setup comprises the definition of the resources that are then used in runtime. To understand how Pulse works, the most important concepts are schemas, plans, rules, TrustScore, and lists.

Schemasâ

Describe the available data fields of a data source, as well as their types. Firstly, they are used to map the fields sent via the RESTful API into a data structure that can be handled by Pulse. Any field that is added to complement the API schema, for example, new metrics such as the amount spent over the last hour, needs to be included in the schema definition. The solution comes pre-packaged with the following schemas:

Check the Pulse Schema page to understand the logic of each pre-packaged schema.

Plansâ

Data Science Plans are a very powerful and generic tool: they are essentially functions that receive data and output a transformation of that data (not necessarily with equal size). TFB includes the following out-of-the-box plans:

Cards, Transfers and Digital Activity plans are used to create derived fields based on the cards, transfers and digital activity API information, respectively. These fields generally comprise metrics such as sums and counts of a given field over a given period. The fields are saved into a new schema named *Workflow Schema*, which is then used as input for rules.

tip

Omnichannel metrics: The Digital Activity metrics can also be used by the Transfers channel. Transfers rules use digital activity metrics to detect fraud scenarios such as Authorized Push Payment (APP) and Account Takeover (ATO). This is possible via Pulse omnichannel, which allows metrics to be built based on events from multiple workflows, such as Transfers.

Check the [Pulse Schemas] page and see the out-of-the-box metrics.

TrustScore

TrustScore is an intelligent scoring system designed to identify and flag potentially fraudulent activity. It's not a single, static model, but rather a sophisticated combination of models that analyze a vast array of data points to provide a holistic view of risk.

TrustScore leverages the federated learning method. This means that instead of centralizing all client data, which can raise privacy concerns, TrustScore uses a federated learning approach to provide a robust, intelligent, and privacy-preserving score.

This allows Feedzai to build expert models that learn from a diverse range of fraud patterns across its entire client base, without ever directly accessing or mixing sensitive transaction data with other clients' data.

Customer data remains strictly segregated and secure within Feedzai's infrastructure. Feedzai builds and trains these models within its machine learning infrastructure to ensure the highest level of data protection.

When a new transaction or event occurs in a client's production environment, it is simultaneously analyzed by all these specialized expert models.

The individual scores from each model are then combined to produce the final TrustScore. This combined score provides a highly accurate assessment of the fraud risk associated with that particular event.

info

Feedzai provides TrustScore out of the box to score all events from **Cards** and **Outbound Transfers**.

TrustScore's `fraud_score` field is present in *Cards Workflow Schema* and *Transfers Workflow Schema*, respectively.

Clients' scoring models and TrustScore

Clients using their own models should validate how to best combine their scoring capabilities with TrustScore within their workflow.

For instance, TrustScore may serve as an additional input to influence a client's existing scoring model.

Learn how to use TrustScore's `fraud_score` field in rules.

Merchant TrustSignalsâ

info

This is an add-on feature. Please contact Feedzai for more information and activation.

Merchant TrustSignals are enriched fields that assess the risk profile of individual merchants within the portfolio of a client by analyzing the merchant's behavior (e.g. number of fraudulent transactions and fraud rate). This collective intelligence helps identify potentially risky merchants and take informed action, while ensuring complete data privacy.

Here's the end-to-end process by which Feedzai enriches TrustSignals fields. You don't need to do anything on your end; once this add-on is activated, the TrustSignals fields are automatically ready for you to use in your rules and models.

note

During the activation process, you might see your production workflows in Pulse being edited. This is a required step perfomed by Feedzai to enable TrustSignals and the workflow will continue behaving as expected after it's published.

1. Feedzai aggregates huge amounts of transactional data across its secure, anonymized client community. This comprehensive dataset allows Feedzai to extract powerful signals tailored to identify patterns indicative of merchant risk. All data from Feedzai's clients remains segregated to ensure complete privacy preservation.
2. Feedzai then pre-calculates and aggregates the risk score based on attributes associated with each merchant. These are derived from observed patterns within the Feedzai community, and help pinpoint risk based on:
    - First Seen Date
    - Churn Rank (Lifecycle Status)
    - Merchant Risk Score
    - Fraud Volume
    - Fraud Amount
    - Transaction Volume
    - Transaction Amount
    - Fraud Volume Rate

    - Fraud Amount Rate
    - Approval Rate
    - Approval Value Rate
3. This intelligence integrates into each client's risk strategy. Merchant TrustSignals are sent via API to Pulse, allowing clients to leverage these real-time insights as payments come in. Once activated, there is no additional integration or technical resources needed to deploy these signals into a client's existing Feedzai environment.

tip

Once activated, Merchant TrustSignals enrich the risk strategy with no additional integration or technical resources required, and are displayed in the **Rules Triggered** widget for each event (if available).

Available Merchant TrustSignals and required fieldsâ

The following TrustSignals are available to use in Pulse [rules]() and require the **merchant's name** to be present during activation:

TrustSignal	Pulse variable	Example		
		Variable level	Case Manager description	Range
First Seen Date: First time the merchant was identified in the network.	`mts_time_since_rank`	3	Level 4 of 4 (> 1 year)	> 1 year
		2	Level 3 of 4 (6 - 12 months)	6 - 12 months
		1	Level 2 of 4 (3 - 6 months)	3 - 6 months
		0	Level 1 of 4 (< 3 months)	< 3 months
Churn Rank (Lifecycle Status): Lifecycle status based on the number of days since the merchant was last seen (e.g. last transaction). It uses 30-day and 45-day thresholds to assign one of four lifecycle status: No data/Null, Active, At risk or Churn."	`mts_churn_rank`	3	Level 4 of 4 (Churn)	> 45 days
		2	Level 3 of 4 (At risk)	30 - 45 days
		1	Level 2 of 4 (Active)	< 30 days
		0	Level 1 of 4 (No data)	No data / NULL
Merchant Risk Score: A rule-based risk category (1â4) combining the Fraud Volume, Churn Rank (Lifecycle Status), and Approval Rate. Possible outputs are: Low, Medium, High, or Critical.	`mts_merchant_risk_score_rank`	4	Level 4 of 4 (Critical)	Critical
		3	Level 3 of 4 (High)	High
		2	Level 2 of 4 (Medium)	Medium
		1	Level 1 of 4 (Low)	Low
Fraud Volume: Number of suspected fraudulent transactions since the First Seen Date.	`mts_fraud_count_rank`	5	Level 6 of 6 (>= 500)	>= 500
		4	Level 5 of 6 (250-499)	250-499
		3	Level 4 of 6 (100-249)	100-249
		2	Level 3 of 6 (70-99)	70-99
		1	Level 2 of 6 (5-70)	5-70
		0	Level 1 of 6 (< 5)	< 5
Fraud Amount: Sum of the amounts from suspected fraudulent transactions since the First Seen Date.	`mts_fraud_value_rank`	4	Level 5 of 5 (>= $42,809.72)	>= $42,809.72
		3	Level 4 of 5 ($4,416.74 - $42,809.72)	$4,416.74 - $42,809.72
		2	Level 3 of 5 ($1922.07 - $4,416.74)	$1922.07 - $4,416.74
		1		$174.73 - $1922.07

TrustSignal	Pulse variable	Example		
		Variable level	Case Manager description	Range
			Level 2 of 5 ($174.73 - $1922.07)	
		0	Level 1 of 5 (< $174.73)	< $174.73
Transaction Volume: Number of transactions from the merchant since the First Seen Date.	`mts_ transaction_count_rank`	4	Level 5 of 5 (>= 6.61K)	>= 6.61K
		3	Level 4 of 5 (2.45K - 6.61K)	2.45K - 6.61K
		2	Level 3 of 5 (695 - 2.45K)	695 - 2.45K
		1	Level 3 of 5 (34 - 694)	34 - 694
		0	Level 1 of 5 (< 34)	< 34
Transaction Amount: Sum of transaction amounts from the merchant since the First Seen Date.	`mts_transaction_value_rank`	4	Level 5 of 5 (>= $106,412.72)	>= $106,412.72
		3	Level 4 of 5 ($47,178.05 - $106,412.72)	$47,178.05 - $106,412.72
		2	Level 3 of 5 ($15,376.55 - $47,178.05)	$15,376.55 - $47,178.05
		1	Level 2 of 5 ($873.76 - $15,376.55)	$873.76 - $15,376.55
		0	Level 1 of 5 (< $873.76)	< $873.76
Fraud Volume Rate: Percentage that the number of fraudulent transactions represents in relation to the merchant's total number of transactions, since the First Seen Date.	`mts_fraud_rate_rank`	5	Level 6 of 6 (>= 20%)	>= 20%
		4	Level 5 of 6 (10% - 20%)	10% - 20%
		3	Level 4 of 6 (5% - 10%)	5% - 10%
		2	Level 3 of 6 (1.8% - 5%)	1.8% - 5%
		1	Level 2 of 6 (0.9% - 1.8%)	0.9% - 1.8%
		0	Level 1 of 6 (< 0.9%)	< 0.9%
Fraud Amount Rate: Percentage that the total amount from fraudulent transactions represents in relation to the merchant's total transactioned amount, since the First Seen Date.	`mts_fraud_value_rate_rank`	5	Level 6 of 6 (>= 20%)	>= 20%
		4	Level 5 of 6 (10% - 20%)	10% - 20%
		3	Level 4 of 6 (5% - 10%)	5% - 10%
		2	Level 3 of 6 (1.8% to 5%)	1.8% to 5%

TrustSignal	Pulse variable	Example		
		Variable level	Case Manager description	Range
		1	Level 2 of 6 (0.9% to 1.8%)	0.9% to 1.8%
		0	Level 1 of 6 (< 0.9%)	< 0.9%
Approval Rate: Percentage that the number of approved transactions represents in relation to the merchant's total number of transactions, since the First Seen Date.	`mts_approval_rate_rank`	5	Level 6 of 6 (>= 80%)	>= 80%
		4	Level 5 of 6 (60% - 80%)	60% - 80%
		3	Level 4 of 6 (40% - 60%)	40% - 60%
		2	Level 3 of 6 (20% - 40%)	20% - 40%
		1	Level 2 of 6 (< 20%)	< 20%
		0	Level 1 of 6 (No transactions)	No transactions
Approval Value Rate: Percentage that the total amount from approved transactions represents in relation to the merchant's total transactioned amount, since the First Seen Date.	`mts_approval_value_rate_rank`	5	Level 6 of 6 (>= 80%)	>= 80%
		4	Level 5 of 6 (60% - 80%)	60% - 80%
		3	Level 4 of 6 (40% - 60%)	40% - 60%
		2	Level 3 of 6 (20% - 40%)	20% - 40%
		1	Level 2 of 6 (< 20%)	< 20%
		0	Level 1 of 6 (No transactions)	No transactions

info

The **Pulse variable** can be used in rules in Pulse, alongside the corresponding **Variable levels**. These refer to the codified value for each of the levels identified in the **Case Manager description**. The CM descriptions are displayed purely for information purposes in the **Rules Triggered** widget.

The following example references the `mts_fraud_count_rank` variable level **4** (equivalent to *Level 5 of 6 (250-499)*) and `mts_transaction_count_rank` variable level **0** (equivalent to *Level 1 of 5 (< 34)*) in a rule that monitors the **Fraud Volume** and **Transaction Volume**.

Rulesâ

A rule defines a condition based on the fields of a data source that produces a binary result for each record that matches the condition. In Pulse, rules are organized into Rules Projects:

The following image shows an example of a pre-packaged rule:

Check the Rules Projects topic for a full list of pre-packaged rules. This page contains conceptual information on how rules are organized in Pulse, naming convention, and nomenclature used in this manual. The Rules Projects pages list all rules within each rules project and provide relevant information, such as description, actions, conditions, and outcomes.

Runtime workflowâ

Workflows represent what Pulse executes in runtime, i.e. contains the resources into a sequence of processing steps that produce a decision in real time. TFB comes with three workflows to process card payments, transfers, and digital activity. The following diagram refers to the cards workflow:

The flow is as follows:

1. The transaction is received by Pulse through the Input Adaptor. At this point, the data is mapped to the input schema.
2. Pulse augments the input schema by calculating metrics defined by the plan using the Plan service. The data structure that holds the new information is the *Workflow Schema*, which is used as input for the rules.
3. The event then flows through the rules which generate outcomes to whether the event is deemed as legitimate or fraudulent. If you want to know more about how the outcomes are generated and which ones prevail, check the Understand How Rules Work page.
4. The *Decision Rules* operator sets up the final outcome as decline, accept or review. If the outcome is set as review, the event then goes into Case Manager for manual revision.

4. Alert revision in Case Managerâ

When a rule with the action Alert is triggered (check the Understand How Rules Work page for more information), the event is displayed on Case Manager's alerts page for manual revision:

The investigation of alerts in card payments is a post-mortem-like analysis because the issuer processes the payment in real time. Credit transfers and direct debits are bank payments subject to longer processing times. Therefore, fraud analysts can manually review alerts generated by these payment types with status as Pending Review.

This manual provides two guides that help fraud analysts and fraud managers to use Case Manager:

- Investigate Alerts: This guide's target audience are fraud analysts. It explains how Case Manager works and how it can be used to analyze alerts that suggest suspicious activity.
- Manage Workload and Rules: Learn how to set up Case Manager to keep the workload of fraud teams organized to prevent fraudulent behavior efficiently.

5. Webhooksâ

Webhooks provide a straightforward way to keep relevant stakeholders up to date with the latest information. They can be enabled within Feedzai's Case Manager to further automate the sending of information to key stakeholders for more effective decision-making. Webhooks empower Feedzai clients to combat fraud with automated responses to downstream systems. Strategically implementing webhooks enables operational efficiency and safeguards banks' customers, according to the established fraud prevention policies.

When something significant happens, such as a transaction or login attempt, the webhook sends an instant notification to the bank's system. Webhooks deliver updates right when they matter.

tip

Contact your Customer Success manager to discuss how Feedzai's webhooks can automate your internal business processes.

Key conceptsâ

- **Webhook**: A webhook is a custom callback that notifies one application (the subject) whenever an event occurs in another application (the observer).
- **Payload**: The data related to the triggering event is sent to the subject's system via an HTTP request.

How webhooks workâ

Imagine you're waiting for an important email. Instead of refreshing your inbox repeatedly, the email server sends you a message as soon as the email arrives. Similarly, webhooks work behind the scenes, quietly notifying your target system when specific events or business conditions are triggered. Webhooks can also automatically notify banks of potential fraudulent transactions on their customers' credit cards. It can also be used to automatically update backend fraud or customer relationship management (CRM) systems.

Technical details

- Using an automation call, âCall Webhook,âCase Manager is able to send HTTP requests to external endpoints.
- The payload of the HTTP request includes details of the webhook trigger source (i.e., an event or case) in JSON format.
- If the webhook was triggered by a case automation, an additional field is sent on the payload with the details of the events attached to the case.

How webhooks improve operational processesâ

Webhooks offer several operational efficiency advantages by automating the flow of crucial information in financial institutions' internal systems:

- **Fraud feedback**: If your bank has multiple systems that consume fraud outcomes (i.e., fraud or not fraud), the traditional approach would be to go into each system and manually update them. This is time consuming and not the best use of your resources. Through webhooks, a Case Manager automation rule will act upon the analyst's decision to provide feedback to all the necessary systems.
- **Updates to CRM systems**: Traditionally, when a bank closes an alert in Case Manager, it must update the CRM with case notes and details of the alert. This means adding the same details and notes to multiple systems. With webhooks, the bank can automatically retrieve backend data with a single click. The necessary data is copied from Case Manager and into the bank's preferred CRM. Both of these scenarios produce 100% task completion in less than half the time and effort. Instead of spending time on administrative tasks, your analysts can focus on strategic decision-making.

How webhooks improve fraud mitigationâ

Webhooks offer several fraud mitigation advantages for financial institutions:

- **Validate event details with their customers**: As your analysts investigate suspicious transactions, you may want to contact the customer to validate the event details or make the customer aware of the event.
- **Update other internal systems**: Based on the analyst's decision, consider triggering actions in other systems, like blocking the card involved in a fraudulent event or blocking the customer's home banking account, preventing additional actions until the issue has been resolved.

Let's explore three scenarios where webhooks play a crucial role in fraud prevention:

Account takeover detectionâ

1. **Scenario**: Based on Feedzai's Transaction Fraud digital activity channels, Feedzai's risk engine can see that a customer's home address recently changed and unusual activity for the customer's profile involving a new card has been detected. Most events occurred using a device located 1,000 miles from normal online locations. All these different behaviors generate an alert in Case Manager.
2. **Webhook use**: Based on a specific rule trigger, Case Manager can send a webhook with selected details for actions from the team.
3. **Action**: Based on the bank's policy, they can send a text message to the customer to validate the address change and the card usage in the identified region. Text messages can be tailored depending on the type of risk identified. Additionally, the bank can request additional authentication in the native banking application before any additional purchase is approved.

Credit card fraudâ

1. **Scenario**: A customer's credit card experiences multiple high-value transactions within a short timeframe.
2. **Webhook use**: Based on Case Manager's fraud decision, banks can send a webhook with selected details for actions from the team.
3. **Action**: Based on the bank's policy, they can send a text message to the customer informing them about the fraudulent activity. Additionally, the bank can use in-app notifications. The workflows are up to the bank to tailor based on their unique risk strategy.

Identity verification failuresâ

1. **Scenario**: During account creation, a user fails identity verification multiple times.
2. **Webhook use**: Based on Case Manager's fraud decision, banks receive notifications when verification attempts fail 2 or more times.
3. **Action**: Based on the bank's policy, they can flag the account for manual review or request additional documentation.

When to use webhooksâ

Webhooks are ideal when real-time updates matter - instances where you need to take immediate actions to prevent the event from being analyzed or stop future use as quickly as possible. This applies to situations involving, for instance, account security.

When not to use webhooksâ

Weigh the pros and cons of using webhooks in the following situations:

- **Customer friction**: The flexibility to validate events with end users in real time can close the feedback loop, but it can also cause customer friction. Evaluate how you want to manage communications with customers to establish the appropriate fraud prevention policies. Taking false-positives into consideration, you can establish a threshold to ensure that webhooks are not over-utilized to minimize friction with customers.
- **Syncs with other internal systems**: There are several methods for synching the data outputs from the Feedzai event processing, and the webhook option should be carefully used for this purpose.
    - **High-frequency polling**: If data updates are frequent, webhooks may not be efficient.
    - **Large-scale data sync**: Consider other methods like APIs for bulk data transfer.
    - **Complex bidirectional communication**: Webhooks are one-way options. APIs may be more suitable for complex interactions.

6. Dashboardsâ

TFB has dashboards out of the box with metrics and real-time charts to provide insights into your business and rule performance. Based on these insights, you can then improve workload management or write self-service rules to quickly block a fraudulent behavior.

Learn about the available dashboards in the View Analytics Dashboard page.

Inbound Payments

Was this page helpful?Â

In response to regulatory requirements and the rise of Authorized Push Payment (APP) scams, Transaction Fraud for Banking (TFB) includes a risk strategy for Inbound Payments. This addition also allows Feedzai to use the inbound payment data for subsequent outbound transactional decisions, especially in scenarios where fraudsters attempt to move the money out of that account to evade detection.

TFB includes a dedicated Inbound Payments channel that operates independently from Outbound transfers. This separation enables the scoring of inbound events and the development of a dedicated risk strategy tailored specifically for inbound transactions.

This channel allows to make decisions at an inbound-event level, as well as investigate other inbound payment events within the transaction history widgets across all payment types.

Moreover, the Inbound Payments API follows the same definition as the Transfers API, ensuring a seamless integration process for clients. This API allows to segregate the strategy for inbound payments and leverage inbound payment data to make risk decisions not only on the legitimacy of received funds but also on any associated risks if these funds are subsequently used for outbound transactions.

Take a look at the available information on Inbound Payments:

Life Cycle

Fraud Scenarios

API

Rules Projects

Pulse Schemas

Learning Paths

Was this page helpful?Â

The learning paths are recommendations on how to access the documentation according to the user's role to get you up and running with the solution quickly.

Introductionâ

If you are a new TFB user, we recommend the following learning paths with the description and key links below:

Solution Basics Pathâ

This path covers basic concepts of how the solution works. We also recommend this path as the starting point for the other paths because you will learn the foundations to use the software to perform more complex tasks.

tip

Audience: Management roles and decision-makers who want to get a basic understanding of the solution. Data scientists, fraud analysts, fraud managers and software engineers can then move on to specific learning paths.

Important topics and learnings:

1. How the Solution Works: Understand how Feedzai generates the recommendation response on whether a given transaction is legitimate or fraudulent.

2. Life Cycles: Understand how the API works. Feedzai provides four data streams, namely Cards, Transfers, Reference Data, and Digital Activity. The life cycle pages get into detail about the endpoint information and how they play a role within the transaction life cycle.
3. Fraud Scenarios and Detection: Get to know the TFB's strategy to fight different fraud schemes using default rules that are provided out-of-the-box. Each page provides a thorough description of the different fraud scenarios associated with the payment method and how TFB prevents them from occurring.
4. References: Learn about the resources that are available out of the box to fight fraud from day 1. These include:
    - Schemas that map directly to the API fields, and computed metrics such as counts and amounts calculated over a time window.
    - Lists that can be used directly in rules.
    - Rules Projects that are used to organize the pre-packaged rules. This page contains conceptual information on how rules are organized in Pulse, naming convention and nomenclature used in this manual. It lists all rules within each rules project and provides relevant information, such as description, actions, conditions, and outcomes. With this reference, data scientists and fraud analysts can have total visibility on what are the tools that are helping them to prevent fraud.

Integration Pathâ

This learning path addresses the needs of those integrating the issuer bank's ecosystem with the Feedzai's API.

tip

Audience: Decision-makers, system architects, and developers working on the API integration process.

Important topics and learnings:

1. Life Cycles: Understand how the API works. Feedzai provides four data streams, Cards, Transfers, Reference Data, and Digital Activity. The life cycle pages get into detail about the endpoint information and how they play a role within the transaction life cycle.
2. Swagger: Download the swagger.json file to generate the client libraries automatically, thus accelerating the integration process.

3. API specification: Map your data catalog to the Feedzai's API by matching the issuer's information with the fields available under:
    - Cards
    - Outbound Transfers
    - Digital Activity
    - Reference Data

Rules and Data Science Path For Pulseâ

This learning path includes how-to guides that help you to use Pulse to work with Rules and execute on Data Science tasks, more specifically Automated Machine Learning. The audience of this path is:

tip

Audience: Data scientists and other roles responsible for working with rules, and machine learning tools.

Important topics and learnings:

1. Work with Rules and Lists: Understand how rules work and learn how to write rules specific to your business, configure your work in the runtime workflow, and test rules using the Transaction Fraud for Banking API. We strongly recommend the How Rules Work page because it explains in detail how the rules outcomes are generated.

2. Use Automated Machine Learning: Learn how to use Pulse's built-in AutoML tool. With this tool, you can automatically generate features for machine learning models and obtain a trained model with a set of optimal parameters. Note that this software functionality is only available under specific use cases.
3. Migrate Pulse Resources: Learn how to export and import resources - Plans, Rules, Model Projects, and List Items - between environments. Note that this software functionality is only available under specific use cases.

Alert path for Case Managerâ

This learning path includes how-to guides that help you to use Case Manager to review alerts. It also addresses Fraud Manager tasks such as operational setups to optimize workload and operationalize fraud teams.

tip

Audience: Fraud analysts and fraud managers responsible for handling events that are considered suspicious, and therefore raise alerts.

Important topics and learnings:

1. Investigate Alerts: This guide's target audience are fraud analysts. It explains how Case Manager works and how it can be used to analyze alerts that suggest suspicious activity.
2. Manage Workload and Rules: Learn how to set up Case Manager to keep the workload of fraud teams organized to prevent fraudulent behavior efficiently.

3. View Analytics Dashboards: Find more about metrics and real-time charts that provide insights into your business and rule performance.

Transaction Fraud for Banking

Was this page helpful?Â

About Transaction Fraud for Banking

Learn the basics to have a solid understanding of how the solution works

Operational work

Explore TFB's practical guides to help data scientists, analysts, and managers use the software

API resources

Learn all the operations you can do with Feedzai's API and how to communicate with its multiple endpoints

References

Learn about rules projects, lists, schemas, schema mappings, among other topics

Admin resources

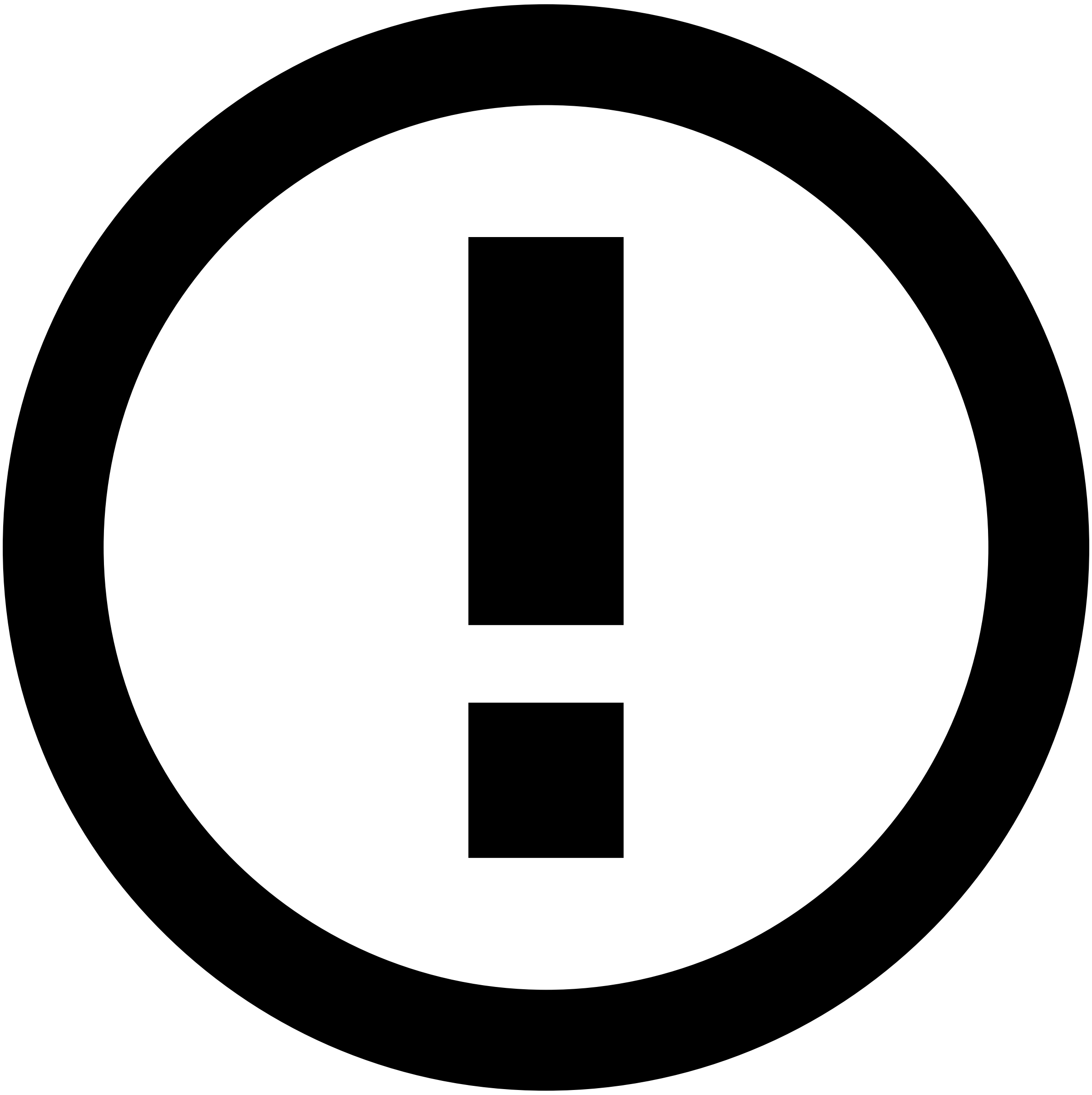
Learn about the default roles and permissions

-
- Admin Resources
- Activate SSO Authentication

On this page

Activate SSO Authentication

Single Sign-On (SSO) is an authentication process that allows you to manage users' access on the Feedzai platform's applications with a single set of login credentials, through an Identity Provider (IdP), such as OKTA, Microsoft Entra ID, OneLogin, among many others.



Get to know your IdP's documentation

As there are countless IdPs in the market, this page provides mostly generic examples. **Always check your IdP's official documentation** or reach out to their own support team for more accurate information about what your IdP may require for the SSO integration to be successful.

Why configure an SSO integrationâ

There are many advantages to integrating SSO authentication with the Feedzai platform, such as:

- Managing users autonomously. For example, upon offboarding, users disabled through the IdP will automatically lose access to the Feedzai applications.
- Allowing your users to enter the corporate credentials they are accustomed to, rather than requiring them to remember an additional username and password.
- Letting you implement multi-factor authentication as already defined by your company's Identity and User Management process, through the IdP.
- Keeping a record of changes done to users stored in your IdP.

caution

Once the SSO integration is set up, **Feedzai will no longer be able to manage your users**. You will manage them **exclusively** through your IdP.

Requirementsâ

To activate SSO authentication for the Feedzai platform, you must have an Identity and User Management process with an IdP compatible with SAML 2.0 protocol.

The SAML attributes you need to configure in your IdP are:

- **firstname**: First name of the user.
- **lastname**: Last name of the user.
- **email**: Email of the user (even if the NameID property is filled in with the email as username).
- **roles**: List of valid roles.
- **tenants**: List of valid tenants (only when applicable).
- **groups**: List of valid groups (only when applicable).

Overview of the SSO integration processâ

Reach out to your Feedzai contact point to request that they start the process and gather the information required for your configuration.

info

The following steps will need to be made **for each environment that needs to be configured**.

1. Obtain environments' dataâ

Upon your request to start the process, Feedzai will share your environments' information with you, including:

- **Identifier (Entity ID)**
- **Reply URL (Assertion Consumer Service URL)**
- **Sign on URL**
- **iss** (= feedzai)

You will use this information to complete the next step.

2. Share your IdP metadata fileâ

Using the information provided by Feedzai as described above, you must retrieve the SAML XML metadata file from your IdP and share it with your Feedzai contact point.

caution

The way to retrieve your metadata file and what its content looks like will be specific to your IdP. However, you should **make sure the attributes configured as part of the Requirements are available in the metadata file**.

Refer to your IdP's official documentation if you have any questions.

Feedzai will use this metadata file to configure the SSO integration on the Feedzai side.

3. Obtain environment URLsâ

Once the configuration is done on the Feedzai side, your contact point will share with you the URL(s) of your environment(s).

Reminder

The SSO integration must be configured for each one of your environments. Therefore, the URL of each environment is only useful to you if steps #1 and #2 have been completed for that environment.

4. Configure SSO on your IdP's sideâ

Once you have the environments' URLs, you can start configuring the SSO integration for each environment on your IdP's side.

You will need to assign the correct roles for each user in the IdP's user profiles.

The role format must always follow the standard:

```
<environment>-<component>-<role_name>
```

You must define a separate role for every combination of environment and component you plan to use. For example, if a user needs access to both the Pulse and Case Manager applications in the `preprod` and `prod` environments, you must create the roles: `preprod-pulse-<role_name>`, `preprod-case-manager-<role_name>`, `prod-pulse-<role_name>`, and `prod-case-manager-<role_name>`.

The available **environments** with SSO are:

- `preprod` - for the testing and validation environment.
- `prod` - for the production environment.
- `perf` - for the performance environment.
- `qa` - for the quality assurance environment.

note

The available environments depend on contracted service, and nomenclature may vary.

The **component** is the element that specifies which application you are configuring the role for. The possible components are:

- `pulse` : For the Pulse application.
- `case-manager` : For the Case Manager application.
- `dsf` : For the Pulse application dedicated to Data Science tests.

info

You do not need to set separate roles for the Genome component, as these are derived from the roles set for Pulse.

The **role** element is specific to each use case. You will find a list of the relevant Pulse and Case Manager roles and permissions applicable to this solution in the Roles and Permissions page. Roles for the DSF environment are provided to you by your Feedzai contact point.

caution

You must ensure that there are **no misspellings** and that you type the roles exactly as presented in the documentation.

Role names are **case-sensitive**. For example, `preprod-pulse-integrator` will be rejected by Pulse. The correct form is `preprod-pulse-Integrator`.

Below are a few examples of the composition of roles to be configured in a generic IdP:

Environment	Component	Role	Role to be configured (for a generic IdP)
preprod	pulse	Fraud Prevention Manager	preprod-pulse-Fraud Prevention Manager
prod	case-manager	Fraud Prevention Agents - L1	prod-case-manager-Fraud Prevention Agents - L1

Varying IdP requirements

Your IdP may only accept a format different from the "generic IdP" example above. As the IdP is managed on your side, you must also make sure to **consult your IdP's official documentation**. Different IdPs often have different requirements.

For example, some IdPs are whitespace-sensitive, so you would have to type `preprod-pulse-Fraud_Prevention_Manager` instead of `preprod-pulse-Fraud Prevention Manager`.

5. Test the configurationâ

After the configuration is finished both on Feedzai's side and your IdP's side, we recommend that you try granting access to a user and ask them to test the access to each environment you configured.

If the configuration is correct, the user will be redirected to the usual IdP login page. Upon logging in, they will be redirected again to the Feedzai platform. Upon accessing the Feedzai application for the first time, the user will be created with the role provided by the IdP.

If the configuration was not done correctly, the user may:

- Not see the IdP's login page, but instead see a Feedzai page asking them to enter their name and email.
- See the IdP's login page and be redirected successfully, but get a "401 - Forbidden" error message when accessing the component. This indicates that something went wrong at the role attribute configuration level.

In these cases, to understand what went wrong with the configuration, Feedzai will ask you to check the SAML response. You must make sure that this SAML response also includes the role attribute for thorough troubleshooting.

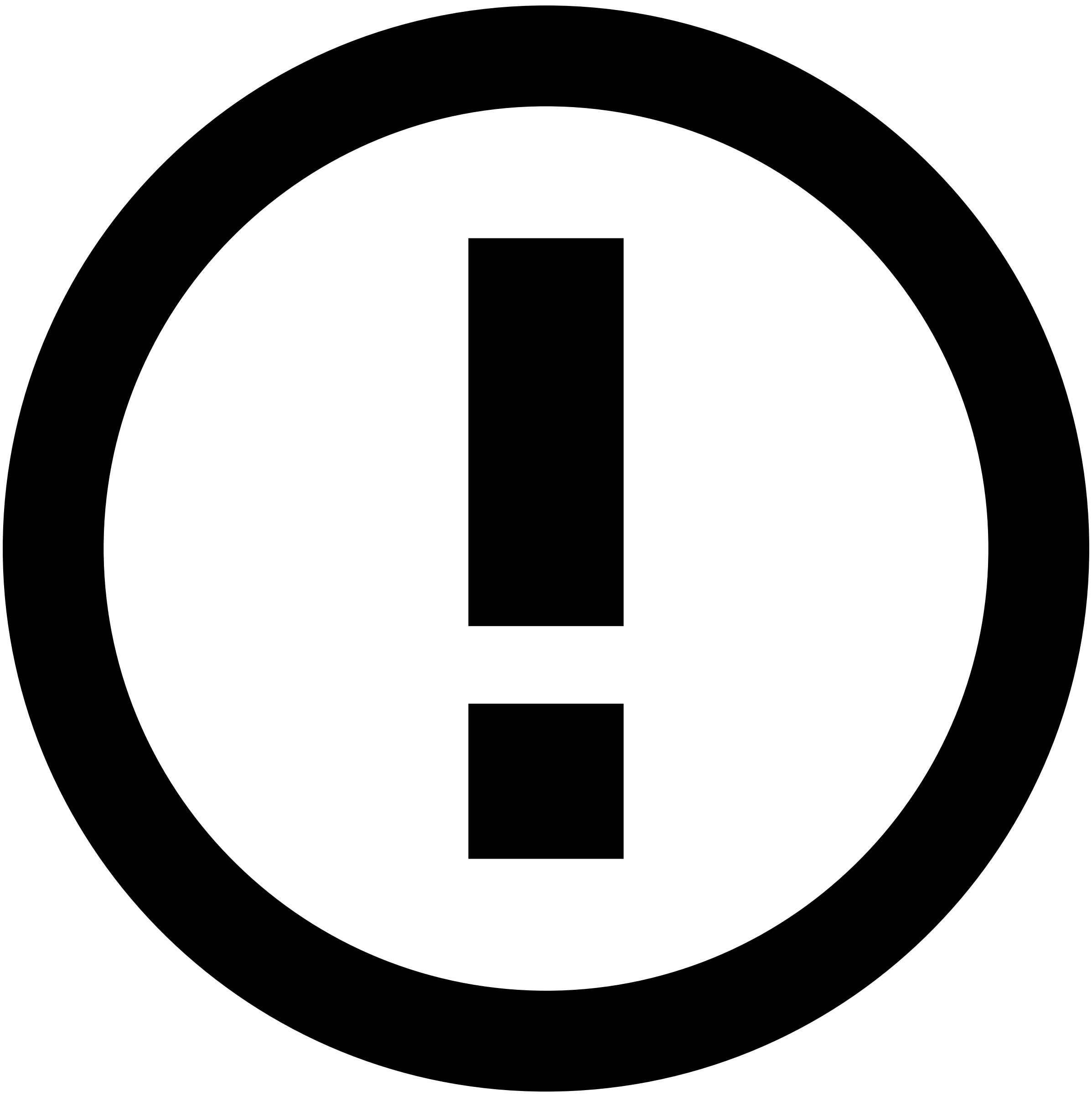
Inspecting the SAML responseâ

Inspecting the SAML response allows you to review the interactions between Feedzai and the IdP to find the issue. There are two SAML responses to take into account:

1. From your IdP into Feedzai's IdP.

1. From Feedzai's IdP into the component.

The **first one is more important to check for troubleshooting purposes**, as it is the one that brings the roles from your IdP's side to Feedzai's side.



How to distinguish them

In the first one (from your IdP into Feedzai's IdP), the role attributes include the prefix with the environment and component (e.g. `prod-pulse-`), followed by the role name. The role names contain underscores (not whitespaces).

In the second one, the roles have no prefix, just a `User.role` format, and they can contain whitespaces.

To obtain the SAML response, you can, for example, use a SAML inspector plug-in on your browser or other internal process you may have for this purpose.

The response includes all the claims and roles Feedzai shared with the IdP, which you can review.

Below is an example of what part of a SAML response may look like:

```
 <saml2:Assertion>
    <saml2:AttributeStatement xmlns:saml2="urn:oasis:names:tc:SAML:2.0:assertion">
        <saml2:Attribute Name="roles"
                         NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified"
                         >
            <saml2:AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >preprod-pulse-Fraud_Strategist</saml2:AttributeValue>
            <saml2:AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >preprod-pulse-Fraud_Prevention_Manager</saml2:AttributeValue>
        </saml2:Attribute>
        <saml2:Attribute Name="email"
                         NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified"
                         >
            <saml2:AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >john.doe@feedzai.com</saml2:AttributeValue>
        </saml2:Attribute>
        <saml2:Attribute Name="firstname"
                         NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified"
                         >
            <saml2:AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >John</saml2:AttributeValue>
        </saml2:Attribute>
        <saml2:Attribute Name="lastname"
                         NameFormat="urn:oasis:names:tc:SAML:2.0:attrname-format:unspecified"
                         >
            <saml2:AttributeValue xmlns:xs="http://www.w3.org/2001/XMLSchema"
                                  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
                                  xsi:type="xs:string"
                                  >Doe</saml2:AttributeValue>
        </saml2:Attribute>
    </saml2:AttributeStatement>
</saml2:Assertion>
```

Admin Resources

Was this page helpful?Â

[Roles and permissions]()

[Learn about the default roles and their associated permissions to understand how Transaction Fraud for Banking manages the access within Pulse, Case Manager, and dashboards]()

[SSO Authentication]()

[Learn about the requirements and necessary steps to activate Single Sign-On (SSO) authentication]()

-

On this page

Analyst Performance

Was this page helpful?Â

Fraud prevention managers need to regularly manage and measure their fraud prevention analysts' performance on an individual level. The Analyst Performance dashboard measures fraud prevention analysts' performance, such as alerts' average resolution time. The Analyst Performance dashboard addresses this need, and consequently allows managers to make decisions that enhance their analysts' performance more efficiently.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by creating queries.

Confirm which fields you can use to create queries for the outputs you need.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An index pattern selects the data to use and allows you to define properties of fields.
2. Add the fields you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. A **Value** field appears after selecting the operator. Enter a value to finalize the creation of your condition.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

- **Quick select**: Shows the "last" or "next" results, considering "now" as the default end or start date, respectively.
- **Commonly used**: Shows the most commonly used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

- **Channel**: The channel that you want to analyze (e.g. inbound payments).
- **Queue**: The different priority categories that each transaction event is assigned to.
- **Assign analyst**: The assigned analyst.
- **Decision analyst**: The analyst who made the decision.
- **Status**: The event's status (e.g. unreviewed, closed).
- **Country code**: The 2-letter code for countries.

Data visualizationâ

- **Average resolution time**: The average time an alert is open until reaching the Fraud or Not Fraud status.

- **Alerts by status**: The number of alerts per status, with each bar corresponding to a status. Hover over each bar to see the number of alerts per status.

- **Average resolution time per day**: The average event resolution time per day, with the y-axis representing the average resolution time in seconds, and the x-axis showing the time in hours. For instance, imagine that you hover over the bar at 9:00 and the information provided says that the average resolution time for that hour is 28 seconds.

- **Average resolution time by assign analyst**: The average time each assigned fraud analyst takes to close an alert.
- **Average resolution time by decision analyst**: The average time each decision-making analyst takes to close an alert.

tip

- The table is sortable. You can arrange data into ascending or descending order according to a column.
- By hovering over each cell, you can use the **Plus** `+` or **Minus** `-` icons to either filter out or filter by that specific value only.
- You can also export the table. Select the **Ellipsis** `...` icon --> **Inspect** --> **Download CSV**.

- The table maps the total number of events and the average resolution time to the corresponding assigned and decision-making analyst per day.

caution

Selecting bars or dragging the cursor over any chart works like a filter. Any of these actions will change the results in all data vizualizations.

- **TFB - Resolution time per analyst over time**: The resolution time (in milliseconds) for events over time, with a threshold line at 300,000 milliseconds. For instance, if the time window is set to 24 hours, the chart displays the amount of time spent on average analyzing alerts per each 10 minutes. The threshold line highlights the maximum acceptable resolution time for events.
- **Team workload**: The team's workload distribution over a defined period, i.e. the number of alerts assigned to each fraud analyst per hour over 9 hours (e.g. 9 am to 6 pm).

- **Analyst accuracy**: The number of events approved by the system, broken down into two categories based on analyst decision, i.e. genuine or fraud.

- **Analyst accuracy per analyst**: Maps the total volume of events classified by each analyst as fraud or genuine to the events approved by the system for each analyst.

tip

- The table is sortable. You can arrange data into ascending or descending order according to a column.
- By hovering over each cell, you can use the **Plus** `+` or **Minus** `-` icons to either filter out or filter by that specific value only.
- You can also export the table. Select the **Ellipsis** `...` icon --> **Inspect** --> **Download CSV**.

- **TFB - Average resolution time per region**: A geographical map that allows you to hover over any country to see the average resolution time in minutes. The darker the region color, the longer the average resolution time.

caution

Selecting a region works like a filter. This action will change the results in all data vizualizations.

caution

Selecting bars or dragging the cursor over any chart works like a filter. Any of these actions will change the results in all data vizualizations.

Analytics Dashboards' Fields

Was this page helpful?Â

This page includes all the relevant fields you can use in each index-pattern to refine the scope of your analysis.

Cardsâ

- `@timestamp`
- `calculated.cardholder_bill_amount_usd`
- `common.country_code`
- `common.queueName`

- `feedback.fraud.history.realOutcome`
- `feedback.fraud.history.type.reportInstant`
- `feedback.fraud.latest.realOutcome`
- `feedback.fraud.latest.reportInstant`
- `fields.cardholder_bill_amount`
- `fields.customer_region`
- `fields.event_occurred_at`
- `fields.event_type`
- `fields. feedback_type`
- `fields.info_type`
- `fields.lifecycle_id`
- `fields.channel_type`
- `fields.customer_segment`
- `fields.transaction_status`
- `psd2_domain.amount`
- `psd2_domain.authentication`
- `psd2_domain.authorization`
- `psd2_domain.channel`
- `psd2_domain.exemption`
- `psd2_domain.initiation`
- `psd2_domain.paymentInstrument`
- `psd2_domain.transactionType`
- `fdz_domain.cm.alert`
- `fdz_domain.cm.assignedAt`
- `fdz_domain.cm.assigneeUserId`
- `fdz_domain.cm.assigneeUsername`
- `fdz_domain.cm.channel`
- `fdz_domain.cm.decisionAt`
- `fdz_domain.cm.TimeSinceAssigned`
- `fdz_domain.cm.TimeSinceCreated`
- `fdz_domain.cm.decisionUserId`
- `fdz_domain.cm.decisionUsername`
- `fdz_domain.cm.eventStates.decision.stateId`
- `fdz_domain.cm.eventStates.operational_status.stateId`
- `fdz_domain.cm.eventStates.status_cards.stateId`
- `fdz_domain.cm.eventStates.status.stateId`
- `fdz_domain.cm.id`
- `fdz_domain.cm.queueId`
- `fdz_domain.cm.reasons.status_cards.reasonName`
- `fdz_domain.cm.reasons.status_cards.reasonId`
- `fdz_domain.cm.reasons.status_cards.stateId`
- `fdz_domain.cm.states`
- `fdz_domain.cm.timestamp`
- `fdz_domain.cm.updatedAt`
- `fdz_domain.cm.userId`
- `fdz_domain.cm.username`
- `fdz_domain.cm.versionId`
- `fdz_domain.cm.versionTimestamp`
- `pulse.id`
- `pulse.instant`
- `pulse.tenantKey`
- `pulse.sourceId`
- `pulse.outcomes.decision.outcomeDecision`
- `pulse.outcomes.decision.score`
- `pulse.outcomes.sca_required.score`
- `pulse.outcomes.sca_required.outcomeDecision`
- `pulse.outcomes.fraud.outcomeDecision`
- `pulse.outcomes.fraud.score`

- `pulse.outcomes.suspicious.outcomeDecision`
- `pulse.outcomes.suspicious.score`
- `pulse.results.result.originalRuleId`
- `pulse.results.result.ruleId`
- `pulse.results.result.targetOutcome`
- `pulse.results.result.targetOutcomeLabel`
- `pulse.results.result.outcome`
- `pulse.results.sourceId`
- `pulse.results.weight`
- `pulse.results.explanations`
- `pulse.triggeredActions.name`
- `pulse.triggeredActions.nestedSourceId`
- `pulse.triggeredActions.sourceId`

Inbound Paymentsâ

- `@timestamp`
- `calculated.cardholder_bill_amount_usd`
- `common.country_code`
- `common.queueName`
- `feedback.fraud.history.realOutcome`
- `feedback.fraud.history.type.reportInstant`
- `feedback.fraud.latest.realOutcome`
- `feedback.fraud.latest.reportInstant`
- `fields.event_type`
- `fields. feedback_type`
- `fields.info_type`
- `fields.payment_type`
- `fields.lifecycle_id`
- `fields.payment_status`
- `fields.channel_type`
- `fields.customer_segment`
- `fields.channel_type`
- `fields.receiver_account_id`
- `fields.sender_account_id`
- `psd2_domain.amount`
- `psd2_domain.authentication`
- `psd2_domain.authorization`
- `psd2_domain.channel`
- `psd2_domain.exemption`
- `psd2_domain.initiation`
- `psd2_domain.paymentInstrument`
- `psd2_domain.transactionType`
- `fdz_domain.cm.alert`
- `fdz_domain.cm.assignedAt`
- `fdz_domain.cm.assigneeUserId`
- `fdz_domain.cm.assigneeUsername`
- `fdz_domain.cm.channel`
- `fdz_domain.cm.decisionAt`
- `fdz_domain.cm.TimeSinceAssigned`
- `fdz_domain.cm.TimeSinceCreated`
- `fdz_domain.cm.decisionUserId`
- `fdz_domain.cm.decisionUsername`
- `fdz_domain.cm.eventStates.decision.stateId`
- `fdz_domain.cm.eventStates.operational_status.stateId`
- `fdz_domain.cm.eventStates.breach_status.stateId`
- `fdz_domain.cm.eventStates.status.stateId`
- `fdz_domain.cm.id`

- `fdz_domain.cm.queueId`
- `fdz_domain.cm.reasons.status_cards.reasonName`
- `fdz_domain.cm.reasons.status_cards.reasonId`
- `fdz_domain.cm.reasons.status_cards.stateId`
- `fdz_domain.cm.states`
- `fdz_domain.cm.timestamp`
- `fdz_domain.cm.updatedAt`
- `fdz_domain.cm.userId`
- `fdz_domain.cm.username`
- `fdz_domain.cm.versionId`
- `fdz_domain.cm.versionTimestamp`
- `pulse.id`
- `pulse.instant`
- `pulse.tenantKey`
- `pulse.sourceId`
- `pulse.outcomes.decision.outcomeDecision`
- `pulse.outcomes.decision.score`
- `pulse.outcomes.sca_required.score`
- `pulse.outcomes.sca_required.outcomeDecision`
- `pulse.outcomes.fraud.outcomeDecision`
- `pulse.outcomes.fraud.score`
- `pulse.outcomes.suspicious.outcomeDecision`
- `pulse.outcomes.suspicious.score`
- `pulse.results.result.originalRuleId`
- `pulse.results.result.ruleId`
- `pulse.results.result.targetOutcome`
- `pulse.results.result.targetOutcomeLabel`
- `pulse.results.result.outcome`
- `pulse.results.sourceId`
- `pulse.results.weight`
- `pulse.results.explanations`
- `pulse.triggeredActions.name`
- `pulse.triggeredActions.nestedSourceId`
- `pulse.triggeredActions.sourceId`

Outbound Transfersâ

- `@timestamp`
- `calculated.cardholder_bill_amount_usd`
- `common.country_code`
- `common.queueName`
- `feedback.fraud.history.realOutcome`
- `feedback.fraud.history.type.reportInstant`
- `feedback.fraud.latest.realOutcome`
- `feedback.fraud.latest.reportInstant`
- `fields.event_type`
- `fields. feedback_type`
- `fields.info_type`
- `fields.payment_type`
- `fields.lifecycle_id`
- `fields.payment_status`
- `fields.channel_type`
- `fields.customer_segment`
- `fields.channel_type`
- `fields.receiver_account_id`
- `fields.sender_account_id`
- `psd2_domain.amount`
- `psd2_domain.authentication`

- `psd2_domain.authorization`
- `psd2_domain.channel`
- `psd2_domain.exemption`
- `psd2_domain.initiation`
- `psd2_domain.paymentInstrument`
- `psd2_domain.transactionType`
- `fdz_domain.cm.alert`
- `fdz_domain.cm.assignedAt`
- `fdz_domain.cm.assigneeUserId`
- `fdz_domain.cm.assigneeUsername`
- `fdz_domain.cm.channel`
- `fdz_domain.cm.decisionAt`
- `fdz_domain.cm.TimeSinceAssigned`
- `fdz_domain.cm.TimeSinceCreated`
- `fdz_domain.cm.decisionUserId`
- `fdz_domain.cm.decisionUsername`
- `fdz_domain.cm.eventStates.decision.stateId`
- `fdz_domain.cm.eventStates.operational_status.stateId`
- `fdz_domain.cm.eventStates.breach_status.stateId`
- `fdz_domain.cm.eventStates.status.stateId`
- `fdz_domain.cm.id`
- `fdz_domain.cm.queueId`
- `fdz_domain.cm.reasons.status_cards.reasonName`
- `fdz_domain.cm.reasons.status_cards.reasonId`
- `fdz_domain.cm.reasons.status_cards.stateId`
- `fdz_domain.cm.states`
- `fdz_domain.cm.timestamp`
- `fdz_domain.cm.updatedAt`
- `fdz_domain.cm.userId`
- `fdz_domain.cm.username`
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- `pulse.outcomes.decision.score`
- `pulse.outcomes.sca_required.score`
- `pulse.outcomes.sca_required.outcomeDecision`
- `pulse.outcomes.fraud.outcomeDecision`
- `pulse.outcomes.fraud.score`
- `pulse.outcomes.suspicious.outcomeDecision`
- `pulse.outcomes.suspicious.score`
- `pulse.results.result.originalRuleId`
- `pulse.results.result.ruleId`
- `pulse.results.result.targetOutcome`
- `pulse.results.result.targetOutcomeLabel`
- `pulse.results.result.outcome`
- `pulse.results.sourceId`
- `pulse.results.weight`
- `pulse.results.explanations`
- `pulse.triggeredActions.name`
- `pulse.triggeredActions.nestedSourceId`
- `pulse.triggeredActions.sourceId`

Business Overview

Was this page helpful?Â

The business overview dashboard gives visibility over your business activity's performance.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by creating queries.

Confirm which fields you can use to create queries for the outputs you need.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An index pattern selects the data to use and allows you to define properties of fields.
2. Add the fields you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. A **Value** field appears after selecting the operator. Enter a value to finalize the creation of your condition.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

- **Quick select**: Shows the "last" or "next" results, considering "now" as the default end or start date, respectively.
- **Commonly used**: Shows the most common used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

- **Region**: The country where the transaction(s) you want to filter by was performed.
- **Channel**: The channel that you want to analyze (e.g. cards).
- **Queue**: The different priority categories that each transaction event is assigned to.
- **Transaction type**: The categories that classify transactions (e.g. direct deposit, bill payment).
- **Payment type**: The payment method that was used (e.g. SEPA, internal).

Data visualizationâ

Business KPIS evaluate the business activity performance with Transaction Fraud for Banking. They provide insightful information such as total processed transaction amounts compared to the avoided loss amounts.

These KPIs evaluate transactional events according to four possible categories - **Alerted**, **Approved**, **Declined**, and **In Review**. These events are measured against:

- **Value analysis**: Displays the transaction amounts grouping them into the aforementioned categories. For instance, the **Alerted** category shows the percentage of transactions flagged as alerts relative to the total transaction amount, along with the corresponding transaction amounts.
- **Volume analysis**: Displays the number of transactions grouping them into the aforementioned categories. For instance, the **Declined** category shows the percentage of declined transactions and corresponding number of transactions.

- **Sum of transactions value over time (breakdown by decision)**: Transaction amounts over time, with the green and red areas representing the approved and declined transactions, respectively.
- **Transactions volume over time (breakdown by decision)**: Sum of transactions over time, with the green and red areas representing the approved and declined transactions, respectively.
- **Sum of loss avoided value over time**: Total loss avoided (in monetary amounts) from transactions over time.
- **Loss avoided volume over time**: Total loss avoided (in number of transactions) over time.

caution

Selecting bars or dragging the cursor over any chart works like a filter. Any of these actions will change the results in all data vizualizations.

Analytics Dashboards

Was this page helpful?Â

The analytics dashboards include metrics that provide you with real-time charts with information about analyst performance, business overview, PSD2 reporting, and rule monitoring. Based on these insights, you can improve workload management, or write self-service rules to quickly block a fraudulent behavior.

This guide provides you with the following information:

- Analyst performance: Measures fraud prevention analysts' performance, such as alerts' average resolution time. It allows managers to make decisions that enhance their analysts' performance more efficiently.
- Business overview: Gives visibility on the business activity including information about transactions, i.e. processed, alerted, approved, declined, and in review, as well as loss avoided.

- PSD2: Presents data associated with the second Payment Services Directive (PSD2), including information about fraud rate, transactions initiated through strong customer authentication (SCA), and transactions under each exemption. Check the PSD Compliance page to learn more about this topic.
- Rules monitoring: Displays rule performance. Based on this analysis, you can write self-service rules to block fraud quickly.

tip

Refer to the Analytics Dashboards' Fields page to see all the relevant fields you can use in each index-pattern to refine the scope of your analysis.

PSD2

Was this page helpful?Â

The PSD2 dashboard allows business analysts to monitor data in real time, act upon alerts, and share the results with the competent authorities such as PSD2 regulators. This dashboard presents information that allows issuer banks to comply with the PSD2 mandate, Article 21 (download the compliance matrix document to learn more about this article).

This dashboard offers:

- Metrics associated with the total of unauthorized or fraudulent payments and the resulting fraud rate.
- Breakdown of transactions initiated through SCA and each of the PSD exemptions.
- Filters to select remote versus non-remote, and electronically versus non-electronically initiated payments.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by creating queries.

Confirm which fields you can use to create queries for the outputs you need.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An index pattern selects the data to use and allows you to define properties of fields.
2. Add the fields you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. A **Value** field appears after selecting the operator. Enter a value to finalize the creation of your condition.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

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- **Commonly used**: Shows the most common used date ranges (e.g. today, this week).
- **Recently used date ranges**: Shows the most recently used date ranges.
- **Refresh every**: Lets you refresh dashboards dynamically. For instance, if you set your start or end date as "now", you can use this option to dynamically apply the changes to reflect the present moment every x number of seconds, minutes, or hours.

caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

- **Initiation**
- **Authentication**
- **Authorization**
- **Exemption**
- **Payment instrument**
- **Type**
- **Channel**

Data visualizationâ

- **Total Amount Processed:** The total financial amount (i.e. transaction amount) processed by the risk engine platform (Pulse).
    - **Unauthorized/Fraudulent:** The financial amount of unauthorized transactions processed by the risk platform and transactions labeled as fraud by the Feedzai's client feedback.
    - **Authorized:** The financial amount of authorized transactions.

- **Average Transaction Value:** The average transaction amount processed by the risk engine platform (Pulse).
- **Unauthorized/Fraudulent Avg:** The average amount for unauthorized transactions processed by the risk platform and transactions labeled as fraud by the Feedzai's client feedback.
- **Authorized Avg:** The average amount for authorized transactions.
- **Number of Transactions Processed:** The number of transactions processed by the risk engine platform (Pulse).
- **Unauthorized/Fraudulent:** The number of unauthorized transactions processed by the risk platform and transactions labeled as fraud by the Feedzai's client feedback.
- **Authorized:** The number of authorized transactions.

This chart shows data associated with authentication for both amounts and number of transactions:

- **Total amount(s) breakdown by authentication:**
  - The graph bar represents the aggregated financial amount of non-authorized transactions and transactions that required SCA in a given period.
  - The line graph represents the trend of the absolute financial amount of non-authorized transactions and transactions that required SCA.
- **Number of transactions breakdown by authentication:**
  - The graph bar represents the aggregated number of non-authorized transactions and transactions that required SCA in a given period.
  - The line graph represents the trend of the absolute number of non-authorized transactions and transactions that required SCA calculated.

These charts show data associated with the exemptions specified in the PSD mandate, Chapter 3 (download the compliance matrix document to learn more about this chapter).

- **Total amount breakdown by exemption article:** The total financial amount of transactions that exempted SCA per article.
- **Number of transactions breakdown by exemption article:** The number of transactions that exempted SCA per article.

- Operational Work
- Fraud Prevention Manager
- Analytics Dashboards
- Rules Monitoring

On this page

Rules Monitoring

Was this page helpful?Â

The rule monitoring dashboards show the rules that were triggered and therefore allows you to control and improve fraud performance. Understanding if rules behave as expected, or quickly discovering which rules need to be adjusted / deleted is critical for improving the fraud prevention system.

Rule monitoring includes two dashboards, one with rule filters, and another one with transaction filters.

Rule Monitoring in RiskOps Studio

Feedzai provides you with the possibility to monitor your rules through the **Rule Monitoring** component in Feedzai RiskOps Studio platform. This platform offers you a centralized dashboard where you can also review and evaluate the performance of your rules.

Check [Rule Monitoring] to know more about this component.

Use filtersâ

Use filters to focus on specific subsets of data, refine your analysis, and extract more meaningful insights. See the available filters below.

Search barâ

The search bar allows you to filter by creating queries.

Confirm which [fields] you can use to create queries for the outputs you need.

Add filtersâ

The **Add filter** option allows you to filter your analysis using conditions.

1. By default, Kibana automatically selects the index pattern containing data relevant to the dashboard you are analyzing. An index pattern selects the data to use and allows you to define properties of fields.
2. Add the fields you want to use to refine your analysis.
3. Select the operator for the condition you're working on (e.g. `is`, `is not`, `is one of`).
4. A **Value** field appears after selecting the operator. Enter a value to finalize the creation of your condition.

Date rangesâ

Use the **Calendar** icon to select the date range you are looking for.

Select one of the following options:

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caution

Note that setting the date as "now" or configuring the dashboards to show the "next" results will not refresh the dashboards automatically. To enable this automation, use the **Refresh every** option.

Other filtersâ

Rule filterâ

The **Rule Monitoring (rule filter)** dashboard provides other filters to refine your search such as:

- **Rule block**: The rule project that contains the rule you want to filter by.
- **Rule owner**: The owner of the rule. In case of an out-of-the-box rule, the owner is Feedzai.
- **Rule type**: Whether it's a self-service rule (SSR) or an out-of-the-box rule (Non-SSR).
- **Rule name**: The name of the rule.

Transaction filterâ

The **Rule Monitoring (transaction filter)** dashboard provides other filters to refine your search such as:

- **Region**: The country where the transaction(s) you want to filter by was performed.
- **Channel**: The channel that contains the rule you want to filter by (e.g. cards).
- **Queue**: The different priority categories that each transaction event is assigned to.
- **Transaction type**: The categories that classify transactions (e.g. direct deposit, bill payment).
- **Payment type**: The payment method that was used (e.g. SEPA, internal).

Data visualizationâ

Rule KPIs are available in the rules monitoring with the transaction filter and evaluate the performance of rules according to four possible outcomes - **False Positives (FP)**, **True Positives (TP)**, **False Negatives (FN)**, and **True Negatives (TN)**. Rule performance is measured against:

- **Value analysis**: Displays the transaction amounts grouping them into the FP, TP, FN, and TN categories. For instance, the **False Positive** category shows the percentage of transactions flagged as false positives relative to the total processed transaction amount, along with the corresponding transaction amounts.
- **Volume analysis**: Displays the number of transactions grouping them into the FP, TP, FN, and TN categories. For instance, the **True Positive** category shows the percentage of transactions flagged as true positives relative to the total processed number of transactions, along with the corresponding number of transactions.

- **Transaction amount over time by triggered rule**: Shows transaction amounts over time (filtered by a specific date range) for each triggered rule. Each dot on the graph represents a rule that was triggered by a transaction amount at a specific timestamp.
- **Number of transactions over time by triggered rule**: Shows the number of transactions over time (filtered by a specific date range) for each triggered rule. Each dot on the graph represents a rule that was triggered by a number of transactions at a specific timestamp.

Hover over each dot for more information regarding the rule's information and effectiveness:

- **Rule type**: Whether it's a self-service rule (SSR) or an out-of-the-box rule.
- **Rule owner**: The owner of the rule. In case of an out-of-the-box rule, the owner is Feedzai.
- **Hitrate count**: Absolute number of processed transactions that triggered the rule.
- **Affected amount**: The value of the transaction amount.
- **Precision**: The percentage of correctly triggered events in relation to all events that were predicted to trigger the rule.
- **Date**: The date the rule was triggered by a transaction of a certain amount.

- **Top 10 rules triggered by transaction amount**: shows the top 10 triggered rules by transaction amount. Each bar corresponds to a specific rule and the transaction amount that triggered it.
- **Top 10 rules triggered by number of transactions**: shows the top 10 triggered rules by number of transactions. Each bar corresponds to a specific rule and the number of transactions that triggered it.
- **Bottom 10 rules triggered by transaction amount**: shows the bottom 10 triggered rules by transaction amount. Each bar corresponds to a specific rule and the transaction amount that triggered it.

- **Bottom 10 rules triggered by number of transactions**: shows the bottom 10 triggered rules by number of transactions. Each bar corresponds to a specific rule and the number of transactions that triggered it.

Hover over each bar for more information regarding:

- **Rule type**: Whether it's a self-service rule (SSR) or an out-of-the-box rule.
- **Rule owner**: The owner of the rule. In case of an out-of-the-box rule, the owner is Feedzai.
- **Hit count**: Absolute number of processed transactions that triggered the rule.
- **Affected amount**: The value of the transaction amount.

-
- [API Resources]
- Additional Resources

Additional Resources

Was this page helpful?Â

Authentication

Authenticate through JWTs

Responses and errors

Learn the successful and error responses, as well as responses for timestamps out of range

Idempotency

Idempotency is a tool designed to ensure the safety and reliability of your API requests

Cards

Includes response scenarios, asynchronous operation, and generated fields

Digital Activity

Includes response scenarios, asynchronous operation, generated fields, and BACEN fraud-specific fields

Outbound Transfers

Includes response scenarios, asynchronous operation, and generated fields

Ad-hoc Investigations

Explains how to send Investigation events to Case Manager's Ad-hoc Customer Investigations channel

List Management

Explains how to use the available API to manage list items

Authentication

Feedzai allows you to authenticate and protect your data through JSON Web Tokens (JWTs). A JSON Web Token (JWT) is a piece of information that is transmitted as a JSON object from clients to Feedzai, and its purpose is to ensure that client requests (e.g. transactions for processing) have not been compromised.

These tokens are generated using the clientâ□□s private key, which can be verified with the corresponding public key. The public key must be shared with Feedzai beforehand to verify that the token sent with the request belongs to the client.

Overviewâ

JWTs protect against different attacks using the following mechanisms:

- **Encryption/signing** so that only authenticated users can access sensitive data.
- **Expiration dates** to protect against replay attacks.
- **JWT revocation** to block exposed JWTs as soon as possible.

You can see more details on integrating JWTs with your software in the sub-sections below, but in general you will:

1. Generate a private-public key pair to sign the JWT.
2. Provide the public key to Feedzai in JSON Web Key Set format.
3. Frequently generate new JWTs with a reduced validity period (e.g. 5 minutes). Each JWT is signed with your private key generated in step 1.
4. For each request to Feedzai, include a JWT in the `Authorization` header.

Given the recommended algorithm and parameters, you can choose the tool that best fits your system. A comprehensive list of tools is available at the libraries section of jwt.io.

danger

Third-party security vulnerabilities: When choosing a library, make sure it has no known security vulnerabilities.

JSON Web Keyâ

A key pair is necessary to cryptographically sign and verify the JWTs sent with your requests.

- You will use the private key to sign your JWTs.
- Feedzai will use the public key to verify the signature, and therefore the authenticity of the requests.

To create a JWK set, you need to choose the signing algorithm and the JWK parameters.

The following algorithms are supported:

- `RS512` (default)
- `RS256`

tip

Note that each algorithm must be specified in both the JWK and JWT json files - e.g. if you send a `RS256` JWT, you must specify `"alg":"RS256"` in the JWK.

Feedzai recommends the `RS512` algorithm (â□□RSASSA-PKCS1-v1_5 using SHA-512â□□) with a 4096-bit key. See the sections below to learn how to generate a JWK set with standard tools.

danger

Beware of private key parameters: Construct the JWK set so that you **do not** send any of the private key parameters.

Your JWK set should look similar to this:

```
{
    "keys": [
        {
```

```
            "e": "AQAB",
            "n": "187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqy
Owf6CBv6xj9ZnqyyScle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z730S4SlYe8rMDiWkFPybt1vfGB
dGqbvm374XNfLRb0sF7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh020iJ91DVvb9JMIFAqRO
HQNyzchbOWgEPUKoRBRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
            "kty": "RSA",
            "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
            "use": "sig"
        }
    ]
}

{
        "kid":"mykey2",
        "kty":"RSA",
        "alg":"RS512",
        "use":"sig",
        "e":"AQAB",
        "n":"k51ZPyu9gocn508aXzob_hSKaIAnJApY4pY3h1oym1Z15-6DN3WxB3YHUyXPRd-iBzF7177XjQAywTzAaz-
PKLZ4bC52noanLx7Ihyf3nQAOwhEv16zaJLI-
HqMEMp6JKPjKJ8LKanVnvxWNI0zUBwcBOwt9cxph0evyVO94cXA0RZxUZxa7C3FuSYkO2zWorMH0ZkC3harkJx0C9Q7nBiGW
PQmB8ZkLx04iWlm8ldyxDJPKl-PC4-
gS3y2mgPfZVAaK_VGod88PUmF4vFax7a03v2VLtn9wm8UWs9SXATxJ5cExgIIvFkdsZAcS3hWnhUe3Of4zAEbKh2Ah5DtfY9
xvtsU8FGlak7wLCwVnubWWglI2hmFvFU7Jsyzb_L-
jD7XCYrg7QuFV8mX2jVcTbsSxnd8GLRNbvmztvsjfaQHXEHraTRJFoGO1Y3JXDiY8Kj40OQXxBNXaFxHda9lBfzWfDkgWvcy
eiTPeWpKc6cS1zcDaQNRzZr3aZR-
gI4U8YTAXcIKDdom2GCQSx9WfmFq3uAj95duL5CgyGfw7L-02c6ckRr6kBAKebURkSsZXOyTO8bdbS8GzPMD59CrRL2SnJJe
qHyrEav8JCu0x3UCXbyqlfqQgzVsXNpFUMIS-sUhe3nyVQOeJ_WoYq7oAbjLpx_Bw5elQJf4zJbJd2Pk"
    }
```

In the aforementioned example, the following parameters are used:

Name	Extended name	Description
`e`	Exponent	RSA public exponent e.
`n`	Modulus	RSA public modulus n.
`kty`	Key Type	Should be `RSA` to symbolize you are using the RS512 algorithm.
`kid`	Key Id	Should specify a key identifier of your choice.
`use`	Public Key Use	Should be `sig` to note this key will be used to verify JWTs.

How to generate a JWK Set (JWKS)â

tip

This section describes how to generate a JWKS using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

Consider the following requirements if you're generating JWKSs in a different way:

- Make sure you store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**

- Manage JWKs in Pulse.

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```

1. Download Feedzai's JWT Generator.

info

Make sure you repeat steps 1 through 3 to create a public-private key pair for each environment.

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
  --mode JWK \
  --generateKeys \
  --privateKey "/path/to/privateKey" \
  --publicKey "/path/to/publicKey" \
  --kid "key_id"
```

   The JWK Set JSON will be displayed on your standard output. Alternatively, you can use the `--jwk` parameter to specify a file where it should be stored instead.
5. Store the generated public and private key files securely as they will be required to generate JWTs. **Never share your private key.**
6. [Manage JWKs](#) in Pulse.

JSON Web Tokensâ

Once you have a private key, you can generate your JWTs. All JWTs are composed of a set of headers, a set of claims and a signature.

JWT Headersâ

Besides the algorithm (`alg`), the only mandatory header for JWTs is the `kid` (the key ID), which indicates which key was used to sign the token and **must** match a valid key in the JWK Set used for validation. The set of headers should be similar to:

```
{
    "kid": "keyID",
    "alg": "RS512"
}
```

JWT Claimsâ

Typically, the following claims should be present on the JWTs:

Claim name	Extended name	Description
`iss`	Issuer name	Identifies the issuer of the JWT. This value will be provided by Feedzai, and it should match the tenant id, if applicable.
`exp`	Expiration Time	Date after which the JWT must be rejected.
`nbf`	Not before Time	Date before which the JWT must be rejected.
`jti`	JWT id	Identifier used to revoke JWTs.
`sub`	Subject	Identifies the user of the JWT.

Your final set of claims should be similar to:

```json
{
    "iss": "feedzai",
    "exp": 1523022355,
    "nbf": 1523022055,
    "jti": "a00cd6dd-4e55-4694-aeb5-4050e7d063b6",
    "sub": "700e55f9-15b0-4f95-8094-960138a0d886"
}
```

How to generate a JWTâ

tip

This section describes how to generate JWTs using Feedzai's JWT Generator. Note that you can use another one at your convenience - the use of Feedzai's JWT Generator **is not mandatory**.

1. Download Feedzai's JWT Generator.
2. Unpack the tar.gz artifact.
3. Check generator options:

```
./bin/generate-asymmetric --help
```

4. Run the JWT Generator using the following command. Adjust the parameter values according to your needs:

```
./bin/generate-asymmetric \
  --mode JWT \
  --privateKey "/path/to/privateKey" \
  --publicKey "/path/to/publicKey" \
  --iss "issuer" \
  --sub "subject" \
  --exp "$((EPOCHSECONDS + 300))" \
  --kid "key_id"
```

The JWT will be displayed on your standard output. Alternatively, you can use the `--jwt` parameter to specify a file where it should be stored instead.

How to send the signed JWTâ

danger

Note that you can only send the signed JWT after the corresponding JWK is uploaded to Pulse (see [Manage JWKs in Pulse]()).

Send the signed JWT in the `Authorization` header of HTTP requests made to Feedzai's API, as follows:

```
Authorization: Bearer <your_jwt>
```

Manage JWKs in Pulseâ

Once you create the JWK set, the following JWK steps must be taken:

1. Log into Pulse. Make sure you are accessing the right environment (i.e. **PROD**) and confirm the current list of Authorization keys. Before revoking any key, make sure you are no longer using it.
2. Add the new key to Pulse.
3. Start signing messages with JWTs based on the new key.
4. Revoke the old key.
5. Delete revoked Authorization keys.

List Authorization keysâ

To list the existing Authorization keys, access the **Administration** page and select the **Authorization Keys** tab on the sidebar.

Select a label name or expand to see the Authorization keys associated with that label:

Add Authorization keysâ

To add a new Authorization key, select a label name and upload the JSON file that contains the new key:

The JSON file should be an object with the key, using the following format:

```
{
    "e": "AQAB",
    "n": "187EWmXQwXi08P8bqSOSeOiyOyJGsojjmeL6qwLLjTBfxxeNgTtivzG8YdCgrLnBQWy4j9FqQ5wbqyOwf6CBv6
xj9ZnqyyScle7ubAKRt4RKtv4yrxVb1g5ZWcfD6yWDIH1jRy5oGQEwWRg9918Z73OS4SlYe8rMDiWkFPybt1vfGBdGqbvm374
XNfLRb0sF7cPCRcBa3Vdzmg5BgoN9kFUXIvFKjXdgZQGEEIonLL8ZSmqq993VBVLOPh02OiJ91DVvb9JMIFAqROHQNyzchbO
WgEPUKoRBRDXSCus9ZSpHuUl7iL7qKLaQdPyO1ffIG0o7kF6allFG6OBcnzqX8r",
    "kty": "RSA",
    "kid": "a28db6df-f2f4-4267-be46-371502768aa1sig",
    "use": "sig"
}
```

info

The new key must be associated with an existing label.

tip

Make sure you start signing the new messages with the new key.

Revoke Authorization keysâ

The following animation illustrates how to add a new key and revoke the current one.

danger

An Authorization key should only be revoked once a new Authorization key has been added (see [previous step]) and JWTs are already being signed with the new key.

Delete revoked Authorization keysâ

By deleting an Authorization key, you are permanently removing it from the list of keys. Only revoked keys can be deleted.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints

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Transaction Fraud for Banking solution

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- API Resources
- Additional Resources
- Historical Data Loading

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Historical Data Loading

Was this page helpful?Â

Historical data loading allows you to send past data that can be used to backfill previous events. This is done via API and does not compromise any event processing that is in progress.

This page explains the historical data loading's end-to-end process.

Purposeâ

Historical data loading is useful in the following situations:

- **Pre-activation**: During activation, historical data loading prepares the system by ingesting events and calculating the initial profiles.
- **Post-activation**:
  - Tenant onboarding: BaaS or multi-tenanted clients may onboard new tenants after the go-live. These tenants bring their own set of historical data that needs to be loaded into an already running environment.
  - Product expansion: Clients may decide to integrate additional use cases from Feedzai's portfolio or support new event types.
- **Recovery**: In case there are integration issues or other external factors, event data may be missing, which may lead to unalerted events. Data can be injected at a later stage as mitigation.

info

Historical data loading can be done manually or in an automated way. Either way, it needs to be performed by your IT team or similar integrator role.

See below what needs to be considered to load historical data.

Data preparationâ

Ensure that the historical data's format is consistent with normal API requests. Convert it accordingly, if necessary.

```
curl --location --request PUT 'http://localhost:8083/api/referenceData/customers/1791457' \
--header 'Content-Type: application/json' \
--header 'Backfill: true' \
--header 'Authorization: Bearer {token}' \
--data-raw '{
    "customer_id": "1791457",
    "customer_active": "true",
    "customer_alias_status": "active",
    "customer_segment": "Retail",
    "customer_name": "Anthony Manoel",
    "customer_email": "[email protected]",
    "customer_phone": "+351276333728",
    "customer_phone_val_code": "T",
    "customer_citizen_id_number": "3254756464",
    "customer_passport_number": "7856875675",
    "customer_driver_license": "76576865",
    "customer_ssn": "543654356",
    "customer_tax_number": "76587656785",
    "customer_registration_date": 1420074061000,
}'
```

Event backfillingâ

Historical data requests are injected through the usual endpoints.

Such requests must include a `Backfill` HTTP request header set to `true`. Note that omitting the header or setting the value to `false` will result in the request being processed in the regular way (i.e. not backfill).

Note that backfilling does not change an event's date and time, i.e. the original event timestamp (`event_occurred_at`) remains unchanged.

Recommendation

Send event data from most recent to oldest for efficiency reasons. This way, the backfilled data can more quickly mitigate gaps by contributing to the short-window profiles first.

To minimize the impact of backfilling data in ongoing processing latencies, historical data loading is configured with the following properties:

1. **Asynchronous processing**: Backfill requests will be processed asynchronously. This is enabled at endpoint level. When enabled, incoming events are placed in a queue, which is then consumed at a controlled, configurable rate. This allows the system to throttle received events.

2. **Backfill rate limiter**: The system will discard backfill requests if the number of requests per second exceeds a defined threshold. Requests exceeding the rate limit receive an HTTP 429 (Too Many Requests) response. This value is determined in each client based on the contracted tier and volume.
3. **Lower priority**: Backfill requests will be assigned a lower priority than regular events.

Historical data loading setup

Reach out to Feedzai for the initial feature configuration, so that the above properties are configured to best suit your system deployment.

Metrics updateâ

Once the data injection has been completed, a Railgun backfill must be manually triggered to update the metrics.

tip

Refer to the Pulse documentation to learn how to manually trigger a Railgun backfill.

Idempotency

caution

Note that idempotency is not enabled out of the box. To enable it, reach out to Feedzai.

This page includes information about the idempotency key feature implemented in the Transaction Fraud for Banking (TFB) RESTful API. The idempotency key is a tool designed to ensure the safety and reliability of your API requests. It consists of a unique identifier that you can include in your API requests to prevent duplicate operations from being performed unintentionally. It allows you to retry requests without worrying about the same operation being executed multiple times. The feature design will enable you to accomplish this if the retry requests occur under short time windows. Consult Feedzai to validate the time window in place for your implementation.

How it worksâ

When you include an idempotency key in your request header under the name "Idempotency-Key," TFB's API system checks if the same key has been used before within a time period. If the key is recognized, the system retrieves the previous response associated with that key and returns it to you without executing the request again.

Benefitsâ

The idempotency key prevents duplicate operations. By using the idempotency key, you can ensure that the same operation is not performed multiple times, even if your request is retried or encounters network issues. You can safely retry them without worrying about duplicate data on metrics or duplicate events.

Guidelines for using idempotency keysâ

- **Unique identifier**: Each idempotency key should be unique for every distinct request you make. Feedzai recommends generating a Universally Unique Identifier (UUID) for each key to ensure uniqueness.

Example Usageâ

To include the idempotency key in your API request, simply add a header named "Idempotency-Key" with a unique identifier for each request. Here's an example using cURL:

```
curl -X POST \
 -H "Content-Type: application/json" \
 -H "Idempotency-Key: YOUR_UNIQUE_ID_HERE" \
 -d '{"example": "data"}' \
 https://api.example.com/resource
```

Conclusionâ

The idempotency key feature is a valuable addition to our API, providing you with the assurance of reliable and safe operations. By following the guidelines outlined in this documentation, you can leverage idempotency keys to enhance the resilience and efficiency of your API integrations.

API Resources

This section provides you with the information you need to work with the API.

Endpoints

Learn all the operations you can do with Feedzai's API and how to communicate with its multiple endpoints.

Additional Resources

Find information about authentication, response and errors, generated fields, among other resources necessary to successfully work with the API.

Responses and Errors

Was this page helpful?Â

Successful responseâ

Successful calls to Feedzai's API result in a response with HTTP 200 code. Please check each endpoint's documentation to see the structure of a successful response.

Warning responseâ

A warning response is a subtype of a successful response which is sent with the HTTP 200 code. The transaction is also processed; however, the response returns the `status` field "warning" and the `warnings` field explaining the reasons for this outcome.

Error responseâ

When a call to Feedzai's API results in an error, the response object returned always has the same fields. In particular, the `status` field is shared with the successful response objects of all endpoints. The **Error Response** object is defined as follows:

Name	Field Type	Description
`code`	String	An indicative code for the error. See possible values below.
`errors`	Array of strings	A list of free text explanations for the error.
`status`	String	Always with value `error` .

Generic Error codes - Possible Valuesâ

Error code	Description	HTTP Code
`authenticationError`	The request could not be authenticated.	`401`
`authorizationError`	The authenticated user does not have permission to access the endpoint.	`403`
`parseError`	The request could not be parsed.	`400`
`invalidParameterError`	The URL parameter provided is invalid.	`400`
`invalidTimestampError`	The timestamp provided in the request is invalid.	`400`
`nonexistentEndpoint`	The endpoint being called does not exist.	`404`
`unsupportedMethod`	The HTTP method to call this endpoint is not supported.	`405`
`tooManyRequests`	The allowed request rate was exceeded.	`429`
`internalError`	Feedzai's system produced an internal error.	`500`
`timeoutError`	Feedzai's system failed to produce a timely response.	`504`

Please check each endpoint's documentation for specific error codes.

Responses for timestamps out of rangeâ

This subsection describes responses for requests with timestamps that are out of the valid range.

Endpoint	Timestamp	Response HTTP code	Error or Warning	Is transaction processed?
Authorization	`event_occurred_at` in the past	400	Error	No
Authorization	`event_occurred_at` in the future	400	Error	No
Info, Clearing, Feedback	`event_occurred_at` in the past	200	Warning	Yes
Info, Clearing, Feedback	`event_occurred_at` in the future	400	Error	No

Error Response Example:

```
{
    "status": "error",
    "code": "invalidTimestampError",
    "errors": [
        "Event has timestamp in the future: <EVENT_OCCURRED_AT> | Now is:
'2018-09-04T16:41:30.774Z'"
    ]
}
```

Warning response example:

```
{
    "lifecycle_id": <LIFECYCLE_ID>,
    "event_external_id": <EVENT_EXTERNAL_ID>,
    "warnings": [
        "Event has timestamp in the past: <EVENT_OCCURRED_AT> | Now is:
'2018-09-04T16:40:40.522Z'"
    ],
    "status": "warning"
}
```

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/fraud-scenarios/cards

Card Payments

Was this page helpful?Â

This page covers the fraud scenarios and the detection strategy for card payments.

Cardingâ

Carding is a term for criminals who attempt to access credit or debit card credentials for their personal interest. To steal a person's card details, fraudsters may resort to methods such as manipulating bots and botnets. These criminals usually sell this information on the dark web with the purpose of increasing their income.

The goal of this detection scenario is to prevent attackers from attempting to access card credentials to perform transactions.

How Feedzai prevents carding

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A card must be declined and flagged to be limited if the number of high amount transactions made by the same card in the past day exceeds the threshold. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A transaction must be declined and card flagged to be limited if count and sum amount in the same acceptor in the last 24 hours exceeds threshold.

Card testingâ

While in the carding scenario criminals use methods to steal card information, in card testing they use the stolen card information to attempt to make transactions. This scenario is common within e-commerce, where 'card testers' make several attempts until they succeed at purchasing goods or services.

The goal of this detection scenario is to prevent card testers from succeeding at purchasing goods or services with stolen card information.

How Feedzai prevents card testing

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A transaction must be declined if a CNP transaction has been performed after several attempts in an MCC known for card testing in the past 24 hours.
- A rule is triggered if the ratio of distinct `card_id` and distinct expiration dates per BIN and Merchant in the last 10 minutes does not exceed the threshold. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A rule is triggered if the ratio of distinct `card_id` and distinct expiration dates per BIN and MCC in the last 10 minutes does not exceed the threshold.

Account takeoverâ

Account takeover is one of the most common forms of credit card fraud. In this scenario, a fraudster gets access to a person's information (e.g. credentials, relevant documents), typically by hacking that person's online accounts (e.g. bank accounts, emails, subscriptions).

These criminals may use this information for several purposes. As an illustration, they may contact that person's credit card company and ask for an address change. The company grants this request because these individuals provide proof of identity, by confirming the person's details. As a result, the company replaces the person's card with a new one and sends it to the new fake address.

How Feedzai prevents account takeover

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A transaction must be declined and card flagged to be limited if the transaction is performed at a high fraud rate acceptor, without card activity in the past 6 months. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the Resources section.
- A rule is triggered if the count of distinct devices per card in the last 24 hours exceeds threshold.
- A rule is triggered if, in the past hour, IP exceeds traveling velocity.

Lost & stolenâ

The lost & stolen scenario occurs when a person is victim of physical theft, i.e. a criminal accesses the victim's pocket or purse and uses the card for unauthorized transactions. These individuals use the card until the cardholder reports to the issuing bank, which blocks the account.

How Feedzai prevents lost & stolen

Since within this type of scenario it is common for criminals to make many transaction in short periods of time, other rules capture user patterns associated with this fraud scenario. Here are some examples:

- A card must be flagged to be limited if two transactions with the same amount were performed in the same acceptor in the past 24 hours. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A transaction must be declined and card flagged to be limited if the amount spent in the past 4 weeks exceeds the threshold.
- A transaction must be declined and card flagged to be limited if the number of distinct devices per card is exceeds the threshold.

Card phishingâ□□

Card phishing is the act of sending fraudulent emails to extract card credential information from the victims. In these emails fraudsters pretend to be from a legitimate source by using an official company's logo and wording. Usually, the content of the email varies in nature (e.g. email compromise, urgent requests, bank account irregular activity). Fraudsters succeed at phishing when the victims access a seemingly legitimate link, which in reality is a cloned version of the official company's link.

Similarly to the lost & stolen scenario, high amounts of transactions are made in a short time.

How Feedzai prevents card phishing

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A transaction must be declined if the device is infected with malware.
- A transaction must be declined and card flagged to be limited if the sum amount in the last 4 weeks exceeds threshold.
- A card must be declined and flagged to be limited if there is a high amount transaction with an unusual currency (CNP). To understand how this is achieved with a rule included in the standard package, check the rule information under the References section.

Skimmingâ□□

Skimming is a type of fraud that consists in stealing a person's card details and using them in a fraudulent card, i.e. electronic copying card information to make cloned cards. Fraudsters may use several methods to do this. For instance, they may use a skimming device that swipes the victims' card information, or simply use the victim's receipts to extract the card details from there. They then proceed to create "fake plastic" cards to store the stolen victim's information.

This scenario is common in touristic countries and may take place in ATMs, gas stations, restaurants, etc. Victims may only notice their card details were stolen when they go back to their home country. They confirm through the bank account movements that their card is being used to purchase goods in a different currency, i.e. the currency from the country they just visited.

How Feedzai prevents skimming

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A transaction must be declined if the device is infected with malware. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A transaction must be declined and card flagged to be limited if the number of distinct devices per card is exceeds the threshold.
- A rule is triggered if the distance between current and previous IP location in the last hour exceeds 1000 km.

Card-not-Present fraudâ

Card-not-present (CNP) fraud is prevalent in transactions made over the internet, but may also take place over the phone. In the CNP scenario, fraudsters are able to purchase goods without a physical card, since they have already managed to access the victim's card credentials. To steal a victim's card details, fraudsters typically resort to methods such as hacking, phishing or skimming. CNP transactions are difficult to prevent due to the lack of a physical card, which makes it impossible for merchants to detect any anomalies in the card (e.g. lack of either a magnetic stripe or a chip).

How Feedzai prevents CNP fraud

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A transaction must be declined and card flagged to be limited if total amount spent by card in the past 24 hours at single merchant (acceptor_id) exceeds the threshold.
- A card must be flagged to be limited if there are multiple CNP transactions in several unique merchants (acceptor_id) in the past 5 minutes. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A transaction must be declined and card flagged to be limited if number of transactions in the past 24 hours exceeds the threshold.

CNP (bot/velocity)â

A bot is a software or configurable script that runs scripts over the internet and attacks e-commerce merchants and financial institutions. This is usually performed by botnets, a group of internet devices running one or more bots. Fraudsters can easily rent bots and adapt botnets (for different malicious purposes) to deploy them fast. Dozens of purchases with the same card in a short period of time (e.g. within 5 minutes) is a good example of a bot attack.

How Feedzai prevents CNP (bot/velocity)

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A transaction must be declined and card flagged to be limited if travel velocity between the last transaction and the current transaction in past hour is unreasonably high (card present). To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A transaction must be declined and card flagged to be limited if the number of transactions and total amount spent by card in the past hour exceeds the threshold.
- A transaction must be declined if the number of transactions in the past 3 minutes exceeds the threshold.

Learn moreâ

Now that you have learned about the most common card payment fraud scenarios, Mapping rules to fraud scenarios provides you with a detailed overview of which card rules are used to detect each of these scenarios.

Device and Behavior

Was this page helpful?Â

This page covers the account takeover scenario and its threats, and how to detect them by using device and behavioral data points.

Account takeoverâ

Account takeover (ATO) is the act of obtaining access to a person's bank account, typically by logging into their online account. Once they have control, fraudsters can easily transfer or wire money out of the victim's account into their own.

The most common threats leading to this fraud scenario are as follows:

Phishing is the act of sending fraudulent emails to extract account credentials from the victims. In these emails fraudsters pretend to be from a legitimate source by using an official company's logo and wording. Usually, the content of the email varies in nature (e.g. email compromise, urgent requests, bank account irregular activity, etc.). Fraudsters succeed at phishing when victims access a seemingly legitimate link, which in reality is a cloned version of the official company's link. In an ATO fraud scenario, the purpose of phishing is to obtain as much information about the victim as possible to take over their account.

Here's an example of a rule that captures user patterns associated with this threat:

- Risky activity with a "phishing" flag and device with high Session Risk Score.

info

Session Risk Score is a score generated by Digital Trust. It considers all the behavioral, device, and network indicators into a single score. The total risk score ranges from 0 (extremely low risk) to 100 (extremely high risk).

Malware is any software intentionally designed to cause damage or perform a fraudulent activity. Malware can get installed into the victimâ□□s device without them noticing. Once installed, criminals can access useful data such as passwords or PII to perform ATO.

Here's an example of a rule that captures user patterns associated with this threat:

- Decline transaction if the device is infected with malware.

Suspicious network activity. Some aspects of network activity provide Feedzai with important information regarding risky users. Typically, suspicious network activity is detected via a combination of anonymous web routing, TOR, Proxy, VPN, or Remote Access Tool (RAT) connections with the right intelligence.

Here's an example of a rule that captures user patterns associated with this threat:

- Login attempt from a dormant account with Session Risk Score and suspicious network activity above threshold.

tip

Feedzai recommends that data scientists combine data points such as phishing, malware, suspicious network, etc, with Session Risk Score to obtain solid detection results with low false positive rates. By combining data points with Session Risk Score, it is possible to confirm, for instance, that an online banking session with data points such as malware and/or VPN is suspicious.

On this page

Digital Activity

Was this page helpful?Â

This page covers the fraud scenarios and the detection strategy for digital activity events.

Account takeoverâ

Account takeover (ATO) is the act of obtaining access to a person's bank account, typically by logging into their online account. Once they have control, fraudsters can easily transfer or wire money out of the victim's account into their own.

The most common threats leading to this fraud scenario type are as follows:

- **Social engineering**, within the ATO scenario, may involve collecting information about the victim from various sources, such as phishing scams, in order to take over the victim's account.
- **Brute force attack** consists of an attacker submitting many passwords with the hope of eventually guessing the correct combination of characters. The perpetrator can also take advantage of the victimâ□□s personally identifiable information (PII) to change the victim's password. After successfully guessing the password, the event becomes an ATO scenario.
- **Phishing** is the act of sending fraudulent emails to extract account credentials from the victims. In these emails fraudsters pretend to be from a legitimate source by using an official company's logo and wording. Usually, the content of the email varies in nature (e.g. email compromise, urgent requests, bank account irregular activity, etc.). Fraudsters succeed at phishing when the victims access a seemingly legitimate link, which in reality is a cloned version of the official company's link. In an ATO fraud scenario, the purpose of phishing is to obtain as much information about the victim as possible to take over their account.
- **Password theft** is a method that may have several variants. For example, cybercriminals can focus on PII to try to match or reset the victimâ□□s password. They may also develop complex tools to decipher passwords, and try them on thousands of sites expecting the same password to be used across multiple accounts.
- **Malware** is any software intentionally designed to cause damage or perform a fraudulent activity. Malware can get installed into the victimâ□□s device without them noticing. Once installed, criminals can access useful data such as passwords or PII to perform ATO.
- **SIM swap**, also known as phone hijacking or port-out scam, aims at succeeding on two-factor authentication mechanisms, where the second authentication happens through an SMS code or mobile phone call. The perpetrator starts by collecting the victim's PII data, either by sending phishing emails or by buying data from other criminals (e.g. dark web). Next, they contact the victimâ□□s mobile telephone provider and use social engineering techniques to convince the company to port the victimâ□□s phone number to the fraudsterâ□□s SIM. For example, this is done by impersonating the victim using the stolen personal details and claiming that they have lost their phone. This will allow the criminal to take over the victimâ□□s account by intercepting any one-time password used during the login or transfer of funds.
- **Fake apps** are Android or iOS applications that mimic legitimate applicationsâ□□look and feel to trick unsuspecting victims into installing them. Once downloaded and installed, similarly to the malware, the fake applications perform various malicious actions benefiting a criminal to perform multiple types of financial crime.

How Feedzai prevents account takeover

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule may be used to block a login attempt originated from a device Id that is listed to be auto declined. To understand how this is achieved with a rule included in the standard package, check Posneg rules under the References section.
- A rule may be used to block a login attempt after a certain number of failed login attempts takes place per account at the last minute. Check this rule information to know more about the standard usage of this rule under the References section.
- An alert is triggered if the number of daily transactions and digital activity (e.g. logins) for a dormant account exceeds a defined threshold.

info

It is crucial for Feedzai to receive digital activity events related to banking sessions in order to detect ATO scenarios. Such events involve login, phone and email address changes, and password resets. For instance, a fraudster may perform a SIM swap scam by using the victim's phone number to reset the security token. Changing the victim's email may also be used to reset a password and take full control of an account.

Authorized push payment fraudâ

Authorized push payment (APP) fraud is the act of criminals tricking their victims into transferring money to their accounts. For instance, these individuals may act as if they are from a legitimate company or entity, such as the victim's bank. To sound convincing, fraudsters mention personal details about the victim, which they have collected from the victim's social media and/or other public spaces that show their profiles. Fraudsters then advise the victims to transfer their money to another account (e.g. a safe one), as their current account has been compromised.

After realizing they transferred money to a scammer, the victims may never recover their stolen money; therefore, it is recommended that they contact their banks as soon as possible.

Consider the following threats, which are the most common methods to perform an APP:

- **Social engineering**, within the APP fraud scenario, is the psychological manipulation fraudsters use to trick victims into giving away money. Take the example of a romance scam, where a fraudster wins the victim's affection and trust. The fraudster then tricks the victim into actively sending money to the fraudster (e.g. for a plane ticket to meet them). As another example, in some instant payment platforms (e.g. Zelle, NPP, PIX, MBWay) a fraudster may try to add several mobile phones/emails in order to find an existing user to perform an Authorized Push Payment fraud scheme via social engineering.
- **Phishing**, in the APP fraud scenario context, is the act of urging the victims to transfer their money to a safe account (i.e. the fraudster account), as their bank account has been compromised. Usually, fraudsters use an urgent tone of voice to convince their target to move quickly and transfer their money.

How Feedzai prevents APP fraud

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule may be used to decline a transaction if the transaction takes place outside the usual time of the day and if the transfer is made to a new receiver within the last 6 months. To understand how this is achieved with a rule included in the standard package, check Fraud Rules under the References section.
- An alert may be triggered if within 90 minutes the count of distinct new receivers from a particular sender is greater than or equal to 3.
- An alert should be raised if, within the last 24 hours, the number of transactions between a sender and a new receiver is greater than or equal to 3 and each amount exceeds USD 5,000.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/fraud-scenarios/inbound-payments

Inbound Payments

Was this page helpful?Â

This page covers the fraud scenarios and the detection strategy for inbound payment events.

Money mulesâ

Money mule is a term for persons or accounts used by criminals to perform money laundering activities such as receiving and transferring funds illegally. These intermediaries are in most cases unemployed people (e.g. students) who accept easy jobs promoted online (e.g. fake job adverts, social media, etc.). The only requirement they get from their 'job agents' is a bank account which is later used for money laundering. They may not know that the money usually comes from fraudulent activities such as phishing, botnet attacks, skimming, etc. They receive the stolen money in their bank accounts and are requested to transfer or wire that money to another account (usually located overseas), thus laundering money knowingly or unknowingly.

Money mules are divided into the following typologies:

- Dedicated mule - an account that was opened purely with intention of becoming a mule.
- Unwitting mule - a genuine customer who was tricked into becoming a mule.
- Account takeover mule - an account that was taken over by fraudsters and used as a mule.

How Feedzai prevents money mules

Transaction Fraud for Banking provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule triggers an alert if the daily funds received in a dormant account exceed the defined threshold and transaction amount exceeds the percentage threshold of the account balance.
- A rule triggers an alert if the ratio of funds sent to received funds is outside a certain range and the amounts transferred exceed the defined threshold.

Scams: authorized push payment fraudâ

Authorized push payment (APP) fraud is a scam that consists of criminals tricking their victims into transferring money to their accounts. They can resort to different scams such as love scam, investment scam, impersonation scam, etc. For instance, these individuals may act as if they are from a legitimate company or entity, such as the victim's bank. To sound convincing, fraudsters mention personal details about the victim, which they have collected from the victim's social media and/or other public spaces that show their profiles. Fraudsters then advise the victims to transfer their money to another account (e.g. a safe one), as their current account has been compromised.

After realizing they transferred money to a scammer, the victims may never recover their stolen money; therefore, it is recommended that they contact their banks as soon as possible.

How Feedzai prevents scams

Transaction Fraud for Banking provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule triggers an alert if funds and number of transactions received from international sources over the last 24 hours exceed the defined thresholds.
- A rule triggers an alert if funds received over the last 24 hours differ significantly from the expected amount considering the previous 6 months.

Fraud Scenarios

Learn about the TFB strategy to fight different fraud schemes using default rules that are provided out-of-the-box. This is a critical step to help fraud teams to identify suspicious behavior and stop attacks before they happen.

TFB is a solution that provides out-of-the-box capabilities to fight multiple payment methods:

- **Card Payments:** The fraud scenarios and detection strategy that applies to card payments accounting for both card-present and card-not-present payments.
- **Transfers:** The fraud scenarios and detection strategy that applies to transfers account for different models such as ACH, Wire, SEPA instant credit, and SEPA direct debit.
- **Digital Activity**: The fraud scenarios and their threats, which apply to transfers related to banking sessions.

- **Device and Behavior**: The fraud scenarios and their threats, which apply to device and behavior patterns in banking sessions.
- **Inbound Payments**: The fraud scenarios and their threats, which apply to inbound payments.

Each page provides a thorough description of the different fraud scenarios associated with the payment method and how TFB prevents them from occurring. The fraud-fighting strategy comes with clear examples of how rules can detect unusual behavior.

For each payment method, the solution provides a set of default rules. Most rules are common to multiple fraud scenarios as they share similarities. For example, in card payments, triggering a rule that looks at a maximum amount per card within 24 hours is one of the traces for carding, card testing, account takeover, skimming, card phishing, among other fraud scenarios. In transfers, the maximum transaction amount per sender or receiver or transactions without sender/receiver information are also an indicator of multiple fraud scenarios, including, for example, account takeover, phishing, money mules and fake invoice fraud.

Learn moreâ

To complement the fraud scenarios and detection strategy, this documentation includes a catalog of default rules containing all the information on how they are structured (such as rule explanations, conditions, actions, etc.). Check the Rules Projects overview to learn more about this.

You can also check the Work with Rules and Lists guide to learn:

- Writing rules specific to your business.
- Configuring your work in the runtime workflow.
- Testing rules using the Transaction Fraud for Banking API.

Outbound Transfers

Was this page helpful?Â

This page covers the fraud scenarios and the detection strategy for outbound transfers.

Account takeoverâ

Account takeover is the act of obtaining access to a person's bank account, typically by logging into their online account. Once they have control, fraudsters can easily transfer or wire money out of the victim's account into their own.

How Feedzai prevents account takeover

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule is triggered if the transaction count made by a particular sender in the last month is considerably greater than the monthly average transaction count from the last 3 months. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A transaction is declined if the count of transactions from a particular sender to new receivers within the last 24 hours exceeds USD 5,000.
- A rule detects if the transaction amount of an instant transfer is greater than or equal to USD 15,000.

Authorized push payment fraudâ□□

Authorized push payment (APP) fraud is the act of criminals tricking their victims into transferring money to their accounts. For instance, these individuals may act as if they are from a legitimate company or entity, such as the victim's bank. To sound convincing, fraudsters mention personal details about the victim, which they have collected from the victim's social media and/or other public spaces that show their profiles. Fraudsters then advise the victims to transfer their money to another account (e.g. a safe one), as their current account has been compromised.

After realizing they transferred money to a scammer, the victims may never recover their stolen money; therefore, it is recommended that they contact their banks as soon as possible.

How Feedzai prevents APP fraud

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A transaction is declined if the count of instant transfers in the past 24 hours is 3 times greater than the count over the last 3 months. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A rule is triggered if, in the past 90 minutes, the count of distinct new receivers from a particular sender is greater than or equal to 3.
- An alert should be raised if, in the last 24 hours, the number of transactions between a sender and a new receiver is greater than or equal to 3 and each amount exceeds USD 5,000.

Phishingâ□□

Phishing is the act of sending fraudulent emails to extract account credentials from the victims. In these emails fraudsters pretend to be from a legitimate source by using an official company's logo and wording. Usually, the content of the email varies in nature (e.g. email compromise, urgent requests, bank account irregular activity, etc.). Fraudsters succeed at phishing when the victims access a seemingly legitimate link, which in reality is a cloned version of the official company's link. As a result, many transactions are made in short time.

How Feedzai prevents phishing

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- An alert should be raised if, in the past 90 minutes, a sender makes transactions to distinct new receivers, exceeding the threshold.
- Transfers with 5 or more distinct currencies from the original bank's currency raise an alert.
- A rule is triggered if a customer makes transactions with persons involved in criminal activities (e.g. people listed to be declined). To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.

Money mulesâ□□

Money mule is a term for persons, who knowingly or unknowingly, are recruited by criminals to perform money laundering activities such as receiving and transferring funds illegally. These intermediaries are in most cases unemployed people who accept easy jobs promoted online (e.g. fake job adverts, social media, etc.). The only requirement they get from their 'job agents' is a bank account which is later used for money laundering. What they may not know is that the money usually comes from fraudulent activities such as phishing, botnet attacks, skimming, etc.

Money mules receive the stolen money in their bank accounts and are requested to transfer or wire that money to another account (usually located overseas). This is a common way to launder money.

How Feedzai prevents money mules

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- If the number of distinct currencies transferred to a particular receiver in the past month is greater than or equal to 4 an alert is raised. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.
- A rule is triggered if 5 or more senders are using shared contact information, such as address, telephone or email, with no reasonable explanation.
- A rule is triggered if, in the past 24 hours, a particular sender made a transaction with an amount 3 times greater than the daily average.

Fake invoice fraudâ

Fake invoice fraud is a form of stealing from people who actively yet unknowingly, transfer money to fraudsters. In a business, a clerk may receive an invoice concerning a product (i.e. a good or a service) that the company 'had ordered'. Since companies pay for several goods and services, these employees may not realize that the invoice is actually a scam. For instance, the invoice may be related to the request of extra telecommunications services or changes in contract, which involves additional costs.

This type of fraud is not exclusive to companies. As an illustration, if a person is buying a house, they may receive an invoice from their 'solicitor' listing extra goods or services they don't recall having requested. However, since the email or letter look genuine (i.e. containing the solicitor's logo and sophisticated wording), they transfer money to the fraudulent account, which is quickly wired to another one, making it difficult to recover.

Usually, criminals committing this type of fraud have access to the victims' information (e.g. through phishing, botnet attacks, etc.), which is why their invoices look genuine.

How Feedzai prevents fake invoice fraud

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule is triggered if the number of bank accounts with shared phone number exceeds the threshold. To understand how this is achieved with a rule included in the standard package, check this rule information under the References section.
- A rule is triggered if the count of distinct receivers from a particular email address in the past month exceeds the threshold.
- The count of transactions greater than or equal to USD 1,000 in the past 24 hours must not be greater than or equal to 5.

Payroll scheme (ghost employee)â

Payroll scheme is a type of fraud that may be performed by employees, payroll staff or even the company itself. This scenario may occur in three ways:

- The employee reports extra worked hours. In this case, they may over-report worked hours in their timesheet or work together with the payroll clerk to forge the payroll record.
- An employee has just left the company but the payroll clerk/staff does not terminate his or her pay, and instead redirects that pay to their own account or splits with someone cooperating in the scam.
- To avoid paying taxes, the company may deliberately misclassify their employees so that they cover for their own taxes.

How Feedzai prevents payroll scheme

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule is triggered if, in direct deposits, the count of distinct senders to the same receivers in the past 3 months is greater than or equal to 9.
- A rule is triggered if, in the past 6 months, the number of transactions made by a particular sender to new distinct receivers exceeds the threshold.
- An alert is raised if, in a direct debit, there's a geographic location mismatch between the sender and the receiver. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.

Mandate fraudâ

In the mandate fraud scenario, organizations are the targeted victims. For instance, a fraudster claims to be a member of an organization (e.g. a supplier) that receives regular payments from the victim's organization. This individual requests the victims to change the payment details to reflect the fraudster's new bank account. To maintain the direct debit payment method, the victims contact their banks to change the transfer mandate to the fraudster account.

Similarly to the APP fraud, fraudsters can easily access their victims' business details (e.g. which suppliers they have contracts with) by researching their social media, website, among other spaces.

How Feedzai prevents mandate fraud

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule is triggered if, in the past 3 months, the count of direct debits from a particular sender to new receivers exceeds the threshold.
- A rule is triggered if, in the past 3 months, the count of direct debit payments made into an account with no activity exceeds the threshold.
- An alert is raised if there's a direct debit from a company in the high-risk direct debit's list. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.

CEO fraudâ

CEO fraud occurs when criminals email the Finance or Accounting department of a company, impersonating as important executives. The content of the email is convincing enough that employees transfer or wire money to the provided bank account. This is called social engineering, a term used for successfully manipulating people into behaving promptly and precipitately as a response to a fraudulent request.

How Feedzai prevents CEO fraud

This solution provides a set of default rules that capture user patterns associated with this fraud scenario. Here are some key examples:

- A rule triggers if in the past 6 months, the number of direct deposits to new distinct receivers per sender is greater than or equal to 25.
- A rule identifies if, in the past 24 hours, a sender made 3 or more transactions with amounts of USD 5,000 each to a new receiver.
- A rule is triggered if a particular sender makes transactions outside their usual business hours. To understand how this is achieved with a rule included in the standard package, check the relevant rule information under the References section.

-
- [Operational Work]
- [Fraud Prevention Agent]
- [Investigate Alerts]
- Access Workload

On this page

Access Workload

Was this page helpful?Â

As a fraud analyst, the first step you should take is to access the workload. Depending on the risk engine's decision, you may find a set of events for review.

This page helps you understand how to access the events that are up for review.

List eventsâ

When events arrive at Case Manager, they can be displayed in more than one page, depending on the fraud risk and the complexity. On Case Manager's sidebar you can access:

- **Search By Event :** Displays all the events in Case Manager. Note that, in a transaction fraud use case, an event may correspond to a transaction associated with a card payment, transfer, or digital activity.
- **Alerts :** Events that trigger a risk engine rule with the `alert` action - flagging a high fraud risk. When such rule is triggered, Case Manager displays the transaction both in the Search by Event and Alerts pages. The Alerts page allows you to focus only on triggered events.
- **Cases :** Events and alerts can be grouped into cases for a more organized action-taking process. This allows you to access events that may be correlated and that require a thorough investigation. Cases offer extra functionalities, such as note attachment. In addition, cases can also be automatically created through **Automation Rules**. The following examples illustrate how both the note attachment and case automatic creation functionalities can help your investigation:
    - A fraudster tests a card in several charity merchants and then performs transactions in other merchants. All these events can be grouped into a case to give you a holistic view of the fraudster behavior. While reviewing this case, you can add notes, such as "send a message to the customer to confirm transaction authenticity".
    - Imagine that a transaction associated with a given Permanent Account Number (PAN) triggers an alert and opens a case automatically. As a consequence, future transactions up to a configurable time range for the same PAN (suspicious or not) are grouped into this case.

The following figure gives you an idea of how events, alerts, and cases relate to each other:

tip

Events versus Alerts: Commonly, fraud analysts use the Alerts page to review suspicious behavior, whereas the Search By Event page is used to search for information. For example, you could use the Search By Event page to find all the transactions performed by the same card brand as a way to obtain extra information to support your investigation.

Since there is a much higher percentage of genuine transactions relative to fraudulent ones, the Search By Event page is not an effective way to access your workload.

Filter eventsâ

Each page allows you to narrow down the workload for your analysis through filters.

To understand how filters play a role in your workload, take the following examples:

- **Queues**: Define different priorities for your workload. Events, alerts, and cases go into a queue, which is associated with a number of fraud analysts who retrieve workload from that queue. Use the **More Actions** button to move events and cases to Queues from the Events and Cases / Case Details pages. Note that TFB already offers out-of-the-box queues that organize events, alerts, and cases in queues automatically. If the default queues do not address your needs, you can create custom queues.
- **Assignee**: Events can be assigned directly to fraud analysts. This means you can filter the events that are already being handled by other fraud analysts.

Access alertsâ

To work with single events, perform the following steps:

1. Using the sidebar, enter the **Alerts** page to access the full list of alerts.
2. The **All Channels** tab is selected by default. You can view the alerts per channel by selecting the corresponding tab.

tip

Case Manager displays alerts originated from card, transfer, and digital activity events using different channels. The channel tab selector is available both on the **Search By Event** and **Alerts** pages.

Note that you also have access to dropdown menus with filter options associated with queues to narrow down your search.

Access casesâ

To work with cases, perform the following steps:

1. Using the sidebar, enter the **Cases** page to access the full list of cases.

Case automatic openedâ

Imagine that you want to retrieve your work from a case that was automatically opened and sent to a queue named *Case Automatic Opened*. To quickly find your case:

1. Select the filter *Case Automatic Opened* under **Queue**.
2. Use the button to filter your search.
3. Open the case to access the list of linked events as well as the activity log. See the [Analyze Events] step to learn how to investigate events.

info

For information on the different event statuses, check the Event Statuses and Investigation Flow page.

Learn how to create and review cases in Manage Cases.

Next stepsâ

In the next step of the fraud analyst journey, you will learn how to analyze events.

On this page

Ad-hoc Customer Investigations

Was this page helpful?Â

Ad-hoc customer investigations provide fraud prevention agents with a generic investigation page to log any intelligence gathered from investigations outside the usual event flow. These events allow agents to leave notes and attachments, track the status of investigations and add them to a case.

About widgets

This page documents the available widgets in a standard order; however, the order in Case Manager may vary depending on your setup. Use the table of contents on the right to navigate to widget-specific documentation and learn more about the information available in each section.

Assign alerts to yourselfâ

Before starting the investigation process, start by assigning a single or a set of alerts to yourself. The product can be configured to prevent fraud prevention agents from performing any action on an alert until it is assigned either to them or to another agent. In turn, this prevents different agents from reviewing the same alert at the same time.

To unlock the event, open the **Unassigned** dropdown in the header and select **Assign to me.**

Assign alerts to other agents

The **Choose Assignee** option is only available to some users upon a specific permission.

In addition, you can use the **Assign All** option to assign all linked alerts to yourself. This enables you to easily act on, for example, all unreviewed alerts from the same customer at the same time.

When assigning all linked alerts, a new widget is added to the **Event Details** page with the list of linked alerts to better support the investigation process.

Depending on the configuration, there may be a limit to the number of linked alerts that can be assigned at once. If a limit is set and the **Assign All** action exceeds that number, a message appears specifying how many alerts were assigned.

The remaining alerts that were not assigned to you during this action will remain available for you or another agent to assign and review.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then the **Assign to me** option. Assigning multiple alerts is helpful to, for example, assign all alerts from a specific queue to yourself.

assign-and-lock rules

Fraud Prevention Agents - L1 and **Fraud Prevention Agents - L2** need to assign alerts to themselves to act on them.

Fraud Prevention Agents - L2 and **Fraud Prevention Managers** can reassign to themselves alerts that have already been assigned to other agents.

Only **Fraud Prevention Managers** can act on alerts assigned to other agents; therefore, they don't need to assign alerts to themselves.

Event headerâ

The following fields provide general information about the event to the fraud prevention agent:

- **Event ID**: The event's ID, which can be copied to share with other agents or track later in the system.
- **Assignee**: Indicates if the event is assigned to an agent. The dropdown button lets you assign the event.
- **Labeling**: Includes the final decisions an agent can take after investigating a customer. The possible options are:
  - **Drop**
  - **Confirm Drop** (in case another agent had already dropped the event)
  - **Action**
- **Alerted**: Indicates whether the ad-hoc event comprises an alert (e.g. triggered one or more alerting rules).
- **Date**: The date and time the ad-hoc investigation was generated for that customer.
- **Customer ID**: The unique ID number of the customer under investigation.

- **Customer Name**: The name of the customer under investigation.
- **Fraud Operational Status**: Indicates the ad-hoc investigation stage. Possible values are:
  - **Unreviewed**: Has not undergone investigation.
  - **In progress**: Under review by an agent.
  - **On hold**: Temporarily paused while waiting for extra information.
  - **Closed**: Final status after making a decision.
- **Analyst Decision**: The decision to either take action or drop the event. The possible values are:
  - **Action taken** (i.e. risky)
  - **Dropped** (i.e. not risky)
- **Reasons**: Indicates the reasons why the agent considered the alert irrelevant (**Dropped**) or needed further investigation (**Action taken**).
- **Queue**: Indicates in which queue the ad-hoc investigation event is, based on the investigation priority. Queues can be configured by a fraud prevention manager.

Summaryâ

Customer Detailsâ

This widget contains information about the customer under investigation, such as the customer ID, customer email and phone number.

You may also come across a situation that requires adding the customer to a list, or reviewing the customer if they are already in a list. Information fields that are eligible for lists have the **Update Lists** icon. Selecting the icon opens the **Update Lists** dialog.

Investigation Detailsâ

This widget shows where the investigation originated from and the reasons for creating the ad-hoc customer investigation.

Transaction Activityâ

Transaction Historyâ

This widget displays the transactional events associated with the alert under investigation. It places alerts in a more general picture, thus allowing agents to act upon them in a more efficient way.

Available data varies depending on the value or combination of values selected to fine-tune your results. For instance, by selecting *Customer Name*, the list shows all transactions made by the same customer as the event under investigation.

note

Note that you must select at least one value for transactional data to be displayed.

Additionally, you can reduce the volume of information by selecting a time window, which allows you to choose the period appropriate to your analysis (e.g. Last year, Last day, Last month).

It helps uncover fraud evidence such as:

Description	Examples and details
Multiple same amount transactions within a small timeframe	Over 10 transactions of the same amount were alerted for the same Customer Name within 30 minutes.
High number of fraudulent past transactions	Customer has had over 15 transactions marked as **Fraud** in the past, indicating there's a high likelihood of suspicious behavior.

tip

On top of the information available on the **Transaction History** widget, you can refer to:

- The **Notes** widget to check if the parties involved were recently contacted and any other information previously left by other agents, for example, when investigating related alerts or the customer.
- Your internal systems to check for other transactions by the same customer that werenâ□□t alerted (as the information in this widget only refers to alerted transactions). This can help you uncover patterns that might otherwise not look fraudulent/suspicious.

Alert Activityâ

Attached Filesâ

This widget also provides a convenient storage for any information that is not currently represented or supported by Case Manager's interface. It allows you to add information that may be useful to support the investigation and for subsequent investigation steps, legal action, or auditing.

You can use this widget to **upload**, **download** and/or **delete** files, as well as **update** files (e.g. to the latest version) by selecting the update icon.

Notesâ

While investigating alerts, you may need to document any steps taken during your investigation to share with other agents that may look into the same alert. To do so:

1. Select **Add Note**.
2. Select what your note refers to (e.g. event, merchant, customer). Note that depending on the option selected, you might need to enter additional information
3. Enter your **Note**.
4. Select **Save**.

All notes added to the alert (either by you or another agent) are then displayed in this widget, allowing you to check, for example:

- If the parties involved were recently contacted.
- If other agents flagged any related alerts or information that might impact your decision.
- If additional information has been requested and hasn't yet been received.

Activity Logâ

This widget displays the different actions performed in relation to the alert. For example, attached files, notes added, updates to lists, etc.

You can use the dropdown at the top right to filter the list by different actions, such as **Event created** or **Status**.

Next stepsâ

- Learn how to manage cases.
- Classify Events as "Fraud" or "Not Fraud".

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/investigate-alerts/alert-activity-widgets

On this page

alert-activity-widgets

Was this page helpful?Â

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1. Select **Add Note**.
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3. Enter your **Note**.
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All notes added to the alert (either by you or another agent) are then displayed in this widget, allowing you to check, for example:

- If the parties involved were recently contacted.
- If other agents flagged any related alerts or information that might impact your decision.
- If additional information has been requested and hasn't yet been received.

Activity Logâ

This widget displays the different actions performed in relation to the alert. For example, attached files, notes added, updates to lists, etc.

You can use the dropdown at the top right to filter the list by different actions, such as **Event created** or **Status**.

Analyze Customers

Was this page helpful?Â

While reviewing alerts, you may want to investigate a specific customer to understand their behavior so that you can better support your decision.

This page explains the available customer widgets and how they can help you to act on alerts.

Access Customer Details pageâ

You can access the Customer Details page via the **Event Search** page, **Fraud Alerts** page and **Search by Customer** page.

The following animation shows how you can access the Customer Details page from the Event Search, Fraud Alerts and Search by Customer pages.

Header and Customer Insightsâ

The header displays general information about the customer, such as risk rating, name and total balance. From here, fraud analysts can easily understand the customer's risk (e.g. when approving a loan or increasing the limit of a card). Additionally, they can add notes and/or upload any files that may be relevant (e.g. for other investigations involving this customer or auditing procedures), as well as create an ad-hoc customer investigation to log intelligence gathered from investigations outside the usual event flow.

You can also collapse or expand the **Customer Insights for the Last 30 Days** section to preview additional details on customer behavior, which combined with other information from the event being investigated helps detect fraud patterns more easily. The insights section displays the customer's historical information regarding accounts, cards, transactions and corresponding amounts, etc. over the last 30 days.

Customer Detailsâ

This widget displays general information about the customer, such as location, contact, and business-related details.

Rules Triggered by Location and Deviceâ

To support the investigation, this widget lists the rules triggered by behaviors associated with the customer's location and device.

Imagine a scenario where a fraudster acquires the customer's card credentials. They then use a VPN with an IP address resolving to New York to perform one test transaction, followed by four larger transactions within 45 minutes.

In this scenario, the system calculates the distance between the last two known transaction IP locations (New York and Paris) and triggers a rule advising fraud analysts to decline transactions due to the distance between the previous and current IP location in the last hour being higher than 1000.0 km.

Check the full list of standard rules to search for other explanations, and different fraud scenarios to better understand how rules help prevent them.

Customer Portfolioâ

This widget displays the different accounts and cards associated with the customer. It provides a baseline to understand the customer's behavior by allowing you to quickly check for the:

- **Status** of all of the customer's accounts and cards.
- **Current Balance** of each account and card.
- **Transaction limits** of each account and card.
- **Type of each card** (e.g. prepaid, debit).

You can also select **Details** to get additional information on each account's credit limit, holders and related cards, as well as each card's issuer bank and credit limit.

While analyzing events, this widget helps uncover fraudulent behavior such as:

Fraud evidence	Example details
Unusually high transaction amount	The transaction amount represents over 50% of the account's current balance.
Dormant account takeover	A prepaid card is associated with one of the customer's dormant accounts and a transfer of 20% of the funds are immediately transferred to the prepaid card.

tip

On top of the information available on the **Customer Portfolio** widget, you can refer to your internal banking system to get additional details on the customer's accounts/cards and crosscheck it with the information available in Case Manager.

Transaction Historyâ

This widget displays all the transactions involving the customer. This places alerts in a more general picture and allows you to act upon them more efficiently.

Similar to what happens in the Event page, you can perform several actions such as exporting, acting on transactions, or managing how the list is displayed to fit your preference.

Notesâ

While investigating customers, you may need to document any steps taken during your investigation to share them with other analysts that may look into a customer. Notes are also useful when escalating alerts to other analysts.

This widget displays the existing notes regarding a customer and also allows you to **Add Notes**.

Attached Filesâ

Similar to notes, you may need to add files to support the investigation. This can be done by selecting **Upload Files** either on the widget or the page's header.

Files may be useful for subsequent investigation steps, legal action, or auditing. This widget also provides a convenient storage for any information that is not currently represented or supported by Case Manager's interface.

On top of uploading files, from this widget you can also **Download**, **Update** or **Delete** any of the available files.

Activity Logâ

This widget displays the different actions performed in the customer page, namely regarding attached files, notes, and updates to lists.

Registered and Mailing Addressâ

These widgets display detailed information about the customer's primary legal address (i.e. registered address) and the preferred address for receiving mail (i.e. mailing address).

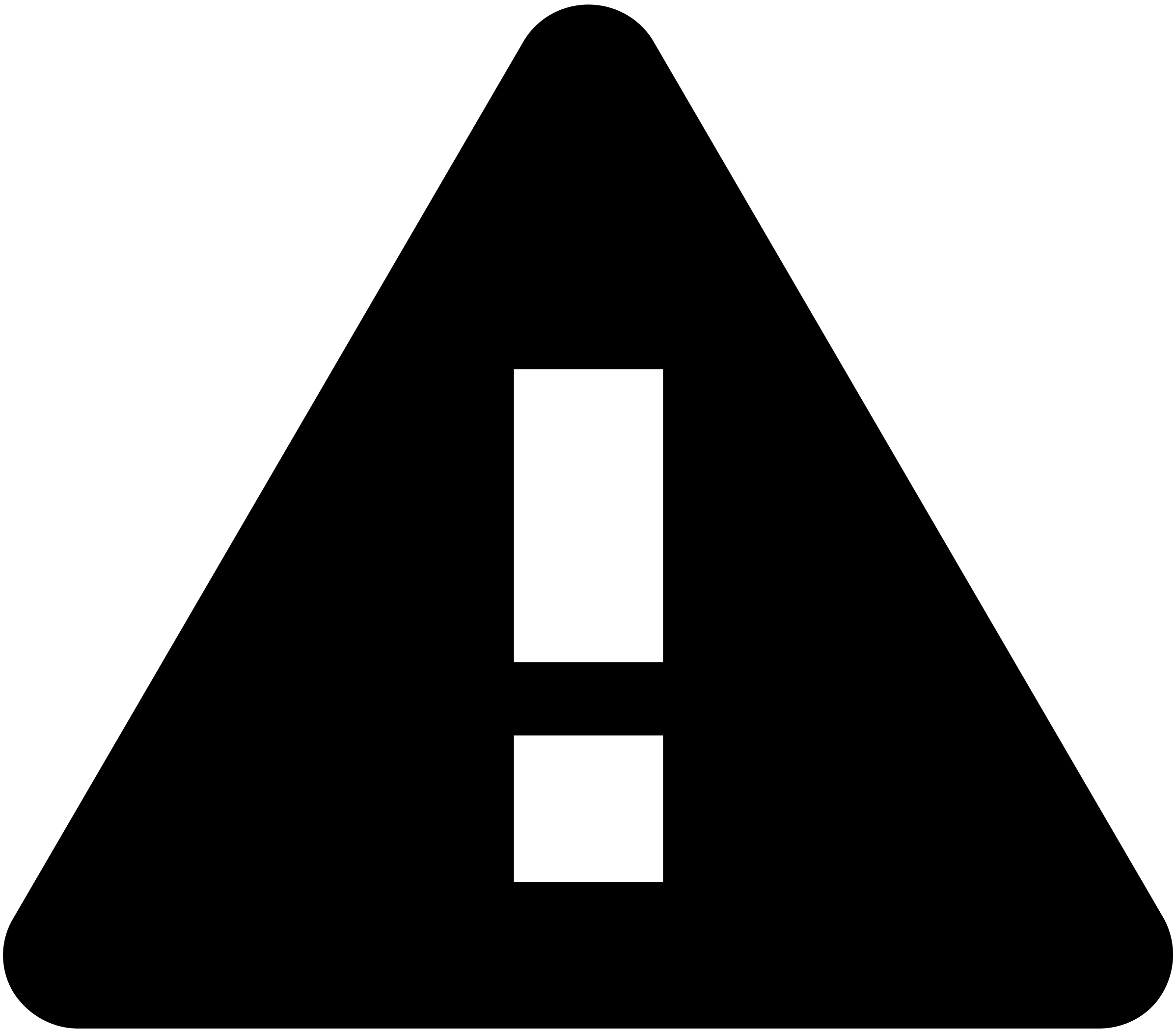
Related Partiesâ

This widget provides an overview of all individuals or entities associated with the customer under review. It helps uncover hidden connections and/or potential fraudulent networks by displaying and scoring each related party based on connected attributes.

Next stepsâ

Now that you understand how to analyze customers, you can return to reviewing alerts or continue your investigation with Genome.

assign-and-lock-rules



assign-and-lock rules

Fraud Prevention Agents - L1 and **Fraud Prevention Agents - L2** need to assign alerts to themselves to act on them.

Fraud Prevention Agents - L2 and **Fraud Prevention Managers** can reassign to themselves alerts that have already been assigned to other agents.

Only **Fraud Prevention Managers** can act on alerts assigned to other agents; therefore, they don't need to assign alerts to themselves.

Cards

When you open an event or an alert, you are presented with information that helps you make your decision. Whether you open an alerted or non-alerted event, you will land on the Cards' **Event** page, which contains information that helps you make your decision.

Note that you won't be able to make a decision based on a single event, or on a specific widget alone. An investigation is a holistic approach, in which you open several events on different windows and search for fraud evidence by looking at multiple widgets.

About widgets

This page documents the available widgets in a standard order; however, the order in Case Manager may vary depending on your setup. Use the table of contents on the right to navigate to widget-specific documentation and learn more about the information available in each.

Assign alerts to yourselfâ

Before starting the investigation process, start by assigning a single or a set of alerts to yourself. The product can be configured to prevent fraud prevention agents from performing any action on an alert until it is assigned either to them or to another agent. In turn, this prevents different agents from reviewing the same alert at the same time.

To unlock the event, open the **Unassigned** dropdown in the header and select **Assign to me.**

Assign alerts to other agents

The **Choose Assignee** option is only available to some users upon a specific permission.

In addition, you can use the **Assign All** option to assign all linked alerts to yourself. This enables you to easily act on, for example, all unreviewed alerts from the same customer at the same time.

When assigning all linked alerts, a new widget is added to the **Event Details** page with the list of linked alerts to better support the investigation process.

Depending on the configuration, there may be a limit to the number of linked alerts that can be assigned at once. If a limit is set and the **Assign All** action exceeds that number, a message appears specifying how many alerts were assigned.

The remaining alerts that were not assigned to you during this action will remain available for you or another agent to assign and review.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then the **Assign to me** option. Assigning multiple alerts is helpful to, for example, assign all alerts from a specific queue to yourself.

assign-and-lock rules

Fraud Prevention Agents - L1 and **Fraud Prevention Agents - L2** need to assign alerts to themselves to act on them.

Fraud Prevention Agents - L2 and **Fraud Prevention Managers** can reassign to themselves alerts that have already been assigned to other agents.

Only **Fraud Prevention Managers** can act on alerts assigned to other agents; therefore, they don't need to assign alerts to themselves.

Event headerâ

The following fields provide general information about the event to the fraud prevention agent:

- **Event ID**: The event's ID, which can be copied to share with other agents or track later in the system.
- **Assignee**: Indicates if the event is assigned to an agent. The dropdown button lets you assign the event.
- **Labeling**: Includes the âFraudâor "Not Fraud" options for a final decision. It also has the âHoldâbutton to pause the investigation while waiting for further information.
- **Score**: The output of Feedzai's TrustScore, or a combination of scoring models and TrustScore. The number ranges from 0 to 1000, 1000 being the most likely for the event to be fraudulent.
- **System Decision**: Generated by rules in the risk engine platform (i.e. a system that works upstream to Case Manager). Possible values are:
    - Approve

    - Decline
- **Fraud Operational Status**: Allows fraud prevention agents to track the alert/event investigation stage. The possible values are:
    - Unreviewed
    - In progress
    - On hold
    - Closed
- **Fraud label**: Indicates whether the agent has reached a final decision. This value can also be populated via API, for example, when the agent receives customer feedback. The possible values are:
    - Fraud
    - Not Fraud
- **Reasons**: Populated when you both mark an event as â□□Fraudâ□□or â□□Not Fraudâ□□and add a reason to clarify the decision.
- **Listed Entities**: Entities that have been added to a list. For example, a Beneficiary Account BIC (Inbound Payments) that has been added to a PosNeg list.
- **Queue**: Indicates if the event is in a queue (e.g. 1 hour to breach).
- **Cases**: Indicates if the event is associated with a case.
- **SCA Required**: Indicates if the risk engine's rules mark the event as requiring strong customer authentication.
- **Fraud Alert**: Indicates whether a rule with the `alert` action was triggered or not.
- **External ID**: The customer ID on the client's end.

Linked alertsâ□□

When assigning alerts, users can choose whether to assign a single alert or all alerts linked to it (see Assign alerts to yourself). This widget displays other existing alerts that are assigned to the fraud prevention agent as the one under review. All linked alerts belong to the same channel and share the same `customer_id`.

note

The **Linked alerts** widget only includes:

- Alerts assigned to the same user as the alert being reviewed.
- Alerts from the past 30 days, by default. To extend this timeframe, reach out to your Feedzai contact.

With this widget, fraud prevention agents can easily act on all linked alerts at the same time, thus improving the efficiency of the investigation process.

You can use the checkboxes to select one or multiple linked alerts and **Move to Queue**, **Choose Assignee** and **Unlink from Case**.

Linked alerts vs. Transaction History

Alerts displayed in the **Linked alerts** widget are flagged alerts that remain open for review, while the alerts displayed in the [Transaction History](#) also include alerts that have already been reviewed.

Preview linked alertsâ

While viewing the linked alerts list, select each alert's row to quickly preview its details.

Rules Triggeredâ

To support the investigation, this widget lists the rules triggered by the event under investigation.

Imagine the following scenario:

- A fraudster tests a card by performing a transaction to a charity merchants. Charity merchants have MCC '8398'.
- This transaction is approved, meaning that the card is valid.
- Then the fraudster performs four other transactions using non-charity merchants to buy goods.

- All these five transactions are performed with the same card.
- The transactions are performed within three minutes. This will trigger a rule that has a limit of five transactions within three minutes. As a result, the fifth transaction generates an alert and opens a case automatically.

The following table details key information used to recreate this scenario:

Transaction number	Merchant	MCC	amount	Fraudster Objective
Transaction #1	Save the Children	8398	$1.47	Check card validity
Transaction #2	Amazon	5691	$4.50	Fraudulently buy goods
Transaction #3	Amazon	5691	$42.20	Fraudulently buy goods
Transaction #4	Amazon	5691	$120.20	Fraudulently buy goods
Transaction #5	Amazon	5691	$826.00	Fraudulently buy goods

Check the full list of standard rules to search for other explanations, and the different fraud scenarios for card payments to better understand how rules help prevent them.

Transactionâ

The following figure shows the transaction details for a card event:

This widget helps uncover fraud evidence such as:

Description	Examples and details
Transaction above customer's normal spending pattern	Amount unusually high or low. High amount transactions followed by low amount transactions (you can better confirm this by opening all events in different tabs or by looking at the Transaction History).
Date/time of transaction outside of the normal business hours	Between 12am and 6am.
Card Pan linked to multiple currencies	When a Card Pan is associated with different currencies in multiple transactions. You can better confirm this by opening all events in different tabs.

Originator Accountâ

This widget includes the originator's personal and account information. Select **View more** to see additional details about the customer.

The **Customer** section helps uncover fraud evidence such as:

Description	Examples and details
Suspicious emails	- A valid customer account profile (name on account: John Hill, email on account: ). - Fraudster posing as a real customer (name on account: John Hill, email on account: ).
Suspicious street address	- A valid address: 85 Lincoln Green Lane, London, UK RG27 5YQ. - Fraudster posing as a real customer: 8.5.0 Lincoln GREEN Lane, London, UK RG27 5YQ.
Phone number that doesn't match the geographic location of the customer	A US customer may have the phone number (901)555-1077 or (404)556-0999. However, they shouldn't have the number 0044 224 79 449, unless it has been added to the profile over six months.

Learn more about how to Analyze Customers and gather more information to support your decision.

Account/Cardâ

The **Account/Card** section helps uncover fraud evidence such as:

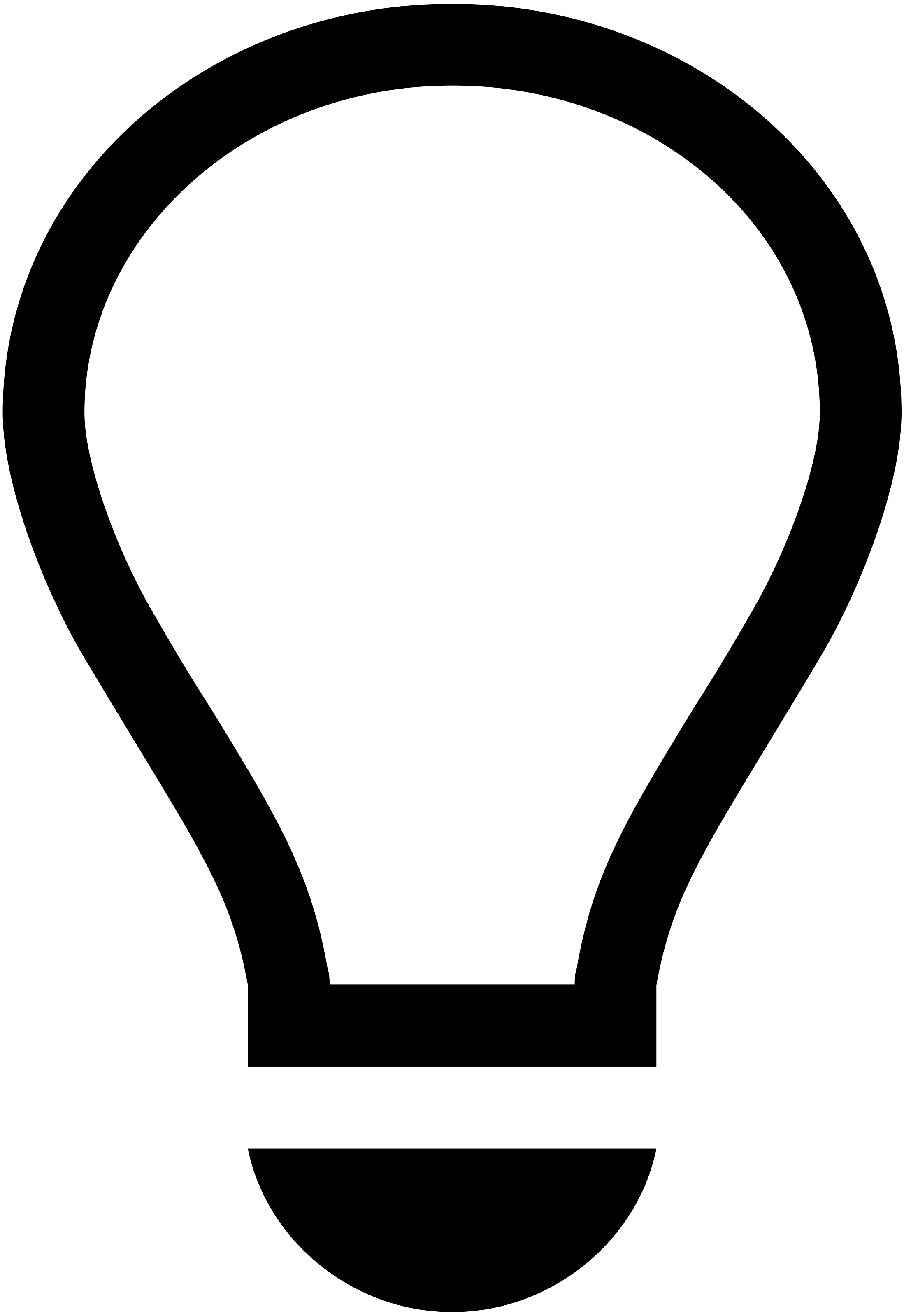
Description	Examples and details
Card recent reissue to a different address on file	When the value in the **Issuing date** field is recent (between 30 and 60 days) and the card is issued to an address other than the one on file (information often available via the banking internal tools), unless itâs been verified with the customer and/or via another valid source (i.e. Lexis Nexis).
âBusting Outâ: Account opening and card issuing dates too close (i.e. less than 30 days) to the first timestamp with unusually high amount transactions	- Example of fraud evidence: A customer opens an account on 8/18/20 and the card is issued on 8/23/20. Unusually high amount transactions occur daily between 8/30/20 and 9/30/20 to max out the card. - Example of genuine behavior evidence: A customer opens an account on 9/28/20 and the card is issued on 10/1/20. The transactions match the customerâs spending history between 10/8/20 and 1/5/21.

Merchantâ

This widget includes the merchant's identification and terminal information. Select **View more** to see additional details about the merchant.

The **Merchant**'s section helps uncover fraud evidence such as:

Description	Examples and details
Risky merchant	Examples of risky merchants: online gambling or casinos, VOIP or telemarketing, pharmaceuticals or drug stores, adult materials, products or services, airline tickets, bitcoins or forex trading, dating services, magazine subscriptions, charity. The **Merchant Risk** field gives an indication of the risk level (1 - lowest risk, 4 - highest risk).
No history with Merchant Name and/or MCC combined with another fraud evidence	The customer never made a purchase on a charity merchant, and suddenly, the transaction history shows various transactions in a short period of time, which suggests a card testing behavior.
Merchant city and/or country associated to high-fraud volumes in retail payments	Examples of retail payment high-risk regions: Russia, Australia, Canada, China, France, Great Britain, Hong Kong, Italy, Korea, Mexico.



tip

On top of the information available in the **Merchant** widget, you can refer to external platforms and websites to get additional information and crosscheck it with the information available in Case Manager. Examples of useful platforms are Google (e.g. to search for the merchant's own website) and government websites (e.g. to confirm a merchant is adequately registered).

Terminalâ

The **Terminal** section helps uncover fraud evidence such as:

Description	Examples and details
Customer has history with merchant but with distinct terminal country and terminal zip	Merchant name seen in customerâs spending history using contactless card (CNP) but flagged due to distinct terminal country, terminal zip, no travel noted on account and done with a âphysicalâcard.
Multiple transactions with the same amount from different Terminal IDs	Same Customer ID (e.g. 24117500) with eight transactions of $85.00 from different terminal IDs in a short period of time.
Region associated with high fraud volumes in retail payments	Examples of high-risk countries: Russia, Australia, Canada, China, France, Great Britain, Hong Kong, Italy, Korea, Mexico.

Last Contactâ

The following widget contains information provided by the issuer. This information is received through the Cards Info endpoint and is specific to card disputes.

Fields	Description	Examples and details	Schema fields
Contact Date	Specifies the date and time the message initiator sends the message	This field is expressed in the Coordinated Universal Time (UTC) (according to ISO 8601), in the format MMDDhhmmss (e.g. 07/16/2018 6:13:57 PM).	`dispute_start_date`
Type of Contact	Type of communication	The type of communication used by the entity that submitted the dispute (e.g. SMS).	`oob_comm_type`
Contact Notes	Contextual information left by the agent at the time of the dispute opening	Dispute.	`dispute_notes`

Dispute Informationâ

The following widget contains information provided by the issuer. This information is received through the Cards Info endpoint and is specific to card disputes.

Fields	Description	Examples and details	Schema fields
Dispute Start Date	Specifies the date and time the message initiator sends the message.	This field is expressed in the Coordinated Universal Time (UTC) (in accordance with ISO 8601), in the format MMDDhhmmss (e.g. 07/16/2018 6:13:57 PM).	`dispute_start_date`
Dispute Reason	Indicates why the dispute was opened.	Fraudulent - Cardholder claims an unauthorized payment. This can happen if the card was lost or stolen and used to make a fraudulent purchase. It can also happen if the cardholder does not recognize the payment as it appears on the billing statement sent by the card issuer.	`dispute_reason`
Dispute Submitted By	Entity that submitted the dispute.	The dispute was submitted by the issuer.	`dispute_submitted_by`
Dispute Notes	Indicates the subtype of the info event.	Dispute.	`dispute_notes`

Client Answerâ

The following widget contains information provided by the issuer. This information is received through the Cards Info endpoint and Feedback endpoints and is specific to card disputes.

Fields	Description	Examples and details	Schema fields
Answer Date	Specifies the date and time the message initiator sends the message.	This field is expressed in the Coordinated Universal Time (UTC) (in accordance with ISO 8601), in the format MMDDhhmmss (e.g. 09/16/2018 3:00:37 PM).	`dispute_start_date`
Answer Type	Type of response.	The type of communication used by the client (e.g. SMS).	`oob_comm_type`
Answer Notes	Disputeâs feedback notes.	The client has called to open a dispute on a transaction that they didnât perform.	`oob_feedback_notes`

Dispute Conclusionâ

The following widget contains information provided by the issuer. This information is received through the Cards Info endpoint and Feedback endpoints and is specific to card disputes.

Fields	Description	Examples and details	Schema fields
Dispute Conclusion Date	Specifies the date and time the message initiator sends the message.	This field is expressed in the Coordinated Universal Time (UTC) (in accordance with ISO 8601), in the format MMDDhhmmss (e.g. 09/16/2016 3:10:00 AM).	`dispute_start_date`
Dispute Conclusion Type	Identifies the different types of feedback.	Chargeback.	`feedback_type`
Dispute Conclusion Notes	Disputeâs conclusion notes.	The client has confirmed that they didnât perform the transaction.	`oob_feedback_notes`

Transaction Activityâ

Transaction Historyâ

This widget displays the transactional events associated with the alert under investigation. It places alerts in a more general picture, thus allowing agents to act upon them in a more efficient way.

Available data varies depending on the value or combination of values selected to fine-tune your results. For instance, by selecting *Customer Name*, the list shows all transactions made by the same customer as the event under investigation.

note

Note that you must select at least one value for transactional data to be displayed.

Additionally, you can reduce the volume of information by selecting a time window, which allows you to choose the period appropriate to your analysis (e.g. Last year, Last day, Last month).

It helps uncover fraud evidence such as:

Description	**Examples and details**
Multiple same amount transactions within a small timeframe	Over 10 transactions of the same amount were alerted for the same Customer Name within 30 minutes.
High number of fraudulent past transactions	Customer has had over 15 transactions marked as **Fraud** in the past, indicating there's a high likelihood of suspicious behavior.

tip

On top of the information available on the **Transaction History** widget, you can refer to:

- The **Notes** widget to check if the parties involved were recently contacted and any other information previously left by other agents, for example, when investigating related alerts or the customer.
- Your internal systems to check for other transactions by the same customer that werenâ□□t alerted (as the information in this widget only refers to alerted transactions). This can help you uncover patterns that might otherwise not look fraudulent/suspicious.

Alert Activityâ

Attached Filesâ

This widget also provides a convenient storage for any information that is not currently represented or supported by Case Manager's interface. It allows you to add information that may be useful to support the investigation and for subsequent investigation steps, legal action, or auditing.

You can use this widget to **upload**, **download** and/or **delete** files, as well as **update** files (e.g. to the latest version) by selecting the update icon.

Notesâ

While investigating alerts, you may need to document any steps taken during your investigation to share with other agents that may look into the same alert. To do so:

1. Select **Add Note**.
2. Select what your note refers to (e.g. event, merchant, customer). Note that depending on the option selected, you might need to enter additional information
3. Enter your **Note**.
4. Select **Save**.

All notes added to the alert (either by you or another agent) are then displayed in this widget, allowing you to check, for example:

- If the parties involved were recently contacted.
- If other agents flagged any related alerts or information that might impact your decision.
- If additional information has been requested and hasn't yet been received.

Activity Logâ

This widget displays the different actions performed in relation to the alert. For example, attached files, notes added, updates to lists, etc.

You can use the dropdown at the top right to filter the list by different actions, such as **Event created** or **Status**.

Next stepsâ

- Learn how to manage cases.
- Classify Events as "Fraud" or "Not Fraud".

-
- [Operational Work]
- [Fraud Prevention Agent]
- [Investigate Alerts]
- Classify Events

On this page

Classify Events

Was this page helpful?Â

After analyzing an event and customer details as described in the previous journey steps, you are ready to take action based on your decision and classify the event as fraud or not fraud.

To classify events and justify your decision:

1. In the event details page, select the options or . You can also select if the investigation is pending or blocked.

note

For **Inbound Payments** and **Outbound Transfers**, you must select before classifying the event.

For **Ad-hoc Customer Investigations**, the available options are or instead of **Fraud** or **Not Fraud**.

2. Select an option in the **Add Reasons** dropdown menu to justify your decision. The available options vary per event type and selected state.
3. Optionally, add a note to provide extra information.
4. Use the button to update the event with the decision and reason.

Submit and Update Lists

The button takes you to the next step: **Update Lists**. The actions performed in this step might have business implications if the information is not properly updated. For safety purposes, this button can be disabled in some use cases. Since it is an action that needs to be done carefully, it is explained as the final step of your user journey.

Once you update the event with decision and reason, this information is displayed in the **System Decision** and **Reasons** fields, respectively, in the event details page.

Review nextâ

You can continue analyzing events by selecting **Get Next** on the page's header. This is particularly useful when you are working with event queues because your goal is to analyze all the events in the queue quickly and with this option, you don't need to go back and stop the event analysis flow.

By selecting **Events Queues** from the dropdown menu, Case Manager will automatically take you to the next event with the **Status** defined as **N/A**.

note

Events that have already been classified as **Fraud** or **Not Fraud** because they have already been reviewed will be skipped by this option.

Alternatively, you can go back to the **Alerts** or **Search by Event** pages to select a new event to review. Check the Access Workload page for more information.

Bulk actionsâ

You can classify more than one event at the same time:

1. Navigate to the events inside a given case, the **Alerts** or **Search by Event** page. See the example below for the **Alerts** page:

info

In the **Alerts** page, you must select an event tab (e.g. Cards or Digital Activity) to classify the events in bulk.

2. Use the checkboxes to select the events that you want to classify.
3. Classify them as or by executing the actions described at the top of this page to justify your decision.

Next stepsâ

Now that you know how to classify events, learn how to update lists with new information. Since this is a sensitive operation, you might require special permissions or additional training to execute this final step.

Create Ad-Hoc Customer Investigations

Learn how to open an ad-hoc investigation into a customer to log any intelligence gathered from investigations outside the usual event flow, even if there are no suspicious transactions detected.

From the Customer Details page header, select **Create investigation** to open an ad-hoc customer investigation.

Fill in the form accordingly:

- **Source**: Use case under which you're creating the investigation.
- **Origin**: Motive for creating the investigation into this customer. For example, 'Transaction monitoring', 'Negative media information' or 'Receipt of a search warrant'.
- **Reason code**: Your organization's internal reason code.
- **Description**: Details regarding the ad-hoc investigation.

After you **Submit** the form, an Ad-hoc Customer Investigation becomes available for you to review, leave notes and attachments, track its status and even add it to a case.

Digital Activity

Was this page helpful?Â

When you open an event or an alert, you are presented with information that helps you make your decision. Whether you open an alerted or non-alerted event, you will land on the Digital Activity's **Event** page, which contains information that helps you make your decision.

Note that you won't be able to make a decision based on a single event, or on a specific widget alone. An investigation is a holistic approach, in which you open several events on different windows and search for fraud evidence by looking at multiple widgets.

About widgets

This page documents the available widgets in a standard order; however, the order in Case Manager may vary depending on your setup. Use the table of contents on the right to navigate to widget-specific documentation and learn more about the information available in each.

Assign alerts to yourselfâ

Before starting the investigation process, start by assigning a single or a set of alerts to yourself. The product can be configured to prevent fraud prevention agents from performing any action on an alert until it is assigned either to them or to another agent. In turn, this prevents different agents from reviewing the same alert at the same time.

To unlock the event, open the **Unassigned** dropdown in the header and select **Assign to me.**

Assign alerts to other agents

The **Choose Assignee** option is only available to some users upon a specific permission.

In addition, you can use the **Assign All** option to assign all linked alerts to yourself. This enables you to easily act on, for example, all unreviewed alerts from the same customer at the same time.

When assigning all linked alerts, a new widget is added to the **Event Details** page with the list of linked alerts to better support the investigation process.

Depending on the configuration, there may be a limit to the number of linked alerts that can be assigned at once. If a limit is set and the **Assign All** action exceeds that number, a message appears specifying how many alerts were assigned.

The remaining alerts that were not assigned to you during this action will remain available for you or another agent to assign and review.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then the **Assign to me** option. Assigning multiple alerts is helpful to, for example, assign all alerts from a specific queue to yourself.

assign-and-lock rules

Fraud Prevention Agents - L1 and **Fraud Prevention Agents - L2** need to assign alerts to themselves to act on them.

Fraud Prevention Agents - L2 and **Fraud Prevention Managers** can reassign to themselves alerts that have already been assigned to other agents.

Only **Fraud Prevention Managers** can act on alerts assigned to other agents; therefore, they don't need to assign alerts to themselves.

Event headerâ

The following fields provide general information about the event to the fraud prevention agent:

- **Event ID**: The event's ID, which can be copied to share with other agents or track later in the system.
- **Assignee**: Indicates if the event is assigned to an agent. The dropdown button lets you assign the event.
- **Labeling**: Includes the âFraudâor "Not Fraud" options for a final decision. It also has the âHoldâbutton to pause the investigation while waiting for further information.
- **Score**: Generated by the model in the risk engine platform (i.e. a system that works upstream to Case Manager). The number ranges from 0 to 1000, 1000 being the most likely for the event to be fraudulent. Note that, if you don't have a model in place, the score will always be 0.

- **System Decision**: Generated by rules in the risk engine platform (i.e. a system that works upstream to Case Manager). Possible values are:
  - Approve
  - Challenge
  - Decline
- **Fraud Operational Status**: Allows fraud prevention agents to track the alert/event investigation stage. The possible values are:
  - Unreviewed
  - In progress
  - On hold
  - Closed
- **Fraud label**: Indicates whether the agent has reached a final decision. This value can also be populated via API, for example, when the agent receives customer feedback. The possible values are:
  - Fraud
  - Not Fraud
- **Reasons**: Populated when you both mark an event as â□□Fraudâ□□or â□□Not Fraudâ□□and add a reason to clarify the decision.
- **Listed Entities**: Entities that have been added to a list. For example, a Beneficiary Account BIC (Inbound Payments) that has been added to a PosNeg list.
- **Queue**: Indicates if the event is in a queue (e.g. 1 hour to breach).
- **Cases**: Indicates if the event is associated with a case.
- **Fraud Alert**: Indicates whether a rule with the `alert` action was triggered or not.

Summaryâ□□

Event Summaryâ□□

This widget helps uncover fraud evidence such as:

Description	Examples and details
Fail reason	A failed authentication due to invalid credentials may not suggest fraud. You can, however, analyze this parameter against other information, such as triggered rules, session alerts, and Session Risk Score's risk score.
Dormant account	The last successful account login was over 3 months ago and in the last 24 hours the user used a proxy to log into the account. Note that you won't be able to confirm this information from this widget alone. You must compare it against other widgets, such as Rules Triggered or Session Alerts.

tip

The **Customer ID** redirects you to the Customer Details page.

Linked alertsâ

When assigning alerts, users can choose whether to assign a single alert or all alerts linked to it (see Assign alerts to yourself). This widget displays other existing alerts that are assigned to the fraud prevention agent as the one under review. All linked alerts belong to the same channel and share the same `customer_id`.

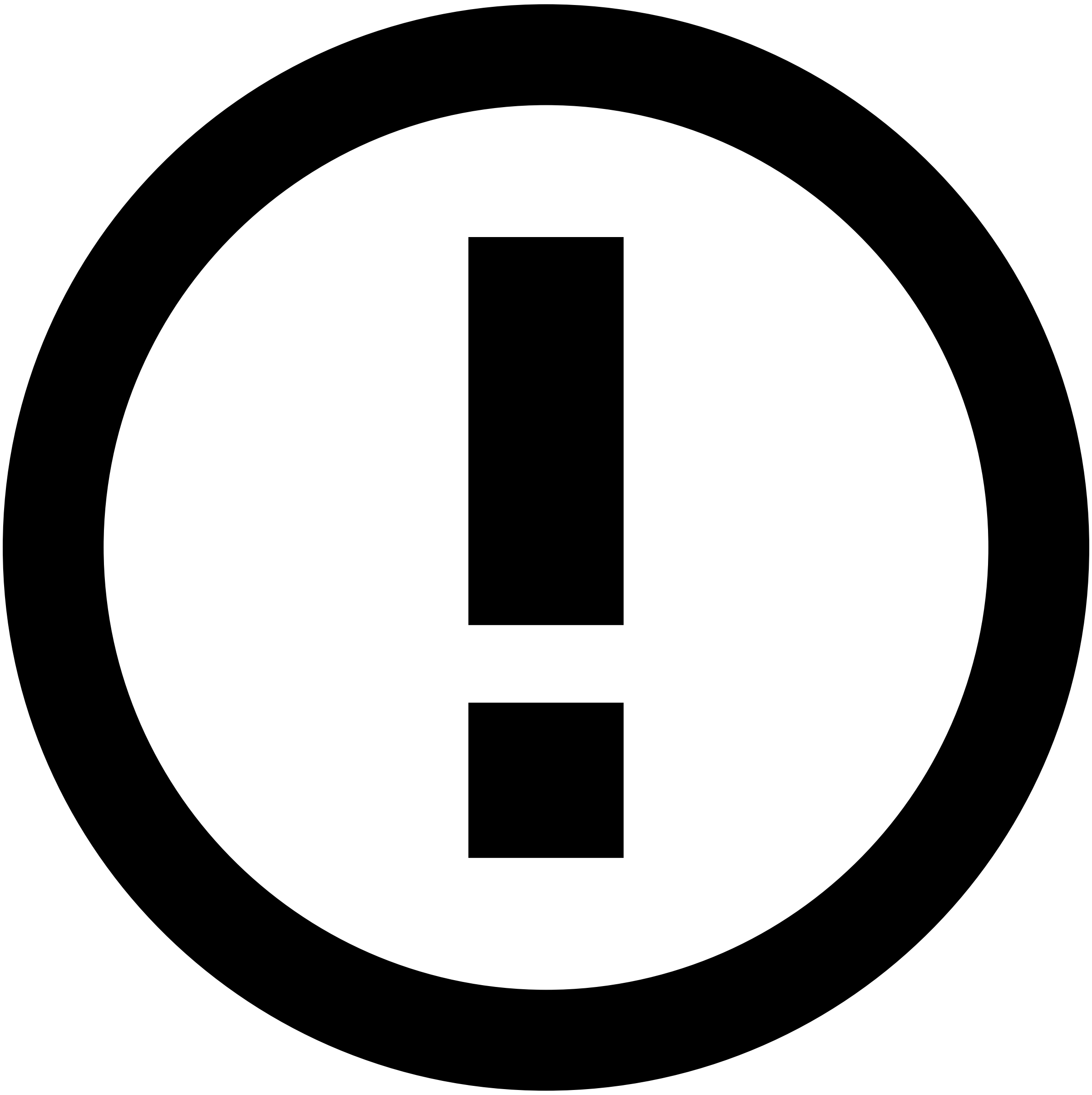
note

The **Linked alerts** widget only includes:

- Alerts assigned to the same user as the alert being reviewed.
- Alerts from the past 30 days, by default. To extend this timeframe, reach out to your Feedzai contact.

With this widget, fraud prevention agents can easily act on all linked alerts at the same time, thus improving the efficiency of the investigation process.

You can use the checkboxes to select one or multiple linked alerts and **Move to Queue**, **Choose Assignee** and **Unlink from Case**.



Linked alerts vs. Transaction History

Alerts displayed in the **Linked alerts** widget are flagged alerts that remain open for review, while the alerts displayed in the [Transaction History] also include alerts that have already been reviewed.

Preview linked alertsâ

While viewing the linked alerts list, select each alert's row to quickly preview its details.

Explanationsâ

This widget explains why an event received a certain score from machine-learning models and how much each parameter contributed to that score. Model parameters can either increase or reduce the risk and, consequently, the score.

The information available allows you to easily identify the biggest risk **Maximizer** and **Minimizer** according to the following thresholds:

- Less than 0 is a Minimizer.
- Up to 10% is a yellow Maximizer.

- Above 10% is a red Maximizer.

Rules Triggeredâ

To support the investigation, this widget lists the rules triggered by the event under investigation.

Imagine the following scenario:

- A customer's account has been inactive for the past year.
- A fraudster obtains access to that account.
- The fraudster logs into the customer's online banking app several times in the same day using a device ID linked to multiple customers.

In this scenario, a rule is triggered by a combination of two factors: maximum number of customers per device and maximum number of logins per device. The event is automatically marked as an alert by the risk engine and sent for investigation in Case Manager.

Check the full list of standard rules to search for other explanations, and the different fraud scenarios for digital activity events to better understand how rules help prevent them.

Digital Fingerprintâ

Networkâ

This widget allows you to see important network details for the event being investigated.

You may also come across a situation that requires adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists (e.g. IP Address) have the **Update Lists** button.

It helps uncover fraud evidence such as:

Description	Examples and details
Proxy server	Fraudsters use proxies to hide their true IP address and geolocation in order to commit fraud without being detected. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner.
IP address	The IP address is associated with previous fraudulent activity and is listed to be declined.

tip

On top of the information available on the **Network** widget, you can refer to external platforms to cross-check specific details with the information available in Case Manager. For example, cross-check the alert's IP address with the one available in the VISA system and use it to assess the distance between the customer's location and the transaction's location.

Sessionâ

The following widget displays the Session Risk Score for the event under analysis.

info

Session Risk Score is a unique digital identifier that is created for each user. The Session Risk Score collects the userâ□□s full context of biometric, behavioral, device and network data points, and continuously analyzes user behavior from login to logout across every interaction.

The score value changes color depending on the risk severity:

- **Blue (0-39)**: Very low likelihood that the event is fraudulent.
- **Green (40-59)**: Low likelihood that the event is fraudulent.
- **Yellow (60-89)**: Medium likelihood that the event is fraudulent.
- **Red (90-100)**: High likelihood that the event is fraudulent.

The following table shows examples of factors that may lead to a high risk score:

Description	Examples and details
Session Risk Score	The device information captured by the Session Risk Score is associated with previous fraudulent activity.

Description	Examples and details
Unusual session	The current session ID is linked to multiple account changes (e.g during this session the user changed their password, email, phone number, and added a new payee before initiating a transfer).

Deviceâ

This widget allows you to see important information about the device that is being used in the event under investigation. Select **View more** to see additional details about the device.

You may also come across a situation that requires adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists (e.g. Device ID) have the **Update Lists** button.

It helps uncover fraud evidence such as:

Description	Examples and details
Risky device	- The device ID, IP, and associated email are listed to be declined. - The email associated with this device has an unknown domain (e.g. ).
Suspicious device OS	Rooted device. This makes it possible for the fraudster to have privileges that are usually not available nor recommended for the end user.

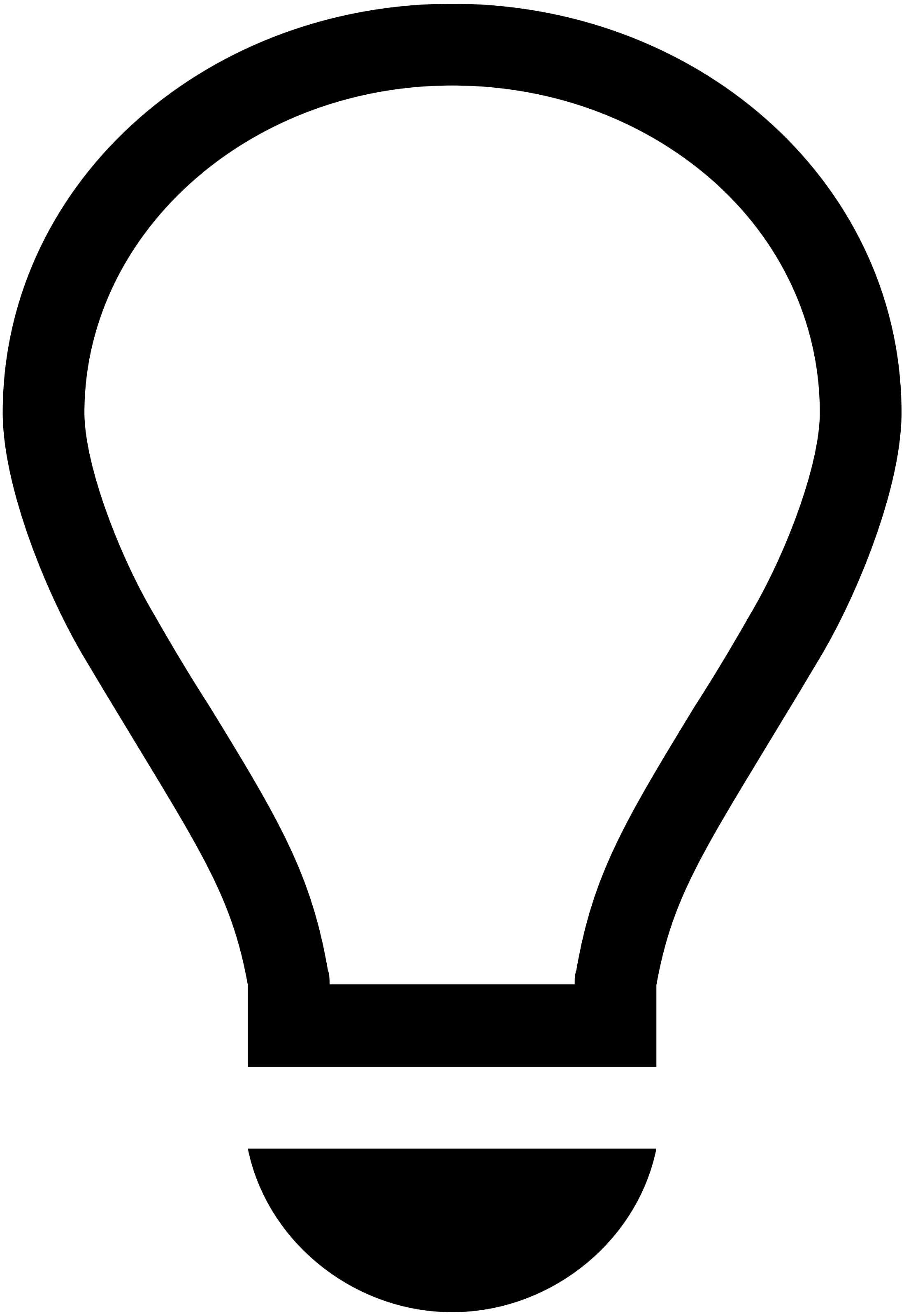
Learn more about device and behavior fraud scenarios and their threats.

Session Alertsâ

The following widget displays the 'Type', 'View' (online session page in which the alert was detected - e.g. login page, profile page, transfers page) and 'Risk Score' of session alerts.

It helps uncover fraud evidence such as:

Description	Examples and details
ATO high risk score	Account Takeover (ATO) high risk score may tell the agents that the session is likely to be ATO.
RAT high risk score	An event with a high score on Remote Access Trojans (RAT) only tells agents that the event is initiated from a remote access tool, which is not necessarily considered fraudulent. However, you can confirm this by looking at the Session Risk Score (see Session widget): - A high RAT score (e.g. 90) and high Session Risk Score (e.g. 95) suggest that the event is likely to be fraud. - A high RAT score (e.g. 90) but low Session Risk Score (e.g. 20) suggest that the event is likely to be genuine.
VPN high risk score	VPN alert detects VPN activity, which is not necessarily a risk. However, if you observe any Proxy detection, TOR connections, or a high Session Risk Score (see Session widget), this means that a fraudster may be trying to mask their location or IP.



Potential Fraud Risk

Pay special attention to the "Potential Fraud Risk" alert, as it is a strong indicator of a fraudulent session.

This alert indicates the detection of patterns or behaviors that closely align with confirmed fraudulent activity.

For more information, read the Digital Trust's [Potential Fraud Risk Alert] documentation.

Transaction Activityâ

Transaction Historyâ

This widget displays the transactional events associated with the alert under investigation. It places alerts in a more general picture, thus allowing agents to act upon them in a more efficient way.

Available data varies depending on the value or combination of values selected to fine-tune your results. For instance, by selecting *Customer Name*, the list shows all transactions made by the same customer as the event under investigation.

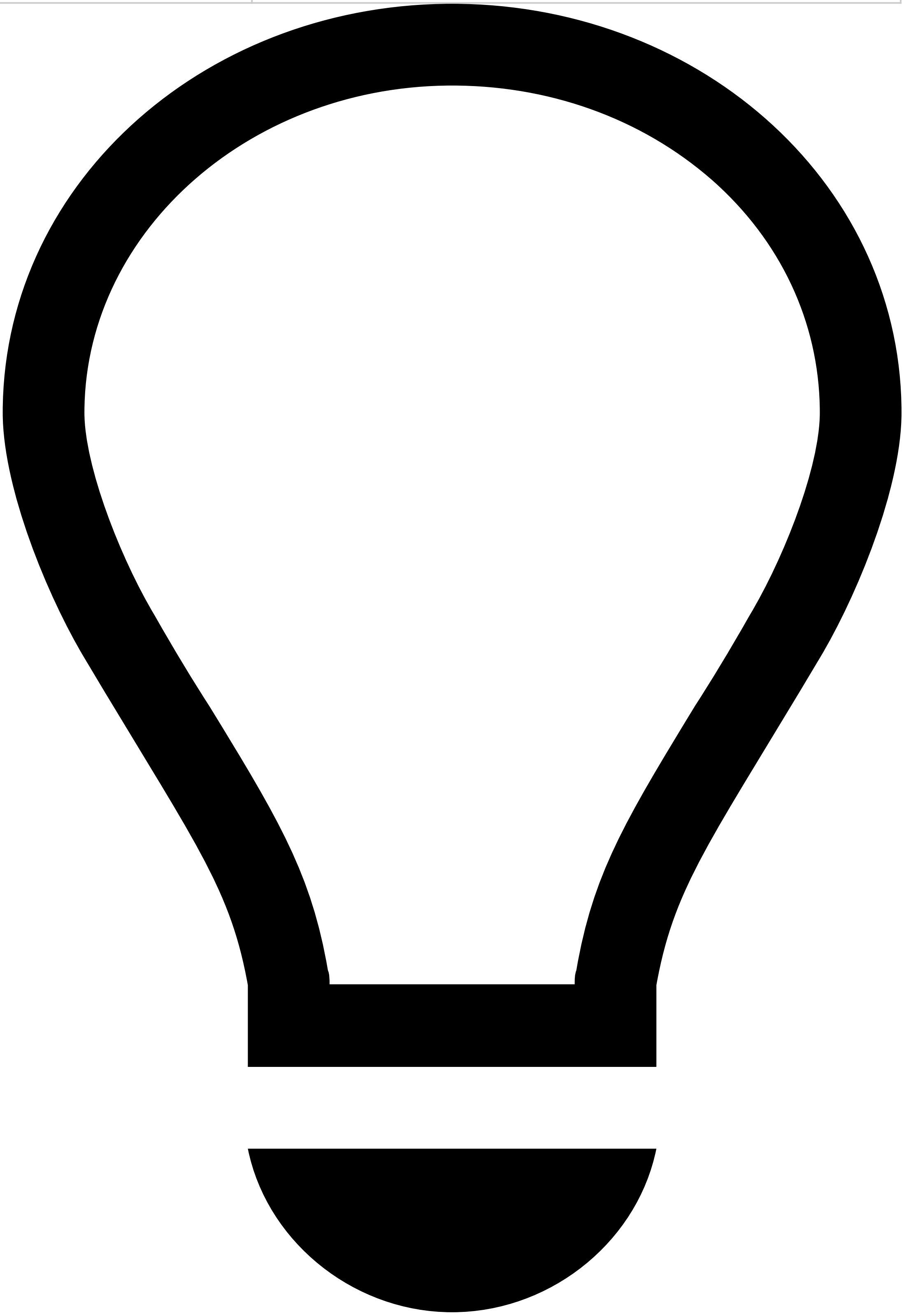
note

Note that you must select at least one value for transactional data to be displayed.

Additionally, you can reduce the volume of information by selecting a time window, which allows you to choose the period appropriate to your analysis (e.g. Last year, Last day, Last month).

It helps uncover fraud evidence such as:

Description	Examples and details
Multiple same amount transactions within a small timeframe	Over 10 transactions of the same amount were alerted for the same Customer Name within 30 minutes.
High number of fraudulent past transactions	Customer has had over 15 transactions marked as **Fraud** in the past, indicating there's a high likelihood of suspicious behavior.



tip

On top of the information available on the **Transaction History** widget, you can refer to:

- The **Notes** widget to check if the parties involved were recently contacted and any other information previously left by other agents, for example, when investigating related alerts or the customer.
- Your internal systems to check for other transactions by the same customer that werenâ t alerted (as the information in this widget only refers to alerted transactions). This can help you uncover patterns that might otherwise not look fraudulent/suspicious.

Alert Activityâ

Attached Filesâ

This widget also provides a convenient storage for any information that is not currently represented or supported by Case Manager's interface. It allows you to add information that may be useful to support the investigation and for subsequent investigation steps, legal action, or auditing.

You can use this widget to **upload**, **download** and/or **delete** files, as well as **update** files (e.g. to the latest version) by selecting the update icon.

Notesâ

While investigating alerts, you may need to document any steps taken during your investigation to share with other agents that may look into the same alert. To do so:

1. Select **Add Note**.
2. Select what your note refers to (e.g. event, merchant, customer). Note that depending on the option selected, you might need to enter additional information
3. Enter your **Note**.
4. Select **Save**.

All notes added to the alert (either by you or another agent) are then displayed in this widget, allowing you to check, for example:

- If the parties involved were recently contacted.
- If other agents flagged any related alerts or information that might impact your decision.
- If additional information has been requested and hasn't yet been received.

Activity Logâ

This widget displays the different actions performed in relation to the alert. For example, attached files, notes added, updates to lists, etc.

You can use the dropdown at the top right to filter the list by different actions, such as **Event created** or **Status**.

Next stepsâ

- Learn how to manage cases.
- Classify Events as "Fraud" or "Not Fraud".

On this page

digital-fingerprint-widgets

Networkâ

This widget allows you to see important network details for the event being investigated.

You may also come across a situation that requires adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists (e.g. IP Address) have the **Update Lists** button.

It helps uncover fraud evidence such as:

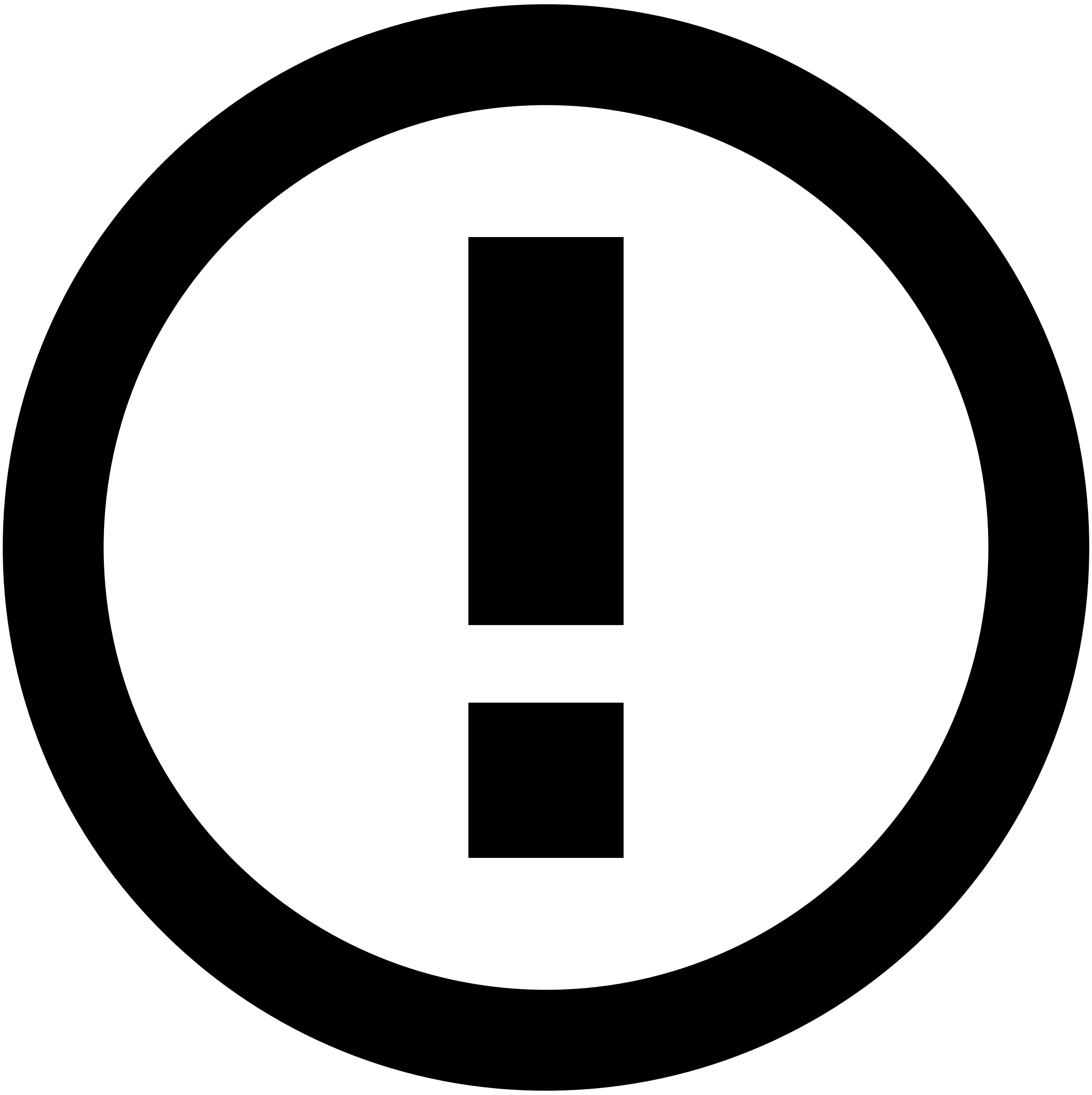
Description	Examples and details
Proxy server	Fraudsters use proxies to hide their true IP address and geolocation in order to commit fraud without being detected. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner.
IP address	The IP address is associated with previous fraudulent activity and is listed to be declined.
Mismatch between IP address location and originator's city	The device's IP address is Lisbon, whereas the originator's city (see Accounts widget) is New York. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner.

tip

On top of the information available on the **Network** widget, you can refer to external platforms to cross-check specific details with the information available in Case Manager. For example, cross-check the alert's IP address with the one available in the VISA system and use it to assess the distance between the customer's location and the transaction's location.

Sessionâ

The following widget displays the Session Risk Score for the event under analysis.



info

Session Risk Score is a unique digital identifier that is created for each user. The Session Risk Score collects the userâ□□s full context of biometric, behavioral, device and network data points, and continuously analyzes user behavior from login to logout across every interaction.

The score value changes color depending on the risk severity:

- **Blue (0-39)**: Very low likelihood that the event is fraudulent.
- **Green (40-59)**: Low likelihood that the event is fraudulent.
- **Yellow (60-89)**: Medium likelihood that the event is fraudulent.
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The following table shows examples of factors that may lead to a high risk score:

Description	Examples and details
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This widget allows you to see important information about the device that is being used in the event under investigation. Select **View more** to see additional details about the device.

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VPN high risk score	VPN alert detects VPN activity, which is not necessarily a risk. However, if you observe any Proxy detection, TOR connections, or a high Session Risk Score (see Session widget), this means that a fraudster may be trying to mask their location or IP.

Potential Fraud Risk

Pay special attention to the "Potential Fraud Risk" alert, as it is a strong indicator of a fraudulent session.

This alert indicates the detection of patterns or behaviors that closely align with confirmed fraudulent activity.

For more information, read the Digital Trust's Potential Fraud Risk Alert documentation.

- Operational Work
- Fraud Prevention Agent
- Investigate Alerts
- Event Statuses and Investigation Flow

On this page

Event Statuses and Investigation Flow

Was this page helpful?Â

As an analyst, understanding how event statuses work is a key step to managing the alert investigation efficiently. Analysts and managers can benefit from an intuitive status flow that helps users track each given alert (e.g. whether the investigation is in progress, on hold, etc.).

In this page, you will also learn how the out-of-the-box queues, automation rules, and SLAs play a role in the investigation flow. These settings integrate an automated process that sorts the alerts by priority for you.

Out-of-the-box statusesâ

Transaction Fraud for Banking (TFB) provides you with the following status categories:

- **System Decision:** Indicates the outcome processed by the Feedzai's risk engine. The possible values are:
    - Accept
    - Decline
    - Review (for transfers with cut-off time)
    - Challenge (for digital activity events)
- **Operational Status:** Allows users to track the alert/event investigation stage. The possible values are:
    - Unreviewed
    - In progress
    - On hold
    - Closed
- **Breach Status:** This status is only available for transfers and allows analysts to manage the investigation and avoid reaching the breached status according to business SLAs. The possible values are:
    - Breached
    - Not Breached
- **Fraud Label:** Indicates whether the analyst has reached a final decision. This value can also be populated via API, for example, when the analyst receives customer feedback. The possible values are:
    - Fraud
    - Not Fraud

Where are statuses displayed?â

In Case Manager, statuses are displayed in the search results of the **Search By Event** and **Alerts** pages, under **System Decision**, **Fraud Label**, and **Status**:

If you select the event's **Date**, notice that the Event page also displays the alert header.

Cardsâ

The following image shows the example of an alert header for cards. Note that the breach status does not apply for card events:

Digital Activityâ

The following image is an example of a digital activity event header. Note that the breach status does not apply for digital activity events:

Inbound Paymentsâ

The following image shows the example of an inbound payment event header:

Outbound Transfersâ

The following image shows the example of an alert header for transfers:

Investigation flowâ

This section focuses on the scenario of a typical investigation flow of an outbound transfer event with the following outcomes:

- **Alerted**: `yes`
- **System Decision**: `review`
- **Analyst Decision** and corresponding **Fraud Label**: `fraud`

1. An alerted transfer event arrives in Case Manager and triggers an automation rule that sets the initial operational and breach status as "Unreviewed" and "Not Breached", respectively.

2. The **System Decision** outcome from the risk engine, which asks the analyst to "Review" the alert, triggers an automation rule to start an initial Service-Level Agreement (SLA) to review the alert. If the alert isn't reviewed over the course of 23 hours, an SLA (i.e. 1h to breach) redirects the alert to a high-priority queue (i.e. "1h to breach") to be analyzed before the 24-hour breach. If the alert isn't reviewed within the 24-hour window, an SLA (i.e. "24h breach") breaches the alert and is then moved to a queue for breached alerts (i.e. "24h breach).

   info

   The **System Decision** "Review" does not change throughout the investigation.

3. An analyst assigns the event to themselves. This action triggers an automation rule (i.e. "handle new assignee"), which starts an SLA to review the alert in one hour and sets the **Operational Status** as "In Progress".

4. If the analyst needs to wait for more information from the customer, they can select the **Hold** button, which triggers an automation rule that starts an SLA. This extends the period of analysis to 48 hours and sets the Operational Status to "On Hold".

5. After collecting the necessary information, the analyst concludes it's a fraudulent transfer. To make a "fraud" decision, they select the **Fraud** button, which triggers an automation rule that removes the alert from the queue, unassigns the alert, sets the **Operational Status** as "Closed", and sets the **Fraud Label** as "Fraud".

tip

Since the status flow is achieved by using pre-configured SLAs, automation rules and queues, look into the Out-of-the-Box Settings page to learn more about these configurations.

Learn moreâ

Learn how to work with Case Manager by completing the following user journey steps:

1. Access Workload: Have a complete view of the events that need to be analyzed using the alert and case pages.

2. Analyze Events: Get to know the information available to analyze the events. The information includes account/ card details, customer information, and transaction history.
3. Analyze Customers: Investigate customers to understand their behavior and to better support your decision.
4. Uncover Patterns with Genome: Find and visualize complex financial crime patterns using link analysis and graph technology.
5. Classify Events: Learn how to mark events as 'fraud' or 'not fraud' and justify your decision, such as suspected or confirmed fraud.
6. Update Lists: Automatically handle situations with the same fraud patterns you just identified, such as a suspicious card ID or an MCC associated with a given card testing activity.

-

- Operational Work
- Fraud Prevention Agent
- Investigate Alerts
- Analyze Events
- Inbound Payments

On this page

Inbound Payments

When you open an event or an alert, you will land on the Inbound Payments' **Event page**, which contains information that helps you make your decision.

Note that you won't be able to make a decision based on a single event, or on a specific widget alone. An investigation is a holistic approach, in which you open several events on different windows and search for fraud evidence by looking at multiple widgets.

About widgets

This page documents the available widgets in a standard order; however, the order in Case Manager may vary depending on your setup. Use the table of contents on the right to navigate to widget-specific documentation and learn more about the information available in each.

Assign alerts to yourselfâ

Before starting the investigation process, start by assigning a single or a set of alerts to yourself. The product can be configured to prevent fraud prevention agents from performing any action on an alert until it is assigned either to them or to another agent. In turn, this prevents different agents from reviewing the same alert at the same time.

To unlock the event, open the **Unassigned** dropdown in the header and select **Assign to me.**

Assign alerts to other agents

The **Choose Assignee** option is only available to some users upon a specific permission.

In addition, you can use the **Assign All** option to assign all linked alerts to yourself. This enables you to easily act on, for example, all unreviewed alerts from the same customer at the same time.

When assigning all linked alerts, a new widget is added to the **Event Details** page with the list of linked alerts to better support the investigation process.

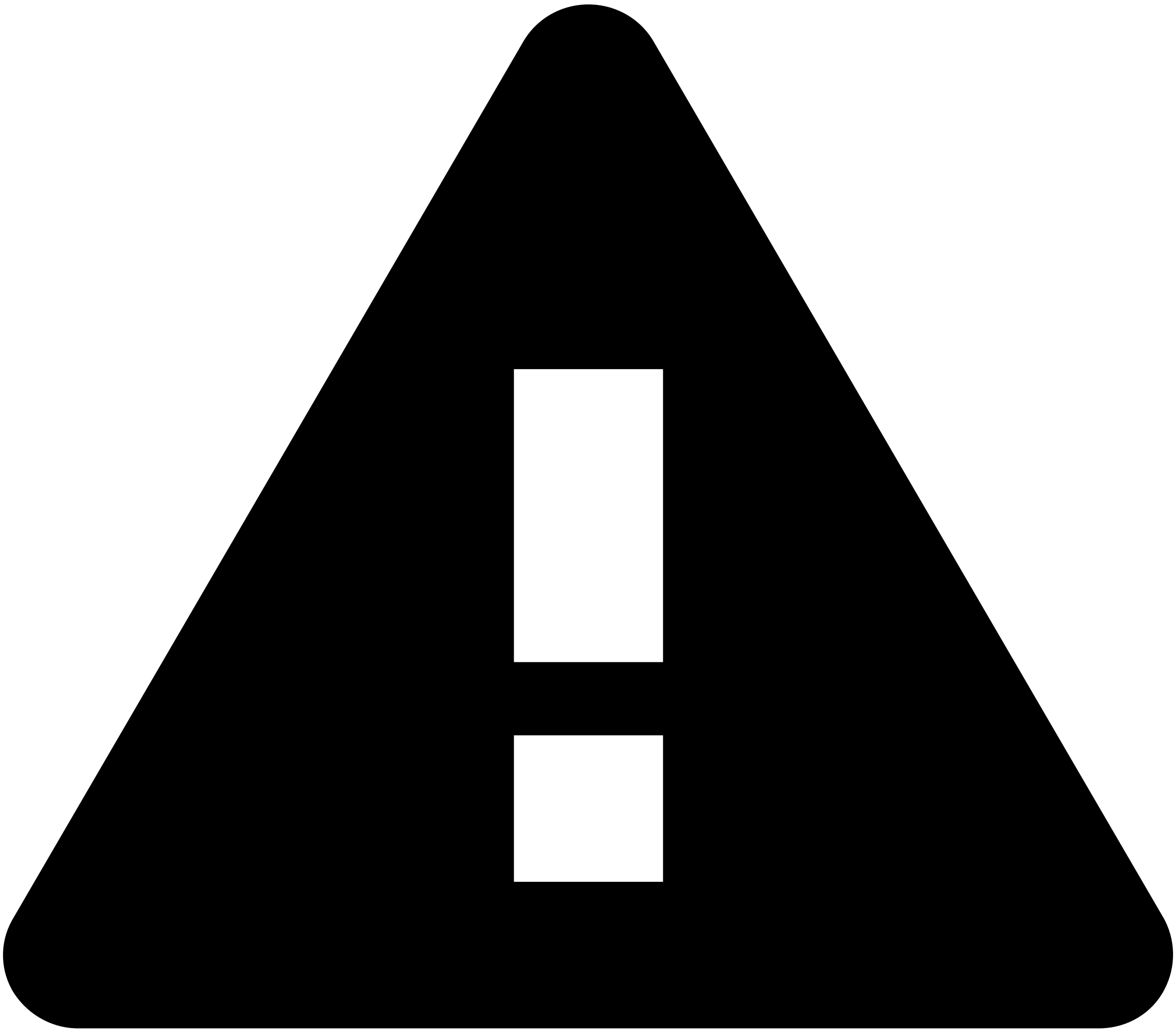
Depending on the configuration, there may be a limit to the number of linked alerts that can be assigned at once. If a limit is set and the **Assign All** action exceeds that number, a message appears specifying how many alerts were assigned.

The remaining alerts that were not assigned to you during this action will remain available for you or another agent to assign and review.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then the **Assign to me** option. Assigning multiple alerts is helpful to, for example, assign all alerts from a specific queue to yourself.



assign-and-lock rules

Fraud Prevention Agents - L1 and **Fraud Prevention Agents - L2** need to assign alerts to themselves to act on them.

Fraud Prevention Agents - L2 and **Fraud Prevention Managers** can reassign to themselves alerts that have already been assigned to other agents.

Only **Fraud Prevention Managers** can act on alerts assigned to other agents; therefore, they don't need to assign alerts to themselves.

Event headerâ

The following fields provide general information about the event to the fraud prevention agent:

- **Event ID**: The event's ID, which can be copied to share with other agents or track later in the system.
- **Assignee**: Indicates if the event is assigned to an agent. The dropdown button lets you assign the event.
- **Labeling**: Includes the âFraudâor "Not Fraud" options for a final decision. It also has the âHoldâbutton to pause the investigation while waiting for further information.
- **Score**: Generated by the model in the risk engine platform (i.e. a system that works upstream to Case Manager). The number ranges from 0 to 1000, 1000 being the most likely for the event to be fraudulent. Note that, if you don't have a model in place, the score will always be 0.

- **System Decision**: Generated by rules in the risk engine platform (i.e. a system that works upstream to Case Manager). Possible values are:
    - Approve
    - Review
    - Decline
- **Fraud Operational Status**: Allows fraud prevention agents to track the alert/event investigation stage. The possible values are:
    - Unreviewed
    - In progress
    - On hold
    - Closed
- **Fraud label**: Indicates whether the agent has reached a final decision. This value can also be populated via API, for example, when the agent receives customer feedback. The possible values are:
    - Fraud
    - Not Fraud
- **Reasons**: Populated when you both mark an event as â□□Fraudâ□□or â□□Not Fraudâ□□and add a reason to clarify the decision.
- **Listed Entities**: Entities that have been added to a list. For example, a Beneficiary Account BIC (Inbound Payments) that has been added to a PosNeg list.
- **Queue**: Indicates if the event is in a queue (e.g. 1 hour to breach).
- **Cases**: Indicates if the event is associated with a case.
- **Alerted**: Indicates that a rule with the `alert` action was triggered.

Linked alertsâ□□

When assigning alerts, users can choose whether to assign a single alert or all alerts linked to it (see Assign alerts to yourself). This widget displays other existing alerts that are assigned to the fraud prevention agent as the one under review. All linked alerts belong to the same channel and share the same `receiver_id`.

note

The **Linked alerts** widget only includes:

- Alerts assigned to the same user as the alert being reviewed.
- Alerts from the past 30 days, by default. To extend this timeframe, reach out to your Feedzai contact.

With this widget, fraud prevention agents can easily act on all linked alerts at the same time, thus improving the efficiency of the investigation process.

You can use the checkboxes to select one or multiple linked alerts and **Move to Queue**, **Choose Assignee** and **Unlink from Case**.

Linked alerts vs. Transaction History

Alerts displayed in the **Linked alerts** widget are flagged alerts that remain open for review, while the alerts displayed in the **Transaction History** also include alerts that have already been reviewed.

Preview linked alertsâ

While viewing the linked alerts list, select each alert's row to quickly preview its details.

Summaryâ

Transaction Detailsâ

The following widget includes the transaction details for an inbound payment. Select **View more** to see additional details about the transaction.

It helps uncover fraud evidence such as:

Fraud evidence	Example details
Date/time of transaction outside the normal business hours	The number of transactions performed by this customer in the last 24 hours exceeds the threshold.
Dormant account	Account last activity (e.g. login or payment) was over 3 months ago and in the last 24 hours the number of transactions exceeds the threshold and/or the transaction amount exceeds the account balance's percentage by 40%.
Transaction rejected	A "Rejected" transaction status is often the first line of defense in a financial system, helping to spot potentially fraudulent behavior at a glance.

Rules and Scores Breakdownâ□□

Rules Triggeredâ□□

To support the investigation, this widget lists the rules triggered by the event under investigation.

Imagine the following scenario:

- A customer's account has been inactive for the past six months.
- Suddenly, the customer receives a transaction amount that exceeds the threshold of the account balance.

In this scenario, a rule is triggered by a combination of two factors: dormant account and an inbound amount that exceeds a certain threshold. The event is automatically marked as an alert by the risk engine, and sent for investigation in Case Manager.

Check the full list of standard rules to search for other explanations, and the different fraud scenarios for inbound payments to better understand how rules help prevent them.

Explanationsâ□□

This widget explains why an event received a certain score from machine-learning models and how much each parameter contributed to that score. Model parameters can either increase or reduce the risk and, consequently, the score.

The information available allows you to easily identify the biggest risk **Maximizer** and **Minimizer** according to the following thresholds:

- Less than 0 is a Minimizer.
- Up to 10% is a yellow Maximizer.
- Above 10% is a red Maximizer.

Accountsâ□□

Originator Accountâ□□

This widget includes account information about the counterparty (i.e. originator or sender). Use the **View more** option to see more details about the counterparty, namely their bank information.

info

Note that the counterparty's account information may be incomplete due to lack of information in Feedzai's system about that counterparty. This may happen if the counterparty belongs to a different bank, in which case Feedzai can update information on an ad-hoc basis through the Reference Data API. If that is the case, reach out to the counterparty's bank and enrich the entity for future events.

It helps uncover fraud evidence such as:

Description	Examples and details
Suspicious emails	- A valid customer account profile (name on account: Joe Doe, email on account: ). - Fraudster posing as a real customer (name on account: Joe Doe, email on account: ).
Last activity	This account is dormant for over three months. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner. However, if you compare this indicator with a device ID that is listed to be declined (see the Device widget details), then there's a strong fraud possibility.

Description	Examples and details
Transaction amount too close to credit limit	Typically, fraudsters make several attempts until they successfully find out the credit limit. For instance, if the credit limit is USD 8,000, a fraudster may attempt a USD 8,500 transfer, then a USD 8,200, and finally a successful USD 7,999 transfer.
Risky bank	- The bank that the originator is using to perform a transfer is listed to be declined. - The bank that the originator is using to perform a transfer has recently started to be associated with fraudulent activities.

Beneficiary Accountâ

This widget includes, account information about the beneficiary (i.e. receiver) under investigation. Use the **View more** option to see more details about the receiver.

It helps uncover fraud evidence such as:

Description	Examples and details
Beneficiary's details listed to be declined	The beneficiary's email, country, city, and/or region are listed to be declined.
Risky account	- For incoming transfers, the originator bank is listed to be declined. - For incoming transfers, the originator bank has recently started to be associated with fraudulent activity.
Mismatch between originator's and beneficiary's currencies	- The beneficiary has an unusual currency. - The beneficiary and originator don't share the same currencies.
Account opening date	The beneficiary has received multiple transfers from different originators a few days after the account opening date.
Risky bank	- The beneficiary bank is listed to be declined. - The beneficiary bank has recently started to be associated with fraudulent activities. - The beneficiary bank is known for receiving transactions from risky banks.

Learn more about how to Analyze Customers and gather additional information to support your decision.

Transaction Activityâ

Account Activityâ

The **Account Activity** visualization shows the distribution of amount or volume of inbound and outbound transactions over a defined period.

With this widget, agents can quickly check for:

- Outliers (e.g. sudden spikes in outbound transactions that deviate from normal patterns).
- Repeated patterns of transactions (e.g. frequent small outbound transactions followed by a large inbound transfer).
- Burst activities (e.g. short periods with a high volume of transactions followed by inactivity).
- An increase in outbound payments, immediately followed by a corresponding increase in inbound transactions. This short lag time is typical of mule behavior.
- Aggregated data, to help spot trends over 7, 30 or 90 days.

Transaction Historyâ

This widget displays the transactional events associated with the alert under investigation. It places alerts in a more general picture, thus allowing agents to act upon them in a more efficient way.

You can use this widget in combination with the **Account Activity** widget, benefiting from seeing related information presented in both a visual, intuitive way (in the Account Activity widget) and as a list (in the Transaction History widget).

info

Note that, for intra-bank transactions, Case Manager receives two different events: one in the outbound transfers channel and another one in the inbound payments channel.

Available data varies depending on the value or combination of values selected to fine-tune your results. For instance, by selecting *Customer Name*, the list shows all transactions made by the same customer as the event under investigation.

note

Note that you must select at least one value for transactional data to be displayed.

Additionally, you can reduce the volume of information by selecting a time window, which allows you to choose the period appropriate to your analysis (e.g. Last year, Last day, Last month).

It helps uncover fraud evidence such as:

Description	**Examples and details**
Multiple same amount transactions within a small timeframe	Over 10 transactions of the same amount were alerted for the same Customer Name within 30 minutes.
High number of fraudulent past transactions	Customer has had over 15 transactions marked as **Fraud** in the past, indicating there's a high likelihood of suspicious behavior.

tip

On top of the information available on the **Transaction History** widget, you can refer to:

- The **Notes** widget to check if the parties involved were recently contacted and any other information previously left by other agents, for example, when investigating related alerts or the customer.
- Your internal systems to check for other transactions by the same customer that werenâ€™t alerted (as the information in this widget only refers to alerted transactions). This can help you uncover patterns that might otherwise not look fraudulent/suspicious.

Digital Fingerprintâ

info

Note that inbound payments can be originated from external banks, and it's likely that this information is not present in those cases.

To update information about counterparties, make sure you send the relevant information to Feedzai via Reference Data API.

Networkâ

This widget allows you to see important network details for the event being investigated.

You may also come across a situation that requires adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists (e.g. IP Address) have the **Update Lists** button.

It helps uncover fraud evidence such as:

Description	Examples and details
Proxy server	Fraudsters use proxies to hide their true IP address and geolocation in order to commit fraud without being detected. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner.
IP address	The IP address is associated with previous fraudulent activity and is listed to be declined.
Mismatch between IP address location and originator's city	The device's IP address is Lisbon, whereas the originator's city (see Accounts widget) is New York. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner.

tip

On top of the information available on the **Network** widget, you can refer to external platforms to cross-check specific details with the information available in Case Manager. For example, cross-check the alert's IP address with the one available in the VISA system and use it to assess the distance between the customer's location and the transaction's location.

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The following widget displays the Session Risk Score for the event under analysis.

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The score value changes color depending on the risk severity:

- **Blue (0-39)**: Very low likelihood that the event is fraudulent.
- **Green (40-59)**: Low likelihood that the event is fraudulent.

- **Yellow (60-89)**: Medium likelihood that the event is fraudulent.
- **Red (90-100)**: High likelihood that the event is fraudulent.

The following table shows examples of factors that may lead to a high risk score:

Description	Examples and details
Session Risk Score	The device information captured by the Session Risk Score is associated with previous fraudulent activity.
Unusual session	The current session ID is linked to multiple account changes (e.g during this session the user changed their password, email, phone number, and added a new payee before initiating a transfer).

Deviceâ

This widget allows you to see important information about the device that is being used in the event under investigation. Select **View more** to see additional details about the device.

You may also come across a situation that requires adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists (e.g. Device ID) have the **Update Lists** button.

It helps uncover fraud evidence such as:

Description	Examples and details
Risky device	- The device ID, IP, and associated email are listed to be declined. - The email associated with this device has an unknown domain (e.g. ).
Suspicious device OS	Rooted device. This makes it possible for the fraudster to have privileges that are usually not available nor recommended for the end user.

Learn more about device and behavior fraud scenarios and their threats.

Session Alertsâ

The following widget displays the 'Type', 'View' (online session page in which the alert was detected - e.g. login page, profile page, transfers page) and 'Risk Score' of session alerts.

It helps uncover fraud evidence such as:

Description	Examples and details
ATO high risk score	Account Takeover (ATO) high risk score may tell the agents that the session is likely to be ATO.
RAT high risk score	An event with a high score on Remote Access Trojans (RAT) only tells agents that the event is initiated from a remote access tool, which is not necessarily considered fraudulent. However, you can confirm this by looking at the Session Risk Score (see Session widget): - A high RAT score (e.g. 90) and high Session Risk Score (e.g. 95) suggest that the event is likely to be fraud. - A high RAT score (e.g. 90) but low Session Risk Score (e.g. 20) suggest that the event is likely to be genuine.
VPN high risk score	VPN alert detects VPN activity, which is not necessarily a risk. However, if you observe any Proxy detection, TOR connections, or a high Session Risk Score (see Session widget), this means that a fraudster may be trying to mask their location or IP.

Potential Fraud Risk

Pay special attention to the "Potential Fraud Risk" alert, as it is a strong indicator of a fraudulent session.

This alert indicates the detection of patterns or behaviors that closely align with confirmed fraudulent activity.

For more information, read the Digital Trust's Potential Fraud Risk Alert documentation.

Alert Activityâ

Attached Filesâ

This widget also provides a convenient storage for any information that is not currently represented or supported by Case Manager's interface. It allows you to add information that may be useful to support the investigation and for subsequent investigation steps, legal action, or auditing.

You can use this widget to **upload**, **download** and/or **delete** files, as well as **update** files (e.g. to the latest version) by selecting the update icon.

Notesâ

While investigating alerts, you may need to document any steps taken during your investigation to share with other agents that may look into the same alert. To do so:

1. Select **Add Note**.
2. Select what your note refers to (e.g. event, merchant, customer). Note that depending on the option selected, you might need to enter additional information
3. Enter your **Note**.
4. Select **Save**.

All notes added to the alert (either by you or another agent) are then displayed in this widget, allowing you to check, for example:

- If the parties involved were recently contacted.
- If other agents flagged any related alerts or information that might impact your decision.
- If additional information has been requested and hasn't yet been received.

Activity Logâ

This widget displays the different actions performed in relation to the alert. For example, attached files, notes added, updates to lists, etc.

You can use the dropdown at the top right to filter the list by different actions, such as **Event created** or **Status**.

Next stepsâ

- Learn how to manage cases.
- Classify Events as "Fraud" or "Not Fraud".

On this page

linked-alerts-widget

Was this page helpful?Â

When assigning alerts, users can choose whether to assign a single alert or all alerts linked to it (see Assign alerts to yourself). This widget displays other existing alerts that are assigned to the fraud prevention agent as the one under review. All linked alerts belong to the same channel and share the same .

note

The **Linked alerts** widget only includes:

- Alerts assigned to the same user as the alert being reviewed.
- Alerts from the past 30 days, by default. To extend this timeframe, reach out to your Feedzai contact.

With this widget, fraud prevention agents can easily act on all linked alerts at the same time, thus improving the efficiency of the investigation process.

You can use the checkboxes to select one or multiple linked alerts and **Move to Queue**, **Choose Assignee** and **Unlink from Case**.

Linked alerts vs. Transaction History

Alerts displayed in the **Linked alerts** widget are flagged alerts that remain open for review, while the alerts displayed in the [Transaction History] also include alerts that have already been reviewed.

Preview linked alertsâ

While viewing the linked alerts list, select each alert's row to quickly preview its details.

Manage Cases

Was this page helpful?Â

Cases allow you to group events when there is a common element that suggests a connection between them. This allows you to save time in your investigation and have a more organized and better-structured action-taking process.

info

Cases can be omnichannel. This means that you can aggregate events from **Cards**, **Digital Activity**, and **Transfers**.

Create a caseâ

You can create cases through the **Cases** page, as well as through the **Event**, **Search**, and **Alerts** pages as follows:

- Cases page
- Search / Alert pages
- Event page

1. Hover over the **Cases** icon and select **New Case**.
2. Fill out the necessary information.

Review casesâ

Once you know which cases are up for review, you can use a case's details page to start the investigation process. This section helps you to understand the different case details displayed and how they can help you to classify cases.

The **Cases / Case Details** page header displays the following information:

- **Name**: The case's name.
- **ID**: The case's alphanumerical ID.
- **Status**: The case's state (i.e. **To be reviewed**, **Review in progress**, **Waiting for info** and **Closed**). Check the classify cases section below for more information about each state and how to update it.
- **Source type**: What originated the investigation (e.g. Referrals from customer-facing employees).
- **Queue**: The queue the case is stored in.
- **Assignee**: The user assigned to the case.

Below this header, contextual information is provided as follows:

- **Case Details**: Description of the case and information about who created it and when.
- **Linked Events**: Events associated with the case under investigation.
- **Attached Files**: This widget allows you to add files that support your investigation.
- **Notes**: This widget displays any notes added for the case and allows you to add new ones.
- **Activity Log**: This widget displays the different actions performed in the case, namely regarding linked events, attached files and notes.

Classify casesâ

There are four different states that help you give visibility of your progress and keep the workload organized during your investigation:

From the case's header you can choose between the following options:

- **To be reviewed**: Default status for new cases pending review. It can also be used if the case was previously classified as one of the options below and now needs to be reviewed again.
- **Review in progress**: When you start the case investigation.
- **Waiting for info**: When you are blocked and need more information.
- **Closed**: When you have fully reviewed the events and alerts within the case.

Advanced use of states

States can be used in automation processes, generally created by fraud managers, to improve workload management. For example, fraud managers can create an automation rule to send to a high priority queue all the cases that are waiting for more information.

Review nextâ

You can continue analyzing cases by selecting **Get Next** on the page's header:

By selecting **Cases Queues** from the dropdown menu, Case Manager will automatically assign you to a new case.

note

You can only move to the next case after classifying the current one using the **Closed** option described in the Classify cases section.

Alternatively, you can go back to the **Cases** page to select a new case to review. Check the Access cases section for more information.

Next stepsâ

Learn how to Classify Events as "Fraud" or "Not Fraud".

About Alert Investigation

Was this page helpful?Â

Learn how to manually review suspicious transactions in order to identify and prevent fraudulent behaviors.

Alerts from different eventsâ

When a rule with the action `alert` is triggered, the event is displayed in Case Manager's **Alerts** page for manual revision. This allows fraud analysts to focus their attention on events that show evidence of suspicious activity. Alerts can be generated from rules that trigger for different event types, such as cards, transfers, and digital activity.

There are two types of analysis:

- Post-mortem analysis for card payments, instant transfers, and digital activity events.
- Manual review for credit transfers and direct debits before they hit the `Breached` status.

Alerts generated by instant transfers follow the same logic as card payment alerts, i.e. they're analyzed after the transaction has occurred. Therefore, the post-mortem analysis is important to:

- Provide feedback to the system on whether the event was correctly accepted or rejected.
- Prevent further transactions, for instance, from the same card, originator, or device.

Credit transfers and direct debits are bank payments subject to longer processing times. Therefore, fraud analysts can manually review alerts generated by these payment types with the Operational Status `Unreviewed` and System Decision `Review`. After analyzing the available information, the analyst can mark the alert as 'Fraud' or 'Not Fraud'. This action triggers the following processes:

- The information is sent back to the risk engine and can be leveraged by rules and models.
- The issuer receives a notification and can make a decision on whether to reject or accept the transaction before the transfer's cut-off time.

The process of investigating an alert is identical for all the event types (card, transfer, and digital activity events). The following section illustrates the user journey.

User journeyâ

As a fraud analyst, you are challenged to review suspicious transactions, understand fraud signals, and identify and block new fraud schemes. At the same time, the volume of transactions in real time that needs to be manually reviewed is huge, and thus, you must act quickly to clear your workload.

This guide helps you to become proficient in reviewing suspicious transactions using Case Manager to make your daily work easier. The following user journey shows the different steps associated with a complete fraud analysis:

Next stepsâ

Before accessing your workload, take a quick look at the Event Statuses and Investigation Flow page.

- Operational Work
- Fraud Prevention Agent
- Investigate Alerts
- Analyze Events

Analyze Events

Was this page helpful?Â

Learn how to analyze events based on the information details, and to decide whether the transaction is genuine or fraudulent through the available channels:

Cards

Digital Activity

Inbound Payments

Outbound Transfers

Ad-hoc Customer Investigations

On this page

transaction-activity-widgets

Was this page helpful?Â

Transaction Historyâ

This widget displays the transactional events associated with the alert under investigation. It places alerts in a more general picture, thus allowing agents to act upon them in a more efficient way.

Available data varies depending on the value or combination of values selected to fine-tune your results. For instance, by selecting *Customer Name*, the list shows all transactions made by the same customer as the event under investigation.

note

Note that you must select at least one value for transactional data to be displayed.

Additionally, you can reduce the volume of information by selecting a time window, which allows you to choose the period appropriate to your analysis (e.g. Last year, Last day, Last month).

It helps uncover fraud evidence such as:

Description	**Examples and details**
Multiple same amount transactions within a small timeframe	Over 10 transactions of the same amount were alerted for the same Customer Name within 30 minutes.
High number of fraudulent past transactions	Customer has had over 15 transactions marked as **Fraud** in the past, indicating there's a high likelihood of suspicious behavior.

tip

On top of the information available on the **Transaction History** widget, you can refer to:

- The **Notes** widget to check if the parties involved were recently contacted and any other information previously left by other agents, for example, when investigating related alerts or the customer.
- Your internal systems to check for other transactions by the same customer that werenâ□□t alerted (as the information in this widget only refers to alerted transactions). This can help you uncover patterns that might otherwise not look fraudulent/suspicious.

-
- [Operational Work]
- [Fraud Prevention Agent]
- [Investigate Alerts]
- [Analyze Events]
- Outbound Transfers

On this page

Outbound Transfers

When you open an event or an alert, you will land on the Outbound Transfers' **Event page**, which contains information that helps you make your decision.

Note that you won't be able to make a decision based on a single event, or on a specific widget alone. An investigation is a holistic approach, in which you open several events on different windows and search for fraud evidence by looking at multiple widgets.

About widgets

This page documents the available widgets in a standard order; however, the order in Case Manager may vary depending on your setup. Use the table of contents on the right to navigate to widget-specific documentation and learn more about the information available in each.

Assign alerts to yourselfâ

Before starting the investigation process, start by assigning a single or a set of alerts to yourself. The product can be configured to prevent fraud prevention agents from performing any action on an alert until it is assigned either to them or to another agent. In turn, this prevents different agents from reviewing the same alert at the same time.

To unlock the event, open the **Unassigned** dropdown in the header and select **Assign to me.**

Assign alerts to other agents

The **Choose Assignee** option is only available to some users upon a specific permission.

In addition, you can use the **Assign All** option to assign all linked alerts to yourself. This enables you to easily act on, for example, all unreviewed alerts from the same customer at the same time.

When assigning all linked alerts, a new widget is added to the **Event Details** page with the list of linked alerts to better support the investigation process.

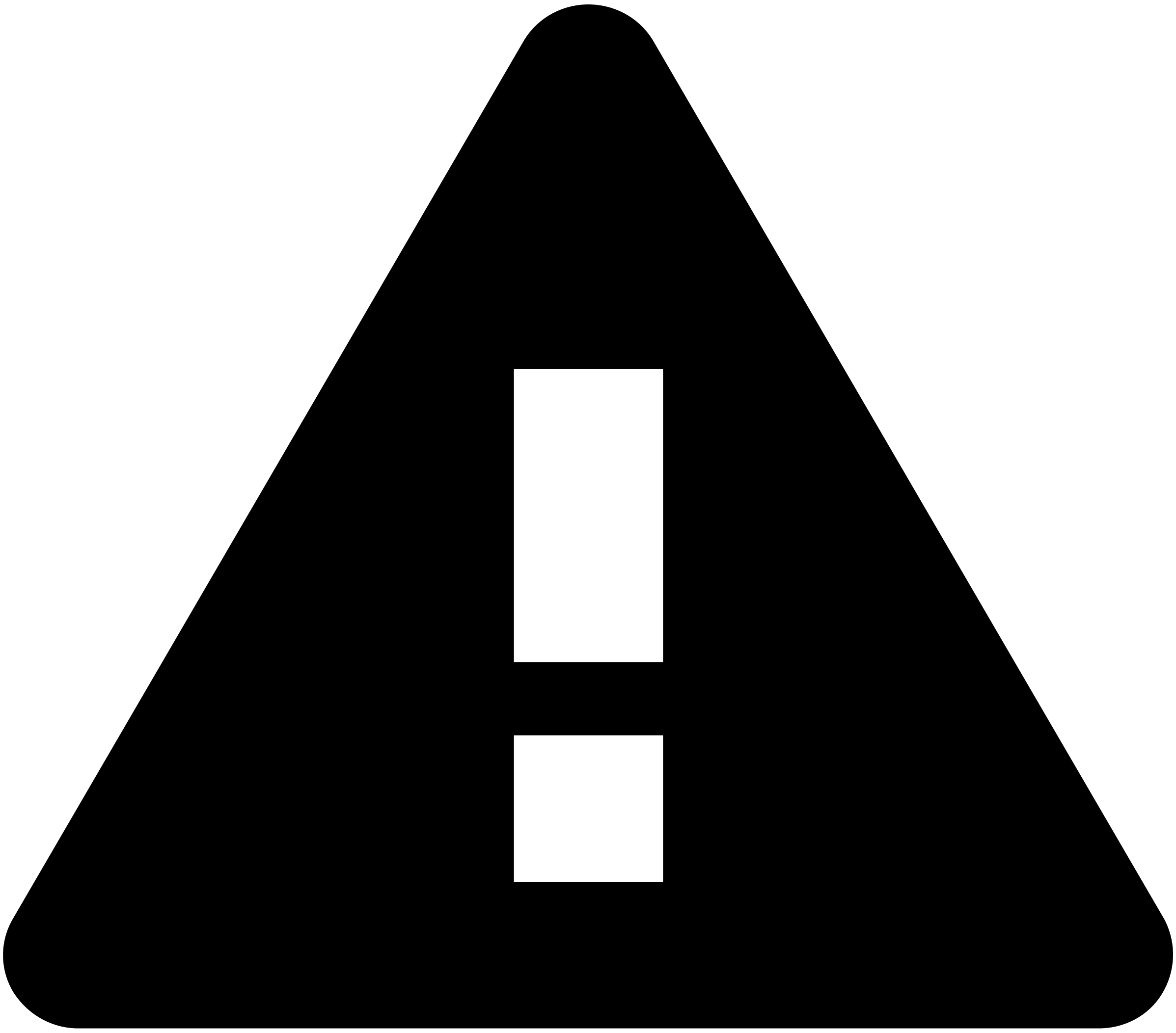
Depending on the configuration, there may be a limit to the number of linked alerts that can be assigned at once. If a limit is set and the **Assign All** action exceeds that number, a message appears specifying how many alerts were assigned.

The remaining alerts that were not assigned to you during this action will remain available for you or another agent to assign and review.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then the **Assign to me** option. Assigning multiple alerts is helpful to, for example, assign all alerts from a specific queue to yourself.



assign-and-lock rules

Fraud Prevention Agents - L1 and **Fraud Prevention Agents - L2** need to assign alerts to themselves to act on them.

Fraud Prevention Agents - L2 and **Fraud Prevention Managers** can reassign to themselves alerts that have already been assigned to other agents.

Only **Fraud Prevention Managers** can act on alerts assigned to other agents; therefore, they don't need to assign alerts to themselves.

Event headerâ

The following fields provide general information about the event to the fraud prevention agent:

- **Event ID**: The event's ID, which can be copied to share with other agents or track later in the system.
- **Assignee**: Indicates if the event is assigned to an agent. The dropdown button lets you assign the event.
- **Labeling**: Includes the âFraudâor "Not Fraud" options for a final decision. It also has the âHoldâbutton to pause the investigation while waiting for further information.
- **Score**: The output of Feedzai's TrustScore, or a combination of scoring models and TrustScore. The number ranges from 0 to 1000, 1000 being the most likely for the event to be fraudulent.
- **System Decision**: Generated by rules in the risk engine platform (i.e. a system that works upstream to Case Manager). Possible values are:
    - Approve

    - Review
    - Decline
- **Fraud Operational Status**: Allows fraud prevention agents to track the alert/event investigation stage. The possible values are:
    - Unreviewed
    - In progress
    - On hold
    - Closed
- **Fraud label**: Indicates whether the agent has reached a final decision. This value can also be populated via API, for example, when the agent receives customer feedback. The possible values are:
    - Fraud
    - Not Fraud
- **Reasons**: Populated when you both mark an event as â□□Fraudâ□□or â□□Not Fraudâ□□and add a reason to clarify the decision.
- **Listed Entities**: Entities that have been added to a list. For example, a Beneficiary Account BIC (Inbound Payments) that has been added to a PosNeg list.
- **Queue**: Indicates if the event is in a queue (e.g. 1 hour to breach).
- **Cases**: Indicates if the event is associated with a case.
- **Fraud Alert**: Indicates whether a rule with the `alert` action was triggered or not.

Linked alertsâ□□

When assigning alerts, users can choose whether to assign a single alert or all alerts linked to it (see Assign alerts to yourself). This widget displays other existing alerts that are assigned to the fraud prevention agent as the one under review. All linked alerts belong to the same channel and share the same `sender_id`.

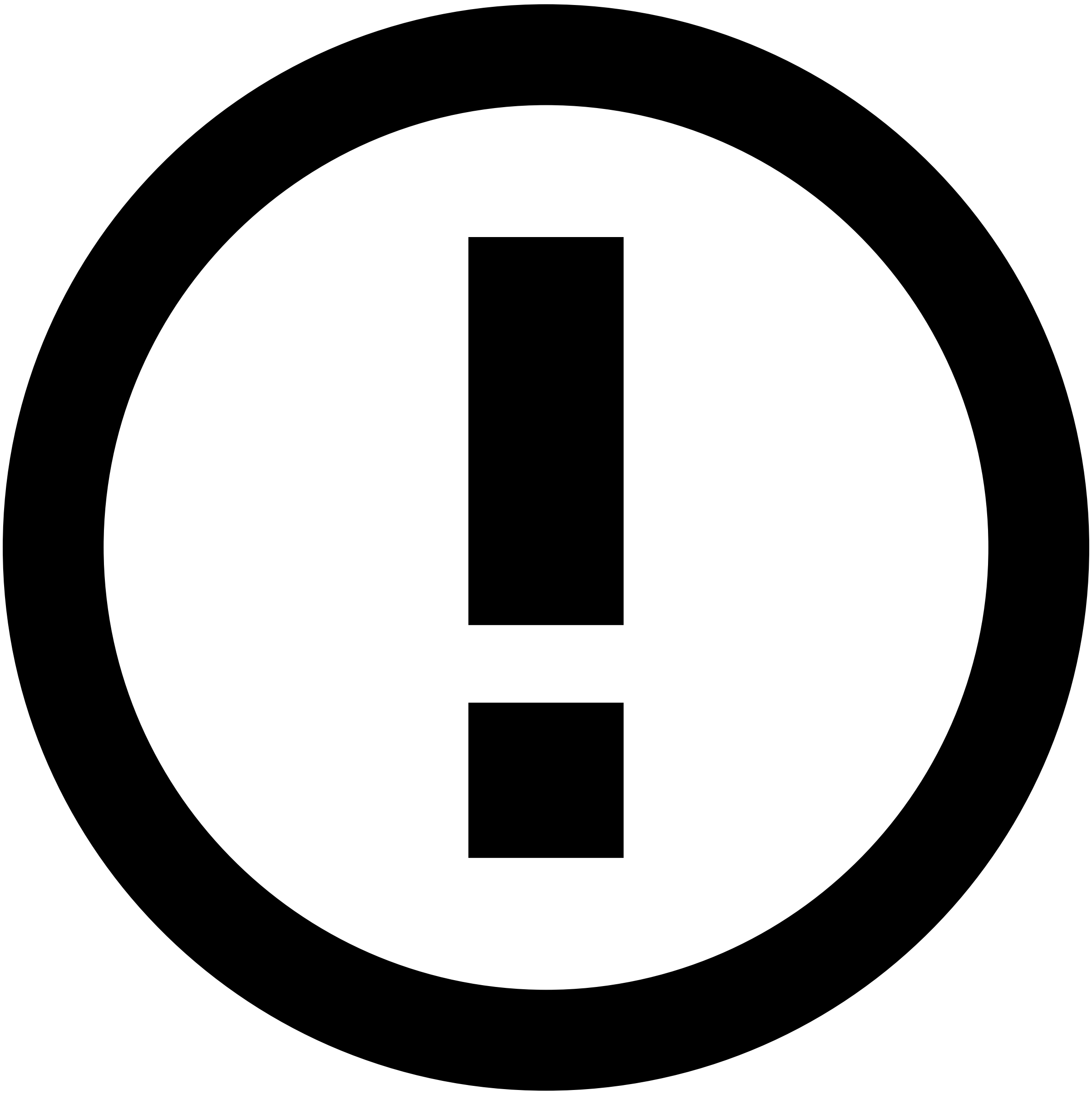
note

The **Linked alerts** widget only includes:

- Alerts assigned to the same user as the alert being reviewed.
- Alerts from the past 30 days, by default. To extend this timeframe, reach out to your Feedzai contact.

With this widget, fraud prevention agents can easily act on all linked alerts at the same time, thus improving the efficiency of the investigation process.

You can use the checkboxes to select one or multiple linked alerts and **Move to Queue**, **Choose Assignee** and **Unlink from Case**.



Linked alerts vs. Transaction History

Alerts displayed in the **Linked alerts** widget are flagged alerts that remain open for review, while the alerts displayed in the [Transaction History] also include alerts that have already been reviewed.

Preview linked alertsâ

While viewing the linked alerts list, select each alert's row to quickly preview its details.

Summaryâ

Transaction Detailsâ

The following figure shows the transaction details for an outbound transfer. Select **View more** to see additional details about the transaction.

This widget helps uncover fraud evidence such as:

Description	Examples and details
Date/time of transaction outside the normal business hours	The number of transactions performed by this customer in the last 24 hours exceeds the threshold.
Dormant account	Account last activity (e.g. login or payment) was over 3 months ago and in the last 24 hours the number of transactions exceeds the threshold and/or the transaction amount exceeds the account balance's percentage by 40%.
Transaction status indicates a "Request for Recall"	Imagine a company that was tricked into paying a large sum of money by a fraudster using a look-alike domain from a regular supplier, only to afterwards realize they were scammed and initiating a payment recall.

Genomeâ

The following graph visualization enables agents to find crime patterns using link analysis and graph technology. Nodes represent entities and edges represent the connection between nodes. To learn how to read Genome graphs, read the Uncover Patterns with Genome page.

To interact with Genome, select the **Expand** icon.

Rules and Scores Breakdownâ

Rules Triggeredâ

To support the investigation, this widget lists the rules triggered by the event under investigation.

Imagine the following scenario:

- A company clerk receives an invoice for a product or service supposedly purchased/contracted by the company.
- The clerk pays the total invoice amount, which exceeds the daily limit of 3k, via transfer to an account based in another country.

In this scenario, a rule is triggered because the maximum amount per day for international transfers was exceeded. This type of event is automatically marked as a high-risk alert by the risk engine, and sent for investigation in Case Manager.

Check the full list of standard rules to search for other explanations, and the different fraud scenarios for outbound transfers to better understand how rules help prevent them.

Explanationsâ

This widget explains why an event received a certain score from machine-learning models and how much each parameter contributed to that score. Model parameters can either increase or reduce the risk and, consequently, the score.

The information available allows you to easily identify the biggest risk **Maximizer** and **Minimizer** according to the following thresholds:

- Less than 0 is a Minimizer.
- Up to 10% is a yellow Maximizer.
- Above 10% is a red Maximizer.

Accountsâ

Sender Accountâ

This widget includes account information about the sender (i.e. originator) under investigation. Use the **View more** option to see more details about the sender, namely their bank information.

It helps uncover fraud evidence such as:

Description	Examples and details
Suspicious emails	- A valid customer account profile (name on account: Joe Doe, email on account: ). - Fraudster posing as a real customer (name on account: Joe Doe, email on account: ).
Last activity	This account is dormant for over three months. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner. However, if you compare this indicator with a device ID that is listed to be declined (see the Device widget details), then there's a strong fraud possibility.
Transaction amount too close to credit limit	Typically, fraudsters make several attempts until they successfully find out the credit limit. For instance, if the credit limit is USD 8,000, a fraudster may attempt a USD 8,500 transfer, then a USD 8,200, and finally a successful USD 7,999 transfer.
Risky bank	- The bank that the originator is using to perform a transfer is listed to be declined. - The bank that the originator is using to perform a transfer has recently started to be associated with fraudulent activities.

Learn more about how to Analyze Customers and gather more information to support your decision.

Receiver Accountâ

This widget includes, account information about the counterparty (i.e. receiver or beneficiary). Use the **View more** option to see more details about the counterparty.

info

Note that the counterparty's account information may be incomplete due to lack of information in Feedzai's system about that counterparty. This may happen if the counterparty belongs to a different bank, in which case Feedzai can update information on an ad-hoc basis through the Reference Data API. If that is the case, reach out to the counterparty's bank and enrich the entity for future events.

It helps uncover fraud evidence such as:

Description	Examples and details
Beneficiary's details listed to be declined	The beneficiary's email, country, city, and/or region are listed to be declined.
Risky account	- For incoming transfers, the originator bank is listed to be declined. - For incoming transfers, the originator bank has recently started to be associated with fraudulent activity.
Mismatch between originator's and beneficiary's currencies	- The beneficiary has an unusual currency. - The beneficiary and originator don't share the same currencies.

Description	Examples and details
Account opening date	The beneficiary has received multiple transfers from different originators a few days after the account opening date.
Risky bank	- The beneficiary bank is listed to be declined. - The beneficiary bank has recently started to be associated with fraudulent activities. - The beneficiary bank is known for receiving transactions from risky banks.

Transaction Activityâ

Account Activityâ

The **Account Activity** visualization shows the distribution of amount or volume of inbound and outbound transactions over a defined period.

With this widget, agents can quickly check for:

- Outliers (e.g. sudden spikes in outbound transactions that deviate from normal patterns).
- Repeated patterns of transactions (e.g. frequent small outbound transactions followed by a large inbound transfer).
- Burst activities (e.g. short periods with a high volume of transactions followed by inactivity).
- An increase in outbound payments, immediately followed by a corresponding increase in inbound transactions. This short lag time is typical of mule behavior.
- Aggregated data, to help spot trends over 7, 30 or 90 days.

Transaction Historyâ

This widget displays the transactional events associated with the alert under investigation. It places alerts in a more general picture, thus allowing agents to act upon them in a more efficient way.

You can use this widget in combination with the **Account Activity** widget, benefiting from seeing related information presented in both a visual, intuitive way (in the Account Activity widget) and as a list (in the Transaction History widget).

info

Note that, for intra-bank transactions, Case Manager receives two different events: one in the outbound transfers channel and another one in the inbound payments channel.

Available data varies depending on the value or combination of values selected to fine-tune your results. For instance, by selecting *Customer Name*, the list shows all transactions made by the same customer as the event under investigation.

note

Note that you must select at least one value for transactional data to be displayed.

Additionally, you can reduce the volume of information by selecting a time window, which allows you to choose the period appropriate to your analysis (e.g. Last year, Last day, Last month).

It helps uncover fraud evidence such as:

Description	**Examples and details**
Multiple same amount transactions within a small timeframe	Over 10 transactions of the same amount were alerted for the same Customer Name within 30 minutes.
High number of fraudulent past transactions	Customer has had over 15 transactions marked as **Fraud** in the past, indicating there's a high likelihood of suspicious behavior.

tip

On top of the information available on the **Transaction History** widget, you can refer to:

- The **Notes** widget to check if the parties involved were recently contacted and any other information previously left by other agents, for example, when investigating related alerts or the customer.
- Your internal systems to check for other transactions by the same customer that werenâ□□t alerted (as the information in this widget only refers to alerted transactions). This can help you uncover patterns that might otherwise not look fraudulent/suspicious.

Digital Fingerprintâ

Networkâ

This widget allows you to see important network details for the event being investigated.

You may also come across a situation that requires adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists (e.g. IP Address) have the **Update Lists** button.

It helps uncover fraud evidence such as:

Description	Examples and details
Proxy server	Fraudsters use proxies to hide their true IP address and geolocation in order to commit fraud without being detected. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner.
IP address	The IP address is associated with previous fraudulent activity and is listed to be declined.
Mismatch between IP address location and originator's city	The device's IP address is Lisbon, whereas the originator's city (see Accounts widget) is New York. Note that you won't be able to confirm whether this is an indicator of fraud if you look at it in an isolated manner.

tip

On top of the information available on the **Network** widget, you can refer to external platforms to cross-check specific details with the information available in Case Manager. For example, cross-check the alert's IP address with the one available in the VISA system and use it to assess the distance between the customer's location and the transaction's location.

Sessionâ

The following widget displays the Session Risk Score for the event under analysis.

info

Session Risk Score is a unique digital identifier that is created for each user. The Session Risk Score collects the userâ□□s full context of biometric, behavioral, device and network data points, and continuously analyzes user behavior from login to logout across every interaction.

The score value changes color depending on the risk severity:

- **Blue (0-39)**: Very low likelihood that the event is fraudulent.
- **Green (40-59)**: Low likelihood that the event is fraudulent.
- **Yellow (60-89)**: Medium likelihood that the event is fraudulent.
- **Red (90-100)**: High likelihood that the event is fraudulent.

The following table shows examples of factors that may lead to a high risk score:

Description	Examples and details
Session Risk Score	The device information captured by the Session Risk Score is associated with previous fraudulent activity.

Description	Examples and details
Unusual session	The current session ID is linked to multiple account changes (e.g during this session the user changed their password, email, phone number, and added a new payee before initiating a transfer).

Deviceâ

This widget allows you to see important information about the device that is being used in the event under investigation. Select **View more** to see additional details about the device.

You may also come across a situation that requires adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists (e.g. Device ID) have the **Update Lists** button.

It helps uncover fraud evidence such as:

Description	Examples and details
Risky device	- The device ID, IP, and associated email are listed to be declined. - The email associated with this device has an unknown domain (e.g. ).
Suspicious device OS	Rooted device. This makes it possible for the fraudster to have privileges that are usually not available nor recommended for the end user.

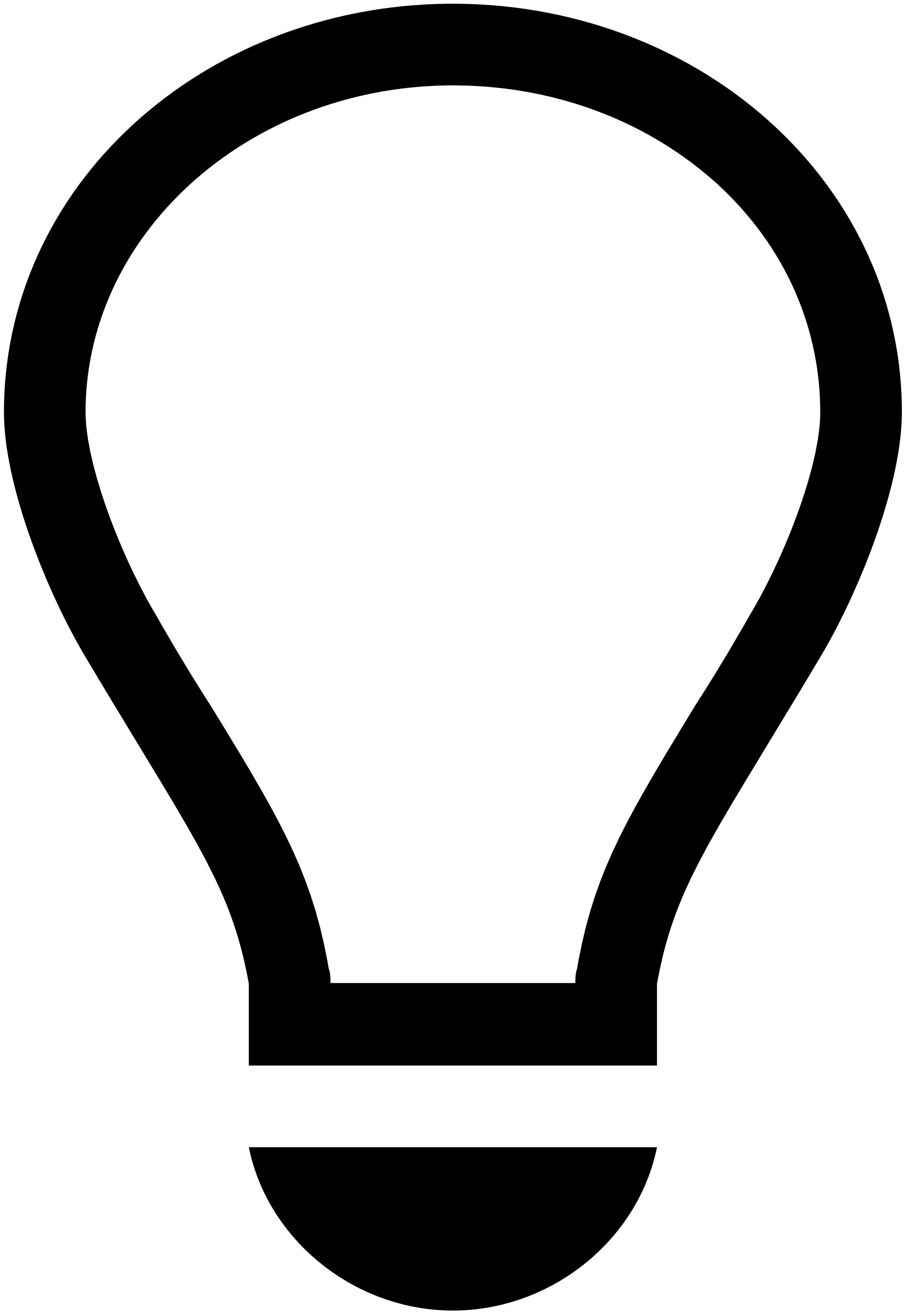
Learn more about device and behavior fraud scenarios and their threats.

Session Alertsâ

The following widget displays the 'Type', 'View' (online session page in which the alert was detected - e.g. login page, profile page, transfers page) and 'Risk Score' of session alerts.

It helps uncover fraud evidence such as:

Description	Examples and details
ATO high risk score	Account Takeover (ATO) high risk score may tell the agents that the session is likely to be ATO.
RAT high risk score	An event with a high score on Remote Access Trojans (RAT) only tells agents that the event is initiated from a remote access tool, which is not necessarily considered fraudulent. However, you can confirm this by looking at the Session Risk Score (see Session widget): - A high RAT score (e.g. 90) and high Session Risk Score (e.g. 95) suggest that the event is likely to be fraud. - A high RAT score (e.g. 90) but low Session Risk Score (e.g. 20) suggest that the event is likely to be genuine.
VPN high risk score	VPN alert detects VPN activity, which is not necessarily a risk. However, if you observe any Proxy detection, TOR connections, or a high Session Risk Score (see Session widget), this means that a fraudster may be trying to mask their location or IP.



Potential Fraud Risk

Pay special attention to the "Potential Fraud Risk" alert, as it is a strong indicator of a fraudulent session.

This alert indicates the detection of patterns or behaviors that closely align with confirmed fraudulent activity.

For more information, read the Digital Trust's [Potential Fraud Risk Alert] documentation.

Alert Activityâ

Attached Filesâ

This widget also provides a convenient storage for any information that is not currently represented or supported by Case Manager's interface. It allows you to add information that may be useful to support the investigation and for subsequent investigation steps, legal action, or auditing.

You can use this widget to **upload**, **download** and/or **delete** files, as well as **update** files (e.g. to the latest version) by selecting the update icon.

Notesâ

While investigating alerts, you may need to document any steps taken during your investigation to share with other agents that may look into the same alert. To do so:

1. Select **Add Note**.
2. Select what your note refers to (e.g. event, merchant, customer). Note that depending on the option selected, you might need to enter additional information
3. Enter your **Note**.
4. Select **Save**.

All notes added to the alert (either by you or another agent) are then displayed in this widget, allowing you to check, for example:

- If the parties involved were recently contacted.
- If other agents flagged any related alerts or information that might impact your decision.
- If additional information has been requested and hasn't yet been received.

Activity Logâ

This widget displays the different actions performed in relation to the alert. For example, attached files, notes added, updates to lists, etc.

You can use the dropdown at the top right to filter the list by different actions, such as **Event created** or **Status**.

Next stepsâ

- Learn how to manage cases.
- Classify Events as "Fraud" or "Not Fraud".

Uncover Patterns with Genome

Feedzai Genome enables analysts to find and visualize complex financial crime patterns using link analysis and graph technology.

The powerful link analysis technology underlying Feedzai Genome, along with the tool's visual capabilities, help analysts make decisions faster and with greater accuracy, thus lowering false positives in manual detection that harm loyal customers. Genome helps fraud analyst to understand the relationship between all the entities participating in a transaction, such as the originator and beneficiary, the bank accounts and the device involved, including the direction of the money flows between them. This enables fraud analysts to detect complex fraud scenarios such as Account Takeover or Authorized Push Payment Fraud, for example, when a customer is transacting from multiple new devices or unusual beneficiaries.

Access Genomeâ

To access Genome, open an event or alert in the Alerts page. The Event Details page displays the Genome widget as follows:

To better understand how you should analyze the links, consider the following examples:

Card transaction

- A user makes a transaction in a merchant. That transaction typically consists of the following entities:
- User
- Device
- Card

Genome displays each of these entities as nodes and the relationship between them as the edges that connect the nodes.

Transfer between originator and beneficiary

A sender (i.e. originator) makes a transfer to the receiver (i.e. beneficiary). That transfer consists of the following entities:

- Originator and beneficiary
- Device (if present)
- Originator and beneficiary banks
- Originator and beneficiary bank accounts

Genome displays each of these entities as nodes and the relationship between them as the edges that connect the nodes. **Nodes (1)** represent entities and **edges (2)** represent the connection between nodes.

info

Genome displays only the nodes that are related to an alert under analysis.

Read Genome graphsâ

Now that you understand what nodes and edges are, read this section to learn how to properly read a Genome graph. Consider the following core concepts:

- Node entities
- Node and edge colors
- Edge gradient and how it affects the money flow direction

Node entities

- Customer corresponds to the sender and/or receiver customers.
- Account corresponds to the sender and/or receiver accounts.
- Device corresponds to the sender and/or receiver devices.

- Phone corresponds to the sender and/or receiver phones.
- Email corresponds to the sender and/or receiver emails.

Node and edge colors

The graph's node and edge colors are visible on the **Genome** lens and correspond to the state of events associated with those nodes and edges:

- Yellow - At least one event in pending state.
- Red - At least one confirmed fraud event.
- Purple - At least one declined event.
- Green - All events are accepted.

Edge gradient

The edge gradient is directly connected to the money flows' functionality. This gradient indicates the money flow direction through the edges' color as follows:

- Grey, with a gradient on the edge's color, indicates who sends and who receives the money. The edge displays a lighter color next to the sender node and darker color next to the receiver node. The money flows from the lightest to the darkest side of the edge.
- Purple is used to indicate that there are transfers in both directions between two entities. For instance, consider two account entities A and B, respectively, where the money would flow from account A to B, as well as from B to A. When there is a history of fraud, the edge displays red instead of purple. Note that bidirectional flows are only displayed in the Genome's default view. To see this gradient color, select the **Expand** icon at the top right corner of Genome's widget on the Event Details page.
- Red indicates that there is a history of fraud from at least one transaction.

info

Note that, when opening a graph, the **Review** lens is displayed by default. To see the edge gradients with the aforementioned colors, move to the **Genome** view.

Read the following fraud use cases, which provide you with step-by-step graph descriptions.

Genuine eventâ

The following event displays a genuine transfer. The nodes that represent the sender (1), email (2), phone (2), and device (2) entities are involved in one transfer. The money settles from the sender's bank account (3) to the receiver's (5) bank account (4). Note that the money flows from the lightest to the darkest side of the edge. This is why the edge connecting to the receiver's bank account is darker. In this case, you should read the graph from left to right.

Fraudulent eventâ

The following event displays a fraudulent transfer. Two senders (1 and 2) share the same phone, device, email, and bank account (3). One of the senders (2) settles money from a second bank account (4) to the receiver's (7) bank account (5).

In addition, money is settling from the bank account (3) shared by both senders (1 and 2) to another bank account (6) owned by the same receiver (7). Note that the edges connecting the sender (2) and the phone, device, email and bank account (3) are thicker. This means that multiple transactions occurred. The thicker the edge, the higher the number of transactions. Furthermore, note that the rules explanation indicates that all these transactions occurred in a short period (less than 90 minutes), which is not a usual behavior.

Learn moreâ

Read the following documentation to learn how to interact with Genome:

- Navigating Genome
- Operations
- Genome and CM Integration

Update Lists

Lists allow Case Manager to automatically classify events by looking at specific entities in each event (e.g. credit card, customer ID) and checking if those entities are associated with a known fraud source or if they belong to a secure source.

You can use the information acquired during your investigation of an event to update lists with the relevant entities.

Implications

Updating lists with new items is an advanced topic because it might have implications for your business. For example, if you add the MCC value of the current event to a PosNeg list with the value **Decline** and there's a specific rule to trigger against this condition, all upcoming transactions with this MCC will be declined.

After classifying an event as or , you are prompted to justify your decision. If you use the button, you have access to the **Update Lists** menu:

This window provides you with an indication of the decision made as the window title (i.e. *Mark as Fraud* or *Mark as Not Fraud*) and allows you to add the available **Entities** to a list with a specific value (Approve, Decline or Limit/Review). For example, you can add the *Card ID* to a list to be declined in future transactions.

note

Checkboxes in gray refer to the current value for each entity in lists.

For each entity added to a list you can also:

- Define a time frame for the entity to remain in the list using the **Validity** columns.
- Add a **Note** (e.g. with extra information to support your decision).

Journey wrap-upâ

The **Investigate Alerts** guide aims to get you up and running with Case Manager from day 1, and able to quickly analyze alerts. Here is a wrap-up of what comprises a typical journey:

1. Access a set of events either by using the **Alerts** or **Cases** pages. In both of these pages, you can further filter the events displayed by selecting queues or events assigned to you.
2. Select an event to analyze by looking at the event details including the header, explanations and other widgets that have information about transaction, account, card, merchant, etc.
3. Classify the event as **Fraud** or **Not Fraud** and justify your decision.
4. Update lists with new entities (advanced topic that might require special permissions or additional training).

Event Life Cycles

Was this page helpful?Â

Learn about the life cycles of each event type and the role each endpoint plays in the different stages of each event.

Cards

Digital Activity

Inbound Payments

Outbound Transfers

Ad-hoc Investigations

Reference Data

Create Self-Service Rules

Self-Service Rules (SSRs) allow you to quickly create a rule, test and deploy it directly in production to improve fraud prevention.

The risk engine platform has a set of standard rules that are used in combination with lists to catch fraud. However, fraudsters are constantly coming up with new workarounds to trick the system. Counteracting new waves of fraud might require tweaking a rule or creating a new one. The rules created in the risk engine platform are normally developed and tested in staging environments, to ensure that the work in development does not affect the work in production. They are also developed by teams that have deep knowledge in this area and often have taken special training. Therefore, there is an inevitable gap between the time that these rules start to be written and the time that they are deployed in production.

SSRs are a Case Manager functionality that allows you to fill this gap, and have a rule live in production in a few minutes. With this capability, you can write a rule using an intuitive interface, test and publish it directly into production. As an example, you can use SSRs to create a rule with a different threshold and publish in production to have it quickly live.

Create an SSRâ

To create SSRs in Case Manager, execute the following steps:

1. Enter the **Self Service Rules** page in Case Manager's **Settings**
2. Use the button to open the SSRs creation panel.

As an example, imagine that you want to define an SSR with a lower threshold than the one defined in the Cards-FR19 rule that is configured in the risk engine (see a full list of Fraud rules. The Cards-FR19 declines transactions and generates an alert when the number of transactions made by a card within three minutes is equal or higher than five. The following settings show you how to create an SSR identical to UB23 but setting a new limit for the number of transactions within three minutes: equal or higher than three.

- **Name:** Limit the number of transactions in the last 3 minutes
- **Explanation:** Decline transaction if the number of transactions made by the card within the last 3 minutes is equal or higher than 3 (updated threshold).

The following image shows you how to define the conditions for this example:

info

The **Schema Entity** is also shared by the risk engine rules. Check the *Workflow Schema <version>* under the Schema Reference page to have access to the full list of fields.

info

When adding the **amount** field to the condition, the value is defined in cents (e.g. 1000 to represent $10.00).

The following image shows you how to define the decision and the status for this example:

Create an SSR with TrustScoreâ

The following image exemplifies an SSR created specifically to auto-decline card events with a score equal or greater than 750.

For Cards and Outbound Transfers, this score is the output of Feedzai's TrustScore, or a combination of scoring models and TrustScore.

Test an SSRâ

You can test a rule after selecting the button, or you can test it later by viewing the rule (this icon is shown when hovering over your list of SSRs). Select a predefined time period or define a 'Custom' start/end date that will be used to perform the test:

The rule is tested using the events that are in Case Manager whose timestamp is within the time window that you select.

Once the test run has ended, you can analyze its results:

- **Hit Rate:** Percentage of events that triggered the rule.
- **Hit Rate Count:** Absolute number of events that triggered the rule.
- **Hit Rate Amount:** Sum of the transaction amount captured by the rule.
- **FPR (False-Positive Rate):** Percentage of events that are not fraud in the real world, and that were classified as fraud by the risk engine.
- **Precision:** Out of all events within the time window that were declined by the risk engine, how many were correctly declined.

Reorder SSRsâ

By clicking and dragging rules, users can establish their priority. This means that rules on the top are prioritized over those that come down below:

Publish SSRsâ

Writing and testing SSRs are stages that are performed offline, and thus, do not impact the processing of live transactions. The SSRs become live only after publishing them.

danger

Publishing rules: Make sure you double check the SSRs and test them carefully before publishing them. Promoting SSRs that do not perform well might have a negative business impact.

To promote SSRs to your live environment, use the button.

tip

Active Rules: when publishing changes, Case Manager will activate the rules that have the state as enabled.

Manage Details

Was this page helpful?Â

Analysts spend most of their time reviewing alerts, cases, or customers. However, the out-of-the-box interface configuration of these pages doesn't always match the processes used by each financial institution which leads analysts to spend more time to make a decision.

Managers can edit the configuration of the Details pages, thus optimizing their team's workflow. For example, managers may want to adjust a field's naming to ensure that all analysts work with company-compliant terminology, or change the order in which the fields are displayed.

This page explains how managers can edit some of the information displayed in the Event, Case, and Customer Details pages.

Start by selecting the **Edit** button to enable the edit mode.

The edit mode allows to add, edit, delete, and change the order of the fields in each widget. Note that some widgets cannot be edited, and therefore, are disabled.

Let's take a look at the available actions using the Customer widget as example.

Change fields' orderâ

To change the order of the fields, drag a field to the new position.

Add fieldsâ

Add a field, for example, to show the customer's account number.

1. Select the **Add Field** button and use the **Add Field** window to configure the field.

2. Define the **Label**, for example, *Account IBAN*.

3. Configure the new field, namely by selecting the:

    - **Field:** Select the field from the data source that corresponds to the information that you want to see, for example, *customer.account.number*.
    - **Format:** Select the field format from the available options: Date, Number, Text, and Translation.

Translation type

Fields such as the customer ID, email, or IBAN are not translatable. However, fields such as the customer or alert status may have a different value according to the interface language.

When adding translatable fields, select the **Translation** type so that the widget displays the value in the correct interface language.

4. Alternatively, select the **Advanced** option to create the field using a *json* format. This option allows you to add a more complex configuration. Learn more about the [advanced mode].

info

This option is only available to some users upon a specific permission.

1. **Preview** the field and then select the **Apply** button to finish the creation.

Edit fieldsâ

Edit existing fields, for example, to change their name or type.

To change a field's name, select the field and type the new name. To apply the changes, press ENTER on your keyboard or click anywhere outside the field.

In addition, you can perform other changes, for example, to change the type of a transaction's **Review State** field from *Text* to *Translation*. This way, analysts can see the state on their preferred language.

1. Hover over the field to display the inline actions.

1. Select the **Edit** option to open the **Edit Field** window.

2. Change the field schema or type.

1. Alternatively, select the **Advanced** option to edit the field using a *json* format. This option allows you to add a more complex configuration.

info

This option is only available to some users upon a specific permission.

1. **Preview** the changes and **Apply** them.

Add actions to widgets or fieldsâ

Add actions, such as copy to clipboard, to widgets or to specific fields.

To add actions to **widgets**, perform the following steps:

1. In the widget's header, select the **actions** option.

1. Select the icons to be displayed on the widget header, namely copy, or add to list.

1. Use the windows displayed to select the fields that should be copied or added to list as well as the corresponding informative tooltip.

info

In the **Add to List** window, you can only select fields that can be added to lists.

1. **Apply** the changes.
2. In the widget's **actions** toolbar, select the **Apply** icon to save the changes.

In addition, you can add actions to specific **fields**:

1. Hover over a field to display the inline actions.

1. Select the **actions** option.

2. Select the icons to be displayed next to the field, namely copy, tooltip with extra information, or links.

- When you add a tooltip, use the **Info Tooltip** window to define the corresponding text. Select **Insert** to save the text.

- When you add links, use the **Social Media** window to select one or multiple links from the available options. **Apply** the selection.

These links redirect users to the website's search page with results matching the value defined for a field. For example, if you add a Facebook link next to the Customer Name field, it will redirect users to Facebook's search page with results matching that customer's name.

1. Select the **Apply** icon to save the changes.

Delete fieldsâ

Delete fields that are no longer useful for analysts during their investigation.

1. Hover over the field to display the inline actions.

1. Select the **Delete** option.

2. In the confirmation window, select the **Delete** button to confirm the deletion.

Advanced modeâ

Use the **Advanced** mode to adjust widgets or specific fields with a *.json* format.

info

This option is only available to some users upon a specific permission.

1. In the widget's header, select the **Advanced** option.

Alternatively, use the **Advanced** option while adding or editing specific fields.

1. In the **Advanced Widget** window, adjust information such as the widget's title or the icons that appear next to a field.

Use the suggestions provided while typing to ensure that you are using the correct properties.

info

Some properties (i.e. `type` , `id` and `width`) are hidden and cannot be changed. If you save the configuration with these properties, they will be ignored. In addition, if you submit an invalid configuration (for example, with an invalid comma or character), the system will save the last valid configuration.

1. Preview the changes and **Apply** them.

The Card component accepts the following properties:

Name	Type	Description	Required
"component"	`string`	The type of component to be rendered. Must be set to "card".	yes
"title"	`string`	The visible title displayed on the card.	yes
"id"	`string`	Unique component identifier.	no
"cssClass"	`string`	A custom CSS class applied to the card component.	no

Name	Type	Description	Required
"type"	`string`	The type of content to be rendered. It can be one of the following: - toplevel - box - map - table - list-box - entity-history - activity - iframe - paginated-widget - notes - genome	no
"layoutColumns"	`number,object`	Specifies the number of columns the component will occupy in the parent grid system. Can be either: - A number: fixed amount across all breakpoints. - An object with optional properties small, medium, and large: different values for each breakpoint.	no
"fieldCutoff"	`number`	Number of fields to display initially before cutoff. If there are more items than the cutoff, a "View more" button will be displayed.	yes
"footer"	`object`	â□□¸□Deprecated Use `fieldCutoff` property instead. Configuration for call-to-action buttons in the card footer.	no

Here's a more practical example for your consideration.

Session widgetâ□□

The **Session** widget displays a link to the Digital Trust's Console through the *Session ID* field.

This link can be changed using the **Advanced** mode.

Depending on your configuration and API versions you may need to update this URL. The URL is composed of the following elements:

```
<protocol>://<domain>/<path>/<company_id>/<session_id>/<timestamp>
```

- `protocol` is the web protocol used in Digital Trust.
- `domain` is the Digital Trust's console hostname.
- `path` is the path of the Digital Trust's console that will give you the session information for that event.
- `company_id` is the company identifier, established by Digital Trust and associated with each specific customer.
- `session_id` is the session ID associated with the event.
- `timestamp` is the event's timestamp.

To update any of these fields, enter the **Edit** mode and select the **Advanced** option.

Considering the information provided in the image above, you can identify:

- The **protocol** as `https` .
- The **domain** as `console.pre.bugaroo.com` .
- The **path** as `console/sessions/explorer/external/details` .
- The **company_id** as `bugbank` .
- The **session_id** as a variable that will fetch the session ID from the API.

If the URL is not matched to the latest version of Digital Trust's API, update the **path** to the correct one, and add the event **timestamp** at the end, as follows:

Besides changing the `headerActions` URL, update the **Session ID** field, as follows:

Learn more about the Digital Trust's integration with Transaction Fraud for Banking.

Custom widgetsâ

Case Manager can also be configured to adjust to your specific needs. To answer these needs, **custom widgets** can be created.

Contact the Customer Success team to create a custom widget.

Save or discard changesâ

Once you're happy with the changes, select the **Save all changes** button to save them and exit the edit mode.

Alternatively, select the **Cancel** button to discard all changes made since you enabled the edit mode, or open the options menu and select the **Restore Page Defaults** option so that all widgets go back to how they were initially configured by the system administrator.

You can also discard only the changes made to specific widgets as follows:

1. Select the **Reset** button displayed in each widget.

1. In the **Reset Changes** window, select the **Reset** button to discard the changes made since you enabled the edit mode. Alternatively, select the **Restore Defaults** button so that the widget goes back to how it was initially configured by the system administrator.

Export and import configurationâ

The edit mode allows you to export and import the page's configuration. This can be helpful, for example, to apply the same configuration between different environments (for example, from QA to Production).

To export the current configuration, select the **Export Current Configuration** option under the more options dropdown menu.

Unsaved changes

If the current edit mode has unsaved changes, you need to save them before exporting.

To import a previously exported configuration:

1. Select the **Import Page Configuration** option under the more options dropdown menu.

2. In the **Import Page Configuration** window, upload or drag the exported file to the **Add Files** area.

1. Select **Apply** to finish the import. The new configuration will replace the current one.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/manage-workload-and-rules/manage-lists

- Operational Work
- Fraud Prevention Manager
- Manage Workload and Rules
- Manage Lists

On this page

Manage Lists

Was this page helpful?Â

List Management is a built-in functionality in Case Manager that allows you to add, remove, and search for items within a list.

To access the List Management panel, hover over the **Settings** option in Case Manager's navigation bar and then select **List Management**.

By default, TFB displays the following lists:

PosNeg lists allow you to store items with the values: approve, decline or review/limit. With simple lists, you can add items directly as entity values. Learn more about Pulse Lists.

From the **List Management** panel you can perform a number of tasks, such as:

- Adding entities to lists
- Searching for entities in lists
- Updating an entity on a list
- Removing an entity from a list

Permissions

The permissions to view, create, edit and delete items from a list are defined by user roles. See the Roles and Permission page to check all the default configurations.

Adding entities to listsâ

There are several different ways to add entities to lists:

- Manually adding an entity to a list

- Adding an entity from an event to a list
- Adding all entities in an event to all relevant lists

note

An entity can be added to multiple lists, but lists only contain one entity type (e.g. a list for countries cannot include account number entities).

Manually adding an entity to a listâ

On the **Lists Management** page, hover over the intended list and select the button. This opens a dialog for you to add an entity to the specific list.

Once you have populated the fields with the relevant information, select to add the entity to that list.

Manually adding entities

Manually adding an entity value to a list removes any automatic control of the value. Review the information carefully to avoid typos, as they will result in an entity mismatch.

Adding an entity from an event to a listâ

While reviewing an event, you can add a specific entity to a list, or alter its configuration if it's already in a list. Information fields eligible for lists have the **Update Lists** button. Select it to open the **Update Lists** dialog.

This dialog allows you to add the entity to a new list or to update its configuration (which action to take, the validity of the action, etc). It can also be used to remove the entity from lists.

Adding all entities in an event to all relevant listsâ

On the **Event** page, select **Update Lists** to view all the event entities that can be included in lists.

This opens the **Update Lists** dialog for you to choose which lists to add the entities to. You can also specify the **Validity** and complement the action with a **Note**. If an entity is already on a list, it can be moved to another list or be **Unlisted**.

Searching for entities in listsâ

You can search for entities that belong to a given list by using the **Search Entities** tab.

Viewing audits in listsâ

Case Manager allows you to view the auditing history of a listed entity.

When you search for an entity in lists, Case Manager displays all audit entries where that entity is present. To view this audit trail, select the **View Audit** button in the **Search entities** tab.

This view allows you to filter the audits by lists. Select the button and choose the intended list(s) from the drop-down menu.

Case Manager supports having multiple audit trails open at the same time. You can go back to the **Search Entities** tab, perform another search and open the audit view for that search. The **Audit Entities** tab remains open with the previous entity audits.

Updating an entity on a listâ

There are two methods for updating an entity on a list. You can either locate the entity through an event (for example, in the **Alerts** page) or through the lists, as described in searching for entities in lists.

On the **Alerts** page, information fields that are eligible for lists have the button. Select it to open the **Update Lists** dialog.

The dialog shows what lists the entity is currently on. Select a list and click on the button to change the configuration of the entity. When all changes are done, click on the button.

Alternatively, if you searched for the entity directly in the lists, hover on the list and select the **Update entity** button.

This opens the **Update Entity** dialog, allowing you to update the entity.

Removing an entity from a listâ

There are two methods for removing an entity from a list. You can either access the entity through an event (for example, in the **Alerts** page) or through the lists, as described in searching for entities in lists.

On the **Alerts** page information, fields that are eligible for lists have the button. Select it to open the **Update Lists** dialog that displays what lists the entity is currently on.

Select the button to remove the entity from all lists or individually update the lists from which you wish to unlist the entity. After selecting all removals, select the button to apply those changes.

Alternatively, if you searched for the entity directly in the lists, click on the button to remove from all lists, or hover on a specific list and select the button.

The button opens the **Update Entity** dialog, allowing you to unlist the entity, removing it from the list.

Learn moreâ

Learn how to create a self-service rule, test and deploy it directly in production to quickly counteract a fraudster scheme.

- Operational Work
- Fraud Prevention Manager
- Manage Workload and Rules
- Organize Workload Manually

On this page

Organize Workload Manually

Was this page helpful?Â

Learn how to manually add events/alerts and cases to a queue as well as to assign them to a team member.

Even with a setup in place to automate the workload management of fraud teams, there are situations in which you need to act manually to correct the workload distribution. For example, you might want to add an alert to a queue that represents new fraud patterns, or you might want to assign an alert that requires a specific skill set to a fraud analyst.

This page describes how to manually perform actions on the workload in Case Manager.

Events and alertsâ

To move events or alerts to a queue or assign them to an analyst:

1. Open the event or alerts pages:
    - **For events:** In the general menu, hover over the **Search** icon, and select the **Search By Event** option.
    - **For alerts:** In the general menu, enter the **Alerts** page.
2. Select one or multiple events/alerts using the checkboxes.
3. Use the button to open more actions.
4. Select the **Move to Queue** option and then choose a **Queue**.
5. Select the **Assign to Analyst** option and then choose an **Analyst**.
6. Alternatively, in the search by events or alerts pages, you can open an event and perform Step 3.

Casesâ

To move cases to a queue or assign them to an analyst:

1. Using the general menu, enter the **Cases** page. You can also access cases by hovering over the **Search** icon and selecting the option **Search By Case**.
2. Select one or multiple cases using the checkboxes.
    - Select the button and then choose an **Analyst**.
    - Select the button and then choose a **Queue**.
3. Alternatively, you can open the case and perform Steps 2(i) and 2(ii) using the button to see those actions:

1. You can also move events already associated with a case, to a queue, and assign them to any analyst, as follows:

- In the **Linked Events** table, select the events using the checkboxes.
- Use the button to open more actions.
    - Select the **Move to Queue** option and then select a **Queue**.
    - Select the **Assign to Analyst** option and then select an **Analyst**.

Learn moreâ

Check the Manage Lists page to learn how to add, remove, and search for items within a list.

Manage Workload and Rules

Was this page helpful?Â

Learn how to set up Case Manager to keep the workload of fraud teams organized, preventing fraudulent behavior efficiently.

Fraud manager tasksâ

As a fraud manager, you need to understand how:

- Organize the workload of fraud teams either by automation processes or manually,
- Act when fraud is not being captured efficiently.

The Manage Workload and Rules guide helps you with these tasks:

Learn moreâ

Learn more about each of the fraud manager's tasks:

- Operationalize fraud teams: Configure queues, automation rules and Service Level Agreements (SLAs) to automate the workload management of fraud teams. This section contains practical examples to quickly configure a standard setup. This setup will enable you to keep the workload organized from day 1.
- Organize Workload Manually: Manually add events/alerts and cases to a queue as well as assign them to a team member. This task complements the previous one enabling you to take care of special situations. For example, to assign an alert that needs to be reviewed by a particular fraud analyst.
- Manage Lists: Learn how to add, remove, and search for items within a list.
- Create Self-Service Rules: Learn how to create a rule, test and deploy it directly in production to quickly counteract a fraudster scheme.

You can also learn more about analytics tools in the View Analytics Dashboards page. These analytic tools provide you big picture of how the business is performing and how the system is fighting fraud with the business overview dashboard and rules monitoring dashboard, respectively.

Configure Resources

Was this page helpful?Â

This page helps you configure your resources in the **new environment**. This process includes the following tasks:

- **Resource versioning**: Guides you through the process of changing the name of your resources for versioning.
- **Resource cleanup**: Helps you remove schemas and plans that are not being used, as well as outdated rules projects.
- **Resource checkup**: Helps you make sure that all the resources are configured as expected.

Resource versioningâ

To version plans in Pulse, execute the following steps:

Duplicate plansâ

1. In the **Data Science** tab, duplicate the most recent version of the plans that you want to version (e.g. assume *Custom Plan m1/d1* as an example, following the nomenclature *project-name mm/dd*).

info

Duplicate Plans: Plan executions are not transferred from the original plan to the duplicate version. The next subsection explains how to add executions to duplicate plans.

1. Rename the plans according to your versioning nomenclature (e.g. assume *Custom Plan m1/d2*).

Add executions to plansâ

1. Select the options menu in the Pulse application where you have duplicated the plans.
2. Select the option.
3. Choose the same resource bundle that you have used to import the plans you are working on, and upload it using the button.
4. In the **Choose Resources** step, open the **Plans** tab and add executions as follows:
5. Use the checkbox to choose the plans that you have duplicated (e.g. *Custom Plan m1/d1*).
6. Choose the **Local Plan Name** that you have duplicated (i.e. *Custom Plan m1/d2*).
7. Make sure you have all the plan executions that you want to keep in the new plans by expanding the collapse button under **Plan Executions**.
8. Continue to the **Next** step.
9. Since you are importing the plan executions to the duplicate plan, the schema should always match and the **Schema Conflicts** step should not be prompted. If after duplicating the plan in the [Duplicate plans] subsection the plan schema is changed, the **Schema Conflicts** step is prompted. Solve the conflicts by choosing **Keep Imported** for each plan execution.
10. Continue to the **Next** step and select the button to complete the import.
11. In the **Data Science** tab, open the executions list using the button in each of the new plans and confirm if they have the correct plan executions.

Create new executionsâ

In your Pulse application, you may have plans that use an existing schema as input and generate a new schema as output. You may also have plans that use the output schema generated from another plan as an input. Start by creating snapshots for the plans that use the existing schema:

1. In the **Data Science** tab, open the plan in which you want to create the new snapshot (e.g. *Custom Plan m1/d2*).

2. In the **More Actions** button, select **Plan Settings** and execute the following steps:

3. In the **Schema** option, choose the schema that applies to the plan (e.g. *Extended Schema [last version]*), and **Confirm** the change using the button in the warning box.

4. In the **Timestamp Field** option, select `event_occurred_at`.

5. **Save** your changes.

6. Select the button and execute the following steps:

7. Rename the execution according to your versioning nomenclature (e.g. *Custom Plan m1/d2 vN*, where N increments the version number).

8. In the **Data Sink** section, tick the export box and rename the new schema (e.g. *Custom Plan m1/d2 vN*). For traceability purposes, Feedzai recommends using the same name for the execution and the schema.

9. the plan execution.

10. Create executions for the plans that input the schema generated in the previous steps (e.g. *Standard Plan m1/d2*):

info

Consider the scenario in which you need to edit a Standard Plan. Repeat Steps 2 and 3, and consider the following:

- **Schema**: In this example, the input of the *Standard Plan d1/m2* is the output schema of the Custom Plan generated in Step 3.
- **Execution name**: Example, *Standard Plan m1/d2 vN*, where N increments the version number.
- **Schema name**: Example, *Standard Plan m1/d2 vN*, where N increments the version number.

For traceability purposes, Feedzai recommends using the same name for the execution and the schema. :::

Version rules projects and snapshotsâ

1. In the **Data Science** tab, select the options button in the **Rules Projects** panel to configure each rules project.
2. Name the rules project according to your versioning nomenclature (e.g. *name_of_rules_project mm/dd*).
3. Select the **Schema** created in Create new executions' step 4.
4. Set the **Target Variable** using the one that applies to your use case (e.g. `decision` in a typical application).
5. **Save** your changes.
6. For each non-self-service rules project (non-SSR):
7. Open the rules project.

8. In the **Snapshots** panel, create a new snapshot.
9. Name the snapshot according to your versioning nomenclature (e.g. *name_of_rules_project mm/dd vN*).
10. the snapshot.

info

The snapshots for self-service rules (SSR) projects will be created in the next subsection.

1. Publish your application using the button.
2. Apply your changes to complete the process using the button.

info

Feedzai recommends that you use the **Rolling Publish** option. This option publishes application changes on one Pulse replica at a time, thus reducing the downtime, as there may be other online replicas still able to handle events.

Update the workflowâ

1. In the **Workflow** tab, open the workflow that you want to update.
2. In the **Schema**, make sure the correct schema is selected (e.g. *Input Schema vN*, where N increments the version number). If you don't have the correct schema, select it from the list and **Confirm** the change using the button in the warning box.
3. In the diagram, edit each operator (e.g. plans and rules) and execute the following actions:
4. In the **Main** tab, select the resource's and snapshot's latest versions.
5. **Save** your changes.
6. For the operator containing SSR projects, select the rules project's latest version. If you have changed the name of the SSR projects as specified in the previous version's step 2, you need to publish the application to create new snapshots. Pulse will automatically recognize the new snapshot and replace the current value of the snapshot field with the new one. To activate this behavior, execute the following:

7. **Save** and exit the workflow.
8. Publish your SSRs using the button.
9. Apply your changes to complete the process using the button.

info

Feedzai recommends that you use the **Rolling Publish** option. This option publishes application changes on one Pulse replica at a time, thus reducing the downtime, as there may be other online replicas still able to handle events.

1. Return to the workflow and make sure that the schema of the SSR project has been updated with the new snapshot.

Resource cleanupâ

1. In the **Data Configuration** tab, delete the schemas that are not being used. The final resources are specific to each use case. As a rule of thumb, you should have an Input Schema and Extended Schema.
2. In the **Data Science** tab:
    1. Delete the data sources that were created when importing the resources.

    2. Delete the plans that are not being used.
    3. Delete the outdated rules projects.

Resource checkupâ

1. In the **Data Configuration** tab, under the **Schema** panel, ensure that the new schemas have the new inserted fields.
2. In the **Data Science** tab, under the **Plans** and **Rules Projects** panels, confirm if:
    1. The version is correct.
    2. The input schema is correct.
    3. The **Target Variable** for rules projects is correct.
    4. The new executions and snapshots have been created for plans and rules.
3. In the **Workflow** tab, ensure that each operator has the latest resource and snapshot versions.
4. In the **Lists** tab, make sure that the lists are correctly configured.

Next stepsâ

Now that you finished configuring your resources:

- Make sure you use the Update the Workflow section as reference for publishing further workflow changes.
- Read the Transfer list items page.

-
- [Operational Work]
- [Data Scientist & Fraud Strategist]
- [Migrate Pulse Resources]
- Export Resources

On this page

Export Resources

Learn how to export resources (i.e. lists, models projects, plans, and rules projects).

danger

Make sure you back up the environment you want to update:

1. Open Pulse in the new environment.
2. Select the options menu in your Pulse application.
3. **Export Application**.

To export resources:

1. Select the **Options** menu in the Pulse application of your current environment.

2. Select the **Export Resources** option:

1. Each resource has a specific tab. Use each tab to select the resources that you want to export.

info

To make sure that all resources are selected, keep a list of all the model projects, models, plans, plan executions, rules projects, and snapshots that you want to export and insert the name of each resource into the corresponding field.

Model projectsâ

1. In the **Export Resources** pop-up window, the **Model Projects** tab is selected by default.
2. Optionally, if you want to include a file with the model object, select the **Include Binary** option.
3. Make sure that you add each resource to the export bundle.

Plansâ

1. Open the **Plans** tab.

2. Select the plans that you want to export.

info

Since non-deployable plans are not directly deployed in a runtime workflow, you can select only deployable plans.

1. If you want to export a specific configuration of the plan, select the plan executions from the **Executions List**. For traceability and history purposes, Feedzai recommends selecting the last plan execution for each exported plan.

2. Make sure that you add each resource to the export bundle.

Rules projectsâ

1. Open the **Rules Projects** tab.

2. Select the rules projects that you want to export.

3. If you want to export a specific configuration of a rules project, select the snapshots from the **Snapshot List**. For traceability and history purposes, Feedzai recommends selecting all existing snapshots.

4. Make sure that you add each resource to the export bundle.

Once all resources are configured, select the **Export Bundle** button.

Listsâ

Lists are simple resources that can be migrated by exporting the list items from the source environment and importing them in the target environment. The resource transfer process allows you to export and import list metadata, but not list items. List items are exported and imported through a CSV file. To export and import list items refer to the Transfer List Items guide.

Next stepsâ

Now that you can export resources from the current environment, learn how to import resources in the new environment.

Import Resources

Was this page helpful?Â

Learn how to **import resources** to upload resources to your application in the new environment.

danger

Make sure you perform a backup of the environment that you want to update. If you already performed this backup in the export resources step, ignore the following steps:

1. Open Pulse in the new environment.
2. Select the options menu in your Pulse application.
3. Select the **Export Application** option to download the file with the application.

To import resources:

1. Open the **Options** menu in the Pulse application of your new environment.
2. Select the **Import Resources** option:

1. Execute the sequence of steps.

Upload fileâ

1. **Drag & Drop** the zip file containing the export bundle to the dashed box. Alternatively, **browse for a file** to upload the zip file.
2. Use the button to go to the next step.

Choose resourcesâ

info

To make sure that all resources are selected, keep a list of all the model projects, models, plans, plan executions, rules projects, and snapshots that you want to import and insert the name of each resource on the corresponding field.

You can choose the resources that you want to import by navigating through each resource tab. Note that the **Lists** resource does not have a specific tab because its configurations are automatically imported as follows:

- Lists that do not exist in the new environment are created without items.
- Lists that already existed in the new environment are updated with the new metadata. However, their items are not affected by the import into the new environment.

Modelsâ

1. In the **Import Resources** pop-up window, the **Models** tab is selected by default.
2. Use the checkbox to select the models that you want to import. Note that in Pulse, it is not possible to bulk import models from a model project, and therefore you should import them individually.
3. For each model, choose the **Local Model Project** to which you want to store the model in the new Pulse application. Note that you must have a model project in the new application in order to import the new model.
4. Optionally, if you want to include a file with the model object, select the **Include Binary** option. This option is only available if you include the binary in the export resources step.

info

Dummy Data Source: When you import a model that does not have a compatible data source available in the new environment, Pulse automatically creates a new data source. This new data source only contains the schema with no data associated.

Plansâ

1. Select the **Plans** tab.

2. Use the checkbox to select the plans that you want to import.

3. Choose the **Local Plan Name** for the plan in the new Pulse application.

info

Local Plan Name:

i. You can save the plan as a new plan by choosing the option *project_name* . The *new* option creates a resource named *resource_name (N)*, where N is a number that increments over the highest value of this resource in your new application. Feedzai recommends that you use this approach when you want to version your resources.

ii. You can replace an existing plan by choosing the same name as the one that you are importing. This keeps your environment clean.

4. Choose the plan executions that you want to import by expanding the menu under the **Plan Executions** column. By default, all the plan executions are selected. For traceability and history purposes, Feedzai recommends that you import all the executions.

tip

Executions persistency: Pulse only keeps the executions that you select. Imagine that you choose option 3 and you only select the most recent plan execution. The current plan in the new application will be replaced with the selected execution (i.e. all the executions in the new environment will be deleted).

info

Dummy Data Source: When importing a plan with executions that do not have a compatible data source available in the new environment, Pulse automatically creates a new data source for each execution that you choose to import. This new data source only contains the schema with no data associated.

Rules projectsâ□□

1. Select the **Rules Projects** tab.
2. Use the checkbox to select the rules projects that you want to import.
3. Choose the **Local Rules Project Name** for the rules project in the new Pulse application:

info

Local Rules Project Name: When you choose the **Local Rules Project Name**, do the same as step 3 for Plans.

4. Select the rule snapshots that you want to import by expanding the menu under the **Snapshots** column. By default, all snapshots are selected. For traceability and history purposes, Feedzai recommends that you import all the executions.

tip

Snapshots persistency: Pulse keeps snapshots as explained in Plans' step 4.

Once you have chosen all the resources that you want to import (i.e. models, plans, and rules projects), go to the **Next** step.

Schema conflictsâ

You are prompted with this step only when the plans' or rules projects' schema previously selected doesn't match the schema of the resources that you currently have in the new environment. Select the schema that you want to keep for each resource:

- **Keep imported**: for the given **Resource Type** and **Resource Name** (i.e. the resource that is being imported), the schema that is selected is kept in the resource that you are importing.
- **Keep local**: for the given **Resource Type** and **Resource Name** (i.e. the resource that is being imported), the schema that is selected is kept in the resource in the new environment.

Once you have chosen the schema for each resource on the list, go to the **Next** step.

Workflowâ

tip

This step is optional since you can also manually update resources in the workflow after importing resources. The workflow's manual update is required in case you decide to perform resource versioning when choosing plans and rules projects in the new environment. This option is recommended if you import a resource that shares the same schema as the one in the new environment (e.g. update a rule that has a new condition).

In this step, associate the resources that you are importing with the new environment's workflow services:

1. Select the **Resource Type** that you want to map to a workflow service.
2. Choose the **Workflow** name that you want to update with the new resource.
3. Choose the **Workflow Service** to which you want to map the resource.
4. After associating the resources with the workflow, select the button.

If the workflow element does not exist in the Workflow yet, you must create the element on the target environment Workflow by taking the following steps:

1. Drag and drop the desired workflow element to match the one from the source environment. Make sure to configure it correctly by selecting the **Edit** icon.

2. Once your workflow element is created, create the connection to the other elements by dragging the **Drag to connect services** icon of the source element to the target element.

   Note that, for data to navigate the workflow from an element to another, schemas have to match. Make sure the schema used in your new element matches the output schema of the previous one. To compare them, select the new element's **Edit** icon and, in the workflow configuration pop-up window, open the **Schemas** tab.

   In case there are schema conflicts, you can adapt the schema of the element -> create a snapshot -> and use it in the new workflow element.

3. The connection between the existing element and the new one can be conditional based on a filter expression. Configure the connection the hovering over the connection arrow and selecting the **Edit** icon.

Next stepsâ

Now that you have imported the new resources in the new application, learn more about the final configurations to make sure that all the resources are working as expected. The next step is also important if you have decided to version your resources.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/migrate-pulse-resources/overview

On this page

Migrate Pulse Resources

Was this page helpful?Â

Rules and Data Science resources such as Plans and Models must be developed, tested, and deployed into Production to score transactions in real time. To keep fraud-fighting tools performant, you need to periodically implement changes that may include:

- Adjusting parameters.
- Incorporating new insights regarding a specific issue.
- Deleting resources because they are no longer relevant.

To ensure that resources that are edited, validated and tested do not affect the runtime in production, the development and testing work are performed in staging environments. This raises the need to transfer your latest developments and combine them with other resources in a new environment. Some of this work is part of the standard product and is not subject to changes whereas, as shown in the following figure, other resources are transferred via a standard procedure named **resource transfer**. You can use resource transfer to autonomously transfer resources between environments.

The Migrate Pulse Resources guide explains how to transfer the following resources:

- Models
- Plans
- Rules
- Lists metadata

Block ownershipâ

While you develop new profiles with plans, re-train models with new data, and/or test new rules using offline environments, your Pulse application (Pulse app or papp) will keep evolving in production. Some resources, such as rules projects are often developed by various teams (i.e. owners). To evolve an application in a collaborative way, you can use the **block ownership**. Each Pulse resource is developed as a block by a single team, within one environment at a time. For example, each rules project should be owned by one team and should be only modified (e.g. adding/removing/editing rules) by the team that owns each rules project.

The block ownership assumes that:

1. Between each deployment, each block evolves in a single environment.
2. For each environment, all the resources evolve using a single Pulse cluster.

By structuring your work with the concept of block ownership, you allow:

1. Multiple teams to work in parallel using different environments.
2. Easy and quick migration of resources between environments.

Flow between environmentsâ

The typical workflow between environments is as follows:

1. **QA**: Run functional tests to verify the effectiveness of the implemented changes. These may include sending transactions to API to validate rules logic or testing profiles with long windows.
2. **Pre-prod**: To ensure the new changes will be successful in Prod, migrate what's in Prod to Pre-prod to run tests with real data.

info

Depending on your service, this environment may not be available out of the box. This is an environment that is commonly used for performance tests.

3. **Prod**: Once tests have successfully run in Pre-prod, update the Prod environment.

This process leverages two Pulse functionalities: exporting and importing resources. The process is completed after you configure the resources in your new environment.

- **Step 1**: Edit resources (e.g. change the plan, or create rules)
- **Step 2**: Export resources.
- **Step 3**: Import resources.
- **Step 4**: Configure resources.

These steps must be followed each time resources are transferred between environments, for instance, between a Pre-prod and a Prod environment.

Next stepsâ

Now that you understand the resource transfer's purpose, you can start exporting resources.

Transfer List Items

Was this page helpful?Â

Lists are simple resources that can be migrated by exporting the list items from the source environment and importing them in the target environment.

In the process of detecting fraud, it is very common to define lists to block transactions from certain cards, or even to send transactions with common parameters for review (e.g. triggering an alarm every time a transaction occurs). Lists allow you to group and manage these values by storing them in items to further use them with rules and plans.

The resource transfer process allows you to export and import lists metadata, but not list items. List items are exported and imported through a CSV file.

Export list itemsâ

To export list items, take the following steps:

1. Go to the **Lists** tab.
2. Open the list that has the items that you want to export.

1. Export all or only the active list items by selecting the download button. A CSV file is downloaded.

Import list itemsâ

To import list items, take the following steps:

1. Go to the **Lists** tab.

2. Make sure the corresponding list items exist in the target environment. If this is not the case, create a new list by selecting the **New List** button.

tip

You can access the List configuration to make sure the lists from both environments match.

3. Open the list where you want to import the items to.
4. Select the button.
5. **Drag & Drop** or browse the CSV file containing the list items and **Save**.

Upload Limit Recommendation

To ensure reliable processing and avoid straining system resources, it is recommended that each uploaded list file contains at most 40,000 items. Note that this figure is a general guideline; the actual threshold for optimal performance can vary depending on the specific contracted resource sizing. If your list is larger than that, split it into multiple files before uploading.

CSV file formattingâ

When importing CSV files, notice that if you try to upload a list with duplicate entries, Pulse will generate a silent error, and therefore won't notify you.

Consider the following formatting to ensure list items are correctly imported:

Example of a simple listâ

- Card listed to be declined

```
## Header: Key, Description, Start Date (UNIX timestamp), End Date (UNIX timestamp), Enabled

"0000-0000-0000-0000","Stolen card, made multiple transactions in different countries separated by minutes",,,"true"
"1111-1111-1111-1111","Duplicated card, attempted several withdraws trying to determine the maximum amount permitted.","1381708800",,"true"
"2222-2222-2222-2222","Two months ban for breaking rule R102","1381708800","1386633600","true"
```

Example of a complex listâ

- PosNeg list

```
## Header: Key, Value, Description, Start Date (UNIX timestamp), End Date (UNIX timestamp), Enabled

"AAA","{"value":"review"}","Card to be reviewed",,,"true"
"BBB","{"value":"decline}","Card to be declined",,,"true"
```

info

By default, if no dates are set in the **Active from** - **To** fields, new list items become enabled and active with no expiration date.

Behavioral Data in ML Models

Was this page helpful?Â

The Revelock enricher (i.e. Digital Trust enricher) is a [Custom Workflow Element] that brings behavioral, device, and network data to the workflow. To detect fraud with the Revelock enricher's data, you must consider using the enricher in at least one of two components: rules or models.

This page presents everything a data scientist has to know to leverage the data provided by Digital Trust for an improved fraud detection.

Timeline for model buildingâ

You can expect a robust model to start scoring payments in production around two to six months. This means you should consider:

1. A minimum of two months (and on average six months) of correctly labeled data to obtain solid results. This period also allows Digital Trust models to mature with a considerable amount of sessions to build a profile.
2. At least two weeks to train and evaluate the model, as usually recommended. This is the typical model creation process for any enricher used.

Rulesâ

Data scientists can write rules from day one in order to detect device and behavior fraud patterns. Typically, one week of data ingestion is enough to evaluate rulesâalert rates. This means that, during this period, data scientists can already test rules in shadow mode to understand which rules they can keep or dismiss.

After obtaining insights on fraud patterns, data scientists should fine-tune rule conditions and thresholds to detect device and behavior fraud patterns more efficiently.

Here are some examples of patterns you can consider when building your own rules:

- Device is infected with malware.
- Risky activity with a "phishing" flag and device with high Session Risk Score.
- Login attempt from a dormant account with Session Risk Score and suspicious network activity above threshold. Suspicious network activity is detected via TOR, VPN or Proxy connections.

Learn about the fraud scenarios detected by device and behavior rules.

info

Session Risk Score is a score generated by Digital Trust. It considers all the behavioral, device, and network indicators into a single score. The total risk score ranges from 0 (extremely low risk) to 100 (extremely high risk). Feedzai recommends that you combine the Session Risk Score with Digital Trustâs threat alerts (account takeover and its threats, i.e. malware, phishing, suspicious network activity) to avoid false positive rates.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/models/day-0-models

-

- [Operational Work]
- [Data Scientist & Fraud Strategist]
- [Use Models]
- Day-0 Models

On this page

Day-0 Models

Was this page helpful?Â

This page contains information about how to use a machine learning model from day 0 with no historical data.

Introductionâ

When kickstarting TFB, rules may trigger numerous, unwanted alerts. To fine-tune them, a machine learning model offers an easy way to control the number of alerts triggered by rules. In some other cases, rules may not exist, and therefore a machine learning model is needed to start scoring transactions. However, it is not always possible to have historical data to train a model.

Pulse includes the Day-0 machine learning model to cover these cases for Cards transactions. It was trained on industry data using the Microsoft LightGBM algorithm. The purpose of this model is to provide support at the start of a project. However, it does not replace a model trained on specific data in the long term, and fraud performance can't be guaranteed. The goal of this tool is to temporarily address the lack of data to train a model.

Day-0 model life cycleâ

The model starts working in shadow mode. Shadow mode means that the model gives a score to each event, but the scores are not impacting the event decision done by Pulse. The model is waiting to be fine-tuned and activated when ready.

In the solution life cycle, take the following steps:

1. After a short period in shadow mode, adjust the model thresholds to fine-tune the output (such as alert rate and, if labels are available during shadow mode, FPR). Copy the data to a Data Science environment, and use a JupyterLab notebook prepared by Feedzai to find the right threshold values.
2. When you are happy with the result, set the thresholds in production, and set the model to live mode. From this point on, the model score is taken into consideration for the decision about the event.
3. Later, when there is enough data and labels, train a new model and run it through the same process (i.e. shadow mode, fine-tuning, live mode).

Data fieldsâ

The model uses fields created by the Cards plan. This plan should be added to the workflow together with the model.

The following table lists the schema fields that are required to be filled to use the 0 Day Model:

Field Name	Field Type	Description
amount	LONG	Input schema field, amount in USD cents
event_occurred_at	LONG	Input schema field, the time stamp of the event
card_pan	STRING	Input schema field, the identifier of the card
mcc	STRING	Input schema field, merchant category code (MCC)
acceptor_country_code	STRING	Input schema field, the country code of the acceptor (merchant)
acceptor_id	STRING	Input schema field, the identifier of the acceptor (merchant)

On top of these raw features, several features are created such as velocities across different entities, categorizations and risky fields.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/models/use-models

Use Models

Was this page helpful?Â

This section explains how you can use models with or without historical data, and highlights what you need to consider when handling behavioral data in Machine Learning (ML) models.

- Automated Machine Learning
- Day-0 Models
- Behavioral Data in ML Models

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/modularized/atm-withdrawals-cards-shared

atm-withdrawals-cards-shared

Was this page helpful?Â

To correctly map ATM withdrawals to card authorizations, and hence score ATM withdrawals, some fields need to be populated to identify ATM withdrawal events as follows:

```
 {
   // Required fields
   "mti": "0110",
   "card_pan": "8955060600182663", // Which card was used
   "ttc": "001", // Type of transaction
   "currency_code": "EUR",
   "amount": 20,
   "cardholder_bill_amount": 20,
   "lifecycle_id": "1df7a7qg78vd7fs",
   "pos_card_read_method": "01",
   "event_occurred_at": 1684492270092,
   "card_bin": "123456",
   "account_iban": "482924bf58091603ed69c5bd1221ad67481bfca8",
   "customer_id": "87b2ee22cb0004722a7265fdbbf7be98c681a3bb",
}
```

Read the Authorization [POST] documentation for more information on other fields you can use in the payload.

cash-and-deposits-ibp-shared

Was this page helpful?Â

To correctly map cash and ATM deposits to inbound payments, and hence score cash and ATM deposits, some fields need to be populated to identify cash or ATM deposit events as follows:

```
    {
       // Required fields
       "event_external_id": "4048611b295e993d1b561d9030c53424bbf2b95e",
       "lifecycle_id": "1df7a7qg78vd7fs",
       "channel_type": "atm", // To indicate it's ATM
       "payment_status": "entered",
       "created_at": 1603207359060,
       "sender_transaction_amount": 1603207359060,
       "sender_transaction_currency": "GBP",
       "sender_transaction_bank_account_number": "5a800244-325a-4086-ab8a-23ecbb927be5",
       "direction": "incoming",
       "event_occurred_at": 1684492270092,

       // Recommended for mapping cash and ATM deposits
       "payment_type": "sepa", // To indicate it's a deposit
       "payment_sub_type": "direct_deposit", // Use the `direct_deposit` value to indicate it's a direct deposit
    }
```

Read the [Inbound Transfer Initiation [POST]] documentation for more information on other fields you can use in the payload.

Operational Work

Was this page helpful?Â

[Data Scientist & Fraud Strategist]

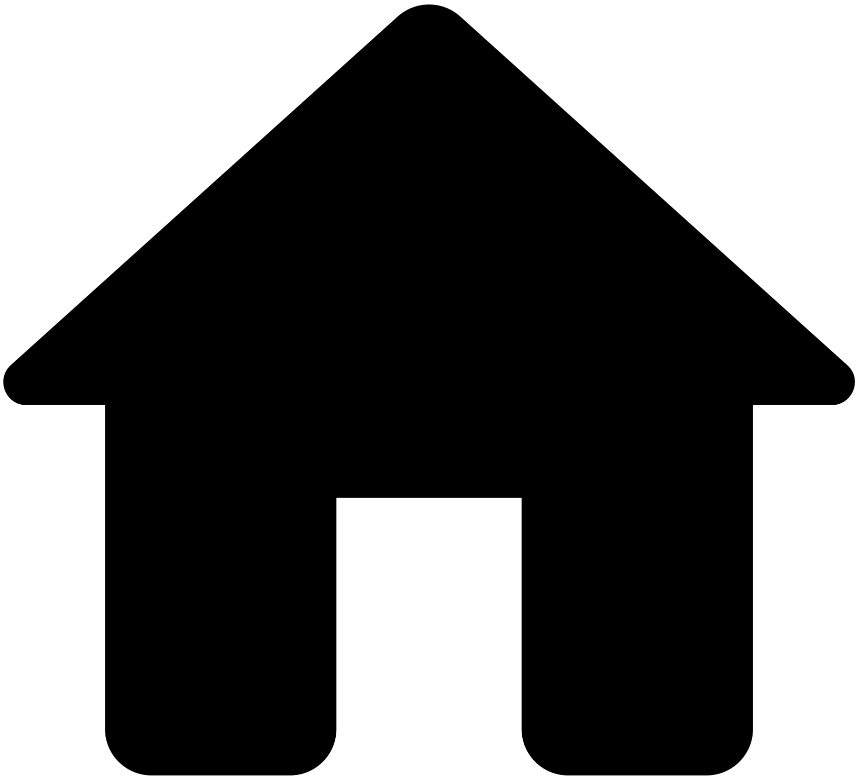
[Learn how to fight fraud with rules and models in Pulse.]

[Fraud Prevention Agent]

[Learn how to work with alerts and/or cases using Case Manager.]

[Fraud Prevention Manager]

[View real-time insights.]

- 

- [About TFB]
- PSD2 Compliance

On this page

PSD2 Compliance

Was this page helpful?Â

Learn how Feedzai's TFB helps financial institutions to be compliant with the second Payment Services Directive, in particular with Transaction Risk Analysis, providing real-time recommendations for strong customer authentication.

Payment Services Directiveâ

The Payment Services Directive (PSD) is an EU directive that regulates payments and payment service providers throughout the European Economic Area.

The second Payment Services Directive (PSD2) defines new rules to:

- Improve individuals protection in online payments.

- Promote the development and use of innovative online and mobile payments, such as open banking.
- Make cross-border European payment services safer.

To protect account holders, PSD2 mandates that e-commerce and bank transfer transactions must meet Strong Customer Authentication (SCA). Transaction Fraud for Banking (TFB) provides real-time recommendations to help issuers decide when to enforce SCA and when to safely exempt SCA.

Strong Customer Authenticationâ

SCA is a method that requires individuals to prove their identity by providing two of the three authentication categories:

- **Knowledge:** Something only the user knows, e.g. a password or a PIN.
- **Possession:** Something only the user possesses, e.g. the card or an authentication code generating device.
- **Inherence:** Something the user is, e.g. the use of a fingerprint or voice recognition.

Note that not all transactions require SCA, i.e. there are out-of-scope conditions that exclude a transaction from PSD2. Additionally, to provide a better user experience âespecially for low-value and low-risk payments âa set of exemptions can be normally applied. These exemptions may be requested upstream by the merchant, acquirer or network, or downstream by the issuer.

Out-of-scope criteriaâ

Out-of-scope transactions include:

- **Interregional transactions:** Payments where the card was issued outside of Europe or where the country you are acquiring from is outside of Europe. Some European issuing banks are expected to require SCA, whether a payment is acquired outside of Europe or not.
- **Cash payment**
- **Payments using a physical card at a POS terminal:** This payment method is already secured via two-factor authentication (i.e. chip and PIN).
- **Merchant-Initiated Transactions (MIT) and Direct Debits:** A payment or a series of payments with fixed or variable amounts that the merchant performs without direct involvement of the shopper. Examples are subscriptions, automatic account top-ups, and installments. The initial transaction should have gone through SCA and the shopper should have agreed to the terms and conditions of the succeeding MITs.
- **Mail Order and Telephone Orders (MOTO):** MOTO transactions are not considered to be electronic payments, so these are out of the scope of the regulation.
- **Anonymous cards:** This type of cards can only be identified by the issuing bank. For example, anonymous prepaid cards.

Exemption criteriaâ

For transactions that are in the scope of PSD2, the merchant or the PSP can request SCA exemption if the transaction meets any criteria in the following list. The issuer decides whether the exemption is granted. Additionally, the issuer can grant an exemption without the merchant or the PSP requesting it.

The exemptions that can be requested are the following:

- **Low Value:** Transactions under 30 euros do not require SCA, but the issuer will keep track of certain counters, such as the number of transactions or the sum of transaction amounts. If the account/card holder exceeds the counter, for example, after five consecutive transactions or if the sum exceeds 100 euros, the issuer will require SCA. This exemption is defined by the PSD2 **Article 16**.
- **Low Risk/Transaction Risk Analysis (TRA):** Issuing banks can consider transactions as low risk based on their average fraud levels, acquirer's fraud levels, or both. The market official measure for fraud level in the PSD2 scope is known as Fraud Rate.This exemption is defined by the PSD2 **Article 18**. This exemption can occur in different ways:
    - **TRA exemption request from merchant or PSP:** The merchant can request the TRA exemption to the issuer if the merchant's fraud levels are below a safe threshold. If the exemption is granted, the chargeback liability stays with the merchant.
    - **TRA exemption from the issuer:** The issuer can exempt TRA even when the merchant or the acquirer does not request it. Sending additional information in the payment request can help to maximize the probability of getting this exemption. The chargeback liability shifts to the issuer.

- **Recurring transactions with exact same amount:** This exemption is expected to be overridden by MIT and might be considered as [out of scope](). This exemption is defined by the PSD2 **Article 14**.
- **Whitelisted merchants or trusted beneficiaries:** After a strongly authenticated payment session, account/card holder can add the merchant to a whitelist for the issuer. The issuing bank will not require SCA on the next payments for the same merchant. However, note that this exemption depends on whether the issuer supports whitelisting. This exemption is defined by the PSD2 **article 13**.
- **Secure corporate payments:** These are payments made through dedicated corporate processes initiated by businesses and not available for an account/card holder. Examples are payments made through central travel accounts, lodged cards, virtual cards, and secure corporate cards, such as those used in a corporate travel management system. This exemption is defined by the PSD2 **Article 17**.
- **Unattended payment terminal for a parking fee:** Payment service providers are allowed to skip SCA when the payer initiates an electronic payment transaction at an unattended payment terminal for the purpose of paying a transport fare or a parking fee. This exemption is defined by the PSD2 **Article 12**.
- **Credit transfers between accounts held by the same natural or legal person:** Payment service providers shall not apply SCA if the payer initiates a credit transfer in circumstances where the payer and the payee are the same natural or legal person, and both payment accounts are held by the same account servicing payment service provider. This exemption is defined by the PSD2 **Article 15**.

Download the compliance matrix to access a comprehensive list of PSD2 articles, including the tools and methodology used by Feedzai to help issuers be PSD2-compliant.

3-D Secureâ

3-D Secure, also known as three-domain secure, is a security protocol for online credit and debit card transactions. The three domains refer to the merchant/acquirer, the issuer, and the interoperability.

TFB manages both the authentication and authorization events present in a 3-D Secure (2.0 or higher) transaction to make an approve, decline or step-up authentication recommendation in a compliant manner and with full risk context.

This means that TFB helps issuer banks to:

- Comply
- Reduce risk
- Reduce friction

How?

- Provides a compliant response with PSD2 regulation that enforces SCA to be conducted, with a number of requirements and exemptions in place.
- Provides a transaction risk assessment that enables accept/decline/step-up decisions by using numerous extra data points such as device details, IP, browser, and malware flag to reduce both friction and risk.
- Makes the decision's outcome and context available to any subsequent authorizations for the same customer, card, and event.

3-D Secure integration flowâ

An authentication is usually the initial stage corresponding to the 3-D Secure message.

However, issuer banks can rely on the `authorization` endpoint to make a decision on the 3-D Secure authentication event (see step 3 on the following diagram).

This means that TFB's `authorization` endpoint is in charge of producing a response that includes both a score and a recommendation:

- An approve/decline decision that considers factors such as device details, IP, browser, and malware flag for transaction risk analysis. This decision also considers the results of previous 3-D Secure challenges, if any.
- A decision on whether the transaction needs SCA (`request_sca`) based on requirements and exemptions from PSD2 articles.

The `info` endpoint (see step 7 on the following diagram) can then be invoked to update and add new information to an existing authentication such as:

- Frictionless transaction: update information on the decision so as not to challenge the transaction with SCA.
- Challenged transaction: update the authentication result, including authentication type, authentication method, etc.

TFB is able to respond to different SCA use cases within the transaction life cycle:

Authorization request with post-SCA

The issuing bank sends an authorization request. The risk engine rules then process the event and generate a recommendation. In this example, consider `sca_required = true`. The bank then asks the customer to authenticate and finally sends the authentication result via the `info` endpoint.

Authorization request with pre-SCA

Feedzai's API can receive an already authenticated transaction. TFB's `authorization` endpoint is prepared to receive the upstream authentication result (e.g. SCA result = true/false, method = biometrics/SMS/phone call). This result is taken into account by the risk engine rules.

How Feedzai's TFB helps with PSD2 complianceâ□□

TFB comes into play within the transaction risk analysis (TRA) by providing a recommendation on whether the issuer bank should request SCA. This recommendation is critical for overall fraud prevention. Note that, even though PSPs can claim the exemption for SCA, the issuer has the final decision and can turn down the request and, consequently, demand SCA.

The TFB contribution to PSD2 compliance is illustrated in the following diagram:

The recommendations generated by TFB can be summarized as follows:

- If `decision = approve`, the issuer should consider the `sca_required` outcome before actually approving the transaction. The `sca_required = true` result means that the payment is safe to be approved, but for compliance reasons the issue should still request SCA before completing the transaction (this case is shown in the above diagram). If `sca_required=false`, the issuer can skip SCA and still be PSD2-compliant. This means that an article of the [exemption criteria] can be applied and the card/account holder is not prompted with SCA.

- If `decision = review` (outcome value available in transfers), the same SCA logic applies, i.e. the issuer should consider the `sca_required` outcome as the TFB recommendation to apply SCA while the transaction is being analyzed by a fraud analyst.

- If `decision = decline`, TFB recommends rejecting the transaction. If the issuer decides not to honor the TFB recommendation:

  - If `sca_required=true`, SCA must be performed to comply with PSD2.
  - If `sca_required=false`, the issuer can skip SCA and still be PSD2 compliant.

The out-of-the-box rules targeting specifically PSD2 compliance are organized into several rules projects. This rules project organization separates rules that trigger under PSD2 out-of-scope transactions from rules that trigger under the exemption criteria. Check the rules project pages for more information about this logic.

Use cases within the transaction life cycleâ□□

TFB is flexible in responding to different SCA use cases within the transaction life cycle:

- **Authorization request with post-SCA:** The issuing bank sends an authorization request. Then the risk engine rules process the event and generate a recommendation - in this example, consider `sca_required = true`. Then the bank asks the customer to authenticate, and finally, the bank sends the authentication result via the info endpoint.
- **Authorization request with pre-SCA:** The transaction can reach Feedzai already authenticated. The TFB's authorization endpoint is prepared to receive the results of the upstream authentication result (e.g. SCA result = true/false, method = biometrics/SMS/phone call). This result is taken into account by the risk engine rules.

Compliance matrix - downloadâ

PSD2 is organized into multiple chapters, and each chapter contains several articles. TFB provides a compliance matrix that maps each article to the tool and methodology used by Feedzai to help issuers to be PSD2-compliant. This matrix is available for download, allowing issuers to demonstrate to PSD2 regulators how their systems are fully compliant with this directive.

COMPLIANCE MATRIX

Fraud rateâ

The fraud rate identifies the fraud level of a given merchant or PSP. Issuer banks use this metric to decide whether to grant exemptions based on TRA as specified in the PSD2 mandate, Article 18 (learn more about this article by downloading the compliance matrix document).

PSD2 states that: "The overall fraud rate for each type of payment instrument should be calculated as the **total value of unauthorised or fraudulent remote transactions**, whether the funds have been recovered or not, **divided by the total value of all remote transactions for the same type of payment instrument**, whether authenticated with the application of strong customer authentication or executed under any relevant exemption in accordance with Articles 13 to 16 on a rolling quarterly basis (90 days)"

The merchants and PSPs that request a TRA exemption must have an overall fraud rate for a specific payment instrument that is equal to, or is lower than, a reference fraud rate defined in the PSD2 document.

The TFB's API specification includes the `reference_fraud_rate` field present in the Cards and Transfers data structures. This field can be populated in different ways:

- The fraud rate is directly sent by the client via the API request.
- The fraud rate is stored in a list (i.e. leveraging the TFB's risk engine platform) and is then used to populate the `reference_fraud_rate` field. This option can also have different scenarios:
    - The value is inserted directly in Pulse by Feedzai's clients.
    - The value is automatically calculated via Insights dashboards, available by default in the TFB's package.

Alertingâ

Even though payment service providers should cease applying the exemption if the â *monitored fraud rate exceeds for two consecutive quarters the reference fraud rate applicable for that payment instrument or type of payment transaction in that exemption threshold rangeâ* , Feedzai alerts when the fraud rate exceeds any of the reference fraud rate in the following Annex table. This will allow the bank to take action and keep fraud levels low before exceeding the reference for two consecutive quarters.

ETV	Remote electronic card-based payments	Remote electronic credit transfers
EUR 500	0,01	0,005
EUR 250	0,06	0,01
EUR 100	0,13	0,015

An email will be sent when the fraud rate for a given payment instrument exceeds any of the above thresholds. The email will indicate under which scope the alert was triggered, thus allowing the payment service providers to take preemptive measures.

Learn moreâ

Now that you know how TFB's rules help issuers to be PSD2 compliant, check the Rules Projects page for the rules provided out of the box.

You can also check the Work with Rules and Lists guide to learn how to:

- Write rules customized for your business needs.

- Configure your work in the runtime workflow.
- Test rules using the TFB API.

Digital Trust Enricher's UDFs

Was this page helpful?Â

The Digital Trust Enricher supplies out-of-the box User Defined Functions (UDFs) to ensure richer usage of Digital Trust data. The following sections describe the various UDFs and examples on how to use them.

User Defined Functionsâ

Alert of type X existsâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and checks if there is an alert of a given type (not case sensitive). It returns `true` if there is an alert of a given type, `false` otherwise.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.

Responseâ

Type	Description
Boolean	Returns true if there is an alert of the given type, false otherwise.

Exampleâ

```
Revelock.existsAlertOfType(event.revelock_alerts ,'Fraudster Behavior') ?? false
```

Get risk of alert of type Xâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and gets the risk factor from an alert of a given type (not case sensitive). If multiple alerts with the same type are found, the highest risk is returned.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.

Responseâ

Type	Description
Double	Returns the highest risk value for the given alert type.

Exampleâ

```
Revelock.getRiskOfAlertType(event.revelock_alerts ,'Fraudster Behavior') ?? 0.0
```

Get date of alert of type Xâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and gets the risk factor from an alert of a given type (not case sensitive). If multiple alerts with the same type are found, the most recent alert date is returned.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.

Responseâ

Type	Description
Long	Returns the most recent alert date value for the alert type given. Null if none is found.

Exampleâ

```
Revelock.getDateOfAlertType(event.revelock_alerts ,'Fraudster Behavior') ?? 0
```

Exists view X of alert of type Yâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and checks if there is an alert of a given type for a given view (not case sensitive).

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.
`view`	String	The alert view to search for.

Responseâ

Type	Description
Boolean	Returns true if there is an alert of a given type for a given view.

Exampleâ

```
Revelock.existsViewOfAlertType(event.revelock_alerts ,'Fraudster Behavior', 'account-details-view') ?? false
```

Get minimum double value of property X from alert of type Yâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and gets a property value from an alert of a given type (not case sensitive). If multiple alerts with the same type are found, the minimum value is returned.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.
`property`	String	The alert property name.

Responseâ

Type	Description
Double	Returns the smallest property value for the alert type given. Null if none is found.

Exampleâ

```
Revelock.getAlertOfTypeDoublePropertyMin(event.revelock_alerts ,'Fraudster Behavior', 'similarity') ?? 0.0
```

Get maximum double value of property X from alert of type Yâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and gets a property value from an alert of a given type (not case sensitive). If multiple alerts with the same type are found, the maximum value is returned.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.
`property`	String	The alert property name.

Responseâ

Type	Description
Double	Returns the biggest property value for the alert type given. Null if none is found.

Exampleâ

```
Revelock.getAlertOfTypeDoublePropertyMax(event.revelock_alerts ,'Fraudster Behavior', 'similarity
') ?? 1.0
```

Get minimum long value of property X from alert of type Yâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and gets a property value from an alert of a given type (not case sensitive). If multiple alerts with the same type are found, the minimum is returned.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.
`property`	String	The alert property name.

Responseâ

Type	Description
Long	Returns smallest property value for the alert type given. Null if none is found.

Exampleâ

```
Revelock.getAlertOfTypeLongPropertyMin(event.revelock_alerts ,'Fraudster Behavior',
'alert_date') ?? 0
```

Get maximum double value of property X from alert of type Yâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and gets a property value from an alert of a given type (not case sensitive). If multiple alerts with the same type are found, the maximum is returned.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.
`property`	String	The alert property name.

Responseâ

Type	Description
Long	Returns biggest property value for the alert type given. Null if none is found.

Exampleâ

```
Revelock.getAlertOfTypeLongPropertyMax(event.revelock_alerts ,'Fraudster Behavior',
'alert_date') ?? 0
```

Property X with value Y of alert of type Z existsâ

Descriptionâ

The UDF parses the list of Digital Trust alerts and gets a property value from an alert of a given type (not case sensitive). Then, it checks if that property value is the same as the value given by the `propertyValue` parameter.

Parametersâ

Name	Type	Description
`alerts`	Object	The list of Digital Trust alerts.
`alertType`	String	The type of the alert to search for.
`property`	String	The alert property name.
`propertyValue`	String	The alert property value to compare.

Responseâ

Type	Description
Boolean	Returns `true` if there is an alert both with a given type and property value.

Exampleâ

```
Revelock.existsAlertOfTypeAndPropertyValue(event.revelock_alerts ,'Fraudster Behavior', 'family_name', 'fraud') ?? false
```

Digital Trust Integration

Was this page helpful?Â

This page encapsulates all the information needed to understand Transaction Fraud for Banking's standard integration with Digital Trust.

Event enrichmentâ

Transaction Fraud for Banking (TFB) allows the enrichment of Transfers and Digital Activity events in real time, with extra data points collected by Digital Trust, which can be used for improved fraud detection.

Note that the `session_id` field is used to link TFB events to Digital Trust's session analysis. This means that `session_id` is a mandatory field for the TFB integration with Digital Trust. For more information on this topic, read Digital Activity's [Lifecycle and Session IDs].

Learn moreâ

- [How it Works]
- [How to write a rule with Digital Trust's enriched fields]
- [Behavioral Data in ML Models]
- [Enriched Fields]
- [Reference Data Life Cycle]
- [Digital Activity]'s API specs
- [Transfers]'s API specs

Multiple-company enrichment with Digital Trust enricherâ

While an event can only be enriched by **one** Digital Trust company, multiple companies can be configured to fetch session details for distinct events.

Take the following steps to enable this in Transfers and Digital Activity's workflows:

1. Open the Transaction Fraud for Banking app.
2. Go to **Workflow** and select the relevant workflow.
3. Select the **Edit Service** icon to edit the **Revelock Pre Plan** operator.
4. On the **Main** tab, select the **Open resource** icon to edit the **Plan** field. This will take you to a new **Edit Plan**.
5. Once in the relevant channel's **Revelock Pre Plan**'s Edit Plan, select the **Edit Operator** icon to edit the **Map Revelock Company** operator.
6. Select the **Fields to Add** tab.
7. The `Revelock Pre Plan` plan defines the `revelock_company` value, and determines which company to use.
   1. If the `revelock_company` field is mapped to an empty value (`''`), use the default companyÂ[1]. Otherwise, you can apply conditions, like the following one, to dynamically assign the company:

```
if (sender_country_code == "UK") then "uk_company" else if (sender_country_code == "PT") then "pt_company " else ''
```

Â[1]The default company is `feedzai`. However, you can customize it in the **Revelock Enricher**'s custom service properties by adding the `enricher.revelock.company` property.

Fraud scenariosâ

Digital Trust's device, network and behavioral data points prevent fraud scenarios such as Account Takeover.

Learn moreâ

- [Device and Behavior]

Case Manager's widgetsâ

Case Manager provides you with Digital Trust's widgets, which help you select and analyze events based on the information details, and decide whether the transaction is genuine or fraudulent.

Learn moreâ

- Transfers' relevant widgets:
    - **Session** widget
    - **Device** widget
    - **Session Alerts** widget, more specifically, the Session Risk Score and the "Potential Fraud Risk" alert type (read more about this alert in Digital Trust's Potential Fraud Risk documentation).
- Digital Activity's relevant widgets:
    - **Network** widget
    - **Session** widget
    - **Device** widget
    - **Session Alerts** widget, more specifically, the Session Risk Score and the "Potential Fraud Risk" alert type (read more about this alert in Digital Trust's Potential Fraud Risk documentation).

Enriched Fields

This page helps you understand:

- The logic behind Reference Data's field enrichment.
- Which Digital Trust's Transfers and Digital Activity fields are enriched.
- How some fields can be enriched by third-party providers.
- Pulse enrichers.

tip

Note that the enrichment only occurs if the target fields are not sent in the payload. This means that the payload fields have precedence over the enrichment.

Reference Data enrichmentâ

Each time an entity is sent to Reference Data, that same entity will be used to enrich further transactional events. This means that the entity data will map directly to the fields in the payload. Consider the following example, in which a Customer entity is sent to Reference Data with the following information:

```
{
  "customer_id": "customer_1",
  "customer_name": "John",
  ...
}
```

From this point onwards, the next card event with the `customer_1` value for `customer_id` will have its `customer_name` field enriched with `John`.

As for the next transfer, since there might be two parties involved (i.e. sender and receiver), the `customer_1` value for `sender_id` or `receiver_id`, will have its respective `sender_name` or `receiver_name` field enriched with `John`. Both the sender's and receiver's information are enriched within the same transaction.

Reference Data entitiesâ

- **Cards** uses Customer (`customer_id`), Account (`account_iban`) and Merchant (`acceptor_id`) entities.
- **Transfers** uses Customer (`sender_id` and `receiver_id`), Account (`sender_account_iban` and `receiver_account_iban`) and Device (`device_id`) entities.
- **Digital Activity** uses Customer (`customer_id`), Account (`account_iban`) and Device (`device_id`) entities.

For more information, refer to the Reference Data Life Cycle page.

Digital Activity endpoints' updates to Reference Dataâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.
- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the Reference Data Operations section to understand the difference between these methods.

In addition, all fields sent with the `new` prefix will update the Reference Data entity. For instance, both `new_device_first_seen` and `device_first_seen` from the Device endpoint are sent, but only `new_device_first_seen` will be updated because it is the new value.

Fields without prefix, like `device_first_seen`, will be enriched by the current Reference Data in case they are not sent in the payload of the update requests.

For more information, check Digital Activity's API documentation.

Digital Trust enrichmentâ

Transfers fieldsâ

The Transfers events' fields that become enriched are as follows:

- `device_browser`
- `device_browser_version`
- `device_browser_language`
- `device_browser_no_tracking`
- `device_browser_private_mode`

- `device_id`
- `device_ip_address`
- `device_ip_address_location`
- `device_ip_country_code`
- `device_ip_region`
- `device_isp`
- `device_latitude`
- `device_longitude`
- `device_operating_system`
- `device_os_version`
- `device_type`
- `device_user_agent`
- `device_android_id`
- `device_cbf`
- `device_model`
- `device_mac_address`
- `device_screen_width`
- `device_screen_height`
- `device_serial_id`
- `session_register_date`
- `device_session_duration`
- `device_network_type`
- `revelock_alerts`, which includes the following:
  - `view`
  - `risk`
  - `alert_date`
  - `profile_quality`
  - `similarity`
  - `user_training`
  - `family_name`
  - `signature_name`
- `revelock_automated_actions`, which includes the following:
  - `action`
  - `timestamp`
- `revelock_score`

Digital Activity fieldsâ

The Digital Activity events' fields that become enriched are the following:

- `device_browser`
- `device_browser_version`
- `device_browser_language`
- `device_browser_no_tracking`
- `device_browser_private_mode`
- `device_id`
- `device_ip_address`
- `device_ip_city`
- `device_ip_country_code`
- `device_ip_region`
- `device_isp`
- `device_latitude`
- `device_longitude`
- `device_os`
- `device_os_version`
- `device_type`
- `device_user_agent`
- `device_android_id`
- `device_cbf`

- `device_model`
- `device_mac_address`
- `device_screen_width`
- `device_screen_height`
- `device_serial_id`
- `session_register_date`
- `device_session_duration`
- `device_network_type`
- `revelock_alerts` [1], which includes the following fields:
  - `view`
  - `risk`
  - `alert_date`
  - `profile_quality`
  - `similarity`
  - `user_training`
  - `family_name`
  - `signature_name`
- `revelock_automated_actions` [2], which includes the following fields:
  - `action`
  - `timestamp`
- `revelock_score` [3]

[1]Find the full list of fields in the Digital Trust documentation.

[2]These fields exist as part of the out-of-the-box fields for Transfers and Digital Activity. If you need to set up new actions for a new rule, read the Digital Trust documentation.

[3]This field exists as part of the out-of-the-box fields in the **Revelock Plan** workflow element for both Transfers and Digital Activity. This field is transformed and enriched by the previous workflow element - **Revelock Enricher**. To set up scores for a new rule - e.g. to populate Case Manager's Session Risk Score, see the Variables example in the How to write a rule with Digital Trust's enriched fields section.

Third-party providersâ

As an extension of the out-of-the-box enrichment, it is possible to enrich IP, Bin, and currency exchange fields by using enrichment functions. Check Pulse's documentation for more information.

Pulse enrichersâ

Bin providersâ

Pulse's current BIN (Bank Identification Number) lookup process uses data from two providers, Binbase and Bindb. Their data is obtained directly from card issuers. While Binbase is the sole source for 8-digit BIN lookups, Bindb remains in use for other purposes, as detailed in the table that follows.

Relying on third-party providers, rather than the card issuers themselves, can result in incomplete or outdated information. This poses a significant risk to the accuracy of Feedzai's data. By going directly to the source, it is possible to bypass this problem and ensure the most current information available.

The following table outlines the information those two providers have.

Field	Meaning	Bin base	BinDB
Bin	Bank identification number of the issuer of the account	X	X
Card brand	Identifies if the brand (e.g. VISA / Mastercard) or brands are associated with the account range	X	X
Issuing organization	Name of the card issuer	X	X

Field	Meaning	Bin base	BinDB
Type of card	Indicates if credit, debit or prepaid	X	X
Category of the card		X	X
Issuing country ISO name		X	X
Issuing country ISO A2 code	Card issuance country	X	X
Issuing country ISO A3 code		X	X
Country ISO number		X	X
Issuing organization website		X	X
Issuing organization phone number		X	X
Maximum PAN length		X	
Bin type	Indicates if consumer Bin or commercial Bin	X	
Bin is regulated	Indicator for regulated Bins	X	

Email, IP, and phone number providersâ

Feedzai cloud uses several sources to enrich transaction data. The following table details each source and the information it provides.

Field	Source	Meaning
Email	adamloving, androidarea51m digitalfaq, ghacks, mailinator_aliases, xenforo	Disposable emails
IPv4	IPligence	IP City IP Country IP Geographical Coordinates
IP	ProxyRSS	List of proxy IPs
IP	Alien Vault	Compromised IPs
IP	TOR List	List of TOR IPs
Phone number	libphonenumber	Library that parses, formats and validates phone numbers

Map API Fields

This page helps you map the API fields used in Cards, Transfers, and Digital Activity rules, among other parameters. Note that the **API Fields** column includes both fields used directly in rules' conditions and fields that generate metrics.

Cardsâ

Rules Project	Rule Code	API Fields	Metrics and Derived Fields	Lists
Feedback Rules	FB1	feedback_type feedback_label	n/a	PosNeg Listed Cards

		feedback_value card_id		
	FB2	feedback_type feedback_label feedback_value card_id	n/a	PosNeg Listed Cards
Decision Rules	DR1	n/a	n/a	Cards - Rules Thresholds
PosNeg Rules	PN1	card_id	posneg_card_value	PosNeg Listed Cards
	PN2	acceptor_zip	posneg_zip_value	PosNeg Listed Address ZIPs
	PN3	acceptor_country_code	posneg_country_value	PosNeg Listed Countries
	PN4	customer_id	posneg_customer_value	PosNeg Listed Customers
	PN5	mcc	posneg_mcc_value	PosNeg Listed MCCs
	PN6	acceptor_id	posneg_acceptor_value	PosNeg Listed Merchants
	PN7	terminal_id	posneg_terminal_value	PosNeg Listed Terminals
	PN8	device_id	posneg_device_id_value	PosNeg Listed Devices
	PN9	card_country_code	posneg_card_country_value	PosNeg Listed Countries
	PN10	device_ip	posneg_device_ip_value	PosNeg Listed IP Addresses
Entity Limit Rules	EL1	info_type transaction_status card_id cardholder_bill_amount currency_code	REVIEW_APPR_AUTH_cardId_sum_billAmount_1d posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
	EL2	info_type transaction_status card_id	REVIEW_APPR_AUTH_cardId_count_1d posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
Fraud Rules	Velocity-1	terminal_latitude terminal_longitude info_type	CP_card_term_vel_1h CP_card_prev_term_dist_1h AUTH_CP_cardid_prev_term_lat_1h	PosNeg Listed Cards

	card_presence_flag card_id	AUTH_CP_cardid_prev_term_long_1h AUTH_CP_cardid_prev_ts_1h card_id_not_blacklisted_nor_whitelisted posneg_card_value	Cards - Rules Thresholds
Velocity-2	card_id info_type	AUTH_cardid_count_1d card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
Velocity-3	info_type transaction_status card_id cardholder_bill_amount currency_code	APPR_AUTH_cardId_count_1h APPR_AUTH_cardId_sum_billAmount_1h sign_cardholder_bill_amount card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
Velocity-4	info_type transaction_status card_id	APPR_AUTH_cardId_count_3m sign_count card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
Velocity-5	info_type card_id card_presence_flag device_ip_latitude device_ip_longitude	AUTH_CNP_cardid_prev_device_ip_lat_1h AUTH_CNP_cardid_prev_device_ip_long_1h card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
Velocity-6	card_id card_bin card_expiration_date acceptor_id	acceptorid_cardbin_cntDist_cardExpDate_10m acceptorid_cardbin_cntDist_cardid_10m acceptorid_cardbin_cur0 acceptorid_cardbin_cur1 acceptorid_cardbin_cur3 acceptorid_cardbin_cur5 acceptorid_cardbin_cntDist_cardid_10m acceptorid_cardbin_cntDist_cardExpDate_10m card_days_to_expire	PosNeg Listed Cards
Velocity-7	card_id card_bin card_expiration_date mcc	mcc_cardbin_cntDist_cardExpDate_10m mcc_cardbin_cntDist_cardid_10m mcc_cardbin_cur0 mcc_cardbin_cur1 mcc_cardbin_cur3 mcc_cardbin_cur5	PosNeg Listed Cards
CNP-1	info_type card_presence_flag card_id acceptor_id	AUTH_CNP_cardId_countDist_acceptor_5m card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards
CNP-2	card_presence_flag card_id currency_code cardholder_bill_amount	card_id_not_blacklisted_nor_whitelisted posneg_card_value is_high_amount	PosNeg Listed Cards Common Currencies
Threshold-1	info_type transaction_status card_id acceptor_id cardholder_bill_amount currency_code	APPR_AUTH_cardId_acceptor_sum_billAmount_1d card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds

	Threshold-2	info_type card_id	AUTH_cardId_count_4w card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
	Threshold-3	card_id currency_code info_type transaction_status cardholder_bill_amount	APPR_AUTH_cardId_sum_billAmount_4w card_id_not_blacklisted_nor_whitelisted posneg_card_value sign_cardholder_bill_amount sign_count is_high_amount	PosNeg Listed Cards Cards - Rules Thresholds
	Threshold-4	card_id currency_code info_type transaction_status cardholder_bill_amount	HIGH_AMOUNT_APPR_AUTH_cardId_count_1d card_id_not_blacklisted_nor_whitelisted posneg_card_value sign_count is_high_amount	PosNeg Listed Cards Cards - Rules Thresholds
	Threshold-5	card_id info_type transaction_status cardholder_bill_amount acceptor_id currency_code	APPR_AUTH_cardId_acceptor_count_1d APPR_AUTH_cardId_acceptor_sum_billAmount_1d card_id_not_blacklisted_nor_whitelisted posneg_card_value sign_count sign_cardholder_bill_amount	PosNeg Listed Cards Cards - Rules Thresholds
	Threshold-6	card_id info_type card_presence_flag device_id	AUTH_CNP_distinct_device_id_per_card_24h card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
	Behavior-1	card_id info_type transaction_status	DECLINE_cardId_count_1d card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Cards - Rules Thresholds
	Behavior-2	cardholder_bill_amount card_id info_type transaction_status acceptor_id amount currency_code	APPR_AUTH_cardId_acceptor_billAmount_count_1d card_id_not_blacklisted_nor_whitelisted posneg_card_value amount_as_str sign_count	PosNeg Listed Cards Cards - Rules Thresholds
	Behavior-3	card_id info_type acceptor_id transaction_status feedback_label feedback_value	APPR_AUTH_acceptorId_count_6M FRAUD_ratio_acceptorId_6M FRAUD_acceptorId_count_6M AUTH_cardId_acceptorId_count_6M sign_count card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards Rules Thresholds
	Behavior-4	card_id malware_flag	card_id_not_blacklisted_nor_whitelisted posneg_card_value	PosNeg Listed Cards
SCA Required	SCAR1	acceptor_country_code card_country_code info_type card_id acceptor_id	AUTH_cardId_acceptorId_count_6M	n/a

	SCAR2	customer_country_code_change_flag	n/a	n/a
	SCAR3	customer_email_change_flag	n/a	n/a
	SCAR4	customer_phone_change_flag	n/a	n/a
	SCAR5	info_type card_id currency_code card_presence_flag mcc	AUTH_CNP_currency_changed_24h AUTH_cardid_mcc_count_6M	n/a
	SCAR6	cardholder_bill_amount card_available_balance	n/a	Cards - Rules Thresholds
	SCAR7	info_type card_id card_presence_flag account_pw_change_flag	AUTH_CNP_cardid_count_pw_changed_24h	Cards - Rules Thresholds
	SCAR8	info_type device_language card_id device_ip_country_code device_id	AUTH_device_language_changed_24h AUTH_cardid_device_ip_country_code_count_6M AUTH_cardid_deviceid_count_6M	n/a
	SCAR9	info_type card_id card_presence_flag mcc	AUTH_CNP_login_session_id_changed_24h AUTH_CNP_auth_type_changed_24h AUTH_cardid_mcc_count_6M	n/a
	SCAR10	cardholder_bill_amount info_type card_id mcc	AUTH_cardid_mcc_avg_amount_1M AUTH_cardid_mcc_count_1M	Cards - Rules Thresholds
	SCAR11	proxy_flag	n/a	n/a
SCA Exemption	SCA11-1	cardholder_bill_amount info_type transaction_status card_id last_sca_at	APPR_AUTH_contactless_last_sca_at_cardid_sum_billAmount_90d APPR_AUTH_contactless_last_sca_at_cardid_sum_billAmount_30h sign_cardholder_bill_amount pos_card_read_method	Cards - Rules Thresholds
	SCA11-2	cardholder_bill_amount pos_card_read_method info_type transaction_status card_id last_sca_at	APPR_AUTH_contactless_last_sca_at_cardid_count_90d APPR_AUTH_contactless_last_sca_at_cardid_count_30h sign_count last_sca_at_as_str	Cards - Rules Thresholds
	SCA12	mcc	n/a	SCA Exempted MCCs
	SCA13	trusted_acceptor	n/a	n/a
	SCA14	first_recurring_payment recurring_payment	n/a	n/a
	SCA16-1	cardholder_bill_amount info_type transaction_status last_sca_at card_id	APPR_AUTH_last_sca_at_cardid_sum_billAmount_90d APPR_AUTH_last_sca_at_cardid_sum_billAmount_30h sign_cardholder_bill_amount	Cards - Rules Thresholds
	SCA16-2	cardholder_bill_amount info_type transaction_status	APPR_AUTH_last_sca_at_cardid_count_90d APPR_AUTH_last_sca_at_cardid_count_30h last_sca_at_as_str	Cards - Rules Thresholds

		card_id last_sca_at		
	SCA17	card_category	n/a	n/a
	SCA18-ETV1	reference_fraud_rate cardholder_bill_amount info_type card_presence_flag card_id	AUTH_CNP_cardid_sum_amount_24h	Fraud Rates Cards - Rules Thresholds
	SCA18-ETV2	reference_fraud_rate cardholder_bill_amount info_type card_presence_flag card_id	AUTH_CNP_cardid_sum_amount_24h	Fraud Rates Cards - Rules Thresholds
	SCA18-ETV3	reference_fraud_rate cardholder_bill_amount info_type card_presence_flag card_id	AUTH_CNP_cardid_sum_amount_24h	Fraud Rates Cards - Rules Thresholds

Transfersâ

Rules Project	Rule Code	API Fields	Metrics and Derived Fields	Lists
Feedback Rules	FB1	feedback_type feedback_label feedback_value sender_transaction_bank_account_number	n/a	PosNeg Listed Accounts
PosNeg Rules	PN1	sender_bank_country_code	n/a	PosNeg Listed Countries
	PN2	receiver_bank_country_code	n/a	PosNeg Listed Countries
	PN3	sender_transaction_bank_account_number	n/a	PosNeg Listed Accounts
	PN4	receiver_transaction_bank_account_number	n/a	PosNeg Listed Accounts
	PN5	sender_bank_bic	n/a	PosNeg Listed BICs
	PN6	receiver_bank_bic	n/a	PosNeg Listed BICs
	PN7	sender_bank_sort_code	n/a	PosNeg Listed Sort Codes
	PN8	receiver_bank_sort_code	n/a	PosNeg Listed Sort Codes
	PN9	sender_transaction_notes receiver_transaction_notes	n/a	PosNeg Listed Keywords

	PN10	receiver_transaction_bank_account_number	n/a	PosNeg Listed Accounts
	PN11	sender_transaction_bank_account_number	n/a	PosNeg Listed Accounts
	PN12	device_ip_address device_ip_address_location	n/a	PosNeg Listed IP Address
	PN13	device_id	n/a	PosNeg Listed Devices
Entity Limit Rules	EL1	info_type sender_transaction_amount sender_transaction_bank_account_number sender_transaction_currency	sum_amount_per_sender_24h	Transfers - Rules Thresholds
	EL2	created_at info_type sender_transaction_amount sender_transaction_bank_account_number sender_transaction_currency	avg_daily_amount_per_sender_1w sum_amount_per_sender_1w distinct_day_week_per_sender_1w	Transfers - Rules Thresholds
	EL3	info_type sender_transaction_bank_account_number created_at	avg_monthly_count_sender_3m_delay_1m count_sender_1m count_sender_3m_delay_1m distinct_month_per_sender_3m_delay_1m month	Transfers - Rules Thresholds
	EL4	sender_transaction_currency, sender_transaction_amount sender_bank_bic receiver_bank_bic	n/a	Transfers - Rules Thresholds
	EL5	created_at info_type receiver_transaction_bank_account_number receiver_transaction_amount receiver_transaction_currency	avg_daily_amount_per_receiver_1w sum_amount_per_receiver_1w distinct_day_week_per_receiver_1w	Transfers - Rules Thresholds
	EL6	info_type sender_transaction_bank_account_number sender_transaction_amount sender_transaction_currency created_at	avg_monthly_amount_per_sender_3m_delay_1m sum_amount_per_sender_1m sum_amount_sender_3m_delay_1m distinct_month_per_sender_3m_delay_1m count_sender_1m month	Transfers - Rules Thresholds
	EL7	info_type receiver_transaction_bank_account_number receiver_transaction_amount receiver_transaction_currency created_at	avg_monthly_amount_per_receiver_3m_delay_1m sum_amount_per_receiver_1m sum_amount_receiver_3m_delay_1m distinct_month_per_receiver_3m_delay_1m count_receiver_1m month	Transfers - Rules Thresholds
	EL8	receiver_transaction_bank_account_number info_type created_at	avg_monthly_count_receiver_3m_delay_1m count_receiver_3m_delay_1m distinct_month_per_receiver_3m_delay_1m count_receiver_1m month	Transfers - Rules Thresholds
	EL9	sender_transaction_currency sender_transaction_amount sender_bank_country_code receiver_bank_country_code	n/a	Transfers - Rules Thresholds

		sender_bank_sort_code receiver_bank_sort_code		
	EL10	sender_transaction_currency sender_transaction_bank_account_number sender_transaction_amount channel_type info_type	sum_amount_per_sender_telephone_24h	Transfers - Rules Thresholds
	EL11	sender_transaction_bank_account_number created_at channel_type info_type	count_sender_telephone_24h avg_monthly_count_sender_telephone_3m_delay_24h	Transfers - Rules Thresholds
	EL12	sender_transaction_bank_account_number created_at channel_type info_type	count_sender_online_24h avg_monthly_count_sender_online_3m_delay_24h	Transfers - Rules Thresholds
Fraud Rules	FR1	sender_transaction_bank_account_number info_type	count_sender_90m	Transfers - Rules Thresholds
	FR2	sender_transaction_bank_account_number info_type	count_sender_24h	Transfers - Rules Thresholds
	FR3	sender_transaction_bank_account_number sender_country_code sender_transaction_country_code info_type	distinct_country_per_sender_24h	Transfers - Rules Thresholds
	FR4	sender_transaction_amount sender_transaction_bank_account_number receiver_transaction_bank_account_number info_type	count_sender_receiver_amount_1k_24h	Transfers - Rules Thresholds
	FR5	sender_transaction_bank_account_number sender_account_open_date created_at info_type sender_transaction_amount sender_account_balance account_id	count_sender_3m_delay_24h count_sender_24h	Transfers - Rules Thresholds
	FR6	sender_transaction_bank_account_number geographic_scope cross_border sender_transaction_amount info_type	count_sender_cross_border_amount_3k_24h	Transfers - Rules Thresholds
	FR7	sender_transaction_bank_account_number geographic_scope cross_border sender_transaction_amount info_type	count_sender_cross_border_amount_3k_1month	Transfers - Rules Thresholds
	FR11	sender_transaction_bank_account_number receiver_transaction_bank_account_number info_type	distinct_receiver_per_sender_24h	Transfers - Rules Thresholds
	FR12	sender_transaction_bank_account_number receiver_transaction_bank_account_number info_type receiver_segment	distinct_sender_per_receiver_24h	Transfers - Rules Thresholds
	FR13	sender_transaction_bank_account_number receiver_name	distinct_receiver_same_name_per_sender_24h	Transfers - Rules Thresholds

		receiver_transaction_bank_account_number info_type		
	FR15	receiver_transaction_bank_account_number sender_account_id sender_transaction_bank_account_number created_at info_type	count_sender_6m count_sender_time_of_day_6m	Transfers - Rules Thresholds
	FR16	receiver_transaction_bank_account_number payment_sub_type sender_transaction_bank_account_number info_type	distinct_sender_per_receiver_direct_deposit_3m	Transfers - Rules Thresholds
	FR18	sender_email sender_transaction_bank_account_number info_type	distinct_sender_per_email_1m	Transfers - Rules Thresholds
	FR19	sender_phone sender_transaction_bank_account_number info_type	distinct_sender_per_phone_1m	Transfers - Rules Thresholds
	FR20	receiver_email receiver_transaction_bank_account_number info_type	distinct_receiver_per_email_1m	Transfers - Rules Thresholds
	FR21	receiver_phone receiver_transaction_bank_account_number info_type	distinct_receiver_per_phone_1m	Transfers - Rules Thresholds
	FR24	sender_transaction_bank_account_number device_id info_type	distinct_sender_per_device_id_90m	Transfers - Rules Thresholds
	FR25	sender_transaction_bank_account_number device_id info_type	distinct_device_id_per_sender_90m	Transfers - Rules Thresholds
	FR27	sender_transaction_bank_account_number info_type payment_status	count_failed_trx_sender_24h	Transfers - Rules Thresholds
	FR28	sender_email	n/a	n/a
	FR30	device_latitude device_longitude sender_transaction_bank_account_number info_type	sender_prev_device_lat_1h sender_prev_device_long_1h	Transfers - Rules Thresholds
	FR32	device_id account_id	distinct_account_per_device_3m	Transfers - Rules Thresholds
Instant Transfers Rules	IT1	sender_transaction_amount sender_transaction_currency	n/a	Transfers - Rules Thresholds
	IT2	receiver_transaction_amount receiver_transaction_currency	n/a	Transfers - Rules Thresholds
	IT3	info_type payment_sub_type sender_transaction_bank_account_number created_at	avg_daily_count_sender_instant_transfer_3m_delay_24h count_sender_instant_transfer_3m_delay_24h distinct_day_year_per_sender_instant_transfer_3m_delay_24h count_sender_instant_transfer_24h day_of_year	Transfers - Rules Thresholds
	IT4	payment_type sender_name sender_street_addr	n/a	n/a

	IT5	payment_type receiver_name receiver_street_addr	n/a	n/a
	IT6	sender_alias	n/a	PosNeg Listed Alias
	IT7	receiver_alias	n/a	PosNeg Listed Alias
	IT8	sender_transaction_notes text_for_receiver sender_transaction_amount	n/a	Transfers - Rules Thresholds
	IT9	receiver_alias receiver_alias_type	n/a	PosNeg Listed Email Domains
	IT10	receiver_alias_last_update_time sender_alias_last_update_time sender_transaction_amount sender_transaction_currency	n/a	Transfers - Rules Thresholds
	IT11	text_for_receiver sender_transaction_amount	n/a	Transfers - Rules Thresholds Whitelisted Strings
	IT12	receiver_alias receiver_alias_type	n/a	PosNeg Listed Phone Numbers
	IT13	info_type payment_sub_type sender_alias_registration_time sender_transaction_currency sender_transaction_amount sender_alias	sum_amt_instant_transfers_per_sender_alias_90m	Transfers - Rules Thresholds
Direct Debit Rules	DD1	info_type created_at payment_sub_type sender_account_open_date sender_transaction_bank_account_number	count_sender_dd_3m_delay_24h count_sender_dd_24h	Transfers - Rules Thresholds
	DD2	receiver_transaction_bank_account_number	n/a	PosNeg Listed DD Companies
	DD3	receiver_transaction_bank_account_number	n/a	PosNeg Listed DD Companies
	DD4	sender_bank_country_code receiver_bank_country_code	n/a	n/a
	DD5	sender_country_code receiver_country_code	n/a	n/a
	DD7	info_type payment_sub_type sender_transaction_bank_account_number payment_status	count_dd_return_per_sender_3m	Transfers - Rules Thresholds
	DD8		count_dd_return_per_sender_24h	

		info_type payment_sub_type sender_transaction_bank_account_number payment_status		Transfers - Rules Thresholds
Bill Payments Rules	BP1	reference_number	n/a	PosNeg Listed Biller IDs
	BP2	sender_transaction_amount sender_transaction_currency reference_number_entry_mode	n/a	Transfers - Rules Thresholds
	BP4	sender_transaction_amount sender_account_id day_of_week created_at info_type account_id day_of_week event_occured_at new_device	sum_amt_bill_payments_per_sender_out_work_days_1m	Transfers - Rules Thresholds
Decision Rules	DR1	enforced_risk_level	n/a	n/a
	DR2	fraud_score payment_sub_type	n/a	Transfers - Rules Thresholds
	DR3	enforced_risk risk_level payment_sub_type	n/a	n/a
	DR4	fraud_score	n/a	Transfers - Rules Thresholds
	DR5	enforced_risk risk_level	n/a	n/a
	DR6	fraud_score	n/a	Transfers - Rules Thresholds
	DR7	enforced_risk risk_level	n/a	n/a
SCA Required	SCAR1	sender_country_code receiver_country_code receiver_transaction_bank_account_number sender_transaction_bank_account_number info_type	count_sender_receiver_6m	Transfers - Rules Thresholds
	SCAR2	sender_country_code_change_flag	n/a	n/a
	SCAR3	sender_email_change_flag	n/a	n/a
	SCAR4	sender_phone_change_flag	n/a	n/a
	SCAR5	sender_account_balance sender_transaction_amount	n/a	Transfers - Rules Thresholds
	SCAR6	sender_account_pw_change_flag sender_transaction_bank_account_number info_type	count_sender_pw_change_24h	Transfers - Rules Thresholds
	SCAR7	sender_device_language_changed device_ip_country_code sender_transaction_bank_account_number	sender_device_language_changed count_sender_device_ip_country count_sender_device_id_6m	n/a

		device_id info_type		
	SCAR8	sender_transacttion_bank_account_number receiver_transaction_bank_account_number auth_type login_session_id info_type sender_transaction_bank_account_number	auth_type_changed_24h login_session_id_changed_24h count_sender_receiver_6m	n/a
	SCAR9	proxy_flag	n/a	n/a
SCA Exemption	SCA13	trusted_receiver	n/a	n/a
	SCA14	first_recurring_payment recurring_payment	n/a	n/a
	SCA15	sender_id receiver_id	n/a	n/a
	SCAR16-1	sender_transaction_amount expired_metrics last_sca_at_as_str sender_transaction_bank_account_number info_type	sum_amount_sender_last_sca_90d sum_amount_sender_last_sca_30h	Transfers - Rules Thresholds
	SCAR16-2	sender_transaction_amount expired_metrics last_sca_at_as_str sender_transaction_bank_account_number info_type	sum_amount_sender_last_sca_90d sum_amount_sender_last_sca_30h	Transfers - Rules Thresholds
	SCA17	segment_channel_type	n/a	n/a
	SCA18-ETV1	reference_fraud_rate sender_transaction_amount fraud_rate decision info_type sender_transaction_bank_account_number	sum_amount_per_sender_24h	Transfers - Rules Thresholds
	SCA18-ETV2	reference_fraud_rate sender_transaction_amount fraud_rate decision info_type sender_transaction_bank_account_number	sum_amount_per_sender_24h	Transfers - Rules Thresholds
	SCA18-ETV3	reference_fraud_rate sender_transaction_amount fraud_rate decision info_type sender_transaction_bank_account_number	sum_amount_per_sender_24h	Transfers - Rules Thresholds
Device and Network Intelligence Rules	Transfers-DNI1	session_id	n/a	n/a
	Transfers-DNI2	device_id	n/a	n/a
	Transfers-DNI3	session_id	n/a	n/a
	Transfers-DNI4	device_id	n/a	n/a
	Transfers-DNI5	device_id	n/a	n/a
		n/a	n/a	n/a

	Transfers-DNI6			
	Transfers-DNI7	created_at sender_account_open_date info_type sender_account_id	count_account_activity_3m	n/a
	Transfers-DNI8	device_ip_country_code device_id	n/a	n/a
	Transfers-DNI9	device_id	n/a	n/a
	Transfers-DNI10	device_id sender_transaction_amount	n/a	n/a
	Transfers-DNI11	created_at payment_sub_type device_id sender_account_id info_type	count_account_activity_out_work_days_3m count_account_activity_3m	n/a

(*) The `account_id` is populated by the Digital Activity API.

(**) These rules are omnichannel and use fields from the Digital Activity API.

Digital Activityâ

Rules Project	Rule Code	API Fields	Scoring Event	Metrics and Derived Fields	Lists
Decision Rules	DR1	enforced_risk_level risk_level	All Digital Events	n/a	n/a
	DR2	fraud_score	All Digital Events	n/a	Digital Activity - Rules Thresholds
	DR3	enforced_risk_level risk_level	All Digital Events	n/a	n/a
	DR4	fraud_score	All Digital Events	n/a	Digital Activity - Rules Thresholds
	DR5	enforced_risk_level risk_level	All Digital Events	n/a	n/a
	DR6	fraud_score	All Digital Events	n/a	Digital Activity - Rules Thresholds
Feedback Rules	FB1	feedback_label feedback_value account_id	All Digital Events	n/a	PosNeg Listed Accounts
	FB2	feedback_label feedback_value account_id	All Digital Events	n/a	PosNeg Listed Accounts
PosNeg Rules	PN1	device_ip_address	All Digital Events	n/a	PosNeg Listed IP Addresses
	PN2	device_id		n/a	

			All Digital Events		PosNeg Listed Devices
	PN3	account_id	All Digital Events	n/a	PosNeg Listed Accounts
	PN4	account_id	All Digital Events	n/a	PosNeg Listed Accounts
	PN5	new_alias_value	Alias	n/a	PosNeg Listed Alias
	PN6	new_customer_email_addr new_customer_email_domain	Profile Update	n/a	PosNeg Listed Email Addresses PosNeg Listed Email Domains
	PN7	new_customer_phone_number	Profile Update	n/a	PosNeg Listed Phone Numbers
Device and Network Intelligence Rules	DNI1	session_id	All Digital Events	n/a	n/a
	DNI2	n/a	All Digital Events	n/a	n/a
	DNI3	session_id	All Digital Events	n/a	n/a
	DNI4	device_id	All Digital Events	n/a	n/a
	DNI5	device_id	All Digital Events	n/a	n/a
	DNI6	device_id	All Digital Events	n/a	n/a
	DNI7	event_status account_age challenge_requested account_id	Authentication	count_account_activity_3M	n/a
	DNI8	challenge_requested device_ip_country_code device_id	Authentication	n/a	n/a
	DNI9	n/a	All Digital Events	n/a	n/a
Fraud Rules	FR1	device_id customer_id challenge_requested	Authentication	distinct_customer_id_per_device_id_1w count_logins_per_device_id_90min	Digital Activity - Rules Thresholds
	FR2	device_ip_address customer_id challenge_requested	Authentication	distinct_customer_id_per_device_ip_addr_1h count_logins_per_device_ip_addr_1h	Digital Activity - Rules Thresholds
	FR3		Authentication		

		device_id challenge_requested event_status account_id		count_succ_logins_per_account_1m count_logins_per_device_id_1m count_failed_logins_per_device_id_90min	Digital Activity - Rules Thresholds
	FR4	device_ip_address challenge_requested event_status account_id	Authentication	count_succ_logins_per_account_1m count_logins_per_device_ip_addr_1m count_failed_logins_per_device_ip_addr_90min	Digital Activity - Rules Thresholds
	FR5	account_id challenge_requested event_status	Authentication	count_failed_logins_per_account_id_1min	Digital Activity - Rules Thresholds
	FR6	customer_region device_ip_region account_id challenge_requested	Authentication	distinct_login_locations_per_account_1h	Digital Activity - Rules Thresholds
	FR7	account_id session_id challenge_requested event_status	Authentication	count_account_changes_per_session_id_1h	Digital Activity - Rules Thresholds
	FR8	account_id challenge_requested	New Payee	count_new_payee_per_account_1min	Digital Activity - Rules Thresholds
	FR9	account_id challenge_requested event_sub_type event_status	New Payee and Profile Update	account_previous_change_is_new_device_24h account_previous_auth_type_changed_24h	n/a
	FR10	event_sub_type account_id challenge_requested event_status	Authentication	account_previous_activity_is_succ_login_24h count_failed_logins_per_account_id_24h	Digital Activity - Rules Thresholds
	FR11	account_id account_age event_sub_type event_status	Profile Update and Credentials Update	account_previous_activity_is_succ_login_24h count_succ_logins_per_account_1m	n/a
	FR12	event_status account_id account_age event_sub_type challenge_requested event_occurred_at	Credentials Update	ount_succ_logins_per_account_time_of_day_3m count_succ_logins_per_account_3m account_previous_activity_is_succ_login_24h	Digital Activity - Rules Thresholds
	FR13	new_device_id event_sub_type account_id account_age challenge_requested device_ip_region customer_region event_status	Device	account_previous_change_is_password_reset_24h count_login_per_account_and_ip_region_1m count_succ_logins_per_account_1m	Digital Activity - Rules Thresholds

(*) The `count_account_activity_3M` metric is omnichannel, and therefore is populated by Cards, Transfers, and Digital Activity APIs.

References

Was this page helpful?Â

Rules projects

Learn how rules are organized in your Pulse application and find out which rules come out of the box

Pulse schemas

A schema describes the available data fields from a data source, as well as their types

Pulse lists

Lists store items, and can be used in rules to check if a transaction field is known for a specific attribute

Enriched fields

Understand how Digital Trust and Transfer fields are enriched

Digital Trust integration

Learn about Transaction Fraud for Banking's standard integration with Digital Trust

Digital Trust Enricher's UDFs

Learn how to use Digital Trust Enricher's UDFs

Map API fields

Map the API fields used in Cards, Transfers, and Digital Activity rules, among other parameters

Schema mappings

Includes comma-separated value (CSV) files that allow you to confirm where API fields are used

Rule explanations

Learn what rule explanations are and how enabling rule explanations affects responses

Model explanations

Learn what model explanations are and how enabling model explanations affects responses

Data Exporting Tool

Learn what the data exporting tool is, how exports are structured and the entities for which they are available

Pulse Lists

Lists store sets of items and are available for rules to check the same pattern against a large number of values (e.g. a list of credit card numbers, a list of merchants).

In the process of fraud detection it's very common to define blacklists to block transactions from certain cards or to define common parameters in transactions that should be reviewed by analysts (triggering an alarm every time such transactions occur). Lists for grouping and managing these values so that they can be used in rules (check the Rules Projects page to learn more about how rules are organized in TFB).

info

Lists regarding sanctions are only relevant if you have Feedzai's Watchlist Transaction Screening solution enabled.

Name	Description
Blacklisted keywords	
Cards - Rules Thresholds	Thresholds used in Cards Rules
Common Currencies	Top 15 most used currencies in payments according to the Society for Worldwide Interbank Financial Telecommunication (SWIFT) as of Feb 2017.
Digital Activity - Rules Thresholds	Thresholds used in Digital Activity Rules
EEA Countries	List of Countries within the European Economic Area (EEA)
Fraud Rates	Fraud Rates internal list for PSD2 monitoring
High Risk Addresses	List of High Risk Addresses
High Risk Counterparty	List of High Risk Counterparties (by either Account Number or Name)

High Risk Geographies	List of High Risk Geographies (by Country Code)
High Risk IP Addresses	List of High Risk IP Addresses
High Risk MCCs	List of High Risk MCCs (Merchant Category Codes)
PosNeg Listed Accounts	PosNeg List for Bank Account Numbers
PosNeg Listed Address ZIPs	PosNeg List for zip codes.
PosNeg Listed Alias	PosNeg List for Alias in an Instant Transfers platform (e.g. phone number, email address, customer ID, etc)
PosNeg Listed BICs	PosNeg List for Bank Identifier Codes for SWIFT payments
PosNeg Listed Biller IDs	PosNeg List for Biller IDs (by reference number) for Bill Payments
PosNeg Listed Cards	PosNeg List for cards by card_id
PosNeg Listed Countries	PosNeg List for country codes.
PosNeg Listed Customers	PosNeg List for customers by customer_id
PosNeg Listed DD Companies	PosNeg List for Direct Debit Companies (by account number)
PosNeg Listed Devices	PosNeg List for Devices
PosNeg Listed Device Types	PosNeg List for Device Types
PosNeg Listed Email Addresses	PosNeg List for Email Addresses
PosNeg Listed Email Domains	PosNeg List for Email Domains
PosNeg Listed IP Addresses	PosNeg List for IPs
PosNeg Listed Keywords	PosNeg List for Keywords in bank transfers' notes
PosNeg Listed MCCs	PosNeg List for MCCs.
PosNeg Listed Merchants	PosNeg List for merchants by acceptor_id
PosNeg Listed Phone Numbers	PosNeg List for Phone Numbers
PosNeg Listed Sort Codes	PosNeg List for Bank Sort Codes
PosNeg Listed Terminals	PosNeg List for terminals.
sanctions_relevant_internal_match_reasons	Relevant internal match reasons for triggering rules conditions
sanctions_relevant_internal_match_types	Relevant internal match types for triggering rules conditions
sanctions_relevant_match_reasons	Relevant match reasons for triggering rules conditions
sanctions_relevant_match_types	Relevant match types for triggering rules conditions
sanctions_review_internal_match_reasons	Review match reasons for triggering rules conditions
sanctions_review_internal_match_types	Review match types for triggering rules conditions
sanctions_review_match_reasons	Review match types for triggering rules conditions
sanctions_review_match_types	Review match types for triggering rules conditions
SCA Exempted MCCs	List of exempted MCCs under PSD2 Article 12
Transfers - Rules Thresholds	Thresholds used in Transfers Rules
Very High Risk Geographies	List of Very High Risk Geographies (by Country Code)
PosNeg Listed BICs Inbound	PosNeg List for Bank Identifier Codes for SWIFT inbound payments
PosNeg Listed DD Companies Inbound	PosNeg List for Direct Debit Companies (by account number) on inbound payments
PosNeg Listed Accounts Inbound	PosNeg List for Bank Account Numbers on inbound payments
Inbound - Rule Thresholds	Thresholds used in Inbound Payments Rules

Model Explanations

Was this page helpful?Â

This page helps you understand:

- What model explanations are.
- How enabling model explanations affects responses.

tip

Note that Model Explanations is an opt-in feature. Contact the Feedzai team to enable this feature.

Each time a scoring event is sent to Feedzai, model evaluations run for that same event. If Model Explanations is enabled, model explanations will be computed in real time, like model evaluations.

The Model Explanations feature consists of returning the configured model explanation. Whenever an event runs through a model and the Model Explanations feature is enabled, a new output field (`model_explanations`) should be expected:

```json
{
    "status": "ok",
    "lifecycle_id": "lifecycle_xyz",
    "event_external_id": "event_external_id_abc",
    "action_codes": [
    ],
    "secondary_action_codes": [],
    "alert": false,
    "model_explanations": [
        {
            "generator_name": "Fraud v2 - explainable model",
            "importance": 12,
            "feature_group_name": "Rapid movement of funds"
        },
        {
            "generator_name": "Fraud v2 - explainable model",
            "description": "The amount of funds transacted to foreign countries",
            "importance": 7,
            "feature_name": "foreign_amount_transacted"
        }
    ],
    "score": 19,
    "decision": "approve",
    "sca_required": false
}
```

Field name	Description
`generator_name`	The name of the model.
`importance`	The impact of each feature on a machine learning model.
`description`	In case the feature is not part of any group, the description of said feature.
`feature_name`	In case the feature is not part of any group, the technical name of said feature.
`feature_group_name`	In case the feature is part of a group, the name of said feature group.

Learn more about:

- Explanations
- Models

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/references/pulse-rule-explanations

Rule Explanations

Was this page helpful?Â

This page helps you understand:

- What rule explanations are.
- How enabling rule explanations affects responses.

tip

Note that Rule Explanations is an opt-in feature. Contact the Feedzai team to enable this feature.

Each time a scoring event is sent to Feedzai, rule evaluations run for that same event. If the Rule Explanations feature is enabled, rule explanations will be computed in real time, like rule evaluations.

Rule Explanations consists of returning the rule explanation configured in the Pulse UI when creating a new rule:

Consider the scenario where you set the same rule configuration as the image above, and enable Rule Explanations. Whenever this rule is triggered by an event, a new output field (`rule_explanations`) is expected:

```
{
    "status": "ok",
    "lifecycle_id": "lifecycle_xyz",
    "event_external_id": "event_external_id_abc",
    "action_codes": [
        "[Cards-PN1]"
    ],
    "secondary_action_codes": [],
    "alert": false,
    "rule_explanations": [
        {
            "rulesProjectName": "Cards - PosNeg Rules",
            "ruleName": "Cards-PN1",
            "explanation": "[Cards-PN1] - Decline - 'card_id' card_123 is listed to be auto
declined."
        }
    ],
    "score": 0,
    "decision": "decline",
    "sca_required": false
}
```

Field name	Description
`rulesProjectName`	The name of the rules project containing the rule triggered.
`ruleName`	The name of the rule triggered.
`explanation`	The explanation configured in the rule triggered.

Learn more about:

- Explanations
- Rules

Pulse Schemas

Was this page helpful?Â

This page contains information about the Pulse schemas that are configured in the standard package. It also includes a list of fields, automatically generated from the source code, for each schema.

Introductionâ

Pulse allows you to deal with different schema's definitions within a Pulse Application. The following schemas are configured by default in the standard package:

- **API Schema <version>:** Contains the fields defined in Pulse that map directly to the ones described in the API Reference.

- **Extended API Schema <version>:** Contains the 'API Schema' plus several fields that are used for internal operations, namely `timestamp` , `fraud` , `fraud_score` , and `decision` fields.
- **Workflow Schema <version>:** Contains the 'Extended API Schema' plus derived metrics and profiles.

See the figure below to understand where these schemas play a role in the Pulse workflow:

Additionally, if you create an AutoML run and then you decide to add the recommended model to the Pulse workflow, you may have to augment the workflow schema. See the Complete the Schema page under the AutoML Guide for more information about this operation.

---

info

The Sanctions Rules Schema is only relevant if you have Feedzai's Watchlist Transaction Screening solution enabled.

> The following information is automatically generated from the source code.

Schemas

Cards API Schema

Field Name	Field Type	Field Description
mti	STRING	
card_pan	STRING	
ttc	STRING	
atc1	STRING	
atc2	STRING	
currency_code	STRING	
currency_unit	INT	
amount	LONG	
cardholder_bill_amount	LONG	
reconciliation_currency_code	STRING	
reconciliation_currency_unit	INT	
reconciliation_amount	LONG	
reconciliation_conv_rate	DOUBLE	
transmission_datetime	STRING	
cardholder_bill_conv_rate	DOUBLE	

stan	STRING	
local_datetime	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
trx_to_rec_conv_date	STRING	
capture_date	STRING	
lifecycle_indicator	STRING	
lifecycle_id	STRING	
lifecycle_seq_number	STRING	
lifecycle_token	STRING	
pos_card_read_method	STRING	
pos_cardholder_ver_method	STRING	
pos_environment	STRING	
pos_security_characteristics	STRING	
card_seq_number	STRING	
function_code	STRING	
reason_code	STRING	
mcc	STRING	
pos_read_cap	STRING	
pos_cardholder_ver_cap	STRING	
pos_approval_code_len	INT	
pos_cardholder_rec_data_len	INT	
pos_acceptor_rec_data_len	INT	
pos_cardholder_disp_data_len	INT	
pos_acceptor_disp_data_len	INT	
pos_icc_data_len	INT	
pos_track3_rw	STRING	
pos_card_capture_cap	STRING	
pos_pin_len_cap	INT	
reconciliation_date	STRING	
reconciliation_indicator	STRING	
original_amount_data	TUPLE	
original_rec_amount_data	TUPLE	
user_format_id	STRING	
acquirer_number	STRING	
julian_processing_date	STRING	
acquirer_seq_number	STRING	
luhn_check_digit	INT	
acquirer_id	STRING	
forwarding_institution_id	STRING	

account_digital_signature	STRING	
cardholder_certificate_number	STRING	
acceptor_certificate_number	STRING	
xid	STRING	
transstain	STRING	
pos_track2_data	STRING	
pos_track3_data	STRING	
retrieval_reference_number	STRING	
approval_code	STRING	
action_code	STRING	
service_code	STRING	
terminal_id	STRING	
acceptor_id	STRING	
acceptor_name	STRING	
acceptor_street_addr	STRING	
acceptor_city	STRING	
acceptor_region	STRING	
acceptor_zip	STRING	
acceptor_country_code	STRING	
acceptor_phone	STRING	
acceptor_cust_service_phone	STRING	
acceptor_add_contact_info	STRING	
acceptor_url	STRING	
acceptor_email	STRING	
pos_track1_data	STRING	
amount_fees	SET	
card_ver_data	STRING	
cardholder_bill_street_addr	STRING	
cardholder_bill_zip	STRING	
cardholder_compress_bill_addr	STRING	
cardholder_add_id_type	STRING	
cardholder_add_id	STRING	
avs_code	STRING	
cardholder_bill_currency_code	STRING	
tvr	STRING	
transaction_seq_counter	LONG	
transaction_category_code	STRING	
auth_agent_id	STRING	
card_presence_flag	BOOLEAN	
file_transfer_ack_code	STRING	

file_transfer_seq_number	STRING	
return_amount	LONG	
destination_id	STRING	
originator_id	STRING	
receiving_institution_id	STRING	
terminal_timezone	STRING	
cvv_val_code	STRING	
secure_3d_val_code	STRING	
card_id	STRING	
card_bin	STRING	
cardholder_name	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_brand	STRING	
card_issuer	STRING	
card_country_code	STRING	
card_last4	STRING	
card_status	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_available_balance	LONG	
card_number_of_cardholders	INT	
account_iban	STRING	
account_bic	STRING	
account_open_date	LONG	
account_balance	LONG	
account_credit_limit	LONG	
account_otb_limit	LONG	
account_available_credit	LONG	
account_debt_balance	LONG	
account_cash_advance_balance	LONG	
account_status	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_associated_cards	SET	
account_debt_payment_method	STRING	

account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
customer_id	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	
customer_tax_number	STRING	
customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_city	STRING	
customer_region	STRING	
customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
terminal_street_addr	STRING	
terminal_city	STRING	
terminal_region	STRING	
terminal_zip	STRING	
terminal_country_code	STRING	
terminal_latitude	DOUBLE	
terminal_longitude	DOUBLE	
terminal_ip	STRING	
terminal_ip_latitude	DOUBLE	
terminal_ip_longitude	DOUBLE	
terminal_ip_zip	STRING	
terminal_ip_city	STRING	
terminal_ip_region	STRING	
terminal_ip_country_code	STRING	
terminal_poc_flag	BOOLEAN	
acceptor_dba_name	STRING	

acceptor_tax_number	STRING	
sic	STRING	
acceptor_open_date	LONG	
acceptor_risky_flag	BOOLEAN	
acceptor_risk	INT	
acceptor_phone_val_code	STRING	
acceptor_latitude	DOUBLE	
acceptor_longitude	DOUBLE	
event_external_id	STRING	
alert_id	STRING	
device_id	STRING	
event_occurred_at	LONG	
transaction_status	STRING	
transaction_status_reason	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_response_type	STRING	
oob_feedback_notes	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_refunded	STRING	
mc_merchant_fraud_code	STRING	
event_received_at	LONG	
customer_segment	STRING	
customer_number_of_cards	INT	
customer_occupation_status	STRING	
customer_risk_rating	STRING	
customer_currency	STRING	
account_type	STRING	
account_currency	STRING	
feedback_type	STRING	
trusted_acceptor	BOOLEAN	
recurring_payment	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	

device_ip	STRING	
device_ip_country_code	STRING	
login_session_id	STRING	
customer_country_code_change_flag	BOOLEAN	
customer_email_change_flag	BOOLEAN	
customer_phone_change_flag	BOOLEAN	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_language	STRING	
last_sca_at	LONG	
merchant_initiated	BOOLEAN	
cardholder_presence_indicator	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
company_id	STRING	
card_expected_payments_frequency	STRING	
application_transaction_counter	INT	
digital_wallet_tokenized_transaction	BOOLEAN	
digital_wallet_token	STRING	
digital_wallet_token_assurance_level	STRING	
digital_wallet_token_requester	STRING	
digital_wallet_token_type	STRING	
cardholder_authentication_verification_value	STRING	
cvv2_present_indicator	STRING	
cardholder_verification_rule	STRING	
business_app_transfer_purpose	STRING	
secure_3d_indicator	STRING	
secure_3d_protocol_version	STRING	
acceptor_visa_risk	INT	
acceptor_mastercard_risk	INT	
customer_accounts	SET	
customer_aum_2m	BOOLEAN	
customer_aum_5m	BOOLEAN	
info_type	STRING	
feedback_value	BOOLEAN	
device_fingerprint	STRING	
session_id	STRING	
card_first_activation_date	LONG	

card_issued_date	LONG	
account_id	STRING	
event_type	STRING	
lists_state	MAP	
feedback_label	STRING	
segment_channel_type	STRING	
secure_3d_val_status	STRING	
card_format	STRING	
mc_fraud_score	INT	
visa_fraud_score	INT	
cvv2_presence_indicator	STRING	
account_limits	SET	
customer_cards	SET	

Cards Extended API Schema

Field Name	Field Type	Field Description
timestamp	LONG	
fraud	BOOLEAN	
fraud_score	DOUBLE	
decision	STRING	
mti	STRING	
card_pan	STRING	
ttc	STRING	
atc1	STRING	
atc2	STRING	
currency_code	STRING	
currency_unit	INT	
amount	LONG	
cardholder_bill_amount	LONG	
reconciliation_currency_code	STRING	
reconciliation_currency_unit	INT	
reconciliation_amount	LONG	
reconciliation_conv_rate	DOUBLE	
transmission_datetime	STRING	
cardholder_bill_conv_rate	DOUBLE	
stan	STRING	
local_datetime	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	

trx_to_rec_conv_date	STRING	
capture_date	STRING	
lifecycle_indicator	STRING	
lifecycle_id	STRING	
lifecycle_seq_number	STRING	
lifecycle_token	STRING	
pos_card_read_method	STRING	
pos_cardholder_ver_method	STRING	
pos_environment	STRING	
pos_security_characteristics	STRING	
card_seq_number	STRING	
function_code	STRING	
reason_code	STRING	
mcc	STRING	
pos_read_cap	STRING	
pos_cardholder_ver_cap	STRING	
pos_approval_code_len	INT	
pos_cardholder_rec_data_len	INT	
pos_acceptor_rec_data_len	INT	
pos_cardholder_disp_data_len	INT	
pos_acceptor_disp_data_len	INT	
pos_icc_data_len	INT	
pos_track3_rw	STRING	
pos_card_capture_cap	STRING	
pos_pin_len_cap	INT	
reconciliation_date	STRING	
reconciliation_indicator	STRING	
original_amount_data	TUPLE	
original_rec_amount_data	TUPLE	
user_format_id	STRING	
acquirer_number	STRING	
julian_processing_date	STRING	
acquirer_seq_number	STRING	
luhn_check_digit	INT	
acquirer_id	STRING	
forwarding_institution_id	STRING	
account_digital_signature	STRING	
cardholder_certificate_number	STRING	
acceptor_certificate_number	STRING	
xid	STRING	

transstain	STRING	
pos_track2_data	STRING	
pos_track3_data	STRING	
retrieval_reference_number	STRING	
approval_code	STRING	
action_code	STRING	
service_code	STRING	
terminal_id	STRING	
acceptor_id	STRING	
acceptor_name	STRING	
acceptor_street_addr	STRING	
acceptor_city	STRING	
acceptor_region	STRING	
acceptor_zip	STRING	
acceptor_country_code	STRING	
acceptor_phone	STRING	
acceptor_cust_service_phone	STRING	
acceptor_add_contact_info	STRING	
acceptor_url	STRING	
acceptor_email	STRING	
pos_track1_data	STRING	
amount_fees	SET	
card_ver_data	STRING	
cardholder_bill_street_addr	STRING	
cardholder_bill_zip	STRING	
cardholder_compress_bill_addr	STRING	
cardholder_add_id_type	STRING	
cardholder_add_id	STRING	
avs_code	STRING	
cardholder_bill_currency_code	STRING	
tvr	STRING	
transaction_seq_counter	LONG	
transaction_category_code	STRING	
auth_agent_id	STRING	
card_presence_flag	BOOLEAN	
file_transfer_ack_code	STRING	
file_transfer_seq_number	STRING	
return_amount	LONG	
destination_id	STRING	
originator_id	STRING	

receiving_institution_id	STRING	
terminal_timezone	STRING	
cvv_val_code	STRING	
secure_3d_val_code	STRING	
card_id	STRING	
card_bin	STRING	
cardholder_name	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_brand	STRING	
card_issuer	STRING	
card_country_code	STRING	
card_last4	STRING	
card_status	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_available_balance	LONG	
card_number_of_cardholders	INT	
account_iban	STRING	
account_bic	STRING	
account_open_date	LONG	
account_balance	LONG	
account_credit_limit	LONG	
account_otb_limit	LONG	
account_available_credit	LONG	
account_debt_balance	LONG	
account_cash_advance_balance	LONG	
account_status	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_associated_cards	SET	
account_debt_payment_method	STRING	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
customer_id	STRING	
customer_name	STRING	

customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	
customer_tax_number	STRING	
customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_city	STRING	
customer_region	STRING	
customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
terminal_street_addr	STRING	
terminal_city	STRING	
terminal_region	STRING	
terminal_zip	STRING	
terminal_country_code	STRING	
terminal_latitude	DOUBLE	
terminal_longitude	DOUBLE	
terminal_ip	STRING	
terminal_ip_latitude	DOUBLE	
terminal_ip_longitude	DOUBLE	
terminal_ip_zip	STRING	
terminal_ip_city	STRING	
terminal_ip_region	STRING	
terminal_ip_country_code	STRING	
terminal_poc_flag	BOOLEAN	
acceptor_dba_name	STRING	
acceptor_tax_number	STRING	
sic	STRING	
acceptor_open_date	LONG	
acceptor_risky_flag	BOOLEAN	

acceptor_risk	INT	
acceptor_phone_val_code	STRING	
acceptor_latitude	DOUBLE	
acceptor_longitude	DOUBLE	
event_external_id	STRING	
alert_id	STRING	
device_id	STRING	
event_occurred_at	LONG	
event_type	STRING	
transaction_status	STRING	
transaction_status_reason	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_response_type	STRING	
oob_feedback_notes	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_refunded	STRING	
mc_merchant_fraud_code	STRING	
event_received_at	LONG	
customer_segment	STRING	
customer_number_of_cards	INT	
customer_occupation_status	STRING	
customer_risk_rating	STRING	
customer_currency	STRING	
account_type	STRING	
account_currency	STRING	
feedback_label	STRING	
feedback_type	STRING	
sca_required	BOOLEAN	
trusted_acceptor	BOOLEAN	
first_recurring_payment	BOOLEAN	
recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
login_session_id	STRING	

session_id	STRING	
customer_country_code_change_flag	BOOLEAN	
customer_email_change_flag	BOOLEAN	
customer_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip	STRING	
last_sca_at	LONG	
merchant_initiated	BOOLEAN	
account_id	STRING	
card_first_activation_date	LONG	
card_issued_date	LONG	
device_fingerprint	STRING	
company_id	STRING	
feedback_value	BOOLEAN	
info_type	STRING	
cardholder_presence_indicator	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
customer_accounts	SET	
secure_3d_val_status	STRING	
card_expected_payments_frequency	STRING	
application_transaction_counter	INT	
digital_wallet_tokenized_transaction	BOOLEAN	
digital_wallet_token	STRING	
digital_wallet_token_assurance_level	STRING	
digital_wallet_token_requester	STRING	
digital_wallet_token_type	STRING	
cardholder_authentication_verification_value	STRING	
cvv2_present_indicator	STRING	
cardholder_verification_rule	STRING	
business_app_transfer_purpose	STRING	
secure_3d_indicator	STRING	
secure_3d_protocol_version	STRING	
acceptor_visa_risk	INT	
acceptor_mastercard_risk	INT	

card_format	STRING	
mc_fraud_score	INT	
visa_fraud_score	INT	
cvv2_presence_indicator	STRING	
account_limits	SET	
customer_cards	SET	

Cards FraudForce Schema v4

Field Name	Field Type	Field Description
timestamp	LONG	
fraud	BOOLEAN	
fraud_score	DOUBLE	
decision	STRING	
mti	STRING	
card_pan	STRING	
ttc	STRING	
atc1	STRING	
atc2	STRING	
currency_code	STRING	
currency_unit	INT	
amount	LONG	
cardholder_bill_amount	LONG	
reconciliation_currency_code	STRING	
reconciliation_currency_unit	INT	
reconciliation_amount	LONG	
reconciliation_conv_rate	DOUBLE	
transmission_datetime	STRING	
cardholder_bill_conv_rate	DOUBLE	
stan	STRING	
local_datetime	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
trx_to_rec_conv_date	STRING	
capture_date	STRING	
lifecycle_indicator	STRING	
lifecycle_id	STRING	
lifecycle_seq_number	STRING	
lifecycle_token	STRING	
pos_card_read_method	STRING	

pos_cardholder_ver_method	STRING	
pos_environment	STRING	
pos_security_characteristics	STRING	
card_seq_number	STRING	
function_code	STRING	
reason_code	STRING	
mcc	STRING	
pos_read_cap	STRING	
pos_cardholder_ver_cap	STRING	
pos_approval_code_len	INT	
pos_cardholder_rec_data_len	INT	
pos_acceptor_rec_data_len	INT	
pos_cardholder_disp_data_len	INT	
pos_acceptor_disp_data_len	INT	
pos_icc_data_len	INT	
pos_track3_rw	STRING	
pos_card_capture_cap	STRING	
pos_pin_len_cap	INT	
reconciliation_date	STRING	
reconciliation_indicator	STRING	
original_amount_data	TUPLE	
original_rec_amount_data	TUPLE	
user_format_id	STRING	
acquirer_number	STRING	
julian_processing_date	STRING	
acquirer_seq_number	STRING	
luhn_check_digit	INT	
acquirer_id	STRING	
forwarding_institution_id	STRING	
account_digital_signature	STRING	
cardholder_certificate_number	STRING	
acceptor_certificate_number	STRING	
xid	STRING	
transstain	STRING	
pos_track2_data	STRING	
pos_track3_data	STRING	
retrieval_reference_number	STRING	
approval_code	STRING	
action_code	STRING	
service_code	STRING	

terminal_id	STRING	
acceptor_id	STRING	
acceptor_name	STRING	
acceptor_street_addr	STRING	
acceptor_city	STRING	
acceptor_region	STRING	
acceptor_zip	STRING	
acceptor_country_code	STRING	
acceptor_phone	STRING	
acceptor_cust_service_phone	STRING	
acceptor_add_contact_info	STRING	
acceptor_url	STRING	
acceptor_email	STRING	
pos_track1_data	STRING	
amount_fees	SET	
card_ver_data	STRING	
cardholder_bill_street_addr	STRING	
cardholder_bill_zip	STRING	
cardholder_compress_bill_addr	STRING	
cardholder_add_id_type	STRING	
cardholder_add_id	STRING	
avs_code	STRING	
cardholder_bill_currency_code	STRING	
tvr	STRING	
transaction_seq_counter	LONG	
transaction_category_code	STRING	
auth_agent_id	STRING	
card_presence_flag	BOOLEAN	
file_transfer_ack_code	STRING	
file_transfer_seq_number	STRING	
return_amount	LONG	
destination_id	STRING	
originator_id	STRING	
receiving_institution_id	STRING	
terminal_timezone	STRING	
cvv_val_code	STRING	
secure_3d_val_code	STRING	
card_id	STRING	
card_bin	STRING	
cardholder_name	STRING	

card_type	STRING	
card_category	STRING	
card_program	STRING	
card_brand	STRING	
card_issuer	STRING	
card_country_code	STRING	
card_last4	STRING	
card_status	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_available_balance	LONG	
card_number_of_cardholders	INT	
account_iban	STRING	
account_bic	STRING	
account_open_date	LONG	
account_balance	LONG	
account_credit_limit	LONG	
account_otb_limit	LONG	
account_available_credit	LONG	
account_debt_balance	LONG	
account_cash_advance_balance	LONG	
account_status	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_associated_cards	SET	
account_debt_payment_method	STRING	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
customer_id	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	

customer_tax_number	STRING	
customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_city	STRING	
customer_region	STRING	
customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
terminal_street_addr	STRING	
terminal_city	STRING	
terminal_region	STRING	
terminal_zip	STRING	
terminal_country_code	STRING	
terminal_latitude	DOUBLE	
terminal_longitude	DOUBLE	
terminal_ip	STRING	
terminal_ip_latitude	DOUBLE	
terminal_ip_longitude	DOUBLE	
terminal_ip_zip	STRING	
terminal_ip_city	STRING	
terminal_ip_region	STRING	
terminal_ip_country_code	STRING	
terminal_poc_flag	BOOLEAN	
acceptor_dba_name	STRING	
acceptor_tax_number	STRING	
sic	STRING	
acceptor_open_date	LONG	
acceptor_risky_flag	BOOLEAN	
acceptor_risk	INT	
acceptor_phone_val_code	STRING	
acceptor_latitude	DOUBLE	
acceptor_longitude	DOUBLE	
event_external_id	STRING	
alert_id	STRING	
device_id	STRING	

event_occurred_at	LONG	
event_type	STRING	
transaction_status	STRING	
transaction_status_reason	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_response_type	STRING	
oob_feedback_notes	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_refunded	STRING	
mc_merchant_fraud_code	STRING	
event_received_at	LONG	
customer_segment	STRING	
customer_number_of_cards	INT	
customer_occupation_status	STRING	
customer_risk_rating	STRING	
customer_currency	STRING	
account_type	STRING	
account_currency	STRING	
feedback_label	STRING	
feedback_type	STRING	
sca_required	BOOLEAN	
trusted_acceptor	BOOLEAN	
first_recurring_payment	BOOLEAN	
recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
login_session_id	STRING	
session_id	STRING	
customer_country_code_change_flag	BOOLEAN	
customer_email_change_flag	BOOLEAN	
customer_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	

proxy_flag	BOOLEAN	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip	STRING	
last_sca_at	LONG	
merchant_initiated	BOOLEAN	
account_id	STRING	
ff_id	STRING	
ff_result	STRING	
ff_tracking_number	LONG	
ff_device_alias	LONG	
ff_device_blackbox_age	INT	
ff_device_blackbox_timestamp	STRING	
ff_device_browser_charset	STRING	
ff_device_browser_cookies_enabled	BOOLEAN	
ff_device_browser_configured_language	STRING	
ff_device_browser_flash_enabled	BOOLEAN	
ff_device_browser_flash_installed	BOOLEAN	
ff_device_browser_flash_storage_enabled	BOOLEAN	
ff_device_browser_flash_version	STRING	
ff_device_browser_language	STRING	
ff_device_browser_type	STRING	
ff_device_browser_timezone	STRING	
ff_device_browser_version	STRING	
ff_device_first_seen	STRING	
ff_device_is_new	BOOLEAN	
ff_device_mobile_app_bundle_id	STRING	
ff_device_mobile_app_debug	BOOLEAN	
ff_device_mobile_app_exe_name	STRING	
ff_device_mobile_app_market_id	STRING	
ff_device_mobile_app_name	STRING	
ff_device_mobile_app_orientation	STRING	
ff_device_mobile_app_proc_name	STRING	
ff_device_mobile_app_signer_id	STRING	
ff_device_mobile_app_vendor_id	STRING	
ff_device_mobile_app_version	STRING	
ff_device_mobile_brand	STRING	
ff_device_mobile_build_device	STRING	
ff_device_mobile_build_product	STRING	
ff_device_mobile_charging	BOOLEAN	

ff_device_mobile_imei	STRING	
ff_device_mobile_location_altitude	DOUBLE	
ff_device_mobile_location_enabled	BOOLEAN	
ff_device_mobile_location_latitude	DOUBLE	
ff_device_mobile_location_longitude	DOUBLE	
ff_device_mobile_location_timestamp	STRING	
ff_device_mobile_location_timezone	STRING	
ff_device_mobile_manufacturer	STRING	
ff_device_mobile_model	STRING	
ff_device_mobile_orientation	STRING	
ff_device_mobile_screen_resolution	STRING	
ff_device_mobile_system_allow_unknown_store	BOOLEAN	
ff_device_mobile_system_android_sdk_level	STRING	
ff_device_mobile_system_build_fingerprint	STRING	
ff_device_mobile_system_carrier	STRING	
ff_device_mobile_system_carrier_country_code	STRING	
ff_device_mobile_system_cellular_network	STRING	
ff_device_mobile_system_hostname	STRING	
ff_device_mobile_system_jailroot_detected	BOOLEAN	
ff_device_mobile_system_locale_currency	STRING	
ff_device_mobile_system_locale_lang	STRING	
ff_device_mobile_system_network_operator_country	STRING	
ff_device_mobile_system_network_operator_name	STRING	
ff_device_mobile_system_os_version	STRING	
ff_device_mobile_system_root_detection_disabled	BOOLEAN	
ff_device_mobile_system_simulator	BOOLEAN	
ff_device_mobile_system_uptime	STRING	
ff_device_mobile_system_voip_allowed	BOOLEAN	
ff_device_os	STRING	
ff_device_registration_result_match_status	STRING	
ff_device_registration_result_measure_of_charge	STRING	
ff_device_screen	STRING	
ff_device_recognition_source	STRING	
ff_device_type	STRING	
ff_real_ip_address	STRING	
ff_real_ip_botnet_score	INT	
ff_real_ip_botnet_intensity	INT	
ff_real_ip_botnet_lastseen	STRING	
ff_real_ip_isp	STRING	
ff_real_ip_loc_city	STRING	

ff_real_ip_loc_country	STRING	
ff_real_ip_loc_country_code	STRING	
ff_real_ip_loc_latitude	DOUBLE	
ff_real_ip_loc_longitude	DOUBLE	
ff_real_ip_loc_region	STRING	
ff_real_ip_org	STRING	
ff_real_ip_proxy	STRING	
ff_real_ip_source	STRING	
ff_stated_ip_address	STRING	
ff_stated_ip_botnet_score	INT	
ff_stated_ip_botnet_intensity	INT	
ff_stated_ip_botnet_lastseen	STRING	
ff_stated_ip_isp	STRING	
ff_stated_ip_loc_city	STRING	
ff_stated_ip_loc_country	STRING	
ff_stated_ip_loc_country_code	STRING	
ff_stated_ip_loc_latitude	DOUBLE	
ff_stated_ip_loc_longitude	DOUBLE	
ff_stated_ip_loc_region	STRING	
ff_stated_ip_org	STRING	
ff_stated_ip_proxy	STRING	
ff_stated_ip_source	STRING	
company_id	STRING	
feedback_value	BOOLEAN	
info_type	STRING	
cardholder_presence_indicator	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
customer_accounts	SET	
secure_3d_val_status	STRING	
card_expected_payments_frequency	STRING	
application_transaction_counter	INT	
digital_wallet_tokenized_transaction	BOOLEAN	
digital_wallet_token	STRING	
digital_wallet_token_assurance_level	STRING	
digital_wallet_token_requester	STRING	
digital_wallet_token_type	STRING	
cardholder_authentication_verification_value	STRING	
cvv2_present_indicator	STRING	
cardholder_verification_rule	STRING	

business_app_transfer_purpose	STRING	
secure_3d_indicator	STRING	
secure_3d_protocol_version	STRING	
acceptor_visa_risk	INT	
acceptor_mastercard_risk	INT	
card_format	STRING	
mc_fraud_score	INT	
visa_fraud_score	INT	
cvv2_presence_indicator	STRING	
account_limits	SET	
customer_cards	SET	

Cards Plan Schema

Field Name	Field Type	Field Description
mti	STRING	
card_pan	STRING	
ttc	STRING	
atc1	STRING	
atc2	STRING	
currency_code	STRING	
currency_unit	INT	
amount	LONG	
cardholder_bill_amount	LONG	
reconciliation_currency_code	STRING	
reconciliation_currency_unit	INT	
reconciliation_amount	LONG	
reconciliation_conv_rate	DOUBLE	
transmission_datetime	STRING	
cardholder_bill_conv_rate	DOUBLE	
stan	STRING	
local_datetime	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
trx_to_rec_conv_date	STRING	
capture_date	STRING	
lifecycle_indicator	STRING	
lifecycle_id	STRING	
lifecycle_seq_number	STRING	
lifecycle_token	STRING	

pos_card_read_method	STRING	
pos_cardholder_ver_method	STRING	
pos_environment	STRING	
pos_security_characteristics	STRING	
card_seq_number	STRING	
function_code	STRING	
reason_code	STRING	
mcc	STRING	
pos_read_cap	STRING	
pos_cardholder_ver_cap	STRING	
pos_approval_code_len	INT	
pos_cardholder_rec_data_len	INT	
pos_acceptor_rec_data_len	INT	
pos_cardholder_disp_data_len	INT	
pos_acceptor_disp_data_len	INT	
pos_icc_data_len	INT	
pos_track3_rw	STRING	
pos_card_capture_cap	STRING	
pos_pin_len_cap	INT	
reconciliation_date	STRING	
reconciliation_indicator	STRING	
original_amount_data	TUPLE	
original_rec_amount_data	TUPLE	
user_format_id	STRING	
acquirer_number	STRING	
julian_processing_date	STRING	
acquirer_seq_number	STRING	
luhn_check_digit	INT	
acquirer_id	STRING	
forwarding_institution_id	STRING	
account_digital_signature	STRING	
xid	STRING	
transstain	STRING	
pos_track2_data	STRING	
pos_track3_data	STRING	
retrieval_reference_number	STRING	
approval_code	STRING	
action_code	STRING	
service_code	STRING	
terminal_id	STRING	

acceptor_id	STRING	
acceptor_name	STRING	
acceptor_street_addr	STRING	
acceptor_city	STRING	
acceptor_region	STRING	
acceptor_zip	STRING	
acceptor_country_code	STRING	
acceptor_phone	STRING	
acceptor_cust_service_phone	STRING	
acceptor_add_contact_info	STRING	
acceptor_url	STRING	
acceptor_email	STRING	
pos_track1_data	STRING	
amount_fees	SET	
card_ver_data	STRING	
cardholder_bill_street_addr	STRING	
cardholder_bill_zip	STRING	
cardholder_compress_bill_addr	STRING	
cardholder_add_id_type	STRING	
cardholder_add_id	STRING	
avs_code	STRING	
cardholder_bill_currency_code	STRING	
tvr	STRING	
transaction_seq_counter	LONG	
transaction_category_code	STRING	
auth_agent_id	STRING	
cardholder_presence_indicator	STRING	
card_presence_flag	BOOLEAN	
file_transfer_ack_code	STRING	
file_transfer_seq_number	STRING	
return_amount	LONG	
destination_id	STRING	
originator_id	STRING	
receiving_institution_id	STRING	
terminal_timezone	STRING	
cvv_val_code	STRING	
secure_3d_val_code	STRING	
card_id	STRING	
card_bin	STRING	
cardholder_name	STRING	

card_type	STRING	
card_category	STRING	
card_program	STRING	
card_brand	STRING	
card_issuer	STRING	
card_country_code	STRING	
card_last4	STRING	
card_status	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_available_balance	LONG	
card_number_of_cardholders	INT	
account_iban	STRING	
account_bic	STRING	
account_open_date	LONG	
account_balance	LONG	
account_credit_limit	LONG	
account_otb_limit	LONG	
account_available_credit	LONG	
account_cash_advance_balance	LONG	
account_status	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_associated_cards	SET	
account_debt_payment_method	STRING	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
customer_id	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	
customer_tax_number	STRING	

customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_city	STRING	
customer_region	STRING	
customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
terminal_street_addr	STRING	
terminal_city	STRING	
terminal_region	STRING	
terminal_zip	STRING	
terminal_country_code	STRING	
terminal_latitude	DOUBLE	
terminal_longitude	DOUBLE	
terminal_ip	STRING	
terminal_ip_latitude	DOUBLE	
terminal_ip_longitude	DOUBLE	
terminal_ip_zip	STRING	
terminal_ip_city	STRING	
terminal_ip_region	STRING	
terminal_ip_country_code	STRING	
terminal_poc_flag	BOOLEAN	
acceptor_dba_name	STRING	
acceptor_tax_number	STRING	
sic	STRING	
acceptor_open_date	LONG	
acceptor_risky_flag	BOOLEAN	
acceptor_risk	INT	
acceptor_visa_risk	INT	
acceptor_mastercard_risk	INT	
acceptor_phone_val_code	STRING	
acceptor_latitude	DOUBLE	
acceptor_longitude	DOUBLE	
event_external_id	STRING	
alert_id	STRING	

device_id	STRING	
event_occurred_at	LONG	
transaction_status	STRING	
transaction_status_reason	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_response_type	STRING	
oob_feedback_notes	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_refunded	STRING	
dispute_feedback_notes	STRING	
mc_merchant_fraud_code	STRING	
event_received_at	LONG	
account_type	STRING	
account_currency	STRING	
customer_number_of_cards	INT	
customer_currency	STRING	
customer_segment	STRING	
customer_risk_rating	STRING	
customer_aum_2m	BOOLEAN	
customer_aum_5m	BOOLEAN	
customer_occupation_status	STRING	
trusted_acceptor	BOOLEAN	
recurring_payment	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
device_ip	STRING	
device_ip_country_code	STRING	
customer_country_code_change_flag	BOOLEAN	
customer_email_change_flag	BOOLEAN	
customer_phone_change_flag	BOOLEAN	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	

malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
last_sca_at	LONG	
login_session_id	STRING	
merchant_initiated	BOOLEAN	
feedback_type	STRING	
feedback_value	BOOLEAN	
info_type	STRING	
device_language	STRING	
cardholder_certificate_number	STRING	
acceptor_certificate_number	STRING	
account_debt_balance	LONG	
customer_accounts	SET	
card_first_activation_date	LONG	
card_issued_date	LONG	
account_id	STRING	
event_type	STRING	
lists_state	MAP	
feedback_label	STRING	
segment_channel_type	STRING	
secure_3d_val_status	STRING	
customer_cards	SET	

Cards PSD2 Schema

Field Name	Field Type	Field Description
timestamp	LONG	
fraud	BOOLEAN	
fraud_score	DOUBLE	
decision	STRING	
mti	STRING	
card_pan	STRING	
ttc	STRING	
atc1	STRING	
atc2	STRING	
currency_code	STRING	
currency_unit	INT	
amount	LONG	
cardholder_bill_amount	LONG	
reconciliation_currency_code	STRING	

reconciliation_currency_unit	INT	
reconciliation_amount	LONG	
reconciliation_conv_rate	DOUBLE	
transmission_datetime	STRING	
cardholder_bill_conv_rate	DOUBLE	
stan	STRING	
local_datetime	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
trx_to_rec_conv_date	STRING	
capture_date	STRING	
lifecycle_indicator	STRING	
lifecycle_id	STRING	
lifecycle_seq_number	STRING	
lifecycle_token	STRING	
pos_card_read_method	STRING	
pos_cardholder_ver_method	STRING	
pos_environment	STRING	
pos_security_characteristics	STRING	
card_seq_number	STRING	
function_code	STRING	
reason_code	STRING	
mcc	STRING	
pos_read_cap	STRING	
pos_cardholder_ver_cap	STRING	
pos_approval_code_len	INT	
pos_cardholder_rec_data_len	INT	
pos_acceptor_rec_data_len	INT	
pos_cardholder_disp_data_len	INT	
pos_acceptor_disp_data_len	INT	
pos_icc_data_len	INT	
pos_track3_rw	STRING	
pos_card_capture_cap	STRING	
pos_pin_len_cap	INT	
reconciliation_date	STRING	
reconciliation_indicator	STRING	
original_amount_data	TUPLE	
original_rec_amount_data	TUPLE	
user_format_id	STRING	
acquirer_number	STRING	

julian_processing_date	STRING	
acquirer_seq_number	STRING	
luhn_check_digit	INT	
acquirer_id	STRING	
forwarding_institution_id	STRING	
account_digital_signature	STRING	
cardholder_certificate_number	STRING	
acceptor_certificate_number	STRING	
xid	STRING	
transstain	STRING	
pos_track2_data	STRING	
pos_track3_data	STRING	
retrieval_reference_number	STRING	
approval_code	STRING	
action_code	STRING	
service_code	STRING	
terminal_id	STRING	
acceptor_id	STRING	
acceptor_name	STRING	
acceptor_street_addr	STRING	
acceptor_city	STRING	
acceptor_region	STRING	
acceptor_zip	STRING	
acceptor_country_code	STRING	
acceptor_phone	STRING	
acceptor_cust_service_phone	STRING	
acceptor_add_contact_info	STRING	
acceptor_url	STRING	
acceptor_email	STRING	
pos_track1_data	STRING	
amount_fees	SET	
card_ver_data	STRING	
cardholder_bill_street_addr	STRING	
cardholder_bill_zip	STRING	
cardholder_compress_bill_addr	STRING	
cardholder_add_id_type	STRING	
cardholder_add_id	STRING	
avs_code	STRING	
cardholder_bill_currency_code	STRING	
tvr	STRING	

transaction_seq_counter	LONG	
transaction_category_code	STRING	
auth_agent_id	STRING	
card_presence_flag	BOOLEAN	
file_transfer_ack_code	STRING	
file_transfer_seq_number	STRING	
return_amount	LONG	
destination_id	STRING	
originator_id	STRING	
receiving_institution_id	STRING	
terminal_timezone	STRING	
cvv_val_code	STRING	
secure_3d_val_code	STRING	
card_id	STRING	
card_bin	STRING	
cardholder_name	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_brand	STRING	
card_issuer	STRING	
card_country_code	STRING	
card_last4	STRING	
card_status	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_available_balance	LONG	
card_number_of_cardholders	INT	
account_iban	STRING	
account_bic	STRING	
account_open_date	LONG	
account_balance	LONG	
account_credit_limit	LONG	
account_otb_limit	LONG	
account_available_credit	LONG	
account_debt_balance	LONG	
account_cash_advance_balance	LONG	
account_status	STRING	
account_last_activity	LONG	

account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_associated_cards	SET	
account_debt_payment_method	STRING	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
customer_id	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	
customer_tax_number	STRING	
customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_city	STRING	
customer_region	STRING	
customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
terminal_street_addr	STRING	
terminal_city	STRING	
terminal_region	STRING	
terminal_zip	STRING	
terminal_country_code	STRING	
terminal_ip	STRING	
terminal_ip_latitude	DOUBLE	
terminal_ip_longitude	DOUBLE	
terminal_ip_zip	STRING	
terminal_ip_city	STRING	
terminal_ip_region	STRING	

terminal_ip_country_code	STRING	
terminal_poc_flag	BOOLEAN	
acceptor_dba_name	STRING	
acceptor_tax_number	STRING	
sic	STRING	
acceptor_open_date	LONG	
acceptor_risky_flag	BOOLEAN	
acceptor_risk	INT	
acceptor_phone_val_code	STRING	
acceptor_latitude	DOUBLE	
acceptor_longitude	DOUBLE	
event_external_id	STRING	
alert_id	STRING	
device_id	STRING	
event_occurred_at	LONG	
event_type	STRING	
transaction_status	STRING	
transaction_status_reason	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_response_type	STRING	
oob_feedback_notes	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_refunded	STRING	
mc_merchant_fraud_code	STRING	
event_received_at	LONG	
customer_segment	STRING	
customer_number_of_cards	INT	
customer_occupation_status	STRING	
customer_risk_rating	STRING	
customer_currency	STRING	
account_type	STRING	
account_currency	STRING	
feedback_label	STRING	
feedback_type	STRING	

trusted_acceptor	BOOLEAN	
first_recurring_payment	BOOLEAN	
recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
login_session_id	STRING	
session_id	STRING	
customer_country_code_change_flag	BOOLEAN	
customer_email_change_flag	BOOLEAN	
customer_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip	STRING	
last_sca_at	LONG	
merchant_initiated	BOOLEAN	
account_id	STRING	
card_first_activation_date	LONG	
card_issued_date	LONG	
posneg_card_country_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
sign_cardholder_bill_amount	LONG	
sign_count	LONG	
event_occurred_at_as_long	LONG	
is_high_amount	BOOLEAN	
card_days_to_expire	LONG	
terminal_latitude	DOUBLE	
terminal_longitude	DOUBLE	
posneg_acceptor_value	STRING	
posneg_card_value	STRING	
posneg_country_value	STRING	
posneg_customer_value	STRING	
posneg_mcc_value	STRING	
posneg_terminal_value	STRING	
posneg_zip_value	STRING	
sca_required	BOOLEAN	

acceptorid_cardbin_cntDist_cardExpDate_10m	LONG	
acceptorid_cardbin_cntDist_cardid_10m	LONG	
APPR_AUTH_acceptorId_count_6M	LONG	
FRAUD_acceptorId_count_6M	LONG	
APPR_AUTH_cardId_acceptor_billAmount_count_1d	LONG	
AUTH_cardId_acceptorId_count_6M	LONG	
APPR_AUTH_cardId_acceptor_sum_billAmount_1d	LONG	
APPR_AUTH_cardId_acceptor_count_1d	LONG	
AUTH_cardid_deviceid_count_6M	LONG	
AUTH_cardid_device_ip_country_code_count_6M	LONG	
APPR_AUTH_last_sca_at_cardid_sum_billAmount_90d	LONG	
APPR_AUTH_last_sca_at_cardid_count_90d	LONG	
APPR_AUTH_contactless_last_sca_at_cardid_sum_billAmount_90d	LONG	
APPR_AUTH_contactless_last_sca_at_cardid_count_90d	LONG	
APPR_AUTH_last_sca_at_cardid_sum_billAmount_30h	LONG	
APPR_AUTH_last_sca_at_cardid_count_30h	LONG	
APPR_AUTH_contactless_last_sca_at_cardid_sum_billAmount_30h	LONG	
APPR_AUTH_contactless_last_sca_at_cardid_count_30h	LONG	
AUTH_cardid_mcc_sum_amount_1M	LONG	
AUTH_cardid_mcc_count_1M	LONG	
AUTH_cardid_mcc_count_6M	LONG	
AUTH_cardId_count_4w	LONG	
APPR_AUTH_cardId_sum_billAmount_4w	LONG	
AUTH_cardId_count_1d	LONG	
HIGH_AMOUNT_APPR_AUTH_cardId_count_1d	LONG	
DECLINE_cardId_count_1d	LONG	
AUTH_CP_cardId_prev_ts_1h	LONG	
AUTH_CP_cardId_prev_term_lat_1h	DOUBLE	
AUTH_CP_cardId_prev_term_long_1h	DOUBLE	
AUTH_CNP_cardid_prev_device_ip_lat_1h	DOUBLE	
AUTH_CNP_cardid_prev_device_ip_long_1h	DOUBLE	
APPR_AUTH_cardId_sum_billAmount_1h	LONG	
APPR_AUTH_cardId_count_1h	LONG	
AUTH_CNP_cardid_count_pw_changed_24h	LONG	
AUTH_CNP_auth_type_changed_24h	INT	
AUTH_CNP_login_session_id_changed_24h	INT	
AUTH_CNP_currency_changed_24h	INT	
AUTH_CNP_cardid_sum_amount_24h	LONG	
AUTH_CNP_distinct_device_id_per_card_24h	LONG	
AUTH_device_language_changed_24h	INT	

REVIEW_APPR_AUTH_cardId_sum_billAmount_1d	LONG	
REVIEW_APPR_AUTH_cardId_count_1d	LONG	
APPR_AUTH_cardId_count_3m	LONG	
AUTH_CNP_cardId_countDist_acceptor_5m	LONG	
mcc_cardbin_cntDist_cardExpDate_10m	LONG	
mcc_cardbin_cntDist_cardid_10m	LONG	
CP_card_prev_term_dist_1h	DOUBLE	
FRAUD_ratio_acceptorId_6M	DOUBLE	
card_id_not_blacklisted_nor_whitelisted	BOOLEAN	
AUTH_cardid_mcc_avg_amount_1M	DOUBLE	
acceptorid_cardbin_cur0	BOOLEAN	
acceptorid_cardbin_cur1	BOOLEAN	
acceptorid_cardbin_cur3	BOOLEAN	
acceptorid_cardbin_cur5	BOOLEAN	
mcc_cardbin_cur0	BOOLEAN	
mcc_cardbin_cur1	BOOLEAN	
mcc_cardbin_cur3	BOOLEAN	
mcc_cardbin_cur5	BOOLEAN	
CP_card_term_vel_1h	DOUBLE	
device_fingerprint	STRING	
company_id	STRING	
feedback_value	BOOLEAN	
info_type	STRING	
cardholder_presence_indicator	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
customer_accounts	SET	
secure_3d_val_status	STRING	
card_expected_payments_frequency	STRING	
application_transaction_counter	INT	
digital_wallet_tokenized_transaction	BOOLEAN	
digital_wallet_token	STRING	
digital_wallet_token_assurance_level	STRING	
digital_wallet_token_requester	STRING	
digital_wallet_token_type	STRING	
cardholder_authentication_verification_value	STRING	
cvv2_present_indicator	STRING	
cardholder_verification_rule	STRING	
business_app_transfer_purpose	STRING	
secure_3d_indicator	STRING	

secure_3d_protocol_version	STRING	
acceptor_visa_risk	INT	
acceptor_mastercard_risk	INT	
card_format	STRING	
mc_fraud_score	INT	
visa_fraud_score	INT	
cvv2_presence_indicator	STRING	
account_limits	SET	
customer_cards	SET	

Cards Workflow Schema

Field Name	Field Type	Field Description
timestamp	LONG	
fraud	BOOLEAN	
fraud_score	DOUBLE	
decision	STRING	
mti	STRING	
card_pan	STRING	
ttc	STRING	
atc1	STRING	
atc2	STRING	
currency_code	STRING	
currency_unit	INT	
amount	LONG	
cardholder_bill_amount	LONG	
reconciliation_currency_code	STRING	
reconciliation_currency_unit	INT	
reconciliation_amount	LONG	
reconciliation_conv_rate	DOUBLE	
transmission_datetime	STRING	
cardholder_bill_conv_rate	DOUBLE	
stan	STRING	
local_datetime	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
trx_to_rec_conv_date	STRING	
capture_date	STRING	
lifecycle_indicator	STRING	
lifecycle_id	STRING	

lifecycle_seq_number	STRING	
lifecycle_token	STRING	
pos_card_read_method	STRING	
pos_cardholder_ver_method	STRING	
pos_environment	STRING	
pos_security_characteristics	STRING	
card_seq_number	STRING	
function_code	STRING	
reason_code	STRING	
mcc	STRING	
pos_read_cap	STRING	
pos_cardholder_ver_cap	STRING	
pos_approval_code_len	INT	
pos_cardholder_rec_data_len	INT	
pos_acceptor_rec_data_len	INT	
pos_cardholder_disp_data_len	INT	
pos_acceptor_disp_data_len	INT	
pos_icc_data_len	INT	
pos_track3_rw	STRING	
pos_card_capture_cap	STRING	
pos_pin_len_cap	INT	
reconciliation_date	STRING	
reconciliation_indicator	STRING	
original_amount_data	TUPLE	
original_rec_amount_data	TUPLE	
user_format_id	STRING	
acquirer_number	STRING	
julian_processing_date	STRING	
acquirer_seq_number	STRING	
luhn_check_digit	INT	
acquirer_id	STRING	
forwarding_institution_id	STRING	
account_digital_signature	STRING	
cardholder_certificate_number	STRING	
acceptor_certificate_number	STRING	
xid	STRING	
transstain	STRING	
pos_track2_data	STRING	
pos_track3_data	STRING	
retrieval_reference_number	STRING	

approval_code	STRING	
action_code	STRING	
service_code	STRING	
terminal_id	STRING	
acceptor_id	STRING	
acceptor_name	STRING	
acceptor_street_addr	STRING	
acceptor_city	STRING	
acceptor_region	STRING	
acceptor_zip	STRING	
acceptor_country_code	STRING	
acceptor_phone	STRING	
acceptor_cust_service_phone	STRING	
acceptor_add_contact_info	STRING	
acceptor_url	STRING	
acceptor_email	STRING	
pos_track1_data	STRING	
amount_fees	SET	
card_ver_data	STRING	
cardholder_bill_street_addr	STRING	
cardholder_bill_zip	STRING	
cardholder_compress_bill_addr	STRING	
cardholder_add_id_type	STRING	
cardholder_add_id	STRING	
avs_code	STRING	
cardholder_bill_currency_code	STRING	
tvr	STRING	
transaction_seq_counter	LONG	
transaction_category_code	STRING	
auth_agent_id	STRING	
card_presence_flag	BOOLEAN	
file_transfer_ack_code	STRING	
file_transfer_seq_number	STRING	
return_amount	LONG	
destination_id	STRING	
originator_id	STRING	
receiving_institution_id	STRING	
terminal_timezone	STRING	
cvv_val_code	STRING	
secure_3d_val_code	STRING	

card_id	STRING	
card_bin	STRING	
cardholder_name	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_brand	STRING	
card_issuer	STRING	
card_country_code	STRING	
card_last4	STRING	
card_status	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_available_balance	LONG	
card_number_of_cardholders	INT	
account_iban	STRING	
account_bic	STRING	
account_open_date	LONG	
account_balance	LONG	
account_credit_limit	LONG	
account_otb_limit	LONG	
account_available_credit	LONG	
account_debt_balance	LONG	
account_cash_advance_balance	LONG	
account_status	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_associated_cards	SET	
account_debt_payment_method	STRING	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
customer_id	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	

customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	
customer_tax_number	STRING	
customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_city	STRING	
customer_region	STRING	
customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
terminal_street_addr	STRING	
terminal_city	STRING	
terminal_region	STRING	
terminal_zip	STRING	
terminal_country_code	STRING	
terminal_ip	STRING	
terminal_ip_latitude	DOUBLE	
terminal_ip_longitude	DOUBLE	
terminal_ip_zip	STRING	
terminal_ip_city	STRING	
terminal_ip_region	STRING	
terminal_ip_country_code	STRING	
terminal_poc_flag	BOOLEAN	
acceptor_dba_name	STRING	
acceptor_tax_number	STRING	
sic	STRING	
acceptor_open_date	LONG	
acceptor_risky_flag	BOOLEAN	
acceptor_risk	INT	
acceptor_phone_val_code	STRING	
acceptor_latitude	DOUBLE	
acceptor_longitude	DOUBLE	
event_external_id	STRING	
alert_id	STRING	

device_id	STRING	
event_occurred_at	LONG	
event_type	STRING	
transaction_status	STRING	
transaction_status_reason	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_response_type	STRING	
oob_feedback_notes	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_refunded	STRING	
mc_merchant_fraud_code	STRING	
event_received_at	LONG	
customer_segment	STRING	
customer_number_of_cards	INT	
customer_occupation_status	STRING	
customer_risk_rating	STRING	
customer_currency	STRING	
account_type	STRING	
account_currency	STRING	
feedback_label	STRING	
feedback_type	STRING	
trusted_acceptor	BOOLEAN	
first_recurring_payment	BOOLEAN	
recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
login_session_id	STRING	
session_id	STRING	
customer_country_code_change_flag	BOOLEAN	
customer_email_change_flag	BOOLEAN	
customer_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	

proxy_flag	BOOLEAN	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip	STRING	
last_sca_at	LONG	
merchant_initiated	BOOLEAN	
account_id	STRING	
card_first_activation_date	LONG	
card_issued_date	LONG	
posneg_card_country_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
sign_cardholder_bill_amount	LONG	
sign_count	LONG	
event_occurred_at_as_long	LONG	
is_high_amount	BOOLEAN	
card_days_to_expire	LONG	
terminal_latitude	DOUBLE	
terminal_longitude	DOUBLE	
posneg_acceptor_value	STRING	
posneg_card_value	STRING	
posneg_country_value	STRING	
posneg_customer_value	STRING	
posneg_mcc_value	STRING	
posneg_terminal_value	STRING	
posneg_zip_value	STRING	
sca_required	BOOLEAN	
acceptorid_cardbin_cntDist_cardExpDate_10m	LONG	
acceptorid_cardbin_cntDist_cardid_10m	LONG	
APPR_AUTH_acceptorId_count_6M	LONG	
FRAUD_acceptorId_count_6M	LONG	
APPR_AUTH_cardId_acceptor_billAmount_count_1d	LONG	
AUTH_cardId_acceptorId_count_6M	LONG	
APPR_AUTH_cardId_acceptor_sum_billAmount_1d	LONG	
APPR_AUTH_cardId_acceptor_count_1d	LONG	
AUTH_cardid_deviceid_count_6M	LONG	
AUTH_cardid_device_ip_country_code_count_6M	LONG	
APPR_AUTH_last_sca_at_cardid_sum_billAmount_90d	LONG	
APPR_AUTH_last_sca_at_cardid_count_90d	LONG	
APPR_AUTH_contactless_last_sca_at_cardid_sum_billAmount_90d	LONG	

APPR_AUTH_contactless_last_sca_at_cardid_count_90d	LONG	
APPR_AUTH_last_sca_at_cardid_sum_billAmount_30h	LONG	
APPR_AUTH_last_sca_at_cardid_count_30h	LONG	
APPR_AUTH_contactless_last_sca_at_cardid_sum_billAmount_30h	LONG	
APPR_AUTH_contactless_last_sca_at_cardid_count_30h	LONG	
AUTH_cardid_mcc_sum_amount_1M	LONG	
AUTH_cardid_mcc_count_1M	LONG	
AUTH_cardid_mcc_count_6M	LONG	
AUTH_cardId_count_4w	LONG	
APPR_AUTH_cardId_sum_billAmount_4w	LONG	
AUTH_cardId_count_1d	LONG	
HIGH_AMOUNT_APPR_AUTH_cardId_count_1d	LONG	
DECLINE_cardId_count_1d	LONG	
AUTH_CP_cardId_prev_ts_1h	LONG	
AUTH_CP_cardId_prev_term_lat_1h	DOUBLE	
AUTH_CP_cardId_prev_term_long_1h	DOUBLE	
AUTH_CNP_cardid_prev_device_ip_lat_1h	DOUBLE	
AUTH_CNP_cardid_prev_device_ip_long_1h	DOUBLE	
APPR_AUTH_cardId_sum_billAmount_1h	LONG	
APPR_AUTH_cardId_count_1h	LONG	
AUTH_CNP_cardid_count_pw_changed_24h	LONG	
AUTH_CNP_auth_type_changed_24h	INT	
AUTH_CNP_login_session_id_changed_24h	INT	
AUTH_CNP_currency_changed_24h	INT	
AUTH_CNP_cardid_sum_amount_24h	LONG	
AUTH_CNP_distinct_device_id_per_card_24h	LONG	
AUTH_device_language_changed_24h	INT	
REVIEW_APPR_AUTH_cardId_sum_billAmount_1d	LONG	
REVIEW_APPR_AUTH_cardId_count_1d	LONG	
APPR_AUTH_cardId_count_3m	LONG	
AUTH_CNP_cardId_countDist_acceptor_5m	LONG	
mcc_cardbin_cntDist_cardExpDate_10m	LONG	
mcc_cardbin_cntDist_cardid_10m	LONG	
CP_card_prev_term_dist_1h	DOUBLE	
FRAUD_ratio_acceptorId_6M	DOUBLE	
card_id_not_blacklisted_nor_whitelisted	BOOLEAN	
AUTH_cardid_mcc_avg_amount_1M	DOUBLE	
acceptorid_cardbin_cur0	BOOLEAN	
acceptorid_cardbin_cur1	BOOLEAN	
acceptorid_cardbin_cur3	BOOLEAN	

acceptorid_cardbin_cur5	BOOLEAN	
mcc_cardbin_cur0	BOOLEAN	
mcc_cardbin_cur1	BOOLEAN	
mcc_cardbin_cur3	BOOLEAN	
mcc_cardbin_cur5	BOOLEAN	
CP_card_term_vel_1h	DOUBLE	
device_fingerprint	STRING	
company_id	STRING	
feedback_value	BOOLEAN	
info_type	STRING	
cardholder_presence_indicator	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
customer_accounts	SET	
secure_3d_val_status	STRING	
card_expected_payments_frequency	STRING	
application_transaction_counter	INT	
digital_wallet_tokenized_transaction	BOOLEAN	
digital_wallet_token	STRING	
digital_wallet_token_assurance_level	STRING	
digital_wallet_token_requester	STRING	
digital_wallet_token_type	STRING	
cardholder_authentication_verification_value	STRING	
cvv2_present_indicator	STRING	
cardholder_verification_rule	STRING	
business_app_transfer_purpose	STRING	
secure_3d_indicator	STRING	
secure_3d_protocol_version	STRING	
acceptor_visa_risk	INT	
acceptor_mastercard_risk	INT	
card_format	STRING	
mc_fraud_score	INT	
visa_fraud_score	INT	
cvv2_presence_indicator	STRING	
account_limits	SET	
customer_cards	SET	

Digital Activity API Schema

Field Name	Field Type	Field Description

event_type	STRING	
event_received_at	LONG	
event_sub_type	STRING	
new_customer_phone_number	STRING	
new_customer_email_addr	STRING	
new_customer_address	STRING	
new_customer_phone_last_update_at	LONG	
new_customer_email_last_update_at	LONG	
new_customer_addr_last_update_at	LONG	
new_account_status_last_update_at	LONG	
customer_email_addr	STRING	
lifecycle_id	STRING	
event_occurred_at	LONG	
event_external_id	STRING	
event_status	STRING	
event_status_fail_reason	STRING	
session_id	STRING	
account_id	STRING	
account_open_date	LONG	
account_age	LONG	
channel_type	STRING	
customer_id	STRING	
challenge_requested	BOOLEAN	
challenge_type	STRING	
challenge_status	STRING	
challenge_status_fail_reason	STRING	
customer_segment	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone_number	STRING	
customer_address	STRING	
customer_country_code	STRING	
customer_preferred_language	STRING	
customer_activity_status	STRING	
customer_occupation	STRING	
customer_dob	STRING	
customer_branch_id	STRING	
customer_zip_code	STRING	
customer_zip_area_code	STRING	
new_device_id	STRING	

new_device_language	STRING	
new_device_email	STRING	
new_device_phone_number	STRING	
new_device_country_code	STRING	
new_device_city	STRING	
new_device_timezone	STRING	
new_device_browser	STRING	
new_device_browser_version	STRING	
new_device_browser_language	STRING	
new_device_model	STRING	
device_id	STRING	
device_account_id	STRING	
fuzzy_device_id	STRING	
device_model	STRING	
device_phone_number	STRING	
device_email_addr	STRING	
device_language	STRING	
device_registration_date	LONG	
device_first_seen	LONG	
device_user_agent	STRING	
device_type	STRING	
device_user_name	STRING	
device_brand	STRING	
device_app_version	STRING	
device_imei	STRING	
device_longitude	DOUBLE	
device_latitude	DOUBLE	
device_country_code	STRING	
device_city	STRING	
device_region	STRING	
device_timezone	STRING	
device_ip_address	STRING	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip_country_code	STRING	
device_ip_city	STRING	
device_ip_region	STRING	
device_ip_first_seen	LONG	
device_ip_proxy	BOOLEAN	
device_ip_proxy_type	STRING	

device_isp	STRING	
device_isp_country_code	STRING	
device_isp_region	STRING	
device_os	STRING	
device_os_language	STRING	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_browser_language	STRING	
device_screen_height	INT	
device_screen_width	INT	
device_java_script_enabled	BOOLEAN	
device_flash_enabled	BOOLEAN	
device_flash_version	STRING	
device_cookies_enabled	BOOLEAN	
device_bot	BOOLEAN	
device_malware	BOOLEAN	
device_rat	BOOLEAN	
device_phishing	BOOLEAN	
device_emulator	BOOLEAN	
device_rooted	BOOLEAN	
device_tor	BOOLEAN	
device_risk_rating	STRING	
device_risk_score	INT	
device_risk_reason	STRING	
device_registered	BOOLEAN	
device_last_update_at	LONG	
new_payee_last_added_at	LONG	
new_payee_bank_account_number	STRING	
new_payee_bank_account_bic	STRING	
new_payee_bank_account_sort_code	STRING	
new_payee_alias_value	STRING	
new_payee_alias_type	STRING	
new_payee_name	STRING	
new_payee_account_type	STRING	
customer_phone_last_update_at	LONG	
customer_email_last_update_at	LONG	
customer_addr_last_update_at	LONG	
account_status_last_update_at	LONG	
new_credentials_last_update_at	LONG	

new_credentials_exp_date	LONG	
new_credentials_attempts_remaining	INT	
credentials_last_update_at	LONG	
credentials_exp_date	LONG	
credentials_attempts_remaining	INT	
alias_value	STRING	
alias_type	STRING	
alias_bank_account_number	STRING	
alias_bank_account_bic	STRING	
alias_bank_account_sort_code	STRING	
alias_registration_date	LONG	
alias_last_added_at	LONG	
new_alias_value	STRING	
new_alias_type	STRING	
new_alias_bank_account_number	STRING	
new_alias_bank_account_bic	STRING	
new_alias_bank_account_sort_code	STRING	
new_alias_registration_date	LONG	
new_alias_last_added_at	LONG	
auth_type	STRING	
auth_password_pasted_flag	BOOLEAN	
auth_username_pasted_flag	BOOLEAN	
auth_level	STRING	
account_last_login_attempt_at	LONG	
account_last_successful_login_at	LONG	
account_login_first_seen	LONG	
previous_auth_result	STRING	
is_scoring	BOOLEAN	
feedback_label	STRING	
feedback_value	BOOLEAN	
customer_username	STRING	
daily_amount_limit	LONG	
new_customer_username	STRING	
new_daily_amount_limit	LONG	
max_daily_amount_limit	LONG	
card_id	STRING	
card_available_balance	LONG	
card_bin	STRING	
card_brand	STRING	
card_country_code	STRING	

card_issuer	STRING	
card_status	STRING	
cardholder_name	STRING	
device_fingerprint	STRING	
device_mobile_system_simulator	BOOLEAN	
device_ip_botnet_intensity	LONG	
device_mobile_system_root_detection_disabled	BOOLEAN	
device_is_new	BOOLEAN	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	
customer_tax_number	STRING	
customer_registration_date	LONG	
customer_city	STRING	
customer_region	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_number_of_cards	INT	
customer_risk_rating	STRING	
customer_currency	STRING	
customer_active	BOOLEAN	
customer_alias	STRING	
customer_alias_type	STRING	
customer_alias_registration_time	LONG	
customer_alias_last_update_time	LONG	
customer_alias_status	STRING	
customer_reported_income	LONG	
customer_cards	SET	
customer_accounts	SET	
session_register_date	LONG	
new_account_bic	STRING	
new_account_active	STRING	
new_account_cash_advance_balance	LONG	
new_account_credit_limit	LONG	
new_account_debt_balance	LONG	
new_account_debt_payment_method	STRING	

new_account_last_activity	LONG	
new_account_last_payment_date	LONG	
new_account_last_payment_amount	LONG	
new_account_number_of_cards	INT	
new_account_otb_limit	LONG	
new_account_available_credit	LONG	
new_account_pw_change_flag	BOOLEAN	
new_account_rewards_used_count	INT	
new_account_status	STRING	
new_account_type	STRING	
new_account_currency	STRING	
new_account_balance	LONG	
new_account_cards	LIST	
new_account_customers	LIST	
new_account_atc1	STRING	
new_account_atc2	STRING	
new_account_limits	SET	
account_iban	STRING	
account_bic	STRING	
account_active	STRING	
account_cash_advance_balance	LONG	
account_credit_limit	LONG	
account_debt_balance	LONG	
account_debt_payment_method	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_otb_limit	LONG	
account_available_credit	LONG	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
account_status	STRING	
account_type	STRING	
account_currency	STRING	
account_balance	LONG	
account_cards	LIST	
account_customers	LIST	
account_atc1	STRING	
account_atc2	STRING	

account_limits	SET	
new_card_bin	STRING	
new_card_pan	STRING	
new_card_effective_date	STRING	
new_card_expiration_date	STRING	
new_card_seq_number	STRING	
new_card_ver_data	STRING	
new_card_type	STRING	
new_card_category	STRING	
new_card_program	STRING	
new_card_brand	STRING	
new_card_issuer	STRING	
new_card_country_code	STRING	
new_card_last4	STRING	
new_card_status	STRING	
new_card_credit_score	INT	
new_card_transaction_amount_limit	LONG	
new_card_credit_limit	LONG	
new_card_available_balance	LONG	
new_card_number_of_cardholders	INT	
new_card_first_activation_date	LONG	
new_card_issued_date	LONG	
new_card_expected_payments_frequency	STRING	
new_card_format	STRING	
card_pan	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
card_seq_number	STRING	
card_ver_data	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_last4	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_number_of_cardholders	INT	
card_first_activation_date	LONG	
card_issued_date	LONG	
card_expected_payments_frequency	STRING	

card_format	STRING	
new_account_open_date	LONG	
new_device_account_id	STRING	
new_fuzzy_device_id	STRING	
new_device_registration_date	LONG	
new_device_first_seen	LONG	
new_device_user_agent	STRING	
new_device_type	STRING	
new_device_user_name	STRING	
new_device_brand	STRING	
new_device_app_version	STRING	
new_device_imei	STRING	
new_device_longitude	DOUBLE	
new_device_latitude	DOUBLE	
new_device_region	STRING	
new_device_ip_address	STRING	
new_device_ip_latitude	DOUBLE	
new_device_ip_longitude	DOUBLE	
new_device_ip_country_code	STRING	
new_device_ip_city	STRING	
new_device_ip_region	STRING	
new_device_ip_first_seen	LONG	
new_device_ip_proxy	BOOLEAN	
new_device_ip_proxy_type	STRING	
new_device_isp	STRING	
new_device_isp_country_code	STRING	
new_device_isp_region	STRING	
new_device_os	STRING	
new_device_os_language	STRING	
new_device_os_version	STRING	
new_device_screen_height	INT	
new_device_screen_width	INT	
new_device_java_script_enabled	BOOLEAN	
new_device_flash_enabled	BOOLEAN	
new_device_flash_version	STRING	
new_device_cookies_enabled	BOOLEAN	
new_device_bot	BOOLEAN	
new_device_malware	BOOLEAN	
new_device_rat	BOOLEAN	
new_device_phishing	BOOLEAN	

new_device_emulator	BOOLEAN	
new_device_rooted	BOOLEAN	
new_device_tor	BOOLEAN	
new_device_risk_rating	STRING	
new_device_risk_score	INT	
new_device_risk_reason	STRING	
new_device_registered	BOOLEAN	
customer_alias_dict_reg_date	LONG	
customer_alias_settle_3d	INT	
customer_alias_settle_30d	INT	
customer_alias_settle_6m	INT	
customer_settle_3d	INT	
customer_settle_30d	INT	
customer_settle_6m	INT	
customer_account_settle_3d	INT	
customer_account_settle_30d	INT	
customer_account_settle_6m	INT	
customer_alias_rep_fraud_3d	INT	
customer_alias_rep_fraud_30d	INT	
customer_alias_rep_fraud_6m	INT	
customer_rep_fraud_3d	INT	
customer_rep_fraud_30d	INT	
customer_rep_fraud_6m	INT	
customer_account_rep_fraud_3d	INT	
customer_account_rep_fraud_30d	INT	
customer_account_rep_fraud_6m	INT	
customer_alias_fraud_3d	INT	
customer_alias_fraud_30d	INT	
customer_alias_fraud_6m	INT	
customer_fraud_3d	INT	
customer_fraud_30d	INT	
customer_fraud_6m	INT	
customer_account_fraud_3d	INT	
customer_account_fraud_30d	INT	
customer_account_fraud_6m	INT	
customer_alias_rejected_3d	INT	
customer_alias_rejected_30d	INT	
customer_alias_rejected_6m	INT	
customer_rejected_3d	INT	
customer_rejected_30d	INT	

customer_rejected_6m	INT	
customer_account_rejected_3d	INT	
customer_account_rejected_30d	INT	
customer_account_rejected_6m	INT	
dict_fields_v2	TUPLE	
customer_type	STRING	
customer_mothersname	STRING	
customer_gender	STRING	
customer_pwd	STRING	
customer_contactoptout	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_address_validated	BOOLEAN	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
card_requested_date	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
new_device_android_id	STRING	
new_device_cbf	STRING	
new_device_mac_address	STRING	
new_device_serial_id	STRING	
new_device_browser_no_tracking	BOOLEAN	
new_device_browser_private_mode	BOOLEAN	
new_device_network_type	STRING	
new_device_session_duration	LONG	
info_type	STRING	

Digital Activity Decision Schema

Field Name	Field Type	Field Description
event_type	STRING	
event_received_at	LONG	
event_sub_type	STRING	

new_customer_phone_number	STRING	
new_customer_email_addr	STRING	
new_customer_address	STRING	
new_customer_phone_last_update_at	LONG	
new_customer_email_last_update_at	LONG	
new_customer_addr_last_update_at	LONG	
new_account_status_last_update_at	LONG	
customer_email_addr	STRING	
lifecycle_id	STRING	
event_occurred_at	LONG	
event_external_id	STRING	
event_status	STRING	
event_status_fail_reason	STRING	
session_id	STRING	
account_id	STRING	
account_open_date	LONG	
account_age	LONG	
channel_type	STRING	
customer_id	STRING	
challenge_requested	BOOLEAN	
challenge_type	STRING	
challenge_status	STRING	
challenge_status_fail_reason	STRING	
customer_segment	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone_number	STRING	
customer_address	STRING	
customer_country_code	STRING	
customer_preferred_language	STRING	
customer_activity_status	STRING	
customer_occupation	STRING	
customer_dob	STRING	
customer_branch_id	STRING	
customer_zip_code	STRING	
customer_zip_area_code	STRING	
new_device_id	STRING	
new_device_language	STRING	
new_device_email	STRING	
new_device_phone_number	STRING	

new_device_country_code	STRING	
new_device_city	STRING	
new_device_timezone	STRING	
new_device_browser	STRING	
new_device_browser_version	STRING	
new_device_browser_language	STRING	
new_device_model	STRING	
device_id	STRING	
device_account_id	STRING	
fuzzy_device_id	STRING	
device_model	STRING	
device_phone_number	STRING	
device_email_addr	STRING	
device_language	STRING	
device_registration_date	LONG	
device_first_seen	LONG	
device_user_agent	STRING	
device_type	STRING	
device_user_name	STRING	
device_brand	STRING	
device_app_version	STRING	
device_imei	STRING	
device_longitude	DOUBLE	
device_latitude	DOUBLE	
device_country_code	STRING	
device_city	STRING	
device_region	STRING	
device_timezone	STRING	
device_ip_address	STRING	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip_country_code	STRING	
device_ip_city	STRING	
device_ip_region	STRING	
device_ip_first_seen	LONG	
device_ip_proxy	BOOLEAN	
device_ip_proxy_type	STRING	
device_isp	STRING	
device_isp_country_code	STRING	
device_isp_region	STRING	

device_os	STRING	
device_os_language	STRING	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_browser_language	STRING	
device_screen_height	INT	
device_screen_width	INT	
device_java_script_enabled	BOOLEAN	
device_flash_enabled	BOOLEAN	
device_flash_version	STRING	
device_cookies_enabled	BOOLEAN	
device_bot	BOOLEAN	
device_malware	BOOLEAN	
device_rat	BOOLEAN	
device_phishing	BOOLEAN	
device_emulator	BOOLEAN	
device_rooted	BOOLEAN	
device_tor	BOOLEAN	
device_risk_rating	STRING	
device_risk_score	INT	
device_risk_reason	STRING	
device_registered	BOOLEAN	
device_last_update_at	LONG	
device_fingerprint	STRING	
device_mobile_system_simulator	BOOLEAN	
device_ip_botnet_intensity	LONG	
device_mobile_system_root_detection_disabled	BOOLEAN	
device_is_new	BOOLEAN	
new_payee_last_added_at	LONG	
new_payee_bank_account_number	STRING	
new_payee_bank_account_bic	STRING	
new_payee_bank_account_sort_code	STRING	
new_payee_alias_value	STRING	
new_payee_alias_type	STRING	
new_payee_name	STRING	
new_payee_account_type	STRING	
customer_phone_last_update_at	LONG	
customer_email_last_update_at	LONG	
customer_addr_last_update_at	LONG	

account_status_last_update_at	LONG	
new_credentials_last_update_at	LONG	
new_credentials_exp_date	LONG	
new_credentials_attempts_remaining	INT	
credentials_last_update_at	LONG	
credentials_exp_date	LONG	
credentials_attempts_remaining	INT	
alias_value	STRING	
alias_type	STRING	
alias_bank_account_number	STRING	
alias_bank_account_bic	STRING	
alias_bank_account_sort_code	STRING	
alias_registration_date	LONG	
alias_last_added_at	LONG	
new_alias_value	STRING	
new_alias_type	STRING	
new_alias_bank_account_number	STRING	
new_alias_bank_account_bic	STRING	
new_alias_bank_account_sort_code	STRING	
new_alias_registration_date	LONG	
new_alias_last_added_at	LONG	
auth_type	STRING	
auth_password_pasted_flag	BOOLEAN	
auth_username_pasted_flag	BOOLEAN	
auth_level	STRING	
account_last_login_attempt_at	LONG	
account_last_successful_login_at	LONG	
account_login_first_seen	LONG	
previous_auth_result	STRING	
posneg_device_type_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
posneg_account_id_value	STRING	
posneg_alias_value	STRING	
posneg_phone_value	STRING	
posneg_email_domain	STRING	
posneg_email_addr_value	STRING	
time_of_day	STRING	
day_of_week	STRING	
mismatch_mobile_settings	INT	

proxy_tor_rooted_flag	INT	
phishing_flag_and_high_risk	INT	
malware_flag_and_high_risk	INT	
high_risk_level	INT	
is_scoring	BOOLEAN	
feedback_label	STRING	
feedback_value	BOOLEAN	
customer_username	STRING	
daily_amount_limit	LONG	
new_customer_username	STRING	
new_daily_amount_limit	LONG	
max_daily_amount_limit	LONG	
card_id	STRING	
card_available_balance	LONG	
card_bin	STRING	
card_brand	STRING	
card_country_code	STRING	
card_issuer	STRING	
card_status	STRING	
cardholder_name	STRING	
customer_city	STRING	
customer_region	STRING	
new_customer_email_domain	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
decision	STRING	
count_login_per_account_and_ip_region_1m	LONG	
count_account_changes_per_session_id_1h	LONG	
count_succ_logins_per_account_time_of_day_3m	LONG	
distinct_login_locations_per_account_1h	LONG	
count_succ_logins_per_account_1m	LONG	
count_succ_logins_per_account_3m	LONG	
count_new_payee_per_account_1min	LONG	
count_failed_logins_per_account_id_1min	LONG	
account_previous_auth_type_changed_24h	INT	
account_previous_change_is_new_device_24h	INT	
account_previous_change_is_password_reset_24h	INT	
account_previous_activity_is_succ_login_24h	INT	
count_failed_logins_per_account_id_24h	LONG	

count_logins_per_device_id_1m	LONG	
distinct_customer_id_per_device_id_1w	LONG	
count_logins_per_device_id_90min	LONG	
count_failed_logins_per_device_id_90min	LONG	
distinct_customer_id_per_device_ip_addr_1h	LONG	
count_logins_per_device_ip_addr_1m	LONG	
count_logins_per_device_ip_addr_1h	LONG	
count_failed_logins_per_device_ip_addr_90min	LONG	
count_account_ip_country_3M	LONG	
count_account_activity_3M	LONG	
count_logins_per_account_3M	LONG	
enforced_risk_level	STRING	
session_register_date	LONG	
revelock_score	INT	
revelock_automated_actions	LIST	
revelock_alerts	LIST	
new_account_bic	STRING	
new_account_active	STRING	
new_account_cash_advance_balance	LONG	
new_account_credit_limit	LONG	
new_account_debt_balance	LONG	
new_account_debt_payment_method	STRING	
new_account_last_activity	LONG	
new_account_last_payment_date	LONG	
new_account_last_payment_amount	LONG	
new_account_number_of_cards	INT	
new_account_otb_limit	LONG	
new_account_available_credit	LONG	
new_account_pw_change_flag	BOOLEAN	
new_account_rewards_used_count	INT	
new_account_status	STRING	
new_account_type	STRING	
new_account_currency	STRING	
new_account_balance	LONG	
new_account_cards	LIST	
new_account_customers	LIST	
new_account_atc1	STRING	
new_account_atc2	STRING	
new_account_limits	SET	
account_iban	STRING	

account_bic	STRING	
account_active	STRING	
account_cash_advance_balance	LONG	
account_credit_limit	LONG	
account_debt_balance	LONG	
account_debt_payment_method	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_otb_limit	LONG	
account_available_credit	LONG	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
account_status	STRING	
account_type	STRING	
account_currency	STRING	
account_balance	LONG	
account_cards	LIST	
account_customers	LIST	
account_atc1	STRING	
account_atc2	STRING	
account_limits	SET	
new_card_bin	STRING	
new_card_pan	STRING	
new_card_effective_date	STRING	
new_card_expiration_date	STRING	
new_card_seq_number	STRING	
new_card_ver_data	STRING	
new_card_type	STRING	
new_card_category	STRING	
new_card_program	STRING	
new_card_brand	STRING	
new_card_issuer	STRING	
new_card_country_code	STRING	
new_card_last4	STRING	
new_card_status	STRING	
new_card_credit_score	INT	
new_card_transaction_amount_limit	LONG	
new_card_credit_limit	LONG	

new_card_available_balance	LONG	
new_card_number_of_cardholders	INT	
new_card_first_activation_date	LONG	
new_card_issued_date	LONG	
new_card_expected_payments_frequency	STRING	
new_card_format	STRING	
card_pan	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
card_seq_number	STRING	
card_ver_data	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_last4	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_number_of_cardholders	INT	
card_first_activation_date	LONG	
card_issued_date	LONG	
card_expected_payments_frequency	STRING	
card_format	STRING	
new_account_open_date	LONG	
new_device_account_id	STRING	
new_fuzzy_device_id	STRING	
new_device_registration_date	LONG	
new_device_first_seen	LONG	
new_device_user_agent	STRING	
new_device_type	STRING	
new_device_user_name	STRING	
new_device_brand	STRING	
new_device_app_version	STRING	
new_device_imei	STRING	
new_device_longitude	DOUBLE	
new_device_latitude	DOUBLE	
new_device_region	STRING	
new_device_ip_address	STRING	
new_device_ip_latitude	DOUBLE	
new_device_ip_longitude	DOUBLE	

new_device_ip_country_code	STRING	
new_device_ip_city	STRING	
new_device_ip_region	STRING	
new_device_ip_first_seen	LONG	
new_device_ip_proxy	BOOLEAN	
new_device_ip_proxy_type	STRING	
new_device_isp	STRING	
new_device_isp_country_code	STRING	
new_device_isp_region	STRING	
new_device_os	STRING	
new_device_os_language	STRING	
new_device_os_version	STRING	
new_device_screen_height	INT	
new_device_screen_width	INT	
new_device_java_script_enabled	BOOLEAN	
new_device_flash_enabled	BOOLEAN	
new_device_flash_version	STRING	
new_device_cookies_enabled	BOOLEAN	
new_device_bot	BOOLEAN	
new_device_malware	BOOLEAN	
new_device_rat	BOOLEAN	
new_device_phishing	BOOLEAN	
new_device_emulator	BOOLEAN	
new_device_rooted	BOOLEAN	
new_device_tor	BOOLEAN	
new_device_risk_rating	STRING	
new_device_risk_score	INT	
new_device_risk_reason	STRING	
new_device_registered	BOOLEAN	
customer_alias_dict_reg_date	LONG	
customer_alias_settle_3d	INT	
customer_alias_settle_30d	INT	
customer_alias_settle_6m	INT	
customer_settle_3d	INT	
customer_settle_30d	INT	
customer_settle_6m	INT	
customer_account_settle_3d	INT	
customer_account_settle_30d	INT	
customer_account_settle_6m	INT	
customer_alias_rep_fraud_3d	INT	

customer_alias_rep_fraud_30d	INT	
customer_alias_rep_fraud_6m	INT	
customer_rep_fraud_3d	INT	
customer_rep_fraud_30d	INT	
customer_rep_fraud_6m	INT	
customer_account_rep_fraud_3d	INT	
customer_account_rep_fraud_30d	INT	
customer_account_rep_fraud_6m	INT	
customer_alias_fraud_3d	INT	
customer_alias_fraud_30d	INT	
customer_alias_fraud_6m	INT	
customer_fraud_3d	INT	
customer_fraud_30d	INT	
customer_fraud_6m	INT	
customer_account_fraud_3d	INT	
customer_account_fraud_30d	INT	
customer_account_fraud_6m	INT	
customer_alias_rejected_3d	INT	
customer_alias_rejected_30d	INT	
customer_alias_rejected_6m	INT	
customer_rejected_3d	INT	
customer_rejected_30d	INT	
customer_rejected_6m	INT	
customer_account_rejected_3d	INT	
customer_account_rejected_30d	INT	
customer_account_rejected_6m	INT	
dict_fields_v2	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
customer_type	STRING	
customer_mothersname	STRING	
customer_gender	STRING	
customer_pwd	STRING	
customer_contactoptout	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_address_validated	BOOLEAN	
card_requested_date	LONG	
device_android_id	STRING	
device_cbf	STRING	

device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
new_device_android_id	STRING	
new_device_cbf	STRING	
new_device_mac_address	STRING	
new_device_serial_id	STRING	
new_device_browser_no_tracking	BOOLEAN	
new_device_browser_private_mode	BOOLEAN	
new_device_network_type	STRING	
new_device_session_duration	LONG	
info_type	STRING	

Digital Activity Extended API Schema

Field Name	Field Type	Field Description
event_type	STRING	
event_received_at	LONG	
event_sub_type	STRING	
new_customer_phone_number	STRING	
new_customer_email_addr	STRING	
new_customer_address	STRING	
new_customer_phone_last_update_at	LONG	
new_customer_email_last_update_at	LONG	
new_customer_addr_last_update_at	LONG	
new_account_status_last_update_at	LONG	
customer_email_addr	STRING	
lifecycle_id	STRING	
event_occurred_at	LONG	
event_external_id	STRING	
event_status	STRING	
event_status_fail_reason	STRING	
session_id	STRING	
account_id	STRING	
account_open_date	LONG	
account_age	LONG	
channel_type	STRING	

customer_id	STRING	
challenge_requested	BOOLEAN	
challenge_type	STRING	
challenge_status	STRING	
challenge_status_fail_reason	STRING	
customer_segment	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone_number	STRING	
customer_address	STRING	
customer_country_code	STRING	
customer_preferred_language	STRING	
customer_activity_status	STRING	
customer_occupation	STRING	
customer_dob	STRING	
customer_branch_id	STRING	
customer_zip_code	STRING	
customer_zip_area_code	STRING	
new_device_id	STRING	
new_device_language	STRING	
new_device_email	STRING	
new_device_phone_number	STRING	
new_device_country_code	STRING	
new_device_city	STRING	
new_device_timezone	STRING	
new_device_browser	STRING	
new_device_browser_version	STRING	
new_device_browser_language	STRING	
new_device_model	STRING	
device_id	STRING	
device_account_id	STRING	
fuzzy_device_id	STRING	
device_model	STRING	
device_phone_number	STRING	
device_email_addr	STRING	
device_language	STRING	
device_registration_date	LONG	
device_first_seen	LONG	
device_user_agent	STRING	
device_type	STRING	

device_user_name	STRING	
device_brand	STRING	
device_app_version	STRING	
device_imei	STRING	
device_longitude	DOUBLE	
device_latitude	DOUBLE	
device_country_code	STRING	
device_city	STRING	
device_region	STRING	
device_timezone	STRING	
device_ip_address	STRING	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip_country_code	STRING	
device_ip_city	STRING	
device_ip_region	STRING	
device_ip_first_seen	LONG	
device_ip_proxy	BOOLEAN	
device_ip_proxy_type	STRING	
device_isp	STRING	
device_isp_country_code	STRING	
device_isp_region	STRING	
device_os	STRING	
device_os_language	STRING	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_browser_language	STRING	
device_screen_height	INT	
device_screen_width	INT	
device_java_script_enabled	BOOLEAN	
device_flash_enabled	BOOLEAN	
device_flash_version	STRING	
device_cookies_enabled	BOOLEAN	
device_bot	BOOLEAN	
device_malware	BOOLEAN	
device_rat	BOOLEAN	
device_phishing	BOOLEAN	
device_emulator	BOOLEAN	
device_rooted	BOOLEAN	

device_tor	BOOLEAN	
device_risk_rating	STRING	
device_risk_score	INT	
device_risk_reason	STRING	
device_registered	BOOLEAN	
device_last_update_at	LONG	
device_fingerprint	STRING	
device_mobile_system_simulator	BOOLEAN	
device_ip_botnet_intensity	LONG	
device_mobile_system_root_detection_disabled	BOOLEAN	
device_is_new	BOOLEAN	
new_payee_last_added_at	LONG	
new_payee_bank_account_number	STRING	
new_payee_bank_account_bic	STRING	
new_payee_bank_account_sort_code	STRING	
new_payee_alias_value	STRING	
new_payee_alias_type	STRING	
new_payee_name	STRING	
new_payee_account_type	STRING	
customer_phone_last_update_at	LONG	
customer_email_last_update_at	LONG	
customer_addr_last_update_at	LONG	
account_status_last_update_at	LONG	
new_credentials_last_update_at	LONG	
new_credentials_exp_date	LONG	
new_credentials_attempts_remaining	INT	
credentials_last_update_at	LONG	
credentials_exp_date	LONG	
credentials_attempts_remaining	INT	
alias_value	STRING	
alias_type	STRING	
alias_bank_account_number	STRING	
alias_bank_account_bic	STRING	
alias_bank_account_sort_code	STRING	
alias_registration_date	LONG	
alias_last_added_at	LONG	
new_alias_value	STRING	
new_alias_type	STRING	
new_alias_bank_account_number	STRING	
new_alias_bank_account_bic	STRING	

new_alias_bank_account_sort_code	STRING	
new_alias_registration_date	LONG	
new_alias_last_added_at	LONG	
auth_type	STRING	
auth_password_pasted_flag	BOOLEAN	
auth_username_pasted_flag	BOOLEAN	
auth_level	STRING	
account_last_login_attempt_at	LONG	
account_last_successful_login_at	LONG	
account_login_first_seen	LONG	
previous_auth_result	STRING	
is_scoring	BOOLEAN	
feedback_label	STRING	
feedback_value	BOOLEAN	
customer_username	STRING	
daily_amount_limit	LONG	
new_customer_username	STRING	
new_daily_amount_limit	LONG	
max_daily_amount_limit	LONG	
card_id	STRING	
card_available_balance	LONG	
card_bin	STRING	
card_brand	STRING	
card_country_code	STRING	
card_issuer	STRING	
card_status	STRING	
cardholder_name	STRING	
customer_city	STRING	
customer_region	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
decision	STRING	
session_register_date	LONG	
new_account_bic	STRING	
new_account_active	STRING	
new_account_cash_advance_balance	LONG	
new_account_credit_limit	LONG	
new_account_debt_balance	LONG	
new_account_debt_payment_method	STRING	

new_account_last_activity	LONG	
new_account_last_payment_date	LONG	
new_account_last_payment_amount	LONG	
new_account_number_of_cards	INT	
new_account_otb_limit	LONG	
new_account_available_credit	LONG	
new_account_pw_change_flag	BOOLEAN	
new_account_rewards_used_count	INT	
new_account_status	STRING	
new_account_type	STRING	
new_account_currency	STRING	
new_account_balance	LONG	
new_account_cards	LIST	
new_account_customers	LIST	
new_account_atc1	STRING	
new_account_atc2	STRING	
new_account_limits	SET	
account_iban	STRING	
account_bic	STRING	
account_active	STRING	
account_cash_advance_balance	LONG	
account_credit_limit	LONG	
account_debt_balance	LONG	
account_debt_payment_method	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_otb_limit	LONG	
account_available_credit	LONG	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
account_status	STRING	
account_type	STRING	
account_currency	STRING	
account_balance	LONG	
account_cards	LIST	
account_customers	LIST	
account_atc1	STRING	
account_atc2	STRING	

account_limits	SET	
new_card_bin	STRING	
new_card_pan	STRING	
new_card_effective_date	STRING	
new_card_expiration_date	STRING	
new_card_seq_number	STRING	
new_card_ver_data	STRING	
new_card_type	STRING	
new_card_category	STRING	
new_card_program	STRING	
new_card_brand	STRING	
new_card_issuer	STRING	
new_card_country_code	STRING	
new_card_last4	STRING	
new_card_status	STRING	
new_card_credit_score	INT	
new_card_transaction_amount_limit	LONG	
new_card_credit_limit	LONG	
new_card_available_balance	LONG	
new_card_number_of_cardholders	INT	
new_card_first_activation_date	LONG	
new_card_issued_date	LONG	
new_card_expected_payments_frequency	STRING	
new_card_format	STRING	
card_pan	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
card_seq_number	STRING	
card_ver_data	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_last4	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_number_of_cardholders	INT	
card_first_activation_date	LONG	
card_issued_date	LONG	
card_expected_payments_frequency	STRING	

card_format	STRING	
new_account_open_date	LONG	
new_device_account_id	STRING	
new_fuzzy_device_id	STRING	
new_device_registration_date	LONG	
new_device_first_seen	LONG	
new_device_user_agent	STRING	
new_device_type	STRING	
new_device_user_name	STRING	
new_device_brand	STRING	
new_device_app_version	STRING	
new_device_imei	STRING	
new_device_longitude	DOUBLE	
new_device_latitude	DOUBLE	
new_device_region	STRING	
new_device_ip_address	STRING	
new_device_ip_latitude	DOUBLE	
new_device_ip_longitude	DOUBLE	
new_device_ip_country_code	STRING	
new_device_ip_city	STRING	
new_device_ip_region	STRING	
new_device_ip_first_seen	LONG	
new_device_ip_proxy	BOOLEAN	
new_device_ip_proxy_type	STRING	
new_device_isp	STRING	
new_device_isp_country_code	STRING	
new_device_isp_region	STRING	
new_device_os	STRING	
new_device_os_language	STRING	
new_device_os_version	STRING	
new_device_screen_height	INT	
new_device_screen_width	INT	
new_device_java_script_enabled	BOOLEAN	
new_device_flash_enabled	BOOLEAN	
new_device_flash_version	STRING	
new_device_cookies_enabled	BOOLEAN	
new_device_bot	BOOLEAN	
new_device_malware	BOOLEAN	
new_device_rat	BOOLEAN	
new_device_phishing	BOOLEAN	

new_device_emulator	BOOLEAN	
new_device_rooted	BOOLEAN	
new_device_tor	BOOLEAN	
new_device_risk_rating	STRING	
new_device_risk_score	INT	
new_device_risk_reason	STRING	
new_device_registered	BOOLEAN	
customer_alias_dict_reg_date	LONG	
customer_alias_settle_3d	INT	
customer_alias_settle_30d	INT	
customer_alias_settle_6m	INT	
customer_settle_3d	INT	
customer_settle_30d	INT	
customer_settle_6m	INT	
customer_account_settle_3d	INT	
customer_account_settle_30d	INT	
customer_account_settle_6m	INT	
customer_alias_rep_fraud_3d	INT	
customer_alias_rep_fraud_30d	INT	
customer_alias_rep_fraud_6m	INT	
customer_rep_fraud_3d	INT	
customer_rep_fraud_30d	INT	
customer_rep_fraud_6m	INT	
customer_account_rep_fraud_3d	INT	
customer_account_rep_fraud_30d	INT	
customer_account_rep_fraud_6m	INT	
customer_alias_fraud_3d	INT	
customer_alias_fraud_30d	INT	
customer_alias_fraud_6m	INT	
customer_fraud_3d	INT	
customer_fraud_30d	INT	
customer_fraud_6m	INT	
customer_account_fraud_3d	INT	
customer_account_fraud_30d	INT	
customer_account_fraud_6m	INT	
customer_alias_rejected_3d	INT	
customer_alias_rejected_30d	INT	
customer_alias_rejected_6m	INT	
customer_rejected_3d	INT	
customer_rejected_30d	INT	

customer_rejected_6m	INT	
customer_account_rejected_3d	INT	
customer_account_rejected_30d	INT	
customer_account_rejected_6m	INT	
dict_fields_v2	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
customer_type	STRING	
customer_mothersname	STRING	
customer_gender	STRING	
customer_pwd	STRING	
customer_contactoptout	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_address_validated	BOOLEAN	
card_requested_date	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
new_device_android_id	STRING	
new_device_cbf	STRING	
new_device_mac_address	STRING	
new_device_serial_id	STRING	
new_device_browser_no_tracking	BOOLEAN	
new_device_browser_private_mode	BOOLEAN	
new_device_network_type	STRING	
new_device_session_duration	LONG	
info_type	STRING	

Digital Activity FraudForce Schema

Field Name	Field Type	Field Description
event_type	STRING	
event_received_at	LONG	
event_sub_type	STRING	

new_customer_phone_number	STRING	
new_customer_email_addr	STRING	
new_customer_address	STRING	
new_customer_phone_last_update_at	LONG	
new_customer_email_last_update_at	LONG	
new_customer_addr_last_update_at	LONG	
new_account_status_last_update_at	LONG	
customer_email_addr	STRING	
lifecycle_id	STRING	
event_occurred_at	LONG	
event_external_id	STRING	
event_status	STRING	
event_status_fail_reason	STRING	
session_id	STRING	
account_id	STRING	
account_open_date	LONG	
account_age	LONG	
channel_type	STRING	
customer_id	STRING	
challenge_requested	BOOLEAN	
challenge_type	STRING	
challenge_status	STRING	
challenge_status_fail_reason	STRING	
customer_segment	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone_number	STRING	
customer_address	STRING	
customer_country_code	STRING	
customer_preferred_language	STRING	
customer_activity_status	STRING	
customer_occupation	STRING	
customer_dob	STRING	
customer_branch_id	STRING	
customer_zip_code	STRING	
customer_zip_area_code	STRING	
new_device_id	STRING	
new_device_language	STRING	
new_device_email	STRING	
new_device_phone_number	STRING	

new_device_country_code	STRING	
new_device_city	STRING	
new_device_timezone	STRING	
new_device_browser	STRING	
new_device_browser_version	STRING	
new_device_browser_language	STRING	
new_device_model	STRING	
device_id	STRING	
device_account_id	STRING	
fuzzy_device_id	STRING	
device_model	STRING	
device_phone_number	STRING	
device_email_addr	STRING	
device_language	STRING	
device_registration_date	LONG	
device_first_seen	LONG	
device_user_agent	STRING	
device_type	STRING	
device_user_name	STRING	
device_brand	STRING	
device_app_version	STRING	
device_imei	STRING	
device_longitude	DOUBLE	
device_latitude	DOUBLE	
device_country_code	STRING	
device_city	STRING	
device_region	STRING	
device_timezone	STRING	
device_ip_address	STRING	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip_country_code	STRING	
device_ip_city	STRING	
device_ip_region	STRING	
device_ip_first_seen	LONG	
device_ip_proxy	BOOLEAN	
device_ip_proxy_type	STRING	
device_isp	STRING	
device_isp_country_code	STRING	
device_isp_region	STRING	

device_os	STRING	
device_os_language	STRING	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_browser_language	STRING	
device_screen_height	INT	
device_screen_width	INT	
device_java_script_enabled	BOOLEAN	
device_flash_enabled	BOOLEAN	
device_flash_version	STRING	
device_cookies_enabled	BOOLEAN	
device_bot	BOOLEAN	
device_malware	BOOLEAN	
device_rat	BOOLEAN	
device_phishing	BOOLEAN	
device_emulator	BOOLEAN	
device_rooted	BOOLEAN	
device_tor	BOOLEAN	
device_risk_rating	STRING	
device_risk_score	INT	
device_risk_reason	STRING	
device_registered	BOOLEAN	
device_last_update_at	LONG	
device_mobile_system_simulator	BOOLEAN	
device_ip_botnet_intensity	LONG	
device_mobile_system_root_detection_disabled	BOOLEAN	
device_is_new	BOOLEAN	
new_payee_last_added_at	LONG	
new_payee_bank_account_number	STRING	
new_payee_bank_account_bic	STRING	
new_payee_bank_account_sort_code	STRING	
new_payee_alias_value	STRING	
new_payee_alias_type	STRING	
new_payee_name	STRING	
new_payee_account_type	STRING	
customer_phone_last_update_at	LONG	
customer_email_last_update_at	LONG	
customer_addr_last_update_at	LONG	
account_status_last_update_at	LONG	

new_credentials_last_update_at	LONG	
new_credentials_exp_date	LONG	
new_credentials_attempts_remaining	INT	
credentials_last_update_at	LONG	
credentials_exp_date	LONG	
credentials_attempts_remaining	INT	
alias_value	STRING	
alias_type	STRING	
alias_bank_account_number	STRING	
alias_bank_account_bic	STRING	
alias_bank_account_sort_code	STRING	
alias_registration_date	LONG	
alias_last_added_at	LONG	
new_alias_value	STRING	
new_alias_type	STRING	
new_alias_bank_account_number	STRING	
new_alias_bank_account_bic	STRING	
new_alias_bank_account_sort_code	STRING	
new_alias_registration_date	LONG	
new_alias_last_added_at	LONG	
auth_type	STRING	
auth_password_pasted_flag	BOOLEAN	
auth_username_pasted_flag	BOOLEAN	
auth_level	STRING	
account_last_login_attempt_at	LONG	
account_last_successful_login_at	LONG	
account_login_first_seen	LONG	
previous_auth_result	STRING	
is_scoring	BOOLEAN	
feedback_label	STRING	
feedback_value	BOOLEAN	
customer_username	STRING	
daily_amount_limit	LONG	
new_customer_username	STRING	
new_daily_amount_limit	LONG	
max_daily_amount_limit	LONG	
card_id	STRING	
card_available_balance	LONG	
card_bin	STRING	
card_brand	STRING	

card_country_code	STRING	
card_issuer	STRING	
card_status	STRING	
cardholder_name	STRING	
customer_city	STRING	
customer_region	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
decision	STRING	
ff_id	STRING	
ff_result	STRING	
ff_tracking_number	LONG	
ff_device_alias	LONG	
ff_device_blackbox_age	INT	
ff_device_blackbox_timestamp	STRING	
ff_device_browser_charset	STRING	
ff_device_browser_cookies_enabled	BOOLEAN	
ff_device_browser_configured_language	STRING	
ff_device_browser_flash_enabled	BOOLEAN	
ff_device_browser_flash_installed	BOOLEAN	
ff_device_browser_flash_storage_enabled	BOOLEAN	
ff_device_browser_flash_version	STRING	
ff_device_browser_language	STRING	
ff_device_browser_type	STRING	
ff_device_browser_timezone	STRING	
ff_device_browser_version	STRING	
ff_device_first_seen	STRING	
ff_device_is_new	BOOLEAN	
ff_device_mobile_app_bundle_id	STRING	
ff_device_mobile_app_debug	BOOLEAN	
ff_device_mobile_app_exe_name	STRING	
ff_device_mobile_app_market_id	STRING	
ff_device_mobile_app_name	STRING	
ff_device_mobile_app_orientation	STRING	
ff_device_mobile_app_proc_name	STRING	
ff_device_mobile_app_signer_id	STRING	
ff_device_mobile_app_vendor_id	STRING	
ff_device_mobile_app_version	STRING	
ff_device_mobile_brand	STRING	

ff_device_mobile_build_device	STRING	
ff_device_mobile_build_product	STRING	
ff_device_mobile_charging	BOOLEAN	
ff_device_mobile_imei	STRING	
ff_device_mobile_location_altitude	DOUBLE	
ff_device_mobile_location_enabled	BOOLEAN	
ff_device_mobile_location_latitude	DOUBLE	
ff_device_mobile_location_longitude	DOUBLE	
ff_device_mobile_location_timestamp	STRING	
ff_device_mobile_location_timezone	STRING	
ff_device_mobile_manufacturer	STRING	
ff_device_mobile_model	STRING	
ff_device_mobile_orientation	STRING	
ff_device_mobile_screen_resolution	STRING	
ff_device_mobile_system_allow_unknown_store	BOOLEAN	
ff_device_mobile_system_android_sdk_level	STRING	
ff_device_mobile_system_build_fingerprint	STRING	
ff_device_mobile_system_carrier	STRING	
ff_device_mobile_system_carrier_country_code	STRING	
ff_device_mobile_system_cellular_network	STRING	
ff_device_mobile_system_hostname	STRING	
ff_device_mobile_system_jailroot_detected	BOOLEAN	
ff_device_mobile_system_locale_currency	STRING	
ff_device_mobile_system_locale_lang	STRING	
ff_device_mobile_system_network_operator_country	STRING	
ff_device_mobile_system_network_operator_name	STRING	
ff_device_mobile_system_os_version	STRING	
ff_device_mobile_system_root_detection_disabled	BOOLEAN	
ff_device_mobile_system_simulator	BOOLEAN	
ff_device_mobile_system_uptime	STRING	
ff_device_mobile_system_voip_allowed	BOOLEAN	
ff_device_os	STRING	
ff_device_registration_result_match_status	STRING	
ff_device_registration_result_measure_of_charge	STRING	
ff_device_screen	STRING	
ff_device_recognition_source	STRING	
ff_device_type	STRING	
ff_real_ip_address	STRING	
ff_real_ip_botnet_score	INT	
ff_real_ip_botnet_intensity	INT	

ff_real_ip_botnet_lastseen	STRING	
ff_real_ip_isp	STRING	
ff_real_ip_loc_city	STRING	
ff_real_ip_loc_country	STRING	
ff_real_ip_loc_country_code	STRING	
ff_real_ip_loc_latitude	DOUBLE	
ff_real_ip_loc_longitude	DOUBLE	
ff_real_ip_loc_region	STRING	
ff_real_ip_org	STRING	
ff_real_ip_proxy	STRING	
ff_real_ip_source	STRING	
ff_stated_ip_address	STRING	
ff_stated_ip_botnet_score	INT	
ff_stated_ip_botnet_intensity	INT	
ff_stated_ip_botnet_lastseen	STRING	
ff_stated_ip_isp	STRING	
ff_stated_ip_loc_city	STRING	
ff_stated_ip_loc_country	STRING	
ff_stated_ip_loc_country_code	STRING	
ff_stated_ip_loc_latitude	DOUBLE	
ff_stated_ip_loc_longitude	DOUBLE	
ff_stated_ip_loc_region	STRING	
ff_stated_ip_org	STRING	
ff_stated_ip_proxy	STRING	
ff_stated_ip_source	STRING	
new_account_bic	STRING	
new_account_active	STRING	
new_account_cash_advance_balance	LONG	
new_account_credit_limit	LONG	
new_account_debt_balance	LONG	
new_account_debt_payment_method	STRING	
new_account_last_activity	LONG	
new_account_last_payment_date	LONG	
new_account_last_payment_amount	LONG	
new_account_number_of_cards	INT	
new_account_otb_limit	LONG	
new_account_available_credit	LONG	
new_account_pw_change_flag	BOOLEAN	
new_account_rewards_used_count	INT	
new_account_status	STRING	

new_account_type	STRING	
new_account_currency	STRING	
new_account_balance	LONG	
new_account_cards	LIST	
new_account_customers	LIST	
new_account_atc1	STRING	
new_account_atc2	STRING	
new_account_limits	SET	
account_iban	STRING	
account_bic	STRING	
account_active	STRING	
account_cash_advance_balance	LONG	
account_credit_limit	LONG	
account_debt_balance	LONG	
account_debt_payment_method	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_otb_limit	LONG	
account_available_credit	LONG	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
account_status	STRING	
account_type	STRING	
account_currency	STRING	
account_balance	LONG	
account_cards	LIST	
account_customers	LIST	
account_atc1	STRING	
account_atc2	STRING	
account_limits	SET	
new_card_bin	STRING	
new_card_pan	STRING	
new_card_effective_date	STRING	
new_card_expiration_date	STRING	
new_card_seq_number	STRING	
new_card_ver_data	STRING	
new_card_type	STRING	
new_card_category	STRING	

new_card_program	STRING	
new_card_brand	STRING	
new_card_issuer	STRING	
new_card_country_code	STRING	
new_card_last4	STRING	
new_card_status	STRING	
new_card_credit_score	INT	
new_card_transaction_amount_limit	LONG	
new_card_credit_limit	LONG	
new_card_available_balance	LONG	
new_card_number_of_cardholders	INT	
new_card_first_activation_date	LONG	
new_card_issued_date	LONG	
new_card_expected_payments_frequency	STRING	
new_card_format	STRING	
card_pan	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
card_seq_number	STRING	
card_ver_data	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_last4	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_number_of_cardholders	INT	
card_first_activation_date	LONG	
card_issued_date	LONG	
card_expected_payments_frequency	STRING	
card_format	STRING	
new_account_open_date	LONG	
new_device_account_id	STRING	
new_fuzzy_device_id	STRING	
new_device_registration_date	LONG	
new_device_first_seen	LONG	
new_device_user_agent	STRING	
new_device_type	STRING	
new_device_user_name	STRING	

new_device_brand	STRING	
new_device_app_version	STRING	
new_device_imei	STRING	
new_device_longitude	DOUBLE	
new_device_latitude	DOUBLE	
new_device_region	STRING	
new_device_ip_address	STRING	
new_device_ip_latitude	DOUBLE	
new_device_ip_longitude	DOUBLE	
new_device_ip_country_code	STRING	
new_device_ip_city	STRING	
new_device_ip_region	STRING	
new_device_ip_first_seen	LONG	
new_device_ip_proxy	BOOLEAN	
new_device_ip_proxy_type	STRING	
new_device_isp	STRING	
new_device_isp_country_code	STRING	
new_device_isp_region	STRING	
new_device_os	STRING	
new_device_os_language	STRING	
new_device_os_version	STRING	
new_device_screen_height	INT	
new_device_screen_width	INT	
new_device_java_script_enabled	BOOLEAN	
new_device_flash_enabled	BOOLEAN	
new_device_flash_version	STRING	
new_device_cookies_enabled	BOOLEAN	
new_device_bot	BOOLEAN	
new_device_malware	BOOLEAN	
new_device_rat	BOOLEAN	
new_device_phishing	BOOLEAN	
new_device_emulator	BOOLEAN	
new_device_rooted	BOOLEAN	
new_device_tor	BOOLEAN	
new_device_risk_rating	STRING	
new_device_risk_score	INT	
new_device_risk_reason	STRING	
new_device_registered	BOOLEAN	
customer_alias_dict_reg_date	LONG	
customer_alias_settle_3d	INT	

customer_alias_settle_30d	INT	
customer_alias_settle_6m	INT	
customer_settle_3d	INT	
customer_settle_30d	INT	
customer_settle_6m	INT	
customer_account_settle_3d	INT	
customer_account_settle_30d	INT	
customer_account_settle_6m	INT	
customer_alias_rep_fraud_3d	INT	
customer_alias_rep_fraud_30d	INT	
customer_alias_rep_fraud_6m	INT	
customer_rep_fraud_3d	INT	
customer_rep_fraud_30d	INT	
customer_rep_fraud_6m	INT	
customer_account_rep_fraud_3d	INT	
customer_account_rep_fraud_30d	INT	
customer_account_rep_fraud_6m	INT	
customer_alias_fraud_3d	INT	
customer_alias_fraud_30d	INT	
customer_alias_fraud_6m	INT	
customer_fraud_3d	INT	
customer_fraud_30d	INT	
customer_fraud_6m	INT	
customer_account_fraud_3d	INT	
customer_account_fraud_30d	INT	
customer_account_fraud_6m	INT	
customer_alias_rejected_3d	INT	
customer_alias_rejected_30d	INT	
customer_alias_rejected_6m	INT	
customer_rejected_3d	INT	
customer_rejected_30d	INT	
customer_rejected_6m	INT	
customer_account_rejected_3d	INT	
customer_account_rejected_30d	INT	
customer_account_rejected_6m	INT	
dict_fields_v2	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
customer_type	STRING	

customer_mothersname	STRING	
customer_gender	STRING	
customer_pwd	STRING	
customer_contactoptout	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_address_validated	BOOLEAN	
card_requested_date	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
new_device_android_id	STRING	
new_device_cbf	STRING	
new_device_mac_address	STRING	
new_device_serial_id	STRING	
new_device_browser_no_tracking	BOOLEAN	
new_device_browser_private_mode	BOOLEAN	
new_device_network_type	STRING	
new_device_session_duration	LONG	
info_type	STRING	

Digital Activity Pre Metrics Schema

Field Name	Field Type	Field Description
event_type	STRING	
event_received_at	LONG	
event_sub_type	STRING	
new_customer_phone_number	STRING	
new_customer_email_addr	STRING	
new_customer_address	STRING	
new_customer_phone_last_update_at	LONG	
new_customer_email_last_update_at	LONG	
new_customer_addr_last_update_at	LONG	
new_account_status_last_update_at	LONG	
customer_email_addr	STRING	
lifecycle_id	STRING	

event_occurred_at	LONG	
event_external_id	STRING	
event_status	STRING	
event_status_fail_reason	STRING	
session_id	STRING	
account_id	STRING	
account_open_date	LONG	
account_age	LONG	
channel_type	STRING	
customer_id	STRING	
challenge_requested	BOOLEAN	
challenge_type	STRING	
challenge_status	STRING	
challenge_status_fail_reason	STRING	
customer_segment	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone_number	STRING	
customer_address	STRING	
customer_country_code	STRING	
customer_preferred_language	STRING	
customer_activity_status	STRING	
customer_occupation	STRING	
customer_dob	STRING	
customer_branch_id	STRING	
customer_zip_code	STRING	
customer_zip_area_code	STRING	
new_device_id	STRING	
new_device_language	STRING	
new_device_email	STRING	
new_device_phone_number	STRING	
new_device_country_code	STRING	
new_device_city	STRING	
new_device_timezone	STRING	
new_device_browser	STRING	
new_device_browser_version	STRING	
new_device_browser_language	STRING	
new_device_model	STRING	
device_id	STRING	
device_account_id	STRING	

fuzzy_device_id	STRING	
device_model	STRING	
device_phone_number	STRING	
device_email_addr	STRING	
device_language	STRING	
device_registration_date	LONG	
device_first_seen	LONG	
device_user_agent	STRING	
device_type	STRING	
device_user_name	STRING	
device_brand	STRING	
device_app_version	STRING	
device_imei	STRING	
device_longitude	DOUBLE	
device_latitude	DOUBLE	
device_country_code	STRING	
device_city	STRING	
device_region	STRING	
device_timezone	STRING	
device_ip_address	STRING	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip_country_code	STRING	
device_ip_city	STRING	
device_ip_region	STRING	
device_ip_first_seen	LONG	
device_ip_proxy	BOOLEAN	
device_ip_proxy_type	STRING	
device_isp	STRING	
device_isp_country_code	STRING	
device_isp_region	STRING	
device_os	STRING	
device_os_language	STRING	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_browser_language	STRING	
device_screen_height	INT	
device_screen_width	INT	
device_java_script_enabled	BOOLEAN	

device_flash_enabled	BOOLEAN	
device_flash_version	STRING	
device_cookies_enabled	BOOLEAN	
device_bot	BOOLEAN	
device_malware	BOOLEAN	
device_rat	BOOLEAN	
device_phishing	BOOLEAN	
device_emulator	BOOLEAN	
device_rooted	BOOLEAN	
device_tor	BOOLEAN	
device_risk_rating	STRING	
device_risk_score	INT	
device_risk_reason	STRING	
device_registered	BOOLEAN	
device_last_update_at	LONG	
device_fingerprint	STRING	
device_mobile_system_simulator	BOOLEAN	
device_ip_botnet_intensity	LONG	
device_mobile_system_root_detection_disabled	BOOLEAN	
device_is_new	BOOLEAN	
new_payee_last_added_at	LONG	
new_payee_bank_account_number	STRING	
new_payee_bank_account_bic	STRING	
new_payee_bank_account_sort_code	STRING	
new_payee_alias_value	STRING	
new_payee_alias_type	STRING	
new_payee_name	STRING	
new_payee_account_type	STRING	
customer_phone_last_update_at	LONG	
customer_email_last_update_at	LONG	
customer_addr_last_update_at	LONG	
account_status_last_update_at	LONG	
new_credentials_last_update_at	LONG	
new_credentials_exp_date	LONG	
new_credentials_attempts_remaining	INT	
credentials_last_update_at	LONG	
credentials_exp_date	LONG	
credentials_attempts_remaining	INT	
alias_value	STRING	
alias_type	STRING	

alias_bank_account_number	STRING	
alias_bank_account_bic	STRING	
alias_bank_account_sort_code	STRING	
alias_registration_date	LONG	
alias_last_added_at	LONG	
new_alias_value	STRING	
new_alias_type	STRING	
new_alias_bank_account_number	STRING	
new_alias_bank_account_bic	STRING	
new_alias_bank_account_sort_code	STRING	
new_alias_registration_date	LONG	
new_alias_last_added_at	LONG	
auth_type	STRING	
auth_password_pasted_flag	BOOLEAN	
auth_username_pasted_flag	BOOLEAN	
auth_level	STRING	
account_last_login_attempt_at	LONG	
account_last_successful_login_at	LONG	
account_login_first_seen	LONG	
previous_auth_result	STRING	
posneg_device_type_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
posneg_account_id_value	STRING	
posneg_alias_value	STRING	
posneg_phone_value	STRING	
posneg_email_domain	STRING	
posneg_email_addr_value	STRING	
time_of_day	STRING	
day_of_week	STRING	
mismatch_mobile_settings	INT	
proxy_tor_rooted_flag	INT	
phishing_flag_and_high_risk	INT	
malware_flag_and_high_risk	INT	
high_risk_level	INT	
is_scoring	BOOLEAN	
feedback_label	STRING	
feedback_value	BOOLEAN	
customer_username	STRING	
daily_amount_limit	LONG	

new_customer_username	STRING	
new_daily_amount_limit	LONG	
max_daily_amount_limit	LONG	
card_id	STRING	
card_available_balance	LONG	
card_bin	STRING	
card_brand	STRING	
card_country_code	STRING	
card_issuer	STRING	
card_status	STRING	
cardholder_name	STRING	
customer_city	STRING	
customer_region	STRING	
new_customer_email_domain	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
decision	STRING	
session_register_date	LONG	
revelock_score	INT	
revelock_automated_actions	LIST	
revelock_alerts	LIST	
new_account_bic	STRING	
new_account_active	STRING	
new_account_cash_advance_balance	LONG	
new_account_credit_limit	LONG	
new_account_debt_balance	LONG	
new_account_debt_payment_method	STRING	
new_account_last_activity	LONG	
new_account_last_payment_date	LONG	
new_account_last_payment_amount	LONG	
new_account_number_of_cards	INT	
new_account_otb_limit	LONG	
new_account_available_credit	LONG	
new_account_pw_change_flag	BOOLEAN	
new_account_rewards_used_count	INT	
new_account_status	STRING	
new_account_type	STRING	
new_account_currency	STRING	
new_account_balance	LONG	

new_account_cards	LIST	
new_account_customers	LIST	
new_account_atc1	STRING	
new_account_atc2	STRING	
new_account_limits	SET	
account_iban	STRING	
account_bic	STRING	
account_active	STRING	
account_cash_advance_balance	LONG	
account_credit_limit	LONG	
account_debt_balance	LONG	
account_debt_payment_method	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_otb_limit	LONG	
account_available_credit	LONG	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
account_status	STRING	
account_type	STRING	
account_currency	STRING	
account_balance	LONG	
account_cards	LIST	
account_customers	LIST	
account_atc1	STRING	
account_atc2	STRING	
account_limits	SET	
new_card_bin	STRING	
new_card_pan	STRING	
new_card_effective_date	STRING	
new_card_expiration_date	STRING	
new_card_seq_number	STRING	
new_card_ver_data	STRING	
new_card_type	STRING	
new_card_category	STRING	
new_card_program	STRING	
new_card_brand	STRING	
new_card_issuer	STRING	

new_card_country_code	STRING	
new_card_last4	STRING	
new_card_status	STRING	
new_card_credit_score	INT	
new_card_transaction_amount_limit	LONG	
new_card_credit_limit	LONG	
new_card_available_balance	LONG	
new_card_number_of_cardholders	INT	
new_card_first_activation_date	LONG	
new_card_issued_date	LONG	
new_card_expected_payments_frequency	STRING	
new_card_format	STRING	
card_pan	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
card_seq_number	STRING	
card_ver_data	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_last4	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_number_of_cardholders	INT	
card_first_activation_date	LONG	
card_issued_date	LONG	
card_expected_payments_frequency	STRING	
card_format	STRING	
new_account_open_date	LONG	
new_device_account_id	STRING	
new_fuzzy_device_id	STRING	
new_device_registration_date	LONG	
new_device_first_seen	LONG	
new_device_user_agent	STRING	
new_device_type	STRING	
new_device_user_name	STRING	
new_device_brand	STRING	
new_device_app_version	STRING	
new_device_imei	STRING	

new_device_longitude	DOUBLE	
new_device_latitude	DOUBLE	
new_device_region	STRING	
new_device_ip_address	STRING	
new_device_ip_latitude	DOUBLE	
new_device_ip_longitude	DOUBLE	
new_device_ip_country_code	STRING	
new_device_ip_city	STRING	
new_device_ip_region	STRING	
new_device_ip_first_seen	LONG	
new_device_ip_proxy	BOOLEAN	
new_device_ip_proxy_type	STRING	
new_device_isp	STRING	
new_device_isp_country_code	STRING	
new_device_isp_region	STRING	
new_device_os	STRING	
new_device_os_language	STRING	
new_device_os_version	STRING	
new_device_screen_height	INT	
new_device_screen_width	INT	
new_device_java_script_enabled	BOOLEAN	
new_device_flash_enabled	BOOLEAN	
new_device_flash_version	STRING	
new_device_cookies_enabled	BOOLEAN	
new_device_bot	BOOLEAN	
new_device_malware	BOOLEAN	
new_device_rat	BOOLEAN	
new_device_phishing	BOOLEAN	
new_device_emulator	BOOLEAN	
new_device_rooted	BOOLEAN	
new_device_tor	BOOLEAN	
new_device_risk_rating	STRING	
new_device_risk_score	INT	
new_device_risk_reason	STRING	
new_device_registered	BOOLEAN	
customer_alias_dict_reg_date	LONG	
customer_alias_settle_3d	INT	
customer_alias_settle_30d	INT	
customer_alias_settle_6m	INT	
customer_settle_3d	INT	

customer_settle_30d	INT	
customer_settle_6m	INT	
customer_account_settle_3d	INT	
customer_account_settle_30d	INT	
customer_account_settle_6m	INT	
customer_alias_rep_fraud_3d	INT	
customer_alias_rep_fraud_30d	INT	
customer_alias_rep_fraud_6m	INT	
customer_rep_fraud_3d	INT	
customer_rep_fraud_30d	INT	
customer_rep_fraud_6m	INT	
customer_account_rep_fraud_3d	INT	
customer_account_rep_fraud_30d	INT	
customer_account_rep_fraud_6m	INT	
customer_alias_fraud_3d	INT	
customer_alias_fraud_30d	INT	
customer_alias_fraud_6m	INT	
customer_fraud_3d	INT	
customer_fraud_30d	INT	
customer_fraud_6m	INT	
customer_account_fraud_3d	INT	
customer_account_fraud_30d	INT	
customer_account_fraud_6m	INT	
customer_alias_rejected_3d	INT	
customer_alias_rejected_30d	INT	
customer_alias_rejected_6m	INT	
customer_rejected_3d	INT	
customer_rejected_30d	INT	
customer_rejected_6m	INT	
customer_account_rejected_3d	INT	
customer_account_rejected_30d	INT	
customer_account_rejected_6m	INT	
dict_fields_v2	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
customer_type	STRING	
customer_mothersname	STRING	
customer_gender	STRING	
customer_pwd	STRING	

customer_contactoptout	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_address_validated	BOOLEAN	
card_requested_date	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
new_device_android_id	STRING	
new_device_cbf	STRING	
new_device_mac_address	STRING	
new_device_serial_id	STRING	
new_device_browser_no_tracking	BOOLEAN	
new_device_browser_private_mode	BOOLEAN	
new_device_network_type	STRING	
new_device_session_duration	LONG	
info_type	STRING	

Digital Activity Revelock Schema

Field Name	Field Type	Field Description
event_type	STRING	
event_received_at	LONG	
event_sub_type	STRING	
new_customer_phone_number	STRING	
new_customer_email_addr	STRING	
new_customer_address	STRING	
new_customer_phone_last_update_at	LONG	
new_customer_email_last_update_at	LONG	
new_customer_addr_last_update_at	LONG	
new_account_status_last_update_at	LONG	
customer_email_addr	STRING	
lifecycle_id	STRING	
event_occurred_at	LONG	
event_external_id	STRING	
event_status	STRING	

event_status_fail_reason	STRING	
session_id	STRING	
account_id	STRING	
account_open_date	LONG	
account_age	LONG	
channel_type	STRING	
customer_id	STRING	
challenge_requested	BOOLEAN	
challenge_type	STRING	
challenge_status	STRING	
challenge_status_fail_reason	STRING	
customer_segment	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone_number	STRING	
customer_address	STRING	
customer_country_code	STRING	
customer_preferred_language	STRING	
customer_activity_status	STRING	
customer_occupation	STRING	
customer_dob	STRING	
customer_branch_id	STRING	
customer_zip_code	STRING	
customer_zip_area_code	STRING	
new_device_id	STRING	
new_device_language	STRING	
new_device_email	STRING	
new_device_phone_number	STRING	
new_device_country_code	STRING	
new_device_city	STRING	
new_device_timezone	STRING	
new_device_browser	STRING	
new_device_browser_version	STRING	
new_device_browser_language	STRING	
new_device_model	STRING	
device_id	STRING	
device_account_id	STRING	
fuzzy_device_id	STRING	
device_model	STRING	
device_phone_number	STRING	

device_email_addr	STRING	
device_language	STRING	
device_registration_date	LONG	
device_first_seen	LONG	
device_user_agent	STRING	
device_type	STRING	
device_user_name	STRING	
device_brand	STRING	
device_app_version	STRING	
device_imei	STRING	
device_longitude	DOUBLE	
device_latitude	DOUBLE	
device_country_code	STRING	
device_city	STRING	
device_region	STRING	
device_timezone	STRING	
device_ip_address	STRING	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip_country_code	STRING	
device_ip_city	STRING	
device_ip_region	STRING	
device_ip_first_seen	LONG	
device_ip_proxy	BOOLEAN	
device_ip_proxy_type	STRING	
device_isp	STRING	
device_isp_country_code	STRING	
device_isp_region	STRING	
device_os	STRING	
device_os_language	STRING	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_browser_language	STRING	
device_screen_height	INT	
device_screen_width	INT	
device_java_script_enabled	BOOLEAN	
device_flash_enabled	BOOLEAN	
device_flash_version	STRING	
device_cookies_enabled	BOOLEAN	

device_bot	BOOLEAN	
device_malware	BOOLEAN	
device_rat	BOOLEAN	
device_phishing	BOOLEAN	
device_emulator	BOOLEAN	
device_rooted	BOOLEAN	
device_tor	BOOLEAN	
device_risk_rating	STRING	
device_risk_score	INT	
device_risk_reason	STRING	
device_registered	BOOLEAN	
device_last_update_at	LONG	
device_fingerprint	STRING	
device_mobile_system_simulator	BOOLEAN	
device_ip_botnet_intensity	LONG	
device_mobile_system_root_detection_disabled	BOOLEAN	
device_is_new	BOOLEAN	
new_payee_last_added_at	LONG	
new_payee_bank_account_number	STRING	
new_payee_bank_account_bic	STRING	
new_payee_bank_account_sort_code	STRING	
new_payee_alias_value	STRING	
new_payee_alias_type	STRING	
new_payee_name	STRING	
new_payee_account_type	STRING	
customer_phone_last_update_at	LONG	
customer_email_last_update_at	LONG	
customer_addr_last_update_at	LONG	
account_status_last_update_at	LONG	
new_credentials_last_update_at	LONG	
new_credentials_exp_date	LONG	
new_credentials_attempts_remaining	INT	
credentials_last_update_at	LONG	
credentials_exp_date	LONG	
credentials_attempts_remaining	INT	
alias_value	STRING	
alias_type	STRING	
alias_bank_account_number	STRING	
alias_bank_account_bic	STRING	
alias_bank_account_sort_code	STRING	

alias_registration_date	LONG	
alias_last_added_at	LONG	
new_alias_value	STRING	
new_alias_type	STRING	
new_alias_bank_account_number	STRING	
new_alias_bank_account_bic	STRING	
new_alias_bank_account_sort_code	STRING	
new_alias_registration_date	LONG	
new_alias_last_added_at	LONG	
auth_type	STRING	
auth_password_pasted_flag	BOOLEAN	
auth_username_pasted_flag	BOOLEAN	
auth_level	STRING	
account_last_login_attempt_at	LONG	
account_last_successful_login_at	LONG	
account_login_first_seen	LONG	
previous_auth_result	STRING	
is_scoring	BOOLEAN	
feedback_label	STRING	
feedback_value	BOOLEAN	
customer_username	STRING	
daily_amount_limit	LONG	
new_customer_username	STRING	
new_daily_amount_limit	LONG	
max_daily_amount_limit	LONG	
card_id	STRING	
card_available_balance	LONG	
card_bin	STRING	
card_brand	STRING	
card_country_code	STRING	
card_issuer	STRING	
card_status	STRING	
cardholder_name	STRING	
customer_city	STRING	
customer_region	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
decision	STRING	
revelock_company	STRING	

revelock_user_id	STRING	
revelock_register_date	LONG	
revelock_score	INT	
revelock_device_id	STRING	
revelock_device_type	STRING	
revelock_device_user_agent	STRING	
revelock_device_os_family	STRING	
revelock_device_os_version	STRING	
revelock_device_browser_family	STRING	
revelock_device_browser_version	STRING	
revelock_network_ip	STRING	
revelock_network_isp	STRING	
revelock_network_country	STRING	
revelock_network_country_code	STRING	
revelock_network_region	STRING	
revelock_network_city	STRING	
revelock_alerts	LIST	
revelock_automated_actions	LIST	
session_register_date	LONG	
revelock_network_latitude	DOUBLE	
revelock_network_longitude	DOUBLE	
new_account_bic	STRING	
new_account_active	STRING	
new_account_cash_advance_balance	LONG	
new_account_credit_limit	LONG	
new_account_debt_balance	LONG	
new_account_debt_payment_method	STRING	
new_account_last_activity	LONG	
new_account_last_payment_date	LONG	
new_account_last_payment_amount	LONG	
new_account_number_of_cards	INT	
new_account_otb_limit	LONG	
new_account_available_credit	LONG	
new_account_pw_change_flag	BOOLEAN	
new_account_rewards_used_count	INT	
new_account_status	STRING	
new_account_type	STRING	
new_account_currency	STRING	
new_account_balance	LONG	
new_account_cards	LIST	

new_account_customers	LIST	
new_account_atc1	STRING	
new_account_atc2	STRING	
new_account_limits	SET	
account_iban	STRING	
account_bic	STRING	
account_active	STRING	
account_cash_advance_balance	LONG	
account_credit_limit	LONG	
account_debt_balance	LONG	
account_debt_payment_method	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_otb_limit	LONG	
account_available_credit	LONG	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
account_status	STRING	
account_type	STRING	
account_currency	STRING	
account_balance	LONG	
account_cards	LIST	
account_customers	LIST	
account_atc1	STRING	
account_atc2	STRING	
account_limits	SET	
new_card_bin	STRING	
new_card_pan	STRING	
new_card_effective_date	STRING	
new_card_expiration_date	STRING	
new_card_seq_number	STRING	
new_card_ver_data	STRING	
new_card_type	STRING	
new_card_category	STRING	
new_card_program	STRING	
new_card_brand	STRING	
new_card_issuer	STRING	
new_card_country_code	STRING	

new_card_last4	STRING	
new_card_status	STRING	
new_card_credit_score	INT	
new_card_transaction_amount_limit	LONG	
new_card_credit_limit	LONG	
new_card_available_balance	LONG	
new_card_number_of_cardholders	INT	
new_card_first_activation_date	LONG	
new_card_issued_date	LONG	
new_card_expected_payments_frequency	STRING	
new_card_format	STRING	
card_pan	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
card_seq_number	STRING	
card_ver_data	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_last4	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_number_of_cardholders	INT	
card_first_activation_date	LONG	
card_issued_date	LONG	
card_expected_payments_frequency	STRING	
card_format	STRING	
new_account_open_date	LONG	
new_device_account_id	STRING	
new_fuzzy_device_id	STRING	
new_device_registration_date	LONG	
new_device_first_seen	LONG	
new_device_user_agent	STRING	
new_device_type	STRING	
new_device_user_name	STRING	
new_device_brand	STRING	
new_device_app_version	STRING	
new_device_imei	STRING	
new_device_longitude	DOUBLE	

new_device_latitude	DOUBLE	
new_device_region	STRING	
new_device_ip_address	STRING	
new_device_ip_latitude	DOUBLE	
new_device_ip_longitude	DOUBLE	
new_device_ip_country_code	STRING	
new_device_ip_city	STRING	
new_device_ip_region	STRING	
new_device_ip_first_seen	LONG	
new_device_ip_proxy	BOOLEAN	
new_device_ip_proxy_type	STRING	
new_device_isp	STRING	
new_device_isp_country_code	STRING	
new_device_isp_region	STRING	
new_device_os	STRING	
new_device_os_language	STRING	
new_device_os_version	STRING	
new_device_screen_height	INT	
new_device_screen_width	INT	
new_device_java_script_enabled	BOOLEAN	
new_device_flash_enabled	BOOLEAN	
new_device_flash_version	STRING	
new_device_cookies_enabled	BOOLEAN	
new_device_bot	BOOLEAN	
new_device_malware	BOOLEAN	
new_device_rat	BOOLEAN	
new_device_phishing	BOOLEAN	
new_device_emulator	BOOLEAN	
new_device_rooted	BOOLEAN	
new_device_tor	BOOLEAN	
new_device_risk_rating	STRING	
new_device_risk_score	INT	
new_device_risk_reason	STRING	
new_device_registered	BOOLEAN	
customer_alias_dict_reg_date	LONG	
customer_alias_settle_3d	INT	
customer_alias_settle_30d	INT	
customer_alias_settle_6m	INT	
customer_settle_3d	INT	
customer_settle_30d	INT	

customer_settle_6m	INT	
customer_account_settle_3d	INT	
customer_account_settle_30d	INT	
customer_account_settle_6m	INT	
customer_alias_rep_fraud_3d	INT	
customer_alias_rep_fraud_30d	INT	
customer_alias_rep_fraud_6m	INT	
customer_rep_fraud_3d	INT	
customer_rep_fraud_30d	INT	
customer_rep_fraud_6m	INT	
customer_account_rep_fraud_3d	INT	
customer_account_rep_fraud_30d	INT	
customer_account_rep_fraud_6m	INT	
customer_alias_fraud_3d	INT	
customer_alias_fraud_30d	INT	
customer_alias_fraud_6m	INT	
customer_fraud_3d	INT	
customer_fraud_30d	INT	
customer_fraud_6m	INT	
customer_account_fraud_3d	INT	
customer_account_fraud_30d	INT	
customer_account_fraud_6m	INT	
customer_alias_rejected_3d	INT	
customer_alias_rejected_30d	INT	
customer_alias_rejected_6m	INT	
customer_rejected_3d	INT	
customer_rejected_30d	INT	
customer_rejected_6m	INT	
customer_account_rejected_3d	INT	
customer_account_rejected_30d	INT	
customer_account_rejected_6m	INT	
dict_fields_v2	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
customer_type	STRING	
customer_mothersname	STRING	
customer_gender	STRING	
customer_pwd	STRING	
customer_contactoptout	BOOLEAN	

customer_email_validated	BOOLEAN	
customer_address_validated	BOOLEAN	
card_requested_date	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
new_device_android_id	STRING	
new_device_cbf	STRING	
new_device_mac_address	STRING	
new_device_serial_id	STRING	
new_device_browser_no_tracking	BOOLEAN	
new_device_browser_private_mode	BOOLEAN	
new_device_network_type	STRING	
new_device_session_duration	LONG	
revelock_device_model	STRING	
revelock_device_android_id	STRING	
revelock_device_cbf	STRING	
revelock_device_mac_address	STRING	
revelock_device_serial_id	STRING	
revelock_device_browser_language	STRING	
revelock_device_browser_no_tracking	BOOLEAN	
revelock_device_browser_private_mode	BOOLEAN	
revelock_network_type	STRING	
revelock_device_screen_resolution	LIST	
revelock_session_duration	LONG	
info_type	STRING	

Digital Activity Workflow Schema

Field Name	Field Type	Field Description
event_type	STRING	
event_received_at	LONG	
event_sub_type	STRING	
new_customer_phone_number	STRING	
new_customer_email_addr	STRING	

new_customer_address	STRING	
new_customer_phone_last_update_at	LONG	
new_customer_email_last_update_at	LONG	
new_customer_addr_last_update_at	LONG	
new_account_status_last_update_at	LONG	
customer_email_addr	STRING	
lifecycle_id	STRING	
event_occurred_at	LONG	
event_external_id	STRING	
event_status	STRING	
event_status_fail_reason	STRING	
session_id	STRING	
account_id	STRING	
account_open_date	LONG	
account_age	LONG	
channel_type	STRING	
customer_id	STRING	
challenge_requested	BOOLEAN	
challenge_type	STRING	
challenge_status	STRING	
challenge_status_fail_reason	STRING	
customer_segment	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone_number	STRING	
customer_address	STRING	
customer_country_code	STRING	
customer_preferred_language	STRING	
customer_activity_status	STRING	
customer_occupation	STRING	
customer_dob	STRING	
customer_branch_id	STRING	
customer_zip_code	STRING	
customer_zip_area_code	STRING	
new_device_id	STRING	
new_device_language	STRING	
new_device_email	STRING	
new_device_phone_number	STRING	
new_device_country_code	STRING	
new_device_city	STRING	

new_device_timezone	STRING	
new_device_browser	STRING	
new_device_browser_version	STRING	
new_device_browser_language	STRING	
new_device_model	STRING	
device_id	STRING	
device_account_id	STRING	
fuzzy_device_id	STRING	
device_model	STRING	
device_phone_number	STRING	
device_email_addr	STRING	
device_language	STRING	
device_registration_date	LONG	
device_first_seen	LONG	
device_user_agent	STRING	
device_type	STRING	
device_user_name	STRING	
device_brand	STRING	
device_app_version	STRING	
device_imei	STRING	
device_longitude	DOUBLE	
device_latitude	DOUBLE	
device_country_code	STRING	
device_city	STRING	
device_region	STRING	
device_timezone	STRING	
device_ip_address	STRING	
device_ip_latitude	DOUBLE	
device_ip_longitude	DOUBLE	
device_ip_country_code	STRING	
device_ip_city	STRING	
device_ip_region	STRING	
device_ip_first_seen	LONG	
device_ip_proxy	BOOLEAN	
device_ip_proxy_type	STRING	
device_isp	STRING	
device_isp_country_code	STRING	
device_isp_region	STRING	
device_os	STRING	
device_os_language	STRING	

device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_browser_language	STRING	
device_screen_height	INT	
device_screen_width	INT	
device_java_script_enabled	BOOLEAN	
device_flash_enabled	BOOLEAN	
device_flash_version	STRING	
device_cookies_enabled	BOOLEAN	
device_bot	BOOLEAN	
device_malware	BOOLEAN	
device_rat	BOOLEAN	
device_phishing	BOOLEAN	
device_emulator	BOOLEAN	
device_rooted	BOOLEAN	
device_tor	BOOLEAN	
device_risk_rating	STRING	
device_risk_score	INT	
device_risk_reason	STRING	
device_registered	BOOLEAN	
device_last_update_at	LONG	
device_fingerprint	STRING	
device_mobile_system_simulator	BOOLEAN	
device_ip_botnet_intensity	LONG	
device_mobile_system_root_detection_disabled	BOOLEAN	
device_is_new	BOOLEAN	
new_payee_last_added_at	LONG	
new_payee_bank_account_number	STRING	
new_payee_bank_account_bic	STRING	
new_payee_bank_account_sort_code	STRING	
new_payee_alias_value	STRING	
new_payee_alias_type	STRING	
new_payee_name	STRING	
new_payee_account_type	STRING	
customer_phone_last_update_at	LONG	
customer_email_last_update_at	LONG	
customer_addr_last_update_at	LONG	
account_status_last_update_at	LONG	
new_credentials_last_update_at	LONG	

new_credentials_exp_date	LONG	
new_credentials_attempts_remaining	INT	
credentials_last_update_at	LONG	
credentials_exp_date	LONG	
credentials_attempts_remaining	INT	
alias_value	STRING	
alias_type	STRING	
alias_bank_account_number	STRING	
alias_bank_account_bic	STRING	
alias_bank_account_sort_code	STRING	
alias_registration_date	LONG	
alias_last_added_at	LONG	
new_alias_value	STRING	
new_alias_type	STRING	
new_alias_bank_account_number	STRING	
new_alias_bank_account_bic	STRING	
new_alias_bank_account_sort_code	STRING	
new_alias_registration_date	LONG	
new_alias_last_added_at	LONG	
auth_type	STRING	
auth_password_pasted_flag	BOOLEAN	
auth_username_pasted_flag	BOOLEAN	
auth_level	STRING	
account_last_login_attempt_at	LONG	
account_last_successful_login_at	LONG	
account_login_first_seen	LONG	
previous_auth_result	STRING	
posneg_device_type_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
posneg_account_id_value	STRING	
posneg_alias_value	STRING	
posneg_phone_value	STRING	
posneg_email_domain	STRING	
posneg_email_addr_value	STRING	
time_of_day	STRING	
day_of_week	STRING	
mismatch_mobile_settings	INT	
proxy_tor_rooted_flag	INT	
phishing_flag_and_high_risk	INT	

malware_flag_and_high_risk	INT	
high_risk_level	INT	
is_scoring	BOOLEAN	
feedback_label	STRING	
feedback_value	BOOLEAN	
customer_username	STRING	
daily_amount_limit	LONG	
new_customer_username	STRING	
new_daily_amount_limit	LONG	
max_daily_amount_limit	LONG	
card_id	STRING	
card_available_balance	LONG	
card_bin	STRING	
card_brand	STRING	
card_country_code	STRING	
card_issuer	STRING	
card_status	STRING	
cardholder_name	STRING	
customer_city	STRING	
customer_region	STRING	
new_customer_email_domain	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
decision	STRING	
count_login_per_account_and_ip_region_1m	LONG	
count_account_changes_per_session_id_1h	LONG	
count_succ_logins_per_account_time_of_day_3m	LONG	
distinct_login_locations_per_account_1h	LONG	
count_succ_logins_per_account_1m	LONG	
count_succ_logins_per_account_3m	LONG	
count_new_payee_per_account_1min	LONG	
count_failed_logins_per_account_id_1min	LONG	
account_previous_auth_type_changed_24h	INT	
account_previous_change_is_new_device_24h	INT	
account_previous_change_is_password_reset_24h	INT	
account_previous_activity_is_succ_login_24h	INT	
count_failed_logins_per_account_id_24h	LONG	
count_logins_per_device_id_1m	LONG	
distinct_customer_id_per_device_id_1w	LONG	

count_logins_per_device_id_90min	LONG	
count_failed_logins_per_device_id_90min	LONG	
distinct_customer_id_per_device_ip_addr_1h	LONG	
count_logins_per_device_ip_addr_1m	LONG	
count_logins_per_device_ip_addr_1h	LONG	
count_failed_logins_per_device_ip_addr_90min	LONG	
count_account_ip_country_3M	LONG	
count_account_activity_3M	LONG	
count_logins_per_account_3M	LONG	
session_register_date	LONG	
revelock_score	INT	
revelock_automated_actions	LIST	
revelock_alerts	LIST	
new_account_bic	STRING	
new_account_active	STRING	
new_account_cash_advance_balance	LONG	
new_account_credit_limit	LONG	
new_account_debt_balance	LONG	
new_account_debt_payment_method	STRING	
new_account_last_activity	LONG	
new_account_last_payment_date	LONG	
new_account_last_payment_amount	LONG	
new_account_number_of_cards	INT	
new_account_otb_limit	LONG	
new_account_available_credit	LONG	
new_account_pw_change_flag	BOOLEAN	
new_account_rewards_used_count	INT	
new_account_status	STRING	
new_account_type	STRING	
new_account_currency	STRING	
new_account_balance	LONG	
new_account_cards	LIST	
new_account_customers	LIST	
new_account_atc1	STRING	
new_account_atc2	STRING	
new_account_limits	SET	
account_iban	STRING	
account_bic	STRING	
account_active	STRING	
account_cash_advance_balance	LONG	

account_credit_limit	LONG	
account_debt_balance	LONG	
account_debt_payment_method	STRING	
account_last_activity	LONG	
account_last_payment_date	LONG	
account_last_payment_amount	LONG	
account_number_of_cards	INT	
account_otb_limit	LONG	
account_available_credit	LONG	
account_pw_change_flag	BOOLEAN	
account_rewards_used_count	INT	
account_status	STRING	
account_type	STRING	
account_currency	STRING	
account_balance	LONG	
account_cards	LIST	
account_customers	LIST	
account_atc1	STRING	
account_atc2	STRING	
account_limits	SET	
new_card_bin	STRING	
new_card_pan	STRING	
new_card_effective_date	STRING	
new_card_expiration_date	STRING	
new_card_seq_number	STRING	
new_card_ver_data	STRING	
new_card_type	STRING	
new_card_category	STRING	
new_card_program	STRING	
new_card_brand	STRING	
new_card_issuer	STRING	
new_card_country_code	STRING	
new_card_last4	STRING	
new_card_status	STRING	
new_card_credit_score	INT	
new_card_transaction_amount_limit	LONG	
new_card_credit_limit	LONG	
new_card_available_balance	LONG	
new_card_number_of_cardholders	INT	
new_card_first_activation_date	LONG	

new_card_issued_date	LONG	
new_card_expected_payments_frequency	STRING	
new_card_format	STRING	
card_pan	STRING	
card_effective_date	STRING	
card_expiration_date	STRING	
card_seq_number	STRING	
card_ver_data	STRING	
card_type	STRING	
card_category	STRING	
card_program	STRING	
card_last4	STRING	
card_credit_score	INT	
card_transaction_amount_limit	LONG	
card_credit_limit	LONG	
card_number_of_cardholders	INT	
card_first_activation_date	LONG	
card_issued_date	LONG	
card_expected_payments_frequency	STRING	
card_format	STRING	
new_account_open_date	LONG	
new_device_account_id	STRING	
new_fuzzy_device_id	STRING	
new_device_registration_date	LONG	
new_device_first_seen	LONG	
new_device_user_agent	STRING	
new_device_type	STRING	
new_device_user_name	STRING	
new_device_brand	STRING	
new_device_app_version	STRING	
new_device_imei	STRING	
new_device_longitude	DOUBLE	
new_device_latitude	DOUBLE	
new_device_region	STRING	
new_device_ip_address	STRING	
new_device_ip_latitude	DOUBLE	
new_device_ip_longitude	DOUBLE	
new_device_ip_country_code	STRING	
new_device_ip_city	STRING	
new_device_ip_region	STRING	

new_device_ip_first_seen	LONG	
new_device_ip_proxy	BOOLEAN	
new_device_ip_proxy_type	STRING	
new_device_isp	STRING	
new_device_isp_country_code	STRING	
new_device_isp_region	STRING	
new_device_os	STRING	
new_device_os_language	STRING	
new_device_os_version	STRING	
new_device_screen_height	INT	
new_device_screen_width	INT	
new_device_java_script_enabled	BOOLEAN	
new_device_flash_enabled	BOOLEAN	
new_device_flash_version	STRING	
new_device_cookies_enabled	BOOLEAN	
new_device_bot	BOOLEAN	
new_device_malware	BOOLEAN	
new_device_rat	BOOLEAN	
new_device_phishing	BOOLEAN	
new_device_emulator	BOOLEAN	
new_device_rooted	BOOLEAN	
new_device_tor	BOOLEAN	
new_device_risk_rating	STRING	
new_device_risk_score	INT	
new_device_risk_reason	STRING	
new_device_registered	BOOLEAN	
customer_alias_dict_reg_date	LONG	
customer_alias_settle_3d	INT	
customer_alias_settle_30d	INT	
customer_alias_settle_6m	INT	
customer_settle_3d	INT	
customer_settle_30d	INT	
customer_settle_6m	INT	
customer_account_settle_3d	INT	
customer_account_settle_30d	INT	
customer_account_settle_6m	INT	
customer_alias_rep_fraud_3d	INT	
customer_alias_rep_fraud_30d	INT	
customer_alias_rep_fraud_6m	INT	
customer_rep_fraud_3d	INT	

customer_rep_fraud_30d	INT	
customer_rep_fraud_6m	INT	
customer_account_rep_fraud_3d	INT	
customer_account_rep_fraud_30d	INT	
customer_account_rep_fraud_6m	INT	
customer_alias_fraud_3d	INT	
customer_alias_fraud_30d	INT	
customer_alias_fraud_6m	INT	
customer_fraud_3d	INT	
customer_fraud_30d	INT	
customer_fraud_6m	INT	
customer_account_fraud_3d	INT	
customer_account_fraud_30d	INT	
customer_account_fraud_6m	INT	
customer_alias_rejected_3d	INT	
customer_alias_rejected_30d	INT	
customer_alias_rejected_6m	INT	
customer_rejected_3d	INT	
customer_rejected_30d	INT	
customer_rejected_6m	INT	
customer_account_rejected_3d	INT	
customer_account_rejected_30d	INT	
customer_account_rejected_6m	INT	
dict_fields_v2	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
customer_type	STRING	
customer_mothersname	STRING	
customer_gender	STRING	
customer_pwd	STRING	
customer_contactoptout	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_address_validated	BOOLEAN	
card_requested_date	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	

device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
new_device_android_id	STRING	
new_device_cbf	STRING	
new_device_mac_address	STRING	
new_device_serial_id	STRING	
new_device_browser_no_tracking	BOOLEAN	
new_device_browser_private_mode	BOOLEAN	
new_device_network_type	STRING	
new_device_session_duration	LONG	
info_type	STRING	

Sanctions Rules Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
payment_type	STRING	
payment_sub_type	STRING	
product_code	STRING	
geographic_scope	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	

sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	

receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_type	STRING	
event_external_id	STRING	
event_received_at	LONG	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	

oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_value	BOOLEAN	
feedback_label	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	

sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	

receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
direction	STRING	

dd_mandate_name	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
receiver_bank_sort_code	STRING	
device_id	STRING	
dd_mandate_number	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
segment_channel_type	STRING	
login_session_id	STRING	
auth_type	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
last_sca_at	LONG	
sender_alias	STRING	
sender_alias_type	STRING	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
text_for_receiver	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_first_seen	LONG	
device_app_version	STRING	

reference_number	STRING	
reference_number_entry_mode	STRING	
sender_alias_status	STRING	
receiver_alias_status	STRING	
instant_transfer_processing_method	STRING	
originator_search_id	LONG	
originator_search_checksum_out	STRING	
originator_search_checksum_in	STRING	
originator_search_timestamp	STRING	
beneficiary_search_id	LONG	
beneficiary_search_checksum_out	STRING	
beneficiary_search_checksum_in	STRING	
beneficiary_search_timestamp	STRING	
originator_total_number_hits	INT	
originator_total_number_non_whitelisted_hits	INT	
originator_max_hit_score	DOUBLE	
originator_max_non_whitelisted_hit_score	DOUBLE	
originator_average_hit_score	DOUBLE	
originator_average_non_whitelisted_hit_score	DOUBLE	
beneficiary_total_number_hits	INT	
beneficiary_total_number_non_whitelisted_hits	INT	
beneficiary_max_hit_score	DOUBLE	
beneficiary_max_non_whitelisted_hit_score	DOUBLE	
beneficiary_average_hit_score	DOUBLE	
beneficiary_average_non_whitelisted_hit_score	DOUBLE	
sanctions_decision	STRING	
originator_match_types	SET	
beneficiary_match_types	SET	
originator_internal_match_types	SET	
beneficiary_internal_match_types	SET	
originator_screening_failed	BOOLEAN	
beneficiary_screening_failed	BOOLEAN	
transaction_notes_search_id	LONG	
transaction_notes_search_checksum_out	STRING	
transaction_notes_search_checksum_in	STRING	
transaction_notes_search_timestamp	STRING	
transaction_notes_total_number_hits	INT	
transaction_notes_max_hit_score	DOUBLE	
transaction_notes_average_hit_score	DOUBLE	
transaction_notes_screening_failed	BOOLEAN	

transaction_notes_match_reasons	SET	
transaction_notes_internal_match_reasons	SET	
screening_skipped	BOOLEAN	
transaction_notes_match_types	SET	
transaction_notes_internal_match_types	SET	
transaction_notes_average_non_whitelisted_hit_score	DOUBLE	
transaction_notes_total_number_non_whitelisted_hits	INT	
transaction_notes_max_non_whitelisted_hit_score	DOUBLE	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
ultimate_debtor_name	STRING	
ultimate_debtor_dob	STRING	
ultimate_debtor_country_code	STRING	
ultimate_creditor_name	STRING	
ultimate_creditor_dob	STRING	
ultimate_creditor_country_code	STRING	
ultimate_debtor_search_id	LONG	
ultimate_debtor_search_checksum_out	STRING	
ultimate_debtor_search_checksum_in	STRING	
ultimate_debtor_search_timestamp	STRING	
ultimate_debtor_total_number_hits	INT	
ultimate_debtor_total_number_non_whitelisted_hits	INT	
ultimate_debtor_max_hit_score	DOUBLE	
ultimate_debtor_max_non_whitelisted_hit_score	DOUBLE	
ultimate_debtor_average_hit_score	DOUBLE	
ultimate_debtor_average_non_whitelisted_hit_score	DOUBLE	
ultimate_debtor_match_types	SET	
ultimate_debtor_internal_match_types	SET	
ultimate_debtor_screening_failed	BOOLEAN	
ultimate_creditor_search_id	LONG	
ultimate_creditor_search_checksum_out	STRING	
ultimate_creditor_search_checksum_in	STRING	
ultimate_creditor_search_timestamp	STRING	
ultimate_creditor_total_number_hits	INT	
ultimate_creditor_total_number_non_whitelisted_hits	INT	
ultimate_creditor_max_hit_score	DOUBLE	
ultimate_creditor_max_non_whitelisted_hit_score	DOUBLE	
ultimate_creditor_average_hit_score	DOUBLE	

ultimate_creditor_average_non_whitelisted_hit_score	DOUBLE	
ultimate_creditor_match_types	SET	
ultimate_creditor_internal_match_types	SET	
ultimate_creditor_screening_failed	BOOLEAN	

Transfers AML Plan Schema

Field Name	Field Type	Field Description
event_occurred_at	LONG	
lifecycle_id	STRING	
bulk_id	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_id	STRING	
sender_dob	STRING	
sender_street_addr	STRING	
sender_zip	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	

sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_id	STRING	
receiver_dob	STRING	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	

device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_external_id	STRING	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_value	BOOLEAN	
sender_correspondent_bank_bic	STRING	

sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_region	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_occupation	STRING	
sender_risk_rating	STRING	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_currency	STRING	

sender_account_bic	STRING	
sender_account_cash_advance_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_debt_balance	LONG	
sender_account_debt_payment_method	STRING	
sender_account_iban	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_number_of_cards	INT	
sender_account_open_date	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_status	STRING	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
receiver_segment	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_region	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_bic	STRING	
receiver_account_cash_advance_balance	LONG	

receiver_account_credit_limit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_debt_payment_method	STRING	
receiver_account_iban	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_number_of_cards	INT	
receiver_account_open_date	LONG	
receiver_account_available_credit	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_status	STRING	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
sender_aum_2m	BOOLEAN	
sender_aum_5m	BOOLEAN	
receiver_aum_2m	BOOLEAN	
receiver_aum_5m	BOOLEAN	
receiver_occupation_status	STRING	
sender_occupation_status	STRING	
sender_kyc_level	STRING	
sender_active	BOOLEAN	
jurisdiction	STRING	
receiver_active	BOOLEAN	
receiver_account_active	BOOLEAN	
sender_account_active	BOOLEAN	
receiver_name	STRING	
receiver_account_otb_limit	LONG	
channel_type	STRING	
dd_mandate_name	STRING	
dd_mandate_number	STRING	
geographic_scope	STRING	
sender_kyc_lvl	STRING	
sender_bank_sort_code	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
sender_account_atc1	STRING	
sender_account_atc2	STRING	

receiver_bank_sort_code	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
device_id	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
login_session_id	STRING	
session_id	STRING	
pass_reset	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
device_ip_country_code	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_language	STRING	
sender_alias	STRING	
sender_alias_type	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
text_for_receiver	STRING	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_app_version	STRING	
reference_number	STRING	
count_blacklist_words	INT	
direction	STRING	
payment_type	STRING	
payment_sub_type	STRING	
segment_channel_type	STRING	
last_sca_at	LONG	

sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
sender_alias_status	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
receiver_alias_status	STRING	
device_first_seen	LONG	
instant_transfer_processing_method	STRING	
reference_number_entry_mode	STRING	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	

Transfers API Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	

payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	

receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_external_id	STRING	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	

dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	

sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	

receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	

receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
device_id	STRING	
channel_type	STRING	
dd_mandate_name	STRING	
dd_mandate_number	STRING	
sender_bank_sort_code	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_sort_code	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
pass_reset	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
device_ip_country_code	STRING	
login_session_id	STRING	
session_id	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_language	STRING	
sender_alias	STRING	
sender_alias_type	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
text_for_receiver	STRING	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_app_version	STRING	
reference_number	STRING	
session_register_date	LONG	

device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	
device_ip_region	STRING	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
sender_aum_2m	BOOLEAN	
sender_aum_5m	BOOLEAN	
receiver_aum_2m	BOOLEAN	
receiver_aum_5m	BOOLEAN	
sender_kyc_level	STRING	
sender_active	BOOLEAN	
jurisdiction	STRING	
receiver_active	BOOLEAN	
receiver_account_active	BOOLEAN	
sender_account_active	BOOLEAN	
device_fingerprint	STRING	
event_received_at	LONG	
event_type	STRING	
feedback_label	STRING	
instant_transfers_review_outcome	BOOLEAN	
receiver_account_id	STRING	
sender_account_id	STRING	
direction	STRING	
payment_type	STRING	
payment_sub_type	STRING	
segment_channel_type	STRING	
last_sca_at	LONG	
sender_alias_registration_time	LONG	

sender_alias_last_update_time	LONG	
sender_alias_status	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
receiver_alias_status	STRING	
device_first_seen	LONG	
instant_transfer_processing_method	STRING	
reference_number_entry_mode	STRING	
transaction_goal	STRING	
transaction_method	STRING	
device_registration_date	LONG	
receiver_alias_settle_3d	INT	
receiver_alias_settle_30d	INT	
receiver_alias_settle_6m	INT	
receiver_settle_3d	INT	
receiver_settle_30d	INT	
receiver_settle_6m	INT	
receiver_account_settle_3d	INT	
receiver_account_settle_30d	INT	
receiver_account_settle_6m	INT	
receiver_alias_rep_fraud_3d	INT	
receiver_alias_rep_fraud_30d	INT	
receiver_alias_rep_fraud_6m	INT	
receiver_rep_fraud_3d	INT	
receiver_rep_fraud_30d	INT	
receiver_rep_fraud_6m	INT	
receiver_account_rep_fraud_3d	INT	
receiver_account_rep_fraud_30d	INT	
receiver_account_rep_fraud_6m	INT	
receiver_alias_fraud_3d	INT	
receiver_alias_fraud_30d	INT	
receiver_alias_fraud_6m	INT	
receiver_fraud_3d	INT	
receiver_fraud_30d	INT	
receiver_fraud_6m	INT	
receiver_account_fraud_3d	INT	
receiver_account_fraud_30d	INT	
receiver_account_fraud_6m	INT	
receiver_alias_rejected_3d	INT	
receiver_alias_rejected_30d	INT	

receiver_alias_rejected_6m	INT	
receiver_rejected_3d	INT	
receiver_rejected_30d	INT	
receiver_rejected_6m	INT	
receiver_account_rejected_3d	INT	
receiver_account_rejected_30d	INT	
receiver_account_rejected_6m	INT	
receiver_alias_portab_date	LONG	
receiver_alias_dict_reg_date	LONG	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_dict_fields_v2	TUPLE	
receiver_dict_fields_v2	TUPLE	
external_fraud_service	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
sender_account_limits	SET	
receiver_account_limits	SET	
is_domestic	BOOLEAN	
ultimate_debtor_name	STRING	
ultimate_debtor_dob	STRING	
ultimate_debtor_country_code	STRING	
ultimate_creditor_name	STRING	
ultimate_creditor_dob	STRING	
ultimate_creditor_country_code	STRING	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	

user_id	STRING	
user_name	STRING	

Transfers Decision Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
payment_type	STRING	
payment_sub_type	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	

sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	

sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_type	STRING	
event_external_id	STRING	
event_received_at	LONG	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_label	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	

sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	

sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	

receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
direction	STRING	
dd_mandate_name	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
receiver_bank_sort_code	STRING	
device_id	STRING	
decision	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
dd_mandate_number	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	

receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
segment_channel_type	STRING	
login_session_id	STRING	
session_id	STRING	
auth_type	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
pass_reset	BOOLEAN	
last_sca_at	LONG	
sender_alias	STRING	
sender_alias_type	STRING	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
text_for_receiver	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_first_seen	LONG	
device_app_version	STRING	
reference_number	STRING	
reference_number_entry_mode	STRING	
sender_alias_status	STRING	
receiver_alias_status	STRING	
instant_transfer_processing_method	STRING	
sender_account_id	STRING	

receiver_account_id	STRING	
receiver_alias_email_domain	STRING	
posneg_sender_bank_country_value	STRING	
posneg_receiver_bank_country_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
time_of_day	STRING	
sca_required	BOOLEAN	
posneg_sender_account_value	STRING	
posneg_receiver_account_value	STRING	
posneg_dd_company_value	STRING	
posneg_sender_alias_value	STRING	
posneg_receiver_alias_value	STRING	
posneg_sender_bank_bic_value	STRING	
posneg_receiver_bank_bic_value	STRING	
posneg_reference_number_value	STRING	
posneg_sender_bank_sort_code_value	STRING	
posneg_receiver_bank_sort_code_value	STRING	
posneg_receiver_alias_phone_value	STRING	
posneg_receiver_alias_email_domain_value	STRING	
distinct_sender_per_device_id_90m	LONG	
sum_amt_instant_transfers_per_sender_alias_90m	LONG	
distinct_sender_per_email_1m	LONG	
distinct_sender_per_phone_1m	LONG	
count_sender_device_id_6m	LONG	
count_sender_device_ip_country_code_count_6m	LONG	
sum_amount_sender_last_sca_90d	LONG	
count_sender_last_sca_90d	LONG	
sum_amount_sender_last_sca_30h	LONG	
count_sender_last_sca_30h	LONG	
distinct_receiver_same_name_per_sender_24h	LONG	
count_sender_receiver_6m	LONG	
count_sender_time_of_day_6m	LONG	
distinct_month_per_sender_3m_delay_1m	LONG	
sum_amount_sender_3m_delay_1m	LONG	
count_sender_3m_delay_1m	LONG	
count_sender_online_3m_delay_24h	LONG	
count_sender_telephone_3m_delay_24h	LONG	
distinct_day_year_per_sender_instant_transfer_3m_delay_24h	LONG	
distinct_month_per_sender_online_3m_delay_24h	LONG	

distinct_month_per_sender_telephone_3m_delay_24h	LONG	
count_sender_dd_3m_delay_24h	LONG	
count_sender_instant_transfer_3m_delay_24h	LONG	
count_sender_6m_delay_24h	LONG	
sum_amt_bill_payments_per_sender_out_work_days_1m	LONG	
count_bill_payments_per_sender_out_work_days_1m	LONG	
count_sender_cross_border_amount_3k_1month	LONG	
sum_amount_per_sender_1m	LONG	
count_sender_1m	LONG	
distinct_day_week_per_sender_1w	LONG	
sum_amount_per_sender_1w	LONG	
count_dd_return_per_sender_3m	LONG	
count_sender_6m	LONG	
sender_prev_device_lat_1h	DOUBLE	
sender_prev_device_long_1h	DOUBLE	
distinct_country_per_sender_24h	LONG	
count_sender_24h	LONG	
count_sender_pw_change_24h	LONG	
auth_type_changed_24h	INT	
count_sender_online_24h	LONG	
count_sender_telephone_24h	LONG	
sender_device_language_changed_24h	INT	
login_session_id_changed_24h	INT	
count_sender_instant_transfer_24h	LONG	
count_sender_dd_24h	LONG	
distinct_receiver_per_sender_24h	LONG	
sum_amount_per_sender_telephone_24h	LONG	
count_sender_cross_border_amount_3k_24h	LONG	
sum_amount_per_sender_24h	LONG	
count_dd_return_per_sender_24h	LONG	
count_failed_trx_sender_24h	LONG	
distinct_device_id_per_sender_90m	LONG	
count_sender_90min	LONG	
avg_daily_amount_per_sender_1w	DOUBLE	
avg_monthly_count_sender_3m_delay_1m	DOUBLE	
avg_monthly_amount_per_sender_3m_delay_1m	DOUBLE	
avg_monthly_count_sender_telephone_3m_delay_24h	DOUBLE	
avg_monthly_count_sender_online_3m_delay_24h	DOUBLE	
avg_daily_count_sender_instant_transfer_3m_delay_24h	DOUBLE	
enforced_risk_level	STRING	

device_fingerprint	STRING	
day_of_week	STRING	
new_device	BOOLEAN	
distinct_account_per_device_3m	LONG	
count_account_activity_out_work_days_3m	LONG	
count_account_activity_time_of_day_3m	LONG	
count_account_activity_3m	LONG	
instant_transfers_review_outcome	BOOLEAN	
count_sender_3m_delay_24h	LONG	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	
device_ip_region	STRING	
revelock_score	INT	
revelock_automated_actions	LIST	
revelock_alerts	LIST	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
transaction_goal	STRING	
transaction_method	STRING	
device_latitude_double	DOUBLE	
device_longitude_double	DOUBLE	
device_latitude	STRING	
device_longitude	STRING	
device_registration_date	LONG	
sender_account_limits	SET	
receiver_account_limits	SET	
sender_transaction_original_amount	LONG	

sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_dict_fields_v2	TUPLE	
receiver_dict_fields_v2	TUPLE	
external_fraud_service	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
new_payees_24h	LONG	
count_inbound_24h	LONG	
sum_amount_per_receiver_7d	LONG	
sum_amount_per_sender_7d	LONG	
count_per_sender_7d	LONG	
count_per_receiver_7d	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	

Transfers FraudForce Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
payment_type	STRING	
payment_sub_type	STRING	
product_code	STRING	
geographic_scope	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	

sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	

receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	

approval_type	STRING	
approval_status	STRING	
event_type	STRING	
event_external_id	STRING	
event_received_at	LONG	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_value	BOOLEAN	
feedback_label	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	

receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	

sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	

receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
direction	STRING	
dd_mandate_name	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
receiver_bank_sort_code	STRING	
device_id	STRING	
decision	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
dd_mandate_number	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
sca_required	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
segment_channel_type	STRING	
login_session_id	STRING	
session_id	STRING	
auth_type	STRING	

sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
pass_reset	BOOLEAN	
last_sca_at	LONG	
sender_alias	STRING	
sender_alias_type	STRING	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
text_for_receiver	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_first_seen	LONG	
device_app_version	STRING	
reference_number	STRING	
reference_number_entry_mode	STRING	
sender_alias_status	STRING	
receiver_alias_status	STRING	
instant_transfer_processing_method	STRING	
sender_account_id	STRING	
receiver_account_id	STRING	
ff_id	STRING	
ff_result	STRING	
ff_tracking_number	LONG	
ff_device_alias	LONG	
ff_device_blackbox_age	INT	
ff_device_blackbox_timestamp	STRING	
ff_device_browser_charset	STRING	
ff_device_browser_cookies_enabled	BOOLEAN	
ff_device_browser_configured_language	STRING	

ff_device_browser_flash_enabled	BOOLEAN	
ff_device_browser_flash_installed	BOOLEAN	
ff_device_browser_flash_storage_enabled	BOOLEAN	
ff_device_browser_flash_version	STRING	
ff_device_browser_language	STRING	
ff_device_browser_type	STRING	
ff_device_browser_timezone	STRING	
ff_device_browser_version	STRING	
ff_device_first_seen	STRING	
ff_device_is_new	BOOLEAN	
ff_device_mobile_app_bundle_id	STRING	
ff_device_mobile_app_debug	BOOLEAN	
ff_device_mobile_app_exe_name	STRING	
ff_device_mobile_app_market_id	STRING	
ff_device_mobile_app_name	STRING	
ff_device_mobile_app_orientation	STRING	
ff_device_mobile_app_proc_name	STRING	
ff_device_mobile_app_signer_id	STRING	
ff_device_mobile_app_vendor_id	STRING	
ff_device_mobile_app_version	STRING	
ff_device_mobile_brand	STRING	
ff_device_mobile_build_device	STRING	
ff_device_mobile_build_product	STRING	
ff_device_mobile_charging	BOOLEAN	
ff_device_mobile_imei	STRING	
ff_device_mobile_location_altitude	DOUBLE	
ff_device_mobile_location_enabled	BOOLEAN	
ff_device_mobile_location_latitude	DOUBLE	
ff_device_mobile_location_longitude	DOUBLE	
ff_device_mobile_location_timestamp	STRING	
ff_device_mobile_location_timezone	STRING	
ff_device_mobile_manufacturer	STRING	
ff_device_mobile_model	STRING	
ff_device_mobile_orientation	STRING	
ff_device_mobile_screen_resolution	STRING	
ff_device_mobile_system_allow_unknown_store	BOOLEAN	
ff_device_mobile_system_android_sdk_level	STRING	
ff_device_mobile_system_build_fingerprint	STRING	
ff_device_mobile_system_carrier	STRING	
ff_device_mobile_system_carrier_country_code	STRING	

ff_device_mobile_system_cellular_network	STRING	
ff_device_mobile_system_hostname	STRING	
ff_device_mobile_system_jailroot_detected	BOOLEAN	
ff_device_mobile_system_locale_currency	STRING	
ff_device_mobile_system_locale_lang	STRING	
ff_device_mobile_system_network_operator_country	STRING	
ff_device_mobile_system_network_operator_name	STRING	
ff_device_mobile_system_os_version	STRING	
ff_device_mobile_system_root_detection_disabled	BOOLEAN	
ff_device_mobile_system_simulator	BOOLEAN	
ff_device_mobile_system_uptime	STRING	
ff_device_mobile_system_voip_allowed	BOOLEAN	
ff_device_os	STRING	
ff_device_registration_result_match_status	STRING	
ff_device_registration_result_measure_of_charge	STRING	
ff_device_screen	STRING	
ff_device_recognition_source	STRING	
ff_device_type	STRING	
ff_real_ip_address	STRING	
ff_real_ip_botnet_score	INT	
ff_real_ip_botnet_intensity	INT	
ff_real_ip_botnet_lastseen	STRING	
ff_real_ip_isp	STRING	
ff_real_ip_loc_city	STRING	
ff_real_ip_loc_country	STRING	
ff_real_ip_loc_country_code	STRING	
ff_real_ip_loc_latitude	DOUBLE	
ff_real_ip_loc_longitude	DOUBLE	
ff_real_ip_loc_region	STRING	
ff_real_ip_org	STRING	
ff_real_ip_proxy	STRING	
ff_real_ip_source	STRING	
ff_stated_ip_address	STRING	
ff_stated_ip_botnet_score	INT	
ff_stated_ip_botnet_intensity	INT	
ff_stated_ip_botnet_lastseen	STRING	
ff_stated_ip_isp	STRING	
ff_stated_ip_loc_city	STRING	
ff_stated_ip_loc_country	STRING	
ff_stated_ip_loc_country_code	STRING	

ff_stated_ip_loc_latitude	DOUBLE	
ff_stated_ip_loc_longitude	DOUBLE	
ff_stated_ip_loc_region	STRING	
ff_stated_ip_org	STRING	
ff_stated_ip_proxy	STRING	
ff_stated_ip_source	STRING	
instant_transfers_review_outcome	BOOLEAN	
transaction_goal	STRING	
transaction_method	STRING	
receiver_alias_settle_3d	INT	
receiver_alias_settle_30d	INT	
receiver_alias_settle_6m	INT	
receiver_settle_3d	INT	
receiver_settle_30d	INT	
receiver_settle_6m	INT	
receiver_account_settle_3d	INT	
receiver_account_settle_30d	INT	
receiver_account_settle_6m	INT	
receiver_alias_rep_fraud_3d	INT	
receiver_alias_rep_fraud_30d	INT	
receiver_alias_rep_fraud_6m	INT	
receiver_rep_fraud_3d	INT	
receiver_rep_fraud_30d	INT	
receiver_rep_fraud_6m	INT	
receiver_account_rep_fraud_3d	INT	
receiver_account_rep_fraud_30d	INT	
receiver_account_rep_fraud_6m	INT	
receiver_alias_fraud_3d	INT	
receiver_alias_fraud_30d	INT	
receiver_alias_fraud_6m	INT	
receiver_fraud_3d	INT	
receiver_fraud_30d	INT	
receiver_fraud_6m	INT	
receiver_account_fraud_3d	INT	
receiver_account_fraud_30d	INT	
receiver_account_fraud_6m	INT	
receiver_alias_rejected_3d	INT	
receiver_alias_rejected_30d	INT	
receiver_alias_rejected_6m	INT	
receiver_rejected_3d	INT	

receiver_rejected_30d	INT	
receiver_rejected_6m	INT	
receiver_account_rejected_3d	INT	
receiver_account_rejected_30d	INT	
receiver_account_rejected_6m	INT	
receiver_alias_portab_date	LONG	
receiver_alias_dict_reg_date	LONG	
device_registration_date	LONG	
sender_account_limits	SET	
receiver_account_limits	SET	
sender_dict_fields_v2	TUPLE	
receiver_dict_fields_v2	TUPLE	
external_fraud_service	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	

Transfers Plan Output Schema

Field Name	Field Type	Field Description
event_occurred_at	LONG	
lifecycle_id	STRING	
bulk_id	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	

sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_id	STRING	
sender_dob	STRING	
sender_street_addr	STRING	
sender_zip	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	

receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_id	STRING	
receiver_dob	STRING	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	

approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_external_id	STRING	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_value	BOOLEAN	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	

receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_region	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_occupation	STRING	
sender_risk_rating	STRING	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_currency	STRING	
sender_account_bic	STRING	
sender_account_cash_advance_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_debt_balance	LONG	
sender_account_debt_payment_method	STRING	
sender_account_iban	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_number_of_cards	INT	
sender_account_open_date	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_status	STRING	

sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
receiver_segment	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_region	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_bic	STRING	
receiver_account_cash_advance_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_debt_payment_method	STRING	
receiver_account_iban	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_number_of_cards	INT	
receiver_account_open_date	LONG	
receiver_account_available_credit	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_status	STRING	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	

sender_aum_2m	BOOLEAN	
sender_aum_5m	BOOLEAN	
receiver_aum_2m	BOOLEAN	
receiver_aum_5m	BOOLEAN	
receiver_occupation_status	STRING	
sender_occupation_status	STRING	
sender_kyc_level	STRING	
sender_active	BOOLEAN	
jurisdiction	STRING	
receiver_active	BOOLEAN	
receiver_account_active	BOOLEAN	
sender_account_active	BOOLEAN	
receiver_name	STRING	
receiver_account_otb_limit	LONG	
channel_type	STRING	
dd_mandate_name	STRING	
dd_mandate_number	STRING	
geographic_scope	STRING	
sender_kyc_lvl	STRING	
sender_bank_sort_code	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_bank_sort_code	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
device_id	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
login_session_id	STRING	
session_id	STRING	
pass_reset	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	

device_ip_country_code	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_language	STRING	
sender_alias	STRING	
sender_alias_type	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
text_for_receiver	STRING	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_app_version	STRING	
reference_number	STRING	
direction	STRING	
payment_type	STRING	
payment_sub_type	STRING	
segment_channel_type	STRING	
last_sca_at	LONG	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
sender_alias_status	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
receiver_alias_status	STRING	
device_first_seen	LONG	
instant_transfer_processing_method	STRING	
reference_number_entry_mode	STRING	
dummy_th	INT	
amount	LONG	
customer_id	STRING	
currency_code	STRING	
account_iban	STRING	
customer_street_addr	STRING	
account_open_date	LONG	
is_risky_geo_sender_country	BOOLEAN	

is_risky_geo_sender_bank	BOOLEAN	
is_risky_geo_sender_correspondent	BOOLEAN	
is_risky_geo_receiv_country	BOOLEAN	
is_risky_geo_receiv_bank	BOOLEAN	
is_risky_geo_receiv_correspondent	BOOLEAN	
is_vhr_geo_sender_country	BOOLEAN	
is_vhr_geo_sender_bank	BOOLEAN	
is_vhr_geo_sender_correspondent	BOOLEAN	
is_vhr_geo_receiv_country	BOOLEAN	
is_vhr_geo_receiv_bank	BOOLEAN	
is_vhr_geo_receiv_correspondent	BOOLEAN	
is_risky_device_ip	BOOLEAN	
is_hr_ip_location	BOOLEAN	
is_risky_cp_sender_name	BOOLEAN	
is_risky_cp_receiv_name	BOOLEAN	
is_risky_cp_sender_account	BOOLEAN	
is_risky_cp_receiv_account	BOOLEAN	
is_risky_cp_sender_addr	BOOLEAN	
is_risky_cp_receiv_addr	BOOLEAN	
is_vhr_ip_location	BOOLEAN	
is_incoming	BOOLEAN	
is_outgoing	BOOLEAN	
is_risky_geo	BOOLEAN	
is_very_high_risk_geo	BOOLEAN	
is_risky_counterparty	BOOLEAN	
is_round_amount	BOOLEAN	
counterparty_identifier	STRING	
is_foreign_terminal	BOOLEAN	
is_risky_ip_location	BOOLEAN	
counterparty_bank_country_code	STRING	
count_blacklist_words	INT	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
original_amount	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	

device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	

Transfers PSD2 Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
payment_type	STRING	
payment_sub_type	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	

sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	

user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_type	STRING	
event_external_id	STRING	
event_received_at	LONG	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_label	STRING	
sender_correspondent_bank_bic	STRING	

sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	

sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	

receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
direction	STRING	
dd_mandate_name	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
receiver_bank_sort_code	STRING	
device_id	STRING	
decision	STRING	

fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
dd_mandate_number	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
segment_channel_type	STRING	
login_session_id	STRING	
session_id	STRING	
auth_type	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
pass_reset	BOOLEAN	
last_sca_at	LONG	
sender_alias	STRING	
sender_alias_type	STRING	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
text_for_receiver	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_first_seen	LONG	
device_app_version	STRING	

reference_number	STRING	
reference_number_entry_mode	STRING	
sender_alias_status	STRING	
receiver_alias_status	STRING	
instant_transfer_processing_method	STRING	
sender_account_id	STRING	
receiver_account_id	STRING	
receiver_alias_email_domain	STRING	
posneg_sender_bank_country_value	STRING	
posneg_receiver_bank_country_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
time_of_day	STRING	
sca_required	BOOLEAN	
posneg_sender_account_value	STRING	
posneg_receiver_account_value	STRING	
posneg_dd_company_value	STRING	
posneg_sender_alias_value	STRING	
posneg_receiver_alias_value	STRING	
posneg_sender_bank_bic_value	STRING	
posneg_receiver_bank_bic_value	STRING	
posneg_reference_number_value	STRING	
posneg_sender_bank_sort_code_value	STRING	
posneg_receiver_bank_sort_code_value	STRING	
posneg_receiver_alias_phone_value	STRING	
posneg_receiver_alias_email_domain_value	STRING	
distinct_sender_per_device_id_90m	LONG	
sum_amt_instant_transfers_per_sender_alias_90m	LONG	
distinct_sender_per_email_1m	LONG	
distinct_sender_per_phone_1m	LONG	
count_sender_device_id_6m	LONG	
count_sender_device_ip_country_code_count_6m	LONG	
sum_amount_sender_last_sca_90d	LONG	
count_sender_last_sca_90d	LONG	
sum_amount_sender_last_sca_30h	LONG	
count_sender_last_sca_30h	LONG	
distinct_receiver_same_name_per_sender_24h	LONG	
count_sender_receiver_6m	LONG	
count_sender_time_of_day_6m	LONG	
distinct_month_per_sender_3m_delay_1m	LONG	

sum_amount_sender_3m_delay_1m	LONG	
count_sender_3m_delay_1m	LONG	
count_sender_online_3m_delay_24h	LONG	
count_sender_telephone_3m_delay_24h	LONG	
distinct_day_year_per_sender_instant_transfer_3m_delay_24h	LONG	
distinct_month_per_sender_online_3m_delay_24h	LONG	
distinct_month_per_sender_telephone_3m_delay_24h	LONG	
count_sender_dd_3m_delay_24h	LONG	
count_sender_instant_transfer_3m_delay_24h	LONG	
count_sender_6m_delay_24h	LONG	
sum_amt_bill_payments_per_sender_out_work_days_1m	LONG	
count_bill_payments_per_sender_out_work_days_1m	LONG	
count_sender_cross_border_amount_3k_1month	LONG	
sum_amount_per_sender_1m	LONG	
count_sender_1m	LONG	
distinct_day_week_per_sender_1w	LONG	
sum_amount_per_sender_1w	LONG	
count_dd_return_per_sender_3m	LONG	
count_sender_6m	LONG	
sender_prev_device_lat_1h	DOUBLE	
sender_prev_device_long_1h	DOUBLE	
distinct_country_per_sender_24h	LONG	
count_sender_24h	LONG	
count_sender_pw_change_24h	LONG	
auth_type_changed_24h	INT	
count_sender_online_24h	LONG	
count_sender_telephone_24h	LONG	
sender_device_language_changed_24h	INT	
login_session_id_changed_24h	INT	
count_sender_instant_transfer_24h	LONG	
count_sender_dd_24h	LONG	
distinct_receiver_per_sender_24h	LONG	
sum_amount_per_sender_telephone_24h	LONG	
count_sender_cross_border_amount_3k_24h	LONG	
sum_amount_per_sender_24h	LONG	
count_dd_return_per_sender_24h	LONG	
count_failed_trx_sender_24h	LONG	
distinct_device_id_per_sender_90m	LONG	
count_sender_90min	LONG	
avg_daily_amount_per_sender_1w	DOUBLE	

avg_monthly_count_sender_3m_delay_1m	DOUBLE	
avg_monthly_amount_per_sender_3m_delay_1m	DOUBLE	
avg_monthly_count_sender_telephone_3m_delay_24h	DOUBLE	
avg_monthly_count_sender_online_3m_delay_24h	DOUBLE	
avg_daily_count_sender_instant_transfer_3m_delay_24h	DOUBLE	
enforced_risk_level	STRING	
device_fingerprint	STRING	
day_of_week	STRING	
new_device	BOOLEAN	
distinct_account_per_device_3m	LONG	
count_account_activity_out_work_days_3m	LONG	
count_account_activity_time_of_day_3m	LONG	
count_account_activity_3m	LONG	
instant_transfers_review_outcome	BOOLEAN	
count_sender_3m_delay_24h	LONG	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	
device_ip_region	STRING	
revelock_score	INT	
revelock_automated_actions	LIST	
revelock_alerts	LIST	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
transaction_goal	STRING	
transaction_method	STRING	
device_latitude_double	DOUBLE	
device_longitude_double	DOUBLE	

device_latitude	STRING	
device_longitude	STRING	
device_registration_date	LONG	
sender_account_limits	SET	
receiver_account_limits	SET	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_dict_fields_v2	TUPLE	
receiver_dict_fields_v2	TUPLE	
external_fraud_service	TUPLE	
new_payees_24h	LONG	
count_inbound_24h	LONG	
sum_amount_per_receiver_7d	LONG	
sum_amount_per_sender_7d	LONG	
count_per_sender_7d	LONG	
count_per_receiver_7d	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	

Transfers Revelock Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
payment_type	STRING	
payment_sub_type	STRING	
product_code	STRING	
geographic_scope	STRING	

payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	

receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	

approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_type	STRING	
event_external_id	STRING	
event_received_at	LONG	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_value	BOOLEAN	
feedback_label	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	

receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	

sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	

receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
direction	STRING	
dd_mandate_name	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
receiver_bank_sort_code	STRING	
device_id	STRING	
decision	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
dd_mandate_number	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
sca_required	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
segment_channel_type	STRING	

login_session_id	STRING	
session_id	STRING	
auth_type	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
pass_reset	BOOLEAN	
last_sca_at	LONG	
sender_alias	STRING	
sender_alias_type	STRING	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
text_for_receiver	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_first_seen	LONG	
device_app_version	STRING	
reference_number	STRING	
reference_number_entry_mode	STRING	
sender_alias_status	STRING	
receiver_alias_status	STRING	
instant_transfer_processing_method	STRING	
sender_account_id	STRING	
receiver_account_id	STRING	
device_fingerprint	STRING	
instant_transfers_review_outcome	BOOLEAN	
revelock_company	STRING	
revelock_user_id	STRING	
revelock_register_date	LONG	
revelock_score	INT	

revelock_device_id	STRING	
revelock_device_type	STRING	
revelock_device_user_agent	STRING	
revelock_device_os_family	STRING	
revelock_device_os_version	STRING	
revelock_device_browser_family	STRING	
revelock_device_browser_version	STRING	
revelock_network_ip	STRING	
revelock_network_isp	STRING	
revelock_network_country	STRING	
revelock_network_country_code	STRING	
revelock_network_region	STRING	
revelock_network_city	STRING	
revelock_alerts	LIST	
revelock_automated_actions	LIST	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	
device_ip_region	STRING	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
revelock_network_latitude	DOUBLE	
revelock_network_longitude	DOUBLE	
transaction_goal	STRING	
transaction_method	STRING	
device_registration_date	LONG	
sender_account_limits	SET	
receiver_account_limits	SET	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	

receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_dict_fields_v2	TUPLE	
receiver_dict_fields_v2	TUPLE	
external_fraud_service	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	
revelock_device_model	STRING	
revelock_device_android_id	STRING	
revelock_device_cbf	STRING	
revelock_device_mac_address	STRING	
revelock_device_serial_id	STRING	
revelock_device_browser_language	STRING	
revelock_device_browser_no_tracking	BOOLEAN	
revelock_device_browser_private_mode	BOOLEAN	
revelock_network_type	STRING	
revelock_device_screen_resolution	LIST	
revelock_session_duration	LONG	

Transfers TFRB Plan Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
payment_type	STRING	
payment_sub_type	STRING	
product_code	STRING	

payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	

receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	

approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_type	STRING	
event_external_id	STRING	
event_received_at	LONG	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_label	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	

receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	

sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	

receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
direction	STRING	
dd_mandate_name	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
receiver_bank_sort_code	STRING	
device_id	STRING	
decision	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
dd_mandate_number	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
sca_required	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
segment_channel_type	STRING	
login_session_id	STRING	

session_id	STRING	
auth_type	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
pass_reset	BOOLEAN	
last_sca_at	LONG	
sender_alias	STRING	
sender_alias_type	STRING	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
text_for_receiver	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_first_seen	LONG	
device_app_version	STRING	
reference_number	STRING	
reference_number_entry_mode	STRING	
sender_alias_status	STRING	
receiver_alias_status	STRING	
instant_transfer_processing_method	STRING	
sender_account_id	STRING	
receiver_account_id	STRING	
device_fingerprint	STRING	
instant_transfers_review_outcome	BOOLEAN	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	

device_ip_region	STRING	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
transaction_goal	STRING	
transaction_method	STRING	
receiver_alias_settle_3d	INT	
receiver_alias_settle_30d	INT	
receiver_alias_settle_6m	INT	
receiver_settle_3d	INT	
receiver_settle_30d	INT	
receiver_settle_6m	INT	
receiver_account_settle_3d	INT	
receiver_account_settle_30d	INT	
receiver_account_settle_6m	INT	
receiver_alias_rep_fraud_3d	INT	
receiver_alias_rep_fraud_30d	INT	
receiver_alias_rep_fraud_6m	INT	
receiver_rep_fraud_3d	INT	
receiver_rep_fraud_30d	INT	
receiver_rep_fraud_6m	INT	
receiver_account_rep_fraud_3d	INT	
receiver_account_rep_fraud_30d	INT	
receiver_account_rep_fraud_6m	INT	
receiver_alias_fraud_3d	INT	
receiver_alias_fraud_30d	INT	
receiver_alias_fraud_6m	INT	
receiver_fraud_3d	INT	
receiver_fraud_30d	INT	
receiver_fraud_6m	INT	
receiver_account_fraud_3d	INT	

receiver_account_fraud_30d	INT	
receiver_account_fraud_6m	INT	
receiver_alias_rejected_3d	INT	
receiver_alias_rejected_30d	INT	
receiver_alias_rejected_6m	INT	
receiver_rejected_3d	INT	
receiver_rejected_30d	INT	
receiver_rejected_6m	INT	
receiver_account_rejected_3d	INT	
receiver_account_rejected_30d	INT	
receiver_account_rejected_6m	INT	
receiver_alias_portab_date	LONG	
receiver_alias_dict_reg_date	LONG	
device_registration_date	LONG	
sender_account_limits	SET	
receiver_account_limits	SET	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_dict_fields_v2	TUPLE	
receiver_dict_fields_v2	TUPLE	
external_fraud_service	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	

Transfers Workflow Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
payment_type	STRING	
payment_sub_type	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	

sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	

approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_type	STRING	
event_external_id	STRING	
event_received_at	LONG	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
feedback_label	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	

sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	

sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	

receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
direction	STRING	
dd_mandate_name	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
receiver_bank_sort_code	STRING	
device_id	STRING	
decision	STRING	
fraud_score	DOUBLE	
fraud	BOOLEAN	
risk_level	STRING	
dd_mandate_number	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	

trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
segment_channel_type	STRING	
login_session_id	STRING	
session_id	STRING	
auth_type	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
device_ip_country_code	STRING	
device_language	STRING	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
pass_reset	BOOLEAN	
last_sca_at	LONG	
sender_alias	STRING	
sender_alias_type	STRING	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
text_for_receiver	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_first_seen	LONG	
device_app_version	STRING	
reference_number	STRING	
reference_number_entry_mode	STRING	
sender_alias_status	STRING	
receiver_alias_status	STRING	
instant_transfer_processing_method	STRING	
sender_account_id	STRING	
receiver_account_id	STRING	
receiver_alias_email_domain	STRING	
posneg_sender_bank_country_value	STRING	

posneg_receiver_bank_country_value	STRING	
posneg_device_ip_value	STRING	
posneg_device_id_value	STRING	
time_of_day	STRING	
sca_required	BOOLEAN	
posneg_sender_account_value	STRING	
posneg_receiver_account_value	STRING	
posneg_dd_company_value	STRING	
posneg_sender_alias_value	STRING	
posneg_receiver_alias_value	STRING	
posneg_sender_bank_bic_value	STRING	
posneg_receiver_bank_bic_value	STRING	
posneg_reference_number_value	STRING	
posneg_sender_bank_sort_code_value	STRING	
posneg_receiver_bank_sort_code_value	STRING	
posneg_receiver_alias_phone_value	STRING	
posneg_receiver_alias_email_domain_value	STRING	
distinct_sender_per_device_id_90m	LONG	
sum_amt_instant_transfers_per_sender_alias_90m	LONG	
distinct_sender_per_email_1m	LONG	
distinct_sender_per_phone_1m	LONG	
count_sender_device_id_6m	LONG	
count_sender_device_ip_country_code_count_6m	LONG	
sum_amount_sender_last_sca_90d	LONG	
count_sender_last_sca_90d	LONG	
sum_amount_sender_last_sca_30h	LONG	
count_sender_last_sca_30h	LONG	
distinct_receiver_same_name_per_sender_24h	LONG	
count_sender_receiver_6m	LONG	
count_sender_time_of_day_6m	LONG	
distinct_month_per_sender_3m_delay_1m	LONG	
sum_amount_sender_3m_delay_1m	LONG	
count_sender_3m_delay_1m	LONG	
count_sender_online_3m_delay_24h	LONG	
count_sender_telephone_3m_delay_24h	LONG	
distinct_day_year_per_sender_instant_transfer_3m_delay_24h	LONG	
distinct_month_per_sender_online_3m_delay_24h	LONG	
distinct_month_per_sender_telephone_3m_delay_24h	LONG	
count_sender_dd_3m_delay_24h	LONG	
count_sender_instant_transfer_3m_delay_24h	LONG	

count_sender_6m_delay_24h	LONG	
sum_amt_bill_payments_per_sender_out_work_days_1m	LONG	
count_bill_payments_per_sender_out_work_days_1m	LONG	
count_sender_cross_border_amount_3k_1month	LONG	
sum_amount_per_sender_1m	LONG	
count_sender_1m	LONG	
distinct_day_week_per_sender_1w	LONG	
sum_amount_per_sender_1w	LONG	
count_dd_return_per_sender_3m	LONG	
count_sender_6m	LONG	
sender_prev_device_lat_1h	DOUBLE	
sender_prev_device_long_1h	DOUBLE	
distinct_country_per_sender_24h	LONG	
count_sender_24h	LONG	
count_sender_pw_change_24h	LONG	
auth_type_changed_24h	INT	
count_sender_online_24h	LONG	
count_sender_telephone_24h	LONG	
sender_device_language_changed_24h	INT	
login_session_id_changed_24h	INT	
count_sender_instant_transfer_24h	LONG	
count_sender_dd_24h	LONG	
distinct_receiver_per_sender_24h	LONG	
sum_amount_per_sender_telephone_24h	LONG	
count_sender_cross_border_amount_3k_24h	LONG	
sum_amount_per_sender_24h	LONG	
count_dd_return_per_sender_24h	LONG	
count_failed_trx_sender_24h	LONG	
distinct_device_id_per_sender_90m	LONG	
count_sender_90min	LONG	
avg_daily_amount_per_sender_1w	DOUBLE	
avg_monthly_count_sender_3m_delay_1m	DOUBLE	
avg_monthly_amount_per_sender_3m_delay_1m	DOUBLE	
avg_monthly_count_sender_telephone_3m_delay_24h	DOUBLE	
avg_monthly_count_sender_online_3m_delay_24h	DOUBLE	
avg_daily_count_sender_instant_transfer_3m_delay_24h	DOUBLE	
device_fingerprint	STRING	
day_of_week	STRING	
new_device	BOOLEAN	
distinct_account_per_device_3m	LONG	

count_account_activity_out_work_days_3m	LONG	
count_account_activity_time_of_day_3m	LONG	
count_account_activity_3m	LONG	
instant_transfers_review_outcome	BOOLEAN	
count_sender_3m_delay_24h	LONG	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	
device_ip_region	STRING	
revelock_score	INT	
revelock_automated_actions	LIST	
revelock_alerts	LIST	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
transaction_goal	STRING	
transaction_method	STRING	
device_latitude_double	DOUBLE	
device_longitude_double	DOUBLE	
device_latitude	STRING	
device_longitude	STRING	
device_registration_date	LONG	
sender_account_limits	SET	
receiver_account_limits	SET	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_dict_fields_v2	TUPLE	

receiver_dict_fields_v2	TUPLE	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
new_payees_24h	LONG	
count_inbound_24h	LONG	
sum_amount_per_receiver_7d	LONG	
sum_amount_per_sender_7d	LONG	
count_per_sender_7d	LONG	
count_per_receiver_7d	LONG	
count_receiver_3m_delay_24h	LONG	
count_digital_3m_delay_24h	LONG	
count_failed_logins_24h	LONG	
last_digital_event_type_24h	STRING	
count_logins_3m	LONG	
count_logins_time_of_day_3m	LONG	
distinct_account_login_per_device_1m	LONG	
count_new_payee_transfers_24h	LONG	
count_activity_3m_delay_24h	LONG	
device_android_id	STRING	
device_cbf	STRING	
device_mac_address	STRING	
device_serial_id	STRING	
device_browser_no_tracking	BOOLEAN	
device_browser_private_mode	BOOLEAN	
device_network_type	STRING	
device_session_duration	LONG	
device_browser_language	STRING	
device_model	STRING	
device_screen_height	INT	
device_screen_width	INT	

Inbound Payments API Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	

cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	

receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	

approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_external_id	STRING	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	

receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	

sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	

receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
device_id	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_sort_code	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
pass_reset	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
device_ip_country_code	STRING	
login_session_id	STRING	
session_id	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_language	STRING	
sender_alias	STRING	
sender_alias_type	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	

text_for_receiver	STRING	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_app_version	STRING	
reference_number	STRING	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	
device_ip_region	STRING	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
transaction_goal	STRING	
transaction_method	STRING	
device_registration_date	LONG	
sender_aum_2m	BOOLEAN	
sender_aum_5m	BOOLEAN	
receiver_aum_2m	BOOLEAN	
receiver_aum_5m	BOOLEAN	
sender_kyc_level	STRING	
sender_active	BOOLEAN	
jurisdiction	STRING	
receiver_active	BOOLEAN	
receiver_account_active	BOOLEAN	
sender_account_active	BOOLEAN	
device_fingerprint	STRING	
event_received_at	LONG	

event_type	STRING	
feedback_label	STRING	
instant_transfers_review_outcome	BOOLEAN	
receiver_account_id	STRING	
sender_account_id	STRING	
direction	STRING	
payment_type	STRING	
payment_sub_type	STRING	
segment_channel_type	STRING	
last_sca_at	LONG	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
sender_alias_status	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
receiver_alias_status	STRING	
device_first_seen	LONG	
instant_transfer_processing_method	STRING	
reference_number_entry_mode	STRING	
is_domestic	BOOLEAN	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_account_limits	SET	
alert_id	STRING	
receiver_account_limits	SET	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
dd_mandate_name	STRING	
dd_mandate_number	STRING	
ultimate_debtor_name	STRING	
ultimate_debtor_dob	STRING	
ultimate_debtor_country_code	STRING	
ultimate_creditor_name	STRING	
ultimate_creditor_dob	STRING	
ultimate_creditor_country_code	STRING	

Inbound Payments Decision Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	
sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	

sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	
receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	

approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_external_id	STRING	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	
oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	

receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	
sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	

sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	
receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	

receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	
receiver_account_atc2	STRING	
device_id	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_sort_code	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
pass_reset	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
device_ip_country_code	STRING	
login_session_id	STRING	
session_id	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	

sender_phone_change_flag	BOOLEAN	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_language	STRING	
sender_alias	STRING	
sender_alias_type	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
text_for_receiver	STRING	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_app_version	STRING	
reference_number	STRING	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	
device_ip_region	STRING	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
transaction_goal	STRING	
transaction_method	STRING	
device_registration_date	LONG	
sender_aum_2m	BOOLEAN	
sender_aum_5m	BOOLEAN	
receiver_aum_2m	BOOLEAN	
receiver_aum_5m	BOOLEAN	

sender_kyc_level	STRING	
sender_active	BOOLEAN	
jurisdiction	STRING	
receiver_active	BOOLEAN	
receiver_account_active	BOOLEAN	
sender_account_active	BOOLEAN	
device_fingerprint	STRING	
event_received_at	LONG	
event_type	STRING	
feedback_label	STRING	
instant_transfers_review_outcome	BOOLEAN	
receiver_account_id	STRING	
sender_account_id	STRING	
direction	STRING	
payment_type	STRING	
payment_sub_type	STRING	
segment_channel_type	STRING	
last_sca_at	LONG	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	
sender_alias_status	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
receiver_alias_status	STRING	
device_first_seen	LONG	
instant_transfer_processing_method	STRING	
reference_number_entry_mode	STRING	
is_domestic	BOOLEAN	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_account_limits	SET	
alert_id	STRING	
receiver_account_limits	SET	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
decision	STRING	
fraud	BOOLEAN	

fraud_score	DOUBLE	
risk_level	STRING	
sca_required	BOOLEAN	
posneg_receiver_account_value	STRING	
posneg_receiver_bank_bic_value	STRING	
received_amount_6m_delay_24h	LONG	
received_transactions_6m_delay_24h	LONG	
received_transactions_cross_border_24h	LONG	
received_amount_cross_border_24h	LONG	
received_amount_24h	LONG	
received_amount_7d	LONG	
received_transactions_7d	LONG	
sent_transactions_7d	LONG	
sent_amount_7d	LONG	

Inbound Payments Dummy Schema

Field Name	Field Type	Field Description
fraud_label	BOOLEAN	

Inbound Payments TFRB Plan Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
bulk_id	STRING	
product_code	STRING	
payment_status	STRING	
created_at	LONG	
cutoff_at	LONG	
sender_initiated_at	LONG	
sender_urgency	BOOLEAN	
requested_execution_at	LONG	
execution_at	LONG	
frequency	STRING	
number_of_payments	INT	
recurring_payment	BOOLEAN	
recurring_payment_day	INT	
recurring_payment_hour	INT	
payment_reference	STRING	
end_to_end_identification	STRING	

sender_city	STRING	
sender_country_code	STRING	
sender_transaction_amount	LONG	
sender_transaction_currency	STRING	
sender_transaction_foreign_exchange_rate	DOUBLE	
sender_transaction_fee_type	STRING	
sender_transaction_fee_amount	LONG	
sender_transaction_fee_currency	STRING	
sender_transaction_category_purpose	STRING	
sender_transaction_purpose	STRING	
sender_transaction_bank_account_number	STRING	
sender_transaction_notes	STRING	
sender_bank_bic	STRING	
sender_bank_name	STRING	
sender_bank_postal_code	STRING	
sender_bank_address	STRING	
sender_bank_city	STRING	
sender_bank_country_code	STRING	
sender_bank_currency	STRING	
sender_bank_foreign_exchange_rate	DOUBLE	
sender_bank_fee_type	STRING	
sender_bank_fee_amount	LONG	
sender_bank_fee_currency	STRING	
receiver_bank_bic	STRING	
receiver_bank_name	STRING	
receiver_bank_postal_code	STRING	
receiver_bank_address	STRING	
receiver_bank_city	STRING	
receiver_bank_country_code	STRING	
receiver_bank_currency	STRING	
receiver_bank_foreign_exchange_rate	DOUBLE	
receiver_bank_fee_type	STRING	
receiver_bank_fee_amount	LONG	
receiver_bank_fee_currency	STRING	
receiver_transaction_amount	LONG	
receiver_transaction_currency	STRING	
receiver_transaction_foreign_exchange_rate	DOUBLE	
receiver_transaction_fee_type	STRING	
receiver_transaction_fee_amount	LONG	
receiver_transaction_fee_currency	STRING	

receiver_settlement_at	LONG	
receiver_transaction_bank_account_number	STRING	
receiver_transaction_notes	STRING	
receiver_city	STRING	
receiver_country_code	STRING	
device_type	STRING	
device_operating_system	STRING	
device_ip_address	STRING	
device_ip_address_location	STRING	
device_longitude	STRING	
device_latitude	STRING	
user_email_address	STRING	
user_phone_number	STRING	
user_phone_number_country_code	STRING	
challenge_risk_score	INT	
challenge_risk_flag	BOOLEAN	
sca_risk_score	INT	
sca_risk_flag	BOOLEAN	
number_of_authentications	INT	
max_number_of_authentications	INT	
approver_login_id	STRING	
approver_customer_id	STRING	
approver_type	STRING	
approval_total_approves	INT	
approval_final_approval	BOOLEAN	
approval_signatures_applied	INT	
approval_current_number	INT	
approval_primary_admin	BOOLEAN	
approval_submitted_at	LONG	
approval_type	STRING	
approval_status	STRING	
event_external_id	STRING	
info_type	STRING	
dispute_start_date	LONG	
dispute_reason	STRING	
dispute_notes	STRING	
dispute_submitted_by	STRING	
dispute_end_date	LONG	
dispute_feedback_notes	STRING	
dispute_refunded	STRING	

oob_start_date	LONG	
oob_comm_type	STRING	
oob_notes	STRING	
oob_end_date	LONG	
oob_feedback_notes	STRING	
oob_response_type	STRING	
feedback_type	STRING	
sender_correspondent_bank_bic	STRING	
sender_correspondent_bank_name	STRING	
sender_correspondent_bank_postal_code	STRING	
sender_correspondent_bank_address	STRING	
sender_correspondent_bank_city	STRING	
sender_correspondent_bank_country_code	STRING	
sender_correspondent_bank_currency	STRING	
sender_correspondent_bank_foreign_exchange_rate	DOUBLE	
sender_correspondent_bank_fee_type	STRING	
sender_correspondent_bank_fee_amount	LONG	
sender_correspondent_bank_fee_currency	STRING	
receiver_correspondent_bank_bic	STRING	
receiver_correspondent_bank_name	STRING	
receiver_correspondent_bank_postal_code	STRING	
receiver_correspondent_bank_address	STRING	
receiver_correspondent_bank_city	STRING	
receiver_correspondent_bank_country_code	STRING	
receiver_correspondent_bank_currency	STRING	
receiver_correspondent_bank_foreign_exchange_rate	DOUBLE	
receiver_correspondent_bank_fee_type	STRING	
receiver_correspondent_bank_fee_amount	LONG	
receiver_correspondent_bank_fee_currency	STRING	
sender_id	STRING	
sender_segment	STRING	
sender_name	STRING	
sender_email	STRING	
sender_phone	STRING	
sender_phone_val_code	STRING	
sender_citizen_id_number	STRING	
sender_passport_number	STRING	
sender_driver_license	STRING	
sender_ssn	STRING	
sender_tax_number	STRING	

sender_registration_date	LONG	
sender_street_addr	STRING	
sender_zip	STRING	
sender_region	STRING	
sender_dob	STRING	
sender_number_of_cards	INT	
sender_number_of_accounts	INT	
sender_total_balance	LONG	
sender_total_credit_limit	LONG	
sender_kyc_lvl	STRING	
sender_occupation	STRING	
sender_occupation_status	STRING	
sender_risk_rating	STRING	
sender_currency	STRING	
sender_account_iban	STRING	
sender_account_bic	STRING	
sender_account_open_date	LONG	
sender_account_type	STRING	
sender_account_currency	STRING	
sender_account_balance	LONG	
sender_account_credit_limit	LONG	
sender_account_otb_limit	LONG	
sender_account_available_credit	LONG	
sender_account_debt_balance	LONG	
sender_account_cash_advance_balance	LONG	
sender_account_status	STRING	
sender_account_last_activity	LONG	
sender_account_last_payment_date	LONG	
sender_account_last_payment_amount	LONG	
sender_account_pw_change_flag	BOOLEAN	
sender_account_rewards_used_count	INT	
sender_account_debt_payment_method	STRING	
sender_account_number_of_cards	INT	
sender_account_atc1	STRING	
sender_account_atc2	STRING	
receiver_id	STRING	
receiver_segment	STRING	
receiver_name	STRING	
receiver_email	STRING	
receiver_phone	STRING	

receiver_phone_val_code	STRING	
receiver_citizen_id_number	STRING	
receiver_passport_number	STRING	
receiver_driver_license	STRING	
receiver_ssn	STRING	
receiver_tax_number	STRING	
receiver_registration_date	LONG	
receiver_street_addr	STRING	
receiver_zip	STRING	
receiver_region	STRING	
receiver_dob	STRING	
receiver_number_of_cards	INT	
receiver_number_of_accounts	INT	
receiver_total_balance	LONG	
receiver_total_credit_limit	LONG	
receiver_kyc_lvl	STRING	
receiver_occupation	STRING	
receiver_occupation_status	STRING	
receiver_risk_rating	STRING	
receiver_currency	STRING	
receiver_account_iban	STRING	
receiver_account_bic	STRING	
receiver_account_open_date	LONG	
receiver_account_type	STRING	
receiver_account_currency	STRING	
receiver_account_balance	LONG	
receiver_account_credit_limit	LONG	
receiver_account_otb_limit	LONG	
receiver_account_available_credit	LONG	
receiver_account_debt_balance	LONG	
receiver_account_cash_advance_balance	LONG	
receiver_account_status	STRING	
receiver_account_last_activity	LONG	
receiver_account_last_payment_date	LONG	
receiver_account_last_payment_amount	LONG	
receiver_account_pw_change_flag	BOOLEAN	
receiver_account_rewards_used_count	INT	
receiver_account_debt_payment_method	STRING	
receiver_account_number_of_cards	INT	
receiver_account_atc1	STRING	

receiver_account_atc2	STRING	
device_id	STRING	
channel_type	STRING	
sender_bank_sort_code	STRING	
sender_bank_routing_number	STRING	
sender_bank_branch_id	STRING	
receiver_bank_sort_code	STRING	
receiver_bank_routing_number	STRING	
receiver_bank_branch_id	STRING	
event_occurred_at	LONG	
pass_reset	BOOLEAN	
trusted_receiver	BOOLEAN	
first_recurring_payment	BOOLEAN	
reference_fraud_rate	DOUBLE	
auth_type	STRING	
device_ip_country_code	STRING	
login_session_id	STRING	
session_id	STRING	
sender_country_code_change_flag	BOOLEAN	
sender_email_change_flag	BOOLEAN	
sender_phone_change_flag	BOOLEAN	
malware_flag	BOOLEAN	
proxy_flag	BOOLEAN	
device_language	STRING	
sender_alias	STRING	
sender_alias_type	STRING	
receiver_alias	STRING	
receiver_alias_type	STRING	
text_for_receiver	STRING	
device_user_agent	STRING	
device_imei_number	STRING	
device_tor_mode	BOOLEAN	
device_cookies	BOOLEAN	
device_app_version	STRING	
reference_number	STRING	
session_register_date	LONG	
device_os_version	STRING	
device_browser	STRING	
device_browser_version	STRING	
device_isp	STRING	

device_ip_region	STRING	
device_is_new	BOOLEAN	
sender_transaction_origin_amount	LONG	
sender_transaction_origin_currency	STRING	
receiver_transaction_origin_amount	LONG	
receiver_transaction_origin_currency	STRING	
instant_transfer_type	STRING	
purchased_amount	LONG	
withdrawal_amount	LONG	
sender_company_id	STRING	
receiver_company_id	STRING	
feedback_value	BOOLEAN	
geographic_scope	STRING	
transaction_goal	STRING	
transaction_method	STRING	
device_registration_date	LONG	
sender_aum_2m	BOOLEAN	
sender_aum_5m	BOOLEAN	
receiver_aum_2m	BOOLEAN	
receiver_aum_5m	BOOLEAN	
sender_kyc_level	STRING	
sender_active	BOOLEAN	
jurisdiction	STRING	
receiver_active	BOOLEAN	
receiver_account_active	BOOLEAN	
sender_account_active	BOOLEAN	
device_fingerprint	STRING	
event_received_at	LONG	
event_type	STRING	
feedback_label	STRING	
instant_transfers_review_outcome	BOOLEAN	
receiver_account_id	STRING	
sender_account_id	STRING	
direction	STRING	
payment_type	STRING	
payment_sub_type	STRING	
segment_channel_type	STRING	
last_sca_at	LONG	
sender_alias_registration_time	LONG	
sender_alias_last_update_time	LONG	

sender_alias_status	STRING	
receiver_alias_registration_time	LONG	
receiver_alias_last_update_time	LONG	
receiver_alias_status	STRING	
device_first_seen	LONG	
instant_transfer_processing_method	STRING	
reference_number_entry_mode	STRING	
is_domestic	BOOLEAN	
sender_transaction_original_amount	LONG	
sender_transaction_original_currency	STRING	
receiver_transaction_original_amount	LONG	
receiver_transaction_original_currency	STRING	
sender_account_limits	SET	
alert_id	STRING	
receiver_account_limits	SET	
network_bssid	STRING	
network_ssid	STRING	
fake_location	STRING	
decision	STRING	
fraud	BOOLEAN	
fraud_score	DOUBLE	
risk_level	STRING	
sca_required	BOOLEAN	

Ad-hoc Customer Investigations API Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
event_external_id	STRING	
reason_code	STRING	
origin	STRING	
description	STRING	
customer_id	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	

customer_ssn	STRING	
customer_tax_number	STRING	
customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_zip_area_code	STRING	
customer_city	STRING	
customer_region	STRING	
customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
customer_active	STRING	
customer_segment	STRING	
customer_number_of_cards	INT	
customer_occupation_status	STRING	
customer_risk_rating	STRING	
customer_currency	STRING	
customer_accounts	SET	
customer_cards	SET	
customer_address_validated	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_contactoptout	BOOLEAN	
customer_pwd	STRING	
customer_gender	STRING	
customer_mothersname	STRING	
customer_type	STRING	
customer_username	STRING	
customer_branch_id	STRING	
customer_preferred_language	STRING	
customer_reported_income	LONG	
customer_alias_status	STRING	
customer_alias_last_update_time	LONG	
customer_alias_registration_time	LONG	
customer_alias	STRING	
customer_alias_type	STRING	
customer_citizenship	STRING	

customer_credit_score	INT	
customer_mailing_city	STRING	
customer_mailing_country_code	STRING	
customer_mailing_street_addr	STRING	
customer_mailing_zip	STRING	
is_client	BOOLEAN	
customer_products_portfolio	LIST	
event_occurred_at	LONG	
event_received_at	LONG	
event_type	STRING	
info_type	STRING	
feedback_value	BOOLEAN	
feedback_label	STRING	
feedback_type	STRING	
source	STRING	

Ad-hoc Customer Investigations Workflow Schema

Field Name	Field Type	Field Description
lifecycle_id	STRING	
event_external_id	STRING	
reason_code	STRING	
origin	STRING	
description	STRING	
customer_id	STRING	
customer_name	STRING	
customer_email	STRING	
customer_phone	STRING	
customer_phone_val_code	STRING	
customer_citizen_id_number	STRING	
customer_passport_number	STRING	
customer_driver_license	STRING	
customer_ssn	STRING	
customer_tax_number	STRING	
customer_registration_date	LONG	
customer_street_addr	STRING	
customer_zip	STRING	
customer_zip_area_code	STRING	
customer_city	STRING	
customer_region	STRING	

customer_country_code	STRING	
customer_dob	STRING	
customer_number_of_accounts	INT	
customer_total_balance	LONG	
customer_total_credit_limit	LONG	
customer_kyc_lvl	STRING	
customer_occupation	STRING	
customer_active	STRING	
customer_segment	STRING	
customer_number_of_cards	INT	
customer_occupation_status	STRING	
customer_risk_rating	STRING	
customer_currency	STRING	
customer_accounts	SET	
customer_cards	SET	
customer_address_validated	BOOLEAN	
customer_email_validated	BOOLEAN	
customer_contactoptout	BOOLEAN	
customer_pwd	STRING	
customer_gender	STRING	
customer_mothersname	STRING	
customer_type	STRING	
customer_username	STRING	
customer_branch_id	STRING	
customer_preferred_language	STRING	
customer_reported_income	LONG	
customer_alias_status	STRING	
customer_alias_last_update_time	LONG	
customer_alias_registration_time	LONG	
customer_alias	STRING	
customer_alias_type	STRING	
customer_citizenship	STRING	
customer_credit_score	INT	
customer_mailing_city	STRING	
customer_mailing_country_code	STRING	
customer_mailing_street_addr	STRING	
customer_mailing_zip	STRING	
is_client	BOOLEAN	
customer_products_portfolio	LIST	
event_occurred_at	LONG	

event_received_at	LONG	
event_type	STRING	
info_type	STRING	
feedback_value	BOOLEAN	
feedback_label	STRING	
feedback_type	STRING	
decision	STRING	
source	STRING	

Cards - Decision Rules

Descriptionâ

Decision Rules are triggered based on the score value produced by a machine learning model. If your application workflow does not have a model, the `fraud_score` field remains empty, and thus rules in inside the *Decision Rules* project are not triggered.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-DR1

Comment

Declines a transaction if it scored above a set threshold.

Actions

- [Cards-DR1]

Variables

```
HIGH_SCORE_THRESHOLD = cards__rules_thresholds["model_high_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
(score ?? 0) >= HIGH_SCORE_THRESHOLD
```

Explanation

[Cards-DR1] - Decline - Model score ({localvar.score}) was above the threshold ({localvar.HIGH_SCORE_THRESHOLD}).

Mark As

The outcome of this rule is: **decline**

Cards - Entity Limit Rules

Was this page helpful?Â

Descriptionâ

Entity Limit Rules allow you to limit the amount and count of transactions for cards that are in PosNeg lists with the value "review". Normally, cards would be added to this list with such value as a result of Fraud rules triggering. Therefore, these rules can be seen as a second layer of protection against suspicious cards, and the actions triggered by these rules decline the card.

Rules Project Metadata

- *Target Variable:* decision

- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-EL1

Comment

Decline transaction and flag card to be auto declined if the amount spent in the last 24h is above a set threshold and the card is listed to be limited . Depends on 'List Management Service' to put the card in the configured list.

Actions

- Auto Decline Card
- [Cards-EL1]

Variables

```
 SUM_AMOUNT_LIMIT = cards__rules_thresholds["max_sum_appr_amout_card_24h"].threshold;
SUM_AMOUNT_LIMIT_CONVERTED = (SUM_AMOUNT_LIMIT * 1.0) / 100;

SUM_AMOUNT_CONVERTED = (event.review_appr_auth_cardid_sum_bi * 1.0) / 100;
```

Condition

```
 // List
event.posneg_card_value == 'review'
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.review_appr_auth_cardid_sum_bi >= SUM_AMOUNT_LIMIT
```

Explanation

[Cards-EL1] - Decline - Amount spent in last 24h ({localvar.SUM_AMOUNT_CONVERTED} {event.currency_code}) was above the threshold for limit listed cards ({localvar.SUM_AMOUNT_LIMIT_CONVERTED} {event.currency_code}). Card {event.card_id} flagged for auto decline.

Mark As

The outcome of this rule is: **decline**

Cards-EL2

Comment

Decline transaction and flag card to be auto declined if the number of transactions made in the last 24h is above a set threshold and the card is listed to be limited . Depends on 'List Management Service' to put the card in the configured list.

Actions

- Auto Decline Card
- [Cards-EL2]

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_count_appr_card_24h"].threshold;
```

Condition

```
 // Lists
event.posneg_card_value == 'review'
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.review_appr_auth_cardid_count_ >= COUNT_LIMIT
```

Explanation

> [Cards-EL2] - Decline - Transactions made in last 24h ({event.review_appr_auth_cardid_count_}) was above the threshold for limit listed cards ({localvar.COUNT_LIMIT}). Card {event.card_id} flagged for auto decline.

Mark As

The outcome of this rule is: **decline**

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/rules-projects/cards-feedback-rules

Cards - Feedback Rules

Was this page helpful?Â

Descriptionâ

Feedback Rules allow you to apply a set of actions to events that were confirmed as fraud or not fraud. Typical scenarios include:

- A cardholder is contacted by the issuer bank and is asked whether the operation is authentic or not. After the cardholderâs confirmation, the issuer sends an API request via the feedback endpoint with `fraud_label` as 'true' (i.e. fraud) or 'false' (not fraud) and `feedback_type` field equal to 'out_of_band'. This `feedback_type` field is mapped to a Pulse internal field `event_subtype` with the same value.

- A fraud analyst reviews an alert in Case Manager and marks it as fraud or not fraud. In this case, an event is automatically sent to Pulse with the `fraud_label` as 'true' (i.e. fraud) or 'false' (not fraud) and `event_subtype` field as 'cm_feedback'.

Both scenarios will trigger feedback rules and apply configurable actions. For example:

- Fraudulent event: Card is blacklisted if the cardholder does not acknowledge the transaction or the analyst marks it as "Fraud".
- Non-fraudulent event: Card is whitelisted if the cardholder acknowledges the transaction or the analyst marks it as "Not Fraud".

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-FB1

Comment

Flag card to be auto declined if customer does not acknowledge transaction or analyst marks as 'Fraud' . Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-FB1]
- Auto Decline Card
- Alert

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback' and (event.feedback_type == "out_of_band" or event.feedback_type
== "cm_feedback")
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == true
and
// Filters
(event.card_id ?? '' != '')
```

Explanation

[Cards-FB1] - Customer did not acknowledge the transaction or analyst marked as 'Fraud'. '{eventname.card_id}' {event.card_id} flagged for auto decline.

Mark As

N/A

Cards-FB2

Comment

Flag card to be auto approved if customer acknowledges transaction or analyst marks as Not 'Fraud' .
Depends on 'List Management Service' to put the card in the configured list.

Actions

- Auto Approve Card
- [Cards-FB2]

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback' and (event.feedback_type == "out_of_band" or event.feedback_type
== "cm_feedback")
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == false
and
// Filters
(event.card_id ?? '' != '')
```

Explanation

[Cards-FB2] - Customer acknowledged the transaction or analyst marked as 'Not Fraud'.
'{eventname.card_id}' {event.card_id} flagged for auto approve.

Mark As

N/A

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/rules-projects/cards-fraud-rules

- [References]
- [Rules Projects]
- Cards - Fraud Rules

On this page

Cards - Fraud Rules

Was this page helpful?Â

Descriptionâ

Contains rules that look for standard fraud patterns. If triggered, most rules included in this rules project do not auto decline a card. Instead, they mark the event as an alert which can then be further investigated by fraud analysts in Case Manager. Examples of typical patterns covered by these rules include:

- High-amount spending and increased number of transactions in a short period of time.
- High-amount spending in the same merchant in a given time frame by card not present.
- Repeated transactions with the same amount.
- Number of transactions above a certain threshold.

- Unexpected spending behavior.
- High cumulative spending amounts.
- Transactions in unusual cities and/or currencies.
- Multiple CVV attempts in a short period of time.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-Velocity-1

Comment

Decline transaction and flag card to be limited if travel velocity between the last transaction within the past 1 hour and the current transaction is unreasonably high. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Velocity-1]
- Limit/Review Card
- Alert

Variables

```
 // maximum velocity in km/h
VELOCITY_LIMIT = cards__rules_thresholds["max_cp_card_term_vel_1h"].threshold;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
event.card_presence_flag ?? false
and
// Limits
event.cp_card_term_vel_1h >= VELOCITY_LIMIT
```

Explanation

[Cards-Velocity-1] - Decline - Card travel velocity ({event.cp_card_term_vel_1h} km/h) between last and current CP transactions in last 1h is higher than limit ({localvar.VELOCITY_LIMIT} km/h). '{eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-2

Comment

Decline transaction and flag card to be limited if number of transactions made by card within the past 24 hours is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Velocity-2]
- Limit/Review Card
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_count_24h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.auth_cardid_count_1d >= COUNT_LIMIT
```

Explanation

[Cards-Velocity-2] - Decline - Card transactions ({event.auth_cardid_count_1d}) in the last 24 hours are above the limit ({localvar.COUNT_LIMIT}). '{eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-3

Comment

Decline transaction and flag card to be limited if the number of transactions and total amount spent by card within the past 1 hour is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Velocity-3]
- Limit/Review Card
- Alert

Variables

```
CNT_LMT = cards__rules_thresholds["max_card_appr_count_1h"].threshold;

SUM_AMT_LMT = cards__rules_thresholds["max_card_sum_appr_amount_1h"].threshold;
SUM_AMT_LMT_CONV = (SUM_AMT_LMT * 1.0) / 100;

SUM_AMT_CONV = (event.appr_auth_cardid_sum_billamosr * 1.0) / 100;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_count_1h >= CNT_LMT
and
event.appr_auth_cardid_sum_billamosr >= SUM_AMT_LMT
```

Explanation

> [Cards-Velocity-3] - Decline - Card transactions ({event.appr_auth_cardid_count_1h}) and spending ({localvar.SUM_AMT_CONV} {event.currency_code}) in the last 1h are over the limits ({localvar.CNT_LMT} and {localvar.SUM_AMT_LMT_CONV} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-4

Comment

> Decline transaction if the number of transactions made by card within the last 3 minutes is above defined threshold.

Actions

- [Cards-Velocity-4]
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_appr_count_3min"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_count_3m >= COUNT_LIMIT
```

Explanation

> [Cards-Velocity-4] - Decline - Card transactions ({event.appr_auth_cardid_count_3m}) in the last 3 minutes is over the limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-5

Comment

Distance between current and previous ip location in the last hour exceeds 1000 km (1 hour)

Actions

- [Cards-Velocity-5]
- Limit/Review Card
- Alert

Variables

```
distance = Location.distFromInKm(event.device_ip_latitude,event.device_ip_longitude,event.auth_cnp_cardid_prev_device_ip,event.auth_cnp_cardid_prev_device_sr) ?? 0.0;

DISTANCE_LIMIT = cards__rules_thresholds["max_distance_ip_location_km_1h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
(event.auth_cnp_cardid_prev_device_ip ?? 0.0 != 0.0) and (event.auth_cnp_cardid_prev_device_sr ?? 0.0 != 0.0)
and
distance >= DISTANCE_LIMIT
```

Explanation

[Cards-Velocity-5] - Decline - Distance between previous and current IP location in the last hour is higher than {localvar.DISTANCE_LIMIT} km.

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-6

Comment

Ratio of distinct card_id and distinct expiry dates per BIN and Merchant in the last 10m is below threshold, indicating a possible BIN attack.

Actions

- [Cards-Velocity-6]
- Alert

Variables

```
// curve3 (th3) is expected to have better results then the others (good compromise between number of alerts and detection rate)
// other curves are still provided in case there is the need to adjust
binAttackThreshold = cards__rules_thresholds["acceptor_cardbin_curve_th"].threshold;
```

```
// acceptor_cardbin_cur = binary fields to split transactions above and below threshold curve
thresholdDecision = match
  if binAttackThreshold == 0 then return event.acceptorid_cardbin_cur0;
  else if binAttackThreshold == 1 then return event.acceptorid_cardbin_cur1;
  else if binAttackThreshold == 3 then return event.acceptorid_cardbin_cur3;
  else if binAttackThreshold == 5 then return event.acceptorid_cardbin_cur5;
  end
end
```

Condition

```
thresholdDecision
```

Explanation

> [Cards-Velocity-6] - Decline - The ratio between distinct card_id ({event.acceptorid_cardbin_cntdist_csr}) and distinct exp_date ({event.acceptorid_cardbin_cntdist_car}) per BIN/Merchant 10m is below {localvar.binAttackThreshold}

Mark As

The outcome of this rule is: **decline**

Cards-Velocity-7

Comment

> Ratio of distinct card_id and distinct expiry dates per BIN and MCC in the last 10m is below threshold, indicating a possible BIN attack.

Actions

- [Cards-Velocity-7]
- Alert

Variables

```
 // curve3 (th3) is expected to have better results then the others (good compromise between
number of alerts and detection rate)
// other curves are still provided in case there is the need to adjust
binAttackThreshold = cards__rules_thresholds["mcc_cardbin_curve_th"].threshold;

// acceptor_cardbin_cur = binary fields to split transactions above and below threshold curve
thresholdDecision = match
  if binAttackThreshold == 0 then return event.mcc_cardbin_cur0;
  else if binAttackThreshold == 1 then return event.mcc_cardbin_cur1;
  else if binAttackThreshold == 3 then return event.mcc_cardbin_cur3;
  else if binAttackThreshold == 5 then return event.mcc_cardbin_cur5;
  end
end
```

Condition

```
thresholdDecision
```

Explanation

[Cards-Velocity-7] - Decline - The ratio between distinct card_id ({event.mcc_cardbin_cntdist_cardid_10m}) and distinct exp_date ({event.mcc_cardbin_cntdist_cardexpdat}) per BIN/MCC 10m is below {localvar.binAttackThreshold}

Mark As

The outcome of this rule is: **decline**

Cards-CNP-1

Comment

Flag card to be limited if it has made multiple CNP transactions in a high number of unique merchants (acceptor_id) in the last 5 minutes. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-CNP-1]
- Limit/Review Card
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_cnp_card_distinct_acceptor_5min"].threshold;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.auth_cnp_cardid_countdist_acce >= COUNT_LIMIT
```

Explanation

[Cards-CNP-1] - Card CNP transactions in unique merchants ({event.auth_cnp_cardid_countdist_acce}) in the last 5 minutes are over the limit ({localvar.COUNT_LIMIT}). '{eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

N/A

Cards-CNP-2

Comment

Decline and flag card to be limited if transaction is CNP and has high amount with an unusual currency. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-CNP-2]
- Limit/Review Card

- Alert

Variables

N/A

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '') and (event.card_presence_flag ?? false == false) and
(event.currency_code ?? '' != '')
and
// Limits
event.is_high_amount and
!common_currencies.contains(event.currency_code)
```

Explanation

[Cards-CNP-2] - Decline - Card CNP transaction with an high amount in an unusual currency.

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-1

Comment

Decline transaction and flag card to be limited if total amount spent by card within the past 24 hours at single Merchant (acceptor_id) is above a defined threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Threshold-1]
- Limit/Review Card
- Alert

Variables

```
 SUM_AMOUNT_LIMIT = cards__rules_thresholds["max_card_acceptor_sum_appr_amount_24h"].threshold;
SUM_AMOUNT_LIMIT_CONV = (SUM_AMOUNT_LIMIT * 1.0) / 100;

SUM_AMOUNT_CONVERTED = (event.appr_auth_cardid_acceptor_sum_ * 1.0) / 100;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_acceptor_sum_ >= SUM_AMOUNT_LIMIT
```

Explanation

[Cards-Threshold-1] - Decline - Card spending ({localvar.SUM_AMOUNT_CONVERTED} {event.currency_code}) in the last 24 hours is above the limit ({localvar.SUM_AMOUNT_LIMIT_CONVERTED} {event.currency_code}). Card {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-2

Comment

Decline transaction if number of transactions made by card within the past 4 weeks is above a set threshold.

Actions

- [Cards-Threshold-2]
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_count_4w"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.auth_cardid_count_4w >= COUNT_LIMIT
```

Explanation

[Cards-Threshold-2] - Decline - Card transactions ({event.auth_cardid_count_4w}) in the last 4 weeks are above the limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-3

Comment

Decline transaction and flag card to be limited if amount spent by card within the past 4 weeks is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Threshold-3]
- Limit/Review Card
- Alert

Variables

```
SUM_AMOUNT_LIMIT = cards__rules_thresholds["max_card_sum_appr_amount_4w"].threshold;
SUM_AMOUNT_LIMIT_CONVERTED = (SUM_AMOUNT_LIMIT * 1.0) / 100;

SUM_AMOUNT_CONVERTED = (event.appr_auth_cardid_sum_billamoun * 1.0) / 100;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_sum_billamoun >= SUM_AMOUNT_LIMIT
```

Explanation

> [Cards-Threshold-3] - Decline - Card spending ({localvar.SUM_AMOUNT_CONVERTED} {event.currency_code}) in the last 4 weeks is above the limit ({localvar.SUM_AMOUNT_LIMIT_CONVERTED} {event.currency_code}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-4

Comment

> Decline and flag card to be limited if number of high amount transactions made by the same card within the past day is above a set threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- [Cards-Threshold-4]
- Limit/Review Card
- Alert

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_appr_count_high_amount_24h"].threshold;
```

Condition

```
// Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits (v1: high amount = 1000 USD, check is_high_amount)
event.high_amount_appr_auth_cardid_c >= COUNT_LIMIT
```

Explanation

> [Cards-Threshold-4] - Decline - Card high amount transactions ({event.high_amount_appr_auth_cardid_c} above 1000.00 {event.currency_code}) in the last 24h are over the limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-5

Comment

> Decline transaction and flag card to be limited if multiple transactions with a high cumulative amount made by the same card in the same merchant (acceptor_id) in the last 24h. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert
- [Cards-Threshold-5]

Variables

```
 CNT_LMT = cards__rules_thresholds["max_card_acceptor_appr_count_24h"].threshold;

SUM_AMNT_LMT = cards__rules_thresholds["max_card_acceptor_sum_appr_billamount_24h"].threshold;
SUM_A_LMT_CONV = (SUM_AMNT_LMT * 1.0) / 100;

SUM_A_CONV = (event.appr_auth_cardid_acceptor_sum_ * 1.0) / 100;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.appr_auth_cardid_acceptor_coun >= CNT_LMT and
event.appr_auth_cardid_acceptor_sum_ >= SUM_AMNT_LMT
```

Explanation

> [Cards-Threshold-5] - Decline - Card transactions ({event.appr_auth_cardid_acceptor_coun}) and total amount ({localvar.SUM_A_CONV} {event.currency_code}) in the same merchant in the last 24h are over the limits ({localvar.CNT_LMT} and {localvar.SUM_A_LMT_CONV}).

Mark As

The outcome of this rule is: **decline**

Cards-Threshold-6

Comment

> Decline transaction and flag card to be limited if the number of distinct devices per card is higher than threshold. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert

- [Cards-Threshold-6]

Variables

```
DISTINCT_LIMIT = cards__rules_thresholds["max_cnp_card_distinct_device_24h"].threshold;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
event.auth_cnp_distinct_device_id_pe >= DISTINCT_LIMIT
```

Explanation

> [Cards-Threshold-6] - Decline - Distinct devices per card in the last 24 hours are above limit ({localvar.DISTINCT_LIMIT}). '{eventname.card_id}' {event.card_id} flagged to be limited.

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-1

Comment

> Decline and flag card to be limited if the card has a high number of failed transactions in the last 24 hours. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert
- [Cards-Behavior-1]

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_declined_count_24h"].threshold;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
// Limits
event.decline_cardid_count_1d >= COUNT_LIMIT
```

Explanation

> [Cards-Behavior-1] - Decline - Card failed transactions ({event.decline_cardid_count_1d}) in the last 24h are over the limit ({localvar.COUNT_LIMIT}).

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-2

Comment

Flag card to be limited if the card had a same amount transaction in the same merchant (acceptor_id) in the last day.

Actions

- Limit/Review Card
- Alert
- [Cards-Behavior-2]

Variables

```
 COUNT_LIMIT = cards__rules_thresholds["max_card_acceptor_amount_count_appr_24h"].threshold;
AMOUNT_LIMIT = cards__rules_thresholds["max_bill_amount"].threshold;

AMOUNT_CONVERTED = ((event.cardholder_bill_amount ?? 0)* 1.0) / 100;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '') and (event.cardholder_bill_amount ?? 0 != 0)
and
// Limits
event.appr_auth_cardid_acceptor_bill >= COUNT_LIMIT and
event.cardholder_bill_amount >= AMOUNT_LIMIT
```

Explanation

[Cards-Behavior-2] - Card transactions with the same amount in the same merchant ({event.appr_auth_cardid_acceptor_bill} of {localvar.AMOUNT_CONVERTED} {event.currency_code}) in the last 24h are over the limit ({localvar.COUNT_LIMIT}).

Mark As

N/A

Cards-Behavior-3

Comment

Decline transaction and flag card to be limited if this transaction is made on very high fraud rate acceptor, previously not used by a card within 6 months. Depends on 'List Management Service' to put the card in the configured list.

Actions

- Limit/Review Card
- Alert
- [Cards-Behavior-3]

Variables

```
 FRAUD_RATE_LIMIT = cards__rules_thresholds["max_acceptor_fraud_rate_6M"].threshold;
COUNT_LIMIT = cards__rules_thresholds["max_acceptor_count_appr_6M"].threshold;
```

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
// Limits
and
event.appr_auth_acceptorid_count_6m >= COUNT_LIMIT and
event.fraud_ratio_acceptorid_6m > FRAUD_RATE_LIMIT and
event.auth_cardid_acceptorid_count_6 == 0
```

Explanation

> [Cards-Behavior-3] - Decline - Card transaction in an acceptor with fraud rate over the limit ({localvar.FRAUD_RATE_LIMIT}) and where the card had no history in the last 6 months.

Mark As

The outcome of this rule is: **decline**

Cards-Behavior-4

Comment

> Decline transaction if the device is infected with malware.

Actions

- [Cards-Behavior-4]
- Limit/Review Card
- Alert

Variables

N/A

Condition

```
 // Lists
event.card_id_not_blacklisted_nor_wh
and
// Filters
(event.card_id ?? '' != '')
and
event.malware_indicator ?? false
```

Explanation

> [Cards-Behavior-4] - Decline - Malware has been detected

Mark As

The outcome of this rule is: **decline**

Cards - PosNeg Rules

Was this page helpful?Â

Descriptionâ

PosNeg Rules allow you to mark transactions that have entities in PosNeg Lists as "fraud" or "not fraud". By default, the solution contains rules that act on the value âdeclinedâand mark the transaction as "fraud". These rules act only on Authorization or Info events with the pre-defined subtype. For each rule, check the filter inside the condition code block to know which entities are being tested.

Rules Project Metadata

- *Target Variable:* decision

- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-PN1

Comment

Decline cards that are listed to be auto declined.

Actions

- [Cards-PN1]

Variables

N/A

Condition

```
 // Filters
(event.card_id ?? '' != '')
and
event.posneg_card_value == 'decline'
```

Explanation

[Cards-PN1] - Decline - '{eventname.card_id}' {event.card_id} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN2

Comment

Decline acceptor ZIPs that are listed to be auto declined.

Actions

- [Cards-PN2]

Variables

N/A

Condition

```
 // Filters
(event.acceptor_zip ?? '' != '')
and
event.posneg_zip_value == 'decline'
```

Explanation

[Cards-PN2] - Decline - '{eventname.acceptor_zip}' {event.acceptor_zip} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN3

Comment

Decline acceptor country codes that are listed to be auto declined.

Actions

- [Cards-PN3]

Variables

N/A

Condition

```
 // Filters
(event.acceptor_country_code ?? '' != '')
and
event.posneg_country_value == 'decline'
```

Explanation

[Cards-PN3] - Decline - '{eventname.acceptor_country_code}' {event.acceptor_country_code} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN4

Comment

Decline customers that are listed to be auto declined.

Actions

- [Cards-PN4]

Variables

N/A

Condition

```
 // Filters
(event.customer_id ?? '' != '')
and
event.posneg_customer_value == 'decline'
```

Explanation

[Cards-PN4] - Decline - '{eventname.customer_id}' {event.customer_id} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN5

Comment

Decline MCCs that are listed to be auto declined.

Actions

- [Cards-PN5]

Variables

N/A

Condition

```
 // Filters
(event.mcc ?? '' != '')
and
event.posneg_mcc_value == 'decline'
```

Explanation

[Cards-PN5] - Decline - '{eventname.mcc}' {event.mcc} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN6

Comment

Decline acceptors that are listed to be auto declined.

Actions

- [Cards-PN6]

Variables

N/A

Condition

```
 // Filters
(event.acceptor_id ?? '' != '')
and
event.posneg_acceptor_value == 'decline'
```

Explanation

[Cards-PN6] - Decline - '{eventname.acceptor_id}' {event.acceptor_id} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN7

Comment

Decline terminals that are listed to be auto declined.

Actions

- [Cards-PN7]

Variables

N/A

Condition

```
 // Filters
(event.terminal_id ?? '' != '')
and
event.posneg_terminal_value == 'decline'
```

Explanation

[Cards-PN7] - Decline - '{eventname.terminal_id}' {event.terminal_id} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN8

Comment

Decline device id that is listed to be auto declined.

Actions

- [Cards-PN8]

Variables

N/A

Condition

```
 (event.device_id ?? '' != '')
and
event.posneg_device_id_value == 'decline'
```

Explanation

[Cards-PN8] - Decline - '{eventname.device_id}' {event.device_id} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN9

Comment

Decline card countries that are in blacklist

Actions

- [Cards-PN9]

Variables

N/A

Condition

```
 (event.card_country_code ?? '' != '')
and
event.posneg_card_country_value == 'decline'
```

Explanation

[Cards-PN9] - Decline - '{eventname.card_country_code}' {event.card_country_code} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN10

Comment

Decline IP that is listed to be auto declined.

Actions

- [Cards-PN10]

Variables

N/A

Condition

```
 (event.device_ip ?? '' != '')
and
event.posneg_device_ip_value == 'decline'
```

Explanation

[Cards-PN10] - Decline - Device IP ({event.device_ip}) is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Cards-PN11

Comment

Approve cards that are listed to be auto approved.

Actions

- [Cards-PN11]

Variables

N/A

Condition

```
 // Filters
(event.card_id ?? '' != '')
and
event.posneg_card_value == 'approve'
```

Explanation

[Cards-PN11] - Approve - '{eventname.card_id}' {event.card_id} is listed to be auto approved.

Mark As

The outcome of this rule is: **approve**

Cards - SCA Exemption Rules

Was this page helpful?Â

Descriptionâ

This rules project is part of the PSD2 compliance strategy offered by TFB. If this regulation does not apply to the customer's jurisdiction, this rules project can be removed from the Pulse workflow without the risk of compromising the fraud detection rate.

The *Cards - SCA Exemption* rules project provides a recommendation on whether Feedzai's customers should request SCA to a consumer performing a card payment.

Rules within **Cards - SCA Exemption** implement the conditions defined by PSD2's Exemption Articles (i.e. Articles 11 to 18) that exempt card payments from SCA. [Download the compliance matrix]() to learn more about these articles.

It is worth highlighting that the Cards-SCA-11-1 and Cards-SCA-11-2 rules consider VISA's codes in the `field pos_card_read_method` field, but other codes may be used if the transaction does not belong to a VISA card.

Rules Project Metadata

- *Target Variable:* sca_required
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-SCA11-1

Comment

PSD2 Article 11 - Contactless Payments at point of sale (sum amount)

Actions

- PSD2-Article-11
- [Cards-SCA11-1]

Variables

```
 AMOUNT_LIMIT = cards__rules_thresholds["psd2_art11_amount_limit"].threshold;
SUM_AMOUNT_LIMIT = cards__rules_thresholds["psd2_art11_sum_amount_limit"].threshold;
METRIC_IS_EXPIRED = event.appr_auth_contactless_last_sca == null;
```

Condition

```
 (event.cardholder_bill_amount ?? 0 != 0) and (event.pos_card_read_method ?? '' != '')
  and !METRIC_IS_EXPIRED
  and event.cardholder_bill_amount <= AMOUNT_LIMIT
  and (event.appr_auth_contactless_last_sca + event.appr_auth_contactless_last_str <=
SUM_AMOUNT_LIMIT)
  and (event.pos_card_read_method == '91' or event.pos_card_read_method == '07') // contactless -
according to VISA's Pan Entry Mode Codes
```

Explanation

[Cards-SCA11-1] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 11

Mark As

The outcome of this rule is: **false**

Cards-SCA11-2

Comment

PSD2 Article 11 - Contactless Payments at point of sale (count transactions)

Actions

- PSD2-Article-11
- [Cards-SCA11-2]

Variables

```
 AMOUNT_LIMIT = cards__rules_thresholds["psd2_art11_amount_limit"].threshold;
COUNT_LIMIT = cards__rules_thresholds["psd2_art11_count_limit"].threshold;
METRIC_IS_EXPIRED = event.appr_auth_contactless_last_ssr == null;
```

Condition

```
 (event.cardholder_bill_amount ?? 0 != 0) and (event.pos_card_read_method ?? '' != '')
  and !METRIC_IS_EXPIRED
  and event.cardholder_bill_amount <= AMOUNT_LIMIT
  and (event.appr_auth_contactless_last_ssr + event.appr_auth_contactless_last_sur <=
COUNT_LIMIT)
  and (event.pos_card_read_method == '91' or event.pos_card_read_method == '07') // contactless -
according to VISA's Pan Entry Mode Codes
```

Explanation

> [Cards-SCA11-2] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 11

Mark As

The outcome of this rule is: **false**

Cards-SCA12

Comment

> PSD2 Article 12 - Unattended terminals for transport fares and parking fees

Actions

- PSD2-Article-12
- [Cards-SCA12]

Variables

N/A

Condition

```
 (event.mcc ?? '' != '')
and
sca_exempted_mccs.contains(event.mcc)
```

Explanation

> [Cards-SCA12] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 12

Mark As

The outcome of this rule is: **false**

Cards-SCA13

Comment

> PSD2 Article 13 - Trusted beneficiaries

Actions

- [Cards-SCA13]
- PSD2-Article-13

Variables

N/A

Condition

```
event.trusted_acceptor ?? false
```

Explanation

> [Cards-SCA13] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 13

Mark As

The outcome of this rule is: **false**

Cards-SCA14

Comment

> PSD2 Article 14 - Recurring transactions

Actions

- PSD2-Article-14
- [Cards-SCA14]

Variables

N/A

Condition

```
(event.first_recurring_payment ?? false == false)
 and (event.recurring_payment ?? false)
```

Explanation

> [Cards-SCA14] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 14

Mark As

The outcome of this rule is: **false**

Cards-SCA16-1

Comment

> PSD2 Article 16 - Low-value transactions (sum amount)

Actions

- [Cards-SCA16-1]

- PSD2-Article-16

Variables

```
 AMOUNT_LIMIT = cards__rules_thresholds["psd2_art16_amount_limit"].threshold;
SUM_AMOUNT_LIMIT = cards__rules_thresholds["psd2_art16_sum_amount_limit"].threshold;
METRIC_IS_EXPIRED = event.appr_auth_last_sca_at_cardid_s == null;
```

Condition

```
 (event.cardholder_bill_amount ?? 0 != 0)
  and !METRIC_IS_EXPIRED
  and event.cardholder_bill_amount <= AMOUNT_LIMIT
  and (event.appr_auth_last_sca_at_cardid_s + event.appr_auth_last_sca_at_cardidsr <=
SUM_AMOUNT_LIMIT)
```

Explanation

> [Cards-SCA16-1] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 16

Mark As

The outcome of this rule is: **false**

Cards-SCA16-2

Comment

> PSD2 Article 16 - Low-value transactions (count transactions)

Actions

- [Cards-SCA16-2]
- PSD2-Article-16

Variables

```
 AMOUNT_LIMIT = cards__rules_thresholds["psd2_art16_amount_limit"].threshold;
COUNT_LIMIT = cards__rules_thresholds["psd2_art16_count_limit"].threshold;
METRIC_IS_EXPIRED = event.appr_auth_last_sca_at_cardid_c == null;
```

Condition

```
 (event.cardholder_bill_amount ?? 0 != 0)
  and !METRIC_IS_EXPIRED
  and event.cardholder_bill_amount <= AMOUNT_LIMIT
  and (event.appr_auth_last_sca_at_cardid_c + event.appr_auth_last_sca_at_cardidtr <=
COUNT_LIMIT)
```

Explanation

> [Cards-SCA16-2] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 16

Mark As

The outcome of this rule is: **false**

Cards-SCA17

Comment

PSD2 Article 17 - Secure corporate payment processes and protocols

Actions

- [Cards-SCA17]
- PSD2-Article-17

Variables

N/A

Condition

```
(event.card_category ?? '' != '')
and
StringUtils.lowerCase(event.card_category) == 'business'
  // if available use channel_type (or another field) instead of card category
  // if you use only card category field, it needs to include all corporate cards
```

Explanation

[Cards-SCA17] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 17

Mark As

The outcome of this rule is: **false**

Cards-SCA18-ETV1

Comment

PSD2 Article 18 - Transaction Risk Analysis for Exemption Threshold Value 100

Actions

- PSD2-Article-18
- [Cards-SCA18-ETV1]

Variables

```
fraud_rate = event.reference_fraud_rate ?? fraud_rates["cards"].value ?? 0;

AMOUNT_LIMIT = cards__rules_thresholds["psd2_art18_etv1_amount_limit"].threshold;

FRAUD_RATE_LIMIT = cards__rules_thresholds["psd2_art18_etv1_fraud_rate_limit"].threshold;

SUM_AMOUNT_LIMIT = cards__rules_thresholds["psd2_art18_etv1_sum_amount_limit"].threshold;
```

Condition

```
(event.cardholder_bill_amount ?? 0 != 0)
and
event.cardholder_bill_amount <= AMOUNT_LIMIT
```

```
  and fraud_rate <= FRAUD_RATE_LIMIT
  and event.decision == 'approve'
  and event.auth_cnp_cardid_sum_amount_24h <= SUM_AMOUNT_LIMIT
```

Explanation

> [Cards-SCA18-ETV1] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 18

Mark As

The outcome of this rule is: **false**

Cards-SCA18-ETV2

Comment

> PSD2 Article 18 - Transaction Risk Analysis for Exemption Threshold Value 100-250

Actions

- PSD2-Article-18
- [Cards-SCA18-ETV2]

Variables

```
 fraud_rate = event.reference_fraud_rate ?? fraud_rates["cards"].value ?? 0;

AMOUNT_MIN = cards__rules_thresholds["psd2_art18_etv2_amount_min"].threshold;
AMOUNT_MAX = cards__rules_thresholds["psd2_art18_etv2_amount_max"].threshold;

FRAUD_RATE_LIMIT = cards__rules_thresholds["psd2_art18_etv2_fraud_rate_limit"].threshold;
SUM_AMOUNT_LIMIT = cards__rules_thresholds["psd2_art18_etv2_sum_amount_limit"].threshold;
```

Condition

```
 (event.cardholder_bill_amount ?? 0 != 0)
and
(event.cardholder_bill_amount > AMOUNT_MIN and event.cardholder_bill_amount <= AMOUNT_MAX)
  and fraud_rate <= FRAUD_RATE_LIMIT
  and event.decision == 'approve'
  and event.auth_cnp_cardid_sum_amount_24h <= SUM_AMOUNT_LIMIT
```

Explanation

> [Cards-SCA18-ETV2] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 18

Mark As

The outcome of this rule is: **false**

Cards-SCA18-ETV3

Comment

> PSD2 Article 18 - Transaction Risk Analysis for Exemption Threshold Value 250-500

Actions

- [Cards-SCA18-ETV3]
- PSD2-Article-18

Variables

```
 fraud_rate = event.reference_fraud_rate ?? fraud_rates["cards"].value ?? 0;

AMOUNT_MIN = cards__rules_thresholds["psd2_art18_etv3_amount_min"].threshold;
AMOUNT_MAX = cards__rules_thresholds["psd2_art18_etv3_amount_max"].threshold;

FRAUD_RATE_LIMIT = cards__rules_thresholds["psd2_art18_etv3_fraud_rate_limit"].threshold;
SUM_AMOUNT_LIMIT = cards__rules_thresholds["psd2_art18_etv3_sum_amount_limit"].threshold;
```

Condition

```
 (event.cardholder_bill_amount ?? 0 != 0)
and
(event.cardholder_bill_amount > AMOUNT_MIN and event.cardholder_bill_amount <= AMOUNT_MAX)
  and fraud_rate <= FRAUD_RATE_LIMIT
  and event.decision == 'approve'
  and event.auth_cnp_cardid_sum_amount_24h <= SUM_AMOUNT_LIMIT
```

Explanation

> [Cards-SCA18-ETV3] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 18

Mark As

The outcome of this rule is: **false**

Cards - SCA Required Rules

Was this page helpful?Â

Descriptionâ

This rules project is part of the PSD2 compliance strategy offered by TFB. If this regulation does not apply to the customer's jurisdiction, this rules project can be removed from the Pulse workflow without the risk of compromising the fraud detection rate.

The *Cards - SCA Required* rules project provides a recommendation on whether Feedzai's customers should request SCA to a consumer performing a card payment.

The rules within this rules project focus on transaction risk analysis defined in the PSD2 Article 18 (download the compliance matrix to learn more about this article). This article establishes that issuer banks are allowed to exempt SCA when the payer initiates a remote electronic payment transaction identified by the payment service provider as low risk level. Therefore, these rules evaluate risk factors defined in Article 18 and generate the `sca_required = true` outcome when the transaction is considered risky.

Note that when a rule within this rules project is triggered, the remaining rules are not evaluated (i.e. the workflow is stopped). This also includes rules within the *Cards - SCA Exemption* rules project - the rules project that follows the *Cards - SCA Required* block in the Pulse workflow. This mechanism prevents exemption rules to trigger on top of an SCA-required rule since the SCA exemption cannot be granted anyway.

Rules Project Metadata

- *Target Variable:* sca_required
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Cards-SCAR1

Comment

Mismatch between acceptor country code and card country code and acceptor is new (6 months)

Actions

- [Cards-SCAR1]

Variables

N/A

Condition

```
 ((event.acceptor_country_code ?? '' != '') and (event.card_country_code ?? '' != ''))
and (event.auth_cardid_acceptorid_count_6 == 0)
and (event.acceptor_country_code != event.card_country_code)
```

Explanation

[Cards-SCAR1] - SCA Required - There's a mismatch between the Acceptor country ({event.acceptor_country_code}) and the Card Country ({event.card_country_code}) and Acceptor is new

Mark As

The outcome of this rule is: **true**

Cards-SCAR2

Comment

Customer country changed

Actions

- [Cards-SCAR2]

Variables

N/A

Condition

```
event.customer_country_code_change_f ?? false
```

Explanation

[Cards-SCAR2] - SCA Required - Customer country changed

Mark As

The outcome of this rule is: **true**

Cards-SCAR3

Comment

Customer email changed

Actions

- [Cards-SCAR3]

Variables

N/A

Condition

```
event.customer_email_change_flag ?? false
```

Explanation

[Cards-SCAR3] - SCA Required - Customer email changed

Mark As

The outcome of this rule is: **true**

Cards-SCAR4

Comment

Customer phone changed

Actions

- [Cards-SCAR4]

Variables

N/A

Condition

```
event.customer_phone_change_flag ?? false
```

Explanation

[Cards-SCAR4] - SCA Required - Customer phone changed

Mark As

The outcome of this rule is: **true**

Cards-SCAR5

Comment

Currency changed and MCC is new (6 months)

Actions

- [Cards-SCAR5]

Variables

N/A

Condition

```
event.auth_cnp_currency_changed_24h == 1 and event.auth_cardid_mcc_count_6m == 0
```

Explanation

[Cards-SCAR5] - SCA Required - The currency changed ({event.currency_code}) and MCC is new ({event.mcc})

Mark As

The outcome of this rule is: **true**

Cards-SCAR6

Comment

Transaction amount is higher than 90% of the card balance

Actions

- [Cards-SCAR6]

Variables

```
 RATIO_AMOUNT_BALANCE_LIMIT = cards__rules_thresholds["max_sca_ratio_amount_balance"].threshold;

formatted_ratio = RATIO_AMOUNT_BALANCE_LIMIT * 100;
```

Condition

```
 event.cardholder_bill_amount ?? 0 != 0
   and event.card_available_balance ?? 0 != 0 and
Maths.safeDivision(event.cardholder_bill_amount,event.card_available_balance) >=
RATIO_AMOUNT_BALANCE_LIMIT
```

Explanation

[Cards-SCAR6] - SCA Required - This transaction exhausted {localvar.formatted_ratio} % or more of the available card balance.

Mark As

The outcome of this rule is: **true**

Cards-SCAR7

Comment

Password changed more than 2 times (24hours)

Actions

- [Cards-SCAR7]

Variables

```
COUNT_LIMIT = cards__rules_thresholds["max_card_count_pw_changed_24h"].threshold;
```

Condition

```
event.auth_cnp_cardid_count_pw_chang >= COUNT_LIMIT
```

Explanation

[Cards-SCAR7] - SCA Required - The password changed {localvar.COUNT_LIMIT} or more times in the last 24 hours

Mark As

The outcome of this rule is: **true**

Cards-SCAR8

Comment

Device language changed and IP country code or device id are new (6 months)

Actions

- [Cards-SCAR8]

Variables

N/A

Condition

```
event.auth_device_language_changed_2 == 1 and (event.auth_cardid_device_ip_country_ == 0 or
event.auth_cardid_deviceid_count_6m == 0)
```

Explanation

[Cards-SCAR8] - SCA Required - In the last 24 hours, device language changed and device ip country or device id are new

Mark As

The outcome of this rule is: **true**

Cards-SCAR9

Comment

Authentication type changed and login session_id changed and MCC is new (6 months)

Actions

- [Cards-SCAR9]

Variables

N/A

Condition

```
event.auth_cnp_login_session_id_chan == 1 and event.auth_cnp_auth_type_changed_24h == 1 and
event.auth_cardid_mcc_count_6m == 0
```

Explanation

[Cards-SCAR9] - SCA Required - In the last 24 hours the authentication type and login session_id changed and MCC is new

Mark As

The outcome of this rule is: **true**

Cards-SCAR10

Comment

Transaction amount higher than average amount per transaction in mcc (1 month)

Actions

- [Cards-SCAR10]

Variables

```
Ratio = Maths.safeDivision((event.cardholder_bill_amount ??
0)*1.0,event.auth_cardid_mcc_avg_amount_1m);
RATIO_LIMIT = cards__rules_thresholds["max_ratio_amount_avg_amount_card_mcc_1m"].threshold;

ratio_for_explanation = (RATIO_LIMIT - 1) * 100;
```

Condition

```
Ratio >= RATIO_LIMIT
```

Explanation

[Cards-SCAR10] - SCA Required - Transaction amount is at least {localvar.ratio_for_explanation} % higher than this card's average transaction amount at this MCC

Mark As

The outcome of this rule is: **true**

Cards-SCAR11

Comment

Proxy has been detected

Actions

- [Cards-SCAR11]

Variables

N/A

Condition

```
event.proxy_flag ?? false
```

Explanation

[Cards-SCAR11] - SCA Required - Proxy has been detected

Mark As

The outcome of this rule is: **true**

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/rules-projects/cards-self-service-rules

- References
- Rules Projects
- Cards - Self Service Rules

On this page

Cards - Self Service Rules

Was this page helpful?Â

Descriptionâ

Standard rules written in Pulse are normally developed and tested in staging environments to ensure that the work in development does not affect the work in production. This type of rules is developed by teams that have deep knowledge in this area and often have taken special training. Therefore, there is an inevitable gap between the time that these rules start to be written and the time that they are deployed in production. At the same time, fraudsters are constantly coming up with new workarounds to trick the system. Counteracting new waves of fraud might require tweaking a rule or creating a new one.

Self-service rules (SSR) allow you to fill this gap, and have an active rule in production in a few minutes. With this capability, you can write a rule in **Case Manager**, test and publish it directly to production. As an example, you can use SSRs to create a rule with a different threshold than the one already defined in a standard rule and publish it in production to have it quickly live.

Therefore, this rules project is empty, and is used in the Pulse workflow to receive the SSRs written in Case Manager.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Digital Activity - Decision Rules

Was this page helpful?Â

Descriptionâ

Decision Rules provide the final recommendation to whether a financial institution should accept, challenge, or decline a given event. This recommendation is sent via the API response of a request evoked by the relevant endpoint (Authentication, Credential Update, Device, New Payee, Profile Update, Alias). Decision Rules take into consideration the following:

- The risk level from previous rules projects.
- The score set by the model. By default, the solution does not provide a standard model in the workflow, and thus, the default score is 0.

The *Decision Rules* apply the following logic:

- If fraud rules classified an event as:
    - low risk, the decision is to accept the digital event
    - high risk, the decision is to decline the digital event
    - medium risk, the decision is to challenge the digital event
- If a model score is:
    - below a certain threshold, the decision is to accept the digital event
    - above a certain threshold, the decision is to decline the digital event
    - between medium and high risk thresholds, the decision is to challenge the digital event

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Digital Activity - DR1

Comment

> Decline a digital event if classified as high risk by previous rules

Actions

- [DA-DR1]
- Alert

Variables

N/A

Condition

```
(event.enforced_risk_level ?? event.risk_level)  == 'high'
```

Explanation

> [DA-DR1] - Decline - Classified as having a high risk of fraud by previous rules.

Mark As

The outcome of this rule is: **decline**

Digital Activity - DR2

Comment

> Decline a digital event if model score is above the threshold

Actions

- [DA-DR2]
- Alert

Variables

```
 HIGH_SCORE_THRESHOLD = digital_activity__rules_thresholds["model_high_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
 (score ?? 0) > HIGH_SCORE_THRESHOLD
```

Explanation

> [DA-DR2] - Decline - Model score ({localvar.score}) is above the threshold: {localvar.HIGH_SCORE_THRESHOLD}

Mark As

The outcome of this rule is: **decline**

Digital Activity - DR3

Comment

> Mark a digital event to Challenge if classified as medium risk by previous rules

Actions

- [DA-DR3]

Variables

N/A

Condition

```
 (event.enforced_risk_level ?? event.risk_level) == 'medium'
```

Explanation

> [DA-DR3] - Challenge - Classified as having a medium risk of fraud by previous rules.

Mark As

The outcome of this rule is: **challenge**

Digital Activity - DR4

Comment

> Mark a digital event to Challenge if model score is between medium and high risky thresholds

Actions

- [DA-DR4]

Variables

```
 HIGH_SCORE_THRESHOLD = digital_activity__rules_thresholds ["model_high_score"].threshold;
LOW_SCORE_THRESHOLD = digital_activity__rules_thresholds ["model_low_score"].threshold;
```

```
score = event.fraud_score * 1000;
```

Condition

```
 (score ?? 0) >= LOW_SCORE_THRESHOLD
and
(score ?? 0) <= HIGH_SCORE_THRESHOLD
```

Explanation

[DA-DR4] - Challenge - Model score ({localvar.score}) is between the low and high model thresholds ({localvar.LOW_SCORE_THRESHOLD} - {localvar.HIGH_SCORE_THRESHOLD})

Mark As

The outcome of this rule is: **challenge**

Digital Activity - DR5

Comment

Approve a digital event if classified as low risk by previous rules

Actions

- [DA-DR5]

Variables

N/A

Condition

```
 (event.enforced_risk_level ?? 'none') == 'low'
```

Explanation

[DA-DR5] - Approve - Classified as having a low risk of fraud by previous rules.

Mark As

The outcome of this rule is: **approve**

Digital Activity - DR6

Comment

Approve a digital event if the model score is below the threshold

Actions

- [DA-DR6]

Variables

```
 LOW_SCORE_THRESHOLD = digital_activity__rules_thresholds["model_low_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
(score ?? 0) < LOW_SCORE_THRESHOLD
```

Explanation

> [DA-DR6] - Approve - Model score ({localvar.score}) is below the threshold: {localvar.LOW_SCORE_THRESHOLD}

Mark As

The outcome of this rule is: **approve**

-
- References
- Rules Projects
- Digital Activity - Device and Network Intelligence Rules

On this page

Digital Activity - Device and Network Intelligence Rules

Was this page helpful?Â

Descriptionâ

Device and Network Intelligence Rules monitor the relevant information about the device and/or network (IP address), as well as user session and biometrics profile that are being used to perform a digital event. Transaction Fraud for Banking collects this information through Feedzai's Revelock enricher (i.e. Digital Trust enricher) by leveraging its indicators out of the box. However, the logic of these rules can be adapted to other providers, if needed.

Rules Project Metadata

- *Target Variable:* risk_level

- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Digital Activity - DNI1

Comment

Suspicious indicators are pointing to a Phishing scenario

Actions

- [DA-DNI1]

Variables

```
is_phishing =
 Revelock.existsAlertOfType(event.revelock_alerts ,'Fraudster Behavior') ?? false
 or Revelock.existsAlertOfType(event.revelock_alerts ,'Suspicious Behavior') ?? false;
```

Condition

```
is_phishing
```

Explanation

[DA-DNI1] - High Risk - Suspicious activity is pointing to a Phishing scenario during this session ({event.session_id})

Mark As

The outcome of this rule is: **high**

Digital Activity - DNI2

Comment

Suspicious indicators are pointing to a Malware scenario

Actions

- [DA-DNI2]

Variables

```
is_malware =
 Revelock.existsAlertOfType(event.revelock_alerts ,'Malware') ?? false
 or Revelock.existsAlertOfType(event.revelock_alerts ,'Suspicious APP') ?? false;
```

Condition

```
is_malware
```

Explanation

[DA-DNI2] - High Risk - Device ({event.device_id}) is suspected to be infected with Malware

Mark As

The outcome of this rule is: **high**

Digital Activity - DNI3

Comment

Suspicious indicators are pointing to an ATO scenario

Actions

- [DA-DNI3]

Variables

```
is_ATO =
  Revelock.existsAlertOfType(event.revelock_alerts ,'ATO') ?? false
  and Revelock.getAlertOfTypeDoublePropertyMin(event.revelock_alerts ,'ATO' ,'similarity') ?? 1 <
0.10
  and
Revelock.getAlertOfTypeDoublePropertyMax(event.revelock_alerts ,'ATO' ,'profile_quality') ?? 0 >
0.85;
```

Condition

```
is_ATO
```

Explanation

[DA-DNI3] - High Risk - Suspicious activity pointing to an ATO scenario during this session ({event.session_id})

Mark As

The outcome of this rule is: **high**

Digital Activity - DNI4

Comment

Suspicious indicators are pointing to an Emulator scenario

Actions

- [DA-DNI4]

Variables

```
is_emulator =
  Revelock.existsAlertOfType(event.revelock_alerts ,'Emulator device') ?? false;
```

Condition

```
is_emulator
```

Explanation

[DA-DNI4] - High Risk - Device ({event.device_id}) is suspected to be running an Emulator

Mark As

The outcome of this rule is: **high**

Digital Activity - DNI5

Comment

Suspicious indicators are pointing to a BOT scenario

Actions

- [DA-DNI5]

Variables

```
is_BOT =
 Revelock.existsAlertOfType(event.revelock_alerts ,'BOT Detected') ?? false;
```

Condition

```
is_BOT
```

Explanation

[DA-DNI5] - High Risk - Device ({event.device_id}) is suspected to be running a BOT software application

Mark As

The outcome of this rule is: **high**

Digital Activity - DNI6

Comment

Suspicious indicators are pointing to a Rooted Device scenario

Actions

- [DA-DNI6]

Variables

```
is_rooted =
 Revelock.existsAlertOfType(event.revelock_alerts ,'Rooted device') ?? false;
```

Condition

```
is_rooted
```

Explanation

[DA-DNI6] - Medium Risk - Device ({event.device_id}) is suspected to be Rooted

Mark As

The outcome of this rule is: **medium**

Digital Activity - DNI7

Comment

Login attempt from a Dormant Account with activity pointing to a\ \ Suspicious Network scenario (use of VPN, TOR or Proxy)

Actions

- [DA-DNI7]

Variables

```
is_dormant_account = event.count_account_activity_3m == 0;

is_suspicious_network =
  Revelock.existsAlertOfType(event.revelock_alerts ,'Tor connection') ?? false
  or Revelock.existsAlertOfType(event.revelock_alerts ,'VPN connection') ?? false
  or Revelock.existsAlertOfType(event.revelock_alerts ,'Suspicious network') ?? false;
```

Condition

```
(event.event_type == 'authentication' and event.challenge_requested != true)
  and is_dormant_account and event.account_age >= 90
  and is_suspicious_network
```

Explanation

[DA-DNI7] - Medium Risk - Login from a dormant account ({event.account_id}) with VPN, TOR or Proxy detected

Mark As

The outcome of this rule is: **medium**

Digital Activity - DNI8

Comment

Login attempt from an unknown IP country and device

Actions

- [DA-DNI8]

Variables

```
is_unknown_country =
  Revelock.existsAlertOfType(event.revelock_alerts ,'Network Anomalies 37') ?? false;

is_unknown_device =
  (Revelock.existsAlertOfType(event.revelock_alerts ,'Device Anomalies 20') ?? false) or
(Revelock.existsAlertOfType(event.revelock_alerts, 'Device Anomalies 21') ?? false);
```

Condition

```
(event.event_type == 'authentication' and event.challenge_requested != true)
  and is_unknown_country
  and is_unknown_device
```

Explanation

[DA-DNI8] - Medium Risk - Login attempt from an unknown IP country ({event.device_ip_country_code}) and unknown device ({event.device_id})

Mark As

The outcome of this rule is: **medium**

Digital Activity - DNI9

Comment

Suspicious indicators are pointing to a RAT scenario

Actions

- [DA-DNI9]

Variables

```
is_RAT =
  (
    Revelock.existsAlertOfType(event.revelock_alerts ,'RAT') ?? false
    or Revelock.existsAlertOfType(event.revelock_alerts ,'RAT TeamViewer') ?? false
  )
  and (event.revelock_score ?? 0 >= 90 and event.revelock_score ?? 0 <= 99)
  ;
```

Condition

```
is_RAT
```

Explanation

[DA-DNI9] - Medium Risk - Device ({event.device_id}) is suspected to be running a RAT

Mark As

The outcome of this rule is: **medium**

Digital Activity - Feedback Rules

Was this page helpful?Â

Descriptionâ

Feedback Rules evaluate events that contain information on whether the digital event was marked as fraudulent or legitimate. This feedback comes from:

- The analyst decision in Case Manager (i.e. `feedback_type` field equal to `cm_feedback`).

It is common to have rules with actions that are preconfigured in the List Management Service leveraging automatic list transitions. For example, this rules project contains a rule that triggers the Auto Decline Account action. It works as follows:

- The rule is triggered when the event is confirmed as fraud (i.e. the `feedback_value` field is equal to `true`). As a result, the action Auto Decline Account is applied. This action is configured in the List Management Service and it adds the account associated with the event to a list of Accounts. Events with accounts within this list are automatically declined in future transactions by *PosNeg Rules*.

Having this example as a reference, you can add rules that leverage different actions preconfigured in the List Management Service, or even write rule conditions that trigger for other feedback values.

Rules Project Metadata

- *Target Variable:* risk_level
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Digital Activity - FB1

Comment

Flag account to be auto declined if customer does not acknowledge transaction or analyst marks as 'Fraud'. Depends on 'List Management Service' to put the account in the configured list.

Actions

- Auto Decline Account
- [DA-FB1]

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback'
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == true
and
// Filters
(event.account_id ?? '' != '')
```

Explanation

[DA-FB1] - Customer does not acknowledge transaction or analyst marks as 'Fraud'. Account Id ({event.account_id}) flagged for auto decline.

Mark As

N/A

Digital Activity - FB2

Comment

Flag account to be auto-approved if customer acknowledges transaction or analyst marks as 'Not Fraud'. Depends on 'List Management Service' to put the account in the configured list.

Actions

- [DA-FB2]
- Auto Approve Account

Variables

N/A

Condition

```
 // Endpoints
event.event_type == 'feedback'
and
// Fraud
event.feedback_label == 'fraud' and event.feedback_value == false
and
// Filters
(event.account_id ?? '' != '')
```

Explanation

[DA-FB2] - Customer acknowledges transaction or analyst marks as 'Not Fraud'. Account Id ({event.account_id}) flagged for auto approve.

Mark As

N/A

Digital Activity - Fraud Rules

Descriptionâ

Fraud Rules look for common fraud patterns in digital events (non-financial events). If triggered, the rules included in this rules project mark the event as `high`, `medium` or `low` risk. Decision Rules block will then provide the final recommendation to whether a financial institution should accept, challenge, or decline a given event, according to the correspondent risk level. Check Decision Rules for more details about Feedzaiâs final recommendation.

info

Note that events marked as `medium` or `low` risk do not trigger an alert in TFBâ□□s pre-packaged resource, but this option is configurable.

Rules Project Metadata

- *Target Variable:* risk_level
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Digital Activity - FR1

Comment

Login attempt from a Device Id linked to multiple customers (last 7d) and with a high number of logins (last 90m)

Actions

- [DA-FR1]

Variables

```
 max_customers_per_device =
digital_activity__rules_thresholds['max_customers_per_device'].threshold;

max_logins_per_device = digital_activity__rules_thresholds['max_logins_per_device'].threshold;
```

Condition

```
 //Filters
(event.device_id ?? '' != '') and (event.customer_id ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Limits
event.distinct_customer_id_per_devic >= max_customers_per_device
and
event.count_logins_per_device_id_90m >= max_logins_per_device
```

Explanation

[DA-FR1] - Medium Risk - Multiple customers per device, last 7d, ({{event.distinct_customer_id_per_devic}}) above [{localvar.max_customers_per_device}] and logins, last 90m, ({event.count_logins_per_device_id_90m}) above [{localvar.max_logins_per_device}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR2

Comment

Login attempt from an IP Address linked to multiple customers and with a high number of login attempts, in the last hour

Actions

- [DA-FR2]

Variables

```
 max_customers_per_ip_addr =
digital_activity__rules_thresholds['max_customers_per_ip_addr'].threshold;
```

```
max_logins_per_ip_addr =
digital_activity__rules_thresholds['max_logins_per_ip_addr'].threshold;
```

Condition

```
 //Filters
(event.device_ip_address ?? '' != '') and (event.customer_id ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Limits
event.distinct_customer_id_per_devsr >= max_customers_per_ip_addr
and
event.count_logins_per_device_ip_asr >= max_logins_per_ip_addr
```

Explanation

> [DA-FR2] - Medium Risk - Multiple customers per IP, last 1h, ({event.distinct_customer_id_per_devsr}) above [{localvar.max_customers_per_ip_addr}] and logins attempts ({event.count_logins_per_device_ip_asr}) above [{localvar.max_logins_per_ip_addr}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR3

Comment

> Login attempt from an unseen device with a high number of failed login attempts, in the last 90m

Actions

- [DA-FR3]

Variables

```
 max_failed_logins_per_device =
digital_activity__rules_thresholds['max_failed_logins_per_device'].threshold;

min_succ_logins_per_account =
digital_activity__rules_thresholds['min_succ_logins_per_account'].threshold; // min of 10 logins
per week
```

Condition

```
 //Filters
(event.device_id ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Account Behavior
event.count_succ_logins_per_accounsr >= min_succ_logins_per_account
and
event.count_logins_per_device_id_1m == 0
and
//Limits
event.count_failed_logins_per_device >= max_failed_logins_per_device
```

Explanation

> [DA-FR3] - Medium Risk - Device Id ({event.device_id}) not seen in the last month and number of failed logins per device, in the last 90m ({event.count_failed_logins_per_device}), is above limit [{localvar.max_failed_logins_per_device}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR4

Comment

> Login attempt from an unseen IP Address with a high number of failed login attempts, in the last 90min

Actions

- [DA-FR4]

Variables

```
 max_failed_logins_per_ip_addr =
digital_activity__rules_thresholds['max_failed_logins_per_ip_addr'].threshold;

min_succ_logins_per_account =
digital_activity__rules_thresholds['min_succ_logins_per_account'].threshold; // min of 5 logins
per week
```

Condition

```
 //Filters
(event.device_ip_address ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Limits
event.count_succ_logins_per_accounsr >= min_succ_logins_per_account
and
event.count_logins_per_device_ip_add == 0
and
event.count_failed_logins_per_devisr >= max_failed_logins_per_ip_addr
```

Explanation

> [DA-FR4] - Medium Risk - IP Address ({event.device_ip_address}) not seen in the last month and number of failed logins per IP Address, in the last 90m ({event.count_failed_logins_per_devisr}), is above limit [{localvar.max_failed_logins_per_ip_addr}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR5

Comment

> High number of failed login attempts per account, in the last minute

Actions

- [DA-FR5]

Variables

```
max_failed_logins_per_account_1min =
digital_activity__rules_thresholds['max_failed_logins_per_account_1min'].threshold;
```

Condition

```
//Filters
(event.account_id ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
//Limits
event.count_failed_logins_per_accoun >= max_failed_logins_per_account_1min
```

Explanation

[DA-FR5] - Medium Risk - Number of failed logins per account ({event.account_id}), in the last minute ({event.count_failed_logins_per_accosr}), is above limit
[{localvar.max_failed_logins_per_account_1min}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR6

Comment

Login attempts from different locations than the account location are above a set threshold, in the last hour

Actions

- [DA-FR6]

Variables

```
max_distinct_locations_per_account =
digital_activity__rules_thresholds['max_distinct_locations_per_account'].threshold;

customer_region_to_lower_case =
com.feedzai.commons.udfs.StringTools.lowerCase(event.customer_region);
device_ip_region_to_lower_case =
com.feedzai.commons.udfs.StringTools.lowerCase(event.device_ip_region);
```

Condition

```
//Filters
(event.account_id ?? '' != '') and (event.customer_region ?? '' != '') and
(event.device_ip_region ?? '' != '')
and
(event.event_type == 'authentication' and event.challenge_requested != true)
and
customer_region_to_lower_case != device_ip_region_to_lower_case
```

```
and
//Limits
event.distinct_login_locations_per_a >= max_distinct_locations_per_account
```

Explanation

[DA-FR6] - Medium Risk - Login attempts ({event.distinct_login_locations_per_a}) from distinct locations than customer location ({event.customer_region}) are above [{localvar.max_distinct_locations_per_account}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR7

Comment

Multiple account changes in one session

Actions

- [DA-FR7]

Variables

```
max_account_changes = digital_activity__rules_thresholds['max_account_changes'].threshold;
```

Condition

```
//Filters
(event.account_id ?? '' != '') and (event.session_id ?? '' != '')
and
event.event_type != 'authentication'
and
event.challenge_requested != true
and
//Limits
event.count_account_changes_per_sess >= max_account_changes
```

Explanation

[DA-FR7] - Medium Risk - Number of account changes ({event.count_account_changes_per_sess}) per session ({event.session_id}) is above limit [{localvar.max_account_changes}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR8

Comment

Number of new payee requests per account is above a set threshold, in the last minute

Actions

- [DA-FR8]

Variables

```
 max_new_payee_per_account_1min =
digital_activity__rules_thresholds['max_new_payee_per_account_1min'].threshold;
```

Condition

```
 //Filters
(event.account_id ?? '' != '')
and
event.event_type == 'new_payee'
and
event.challenge_requested != true
and
//Limits
event.count_new_payee_per_account_1m >= max_new_payee_per_account_1min
```

Explanation

> [DA-FR8] - Medium Risk - Number of new payee requests per account ({event.account_id}), in the last minute ({event.count_new_payee_per_account_1m}), is above limit [{localvar.max_new_payee_per_account_1min}]

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR9

Comment

> New trusted device added followed by an account change request or new payee request, in the last 24h

Actions

- [DA-FR9]

Variables

N/A

Condition

```
 //Filters
(event.account_id ?? '' != '')
and
 (event.event_type == 'profile_update' or event.event_type == 'new_payee')
and
 event.challenge_requested != true
and
//Previous event
 event.account_previous_change_is_new == 1
```

Explanation

> [DA-FR9] - Medium Risk - New trusted device added followed by account change request ({event.event_type})

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR10

Comment

Failed login attempts per account followed by a successful login and an account change request, in the last 24h

Actions

- [DA-FR10]

Variables

```
max_failed_logins_per_account_24h =
digital_activity__rules_thresholds['max_failed_logins_per_account_24h'].threshold;
```

Condition

```
//Filters
(event.account_id ?? '' != '')
and
event.event_type != 'authentication'
and
event.challenge_requested != true
and
//Previous Event
event.account_previous_activity_is_s == 1
and
//Limits
event.count_failed_logins_per_accosr >= max_failed_logins_per_account_24h
```

Explanation

[DA-FR10] - Medium Risk - Failed login attempts per account ({event.count_failed_logins_per_accosr}) followed by a successful login and an account change ({event.event_type}; {event.event_sub_type}), in the last 24 hours

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR11

Comment

First successful login in the last month, followed by credentials or profile update request, in the last 24h

Actions

- [DA-FR11]

Variables

N/A

Condition

```
//Filters
((event.account_id ?? '' != '') and event.account_age ?? 0 > 30)
and
(event.event_type == 'profile_update' or event.event_type == 'credentials_update')
```

```
and
//Previous event
event.account_previous_activity_is_s == 1
and
//Account historical behavior
event.count_succ_logins_per_accounsr == 0
```

Explanation

[DA-FR11] - Medium Risk - First successful login in 1 month followed by an account update ({event.event_type}; {event.event_sub_type}), in the last hour

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR12

Comment

Login followed by a password reset at unusual hours, in the last 24h

Actions

- [DA-FR12]

Variables

```
 max_account_activity_time_of_day_3m_th =
digital_activity__rules_thresholds['max_account_activity_time_of_day_3m_th'].threshold; // max of
5% logins at this time of day

min_account_activity_3m_th =
digital_activity__rules_thresholds['min_account_activity_3m_th'].threshold; // min 10 logins per
week = 120 logins in 3 months

percentage_account_activity_time_of_day_3m =
Maths.safeDivision(event.count_succ_logins_per_account_, event.count_succ_logins_per_accountr);
```

Condition

```
 //Filters
(event.account_id ?? '' != '') and (event.account_age ?? 0 > 90)
and
(event.event_type == 'credentials_update' and event.event_sub_type == 'password_reset')
and
event.challenge_requested != true
and
//Previous event
event.account_previous_activity_is_s == 1
and
//Account historical behavior
percentage_account_activity_time_of_day_3m <= max_account_activity_time_of_day_3m_th
and
event.count_succ_logins_per_accountr >= min_account_activity_3m_th
```

Explanation

[DA-FR12] - Medium Risk - Login followed by a password reset at unusual hours ({event.event_occurred_at}).

Mark As

The outcome of this rule is: **medium**

Digital Activity - FR13

Comment

> Password change followed by new device request from an unusual location, different than customer location, in the last 24h

Actions

- [DA-FR13]

Variables

```
min_account_activity_1m =
digital_activity__rules_thresholds['min_account_activity_1m'].threshold; // min 10 logins per
week = 40 logins in 1 month

customer_region_to_lower_case =
com.feedzai.commons.udfs.StringTools.lowerCase(event.customer_region);
device_ip_region_to_lower_case =
com.feedzai.commons.udfs.StringTools.lowerCase(event.device_ip_region);
```

Condition

```
//Filters
((event.account_id ?? '' != '') and event.account_age ?? 0 > 30)
and
(event.event_type == 'device' and event.challenge_requested != true)
and
customer_region_to_lower_case != device_ip_region_to_lower_case
and
//Previous event
event.account_previous_change_is_pas == 1
and
//Account historical behavior
event.count_login_per_account_and_ip == 0
and
event.count_succ_logins_per_accounsr >= min_account_activity_1m
```

Explanation

> [DA-FR13] - Medium Risk - Password change followed by new device request ({event.new_device_id}) from an unusual ({event.device_ip_region}) different than customer location ({event.customer_region}), in the last 24 hours

Mark As

The outcome of this rule is: **medium**

Digital Activity - PosNeg Rules

Was this page helpful?Â

Descriptionâ

PosNeg Rules mark digital activities as `decline` or `accept` when the event has entities that are in Pulse lists. Examples of entities include IP, device ID, account ID, and Alias ID. PosNeg Rules act only on requests evoked by the Digital Activity endpoints.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Digital Activity - PN1

Comment

Digital activity from an IP that is listed to be auto declined.

Actions

- [DA-PN1]

Variables

N/A

Condition

```
 //Filters
(event.device_ip_address ?? '' != '')
and
event.posneg_device_ip_value == 'decline'
```

Explanation

[DA-PN1] - Decline - IP address {{event.device_ip_address}} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Digital Activity - PN2

Comment

Digital activity from a Device Id that is listed to be auto declined.

Actions

- [DA-PN2]

Variables

N/A

Condition

```
 //Filters
(event.device_id ?? '' != '')
and
event.posneg_device_id_value == 'decline'
```

Explanation

[DA-PN2] - Decline - Device ID {{event.device_id}} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Digital Activity - PN3

Comment

Digital activity from an Account Id that is listed to be auto declined.

Actions

- [DA-PN3]

Variables

N/A

Condition

```
 //Filters
(event.account_id ?? '' != '')
and
event.posneg_account_id_value == 'decline'
```

Explanation

[DA-PN3] - Decline - Account ID {{event.account_id}} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Digital Activity - PN4

Comment

Digital activity from an Account Id that is listed to be auto accepted.

Actions

- [DA-PN4]

Variables

N/A

Condition

```
 //Filters
(event.account_id ?? '' != '')
and
event.posneg_account_id_value == 'approve'
```

Explanation

[DA-PN4] - Approve - Account ID {{event.account_id}} is listed to be auto accepted.

Mark As

The outcome of this rule is: **approve**

Digital Activity - PN5

Comment

Creating or changing an Alias Id to one that is listed to be auto declined.

Actions

- [DA-PN5]

Variables

N/A

Condition

```
 //Endpoint
event.event_type == 'alias'
and
//Filters
(event.new_alias_value ?? '' != '')
and
event.posneg_alias_value == 'decline'
```

Explanation

[DA-PN5] - Decline - Alias {{event.new_alias_value}} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Digital Activity - PN6

Comment

Profile Update request to an email address or email domain that is listed to be auto declined.

Actions

- [DA-PN6]

Variables

N/A

Condition

```
 //Endpoint
event.event_type == 'profile_update'
and
//Filters
((event.new_customer_email_addr ?? '' != '') or (event.new_customer_email_domain ?? '' != ''))
 and
((event.posneg_email_addr_value == 'decline') or (event.posneg_email_domain == 'decline'))
```

Explanation

[DA-PN6] - Decline - Profile Update request to an email address {{event.new_customer_email_addr}} or email domain {{event.new_customer_email_addr}} listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Digital Activity - PN7

Comment

> Profile Update request to a phone number that is listed to be auto declined.

Actions

- [DA-PN7]

Variables

N/A

Condition

```
 //Endpoint
event.event_type == 'profile_update'
and
//Filters
(event.new_customer_phone_number ?? '' != '')
and
event.posneg_phone_value == 'decline'
```

Explanation

> [DA-PN7] - Decline - Profile Update request to a phone number {{event.new_customer_phone_number}} listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Digital Activity - Self Service Rules

Was this page helpful?Â

Descriptionâ

Standard rules written in Pulse are normally developed and tested in staging environments to ensure that the work in development does not affect the work in production. This type of rules is developed by teams that have deep knowledge in this area and often have taken special training. Therefore, there is an inevitable gap between the time that these rules start to be written and the time that they are deployed in production. At the same time, fraudsters are constantly coming up with new workarounds to trick the system. Counteracting new waves of fraud might require tweaking a rule or creating a new one.

Self-service rules (SSR) allow you to fill this gap, and have an active rule in production in a few minutes. With this capability, you can write a rule in **Case Manager**, test and publish it directly to production. As an example, you can use SSRs to create a rule with a different threshold than the one already defined in a standard rule and publish it in production to have it quickly live.

Therefore, this rules project is empty, and is used in the Pulse workflow to receive the SSRs written in Case Manager.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Inbound Payments Rules

Was this page helpful?Â

Descriptionâ

The *Inbound Payments* rules project includes a set of rules that are triggered by risky inbound payments, focusing on volume/velocity of fund movements and behavioral indicators.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Inbound - PosNeg 1

Comment

Decline transfer to an account which is listed to be auto declined

Actions

- Inbound-PN1
- Alert

Variables

N/A

Condition

```
 (event.receiver_transaction_bank_acco ?? '' != '')
and
event.posneg_receiver_account_value == 'decline'
```

Explanation

[Inbound-PN1] - Decline - Transfer is to an account ({event.receiver_transaction_bank_acco}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Inbound - PosNeg 2

Comment

Decline transfer to a Bank Identifier Code (BIC) which is listed to be auto declined

Actions

- Alert
- Inbound-PN2

Variables

N/A

Condition

```
 (event.receiver_bank_bic ?? '' != '')
and
event.posneg_receiver_bank_bic_value == 'decline'
```

Explanation

[Inbound-PN2] - Decline - Transfer is to a Bank Identifier Code ({event.receiver_bank_bic}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Inbound - Model Decision 1

Comment

Decline transaction if the model score is above a certain threshold

Actions

- Inbound-MD1
- Alert

Variables

```
 score_th = inbound__rule_thresholds["model_score_th"].threshold;

score = (event.fraud_score ?? 0) * 1000;
```

Condition

```
 score > score_th
```

Explanation

[Inbound-MD1] - Decline - Model score ({localvar.score}) was above the threshold ({localvar.score_th}).

Mark As

The outcome of this rule is: **decline**

Inbound - Scam Prevention 1

Comment

Funds and number of transactions received from international sources in the last 24 hours exceeds defined thresholds

Actions

- Alert
- Inbound-SP1

Variables

```
 count_th = inbound__rule_thresholds["sp1_count_th"].threshold;
amount_th = inbound__rule_thresholds["sp1_amount_th"].threshold;
```

Condition

```
 event.received_transactions_cross_bo >= count_th
and
event.received_amount_cross_border_2 >= amount_th
```

Explanation

[Inbound-SP1] - Alert - Number of cross-border transactions exceeds threshold ({localvar.count_th}) as well as the amount transactacted ({localvar.amount_th}).

Mark As

N/A

Inbound - Scam Prevention 2

Comment

Funds received in the last 24 hours differ significantly from the expectation considering the previous 6 months

Actions

- Alert
- Inbound-SP2

Variables

```
 percent_th = inbound__rule_thresholds["sp2_percent_th"].threshold;
amount_th = inbound__rule_thresholds["sp2_amount_th"].threshold;

account_age_months = (event.event_occurred_at - (event.receiver_account_open_date ?? 0)) /
2678400000.0;  // 1000 * 60 * 60 * 24 * 31
account_age_maximum = Math.min(account_age_months, 6.0);

average_received_amount = Maths.safeDivision(event.received_amount_6m_delay_24h * 1.0,
account_age_maximum);
relative_difference = Maths.safeDivision(Math.abs(event.received_amount_24h -
event.received_amount_6m_delay_24h), event.received_amount_6m_delay_24h);
```

Condition

```
 account_age_maximum >= 6
and
event.received_amount_24h >= amount_th
and
relative_difference >= percent_th
```

Explanation

[Inbound-SP2] - Alert - Funds received exceed {localvar.percent_th} times the 6 months average for an account opened at least {localvar.account_age_maximum} months ago.

Mark As

N/A

Inbound - Money Mules 1

Comment

Daily funds received in a dormant account exceeds threshold and transaction amount exceeds percentage threshold of account balance

Actions

- Inbound-MM1
- Alert

Variables

```
 amount_th = inbound__rule_thresholds["mm1_amount_th"].threshold;
percent_th = inbound__rule_thresholds["mm1_percent_th"].threshold;

account_age_months = ((event.event_occurred_at ?? 0) - (event.receiver_account_open_date ?? 0)) /
2678400000.0; // 1000 * 60 * 60 * 24 * 31

balance_percentage = Maths.safeDivision((event.sender_transaction_amount ?? 0),
(event.receiver_account_balance ?? 0));
```

Condition

```
 event.received_transactions_6m_delay == 0
and
event.received_amount_24h >= amount_th
and
account_age_months >= 6
and
balance_percentage >= percent_th
```

Explanation

[Inbound-MM1] - Alert - Daily funds received exceed threshold ({localvar.amount_th}) as well as the balance percentage ({localvar.percent_th}) for a dormant account.

Mark As

N/A

Inbound - Money Mules 2

Comment

Ratio of funds received to funds sent is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- Inbound-MM2
- Alert

Variables

```
 ratio_th = inbound__rule_thresholds["mm2_ratio_th"].threshold;
total_th = inbound__rule_thresholds["mm2_total_th"].threshold;

ratio = Maths.safeDivision( event.received_amount_7d, event.sent_amount_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
 event.received_amount_7d >= total_th
and
event.sent_amount_7d >= total_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Inbound-MM2] - Alert - Ratio of funds received to funds sent falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

N/A

Inbound - Money Mules 3

Comment

Ratio of funds sent to funds received is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- Inbound-MM3
- Alert

Variables

```
 ratio_th = inbound__rule_thresholds["mm3_ratio_th"].threshold;
total_th = inbound__rule_thresholds["mm3_total_th"].threshold;

ratio = Maths.safeDivision(event.sent_amount_7d, event.received_amount_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
 event.received_amount_7d >= total_th
and
event.sent_amount_7d >= total_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Inbound-MM3] - Alert - Ratio of funds sent to funds received falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

N/A

Inbound - Money Mules 4

Comment

Ratio of transactions sent to transactions received is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- Alert
- Inbound-MM4

Variables

```
 ratio_th = inbound__rule_thresholds["mm4_ratio_th"].threshold;
count_th = inbound__rule_thresholds["mm4_count_th"].threshold;

ratio = Maths.safeDivision(event.sent_transactions_7d, event.received_transactions_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
 event.received_transactions_7d >= count_th
and
event.sent_transactions_7d >= count_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

> [Inbound-MM4] - Alert - Ratio of transactions sent to transactions received falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

N/A

Inbound - Money Mules 5

Comment

> Ratio of transactions received to transactions sent is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- Alert
- Inbound-MM5

Variables

```
 ratio_th = inbound__rule_thresholds["mm5_ratio_th"].threshold;
count_th = inbound__rule_thresholds["mm5_count_th"].threshold;

ratio = Maths.safeDivision(event.received_transactions_7d, event.sent_transactions_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
 event.received_transactions_7d >= count_th
and
event.sent_transactions_7d >= count_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Inbound-MM5] - Alert - Ratio of transactions received to transactions sent falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

N/A

Rules Projects

Was this page helpful?Â

Learn how rules are organized in your Pulse application and find out which rules come out of the box.

Rules project organizationâ

In TFB, rules are organized in a "folder-like system" referred to as rules projects. This allows you to keep your application tidy and apply filter conditions in the workflow so that certain events flow through certain rules projects. TFB includes rules that cover the following use cases:

- **Cards**: Card-present and card-not-present transactions.
- **Digital Activity**: Non-financial digital events, such as login attempts, changes to profile data (e.g. email, phone number) happening in an online banking session.

- **Transfers**: Traditional credit transfers (also called bank transfers or credit transfers), direct debits, and instant transfers (also known as instant payments or real-time payments).

Rules projects are not organized by fraud scenarios because most of the rules apply to more than one fraud scenario (learn more about fraud scenarios and detection). Instead, the out-of-the-box rules are organized by:

- **Payment type specificities**: Some payment types inherit unique fraud patterns. For example, for direct debits, a mismatch between the countries of the sender and receiver bank can be considered suspicious. As another example, in instant transfers, the thresholds on the transfer amount need to be more strict to prevent fraud more efficiently. Separating these rules into different rules projects provides you a logical way of keeping the application clean, making it easier to maintain.
- **Fraud traces and functional purpose**: There are common fraud traces that are present in both card payments and transfers. Examples include velocity rules and repeated transactions. In this case, these rules are organized under *Fraud Rules*. Also, it is common to define rules that trigger when the transaction amount or the number of transactions is above a certain threshold. In this case, these rules are organized under *Entity Limit Rules*.
- **Risk engine processing**: Another organization strategy is to place all the rules that perform certain logic under the same rules project. For example, rules that trigger when the analysts make a decision in Case Manager are under the *Feedback Rules* project, whereas rules that trigger when a certain entity is within a Pulse list are under the *PosNeg Rules* project.

These are just a few examples; the key take away is that a rules project gives you full flexibility to organize rules according to your business needs.

Rules project order and self-serviceabilityâ

The order of the rules projects in the workflow and the order of the rules in a rules project impact how the risk engine generates the final recommendation. This is described in more detail in the How Rules Work page.

For Cards, Transfers, and Digital Activity workflows, the self-service rules project - the block that contains a snapshot of the rules written in Case Manager - is placed before other standard rules project elements. The following workflow belongs to Cards:

With this rules projects' order, only the *Feedback Rules* and *PosNeg Rules* rules projects take precedence over the self-service rules project because they contain rules with a high certainty of risk detection. This allows users to define rules in Case Manager that take precedence over the remaining rules projects, thus promoting self-serviceability. Check the Create Self-Service Rules page to learn about creating rules in Case Manager.

Naming conventionâ

Rules project names follow the naming convention:

< payment type > - < rules project name >

where the payment type is replaced by Cards or Transfers and the rules project name corresponds to the logic explained in the previous section.

Mapping rules to fraud scenariosâ

The following section helps you understand which rules detect each fraud scenario.

Card paymentsâ

Rule Project	Rule	ATO	Card Testing	Carding	CNP (bot/ velocity)	CNP fraud	Lost & Stolen	Phishing	Skimming
Entity Limit Rules	Cards-EL1	x							
	Cards-EL2	x							
Fraud Rules	Cards-Behavior-1		x						

	Cards-Behavior-2						x		
	Cards-Behavior-3	x							
	Cards-Behavior-4							x	x
	Cards-CNP-1				x	x			
	Cards-CNP-2							x	
	Cards-Threshold-1					x			
	Cards-Threshold-2						x	x	
	Cards-Threshold-3						x	x	
	Cards-Threshold-4			x					
	Cards-Threshold-5			x				x	
	Cards-Threshold-6	x					x	x	x
	Cards-Velocity-1				x				x
	Cards-Velocity-2					x			
	Cards-Velocity-3				x				
	Cards-Velocity-4				x				
	Cards-Velocity-5	x							x
	Cards-Velocity-6		x						
	Cards-Velocity-7		x						

Nomenclature in rules project pagesâ

This section describes the naming convention that is used within each rules project page. Each page begins with a general section that is common to all the rules within the rules project, and then provides a list of rules with detailed information. The nomenclature used in the rules project pages is the following:

Information that applies to all the rules:

- **Description**: Rules project description providing details that are common to all the rules within the project.
- **Rules project metadata**: The outcome to which the rules contribute (i.e. the **Target Variable**), and the Tenancy type with the possible values single tenant and multi-tenancy.

Information that applies to each rule:

- **Comment**: Rule description.
- **Condition**: The condition expression must return a boolean value, and that dictates whether the event triggers a rule.
- **Explanation**: Allows you to understand why a rule is triggered in a human-readable way. The explanation is displayed in Case Manager if an event triggers the rule giving extra insights to fraud analysts.
- **Actions**: Operations that are performed when the rule is triggered.
- **Mark As**: The rule outcome value that is written into the field defined by the target variable within the rules project metadata. See the How Rules Work page to understand the logic behind outcome generation.

See an example of how a rule is written in Pulse:

Remarks on rule outcomes and recommendation generationâ

For card payments, all the rules contribute to the `decision` outcome. The first rule triggering sets the final value. Learn more about the default rule precedence.

For transfers, all the rules but those in the *Transfers - Decision Rules* project contribute the `risk_level` outcome. This outcome is used as an intermediate outcome which is then evaluated by the *Transfers - Decision Rules*. The decision rules evaluate the risk level and write a final recommendation to the `decision` outcome.

Scorable events retrieve an API response with the `decision` outcome so that financial institutions can take an informed decision on whether to decline or accept the transaction. See the How Rules Work page for more information about this logic.

Learn moreâ

Check the list of out-of-the-box rules for each rules project:

- Cards
  - Feedback Rules
  - Posneg Rules
  - Entity Limit Rules
  - Fraud Rules
  - Self-Service Rules
  - Decision Rules
  - SCA Required Rules
  - SCA Exemption Rules
- Digital Activity
  - Feedback Rules
  - Posneg Rules
  - Device and Network Intelligence Rules
  - Fraud Rules
  - Self-Service Rules
  - Decision Rules
- Inbound Payments
  - Inbound Payments Rules
- Transfers
  - Feedback Rules
  - Posneg Rules
  - Entity Limit Rules
  - Fraud Rules
  - Instant Transfers Rules
  - Bill Payments
  - Direct Debit Rules
  - Self-Service Rules
  - Decision Rules
  - Device and Network Intelligence Rules
  - SCA Required
  - SCA Exemption

info

A rules project named Sanctions Rules Project may also be included. Note that it is only relevant and active if you have Feedzai's Watchlist Transaction Screening solution enabled. Otherwise, this rules project is disabled, even if visible in the interface.

Additionally, the following pages have rules-related information to help fight fraud more efficiently:

- Fraud Scenarios and Detection page contains information about the TFB's strategy to fight different fraud schemes using default rules that are provided out of the box.
- Work with Rules and Lists guide helps you manage and create new rules. In particular, check the How Rules Work to understand how the Pulse generates a final recommendation.

Transfers - Bill Payments Rules

Was this page helpful?Â

Descriptionâ

This rules project contains rules that prevent fraud associated with bill payments like Boleto and BPay.

The authorized push payment is an example of a common fraud scenario covered by these rules. In this scenario, criminals trick their victims into transferring money to their accounts. For instance, these individuals may act as if they were from a legitimate company or entity, such as the victim's bank. To sound convincing, fraudsters mention personal details about the victim, collected from the victim's social media and/or other public spaces that show their profiles. Fraudsters then advise the victims to transfer their money to another account (e.g. a safe one), as their current account has been compromised.

Other examples comprise payments associated with Boleto Malware or 'Bolware' fraud (specific for Boleto). Fake Boletos are generated by modifying the Boleto's bar code and ID number. These fake Boletos look genuine, but the payment will be processed to a fraudulent account instead.

The bill payment rules check against key information such as whether the payment reference number is in a decline list, payment amount and volume, and payments performed outside of the working hours.

Rules Project Metadata

- *Target Variable:* risk_level
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-BP1

Comment

Payment to a reference number that is listed to be auto declined.

Actions

- [Transfers-BP1]

Variables

N/A

Condition

```
 (event.reference_number ?? '' != '')
and event.posneg_reference_number_value == 'decline'
```

Explanation

[Transfers-BP1] - High Risk - Payment is to a reference number ({event.reference_number}) that is listed to be auto declined.

Mark As

The outcome of this rule is: **high**

Transfers-BP2

Comment

Payment ID was entered manually and the payment amount is above a set threshold

Actions

- [Transfers-BP2]

Variables

```
 max_amt_bp_threshold = transfers__rules_thresholds['max_amt_bp'].threshold;

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency,
max_amt_bp_threshold);
```

Condition

```
 (event.reference_number_entry_mode ?? '' != '') and (event.sender_transaction_amount ?? 0 != 0)
and
event.reference_number_entry_mode == 'manual'
  and event.sender_transaction_amount >= max_amt_bp_threshold
```

Explanation

> [Transfers-BP2] - High Risk - Payment ID was entered manually and the payment amount ({localvar.formatted_amount}) is above {localvar.formatted_th}

Mark As

The outcome of this rule is: **high**

Transfers-BP4

Comment

> Limit unusual payment outside working days with amount above threshold, considering the account behavior of the last 3 months

Actions

- [Transfers-BP4]

Variables

```
 max_account_activity_out_work_days_3m_th =
transfers__rules_thresholds['max_account_activity_out_work_days_3m_th'].threshold;

min_account_activity_3m_th = transfers__rules_thresholds['min_account_activity_3m_th'].threshold;

percentage_account_activity_out_work_days_3m =
Maths.safeDivision(event.count_account_activity_out_wor, event.count_account_activity_3m);

outside_work_days = event.day_of_week == '6' or event.day_of_week == '7';

amount_limit = transfers__rules_thresholds['payment_amount_limit'].threshold;
```

Condition

```
 outside_work_days
and percentage_account_activity_out_work_days_3m <= max_account_activity_out_work_days_3m_th
and event.count_account_activity_3m >= min_account_activity_3m_th
and event.sender_transaction_amount ?? 0 >= amount_limit
```

Explanation

> [Transfers-BP4] - Medium Risk - Unusual payment outside working days with a payment amount ({event.sender_transaction_amount}) above {localvar.amount_limit}, considering the account ({event.sender_account_id}) behavior of the last 3 months

Mark As

The outcome of this rule is: **medium**

Transfers - Decision Rules

Descriptionâ

Decision Rules provide the final recommendation to whether a financial institution should accept, review, or decline a given transaction. This recommendation is sent via the API response of a request evoked by a scoring endpoint (Transfer Initiation). *Decision Rules* take into consideration the following:

- Rules in preceding rules projects in the workflow that triggered.
- The API field `payment_sub_type` to apply specific conditions to credit transfers, direct debits and instant transfers. For example, to only accept or decline events with the `payment_sub_type` field equal to `instant_transfers` (i.e. never send them to review).

- The score set by the model. By default, the solution does not provide a standard model in the workflow, and thus, the default score is 0.

The *Decision Rules* apply the following logic:

- If the score given by a model is above the review or decline thresholds, the transfer is marked as `review` or `declined`, respectively.
- All the rules in the Pulse workflow contribute to the `risk_level` outcome. This outcome is then taken into account by the *Decision Rules* to define the final recommendation using the `decision` outcome. If the previous rules mark the `risk_level` outcome as:
    - `low`: the *Decision Rules* mark the event as `approved`.
    - `medium`: the *Decision Rules* mark the event as `review`.
    - `high`: the *Decision Rules* mark the event as `decline`.
- The `enforced_risk_level` field is taken into account by the *Decision Rules*, and might differ from the `risk_level` outcome if more than one rule is triggered. The purpose of this field is to make sure that the risk level taken into account is the one with the highest priority.
- The *Decision Rules* also assure that if the `payment_sub_type` API field is `instant transfer`, the event is never sent to review. Instead, it is only `declined` or `accepted`, depending on the rules that triggered and the score given by the model.

Check the [Understand How Rules Work] page for more information about this rule logic.

About the review recommendationâ

Feedzai's customers can use the review outcomes according to the business needs. Here are a few examples of how to leverage this outcome value:

- Exclude the transfer from being automatically sent in the daily batch until the analyst takes the decision to manually include it (by labeling as not fraud).
- Include the transfer anyway, and continue its processing unless the analysts mark it as fraud before the cut-off time.

About the rules in this rules projectâ

There are two rules for each final recommendation decision (i.e. accept, review, decline), one rule is associated with the outcome generated by previous rules whereas the other rule is associated with the score generated by the model.

The order of the rules is also important to generate the right recommendation. The first pair of rules set the `decision` outcome as `decline`, the second pair generate `review`, and the last pair generate `accept`.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-DR1

Comment

Approve a Transfer if a whitelist rule was triggered

Actions

- [Transfers-DR1]

Variables

N/A

Condition

```
(event.enforced_risk_level ?? 'none') == 'low'
```

Explanation

[Transfers-DR7] - Approve - Classified as having a low risk of fraud by a whitelist rule.

Mark As

The outcome of this rule is: **approve**

Transfers-DR2

Comment

Decline an Instant Transfer if model score is above the threshold

Actions

- [Transfers-DR2]
- Alert

Variables

```
HIGH_SCORE_THRESHOLD = transfers__rules_thresholds["model_high_score"].threshold;
LOW_SCORE_THRESHOLD = transfers__rules_thresholds["model_low_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
(score ?? 0) >= HIGH_SCORE_THRESHOLD
  or
((score ?? 0) > LOW_SCORE_THRESHOLD and event.payment_sub_type == 'instant_transfer' and
event.instant_transfers_review_outco == false)
```

Explanation

[Transfers-DR2] - Decline - Model score ({localvar.score}) was above the threshold: {localvar.HIGH_SCORE_THRESHOLD} ({localvar.LOW_SCORE_THRESHOLD} for Instant Transfers).

Mark As

The outcome of this rule is: **decline**

Transfers-DR3

Comment

Decline a Transfer if classified as high risk by previous rules

Actions

- [Transfers-DR3]
- Alert

Variables

N/A

Condition

```
 (event.enforced_risk_level ?? event.risk_level)  == 'high'
  or
((event.enforced_risk_level ?? event.risk_level) == 'medium' and event.payment_sub_type ==
'instant_transfer' and event.instant_transfers_review_outco == false)
```

Explanation

[Transfers-DR3] - Decline - Classified as having a high risk of fraud by the rules.

Mark As

The outcome of this rule is: **decline**

Transfers-DR4

Comment

Mark a Transfer to Pending Review if model score is between medium and high risky thresholds

Actions

- Alert
- [Transfers-DR4]

Variables

```
 HIGH_SCORE_THRESHOLD = transfers__rules_thresholds["model_high_score"].threshold;
LOW_SCORE_THRESHOLD = transfers__rules_thresholds["model_low_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
 (score ?? 0) > LOW_SCORE_THRESHOLD
```

Explanation

[Transfers-DR4] - Review - Model score ({localvar.score}) was between the review and risky thresholds ({localvar.LOW_SCORE_THRESHOLD} - {localvar.HIGH_SCORE_THRESHOLD})

Mark As

The outcome of this rule is: **review**

Transfers-DR5

Comment

Mark a Transfer to Pending Review if classified as medium risk by previous rules

Actions

- Alert
- [Transfers-DR5]

Variables

N/A

Condition

```
(event.enforced_risk_level ?? event.risk_level) == 'medium'
```

Explanation

[Transfers-DR5] - Review - Classified as having a medium risk of fraud by the rules.

Mark As

The outcome of this rule is: **review**

Transfers-DR6

Comment

Approve a Transfer if model score is below the threshold

Actions

- [Transfers-DR6]

Variables

```
LOW_SCORE_THRESHOLD = transfers__rules_thresholds["model_low_score"].threshold;

score = event.fraud_score * 1000;
```

Condition

```
(score ?? 0) < LOW_SCORE_THRESHOLD
```

Explanation

[Transfers-DR6] - Approve - Model score ({localvar.score}) was below the threshold: {localvar.LOW_SCORE_THRESHOLD}

Mark As

The outcome of this rule is: **approve**

Transfers-DR7

Comment

Approve a Transfer if classified as low risk by previous rules

Actions

- [Transfers-DR7]

Variables

N/A

Condition

```
(event.enforced_risk_level ?? event.risk_level) == 'low'
```

Explanation

[Transfers-DR7] - Approve - Classified as having a low risk of fraud by the rules.

Mark As

The outcome of this rule is: **approve**

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/rules-projects/transfers-device-and-network-intelligence-rules

Transfers - Device and Network Intelligence Rules

Was this page helpful?Â

Descriptionâ

Device and Network Intelligence Rules monitor the relevant information about the device and/or network (IP address), as well as user session and biometrics profile that are being used to perform a transfer event. Transaction Fraud for Banking collects this information through Feedzai's Revelock enricher (i.e. Digital Trust enricher) by leveraging its indicators out of the box. However, the logic of these rules can be adapted to other providers, if needed.

Rules Project Metadata

- *Target Variable:* risk_level

- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers - DNI1

Comment

Suspicious indicators are pointing to a Phishing scenario

Actions

- [Transfers-DNI1]

Variables

```
is_phishing =
 Revelock.existsAlertOfType(event.revelock_alerts ,'Fraudster Behavior') ?? false
 or Revelock.existsAlertOfType(event.revelock_alerts ,'Suspicious Behavior') ?? false;
```

Condition

```
is_phishing
```

Explanation

[Transfers-DNI1] - High Risk - Suspicious activity is pointing to a Phishing scenario during this session ({event.session_id}).

Mark As

The outcome of this rule is: **high**

Transfers - DNI2

Comment

Suspicious indicators are pointing to a Malware scenario

Actions

- [Transfers-DNI2]

Variables

```
is_malware =
 Revelock.existsAlertOfType(event.revelock_alerts ,'Malware') ?? false
 or Revelock.existsAlertOfType(event.revelock_alerts ,'Suspicious APP') ?? false;
```

Condition

```
is_malware
```

Explanation

[Transfers-DNI2] - High Risk - Device ({event.device_id}) is suspected to be infected with Malware

Mark As

The outcome of this rule is: **high**

Transfers - DNI3

Comment

Suspicious indicators are pointing to an ATO scenario

Actions

- [Transfers-DNI3]

Variables

```
is_ATO =
  Revelock.existsAlertOfType(event.revelock_alerts ,'ATO') ?? false
  and Revelock.getAlertOfTypeDoublePropertyMin(event.revelock_alerts ,'ATO' ,'similarity') ?? 1 <
0.10
  and
Revelock.getAlertOfTypeDoublePropertyMax(event.revelock_alerts ,'ATO' ,'profile_quality') ?? 0 >
0.85;
```

Condition

```
is_ATO
```

Explanation

[Transfers-DNI3] - High Risk - Suspicious activity pointing to an ATO scenario during this session ({event.session_id})

Mark As

The outcome of this rule is: **high**

Transfers - DNI4

Comment

Suspicious indicators are pointing to an Emulator scenario

Actions

- [Transfers-DNI4]

Variables

```
is_emulator =
  Revelock.existsAlertOfType(event.revelock_alerts ,'Emulator device') ?? false;
```

Condition

```
is_emulator
```

Explanation

[Transfers-DNI4] - High Risk - Device ({event.device_id}) is suspected to be running an Emulator

Mark As

The outcome of this rule is: **high**

Transfers - DNI5

Comment

Suspicious indicators are pointing to a BOT scenario

Actions

- [Transfers-DNI5]

Variables

```
is_BOT =
 Revelock.existsAlertOfType(event.revelock_alerts ,'BOT Detected') ?? false;
```

Condition

```
is_BOT
```

Explanation

[Transfers-DNI5] - High Risk - Device ({event.device_id}) is suspected to be running a BOT software application

Mark As

The outcome of this rule is: **high**

Transfers - DNI6

Comment

Suspicious indicators are pointing to a Rooted Device scenario

Actions

- [Transfers-DNI6]

Variables

```
is_rooted =
 Revelock.existsAlertOfType(event.revelock_alerts ,'Rooted device') ?? false;
```

Condition

```
is_rooted
```

Explanation

[Transfers-DNI6] - Medium Risk - Device ({event.device_id}) is suspected to be Rooted

Mark As

The outcome of this rule is: **medium**

Transfers - DNI7

Comment

Transfer from a Dormant Account with activity pointing to a Suspicious\ \ Network scenario (use of VPN, TOR or Proxy)

Actions

- [Transfers-DNI7]

Variables

```
 account_age_months = ((event.created_at ?? 0) - (event.sender_account_open_date ?? 0)) /
2678400000.0; // 1000 * 60 * 60 * 24 * 31

is_dormant_account = event.count_activity_3m_delay_24h == 0;

is_suspicious_network =
  Revelock.existsAlertOfType(event.revelock_alerts ,'Tor connection') ?? false
  or Revelock.existsAlertOfType(event.revelock_alerts ,'VPN connection') ?? false
  or Revelock.existsAlertOfType(event.revelock_alerts ,'Suspicious network') ?? false;
```

Condition

```
 is_dormant_account
  and account_age_months >= 3
  and is_suspicious_network
```

Explanation

[Transfers-DNI7] - Medium Risk - Transfer initiation from a dormant account ({eventname.sender_transaction_bank_accoun}) with VPN, TOR or Proxy detected

Mark As

The outcome of this rule is: **medium**

Transfers - DNI8

Comment

Suspicious activity from an unknown IP country and device

Actions

- [Transfers-DNI8]

Variables

```
 is_unknown_country =
  Revelock.existsAlertOfType(event.revelock_alerts ,'Network Anomalies 37') ?? false;

is_unknown_device =
  (Revelock.existsAlertOfType(event.revelock_alerts ,'Device Anomalies 20') ?? false) or
(Revelock.existsAlertOfType(event.revelock_alerts, 'Device Anomalies 21') ?? false);
```

Condition

```
 is_unknown_country
  and is_unknown_device
```

Explanation

[Transfers-DNI8] - Medium Risk - Suspicious activity detected from an unknown IP country ({event.device_ip_country_code}) and unknown device ({event.device_id})

Mark As

The outcome of this rule is: **medium**

Transfers - DNI9

Comment

Suspicious indicators are pointing to a RAT scenario

Actions

- [Transfers-DNI9]

Variables

```
is_RAT =
  (
    Revelock.existsAlertOfType(event.revelock_alerts ,'RAT') ?? false
    or Revelock.existsAlertOfType(event.revelock_alerts ,'RAT TeamViewer') ?? false
  )
  and (event.revelock_score ?? 0 >= 90 and event.revelock_score ?? 0 <= 99)
  ;
```

Condition

```
is_RAT
```

Explanation

[Transfers-DNI9] - Medium Risk - Device ({event.device_id}) is suspected to be running a RAT

Mark As

The outcome of this rule is: **medium**

Transfers - DNI10

Comment

High-risk indicators are pointing to a RAT scenario,\ \ and the user typed a high-amount transaction

Actions

- [Transfers-DNI10]

Variables

```
amount_threshold = transfers__rules_thresholds['transfers_dni10_high_amount'].threshold;
   is_RAT_and_high_amount_typed =
  (
    Revelock.existsAlertOfType(event.revelock_alerts ,'RAT') ?? false
    or Revelock.existsAlertOfType(event.revelock_alerts ,'RAT TeamViewer') ?? false
  )
  and (event.sender_transaction_amount >= amount_threshold)
  ;
```

Condition

```
is_RAT_and_high_amount_typed
```

Explanation

> [Transfers-DNI10] - High Risk - Device ({event.device_id}) is suspected to be running RAT and user typed in a high-amount transaction

Mark As

The outcome of this rule is: **high**

Transfers - DNI11

Comment

> Limit bill payments with unknown device outside working days,\ \ considering the account behavior in the last 3 months

Actions

- [Transfers-DNI11]

Variables

```
 max_account_activity_out_work_days_3m_th =
transfers__rules_thresholds['max_account_activity_out_work_days_3m_th'].threshold;
min_account_activity_3m_th = transfers__rules_thresholds['min_account_activity_3m_th'].threshold;
percentage_account_activity_out_work_days_3m =
Maths.safeDivision(event.count_account_activity_out_wor, event.count_account_activity_3m);

outside_work_days = event.day_of_week == '6' or event.day_of_week == '7';

is_unknown_device = Revelock.existsAlertOfType(event.revelock_alerts,'Unknown device') ??
false;
```

Condition

```
 event.payment_sub_type == 'bill_payment'
and outside_work_days
and percentage_account_activity_out_work_days_3m <= max_account_activity_out_work_days_3m_th
and event.count_account_activity_3m >= min_account_activity_3m_th
and is_unknown_device
```

Explanation

> [Transfers-DNI11] - Medium Risk - Unusual payment with an unknown device ({event.device_id}) outside working days, considering the account ({event.sender_account_id}) behavior in the last 3 months

Mark As

The outcome of this rule is: **medium**

- References
- Rules Projects
- Transfers - Direct Debit Rules

On this page

Transfers - Direct Debit Rules

Was this page helpful?Â

Descriptionâ

TFB provides a **Direct Debit** rules project to fight fraud schemes that are associated with this payment type. This is important because this payment type has unique patterns that might be considered normal in other payment types. For example, a mismatch between the countries of the sender and receiver bank can be considered suspicious in direct debit, making it worth to send the event to review.

In the workflow, this rule block is configured with a filter so that only events with `payment_sub_type` equal to `direct_debit` are evaluated by **Direct Debit** rules.

Rules Project Metadata

- *Target Variable:* risk_level
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-DD1

Comment

Number of daily Direct Debits for a dormant account exceeds threshold

Actions

- [Transfers-DD1]

Variables

```
 min_count_dd_sender_dormant_th
=transfers__rules_thresholds["min_count_dd_sender_dormant_th"].threshold;

account_age_months = ((event.created_at ?? 0) - (event.sender_account_open_date ?? 0)) /
2678400000.0; // 1000 * 60 * 60 * 24 * 31
```

Condition

```
 event.count_activity_3m_delay_24h == 0
and
event.count_sender_dd_24h >= min_count_dd_sender_dormant_th
and
account_age_months >= 3
```

Explanation

[Transfers-DD1] - High Risk - This sender has no associated transactions in the last 3 months, but in the last 24 hours it suddenly has ({event.count_sender_dd_24h}) associated transactions.

Mark As

The outcome of this rule is: **high**

Transfers-DD2

Comment

Direct debit is to a company which is listed to be auto declined.

Actions

- [Transfers-DD2]

Variables

N/A

Condition

```
 (event.receiver_transaction_bank_acco ?? '' != '')
and
event.posneg_dd_company_value == 'decline'
```

Explanation

[Transfers-DD2] - High Risk - Direct debit is to a company which is listed to be auto declined ({event.receiver_transaction_bank_acco}).

Mark As

The outcome of this rule is: **high**

Transfers-DD3

Comment

Direct debit is to a company which is listed to be auto approved.

Actions

- [Transfers-DD3]

Variables

N/A

Condition

```
 (event.receiver_transaction_bank_acco ?? '' != '')
and
event.posneg_dd_company_value == 'approve'
```

Explanation

[Transfers-DD3] - Low Risk - Direct debit is to a company which is listed to be auto approved ({event.receiver_transaction_bank_acco}).

Mark As

The outcome of this rule is: **low**

Transfers-DD4

Comment

Mismatch between Sender bank's country and Receiver bank's country

Actions

- [Transfers-DD4]

Variables

N/A

Condition

```
 (event.sender_bank_country_code ?? '' != '') and (event.receiver_bank_country_code ?? '' != '')
and
event.sender_bank_country_code != event.receiver_bank_country_code
```

Explanation

[Transfers-DD4] - Medium Risk - There's a mismatch between the Sender's bank country ({event.sender_bank_country_code}) and the Receiver's bank country ({event.receiver_bank_country_code})

Mark As

The outcome of this rule is: **medium**

Transfers-DD5

Comment

Mismatch between Sender's country and Receiver's country

Actions

- [Transfers-DD5]

Variables

N/A

Condition

```
 (event.sender_country_code ?? '' != '') and (event.receiver_country_code ?? '' != '')
  and
event.sender_country_code != event.receiver_country_code
```

Explanation

[Transfers-DD5] - Medium Risk - There's a mismatch between the Sender country ({event.sender_country_code}) and the Receiver country ({event.receiver_country_code})

Mark As

The outcome of this rule is: **medium**

Transfers-DD7

Comment

Returned direct debits from this Sender in the last 3 months exceeds threshold

Actions

- [Transfers-DD7]

Variables

```
 min_count_dd_return_per_sender_3m_th=transfers__rules_thresholds['min_count_dd_return_per_sender_
3m_th'].threshold;
```

Condition

```
event.count_dd_return_per_sender_3m >= min_count_dd_return_per_sender_3m_th
```

Explanation

[Transfers-DD7] - Medium Risk - Count of returned direct debits from this Sender in the last 3 months ({event.count_dd_return_per_sender_3m}) is above {localvar.min_count_dd_return_per_sender_3m_th}.

Mark As

The outcome of this rule is: **medium**

Transfers-DD8

Comment

Daily returned direct debits from this Sender exceeds threshold

Actions

- [Transfers-DD8]

Variables

```
min_count_dd_return_per_sender_24h_th =
transfers__rules_thresholds['min_count_dd_return_per_sender_24h_th'].threshold;
```

Condition

```
event.count_dd_return_per_sender_24h >= min_count_dd_return_per_sender_24h_th
```

Explanation

[Transfers-DD8] - Medium Risk - Count of returned direct debits from this Sender in the last 24 hours ({event.count_dd_return_per_sender_24h}) is above {localvar.min_count_dd_return_per_sender_24h_th}

Mark As

The outcome of this rule is: **medium**

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/rules-projects/transfers-entity-limit-rules

Transfers - Entity Limit Rules

Was this page helpful?Â

Descriptionâ

Entity Limit rules have the purpose of limiting the financial amount and number of transactions. The majority of these rules compare the recent user account behavior with past, long-term behaviors and define trigger conditions based on thresholds. These thresholds compare how much the transaction amount or number of transactions has increased compared to the past.

Rules Project Metadata

- *Target Variable:* risk_level

- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-EL1

Comment

Daily amount per sender exceeds threshold

Actions

- [Transfers-EL1]

Variables

```
 min_sum_amount_sender_24h_th
=transfers__rules_thresholds["min_sum_amount_sender_24h_th"].threshold;

formatted_sum = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sum_amount_per_sender_24h);

formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency,
min_sum_amount_sender_24h_th);
```

Condition

```
 event.sum_amount_per_sender_24h >= min_sum_amount_sender_24h_th
```

Explanation

[Transfers-EL1] - Medium Risk - Sum amount ({localvar.formatted_sum}) by this Sender in last 24h is above {localvar.formatted_th}.

Mark As

The outcome of this rule is: **medium**

Transfers-EL2

Comment

Sender's amount is significantly higher than daily average amount (considering last week's behavior)

Actions

- [Transfers-EL2]

Variables

```
 min_ratio_sender_trx_amount_sender_daily_avg_th
=transfers__rules_thresholds["min_ratio_sender_trx_amount_sender_daily_avg_th"].threshold;

ratio=Maths.safeDivision((event.sender_transaction_amount ?? 0)*1.0,
event.avg_daily_amount_per_sender_1w);

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
```

Condition

```
ratio >= min_ratio_sender_trx_amount_sender_daily_avg_th
```

Explanation

> [Transfers-EL2] - High Risk - Transaction amount ({localvar.formatted_amount}) is significantively higher ({localvar.min_ratio_sender_trx_amount_sender_daily_avg_th} x) than Sender's daily average amount (considering last week's behavior).

Mark As

The outcome of this rule is: **high**

Transfers-EL3

Comment

> Sender's monthly number of transactions is extremely higher than average number of transaction (considering behavior from previous 3 months)

Actions

- [Transfers-EL3]

Variables

```
min_ratio =
transfers__rules_thresholds["min_ratio_sender_trx_count_sender_monthly_avg_th"].threshold;
min_count = transfers__rules_thresholds["min_count_trx_sender_1m_th"].threshold;

ratio = match
        if (event.avg_monthly_count_sender_3m_de == 0.0) then
                return event.count_sender_1m;
        else
                return
Maths.safeDivision(event.count_sender_1m*1.0,event.avg_monthly_count_sender_3m_de);
        end
end
```

Condition

```
event.count_sender_1m >= min_count
and
ratio >= min_ratio
```

Explanation

> [Transfers-EL3] - High Risk - Transaction count ({event.count_sender_1m}) for this Sender in the last month is significantively higher ({localvar.min_ratio} x) than Sender's monthly average count, considering behavior in the last 3 months

Mark As

The outcome of this rule is: **high**

Transfers-EL4

Comment

Mismatch between sender's and receiver's BICs for transfers with amount higher than a established threshold

Actions

- [Transfers-EL4]

Variables

```
 min_amount_th =transfers__rules_thresholds["min_amount_th"].threshold;

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency, min_amount_th);
```

Condition

```
 (event.sender_transaction_amount ?? 0 != 0) and (event.sender_bank_bic ?? '' != '') and
(event.receiver_bank_bic ?? '' != '')
and
event.sender_transaction_amount>=min_amount_th
and
event.sender_bank_bic != event.receiver_bank_bic
```

Explanation

[Transfers-EL4] - High Risk - Transaction amount ({localvar.formatted_amount}) exceeds the established limit ({localvar.formatted_th}) and sender's and receiver's BICs mismatch.

Mark As

The outcome of this rule is: **high**

Transfers-EL6

Comment

Sender's monthly total amount is extremely higher than average total amount (considering behavior from previous 3 months)

Actions

- [Transfers-EL6]

Variables

```
 min_ratio =
transfers__rules_thresholds["min_ratio_sender_trx_amount_sender_monthly_avg_th"].threshold;
min_avg_amount = transfers__rules_thresholds["min_avg_monthly_amount_sender_th"].threshold;
min_count = transfers__rules_thresholds["min_count_sender_1m_th"].threshold;

ratio = match
  if (event.avg_monthly_amount_per_sender_ < min_avg_amount) then
        return -1;
  else
    return Maths.safeDivision((event.sum_amount_per_sender_1m ?? 0)*1.0,
event.avg_monthly_amount_per_sender_);
```

```
  end
end

formatted_sum = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sum_amount_per_sender_1m);
```

Condition

```
 event.count_sender_1m >= min_count
  and
ratio >= min_ratio
```

Explanation

> [Transfers-EL6] - High Risk - Sum amount ({localvar.formatted_sum}) for this sender in the last month is significantively higher ({localvar.min_ratio} x) than Sender's monthly average amount (considering past 3 months)

Mark As

The outcome of this rule is: **high**

Transfers-EL9

Comment

> Mismatch between sender's and receiver's Sort Code for transfers with amount higher than a established threshold

Actions

- [Transfers-EL9]

Variables

```
 min_amount_sort_code_th=transfers__rules_thresholds['min_amount_sort_code_th'].threshold;

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
```

Condition

```
 (event.sender_bank_country_code ?? '' != '') and (event.receiver_bank_country_code ?? '' != '')
and
(event.sender_bank_sort_code ?? '' != '') and (event.receiver_bank_sort_code ?? '' != '')
and
(event.sender_transaction_amount ?? 0 != 0)
and
((event.sender_bank_country_code == 'GB' or event.sender_bank_country_code == 'IE')
and
(event.receiver_bank_country_code == 'GB' or event.receiver_bank_country_code == 'IE'))
and
event.sender_transaction_amount >= min_amount_sort_code_th
and
event.sender_bank_sort_code != event.receiver_bank_sort_code
```

Explanation

> [Transfers-EL9] - High Risk - Transaction amount ({localvar.formatted_amount}) exceeds the established limit ({localvar.min_amount_sort_code_th}) and sender's and receiver's Sort Codes mismatch.

Mark As

The outcome of this rule is: **high**

Transfers-EL10

Comment

Daily amount of transactions made by telephone per Sender exceeds threshold

Actions

- [Transfers-EL10]

Variables

```
min_sum_amount_per_sender_telephone_24h_th=transfers__rules_thresholds['min_sum_amount_per_sende
r_telephone_24h_th'].threshold;

formatted_sum = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sum_amount_per_sender_telephon);

formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency,
min_sum_amount_per_sender_telephone_24h_th);
```

Condition

```
event.sum_amount_per_sender_telephon>= min_sum_amount_per_sender_telephone_24h_th
```

Explanation

[Transfers-EL10] - High Risk - Transaction daily amount for this Sender ({localvar.formatted_sum}) when channel type is telephone exceeds the established limit of {localvar.formatted_th}.

Mark As

The outcome of this rule is: **high**

Transfers-EL11

Comment

Daily transactions per sender made by telephone is extremely higher than monthly average transaction number (considering behavior from previous 3 months)

Actions

- [Transfers-EL11]

Variables

```
min_ratio =
transfers__rules_thresholds['min_ratio_sender_trx_count_channel_telephone_sender_monthly_avg_th']
.threshold;

min_count_trx_sender_telephone_24h_th =
transfers__rules_thresholds['min_count_trx_sender_telephone_24h_th'].threshold;

ratio = match
    if event.avg_monthly_count_sender_telep == 0.0 then
       return event.count_sender_telephone_24h;
```

```
    else
        return ( event.count_sender_telephone_24h/event.avg_monthly_count_sender_telep);
    end
end
```

Condition

```
 ratio >= min_ratio
and
event.count_sender_telephone_24h >= min_count_trx_sender_telephone_24h_th
```

Explanation

[Transfers-EL11] - High Risk - Transaction count ({event.count_sender_telephone_24h}) for this Sender when channel is telephone in last 24h is significantively higher ({localvar.min_ratio} x) than Sender's monthly average count

Mark As

The outcome of this rule is: **high**

Transfers-EL12

Comment

Daily transactions per sender made online is extremely higher than monthly average transaction number (considering behavior from previous 3 months)

Actions

- [Transfers-EL12]

Variables

```
 min_ratio =
transfers__rules_thresholds['min_ratio_sender_trx_count_channel_online_sender_monthly_avg_3m_th']
.threshold;
min_count_trx_sender_online_24h_th =
transfers__rules_thresholds['min_count_trx_sender_online_24h_th'].threshold;

ratio = match
    if event.avg_monthly_count_sender_onlin == 0.0 then
        return event.count_sender_online_24h;
    else
        return ( event.count_sender_online_24h/event.avg_monthly_count_sender_onlin);
    end
end
```

Condition

```
 ratio >= min_ratio
and
event.count_sender_online_24h >= min_count_trx_sender_online_24h_th
```

Explanation

[Transfers-EL12] - High Risk - Transaction count ({event.count_sender_online_24h}) for this Sender when channel is online in last 24h is significantively higher ({localvar.min_ratio} x) than Sender's monthly average count

Mark As

The outcome of this rule is: **high**

Transfers - Feedback Rules

Descriptionâ

Feedback Rules evaluate events that contain information on whether the transaction is fraudulent or legitimate. This feedback comes from:

- The issuer via the Feedback endpoint (i.e. `feedback_type` field equal to `out_of_band`).
- The analyst decision in Case Manager (i.e. `feedback_type` field equal to `cm_feedback`).

It is common to have rules with actions that are preconfigured in the list management service leveraging automatic list transitions. For example, this rules project contains a rule that triggers the Auto Decline Account action. It work as follows:

The rule is triggered when the event is confirmed as fraud (i.e. the `feedback_value` field is equal to `true`). As a result, the action Auto Decline Account is applied. This action is configured in the [List Management Service]() and it adds the account associated with the event to a list of Accounts. Events with accounts within this list are automatically declined in future transactions by *PosNeg Rules*.

Having this example as a reference, you can add rules that leverage different actions preconfigured in the List Management Service, or even write rule conditions that trigger for other feedback values.

Rules Project Metadata

- *Target Variable:* risk_level
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-FB1

Comment

> Flag account to be auto declined if customer does not acknowledge transaction or analyst marks as 'Fraud' . Depends on 'List Management Service' to put the account in the configured list.

Actions

- [Transfers-FB1]
- Auto Decline Account

Variables

N/A

Condition

```
 // Endpoints
(event.feedback_type == "out_of_band" or event.feedback_type == "cm_feedback")
and
// Fraud
event.feedback_label == "fraud" and event.feedback_value == true
and
// Filters
(event.sender_transaction_bank_accoun ?? '' != '')
```

Explanation

> [Transfers-FB1] - Customer did not acknowledge the transaction or analyst marked as 'Fraud'. '{eventname.sender_transaction_bank_accoun}' {event.sender_transaction_bank_accoun} flagged for auto decline.

Mark As

N/A

Transfers-FB2

Comment

> Flag account to be auto approved if customer acknowledges transaction or analyst marks as 'Not Fraud' . Depends on 'List Management Service' to put the account in the configured list.

Actions

- Auto Approve Account
- [Transfers-FB2]

Variables

N/A

Condition

```
 // Endpoints
(event.feedback_type == "out_of_band" or event.feedback_type == "cm_feedback")
and
// Fraud
event.feedback_label == "fraud" and event.feedback_value == false
and
// Filters
(event.sender_transaction_bank_accoun ?? '' != '')
```

Explanation

[Transfers-FB2] - Customer acknowledged the transaction or analyst marked as 'Not Fraud'. '{eventname.sender_transaction_bank_accoun}' {event.sender_transaction_bank_accoun} flagged for auto approve.

Mark As

N/A

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/rules-projects/transfers-fraud-rules

Transfers - Fraud Rules

Was this page helpful?Â

Descriptionâ

Fraud Rules look for common fraud patterns in transfers.Â If triggered, some rules included in this rules project mark the event as `high` risk, which is then marked as `decline` by the *Decision Rules* (i.e. if the `enforced_risk_level` does not override the risk outcome). Other rules mark the event as `medium` risk, and then the *Decision Rules* can trigger an alert. In the case of generating an alert, the event can be further investigated by fraud analysts in Case Manager.

Rules Project Metadata

- *Target Variable:* risk_level

- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-FR1

Comment

Number of transactions per sender in 90 minutes exceeds threshold

Actions

- [Transfers-FR1]

Variables

```
min_count_trx_sender_90min_th =
transfers__rules_thresholds["min_count_trx_sender_90min_th"].threshold;
```

Condition

```
event.count_sender_90min >= min_count_trx_sender_90min_th
```

Explanation

[Transfers-FR1] - High Risk - Count of transactions by this Sender ({event.count_sender_90min}) in the last 90min is above {localvar.min_count_trx_sender_90min_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR2

Comment

Number of transactions per sender in 24 hours exceeds threshold

Actions

- [Transfers-FR2]

Variables

```
min_count_trx_sender_24h_th =
transfers__rules_thresholds['min_count_trx_sender_24h_th'].threshold;
```

Condition

```
event.count_sender_24h >= min_count_trx_sender_24h_th
```

Explanation

[Transfers-FR2] - High Risk - Count of transactions by this Sender ({event.count_sender_24h}) in the last 24h is above {localvar.min_count_trx_sender_24h_th}.

Mark As

The outcome of this rule is: **high**

Transfers-FR3

Comment

Daily limit on the number of different country codes per sender

Actions

- [Transfers-FR3]

Variables

```
min_distinct_country_code_per_sender_24h_th
=transfers__rules_thresholds["min_distinct_country_code_per_sender_24h_th"].threshold;
```

Condition

```
event.distinct_country_per_sender_24 >= min_distinct_country_code_per_sender_24h_th
```

Explanation

[Transfers-FR3] - High Risk - Count of distinct country codes ({event.distinct_country_per_sender_24}) per Sender in 24h is above {localvar.min_distinct_country_code_per_sender_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR5

Comment

Number of daily transactions for a dormant account exceeds threshold and transaction amount exceeds percentage threshold of account balance.

Actions

- [Transfers-FR5]

Variables

```
min_count_trx_sender_dormant_th =
transfers__rules_thresholds["min_count_trx_sender_dormant_th"].threshold;
min_amount_account_balance_percentage_th =
transfers__rules_thresholds["min_amount_account_balance_percentage_th"].threshold;

account_age_months = ((event.created_at ?? 0) - (event.sender_account_open_date ?? 0)) /
2678400000.0; // 1000 * 60 * 60 * 24 * 31

amount_account_balance_percentage = Maths.safeDivision((event.sender_transaction_amount ?? 0),
(event.sender_account_balance ?? 0));
```

Condition

```
event.count_activity_3m_delay_24h == 0
and
event.count_sender_24h >= min_count_trx_sender_dormant_th
and
account_age_months >= 6
and
amount_account_balance_percentage >= min_amount_account_balance_percentage_th
```

Explanation

[Transfers-FR5] - High Risk - This sender has no associated transactions in the last 3 months, but in the last 24 hours it suddenly has {event.count_sender_24h} associated transactions.

Mark As

The outcome of this rule is: **high**

Transfers-FR6

Comment

Daily limit on international transactions over 3K per sender

Actions

- [Transfers-FR6]

Variables

```
min_count_trx_sender_cross_border_amount_3k_24h_th =
transfers__rules_thresholds["min_count_trx_sender_cross_border_amount_3k_24h_th"].threshold;
```

Condition

```
event.count_sender_cross_border_amsr >= min_count_trx_sender_cross_border_amount_3k_24h_th
```

Explanation

[Transfers-FR6] - High Risk - Count of international transactions by this Sender ({event.count_sender_cross_border_amsr}) in the last 24h in is above {localvar.min_count_trx_sender_cross_border_amount_3k_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR7

Comment

Monthly limit on international transactions over 3K per sender

Actions

- [Transfers-FR7]

Variables

```
min_count_trx_sender_cross_border_amount_3k_1m_th =
transfers__rules_thresholds["min_count_trx_sender_cross_border_amount_3k_1m_th"].threshold;
```

Condition

```
event.count_sender_cross_border_amou >= min_count_trx_sender_cross_border_amount_3k_1m_th
```

Explanation

[Transfers-FR7] - High Risk - Count of international transactions by this Sender ({event.count_sender_cross_border_amou}) in the last month in is above {localvar.min_count_trx_sender_cross_border_amount_3k_1m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR11

Comment

Daily limit on different receivers per sender

Actions

- [Transfers-FR11]

Variables

```
min_distinct_receiver_per_sender_24h_th =transfers__rules_thresholds
["min_distinct_receiver_per_sender_24h_th"].threshold;
```

Condition

```
event.distinct_receiver_per_sender_2 >=min_distinct_receiver_per_sender_24h_th
```

Explanation

[Transfers-FR11] - High Risk - Count of distinct receivers ({event.distinct_receiver_per_sender_2}) per Sender in 24 hours is above {localvar.min_distinct_receiver_per_sender_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR13

Comment

Daily limit on different receivers with the same name per sender

Actions

- [Transfers-FR13]

Variables

```
min_distinct_receiver_same_name_per_sender_24h_th
=transfers__rules_thresholds["min_distinct_receiver_same_name_per_sender_24h_th"].threshold;
```

Condition

```
event.distinct_receiver_same_name_pe >= min_distinct_receiver_same_name_per_sender_24h_th
```

Explanation

[Transfers-FR13] - High Risk - The number of distinct Receivers with the same name that received transfers from this Sender ({event.distinct_receiver_same_name_pe}) in the last 24h is above {localvar.min_distinct_receiver_same_name_per_sender_24h_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR15

Comment

Limit on transfers outside sender's usual hours and is to a new receiver (6 months)

Actions

- [Transfers-FR15]

Variables

```
max_logins_time_of_day_th =
transfers__rules_thresholds['max_account_activity_time_of_day_3m_th'].threshold;
min_logins_th = transfers__rules_thresholds['min_account_activity_3m_th'].threshold;

percentage_login_time_of_day = Maths.safeDivision(event.count_logins_time_of_day_3m,
event.count_logins_3m);
```

Condition

```
percentage_login_time_of_day <= max_logins_time_of_day_th
and event.count_logins_3m >= min_logins_th
and event.count_sender_receiver_6m == 0
```

Explanation

[Transfers-FR15] - High Risk - Transaction is outside of usual time of day for this account and is to a new receiver (last 6 months)

Mark As

The outcome of this rule is: **high**

Transfers-FR18

Comment

Monthly limit on different senders per email address

Actions

- [Transfers-FR18]

Variables

```
min_distinct_sender_per_email_1m_th
=transfers__rules_thresholds['min_distinct_sender_per_email_1m_th'].threshold;
```

Condition

```
event.distinct_sender_per_email_1m >= min_distinct_sender_per_email_1m_th
```

Explanation

[Transfers-FR18] - High Risk - Count of distinct senders ({event.distinct_sender_per_email_1m}) for this email address ({event.sender_email}) in the last month is above {localvar.min_distinct_sender_per_email_1m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR19

Comment

Monthly limit on different senders per phone number

Actions

- [Transfers-FR19]

Variables

```
min_distinct_sender_per_phone_1m_th=transfers__rules_thresholds['min_distinct_sender_per_phone_1m_th'].threshold;
```

Condition

```
event.distinct_sender_per_phone_1m>= min_distinct_sender_per_phone_1m_th
```

Explanation

[Transfers-FR19] - High Risk - Count of distinct senders ({event.distinct_sender_per_phone_1m}) for this phone number({event.sender_phone}) in the last month is above {localvar.min_distinct_sender_per_phone_1m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR22

Comment

Multiple failed login attempts followed by transaction to a new added payee

Actions

- [Transfers-FR22]

Variables

```
failed_attempts_th = transfers__rules_thresholds["fr22_failed_attempts_th"].threshold;
new_payee_transfers_th = transfers__rules_thresholds["fr22_new_payee_transfers_th"].threshold;
```

Condition

```
 event.count_failed_logins_24h >= failed_attempts_th
and
event.count_new_payee_transfers_24h >= new_payee_transfers_th
```

Explanation

> [Transfers-FR22] - Medium Risk - Transfers to new payees ({event.count_new_payee_transfers_24h}) exceeded {localvar.new_payee_transfers_th} for an account with multiple failed login attempts ({event.count_failed_logins_24h} > {localvar.failed_attempts_th}).

Mark As

The outcome of this rule is: **medium**

Transfers-FR23

Comment

> Credentials Update or Profile Update followed by transferring funds that exceeds a percentage threshold of the account balance

Actions

- [Transfers-FR23]

Variables

```
 percentage_th = transfers__rules_thresholds["fr23_percentage_th"].threshold;
```

Condition

```
 (event.last_digital_event_type_24h == "credentials_update"
or
event.last_digital_event_type_24h == "profile_update")
and
event.sum_amount_per_sender_24h >= percentage_th * event.sender_account_balance
```

Explanation

> [Transfers-FR23] - Medium Risk - Sender tried to transfer at least {localvar.percentage_th} x their account balance after having updated their credentials or profile.

Mark As

The outcome of this rule is: **medium**

Transfers-FR24

Comment

> Number of different Senders per Device ID in 90 minutes exceeds threshold

Actions

- [Transfers-FR24]

Variables

```
min_distinct_sender_per_device_id_90m_th=transfers__rules_thresholds['min_distinct_sender_per_de
vice_id_90m_th'].threshold;
```

Condition

```
event.distinct_sender_per_device_id_ >= min_distinct_sender_per_device_id_90m_th
```

Explanation

[Transfers-FR24] - High Risk - Count of distinct senders ({event.distinct_sender_per_device_id_}) for this Device Id in last 90 minutes is above {localvar.min_distinct_sender_per_device_id_90m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR25

Comment

Number of different Device IDs per Sender in 90 minutes exceeds threshold

Actions

- [Transfers-FR25]

Variables

```
min_distinct_device_id_per_sender_90m_th=transfers__rules_thresholds['min_distinct_device_id_per_
sender_90m_th'].threshold;
```

Condition

```
event.distinct_device_id_per_sender_>=min_distinct_device_id_per_sender_90m_th
```

Explanation

[Transfers-FR25] - High Risk - Count of distinct devices ({event.distinct_device_id_per_sender_}) per Sender in last 90m is above {localvar.min_distinct_device_id_per_sender_90m_th}

Mark As

The outcome of this rule is: **high**

Transfers-FR27

Comment

Daily number of failed transactions per sender exceeds threshold

Actions

- [Transfers-FR27]

Variables

```
min_count_failed_trx_per_sender_24h_th=transfers__rules_thresholds['min_count_failed_trx_per_sen
der_24h_th'].threshold;
```

Condition

```
event.count_failed_trx_sender_24h >= min_count_failed_trx_per_sender_24h_th
```

Explanation

[Transfers-FR27] - High Risk - Count of failed transactions from this Sender in the last 24hours ({event.count_failed_trx_sender_24h}) is above {localvar.min_count_failed_trx_per_sender_24h_th}.

Mark As

The outcome of this rule is: **high**

Transfers-FR28

Comment

Decline transaction if email domain is gmail and contains a "_" or a "+"

Actions

- [Transfers-FR28]

Variables

N/A

Condition

```
StringUtils.substringAfterLast(event.sender_email ?? '','@') ?? '' == 'gmail.com'
  and
(StringUtils.contains(event.sender_email ?? '', '+') or
StringUtils.contains(event.sender_email ?? '', '_'))
```

Explanation

[Transfers-FR28] - High Risk - Email domain is gmail and contains invalid characters ("_" or "+")

Mark As

The outcome of this rule is: **high**

Transfers-FR30

Comment

Distance between current and previous ip location in the last hour exceeds 1000 km (1 hour)

Actions

- [Transfers-FR30]

Variables

```
min_sender_distance_km_1h_th =
transfers__rules_thresholds["min_sender_distance_km_1h_th"].threshold;

dist =
Location.distFromInKm(event.device_latitude_double,event.device_longitude_double,event.sender_pr
ev_device_lat_1h,event.sender_prev_device_long_1h) ?? 0.0;
```

Condition

```
 (event.sender_prev_device_lat_1h ?? 0.0 != 0.0) and (event.sender_prev_device_long_1h ?? 0.0 !=
0.0)
and
dist >= min_sender_distance_km_1h_th
```

Explanation

[Transfers-FR30] - High Risk - Distance between previous and current IP location in the last hour is higher than {localvar.min_sender_distance_km_1h_th} km

Mark As

The outcome of this rule is: **high**

Transfers-FR32

Comment

Transaction from a device_id with login activity linked to multiple accounts, in the last 3 months

Actions

- [Transfers-FR32]

Variables

```
count_th = transfers__rules_thresholds['distinct_account_per_device_3m_th'].threshold;
```

Condition

```
event.distinct_account_login_per_dev >= count_th
```

Explanation

[Transfers-FR32] - Medium Risk - The number of logins with distinct account id per device ({event.device_id}) in the last 3m ({event.distinct_account_login_per_dev}) is above {localvar.count_th}

Mark As

The outcome of this rule is: **medium**

Transfers-FR33

Comment

Funds sent exceed daily average amount after adding a number of new payees that exceeds a certain threshold

Actions

- [Transfers-FR33]

Variables

```
 ratio_th = transfers__rules_thresholds["fr33_ratio_th"].threshold;
count_th = transfers__rules_thresholds["fr33_count_th"].threshold;

ratio = Maths.safeDivision((event.sender_transaction_amount ?? 0) * 1.0,
event.avg_daily_amount_per_sender_1w);
```

```
formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
```

Condition

```
 event.new_payees_24h > count_th
and
ratio >= ratio_th
```

Explanation

[Transfers-FR33] - High Risk - Transaction amount ({localvar.formatted_amount}) exceeds ({localvar.ratio_th} x) the sender's daily average after adding at least {localvar.count_th} new payees in the last 24 hours.

Mark As

The outcome of this rule is: **high**

Transfers-FR34

Comment

Number of inbound transactions in the last 24 hours exceed a certain threshold for an account with many new beneficiaries in the last 24 hours when using a VPN or Tor connection

Actions

- [Transfers-FR34]

Variables

```
 sender_count_th = transfers__rules_thresholds["fr34_sender_count_th"].threshold;
transaction_count_th = transfers__rules_thresholds["fr34_transaction_count_th"].threshold;

is_suspicious_network =
  Revelock.existsAlertOfType(event.revelock_alerts ,'Tor connection') ?? false
  or Revelock.existsAlertOfType(event.revelock_alerts ,'VPN connection') ?? false;
```

Condition

```
 event.count_inbound_24h >= transaction_count_th
and
event.distinct_receiver_per_sender_2 >= sender_count_th
and
is_suspicious_network
```

Explanation

[Transfers-FR34] - High Risk - Inbound transactions in the last 24 hours ({event.count_inbound_24h}) exceed threshold ({localvar.transaction_count_th}) for an account with {event.distinct_receiver_per_sender_2} new beneficiaries in the last 24 hours.

Mark As

The outcome of this rule is: **high**

Transfers-FR35

Comment

Ratio of funds received to funds sent is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR35]

Variables

```
ratio_th = transfers__rules_thresholds["fr35_ratio_th"].threshold;
total_th = transfers__rules_thresholds["fr35_total_th"].threshold;

ratio = Maths.safeDivision(event.sum_amount_per_receiver_7d, event.sum_amount_per_sender_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
event.sum_amount_per_receiver_7d >= total_th
and
event.sum_amount_per_sender_7d >= total_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Transfers-FR35] - High Risk - Ratio of funds received to funds sent falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**

Transfers-FR36

Comment

Ratio of funds sent to funds received is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR36]

Variables

```
ratio_th = transfers__rules_thresholds["fr36_ratio_th"].threshold;
total_th = transfers__rules_thresholds["fr36_total_th"].threshold;

ratio = Maths.safeDivision(event.sum_amount_per_sender_7d, event.sum_amount_per_receiver_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
 event.sum_amount_per_receiver_7d >= total_th
and
event.sum_amount_per_sender_7d >= total_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

> [Transfers-FR36] - High Risk - Ratio of funds sent to funds received falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**

Transfers-FR37

Comment

> Ratio of transactions sent to transactions received is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR37]

Variables

```
 ratio_th = transfers__rules_thresholds["fr37_ratio_th"].threshold;
count_th = transfers__rules_thresholds["fr37_count_th"].threshold;

ratio = Maths.safeDivision(event.count_per_sender_7d, event.count_per_receiver_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
 event.count_per_receiver_7d >= count_th
and
event.count_per_sender_7d >= count_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

> [Transfers-FR37] - High Risk - Ratio of transactions sent to transactions received falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**

Transfers-FR38

Comment

Ratio of transactions received to transactions sent is outside a certain range and the amounts transferred exceed a certain threshold

Actions

- [Transfers-FR38]

Variables

```
 ratio_th = transfers__rules_thresholds["fr38_ratio_th"].threshold;
count_th = transfers__rules_thresholds["fr38_count_th"].threshold;

ratio = Maths.safeDivision(event.count_per_receiver_7d, event.count_per_sender_7d);
ratio_lower_limit = 1 - ratio_th;
ratio_upper_limit = 1 + ratio_th;
```

Condition

```
 event.count_per_receiver_7d >= count_th
and
event.count_per_sender_7d >= count_th
and
ratio >= ratio_lower_limit
and
ratio <= ratio_upper_limit
```

Explanation

[Transfers-FR38] - High Risk - Ratio of transactions received to transactions sent falls within the pre-set range ({localvar.ratio_lower_limit} to {localvar.ratio_upper_limit}).

Mark As

The outcome of this rule is: **high**

Transfers - Instant Transfers Rules

Descriptionâ

TFB provides **Instant Transfers Rules** to account for fraud schemes that are associated with this payment type. This is important because instant transfers have unique patterns that might be considered normal in other payment types. For example, thresholds on the transfer's amount are more strict on instant transfers than on other transfer payment types. Therefore, this rules project includes rules that limit the transaction amount with lower thresholds.

Additionally, instant transfers do not allow for manual revision due to the instantaneous nature of this payment type (i.e. they are accepted or decline). Therefore, this rules project is built-in with outcome logic to not trigger alerts in Case Manager. See the Understand How Rules Work page for more information on how rules generate a final recommendation.

In the workflow, this rule block is configured with a filter so that only events with `payment_sub_type` equal to `instant_transfer` are evaluated by **Instant Transfer Rules**.

Rules Project Metadata

- *Target Variable:* risk_level
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-IT1

Comment

Transaction amount exceeds sender threshold

Actions

- [Transfers-IT1]

Variables

```
max_trx_amount_sender_th=transfers__rules_thresholds["max_trx_amount_sender_th"].threshold;

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency,
max_trx_amount_sender_th);
```

Condition

```
(event.sender_transaction_amount ?? 0 != 0)
and
event.sender_transaction_amount >= max_trx_amount_sender_th
```

Explanation

[Transfers-IT1] - High Risk - Transaction amount ({localvar.formatted_amount}) for this sender exceeds the established limit ({localvar.formatted_th}) for this type of transfer.

Mark As

The outcome of this rule is: **high**

Transfers-IT2

Comment

Transaction amount exceeds receiver threshold

Actions

- [Transfers-IT2]

Variables

```
max_trx_amount_receiver_th=transfers__rules_thresholds["max_trx_amount_receiver_th"].threshold;

formatted_amount = ValueFormatTools.formatMoney(event.receiver_transaction_currency,
event.receiver_transaction_amount);
```

```
formatted_th = ValueFormatTools.formatMoney(event.receiver_transaction_currency,
max_trx_amount_receiver_th);
```

Condition

```
 (event.receiver_transaction_amount ?? 0 != 0)
and
event.receiver_transaction_amount >= max_trx_amount_receiver_th
```

Explanation

> [Transfers-IT2] - High Risk - Transaction amount ({localvar.formatted_amount}) for this receiver exceeds the established limit ({localvar.formatted_th}) for this type of transfer.

Mark As

The outcome of this rule is: **high**

Transfers-IT3

Comment

> Sender's daily number of transactions is extremely higher than daily average number of transaction (considering behavior from previous 3 months)

Actions

- [Transfers-IT3]

Variables

```
 min_ratio =
transfers__rules_thresholds["min_ratio_sender_instant_transfer_count_sender_daily_avg_3m_th"].th
reshold;

min_count_trx_sender_instant_transfer_24h =
transfers__rules_thresholds["min_count_trx_sender_instant_transfer_24h"].threshold;

ratio = match
        if (event.avg_daily_count_sender_instant == 0.0) then
                return event.count_sender_instant_transfesr;
        else
                return
Maths.safeDivision(event.count_sender_instant_transfesr*1.0,event.avg_daily_count_sender_instant)
;
        end
end
```

Condition

```
 ratio >= min_ratio
and
event.count_sender_instant_transfesr>=min_count_trx_sender_instant_transfer_24h
```

Explanation

> [Transfers-IT3] - High Risk - Transaction count ({event.count_sender_instant_transfesr}) for this sender is significantively higher ({localvar.min_ratio} x) than Sender's daily average count over the last 3 months

Mark As

The outcome of this rule is: **high**

Transfers-IT4

Comment

Wire transaction with no Sender's information

Actions

- [Transfers-IT4]

Variables

```
```

Condition

```
 event.payment_type == 'wire'
  and
((event.sender_name ?? '' == '') or (event.sender_street_addr ?? '' == ''))
```

Explanation

[Transfers-IT4] - High Risk - Wire transaction with no Sender's information.

Mark As

The outcome of this rule is: **high**

Transfers-IT5

Comment

Wire transaction with no Receiver's information

Actions

- [Transfers-IT5]

Variables

N/A

Condition

```
 event.payment_type == 'wire'
  and
((event.receiver_name ?? '' == '') or (event.receiver_street_addr ?? '' == ''))
```

Explanation

[Transfers-IT5] - High Risk - Wire transaction with no Receiver's information.

Mark As

The outcome of this rule is: **high**

Transfers-IT6

Comment

Sender alias is listed to be auto declined.

Actions

- [Transfers-IT6]

Variables

N/A

Condition

```
(event.sender_alias ?? '' != '')
and event.posneg_sender_alias_value == 'decline'
```

Explanation

[Transfers-IT6] - High Risk - Sender alias ({event.sender_alias}) is listed to be auto declined.

Mark As

The outcome of this rule is: **high**

Transfers-IT7

Comment

Receiver alias is listed to be auto declined.

Actions

- [Transfers-IT7]

Variables

N/A

Condition

```
(event.receiver_alias ?? '' != '')
and event.posneg_receiver_alias_value == 'decline'
```

Explanation

[Transfers-IT7] - High Risk - Receiver alias ({event.receiver_alias}) is listed to be auto declined.

Mark As

The outcome of this rule is: **high**

Transfers-IT8

Comment

Instant payments without any description and above amount threshold

Actions

- [Transfers-IT8]

Variables

```
 text_for_self_empty = event.sender_transaction_notes ?? '' == '';
text_for_receiver_empty = event.text_for_receiver ?? '' == '';
max_amt_threshold_empty_text =
transfers__rules_thresholds['max_amt_threshold_empty_text'].threshold;
```

Condition

```
 text_for_self_empty
and text_for_receiver_empty
and (event.sender_transaction_amount ?? 0)>= max_amt_threshold_empty_text
```

Explanation

[Transfers-IT8] - High Risk - Payments without any description and payment amount ({event.sender_transaction_amount}) is above [{localvar.max_amt_threshold_empty_text}]

Mark As

The outcome of this rule is: **high**

Transfers-IT9

Comment

Receiver alias type is email address and the email domain is listed to be auto declined.

Actions

- [Transfers-IT9]

Variables

```

```

Condition

```
 (event.receiver_alias_type ?? '' == 'email') and (event.receiver_alias_email_domain ?? '' != '')
  and event.posneg_receiver_alias_email_do == 'decline'
```

Explanation

[Transfers-IT9] - High Risk - Receiver alias email domain is listed to be auto declined.

Mark As

The outcome of this rule is: **high**

Transfers-IT10

Comment

Instant payment above amount threshold to a receiver alias if alias_last_update_time was less than 24h ago

Actions

- [Transfers-IT10]

Variables

```
 receiver_alias_last_update_hours = (event.event_occurred_at - 
(event.receiver_alias_last_update_tim ?? 0)) / (1000*60*60);
sender_alias_last_update_hours = (event.event_occurred_at - 
(event.sender_alias_last_update_time ?? 0)) / (1000*60*60);
max_amt_threshold_last_update = 
transfers__rules_thresholds['max_amt_threshold_last_update'].threshold;

formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency, 
max_amt_threshold_last_update);
formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency, 
event.sender_transaction_amount);
```

Condition

```
 (
  receiver_alias_last_update_hours < 24
  or sender_alias_last_update_hours < 24
)
and (event.sender_transaction_amount ?? 0) >= max_amt_threshold_last_update
```

Explanation

> [Transfers-IT10] - High Risk - Payment amt ({localvar.formatted_amount}) is above {localvar.formatted_th} and receiver or sender alias last update was less than 24h ago

Mark As

The outcome of this rule is: **high**

Transfers-IT11

Comment

> Instant payments with malformed text (more than 2 special characters, odd keyboard distance) and above amount threshold

Actions

- [Transfers-IT11]

Variables

```
 text_regex = com.feedzai.commons.udfs.StringTools.regexReplaceAll('[^a-zA-
Z0-9.]',event.text_for_receiver,'%');

text_keyboard_distance = com.feedzai.cloud.udf.Strings.keyboardDistance(event.text_for_receiver, 
1);
keyboard_distance_threshold = 
transfers__rules_thresholds['keyboard_distance_threshold'].threshold;

max_amt_threshold_malformed_text = 
transfers__rules_thresholds['max_amt_threshold_malformed_text'].threshold;
```

Condition

```
 (event.text_for_receiver ?? '' != '') and (!
ListedKeywords.containsKeyword(posneg_listed_keywords, "accept", event.text_for_receiver, true))
and
(
  com.feedzai.commons.udfs.StringTools.contains(text_regex ?? '', '%%')
  or text_keyboard_distance < keyboard_distance_threshold
)
and
(event.sender_transaction_amount ?? 0)>= max_amt_threshold_malformed_text
```

Explanation

> [Transfers-IT11] - High Risk - Instant payments with malformed text (more than 2 special characters, odd keyboard distance) and above amount threshold

Mark As

The outcome of this rule is: **high**

Transfers-IT12

Comment

> Receiver alias type is 'phone' and this number is listed to be auto declined.

Actions

- [Transfers-IT12]

Variables

```

```

Condition

```
 (event.receiver_alias_type ?? '') == 'phone'
  and event.receiver_alias ?? '' != ''
  and event.posneg_receiver_alias_phone_va == 'decline'
```

Explanation

> [Transfers-IT12] - High Risk - Receiver alias phone number ({event.receiver_alias}) is listed to be auto declined.

Mark As

The outcome of this rule is: **high**

Transfers-IT13

Comment

> Total amount of instant payments per sender alias in 90 minutes, after a new alias registration, is above a set threshold

Actions

- [Transfers-IT13]

Variables

```
 sender_alias_new_registration = (event.event_occurred_at -
(event.sender_alias_registration_time ?? 0)) / (1000*60);
max_total_amt_new_reg_threshold =
transfers__rules_thresholds['max_total_amt_new_reg_threshold'].threshold;

formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency,
max_total_amt_new_reg_threshold);
formatted_sum = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sum_amt_instant_transfers_per_);
```

Condition

```
 sender_alias_new_registration <= 90
and event.sum_amt_instant_transfers_per_ >= max_total_amt_new_reg_threshold
```

Explanation

> [Transfers-IT13] - High Risk - Total amount per sender alias in the last 90min ({localvar.formatted_sum}) is above {localvar.formatted_th}, after a new alias registration ({event.sender_alias})

Mark As

The outcome of this rule is: **high**

-
- [References]
- [Rules Projects]
- Transfers - PosNeg Rules

On this page

Transfers - PosNeg Rules

Was this page helpful?Â

Descriptionâ

PosNeg Rules implement decisions based on listed entity values. This means that these rules check if entities are listed to be approved / declined, and, based on that, PosNeg rules set the respective approve / decline decision. Examples of entities include accounts, countries, and even blacklisted key-words that come in the transaction notes.

PosNeg Rules act only on requests evoked by the Transfer Initiation endpoint.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-PN1

Comment

Transfer is from a country which is listed to be auto declined.

Actions

- [Transfers-PN1]

Variables

N/A

Condition

```
 (event.sender_bank_country_code ?? '' != '')
and
event.posneg_sender_bank_country_val == 'decline'
```

Explanation

[Transfers-PN1] - Decline - Transfer is from a country ({event.sender_bank_country_code}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN2

Comment

Transfer is to a country which is listed to be auto declined.

Actions

- [Transfers-PN2]

Variables

N/A

Condition

```
 (event.receiver_bank_country_code ?? '' != '')
and
event.posneg_receiver_bank_country_v == 'decline'
```

Explanation

[Transfers-PN2] - Decline - Transfer is to a country ({event.receiver_bank_country_code}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN3

Comment

Transfer is from an account which is listed to be auto declined.

Actions

- [Transfers-PN3]

Variables

N/A

Condition

```
 (event.sender_transaction_bank_accoun ?? '' != '')
and
event.posneg_sender_account_value == 'decline'
```

Explanation

[Transfers-PN3] - Decline - Transfer is from an account ({event.sender_transaction_bank_accoun}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN4

Comment

Transfer is to an account which is listed to be auto declined.

Actions

- [Transfers-PN4]

Variables

N/A

Condition

```
 (event.receiver_transaction_bank_acco ?? '' != '')
and
event.posneg_receiver_account_value == 'decline'
```

Explanation

[Transfers-PN4] - Decline - Transfer is to an account ({event.receiver_transaction_bank_acco}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN5

Comment

Transfer is from a Bank Identifier Code (BIC) which is listed to be auto declined.

Actions

- [Transfers-PN5]

Variables

N/A

Condition

```
 (event.sender_bank_bic ?? '' != '')
and
event.posneg_sender_bank_bic_value == 'decline'
```

Explanation

[Transfers-PN5] - Decline - Transfer is from a Bank Identifier Code ({event.sender_bank_bic}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN6

Comment

Transfer is to a Bank Identifier Code (BIC) which is listed to be auto declined.

Actions

- [Transfers-PN6]

Variables

N/A

Condition

```
 (event.receiver_bank_bic ?? '' != '')
and
event.posneg_receiver_bank_bic_value == 'decline'
```

Explanation

[Transfers-PN6] - Decline - Transfer is to a Bank Identifier Code ({event.receiver_bank_bic}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN7

Comment

Transfer is from a Sort Code which is listed to be auto declined.

Actions

- [Transfers-PN7]

Variables

N/A

Condition

```
 (event.sender_bank_sort_code ?? '' != '')
and
event.posneg_sender_bank_sort_code_v == 'decline'
```

Explanation

[Transfers-PN7] - Decline - Transfer is from a Sort Code ({event.sender_bank_sort_code}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN8

Comment

Transfer is to a Sort Code which is listed to be auto declined.

Actions

- [Transfers-PN8]

Variables

N/A

Condition

```
 (event.receiver_bank_sort_code ?? '' != '')
and
event.posneg_receiver_bank_sort_code == 'decline'
```

Explanation

[Transfers-PN8] - Decline - Transfer is to a Sort Code ({event.receiver_bank_sort_code}) which is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN9

Comment

Transaction notes contain keywords which are listed to be auto declined

Actions

- [Transfers-PN9]

Variables

N/A

Condition

```
 ((event.sender_transaction_notes ?? '' != '') and
  ListedKeywords.containsKeyword(posneg_listed_keywords, "decline",
event.sender_transaction_notes, true) )
or
((event.receiver_transaction_notes ?? '' != '') and
ListedKeywords.containsKeyword(posneg_listed_keywords, "decline",
event.receiver_transaction_notes, true) )
```

Explanation

[Transfers-PN9] - Decline - Transfer notes contain keywords that are listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN10

Comment

Transfer is to an account which is listed to be auto approved.

Actions

- [Transfers-PN10]

Variables

N/A

Condition

```
 (event.receiver_transaction_bank_acco ?? '' != '')
and
event.posneg_receiver_account_value == 'approve'
```

Explanation

[Transfers-PN10] - Low Risk - Transfer is to an account ({event.receiver_transaction_bank_acco}) which is listed to be auto approved.

Mark As

The outcome of this rule is: **approve**

Transfers-PN11

Comment

Transfer is from an account which is listed to be auto approved.

Actions

- [Transfers-PN11]

Variables

N/A

Condition

```
 (event.sender_transaction_bank_accoun ?? '' != '')
and
event.posneg_sender_account_value == 'approve'
```

Explanation

[Transfers-PN11] - Low Risk - Transfer is from an account ({event.sender_transaction_bank_accoun}) which is listed to be auto approved.

Mark As

The outcome of this rule is: **approve**

Transfers-PN12

Comment

Decline IP that is listed to be auto declined.

Actions

- [Transfers-PN12]

Variables

N/A

Condition

```
 (event.device_ip_address ?? '' != '')
and
event.posneg_device_ip_value == 'decline'
```

Explanation

[Transfers-PN12] - Decline - Device IP {{event.device_ip_address_location}} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers-PN13

Comment

Decline Device Id that is listed to be auto declined.

Actions

- [Transfers-PN13]

Variables

N/A

Condition

```
 (event.device_id ?? '' != '')
and
event.posneg_device_id_value == 'decline'
```

Explanation

[Transfers-PN13] - Decline - '{eventname.device_id}' {{event.device_id}} is listed to be auto declined.

Mark As

The outcome of this rule is: **decline**

Transfers - SCA Exemption Rules

Was this page helpful?Â

Descriptionâ

This rules project is part of the PSD2 compliance strategy offered by TFB. If this regulation does not apply to the customer's jurisdiction, this rules project can be removed from the Pulse workflow without the risk of compromising the fraud detection rate.

The *Transfer - SCA Exemption* rules project provides a recommendation on whether Feedzai's customers should request SCA to a transfer originator.

Rules within **Cards - SCA Exemption** implement the conditions defined by PSD2's Exemption Articles (i.e. Articles 11 to 18) that exempt transfers from SCA. Download the compliance matrix to learn more about these articles.

Rules Project Metadata

- *Target Variable:* sca_required
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-SCA13

Comment

PSD2 Article 13 - Trusted beneficiaries

Actions

- [Transfers-SCA13]
- PSD2-Article-13

Variables

N/A

Condition

```
event.trusted_receiver ?? false
```

Explanation

[Transfers-SCA13] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 13

Mark As

The outcome of this rule is: **false**

Transfers-SCA14

Comment

PSD2 Article 14 - Recurring transactions

Actions

- [Transfers-SCA14]
- PSD2-Article-14

Variables

N/A

Condition

```
(event.first_recurring_payment ?? false == false)
 and (event.recurring_payment ?? false)
```

Explanation

[Transfers-SCA14] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 14

Mark As

The outcome of this rule is: **false**

Transfers-SCA15

Comment

PSD2 Article 15 - Credit transfers between accounts held by the same natural or legal person

Actions

- [Transfers-SCA15]
- PSD2-Article-15

Variables

N/A

Condition

```
 (event.sender_id ?? '' != '') and (event.receiver_id ?? '' != '')
  and
(event.sender_id == event.receiver_id)

 // depending on the bank, personal ids can be used instead, like: sender_citizen_id_number ==
receiver_citizen_id_number
```

Explanation

[Transfers-SCA15] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 15

Mark As

The outcome of this rule is: **false**

Transfers-SCA16-1

Comment

PSD2 Article 16 - Low value transactions (sum amount)

Actions

- [Transfers-SCA16-1]
- PSD2-Article-16

Variables

```
 AMOUNT_LIMIT = transfers__rules_thresholds["psd2_art16_amount_limit"].threshold;
SUM_AMOUNT_LIMIT = transfers__rules_thresholds["psd2_art16_sum_amount_limit"].threshold;
METRIC_IS_EXPIRED = event.sum_amount_sender_last_sca_90d == null;
```

Condition

```
(event.sender_transaction_amount ?? 0 != 0)
  and !METRIC_IS_EXPIRED
  and event.sender_transaction_amount <= AMOUNT_LIMIT
  and (event.sum_amount_sender_last_sca_90d + event.sum_amount_sender_last_sca_30h <=
SUM_AMOUNT_LIMIT)
```

Explanation

> [Transfers-SCA16-1] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 16

Mark As

The outcome of this rule is: **false**

Transfers-SCA16-2

Comment

> PSD2 Article 16 - Low-value transactions (count transactions)

Actions

- [Transfers-SCA16-2]
- PSD2-Article-16

Variables

```
AMOUNT_LIMIT = transfers__rules_thresholds["psd2_art16_amount_limit"].threshold;
COUNT_LIMIT = transfers__rules_thresholds["psd2_art16_count_limit"].threshold;
METRIC_IS_EXPIRED = event.count_sender_last_sca_90d == null;
```

Condition

```
(event.sender_transaction_amount ?? 0 != 0)
  and !METRIC_IS_EXPIRED
  and event.sender_transaction_amount <= AMOUNT_LIMIT
  and (event.count_sender_last_sca_90d + event.count_sender_last_sca_30h <= COUNT_LIMIT)
```

Explanation

> [Transfers-SCA16-2] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 16

Mark As

The outcome of this rule is: **false**

Transfers-SCA17

Comment

> PSD2 Article 17 - Secure corporate payment processes and protocols

Actions

- PSD2-Article-17
- [Transfers-SCA17]

Variables

N/A

Condition

```
 (event.segment_channel_type ?? '' != '')
and
event.segment_channel_type == 'business_banking'
```

Explanation

> [Transfers-SCA17] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 17

Mark As

The outcome of this rule is: **false**

Transfers-SCA18-ETV1

Comment

> PSD2 Article 18 - Transaction Risk Analysis for Exemption Threshold Value 100

Actions

- PSD2-Article-18
- [Transfers-SCA18-ETV1]

Variables

```
 fraud_rate = event.reference_fraud_rate ?? fraud_rates["transfers"].value ?? 0;

AMOUNT_LIMIT = transfers__rules_thresholds["psd2_art18_etv1_amount_limit"].threshold;

FRAUD_RATE_LIMIT = transfers__rules_thresholds["psd2_art18_etv1_fraud_rate_limit"].threshold;

SUM_AMOUNT_LIMIT = transfers__rules_thresholds["psd2_art18_etv1_sum_amount_limit"].threshold;
```

Condition

```
 (event.sender_transaction_amount ?? 0 != 0)
and
event.sender_transaction_amount <= AMOUNT_LIMIT
and fraud_rate <= FRAUD_RATE_LIMIT
  and event.decision == 'approve'
  and event.sum_amount_per_sender_24h <= SUM_AMOUNT_LIMIT
```

Explanation

> [Transfers-SCA18-ETV1] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 18

Mark As

The outcome of this rule is: **false**

Transfers-SCA18-ETV2

Comment

PSD2 Article 18 - Transaction Risk Analysis for Exemption Threshold Value 100-250

Actions

- [Transfers-SCA18-ETV2]
- PSD2-Article-18

Variables

```
 fraud_rate = event.reference_fraud_rate ?? fraud_rates["transfers"].value ?? 0;

AMOUNT_MIN = transfers__rules_thresholds["psd2_art18_etv2_amount_min"].threshold;
AMOUNT_MAX = transfers__rules_thresholds["psd2_art18_etv2_amount_max"].threshold;

FRAUD_RATE_LIMIT = transfers__rules_thresholds["psd2_art18_etv2_fraud_rate_limit"].threshold;
SUM_AMOUNT_LIMIT = transfers__rules_thresholds["psd2_art18_etv2_sum_amount_limit"].threshold;
```

Condition

```
 (event.sender_transaction_amount ?? 0 != 0)
and
(event.sender_transaction_amount > AMOUNT_MIN and event.sender_transaction_amount <= AMOUNT_MAX)
and fraud_rate <= FRAUD_RATE_LIMIT
  and event.decision == 'approve'
  and event.sum_amount_per_sender_24h <= SUM_AMOUNT_LIMIT
```

Explanation

[Transfers-SCA18-ETV2] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 18

Mark As

The outcome of this rule is: **false**

Transfers-SCA18-ETV3

Comment

PSD2 Article 18 - Transaction Risk Analysis for Exemption Threshold Value 250-500

Actions

- [Transfers-SCA18-ETV3]
- PSD2-Article-18

Variables

```
 fraud_rate = event.reference_fraud_rate ?? fraud_rates["transfers"].value ?? 0;

AMOUNT_MIN = transfers__rules_thresholds["psd2_art18_etv3_amount_min"].threshold;
AMOUNT_MAX = transfers__rules_thresholds["psd2_art18_etv3_amount_max"].threshold;

FRAUD_RATE_LIMIT = transfers__rules_thresholds["psd2_art18_etv3_fraud_rate_limit"].threshold;
SUM_AMOUNT_LIMIT = transfers__rules_thresholds["psd2_art18_etv3_sum_amount_limit"].threshold;
```

Condition

```
 (event.sender_transaction_amount ?? 0 != 0)
and
(event.sender_transaction_amount > AMOUNT_MIN and event.sender_transaction_amount <= AMOUNT_MAX)
and fraud_rate <= FRAUD_RATE_LIMIT
  and event.decision == 'approve'
  and event.sum_amount_per_sender_24h <= SUM_AMOUNT_LIMIT
```

Explanation

> [Transfers-SCA18-ETV3] - Not SCA Required - This transaction is exempted from Strong Customer Authentication according to Article 18

Mark As

The outcome of this rule is: **false**

Transfers - SCA Required Rules

Was this page helpful?Â

Descriptionâ

This rules project is part of the PSD2 compliance strategy offered by TFB. If this regulation does not apply to the customer's jurisdiction, this rules project can be removed from the Pulse workflow without the risk of compromising the fraud detection rate.

The *Transfer - SCA Required* rules project provides a recommendation on whether Feedzai's customers should request SCA to a transfer originator.

The rules within this rules project focus on transaction risk analysis defined in the PSD2 Article 18 (download the compliance matrix to learn more about this article). This article establishes that issuer banks are allowed to exempt SCA when the payer initiates a remote electronic payment transaction identified by the payment service provider as low risk level. Therefore, these rules evaluate risk factors defined in Article 18 and generate the `sca_required = true` outcome when the transaction is considered risky.

Note that when a rule within this rules project is triggered, the remaining rules are not evaluated (i.e. the workflow is stopped). This also includes rules within the *Transfer - SCA Exemption* rules project - the rules project that follows the *Transfer - SCA Required* block in the Pulse workflow. This mechanism prevents exemption rules to trigger on top of an SCA-required rule since the SCA exemption cannot be granted anyway.

Rules Project Metadata

- *Target Variable:* sca_required
- *Tenancy Type:* SINGLE_TENANT

Rules in this rules project:

Transfers-SCAR1

Comment

> Mismatch between sender and receiver country code and receiver is new (6 months)

Actions

- [Transfers-SCAR1]

Variables

N/A

Condition

```
 ((event.sender_country_code ?? '' != '') and (event.receiver_country_code ?? '' != ''))
and (event.count_sender_receiver_6m == 0)
and (event.sender_country_code != event.receiver_country_code)
```

Explanation

> [Transfers-SCAR1] - SCA Required - There's a mismatch between the Sender country ({event.sender_country_code}) and the Receiver country ({event.receiver_country_code}) and Receiver is new

Mark As

The outcome of this rule is: **true**

Transfers-SCAR2

Comment

> Sender country changed

Actions

- [Transfers-SCAR2]

Variables

N/A

Condition

```
event.sender_country_code_change_fla ?? false
```

Explanation

[Transfers-SCAR2] - SCA Required - Sender country changed

Mark As

The outcome of this rule is: **true**

Transfers-SCAR3

Comment

Sender email changed

Actions

- [Transfers-SCAR3]

Variables

N/A

Condition

```
event.sender_email_change_flag ?? false
```

Explanation

[Transfers-SCAR3] - SCA Required - Sender email changed

Mark As

The outcome of this rule is: **true**

Transfers-SCAR4

Comment

Sender phone changed

Actions

- [Transfers-SCAR4]

Variables

N/A

Condition

```
event.sender_phone_change_flag ?? false
```

Explanation

[Transfers-SCAR4] - SCAR Required - Sender phone changed

Mark As

The outcome of this rule is: **true**

Transfers-SCAR5

Comment

Transaction amount is higher than 90% of the account balance

Actions

- [Transfers-SCAR5]

Variables

```
 RATIO_AMOUNT_BALANCE_LIMIT =
transfers__rules_thresholds["max_sca_ratio_amount_balance"].threshold;

formatted_ratio = RATIO_AMOUNT_BALANCE_LIMIT * 100;
```

Condition

```
 event.sender_transaction_amount ?? 0 != 0
   and event.sender_account_balance ?? 0 != 0 and
Maths.safeDivision(event.sender_transaction_amount,event.sender_account_balance) >=
RATIO_AMOUNT_BALANCE_LIMIT
```

Explanation

[Transfers-SCAR5] - SCA Required - This transaction exhausted {localvar.formatted_ratio} % or more of the available account balance.

Mark As

The outcome of this rule is: **true**

Transfers-SCAR6

Comment

Password changed 3 or more times (24hours)

Actions

- [Transfers-SCAR6]

Variables

```
COUNT_LIMIT = transfers__rules_thresholds["max_sender_count_pw_changed_24h"].threshold;
```

Condition

```
event.count_sender_pw_change_24h >= COUNT_LIMIT
```

Explanation

[Transfers-SCAR6] - SCA Required - The password changed {localvar.COUNT_LIMIT} or more times in the last 24 hours

Mark As

The outcome of this rule is: **true**

Transfers-SCAR7

Comment

Device language changed and IP country code or device id are new (6 months)

Actions

- [Transfers-SCAR7]

Variables

N/A

Condition

```
event.sender_device_language_changed == 1 and (event.count_sender_device_ip_country == 0 or
event.count_sender_device_id_6m == 0)
```

Explanation

[Transfers-SCAR7] - SCA Required - In the last 24 hours, device language changed and device ip country or device id are new

Mark As

The outcome of this rule is: **true**

Transfers-SCAR8

Comment

Authentication type and login session id changed and receiver is new (6 months)

Actions

- [Transfers-SCAR8]

Variables

N/A

Condition

```
event.auth_type_changed_24h == 1 and event.login_session_id_changed_24h == 1 and
event.count_sender_receiver_6m == 0
```

Explanation

[Transfers-SCAR8] - SCA Required - Authentication type and login session id changed and receiver is new (6 months)

Mark As

The outcome of this rule is: **true**

Transfers-SCAR9

Comment

Proxy has been detected

Actions

- [Transfers-SCAR9]

Variables

N/A

Condition

```
event.proxy_flag ?? false
```

Explanation

[Transfers-SCAR9] - SCA Required - Proxy has been detected

Mark As

The outcome of this rule is: **true**

Transfers - Self Service Rules

Was this page helpful?Â

Descriptionâ

Standard rules written in Pulse are normally developed and tested in staging environments to ensure that the work in development does not affect the work in production. This type of rules is developed by teams that have deep knowledge in this area and often have taken special training. Therefore, there is an inevitable gap between the time that these rules start to be written and the time that they are deployed in production. At the same time, fraudsters are constantly coming up with new workarounds to trick the system. Counteracting new waves of fraud might require tweaking a rule or creating a new one.

Self-service rules (SSR) allow you to fill this gap, and have an active rule in production in a few minutes. With this capability, you can write a rule in **Case Manager**, test and publish it directly to production. As an example, you can use SSRs to create a rule with a different threshold than the one already defined in a standard rule and publish it in production to have it quickly live.

Therefore, this rules project is empty, and is used in the Pulse workflow to receive the SSRs written in Case Manager.

Rules Project Metadata

- *Target Variable:* decision
- *Tenancy Type:* SINGLE_TENANT

Schema Mappings

Was this page helpful?Â

This page contains comma-separated value (CSV) files that allow you to confirm where API fields are used. Among other parameters, these CSVs contain rules, metrics, and variables from Cards and Transfers.

As a business analyst, you need to understand the impact of missing data fields on your client's data. As such, the following files allow you to:

- Validate the API fields required to run each rule (e.g. fields in a rule's condition and metrics).
- Confirm where an API field is being used in the whole solution (e.g. in which metrics, rule conditions, or variables).

Cards

Map API Fields to Rules Â Â API Fields Usage

Digital Activity

Map API Fields to Rules Â Â API Fields Usage

Inbound Payments

Map API Fields to Rules Â Â API Fields Usage

Transfers

Map API Fields to Rules Â Â API Fields Usage

final-recommendation-for-transfers

Was this page helpful?Â

If	Then
`enforced_risk_level` is `high` OR `enforced_risk_level` is `medium` AND `payment_sub_type` is `instant_transfer`	`decision` field is set as `decline`
`enforced_risk_level` is `medium`	`decision` field is set as `review`. The event is displayed on the alerts page in Case Manager
`enforced_risk_level` is `low`	`decision` field is set as `approve`

How Rules Work

Was this page helpful?Â

This page provides you with an overview of how rules work to detect and prevent fraud with the resources that come out of the box with your application.

API response with Feedzai's recommendationâ

When a financial institution invokes Feedzai's API to score a request via the card authorization, transfer initiation or digital activity device intelligence endpoints, Feedzai sends an API response containing a decision recommendation. This field helps you decide whether the transaction is fraudulent or legitimate:

Behind the scenes, the event is processed by the risk engine platform, Pulse, which uses data science resources to evaluate the event. If you look at the result generated by rules, the API response contains a `decision` field with the recommendation values `approve` , `decline` or `review` , and the boolean `alert` field. The following section explains how both decision outcomes and alerts are generated.

How is Feedzai's recommendation generated?â

It is important to understand how rules process events and contribute to the API response to detect and prevent fraud efficiently. This section breaks down this life cycle in the following conceptual steps:

1. Mapping from API fields to schema fields
2. Rule options in the Pulse application
3. Rule blocks and outcomes in the Pulse workflow
4. Manual revision in Case Manager
5. API response

1. Mapping from API to Pulse schemaâ

The fields in the API request are mapped to a Pulse schema named *API Schema*. This schema is then extended with internal fields, including the ones that are needed to write the rule outcomes (e.g. `decision` field). This extension is named *Extended API Schema*. See the Map API Fields page to access the schemas and their associated fields.

The internal fields are used to register the outcome generated by a rule. At the end of the Pulse workflow, the decision rules generate a final outcome, which is then sent in the API response in the `decision` field. This concept is further explained in the following section.

2. Rule options in the Pulse applicationâ

Before getting into the specific logic behind each event, let's have a closer look at two important rule settings. Each rule is configured with **Actions** and **Mark As**:

When a rule with the action **Alert** is triggered, the event is displayed in the Case Manager alerts page for manual revision. This also sets the `alert` field as `true` in the API response.

In the **Mark As** option, you can select the outcome values that are configured beforehand in the Pulse workflow. When a rule is triggered, the **Mark As** value is written to the schema field that is associated with this rule outcome. For example, in this case, the **Decline** value is written in the `decision` internal field.

Different rules are defined with different outcomes according to the specificities of each event. The position of rules projects in the Pulse workflow, and the additional custom services dictate how the final recommendation is generated. This is further discussed in the next step.

3. Rule blocks and outcomes in the Pulse workflowâ

Each set of APIs contains a Pulse workflow (e.g. *Cards Workflow*, *Transfers Workflow*, *Digital Activity Workflow*). The possible rule outcomes and the rules' processing order are defined on each workflow:

How does the Cards Workflow generate the final outcome?

The *Cards Workflow* is configured with **decision**, **fraud**, and **sca_required** outcomes. The fraud outcome is generally configured to be used by machine learning models. This means that each rule contributes to the `decision` field, and the first rule triggering sets the final value. Learn more about the default rule precedence.

How do Transfers and Digital Activity Workflows generate the final outcome?

tip

The *Transfers Workflow* contains a slightly more complex logic:

- Transfers payment types
  - Traditional credit transfers (also called bank transfers or credit transfers)
  - Direct debits
  - Instant transfers (also known as instant payments or real-time payments)
- Rule's importance with a service enforcer to define priority (i.e. the default rule precedence does not prevail in this case)

Transfers and Digital Activity workflows are configured with the **decision**, **fraud**, **risk_level**, and **sca_required** outcomes. The latter is specific to financial institutions under the PSD2 regulation. Let's have a closer look at the conceptual logic behind the recommendation generation in transfers and digital activity. This logic comprises the following contributions:

The `risk_level` outcome contains the possible values **high**, **low**, and **medium**. Each rule contributes to the `risk_level` outcome, except the rules in the *Decision Rules* project, which contribute to the `decision` outcome. Therefore, when the event flows through the rules project preceding the *Decision Rules* block, rules write to the `risk_level` internal field.

Right before the *Decision Rules* block, the workflow includes an out-of-the-box custom service. When the event reaches this element, the custom service looks at the outcome value of each triggered rule (i.e. **high**, **low**, and **medium**) to enforce a new outcome. This new outcome is written to the `enforced_risk_level` field.

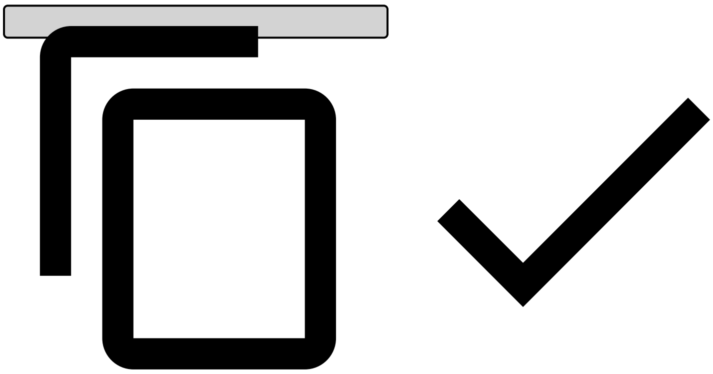
The custom service provides the following logic:

- **Rule order**: The service can be configured with a list of rules ordered by priority. The rule triggered with the highest priority defines the enforced outcome value.
- **Outcome value order**: The service can be configured with risk level order (e.g. medium - low - high). This means that the defined risk level order prevails over the order in which the rules triggered. In the medium - low - high example, if a rule with medium risk level is triggered, then this risk level is enforced.

Outcome Priority Enforcer Service in more detail

Let's assume the Outcome Priority Enforcer Service is configured in a workflow with the following configuration:

```
 pulse.global.services.outcomepriorityenforcer.output=enforced_risk_level
pulse.global.services.outcomepriorityenforcer.rules=WhitelistRuleId
pulse.global.services.outcomepriorityenforcer.order=high_risk,medium_risk,low_risk
```



To describe possible scenarios, let's use the notation `ruleId` , `outcomeValue` to indicate that the rule with ID `ruleId` triggered and set the outcome to `outcomeValue` .

Scenario 1: If an event triggers (`rule1` , `low_risk`) and (`rule2` , `medium_risk`), then the service will output `medium_risk` to the schema field `enforced_risk_level` . This happens because none of the rules is defined in the rule list, and *medium_risk* appears first on the order list, thus having higher priority.

Scenario 2: If an event triggers (`rule1` , `high_risk`) and (`WhitelistRuleId` , `low_risk`), then the service will output `low_risk` to the schema field `enforced_risk_level` . This happens because the rule `WhitelistRuleId` is configured in the rule's configuration property. Note that, in this scenario, the order's property is ignored since the rule configuration takes precedence over the order configuration.

Scenario 3: If an event triggers (`rule1` , `super_high_risk`), then the service will not output any value to the schema field `enforced_risk_level` . This happens because the service does not recognize `super_high_risk` as a possible value for both rule ID or outcome value.

Finally, the decision rules are defined with the **Decision** outcome. These rules look at the `enforced_risk_level` field and make a final recommendation as follows:

If	**Then**
`enforced_risk_level` is `high` OR `enforced_risk_level` is `medium` AND `payment_sub_type` is `instant_transfer`	`decision` field is set as `decline`

If	Then
`enforced_risk_level` is `medium`	`decision` field is set as `review`. The event is displayed on the alerts page in Case Manager
`enforced_risk_level` is `low`	`decision` field is set as `approve`

info

Instant transfers: For rules that act on the transfers payment method, only the decision block contains rules that can trigger an alert. Also, the [Transfers - Decision Rules]() are defined by the order presented above. This means that instant transfers with `enforced_risk_level` field as `high` or `medium` are caught by the first rule and, therefore, automatically declined. This is important because instant transfers are not subject to manual revision due to their instant processing nature. Only credit transfers and direct debits can trigger an alert action which is later sent for review.

4. Manual revision in Case Managerâ

The following events generate alerts as follows:

- **Card events**: Various rules across rules projects contain the alert action. When one of these rules is triggered, the event is displayed in the Case Manager alerts page. See the rules' actions by checking each *Cards* rules project within the Rules Projects section.

- **Transfer events**: Only decision rules contain the **alert** action following the logic described in the previous section. Note that instant transfers is a payment type that is processed instantaneously. When these events are declined, they also generate alerts. They are manually reviewed with the sole purpose of giving feedback to the risk engine because the final recommendation has already been provided.

- **Digital events**: Rules track digital events to identify the user behavior during a banking session. If a digital event has signs of suspicious activity, it will trigger an alert which will be visible in Case Manager.

Analysts investigate alerts in Case Manager and decide whether the event is fraudulent or legitimate. This decision is registered in the `feedback_value` field as `true` or `false` for fraud and not fraud, respectively. Then, the *Feedback Rules* project evaluates this result and makes a new recommendation for card, transfer and digital events.

5. API responseâ

At this point, you already understand a scoring event's API response, which allows you to benefit from the recommendation provided by Feedzai. Note that, whether you send an API request via Cards, Transfers or Digital Activity streams using one of their scoring endpoints, you will get an API response with the following key fields:

- `decision` : The final outcome generated by rules according to the concepts explained in the Rule blocks and outcomes in the Pulse workflow section.
- `alert` : Whether the event generated an alert that is visible on the alerts page in Case Manager.
- `score` : The numeric value a machine learning model generates to evaluate an event. Possible values range from 0 to 1000.
- `action_codes` : Every rule has an array of Actions including the name of the rule (e.g. fraud rule Cards-FR1 contains a Cards-FR1 Action). The `action_codes` array includes all the Actions of triggered rules to further understand which rules have triggered for each event.
- `sca_required` : Strong Customer Authentication is required or not, based on compliance and risk assessment.

Learn moreâ

Check the Create and Manage Rules page to assist you in adding new rules to your Pulse Application. Additionally, to learn more about fraud scenarios or access a list of rules and metadata that are provided out of the box, check the following pages:

- Fraud Scenarios and Detection.
- Rules Projects.

- [Operational Work]
- [Data Scientist & Fraud Strategist]
- Work with Rules

On this page

Work with Rules

Was this page helpful?Â

Learn how to fully manage your rules, including:

- Understanding how rules work.
- Writing rules specific to your business.
- Configuring your work in the runtime workflow.
- Testing rules using the Transaction Fraud for Banking API.

User journeyâ

As a data scientist or a fraud analyst, you can leverage rules to detect and mitigate fraud events at scale in real time. The following user journey shows the different stages required to set up your rules and test their behavior with transaction requests:

Next stepsâ

Start by understanding **how rules work** to learn the basic concepts behind this functionality. If you are already familiar with rules in Pulse, proceed to the user journey steps:

- Create and Manage Rules
- Migrate Pulse Resources
- Publish Rules to Workflow
- Send Test Transactions via API

Generated Fields

The Ad-hoc Investigations API generates some fields which will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Investigation	investigation
Info	info
Feedback	feedback

event_received_atâ

The `event_received_at` represents the time the request reached the Ad-hoc Investigations API as a timestamp, in milliseconds.

Ad-hoc Customer Investigations

Was this page helpful?Â

This page provides you with the necessary fields to include in the payload when sending an ad-hoc investigation's `investigation` or `info` request.

Send `investigation POST` requestâ

To generate an ad-hoc investigation event in Case Manager, send an `investigation` (`POST`) request to Feedzai. The request must contain the `customer_id` to fetch the customer data from the Reference Data Service and populate the customer event in Case Manager. Consider the following payload when sending the request:

```
{
    "lifecycle_id":"<lifecycle_id>",
    "event_external_id":"<event_external_id>",
    "event_occurred_at": <event_occurred_at>, // event timestamp
    "customer_id":"<customer_id>", // customer_id
    "reason_code": "reason_code",
    "origin": "media",
    "description":"Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed interdum
tristique turpis cursus venenatis.",
    "customer_name": "John Doe",
... // Rest of the Customer payload
}
```

Send `info PUT` requestâ

To update the previously generated ad-hoc investigation event, e.g. to signal that the investigation can be closed, send an `info PUT` request to Feedzai. Consider the following payload:

```
{
    "lifecycle_id": "<lifecycle_id>",
    "event_external_id": "<event_external_id>",
    "event_occurred_at": <event_occurred_at>, // event timestamp
    "customer_id": "<customer_id>", // customer_id
    "info_type": additional_information",
    "reason_code": "reason_code",
    "origin": "media",
    "description":"Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed interdum
tristique turpis cursus venenatis.",
    "customer_name": "John Doe",
... // Rest of the customer payload
}
```

Send `feedback PUT` requestâ

To update a previously decided outcome, e.g. change from `feedback_value` as `false` (i.e. "dropped") to `feedback_value` as `true` ("action taken"), send the following `feedback PUT` request:

```
{
    "event_external_id": "event_external_id_{{ID}}",
    "event_occurred_at": {{nowTimestamp}},
    "feedback_type": "feedback",
    "feedback_value": true
}
```

Responseâ

In both scenarios, i.e. `investigation POST` and `info PUT`, a successful response should be as follows:

```
{
    "status": "ok",
    "lifecycle_id": "<lifecycle_id>",
    "event_external_id": "<event_external_id>"
}
```

Find more information in Responses and Errors.

Nextâ

- Ad-hoc Investigations API's endpoints.
- Ad-hoc Investigations' life cycles.

Ad-Hoc Customer Investigations

Was this page helpful?Â

The Investigations API provides a mechanism for creating ad-hoc investigation points for customers initiated from the client's side, independent of Cards, Transfers or Customer Assessment Alerts. This functionality enables external systems to trigger investigations, facilitating proactive analysis and assessments for specific Customers.

ðï¸ Ad-hoc Customer Investigation Info

The Info endpoint enables you to update and add new information to a previously created investigation, e.g. signalling that the investigation can be closed.

ð□□□ï¸□Ad-hoc Customer Investigation Feedback

The Feedback endpoint allows you to label customers as 'fraud' or 'not fraud'The labelling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

ð□□□ï¸□Ad-hoc Customer Investigation

Creates a new investigation event. This event will be the investigation unit in Case Manager.

Reference Data: accounts

PUT

/api/referenceData/accounts/:key

The Reference Data Provisioning endpoint allows you to send 'accounts' data to Feedzai that will enrich the real time scoring events.

Requestâ

Responsesâ

- 200
- 401

OK

-
- [API Resources]
- [Endpoints]
- [Reference Data]
- Reference Data: aliases

Was this page helpful?Â

Reference Data: aliases

PUT

```
/api/referenceData/aliases/:key
```

The Reference Data Provisioning endpoint allows you to send 'aliases' data to Feedzai that will enrich the real time scoring events.

Requestâ

Responsesâ

- 200
- 401

OK

- API Resources
- Endpoints
- Reference Data
- Reference Data: cards

Was this page helpful?Â

Reference Data: cards

PUT

`/api/referenceData/cards/:key`

The Reference Data Provisioning endpoint allows you to send 'cards' data to Feedzai that will enrich the real time scoring events.

Requestâ

Responsesâ

- 200
- 401

OK

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/add-credentials-reference-data

Was this page helpful?Â

Reference Data: credentials

PUT

```
/api/referenceData/credentials/:key
```

The Reference Data Provisioning endpoint allows you to send 'credentials' data to Feedzai that will enrich the real time scoring events.

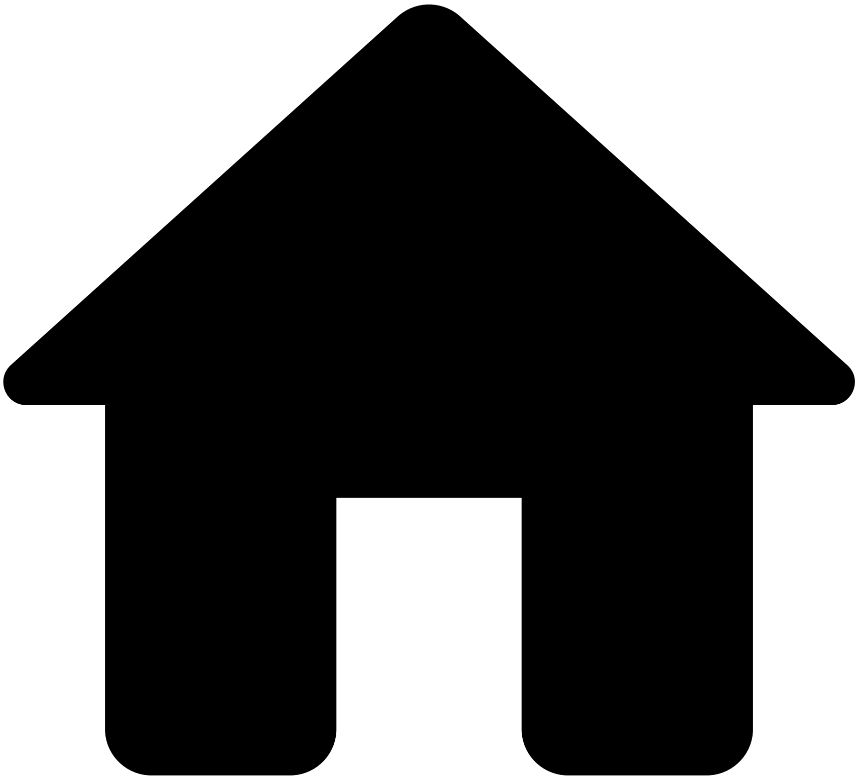
Requestâ

Responsesâ

- 200
- 401

OK

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/add-customers-reference-data

- 
- [API Resources]
- [Endpoints]
- [Reference Data]
- Reference Data: customers

Was this page helpful?Â

Reference Data: customers

PUT

`/api/referenceData/customers/:key`

The Reference Data Provisioning endpoint allows you to send 'customers' data to Feedzai that will enrich the real time scoring events.

Requestâ

Responsesâ

- 200
- 401

OK

Reference Data: devices

```
PUT
```

/api/referenceData/devices/:key

The Reference Data Provisioning endpoint allows you to send 'devices' data to Feedzai that will enrich the real time scoring events.

Requestâ

Responsesâ

- 200
- 401

OK

Was this page helpful?Â

Reference Data: merchants

PUT

/api/referenceData/merchants/:key

The Reference Data Provisioning endpoint allows you to send 'merchants' data to Feedzai that will enrich the real time scoring events.

Requestâ

Responsesâ

- 200
- 401

OK

Reference Data: payees

PUT

`/api/referenceData/payees/:key`

The Reference Data Provisioning endpoint allows you to send 'payees' data to Feedzai that will enrich the real time scoring events.

Requestâ

Responsesâ

- 200
- 401

OK

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/cards

-

- [API Resources]
- [Endpoints]
- CardsÂ

Cards

Was this page helpful?Â

The Cards API provides various endpoints to allow you to send information during the whole transaction's life cycle. You can send data related to authorization, transaction updates, clearings, and fraud feedback actions, corresponding to API requests to the Authorization, Info, Clearing and Fraud-Feedback endpoints, respectively.

ðï¸Feedback

The Feedback Endpoint allows you to label a transaction, for example as 'Fraud' or 'Not Fraud'.

ð□□ï¸□Responses

The Responses Get Endpoint allows you to retrieve the result of events that were processed asynchronously.

ð□□ï¸□Authorization

An Authorization request allows you to leverage Feedzai's data science capabilities to score a transaction according to its fraud risk.

ð□□ï¸□Transaction

The Transaction Get Endpoint allows you to retrieve a transaction previously sent to Feedzai.

ð□□ï¸□Scorable Info

The Info Endpoint enables you to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream.

ð□□ï¸□Info

The Info Endpoint enables you to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream.

ð□□ï¸□Clearing

The Clearing Endpoint allows you to inform Feedzai when a transaction is settled.

- API Resources
- Endpoints
- Reference Data
- Delete Reference Data: accounts

Was this page helpful?Â

Delete Reference Data: accounts

```
DELETE

/api/referenceData/accounts/:key
```

The Reference Data Delete endpoint allows you to delete 'accounts' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Delete Reference Data: aliases

```
DELETE

/api/referenceData/aliases/:key
```

The Reference Data Delete endpoint allows you to delete 'aliases' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Delete Reference Data: cards

```
DELETE

/api/referenceData/cards/:key
```

The Reference Data Delete endpoint allows you to delete 'cards' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Delete Reference Data: credentials

```
DELETE
/api/referenceData/credentials/:key
```

The Reference Data Delete endpoint allows you to delete 'credentials' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Was this page helpful?Â

Delete Reference Data: customers

```
DELETE
/api/referenceData/customers/:key
```

The Reference Data Delete endpoint allows you to delete 'customers' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Delete Reference Data: devices

```
DELETE

/api/referenceData/devices/:key
```

The Reference Data Delete endpoint allows you to delete 'devices' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Was this page helpful?Â

Delete Reference Data: merchants

```
DELETE

/api/referenceData/merchants/:key
```

The Reference Data Delete endpoint allows you to delete 'merchants' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Delete Reference Data: payees

```
DELETE

/api/referenceData/payees/:key
```

The Reference Data Delete endpoint allows you to delete 'payees' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401

OK

Delete from list

```
DELETE

/api/entitytagging/:appSlug/:listName/:key
```

Allows you to remove an item from a list

Requestâ

Responsesâ□□

- 200
- 400
- 401
- 404
- 500

OK

Digital Activity

Was this page helpful?Â

The Digital Activity API provides various endpoints that allow you to send information about digital activity. You can send data related to authentication attempts, credential and profile updates, devices, new payees and aliases.

ð□□ï¸□Device

The customer attempts to register, change, or delete a device associated with the bank's platform.

ð□□ï¸□Device

The customer attempts to register, change, or delete a device associated with the bank's platform.

ð□□ï¸□Info Alias

The Info Alias endpoint enables you to update and add new information to an existing account alias.

ð□□ï¸□Info Card Update

The Info Card Update endpoint enables you to update and add new information to an existing card update event.

ð□□ï¸□Info Authentication

The Info Authentication endpoint enables you to update and add new information to an existing authentication.

ð□□ï¸□Info Device

The Info Device endpoint enables you to update and add new information to an existing device registration.

ð□□ï¸□Account Update

The customer attempts to change their account, e.g. their account limits.

ð□□ï¸□Account Update

The customer attempts to change their account, e.g. their account limits.

ð□□ï¸□Card Update

The customer attempts to change their card, e.g. define new card format as physical or virtual

ð□□ï¸□Card Update

The customer attempts to change their card, e.g. classify it as real or virtual card.

ð□□ï¸□Info Account Update

The Account Update Info endpoint enables you to update and add new information to an existing account update.

ð□□ï¸□Info Credentials Update

The Info Credentials Update endpoint enables you to update and add new information to an existing credentials update.

ð□□ï¸□Info Profile Update

The Info Profile endpoint enables you to update and add new information to an existing profile.

ð□□ï¸□New Payee

The customer attempts to add a new trusted payee to their account or attempts an alias resolution request. For instance, searching for random mobile phone numbers and adding them as a contact in the banking app.

ð□□ï¸□New Payee

The customer attempts to add a new trusted payee to their account or attempts an alias resolution request. For instance, searching for random mobile phone numbers and adding them as a contact in the banking app.

ð□□ï¸□Authentication

The customer attempts to authenticate or log in to the banking system via web home banking or mobile app.

ð□□ï¸□Authentication

The customer attempts to authenticate or log in to the banking system via web home banking or mobile app.

ð□□ï¸□Profile Update

The customer attempts to change their profile, e.g. their address.

ð□□ï¸□Profile Update

The customer attempts to change their profile, e.g. their address.

ð□□ï¸□Alias

The customer attempts to register or change the default account alias. In some payment methods (e.g. Zelle, PIX, NPP) an alias is the phone number, email or QR code associated with the bank account.

ð□□ï¸□Alias

The customer attempts to register or change the default account alias. In some payment methods (e.g. Zelle, PIX, NPP) an alias is the phone number, email or QR code associated with the bank account.

ð□□ï¸□Credentials Update

The customer attempts to change their credentials, e.g. their username or password.

ð□□□ï¸□Credentials Update

The customer attempts to change their credentials, e.g. their username or password.

ð□□□ï¸□Feedback

The Feedback endpoint allows you to label digital activity events, for instance, as 'Fraud' or 'Not Fraud'. The labeling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

ð□□□ï¸□Info Card Issuing

The Info Card Issuing endpoint enables you to update and add new information to an existing card issuing event.

ð□□□ï¸□Info New Payee

The Info New Payee endpoint enables you to update and add new information to an existing payee event.

ð□□□ï¸□Card Issuing

The customer attempts to issue a card

ð□□□ï¸□Card Issuing

The customer attempts to issue a card

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/entity-tagging

-

Entity Tagging

Was this page helpful?Â

The Entity Tagging API brings a new way of managing the contents of lists, i.e. new entities or fragments can be easily added to lists without requiring access to Case Manager or Pulse.

ð□□ï¸□Fragment Tagging

The Fragment Tagging endpoint allows you to give feedback on certain fragments, like Card PAN's, Tokens, etc., and flag that fragment in a Whitelist / Blacklist / Risky List.

ð□□ï¸□Untagging

The Untagging endpoint allows you to unflag a specific Entity / Fragment from a Whitelist / Blacklist / Risky List.

ð□□ï¸□Entity Tagging

The Entity Tagging endpoint allows you to give feedback on certain entities, like Cards, Customers, Accounts, etc., and flag that entity in a Whitelist / Blacklist / Risky List.

ð□□ï¸□Entity Tagging Possible Tags

The Get Entity Tagging Possible Tags Endpoint allows you to know what are the possible tags available to use for a particular entity type. You only need to include the Entity Type in the URL (cards, for example), and the response would include all the possible tags.

ð□□ï¸□Searching

The Search Fragment Tag endpoint allows you to search for tags of a fragment.

ð□□ï¸□Entity Tagging

The Get Entity Tagging Endpoint returns all the tags for a given entityType, scope, category, and tenant or tenant group (if the scope is tenant or group, respectively). The endpoint is additionally setup for pagination with optional offset and limit query parameters.

Get Reference Data: accounts

GET

`/api/referenceData/accounts/:key`

The Reference Data Get endpoint allows you to obtain 'accounts' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

-
- API Resources
- Endpoints
- Reference Data
- Get Reference Data: aliases

Was this page helpful?Â

Get Reference Data: aliases

```
GET

/api/referenceData/aliases/:key
```

The Reference Data Get endpoint allows you to obtain 'aliases' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

Was this page helpful?Â

Get Reference Data: cards

```
GET

/api/referenceData/cards/:key
```

The Reference Data Get endpoint allows you to obtain 'cards' data from Feedzai.

Requestâ

Responsesâ¶

- 200
- 401
- 404

OK

Was this page helpful?Â

Get Reference Data: credentials

```
GET

/api/referenceData/credentials/:key
```

The Reference Data Get endpoint allows you to obtain 'credentials' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

Get Reference Data: customers

```
GET

/api/referenceData/customers/:key
```

The Reference Data Get endpoint allows you to obtain 'customers' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

-
- [API Resources]
- [Endpoints]
- [Reference Data]
- Get Reference Data: devices

Was this page helpful?Â

Get Reference Data: devices

```
GET

/api/referenceData/devices/:key
```

The Reference Data Get endpoint allows you to obtain 'devices' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

Entity Tagging

```
GET
/api/entitytagging/:entityType/category/:category
```

The Get Entity Tagging Endpoint returns all the tags for a given entityType, scope, category, and tenant or tenant group (if the scope is tenant or group, respectively). The endpoint is additionally setup for pagination with optional offset and limit query parameters.

Requestâ

Responsesâ

- 200
- 400
- 401
- 404
- 500

OK

Was this page helpful?Â

Entity Tagging Possible Tags

GET

`/api/entitytagging/:entityType/tags/categories`

The Get Entity Tagging Possible Tags Endpoint allows you to know what are the possible tags available to use for a particular entity type. You only need to include the Entity Type in the URL (cards, for example), and the response would include all the possible tags.

Requestâ

Responsesâ

- 200
- 400
- 401
- 404
- 500

OK

InboundTransfer

```
GET

/api/inbound-transfers/:lifecycle_id
```

The Inbound Transfer Endpoint allows you to retrieve a transfer previously sent to Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404
- 500

OK

Get Reference Data: merchants

GET

/api/referenceData/merchants/:key

The Reference Data Get endpoint allows you to obtain 'merchants' data from Feedzai.

Requestâ

Responsesâ¶

- 200
- 401
- 404

OK

Get Reference Data: payees

```
GET

/api/referenceData/payees/:key
```

The Reference Data Get endpoint allows you to obtain 'payees' data from Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

Get items from list

```
GET

/api/entitytagging/:appSlug/:listName
```

Returns all items in a list. The endpoint is additionally set up for pagination with optional offset and limit query parameters.

Requestâ

Responsesâ

- 200
- 400
- 401
- 404
- 500

OK

Responses

```
GET

/api/transactions/async/responses
```

The Responses Get Endpoint allows you to retrieve the result of events that were processed asynchronously.

Responsesâ

- 200

- 401
- 500

OK

Was this page helpful?Â

Transaction

```
GET

/api/transactions/:lifecycle_id
```

The Transaction Get Endpoint allows you to retrieve a transaction previously sent to Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 500

OK

Was this page helpful?Â

Responses

GET

/api/transfers/async/responses

The Responses Get Endpoint allows you to retrieve the result of events that were processed asynchronously.

Responsesâ

- 200

- 401
- 500

OK

Was this page helpful?Â

Transfer

```
GET

/api/transfers/:lifecycle_id
```

The Transfer Endpoint allows you to retrieve a transfer previously sent to Feedzai.

Requestâ

Responsesâ

- 200
- 401
- 404
- 500

OK

-
- [API Resources]
- [Endpoints]
- Inbound PaymentsÂ

Inbound Payments

Was this page helpful?Â

The Inbound Payments API provides various endpoints to allow you to send information during the whole inbound transfer's life cycle. You can send data related to inbound transfer initiation, inbound transfer updates, inbound settlements, and fraud feedback actions, corresponding to API requests to the Inbound Transfer Initiation, Info, Settlement and Feedback endpoints, respectively.

ð□□ï¸□Settlement

The Settlement endpoint allows you to inform Feedzai when a inbound transfer is settled.

ð□□ï¸□Inbound Transfer initiation

The Inbound Transfer Initiation endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

ð□□ï¸□InboundTransfer

The Inbound Transfer Endpoint allows you to retrieve a transfer previously sent to Feedzai.

ð□□ï¸□Scorable Info

The Inbound Transfer Scorable Info endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

ð□□ï¸□Info

The Inbound Transfer Info endpoint enables you to update and add new information to an existing inbound transfer.

ð□□ï¸□Feedback

The Feedback endpoint allows you to label inbound transfers, for instance, as 'Fraud' or 'Not Fraud'. The labeling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

Info Account Update

```
PUT

/api/activity/account_update/:lifecycle_id/info
```

The Account Update Info endpoint enables you to update and add new information to an existing account update.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Info Alias

PUT

`/api/activity/alias/:lifecycle_id/info`

The Info Alias endpoint enables you to update and add new information to an existing account alias.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Info Authentication

PUT

`/api/activity/authentication/:lifecycle_id/info`

The Info Authentication endpoint enables you to update and add new information to an existing authentication.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Info Card Issuing

PUT

/api/activity/card_issuing/:lifecycle_id/info

The Info Card Issuing endpoint enables you to update and add new information to an existing card issuing event.

Requestâ

Responsesâ¶

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/info-card-update

Was this page helpful?Â

Info Card Update

PUT

```
/api/activity/card_update/:lifecycle_id/info
```

The Info Card Update endpoint enables you to update and add new information to an existing card update event.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Info Credentials Update

PUT

/api/activity/credentials_update/:lifecycle_id/info

The Info Credentials Update endpoint enables you to update and add new information to an existing credentials update.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Info Device

PUT

/api/activity/device/:lifecycle_id/info

The Info Device endpoint enables you to update and add new information to an existing device registration.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Info New Payee

```
PUT

/api/activity/new_payee/:lifecycle_id/info
```

The Info New Payee endpoint enables you to update and add new information to an existing payee event.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Info Profile Update

PUT

/api/activity/profile_update/:lifecycle_id/info

The Info Profile endpoint enables you to update and add new information to an existing profile.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

List Management

Was this page helpful?Â

The List Management API provides an easy way to directly manage list in Pulse, leveraging the entity tagging connector.

ð□□ï¸□Delete from list

Allows you to remove an item from a list

ð□□ï¸□Add to List

Allows you to add an item to a list.

ð□□ï¸□Update list item

Allows you to update an existing item on a list.

ð□□ï¸□Get items from list

Returns all items in a list. The endpoint is additionally set up for pagination with optional offset and limit query parameters.

Outbound Transfers

Was this page helpful?Â

The transfer API provides various endpoints to allow you to send information during the whole transfer's life cycle. You can send data related to transfer initiation, transfer updates, settlements, and fraud feedback actions, corresponding to API requests to the Transfer Initiation, Info, Settlement and Feedback endpoints, respectively.

ð□□ï¸□Responses

The Responses Get Endpoint allows you to retrieve the result of events that were processed asynchronously.

ð□□ï¸□Transfer

The Transfer Endpoint allows you to retrieve a transfer previously sent to Feedzai.

ð□□ï¸□Scorable Info

The Transfer Info Scorable endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

ð□□ï¸□Info

The Info endpoint enables you to update and add new information to an existing transfer.

ð□□ï¸□Settlement

The Settlement endpoint allows you to inform Feedzai when a transfer is settled.

ð□□ï¸□Settlement

The Settlement endpoint allows you to inform Feedzai when a transfer is settled.

ð□□ï¸□Transfer initiation

The Transfer Initiation endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

ð□□ï¸□Feedback

The Feedback endpoint allows you to label transfers, for instance, as 'Fraud' or 'Not Fraud'. The labeling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/patch-accounts-reference-data

Was this page helpful?Â

Patch Reference Data: accounts

PATCH

`/api/referenceData/accounts/:key`

The Reference Data Patch endpoint allows you to send partial 'accounts' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

Was this page helpful?Â

Patch Reference Data: aliases

```
PATCH

/api/referenceData/aliases/:key
```

The Reference Data Patch endpoint allows you to send partial 'aliases' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

-
- [API Resources]
- [Endpoints]
- [Reference Data]
- Patch Reference Data: cards

Was this page helpful?Â

Patch Reference Data: cards

```
PATCH

/api/referenceData/cards/:key
```

The Reference Data Patch endpoint allows you to send partial 'cards' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

-

- [API Resources]
- [Endpoints]
- [Reference Data]
- Patch Reference Data: credentials

Was this page helpful?Â

Patch Reference Data: credentials

```
PATCH

/api/referenceData/credentials/:key
```

The Reference Data Patch endpoint allows you to send partial 'credentials' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ□□

- 200
- 401
- 404

OK

Patch Reference Data: customers

PATCH

```
/api/referenceData/customers/:key
```

The Reference Data Patch endpoint allows you to send partial 'customers' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

- API Resources
- Endpoints
- Reference Data
- Patch Reference Data: devices

Was this page helpful?Â

Patch Reference Data: devices

```
PATCH

/api/referenceData/devices/:key
```

The Reference Data Patch endpoint allows you to send partial 'devices' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

Was this page helpful?Â

Patch Reference Data: merchants

```
PATCH

/api/referenceData/merchants/:key
```

The Reference Data Patch endpoint allows you to send partial 'merchants' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ¶

- 200
- 401
- 404

OK

Patch Reference Data: payees

```
PATCH

/api/referenceData/payees/:key
```

The Reference Data Patch endpoint allows you to send partial 'payees' data to Feedzai that will be merged with previous entity.

Requestâ

Responsesâ

- 200
- 401
- 404

OK

-
- API Resources
- Endpoints
- Digital ActivityÂ
- Account Update

Was this page helpful?Â

Account Update

```
POST

/api/activity/account_update
```

The customer attempts to change their account, e.g. their account limits.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Ad-hoc Customer Investigation

```
POST

/api/customers/investigations
```

Creates a new investigation event. This event will be the investigation unit in Case Manager.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Alias

```
POST

/api/activity/alias
```

The customer attempts to register or change the default account alias. In some payment methods (e.g. Zelle, PIX, NPP) an alias is the phone number, email or QR code associated with the bank account.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Authentication

```
POST

/api/activity/authentication
```

The customer attempts to authenticate or log in to the banking system via web home banking or mobile app.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-

- [API Resources]
- [Endpoints]
- [CardsÂ]
- Authorization

Was this page helpful?Â

Authorization

POST

`/api/transactions/authorization`

An Authorization request allows you to leverage Feedzai's data science capabilities to score a transaction according to its fraud risk.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Clearing

```
POST

/api/transactions/clearing
```

The Clearing Endpoint allows you to inform Feedzai when a transaction is settled.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Scorable Info

```
POST

/api/transactions/:lifecycle_id/info
```

The Info Endpoint enables you to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-
- [API Resources]
- [Endpoints]
- [Digital ActivityÂ]
- Card Issuing

Was this page helpful?Â

Card Issuing

```
POST

/api/activity/card_issuing
```

The customer attempts to issue a card

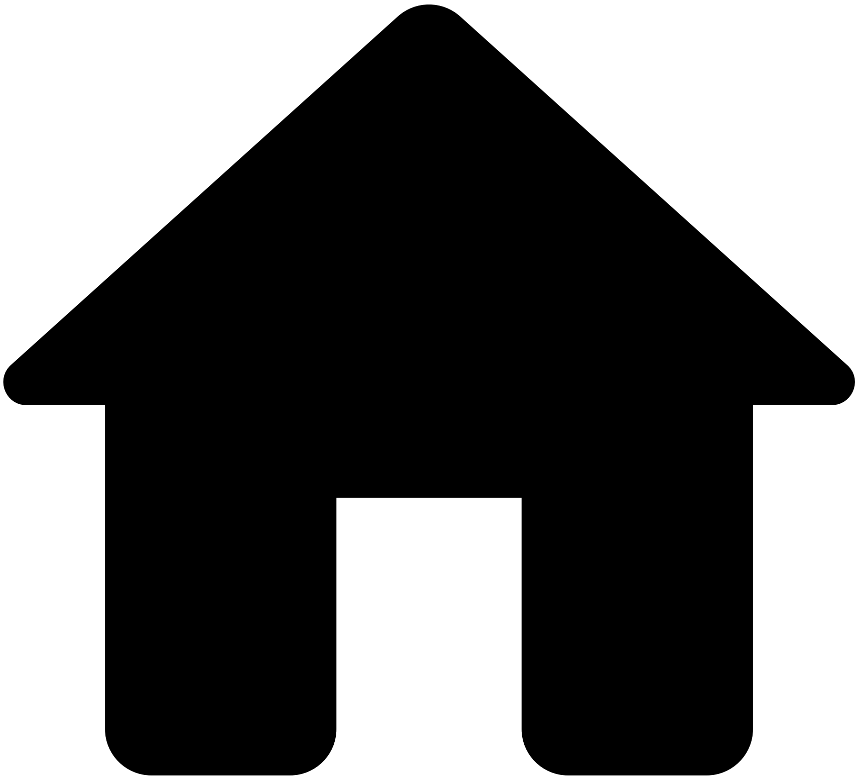
Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/post-card-update

- 

Was this page helpful?Â

Card Update

```
POST

/api/activity/card_update
```

The customer attempts to change their card, e.g. define new card format as physical or virtual

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/post-credentials-update

Credentials Update

```
POST

/api/activity/credentials_update
```

The customer attempts to change their credentials, e.g. their username or password.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Device

```
POST

/api/activity/device
```

The customer attempts to register, change, or delete a device associated with the bank's platform.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Entity Tagging

```
POST
```

`/api/entitytagging/entities/:entityType/:key/tags`

The Entity Tagging endpoint allows you to give feedback on certain entities, like Cards, Customers, Accounts, etc., and flag that entity in a Whitelist / Blacklist / Risky List.

Requestâ

Responsesâ

- 200
- 400
- 401
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-
- [API Resources]
- [Endpoints]
- [Entity Tagging]
- Fragment Tagging

Was this page helpful?Â

Fragment Tagging

```
POST
/api/entitytagging/fragments/:fragmentType/tags
```

The Fragment Tagging endpoint allows you to give feedback on certain fragments, like Card PAN's, Tokens, etc., and flag that fragment in a Whitelist / Blacklist / Risky List.

Requestâ

Responsesâ

- 200
- 400
- 401
- 500

The request has been processed successfully. This response is sent for synchronous requests.

- API Resources
- Endpoints
- Inbound PaymentsÂ
- Scorable Info

Was this page helpful?Â

Scorable Info

POST

```
/api/inbound-transfers/:lifecycle_id
```

The Inbound Transfer Scorable Info endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Inbound Transfer initiation

```
POST

/api/inbound-transfers
```

The Inbound Transfer Initiation endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

New Payee

```
POST

/api/activity/new_payee
```

The customer attempts to add a new trusted payee to their account or attempts an alias resolution request. For instance, searching for random mobile phone numbers and adding them as a contact in the banking app.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Add to List

POST

/api/entitytagging/

Allows you to add an item to a list.

Requestâ

Responsesâ

- 200
- 400
- 401
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Profile Update

```
POST

/api/activity/profile_update
```

The customer attempts to change their profile, e.g. their address.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Searching

```
POST
/api/entitytagging/fragments/:fragmentType/tags/search
```

The Search Fragment Tag endpoint allows you to search for tags of a fragment.

Requestâ

Responsesâ

- 200
- 400
- 401
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-
- API Resources
- Endpoints
- Outbound TransfersÂ
- Scorable Info

Was this page helpful?Â

Scorable Info

POST

/api/transfers/:lifecycle_id

The Transfer Info Scorable endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

Requestâ

Responsesâ¶

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Transfer initiation

```
POST

/api/transfers
```

The Transfer Initiation endpoint allows you to leverage Feedzai's data science capabilities to score a transaction according to its risk.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Settlement

```
POST

/api/transfers/:lifecycle_id/settlement
```

The Settlement endpoint allows you to inform Feedzai when a transfer is settled.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-

- [API Resources]
- [Endpoints]
- [Entity Tagging]
- Untagging

Was this page helpful?Â

Untagging

```
POST
/api/entitytagging/:entityOrfragmentType/untag
```

The Untagging endpoint allows you to unflag a specific Entity / Fragment from a Whitelist / Blacklist / Risky List.

Requestâ

Responsesâ

- 200
- 400
- 401
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Account Update

```
PUT

/api/activity/account_update
```

The customer attempts to change their account, e.g. their account limits.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Ad-hoc Customer Investigation Feedback

PUT

/api/customers/investigations/:lifecycle_id/feedback/:feedback_label

The Feedback endpoint allows you to label customers as "fraud" or "not fraud"The labelling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 404
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Ad-hoc Customer Investigation Info

PUT

/api/customers/investigations/:lifecycle_id/info

The Info endpoint enables you to update and add new information to a previously created investigation, e.g. signalling that the investigation can be closed.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Alias

PUT

`/api/activity/alias`

The customer attempts to register or change the default account alias. In some payment methods (e.g. Zelle, PIX, NPP) an alias is the phone number, email or QR code associated with the bank account.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Authentication

```
PUT

/api/activity/authentication
```

The customer attempts to authenticate or log in to the banking system via web home banking or mobile app.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Feedback

PUT

```
/api/transactions/:lifecycle_id/feedback/:feedback_label
```

The Feedback Endpoint allows you to label a transaction, for example as 'Fraud' or 'Not Fraud'.

Requestâ

Responsesâ□□

- 200
- 202
- 400
- 401
- 403
- 404
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Info

PUT

`/api/transactions/:lifecycle_id/info`

The Info Endpoint enables you to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream.

Requestâ

Responsesâ□□

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/put-card-issuing

-

- [API Resources]
- [Endpoints]
- [Digital ActivityÂ]
- Card Issuing

Was this page helpful?Â

Card Issuing

```
PUT

/api/activity/card_issuing
```

The customer attempts to issue a card

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-
- [API Resources]
- [Endpoints]
- [Digital ActivityÂ]
- Card Update

Was this page helpful?Â

Card Update

```
PUT

/api/activity/card_update
```

The customer attempts to change their card, e.g. classify it as real or virtual card.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Credentials Update

```
PUT

/api/activity/credentials_update
```

The customer attempts to change their credentials, e.g. their username or password.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

- [API Resources]
- [Endpoints]
- [Digital ActivityÂ]
- Device

Was this page helpful?Â

Device

```
PUT

/api/activity/device
```

The customer attempts to register, change, or delete a device associated with the bank's platform.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Feedback

PUT

/api/activity/digital/:lifecycle_id/feedback/:feedback_label

The Feedback endpoint allows you to label digital activity events, for instance, as 'Fraud' or 'Not Fraud'. The labeling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 404
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-

- API Resources
- Endpoints
- Inbound PaymentsÂ
- Feedback

Was this page helpful?Â

Feedback

PUT

/api/inbound-transfers/:lifecycle_id/feedback/:feedback_label

The Feedback endpoint allows you to label inbound transfers, for instance, as 'Fraud' or 'Not Fraud'. The labeling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 404
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/endpoints/put-inbound-transfer-info

- API Resources
- Endpoints
- Inbound PaymentsÂ
- Info

Was this page helpful?Â

Info

PUT

/api/inbound-transfers/:lifecycle_id

The Inbound Transfer Info endpoint enables you to update and add new information to an existing inbound transfer.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

-
- [API Resources]
- [Endpoints]
- [Inbound PaymentsÂ]
- Settlement

Was this page helpful?Â

Settlement

PUT

/api/inbound-transfers/:lifecycle_id/settlement

The Settlement endpoint allows you to inform Feedzai when a inbound transfer is settled.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

New Payee

```
PUT

/api/activity/new_payee
```

The customer attempts to add a new trusted payee to their account or attempts an alias resolution request. For instance, searching for random mobile phone numbers and adding them as a contact in the banking app.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Update list item

PUT

/api/entitytagging/

Allows you to update an existing item on a list.

Requestâ

Responsesâ

- 200
- 400
- 401
- 500

OK

-

- [API Resources]
- [Endpoints]
- [Digital ActivityÂ]
- Profile Update

Was this page helpful?Â

Profile Update

```
PUT

/api/activity/profile_update
```

The customer attempts to change their profile, e.g. their address.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Feedback

```
PUT

/api/transfers/:lifecycle_id/feedback/:feedback_label
```

The Feedback endpoint allows you to label transfers, for instance, as 'Fraud' or 'Not Fraud'. The labeling of an event is highly recommended since it gives feedback so that Feedzai's risk engine can learn and consolidate fraud patterns

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 404
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Was this page helpful?Â

Info

```
PUT

/api/transfers/:lifecycle_id
```

The Info endpoint enables you to update and add new information to an existing transfer.

Requestâ

Responsesâ

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Settlement

PUT

`/api/transfers/:lifecycle_id/settlement`

The Settlement endpoint allows you to inform Feedzai when a transfer is settled.

Requestâ

Responsesâ□□

- 200
- 202
- 400
- 401
- 403
- 500

The request has been processed successfully. This response is sent for synchronous requests.

Reference Data

Was this page helpful?Â

The Reference Data API provides an endpoint to provision Reference Data entities to Feedzai.

ð□□ï¸□Get Reference Data: credentials

The Reference Data Get endpoint allows you to obtain 'credentials' data from Feedzai.

ð□□ï¸□Reference Data: credentials

The Reference Data Provisioning endpoint allows you to send 'credentials' data to Feedzai that will enrich the real time scoring events.

ð□□ï¸□Delete Reference Data: credentials

The Reference Data Delete endpoint allows you to delete 'credentials' data from Feedzai.

ð□□ï¸□Patch Reference Data: credentials

The Reference Data Patch endpoint allows you to send partial 'credentials' data to Feedzai that will be merged with previous entity.

ð□□ï¸□Get Reference Data: cards

The Reference Data Get endpoint allows you to obtain 'cards' data from Feedzai.

ð□□ï¸□Reference Data: cards

The Reference Data Provisioning endpoint allows you to send 'cards' data to Feedzai that will enrich the real time scoring events.

ð□□ï¸□Delete Reference Data: cards

The Reference Data Delete endpoint allows you to delete 'cards' data from Feedzai.

ð□□ï¸□Patch Reference Data: cards

The Reference Data Patch endpoint allows you to send partial 'cards' data to Feedzai that will be merged with previous entity.

ð□□ï¸□Get Reference Data: accounts

The Reference Data Get endpoint allows you to obtain 'accounts' data from Feedzai.

ð□□ï¸□Reference Data: accounts

The Reference Data Provisioning endpoint allows you to send 'accounts' data to Feedzai that will enrich the real time scoring events.

ð□□ï¸□Delete Reference Data: accounts

The Reference Data Delete endpoint allows you to delete 'accounts' data from Feedzai.

ð□□□ï¸□Patch Reference Data: accounts

The Reference Data Patch endpoint allows you to send partial 'accounts' data to Feedzai that will be merged with previous entity.

ð□□□ï¸□Get Reference Data: payees

The Reference Data Get endpoint allows you to obtain 'payees' data from Feedzai.

ð□□□ï¸□Reference Data: payees

The Reference Data Provisioning endpoint allows you to send 'payees' data to Feedzai that will enrich the real time scoring events.

ð□□□ï¸□Delete Reference Data: payees

The Reference Data Delete endpoint allows you to delete 'payees' data from Feedzai.

ð□□□ï¸□Patch Reference Data: payees

The Reference Data Patch endpoint allows you to send partial 'payees' data to Feedzai that will be merged with previous entity.

ð□□□ï¸□Get Reference Data: devices

The Reference Data Get endpoint allows you to obtain 'devices' data from Feedzai.

ð□□□ï¸□Reference Data: devices

The Reference Data Provisioning endpoint allows you to send 'devices' data to Feedzai that will enrich the real time scoring events.

ð□□□ï¸□Delete Reference Data: devices

The Reference Data Delete endpoint allows you to delete 'devices' data from Feedzai.

ð□□□ï¸□Patch Reference Data: devices

The Reference Data Patch endpoint allows you to send partial 'devices' data to Feedzai that will be merged with previous entity.

ð□□□ï¸□Get Reference Data: customers

The Reference Data Get endpoint allows you to obtain 'customers' data from Feedzai.

ð□□ï¸□Reference Data: customers

The Reference Data Provisioning endpoint allows you to send 'customers' data to Feedzai that will enrich the real time scoring events.

ð□□ï¸□Delete Reference Data: customers

The Reference Data Delete endpoint allows you to delete 'customers' data from Feedzai.

ð□□ï¸□Patch Reference Data: customers

The Reference Data Patch endpoint allows you to send partial 'customers' data to Feedzai that will be merged with previous entity.

ð□□ï¸□Get Reference Data: aliases

The Reference Data Get endpoint allows you to obtain 'aliases' data from Feedzai.

ð□□ï¸□Reference Data: aliases

The Reference Data Provisioning endpoint allows you to send 'aliases' data to Feedzai that will enrich the real time scoring events.

ð□□ï¸□Delete Reference Data: aliases

The Reference Data Delete endpoint allows you to delete 'aliases' data from Feedzai.

ð□□ï¸□Patch Reference Data: aliases

The Reference Data Patch endpoint allows you to send partial 'aliases' data to Feedzai that will be merged with previous entity.

ð□□ï¸□Get Reference Data: merchants

The Reference Data Get endpoint allows you to obtain 'merchants' data from Feedzai.

ð□□ï¸□Reference Data: merchants

The Reference Data Provisioning endpoint allows you to send 'merchants' data to Feedzai that will enrich the real time scoring events.

ð□□ï¸□Delete Reference Data: merchants

The Reference Data Delete endpoint allows you to delete 'merchants' data from Feedzai.

ð□□ï¸□Patch Reference Data: merchants

The Reference Data Patch endpoint allows you to send partial 'merchants' data to Feedzai that will be merged with previous entity.

-
- API Resources
- Endpoints
- Introduction

Was this page helpful?Â

Introduction

Export

- OpenAPI Spec

Feedzai's APIs are organized according to REST and follow the RESTful design. This means that their resource URLs focus on representing data as resources (e.g. cards, inbound payments) and using HTTP methods (e.g. `POST`, `PUT`) to perform operations on those resources. They also return JSON-encoded responses, and use standard HTTP response codes.

Use casesâ

APIs such as Cards, Inbound Payments, Outbound Transfers, and Digital Activity help automate the risk strategy by scoring events (0 - 100) and providing real-time decisions (e.g. approve, decline) based on scoring models and rules, respectively.

Scoring logicâ

Method	Description
`POST`	Scores and returns a decision to the event
`PUT`	Does not score or return a decision to the event

Authenticationâ

- API Key: Authorization

Security Scheme Type:	apiKey
Header parameter name:	Authorization

Find all authentication-related information in the Authentication page.

- [API Resources]
- [Additional Resources]
- Inbound Payments
- Cash and ATM Deposits

Cash and ATM Deposits

To correctly map cash and ATM deposits to inbound payments, and hence score cash and ATM deposits, some fields need to be populated to identify cash or ATM deposit events as follows:

```
{
   // Required fields
   "event_external_id": "4048611b295e993d1b561d9030c53424bbf2b95e",
   "lifecycle_id": "1df7a7qg78vd7fs",
   "channel_type": "atm", // To indicate it's ATM
   "payment_status": "entered",
   "created_at": 1603207359060,
   "sender_transaction_amount": 1603207359060,
   "sender_transaction_currency": "GBP",
```

```
        "sender_transaction_bank_account_number": "5a800244-325a-4086-ab8a-23ecbb927be5",
        "direction": "incoming",
        "event_occurred_at": 1684492270092,

        // Recommended for mapping cash and ATM deposits
        "payment_type": "sepa", // To indicate it's a deposit
        "payment_sub_type": "direct_deposit", // Use the `direct_deposit` value to indicate it's a direct deposit
    }
```

Read the [Inbound Transfer Initiation [POST]] documentation for more information on other fields you can use in the payload.

On this page

Generated Fields

Was this page helpful?Â

The Inbound Payments API generates some fields which will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Initiation	???
Info	info
Settlement	settlement
Feedback	feedback

event_received_atâ

The `event_received_at` represents the time the request reached the Transfers API as a timestamp, in milliseconds.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/inbound-payments/ibp-transfer-initiation-response-scenarios

On this page

Inbound Transfer Initiation Response Scenarios

Was this page helpful?Â

This section presents several scenarios that might occur in the Inbound Transfer Initiation response, depending on the rules that are triggered in the Pulse workflow. Check the Inbound Payments' Inbound Transfer Initiation Endpoint for more information about detailed request and response examples.

When no rules are triggered, the following fields and default values are shown in the Inbound Payments' Inbound Transfer Initiation response:

- `"alert": false` This field is changed to 'true' when any rule with the action 'alert' is triggered. See examples of possible scenarios below.
- `"action_codes": []` This field is populated with the actions of the first triggered rule, other than the 'alert' action. Note that the 'alert' action is represented by the `alert` field.
- `"secondary_action_codes":[]` This field is populated with all the non-'alert' actions of every triggered rule, except the first triggered rule.

When rules are triggered, they match the following scenarios:

Scenario 1â

- **The first triggered rule:**
  - Action field - Alert
- **Excerpt from the Inbound Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = []
}
```

Scenario 2â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Transfer Initiation response:**

```
 {
"alert": true,
```

```
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = []
}
```

Scenario 3â

- **The first triggered rule:**
    - Action field - Alert
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Scenario 4â

- **The first triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Scenario 5â

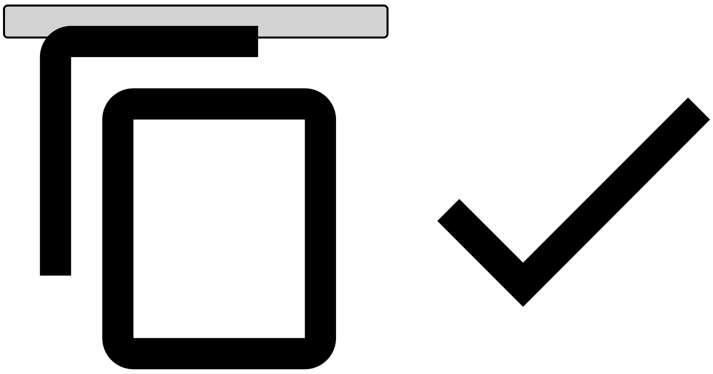
- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The third triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 4
- **Excerpt from the Inbound Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3", "Action 4"]
}
```

Scenario 6â

- **The first triggered rule:**
  - This rule doesn't have actions.
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Inbound Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```



-
- API Resources
- Additional Resources
- List Management

On this page

List Management

Was this page helpful?Â

Data scientists or fraud prevention agents can insert, update or delete items for simple and complex lists via Pulse's user interface (UI). To give them the autonomy to manage list items through different systems other than Pulse (e.g. via an integration between systems), the List Management API can be used.

The List Management API allows you to insert items into simple and complex lists as global, by tenant, or by tenant group. Learn the different requests you can send to manage list items end to end.

Insert list items `[POST]` â

caution

To insert a list item, ensure the list has been created via UI beforehand.

Note that you can only insert list items individually (i.e. bulk action is not possible).

Simple listâ

The following payload includes the fields required for a simple list. Send a request through `POST <>/api/entitytagging` as follows:

```
{
   // required fields for a complex list
   "appSlug": "tfrb", // the Pulse App where the list belongs to
   "listName": "my-travel-abroad-list", // the name of the list you're inserting this item into
   "scope": "global", // must match the scope configured in the list, i.e. global, by tenant or
by tenant group
   "key": "4174337685469288", // the entity's ID, e.g. card number
   // optional fields that provide more context about this item:
   "note": "The client will be travelling abroad next weekend",
   "startTime": "2025-01-25T00:00:00+00:00", // specifies the period in which the item will
become enabled
```

```
  "endTime": "2025-01-27T23:59:59+00:00", // specifies the period in which the item will become disabled
}
```

Complex listâ

A complex list includes adding a field called `value`. If this field is not set, you won't be able to insert items to that list via API.

The following payload includes the fields required for a complex list. Send a request through `POST <>/api/entitytagging` as follows:

```
{
  // required fields for a complex list
  "appSlug": "tfrb", // the Pulse App where the list belongs to
  "listName": "my-travel-abroad-list", // the name of the list you're inserting this item into
  "scope": "tenant", // must match the scope configured in the list, i.e. global, by tenant or by tenant group
  "tenant": "feedzai", // if the scope's value is "tenant" or "group", a `tenant` or `group` field, respectively, is required with the corresponding `tenantId`
  "key": "4174337685469288", // the entity's ID, e.g. card number
  "value": "true", // the accepted value for the list item (e.g. true means `decline`)
  // optional fields that provide more context about this item:
  "note": "The client will be travelling abroad next weekend",
  "startTime": "2025-01-25T00:00:00+00:00", // specifies the period in which the item will become enabled
  "endTime": "2025-01-27T23:59:59+00:00", // specifies the period in which the item will become disabled
}
```

warning

Note that, if any of the required fields is not included in the payload, you will get a `400 Bad Request` error response.

items' time to live

Consider the following timeframes when inserting list items:

- **Defined `startTime`** : enables the item at the specified `startTime` .
- **No `startTime` defined**: enables the item as soon as the item is added to the list.
- **Defined `endTime`** : disables the item at the specified `endTime` .
- **No `endTime` defined**: does not disable the item, i.e. the item will be enabled indefinitely unless you set an `endTime` or remove it from the list.

Fetch list item `[GET]` â

To send the inserted item to the UI for any list, send a request through `GET <>/api/entitytagging/:appSlug/:listName?scope=tenant&tenant=feedzai&offset=0&limit=10` . Ensure the URL includes all required path and query parameters.

Required path parametersâ

- `:appSlug`
- `:listName`

tip

Replace the aforementioned variables with their corresponding values.

Required query parametersâ

The following query parameters must be used according to the scope defined on the list via UI, i.e. **Global**, **By Tenant**, or **By Tenant Group**:

- `scope=global`

  OR

- `scope=tenant&tenant=tenantID`

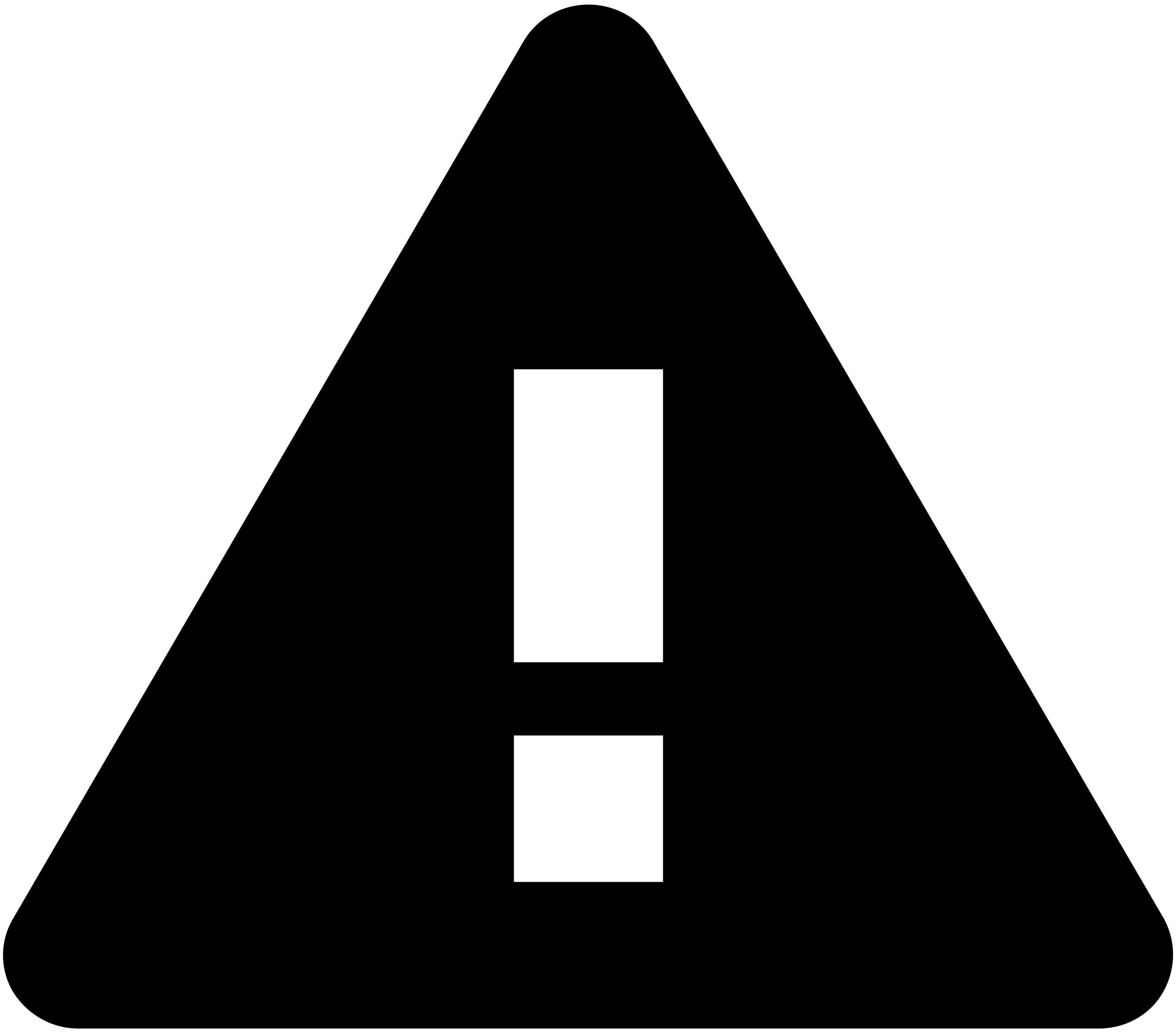
  OR

- `scope=group&group=groupID`

Optional query parametersâ

If you have a list of `100` items, and you want to fetch items `11` to `20`, set:

- `offset=10` to skip the first `10` items.
- `limit=10` to delete the next `10` items.



warning

Note that, if any of the required path or query parameters are not included in the URL, you will get the following error responses:

- **Missing path parameter**: `404 Transaction was not found`
- **Missing query parameter**: `400 Bad Request`

Update list items `[PUT]` â

Simple listâ

To update items on a simple list, send a request through `PUT <>/api/entitytagging` with the updated values as follows:

```
{
    // required fields
    "appSlug": "tfrb", // the app where the list belongs to
    "listName": "my-travel-abroad-list", // the name of the list where you're updating this item
    "scope": "tenant", // must match the scope configured in the list, i.e. global, by tenant or by tenant group
    "tenant": "feedzai",  // if the scope's value is "tenant" or "group", a `tenant` or `group` field, respectively, is required with the corresponding `tenantId`
    "key": "4174337685469288", // the entity's ID, e.g. card number
    // optional fields
    "note": "The client will be travelling abroad next weekend",
    "startTime": "2025-01-25T00:00:00+00:00", // specifies the period in which the item will become enabled
    "endTime": "2025-01-27T23:59:59+00:00", // specifies the period in which the item will become disabled
}
```

warning

Note that, if any of the required fields is not included in the payload, you will get a `400 Bad Request` error response.

Complex listâ

To update items on a complex list, send a request through `PUT <>/api/entitytagging` with the updated values as follows:

```
{
    // required fields
    "appSlug": "tfrb", // the app where the list belongs to
    "listName": "my-travel-abroad-list", // the name of the list where you're updating this item
    "scope": "global", // must match the scope configured in the list, i.e. global, by tenant or by tenant group
    "key": "4174337685469288", // the entity's ID, e.g. card number
    "value": "true", // the accepted value for the list item (e.g. true means `decline`)
    // optional fields
    "note": "The client will be travelling abroad next weekend",
    "startTime": "2025-01-25T00:00:00+00:00", // specifies the period in which the item will become enabled
    "endTime": "2025-01-27T23:59:59+00:00", // specifies the period in which the item will become disabled
}
```

warning

Note that, if any of the required fields is not included in the payload, you will get a `400 Bad Request` error response.

Delete list items `[DEL]` â

To delete items on any list, send a request through `DEL <>/api/entitytagging/:appSlug/:listName/:key?scope=tenant&tenant=feedzai` with the removed item. Ensure the URL includes all required path and query parameters.

Required path parametersâ

- `:appSlug`
- `:listName`
- `:key`

tip

Replace the aforementioned variables with their corresponding values.

Required query parametersâ

The following query parameters must be used according to the scope defined on the list via UI, i.e. **Global**, **By Tenant**, or **By Tenant Group**.

- `scope=global`

  OR

- `scope=tenant&tenant=tenantID`

  OR

- `scope=group&group=groupID`

---

warning

Note that, if any of the required path or query parameters are not included in the URL, you will get the following error responses:

- **Missing path parameter**: `404 Transaction was not found`
- **Missing query parameter**: `400 Bad Request`

Read moreâ

- Learn about the available endpoints in the List Management API.
- Understand how lists impact system decisions.

-
- [About TFB]
- [Event Life Cycles]
- Ad-hoc Customer Investigations

On this page

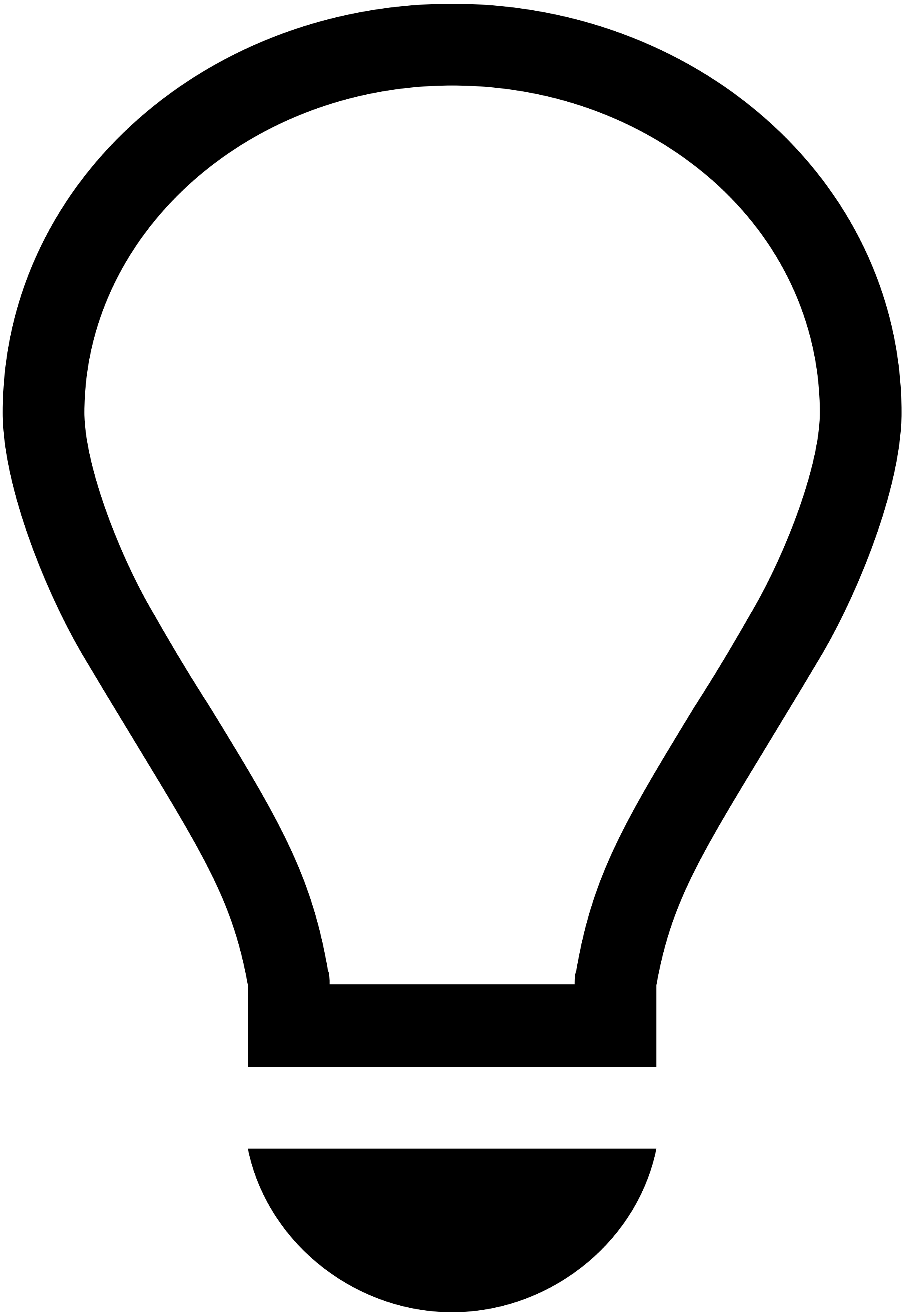
Ad-hoc Customer Investigations

Was this page helpful?Â

Learn how the Ad-hoc Investigations API is used in the different event stages.

Investigation (`POST`)â

1. An Investigation request (POST) is sent to Feedzai.
2. The investigation event is created in Case Manager's Ad-hoc Investigations channel.



Event creation in Case Manager

To ensure investigation events are generated in Case Manager's Ad-hoc Customer Investigations channel, reach out to the person responsible for API integrations.

More information on how to send Investigation event requests in this page.

Info (PUT)â

The `info` endpoint allows you to update and add new information to a previously created investigation, e.g. to signal that the investigation can be closed.

Feedback (`PUT`)â

The `feedback` endpoint allows you to label the outcome of an ad-hoc investigation as **risky** (i.e. take action on the event) or **not risky** (i.e. drop the event). The labelling of an event is highly recommended, as it gives feedback to Feedzai's risk engine, which uses this information to learn and consolidate fraud patterns.

Learn moreâ

- Learn how to send ad-hoc investigation requests using the API.

Authorization Endpoint

Was this page helpful?Â

An authorization is often the initial message on your payment process chain, corresponding to x1xx or x2xx ISO 8583 messages. The request sent to the Authorization endpoint is processed by Feedzai's risk engine using a combination of machine learning models and rules. Feedzai produces a response that includes:

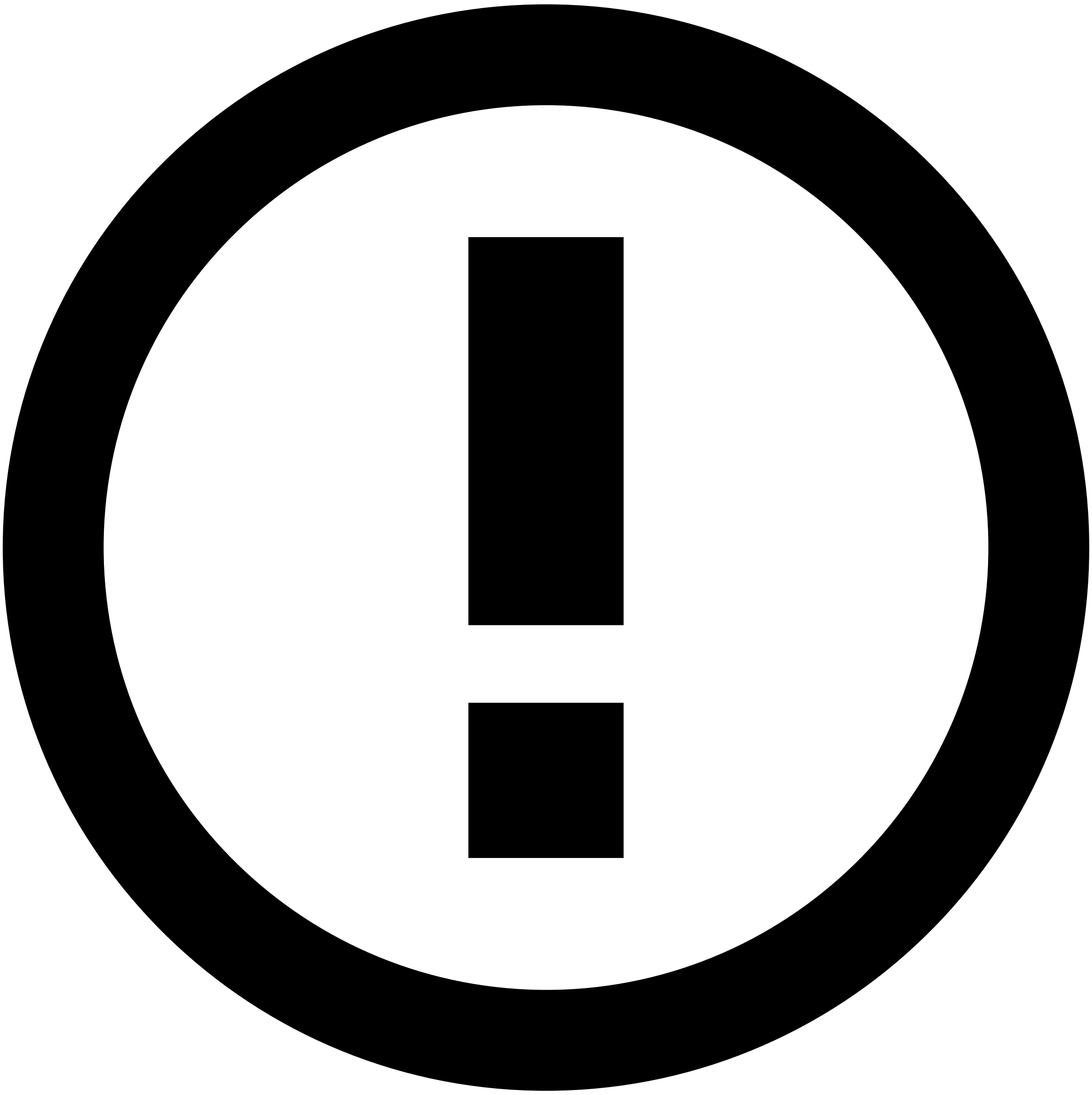
- **A score**: Represents the fraud probability.
- **A decision**: Represents the Feedzai recommendation to approve or decline a transaction.

Based on the response of this endpoint, you can decide on whether or not to accept the transaction. It also helps you to take further actions, such as contacting a client to confirm the authenticity of the transaction (check the [Out of Band Authorization](#) section to learn more about this scenario).

The following figure illustrates the transaction life cycle for an Authorization message:

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (Client, Account, Card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.



reference data enrichment

The event may become enriched with data fetched from the [Reference Data Storage](#) if any of the event's entities (e.g. Card, Customer, or Account) have a match in the Reference Data Storage.

info

Note that any new data from card authorization events (e.g. new `card_category`) will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle] page to understand the logic behind reference data's entity-based enrichment.

1. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
2. Case Manager is provided with the field values that are sent via the Authorization endpoint.
3. Authorization response: score, decision (**decline** or **approve**), actions and alert.
4. The issuer may honor or not the decision recommendation: **reject** or **accept** transaction (with a reason code).
5. ISO 8583 authorization (1xx) or presentment (2xx) message response.
6. Authorization message response (ISO 8583 or network proprietary message).
7. Payment is **rejected** or **accepted**.

Recommendations on ATM withdrawalsâ

To correctly map ATM withdrawals to card authorizations, and hence score ATM withdrawals, some fields need to be populated to identify ATM withdrawal events as follows:

```
{
    // Required fields
    "mti": "0110",
    "card_pan": "8955060600182663", // Which card was used
    "ttc": "001", // Type of transaction
    "currency_code": "EUR",
    "amount": 20,
    "cardholder_bill_amount": 20,
    "lifecycle_id": "1df7a7qg78vd7fs",
    "pos_card_read_method": "01",
    "event_occurred_at": 1684492270092,
    "card_bin": "123456",
    "account_iban": "482924bf58091603ed69c5bd1221ad67481bfca8",
    "customer_id": "87b2ee22cb0004722a7265fdbbf7be98c681a3bb",
}
```

Read the [Authorization [POST]] documentation for more information on other fields you can use in the payload.

Learn moreâ

Learn more about the detailed life cycle of other endpoints:

- Info
- Clearing
- Feedback

Clearing Endpoint

The Clearing endpoint allows you to inform Feedzai when a transaction is settled, corresponding to x1xx or x2xx ISO 8583 messages. Note that this endpoint is non-scorable, i.e. it does not flow through rules in order to generate a decision. However, the information is used by the Feedzai's risk engine platform to calculate metrics and profiles, and thus, impacts subsequent transfers associated with the same customer. It is also shown in the fraud analyst interface (Case Manager) as a new event to support the alert investigation. Check the Transaction Life Cycle initial page to understand where this endpoint fits within the transaction life cycle.

A clearing might result from a dispute initiated by the cardholder. In this case, the information is sent to Feedzai via the Info endpoint as described in the Dispute use case.

If the process of dispute leads to a chargeback, the information is communicated to Feedzai as described in the Feedback use case.

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
8. Case Manager is provided with the field values that are sent via the Authorization endpoint.
9. Authorization response: score, decision (**decline** or **approve**), actions and alert.
10. The issuer **accepts** the transaction (with a reason code).
11. ISO 8583 authorization (1xx) or presentment (2xx) message response.
12. Authorization message response (ISO 8583 or network proprietary message).
13. Payment is **accepted**.
14. Reconciliation of the day amounts.
15. ISO 8583 reconciliation request (5xx).
16. Internal validations (client, account, card, etc). Uncleared funds are changed to cleared.
17. Settlement of the cardholder funds.
18. ISO 8583 reconciliation response (5xx).
19. Settlement of the merchant/ATM account.
20. Clearing request.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities (e.g. Card, Customer, or Account) have a match in the Reference Data Storage.

info

Note that any new data from card clearing events (e.g. new `card_category`) will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle] page to understand the logic behind reference data's entity-based enrichment.

1. Pulse receives the Clearing request, and updates the transaction information in Case Manager.

- Key: Life cycle ID.
- Adds the clearing information.

1. Case Manager is updated with the field values that are sent via the Clearing endpoint.
2. Clearing response.
3. Message acknowledged.

Learn moreâ□□

Learn more about the detailed life cycle of other endpoints:

- Authorization
- Info
- Feedback

-
- About TFB
- Event Life Cycles
- Cards
- Feedback Endpoint

On this page

Feedback Endpoint

The Feedback endpoint allows you to label a transaction on whether the transaction is legitimate or suspicious. The labeling of an event is highly recommended since it gives feedback to Feedzai's risk engine enabling it to learn or consolidate fraud patterns.

Chargebackâ

This use case applies when a cardholder decides to dispute a transaction. First, send to Feedzai the Info request under Dispute. Then send the final dispute result to Feedzai using the Feedback endpoint. Note that this use case normally applies after Clearing. Check the Transaction Life Cycle initial page to understand where this endpoint fits within the customer journey of a transaction.

Under this use case, the cardholder can either dispute the transaction directly with the issuer (referred to as scenario 1) or with the acquirer/network (scenario 2).

Scenario 1 - cardholder disputes with the issuerâ

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (Client, Account, Card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
8. Case Manager is provided with the field values that are sent via the Authorization endpoint.
9. Authorization response: score, decision (**decline** or **approve**), actions and alert.
10. The issuer **accepts** the transaction (with a reason code).
11. ISO 8583 authorization (1xx) or presentment (2xx) message response.
12. Authorization message response (ISO 8583 or network proprietary message).
13. Payment is **accepted**.
14. Cardholder disputes the transaction authenticity.
15. Info request (Dispute): e.g. dispute related information is sent.
16. Pulse receives the Info request, and updates the transaction information in Case Manager.

- Key: Life cycle ID.
- Adds the dispute information.

1. Case Manager is updated with the field values sent via the Info endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Dispute` functional block label).
2. Info response (Dispute).
3. Message acknowledged.
4. After being contacted by the cardholder, the issuer bank opens a dispute.
5. ISO 8583 reversal or chargeback request message(4xx).
6. Network opens a case dispute (life cycle ID, Dispute ID).
7. ISO 8583 reversal or chargeback response message(4xx).
8. Dispute conclusion - ISO 8583 reversal or chargeback request message (4xx).
9. Feedback request (Chargeback): information related with the cardholder refunds is sent.
10. Pulse updates the transaction label.
11. Case Manager is updated with the field values sent via the Feedback endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Labeling` functional block label).
12. Feedback response (Chargeback).
13. Cardholder is informed of the dispute conclusion.

Scenario 2 - cardholder disputes with the acquirer/networkâ

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
8. Case Manager is provided with the field values that are sent via the Authorization endpoint.
9. Authorization response: score, decision (**decline** or **approve**), actions and alert.
10. The issuer **accepts** the transaction (with a reason code).
11. ISO 8583 authorization (1xx) or presentment (2xx) message response.

12. Authorization message response (ISO 8583 or network proprietary message).
13. Payment is **accepted**.
14. Cardholder disputes the transaction authenticity with the merchant.
15. After being contacted by the cardholder, the merchant opens a dispute with acquirer/network.
16. ISO 8583 reversal or chargeback request message(4xx).
17. Info request (Dispute): e.g. dispute related information is sent.
18. Pulse receives the info request, and updates the transaction information in Case Manager.

- Key: Life cycle ID.
- Adds the dispute information.

1. Case Manager is updated with the field values sent via the Info endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Dispute` functional block label).
2. Info response (Dispute).
3. Message acknowledged.
4. Dispute conclusion - ISO 8583 reversal or chargeback request message (4xx).
5. Feedback request (Chargeback): information related with the cardholder refunds is sent.
6. Pulse updates the transaction label.
7. Case Manager is updated with the field values sent via the Feedback endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Labeling` functional block label).
8. Feedback response (Chargeback).
9. Cardholder is informed of the dispute conclusion.

Out of Band Authorizationâ

This use case applies when you want to communicate to Feedzai the outcome of the contact that you established with the cardholder on whether the transaction is legitimate or suspicious

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
8. Case Manager is provided with the field values that are sent via the Authorization endpoint.
9. Authorization response: score, decision (**decline** or **approve**), actions and alert.
10. The issuer **accepts** the transaction (with a reason code).
11. SMS/Call from issuer to cardholder to verify the transaction authenticity.
12. ISO 8583 authorization (1xx) or presentment (2xx) message response.
13. Authorization message response (ISO 8583 or network proprietary message).
14. Payment is **accepted**.
15. Info request (Out of Band): e.g. with the type of contact made.
16. Pulse receives the Info request, and updates the transaction information in Case Manager.

- Key: Life cycle ID.
- Adds the contact Information.

1. Case Manager is updated with the field values sent via the Info endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Out of Band` functional block label).
2. Info response (Out of Band): e.g. with the type of contact made.
3. Message acknowledged.
4. The cardholder replies with the confirmation of the transaction.
5. Feedback request (Out of Band Authorization): with the type of contact made and the cardholder's reply.
6. Pulse updates the transaction label.
7. Case Manager is updated with the field values sent via the Feedback endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Labeling` functional block label).
8. Feedback response (Out of Band Authorization).

9. Message acknowledged.

Learn moreâ

Learn more about the detailed life cycle of other endpoints:

- Authorization Endpoint
- Info Endpoint
- Clearing Endpoint

Info Endpoint

The Info endpoint enables you to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream (see Pre-decided Auths).

Sending information to Feedzai with this endpoint is important to:

- Enrich profiles that are used to build features for machine learning models.
- Write rules that test specific information.
- Enrich alerts so that fraud analysts have extra information, and thus, gain a holistic view of the alert and the case under investigation.

info

When you use the Info endpoint, you need to send the original fields (i.e. even if they are not being updated with new information) in addition to the fields containing new information.

This endpoint applies to the use cases described below.

Additional Authorization Informationâ

In this use case, you would typically call the Info endpoint to update information sent in a previous request.

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).

6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
8. Case Manager is provided with the field values that are sent via the Authorization endpoint.
9. Authorization response: score, decision (**decline** or **approve**), actions and alert.
10. The issuer may honor or not the decision recommendation: **reject** or **accept** transaction (with a reason code).
11. ISO 8583 authorization (1xx) or presentment (2xx) message response.
12. Authorization message response (ISO 8583 or network proprietary message).
13. Payment is **rejected** or **accepted**.
14. Info request (Additional Authorization Information).

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities (e.g. Card, Customer, or Account) have a match in the Reference Data Storage.

info

Note that any new data from card info events (e.g. new `card_category`) will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle] page to understand the logic behind reference data's entity-based enrichment.

1. Pulse receives the info request, and updates the transaction information in Case Manager:

- Key: Life cycle ID.
- Adds additional information.

1. Case Manager is updated with the field values sent via the Info endpoint.
2. Info response (Additional Authorization Information).
3. Message acknowledged.

Pre-decided Authsâ

Pre-decided Auths are payments that have been either rejected or accepted before arriving to Feedzai. Typical use cases include a payment rejected at the network level before it arrives at the issuer, or a payment rejected on the issuer side for example for lack of account balance, or account blocked.

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).
6. ISO 8583 authorization (1xx) or presentment (2xx) message response.
7. Authorization message response (ISO 8583 or network proprietary message).
8. Payment is **accepted** or **rejected**.
9. Info request (Pre-decided Auth): ISO 8583, additional info, decision (**accepted** or **rejected**) and reason.
10. Pulse receives the Info request, and updates the transaction information in Case Manager.
11. Case Manager is provided with the field values sent via the Info endpoint.
12. Info response (Pre-decided Auth).
13. Message acknowledged.

Not-honorâ

The Not-honor use case occurs when you choose to take a different decision than the one recommended by Feedzai. For example, Feedzai's response is approve, and you decide to reject the transaction.

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation:

- **Decline** transaction.
- **Approve** transaction.

1. Case Manager is provided with the field values that are sent via the Authorization endpoint.
2. Authorization response: score, decision (**approve** or **decline**), actions and alert.
3. The issuer may honor or not the decision recommendation:

- **Accept** transaction (with a reason code): not honoring the **decline** recommendation.
- **Reject** transaction (with a reason code): not honoring the **approve** recommendation.

1. ISO 8583 authorization (1xx) or presentment (2xx) message response.
2. Authorization message response (ISO 8583 or network proprietary message).
3. Payment is **accepted** or **rejected**.
4. Info request (Not-honor): **accepted** or **rejected** (with reason code).
5. Pulse receives the info request, and updates the transaction information in Case Manager:

- Key: Life cycle ID.
- Adds not-honor information.

1. Case Manager is updated with the field values sent via the Info endpoint.
2. Info response (Not-honor).
3. Message acknowledged.

Out of Bandâ

You can define policies for advising downstream systems to contact a cardholder (e.g. via SMS or Call) to validate a transaction. These policies can be defined, for example, based on the response provided by Feedzai's model score, another output parameter - an action triggered by a rule - or any external reason.

This endpoint allows you to notify Feedzai that the cardholder is being contacted. If the response provided by the cardholder leads to a decision whether the transaction is fraud or legitimate, then this information is sent via the Feedback use case.

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
8. Case Manager is provided with the field values that are sent via the Authorization endpoint.
9. Authorization response: score, decision (**decline** or **approve**), actions and alert.
10. The issuer may honor or not the decision recommendation: **reject** or **accept** transaction (with a reason code).
11. SMS/Call from issuer to cardholder to verify the transaction authenticity.
12. ISO 8583 authorization (1xx) or presentment (2xx) message response.
13. Authorization message response (ISO 8583 or network proprietary message).
14. Payment is **rejected** or **accepted**.
15. Info request (Out of Band): e.g. with the type of contact made.
16. Pulse receives the Info request, and updates the transaction information in Case Manager.

- Key: Life cycle ID.
- Adds the contact Information.

1. Case Manager is updated with the field values sent via the Info endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Out of Band` functional block label).
2. Info response (Out of Band).
3. Message acknowledged.

Disputeâ

A cardholder can contact a bank to file a complaint for different reasons, such as:

- **Payment related reasons:** Unauthorized payment.
- **Non-payment related reasons:** A purchased item that was not delivered.

In this case, the bank and the cardholder may initiate a dispute. This endpoint can be used to inform Feedzai about the details of this process. If the process results in a chargeback, you can use the Feedback use case to inform Feedzai about fraud related reasons.

The numbers in the figure are described below:

1. Cardholder makes a purchase or withdrawal.
2. Merchant or ATM receives cardholder information.
3. Authorization message (ISO 8583 or network proprietary message).
4. ISO 8583 authorization (1xx) or presentment (2xx) message request.
5. Internal validations (client, account, card, etc).
6. Authorization request - ISO 8583 authorization (1xx) or presentment (2xx) message.
7. Pulse scores and sets a decision recommendation: **decline** or **approve** transaction.
8. Case Manager is provided with the field values that are sent via the Authorization endpoint.
9. Authorization response: score, decision (**decline** or **approve**), actions and alert.
10. The issuer **accepts** the transaction (with a reason code).
11. ISO 8583 authorization (1xx) or presentment (2xx) message response.

12. Authorization message response (ISO 8583 or network proprietary message).
13. Payment is **accepted**.
14. Cardholder disputes the transaction authenticity.
15. Info request (Dispute): e.g. dispute related information is sent.
16. Pulse receives the Info request, and updates the transaction information in Case Manager.

- Key: Life cycle ID.
- Adds the dispute information.

1. Case Manager is updated with the field values sent via the Info endpoint (the fields that are specific to this scenario are marked in the endpoint specification with the `Dispute` functional block label).
2. Info response (Dispute).
3. Message acknowledged.

Learn moreâ

Learn more about the detailed life cycle of other endpoints:

- Authorization
- Clearing
- Feedback

- About TFB
- Event Life Cycles
- Cards

On this page

Cards

Learn about the transaction life cycle and understand how the Cards API is used in the different stages of a transaction, e.g. authorizations, disputes, and clearings.

Transaction life cycleâ

This provides various endpoints to process transactions - Authorization, Info, Clearing, and Feedback - designed to map directly with version 3 of the ISO 8583. The following diagram shows how these endpoints come into play within a typical transaction use case:

These endpoints allow users to send information at different stages of a transaction life cycle as follows:

- **Authorization (x1xx and x2xx messages):** Allows Feedzai's customers to leverage Feedzai's data science capabilities to score a transaction according to its risk. The x1xx messages refer to the transaction authorization, and the x2xx messages refer to funds requests.
- **Info (x1xx and x2xx messages):** Allows Feedzai's customers to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream. The x1xx messages refer to the transaction authorization, and the x2xx messages refer to funds requests.
- **Clearing (x5xx messages):** Allows Feedzai's customers to inform Feedzai when a transaction is settled. Note that this endpoint is non-scorable, i.e. it does not flow through rules in order to generate a decision. However, the information is used by the Feedzai's risk engine platform to calculate metrics and profiles, and thus, impacts subsequent transfers associated with the same customer. It is also shown in the fraud analyst interface (Case Manager) as a new event to support the alert investigation. The x5xx messages communicate clearing information.
- **Feedback (x4xx messages):** Allows Feedzai's customers to label a transaction using pre-configured options that classify the transfer according to its risk. The labeling of an event is highly recommended since it gives feedback to Feedzai's platform enabling it to learn or consolidate illegal activity patterns. The x4xx messages refer to the dispute cases and reverse the action of a previous authorization.

Learn moreâ

Learn more about the detailed life cycle of each endpoint within a transaction:

- Authorization
- Info
- Clearing
- Feedback

Lifecycle and Session IDs

Was this page helpful?Â

A Digital Activity (DA) banking session may include events other than DA-related, such as Transfers, for instance.

This page provides you with an example of a banking session that includes both DA and Transfers events, while also showing you how monitoring the digital footprint of a session can further benefit the analyst's fraud detection capabilities.

Digital Activity and Transfers session use caseâ

A customer may start an online session, perform digital actions, make an online transfer and then log out. Although within the same session, we are looking at different events that are being monitored, with different APIs being invoked.

The image above illustrates the following flow:

1. The customer makes an online login attempt in the banking system.
    1. The issuer bank invokes Feedzai's Authentication endpoint for the login attempt.
    2. The issuer bank invokes Feedzai's Info Authentication endpoint to update Feedzai with the MFA result, using the same `lifecycle_id` as the previous event.
2. The customer makes a profile update attempt.
    1. The issuer bank invokes Feedzai's Profile Update endpoint.
3. The customer attempts to initiate a transfer.
    1. The issuer bank invokes Feedzai's Transfer Initiation endpoint.

Note that this single session sees endpoints from different APIs being invoked - for the customer login and profile update attempts, the DA API is invoked, while for the transfer initiation attempt, the Transfer API is invoked.

Lifecycles and sessionsâ

As the diagram illustrates, the session starts when the user attempts to log in and ends at session logout. The same session consists of different lifecycle events - a session is, therefore, the ensemble of operations performed from login to logout. Each session is given an ID, i.e., an identifier string.

Each of the operations performed - in this case, authentication, profile update, and transfer initiation - are independent events, identifiable by an ID. Note that a unique event can have more than one stage, as you can see in `Lifecycle_id_1`, which includes the login and MFA (Multiple Factor Authentication) actions.

Digital footprint collectionâ

Having a session ID enables Feedzai to collect the customerâs digital footprint, namely, to map and analyze the events that are part of the same banking session.

By leveraging data about all the session events, fraud analysts can have a better understanding of the risk level associated with that customer. Imagine the scenario of a customer changing their email address right before attempting to make a transfer transaction - this scenario could be an indicator of an account takeover.

Collecting the userâs digital footprint improves Feedzai's fraud detection capabilities. Analysts can use the collected information to refine their decision when manually reviewing an alert, and this information can also be used to further develop metrics, rules, etc.

- About TFB
- Event Life Cycles
- Digital Activity
- Account Update Endpoint

On this page

Account Update Endpoint

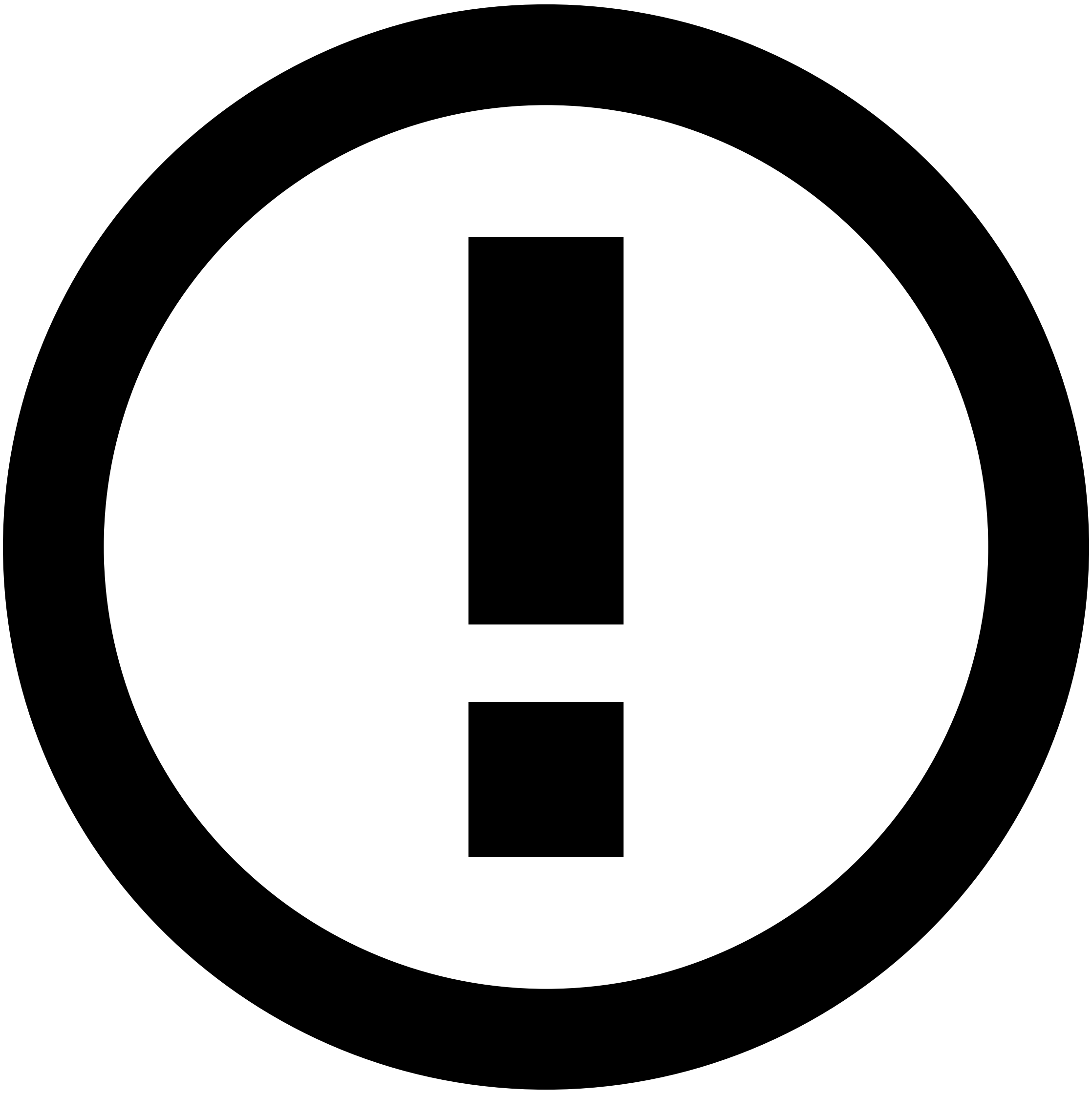
Was this page helpful?Â

The Digital Activity's Account Update endpoint leverages account change attempts, such as account limits via homebanking or mobile app.

The following image shows the Account Update endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The user attempts to make changes to their account, e.g., change account limit.

2. The issuer bank invokes Feedzai's Account Update endpoint.



reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from account update events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle]() page to understand the logic behind reference data's entity-based enrichment.

3. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the account change event.
4. The customer's bank makes an automated decision based on Feedzai's recommendation.
5. The customer is or isn't allowed to change their account.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.

- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the [Reference Data Operations](#) section to understand the difference between these methods.

Learn moreâ

Read the [Device Endpoint](#) page to continue learning about the endpoints' life cycles.

Alias Endpoint

The Digital Activity's Alias endpoint leverages account alias creation or change attempts. An alias is an element that associates the user account in a banking system with their bank account - for instance, a mobile phone, username, or QR code.

The following image shows the Alias endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The user attempts to create or change a bank account alias.
2. The issuer bank invokes Feedzai's Alias endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from alias events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle](#) page to understand the logic behind reference data's entity-based enrichment.

1. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the bank account alias change.
2. The customer's bank makes an automated decision based on Feedzai's recommendation.
3. The customer is or isn't allowed to their bank account alias.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.

- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the [Reference Data Operations](#) section to understand the difference between these methods.

Learn moreâ

Read the [Authentication Endpoint](#) page to continue learning about the endpoints' life cycles.

Authentication Endpoint

Was this page helpful?Â

The Digital Activity's Authentication endpoint leverages the customers' authentication or login change attempts via homebanking or mobile app.

The following image shows the Authentication endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The customer makes a login attempt in the banking system, through the mobile app for instance.
2. The issuer bank invokes Feedzai's Authentication endpoint.

3. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the authentication event.
4. The customer's bank makes an automated decision based on Feedzai's recommendation.
5. The customer is or isn't allowed to authenticate.

Learn moreâ

Read the Card Issuing Endpoint page to continue learning about the endpoints' life cycles.

Digital Activity

Was this page helpful?Â

Learn about the banking session life cycle and understand how the Digital Activity API is used in the different stages of a banking session. For example, when the customer changes their credentials, the bank invokes the Digital Activity's credentials update endpoint. Feedzai then receives all these events and analyzes them in order to generate highly-accurate profiles about the customer's digital behavior.

Banking session life cycleâ

1. The **customer** performs a digital action in a banking system via web home banking or mobile app.

2. The **issuer bank** invokes one of the Digital Activity connector's endpoints. The invoked endpoint depends on the digital action performed. For instance, if the customer has attempted to authenticate, the issuer bank will invoke the Authentication endpoint.
3. **Feedzai** conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the event.
4. The **customer's bank** makes an automated decision based on Feedzai's recommendation.
5. The **customer** is or isn't allowed to perform the digital action.

Endpoint summaryâ

The following endpoints allow users to send information at different stages of a banking session life cycle as follows:

- **Authentication**: The customer attempts to authenticate or log in to the banking system via web home banking or mobile app.
- **Alias**: The customer attempts to register or change the default account alias. In some payment methods (e.g. Zelle, PIX, NPP), an alias is the phone number, email or QR code associated with the bank account.
- **Account update**: Leverages account change attempts, such as account limits via homebanking or mobile app.
- **Card issuing**: The customer attempts to register a new payment card (i.e. credit card or debit card) with, for instance, a mobile wallet (e.g. Google Pay, Apple Pay), or a subscription service (e.g. Netflix).
- **Card update**: Leverages card change attempts, such as card PIN, transaction limits, or new card format as physical / virtual via homebanking or mobile app.
- **Credentials update**: The customer attempts to change their credentials, e.g. their username or password.
- **Device**: The customer attempts to register, change, or delete a device associated with the bank's platform.
- **New payee**: The customer attempts to add a new trusted payee to their account or attempts an alias resolution request. For instance, searching for random mobile phone numbers and adding them as a contact in the banking app.
- **Profile update**: The customer attempts to change their profile, e.g. their address.
- **Info endpoints**: Info endpoints send additional information for the last sent `POST` or `PUT` request for each aforementioned endpoint.

Scoring logicâ

Digital Activity's scoring logic is as follows:

- `POST` events are scored.
- `PUT` events are not scored. This also applies to all Info endpoints.

Learn moreâ

Learn how each endpoint is used along the banking session life cycle:

- Authentication
- Alias
- Account update
- Card issuing
- Card update
- Credentials update
- Device endpoint
- New payee
- Profile update
- Info

Card Issuing Endpoint

Was this page helpful?Â

Card issuing, also known as card enrollment, is the process of registering a new payment card (i.e. credit card or debit card) with, for instance, a mobile wallet (e.g. Google Pay, Apple Pay), or a subscription service (e.g. Netflix). It involves a process similar to the following:

1. The customer adds a card to a Wallet, such as ApplePay, by providing the card details, which usually include the card number, expiration date, and CVV (the 3-digit number on the back of the card).
2. After confirming that the card details are correct, the service asks the customer to confirm their identity, for instance, by asking them for an extra layer of security (e.g. 3D Secure).

3. The service sends the enrollment request to the customer's issuing bank. The bank then confirms that everything is accordingly and approves the enrollment.
4. The customer receives a confirmation from the service (e.g. by email) that their card was successfully enrolled.
5. Certain services like Apple Pay, replace the customer's card number with a unique digital identifier or token, which will be used for transactions, instead, thus enhancing security.

This page explains how Transaction Fraud for Banking integrates the card enrollment process for fraud prevention.

Endpoint life cycleâ

1. The customer adds a card to a service (e.g. Apple Pay).

2. The issuing bank receives the request from the mobile wallet and invokes Feedzai's Digital Activity card issuing endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from new payee events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle] page to understand the logic behind reference data's entity-based enrichment.

3. Feedzai conducts real-time data analysis and makes a decision on whether to approve or decline the card issuing event.

4. The issuing bank makes an automated decision based on Feedzai's recommendation.

5. If the issuing bank approves the event, the card is successfully enrolled.

6. Optionally, the issuing bank calls Feedzai as soon as they get the token from the Token Service Provider (TSP). The Card Update endpoint with `card_status = active` will tell Feedzai that the token has become active, and that subsequent Authorizations with that token can be scored. At this point, the customer is allowed to perform transactions. Any new card event requires the issuing bank to call Feedzai's [Cards API] to prevent a fraudulent transaction.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.
- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the [Reference Data Operations] section to understand the difference between these methods.

Learn moreâ

Read the Credentials Update Endpoint page to continue learning about the endpoints' life cycles.

Card Update Endpoint

The Digital Activity's Card Update endpoint leverages card change attempts, such as card PIN, transaction limits, or new card format as physical / virtual via homebanking or mobile app.

The following image shows the Cards Update endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The user attempts to make changes to their card, e.g., new card format as physical.

2. The issuer bank invokes Feedzai's Card Update endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from card update events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle]() page to understand the logic behind reference data's entity-based enrichment.

3. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the card change event.
4. The customer's bank makes an automated decision based on Feedzai's recommendation.
5. The customer is or isn't allowed to change their card.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.

- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the [Reference Data Operations](#) section to understand the difference between these methods.

Learn moreâ

Read the [Device Endpoint](#) page to continue learning about the endpoints' life cycles.

Credentials Update Endpoint

Was this page helpful?Â

The Digital Activity's Credentials Update endpoint leverages username, password, face ID and fingerprint change attempts via homebanking or mobile app.

The following image shows the Credentials Update endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The user attempts to change their credentials - username, password, etc.
2. The issuer bank invokes Feedzai's Credentials Update endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from credential update events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle] page to understand the logic behind reference data's entity-based enrichment.

1. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the credentials change event.
2. The customer's bank makes an automated decision based on Feedzai's recommendation.
3. The customer is or isn't allowed to change their credentials.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.

- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the [Reference Data Operations](#) section to understand the difference between these methods.

Learn moreâ

Read the [Device Endpoint](#) page to continue learning about the endpoints' life cycles.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/life-cycles/digital-activity/device

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- [About TFB]
- [Event Life Cycles]
- [Digital Activity]
- Device Endpoint

On this page

Device Endpoint

Was this page helpful?Â

The Digital Activity's Device endpoint leverages registration, changes or deletion of devices associated with the bank's platform. For instance, attempting to associate a new device with a banking app is one of the scenarios where a bank will invoke Feedzai's Device endpoint.

The following image shows the Device endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The user attempts to associate a new device with a banking app or to change/delete a device associated with a banking app.

2. The issuer bank invokes Feedzai's Device endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from device events will take precedence over what's in the Reference Data Storage. Read the Reference Data Life Cycle page to understand the logic behind reference data's entity-based enrichment.

3. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the device-related event.
4. The customer's bank makes an automated decision based on Feedzai's recommendation.
5. The customer is or isn't allowed to perform the device-related event.

Warning

Each event must have a unique `lifecycle_id`, as these events don't support updates.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.
- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the [Reference Data Operations](#) section to understand the difference between these methods.

Learn moreâ

Read the [New Payee Endpoint](#) page to continue learning about the endpoints' life cycles.

-
- About TFB
- Event Life Cycles
- Digital Activity
- Feedback Endpoint

On this page

Feedback Endpoint

The Digital Activity's Feedback endpoint allows you to label digital activity events as "Fraud" or "Not Fraud". The labeling of an event is highly recommended, as Feedzai's risk engine can learn from it and consolidate fraud patterns.

Consider the following diagram, which shows the **Feedback** endpoint's end-to-end process in an Account Update event.

1. The user attempts to make changes to their account, e.g., change account limit.

2. The issuer bank invokes Feedzai's Account Update endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from account update events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle]() page to understand the logic behind reference data's entity-based enrichment.

3. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the account change event.
4. The customer's bank makes an automated decision based on Feedzai's recommendation.

   At this point, the bank can use the **Feedback** endpoint to send its final decision to Feedzai, using the account update event's `lifecycle_id`. This is highly recommended as Feedzai can use the bank's feedback to train its models and provide more accurate decisions in the future.
5. The customer is or isn't allowed to change their account.

Learn moreâ

Now that you understand each endpoint's life cycle, take a look at Digital Activity's API documentation to have a complete overview of each endpoint's schema and respective fields.

Info Endpoints

This page explains how Info endpoints play a role in any digital activity event life cycle.

The Digital Activity API includes Info endpoints to send additional information for the last sent `POST` or `PUT` request. Consider the following diagram, which shows an Info endpoint's end-to-end process in a Profile Update event.

1. The user attempts to change their profile, e.g. their address.

2. The issuer bank invokes Feedzai's Profile Update endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from profile update events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle]() page to understand the logic behind reference data's entity-based enrichment.

3. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the account change event.

4. The bank makes an automated decision based on Feedzai's recommendation.

5.
    - 5a. The customer is or isn't allowed to update their profile.
    - 5b. The bank invokes the Profile Update Info endpoint with additional information to the previously sent event, using the same `lifecycle_id` .

info

By default, Feedzai does not store additional information sent through Info Endpoints in the Reference Data Storage.

Learn moreâ

Read the Feedback Endpoint page to continue learning about the endpoints' life cycles.

-

- [About TFB]
- [Event Life Cycles]
- [Digital Activity]
- New Payee Endpoint

On this page

New Payee Endpoint

The Digital Activity's New Payee endpoint leverages the users' attempts to add a new payee to a payment app. Consider the scenario where a user tries to add random mobile phone numbers to a payment app (e.g. Zelle, PIX, NPP, etc) until they find an existing one. When they try to add that number to the payment app, the bank invokes Feedzai's New Payee endpoint. Note that this type of scams usually comes with social engineering techniques to lead the victim into actively sending money to the fraudster's account. In such a scenario, the payee becomes the payer.

The following image shows the New Payee endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The user attempts to add a random mobile phone number to a payment app, such as Zelle.
2. The issuer bank invokes Feedzai's New Payee endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from new payee events will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle] page to understand the logic behind reference data's entity-based enrichment.

1. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the adding a new payee event.
2. The customer's bank makes an automated decision based on Feedzai's recommendation.
3. The customer is or isn't allowed to add a new payee to the payment app.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.

- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the Reference Data Operations section to understand the difference between these methods.

Learn moreâ

Read the Profile Update Endpoint page to continue learning about the endpoints' life cycles.

Profile Update Endpoint

Was this page helpful?Â

The Digital Activity's Profile Update endpoint leverages users' attempts to change their profiles via homebanking or mobile app. For instance, when a user tries to change their address or phone number, the bank invokes Feedzai's Profile Update endpoint.

The following image shows the Profile Update endpoint's end-to-end process. Note that it applies to a simplified processing model.

1. The user attempts to change their profile, their address for instance.
2. The issuer bank invokes Feedzai's Profile Update endpoint.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities have a match in the Reference Data Storage.

data precedence in reference data storage

Note that any new data from profile update events will take precedence over what's in the Reference Data Storage. Read the Reference Data Life Cycle page to understand the logic behind reference data's entity-based enrichment.

1. Feedzai conducts real-time data analysis and makes a decision on whether to approve, decline or challenge the profile change.
2. The customer's bank makes an automated decision based on Feedzai's recommendation.
3. The customer is or isn't allowed to perform the change to their profile.

Reference Data updatesâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked`. If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.

- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654`. If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve`.
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the [Reference Data Operations](#) section to understand the difference between these methods.

Learn moreâ

Read the [Info Endpoints](#) page to continue learning about the endpoints' life cycles.

Inbound Payments

Was this page helpful?Â

Learn about the inbound payments' transaction life cycle and understand how the Inbound Payments API is used in the different stages of a transaction, i.e. inbound transfer initiation, info, feedback, and settlement.

Transaction life cycleâ

The following diagram shows how the Inbound Payments API endpoints come into play in the course of a typical transaction life cycle:

The following steps describe the flow depicted in the diagram above, which represents an integration between the receiver's bank and Feedzai.

1. The receiver's bank receives an inbound payment request.
2. The receiver's bank invokes Feedzai's **Inbound Transfer Initiation** endpoint with the payload related to the given inbound payment. In some situations, the **Info** endpoint is also used at this point, for example, to communicate pre-decided payments.

   info

   For transactions that occur between customers from the same bank, two events should be sent to Feedzai, one via **Inbound Payments API**, and another one via **Outbound Transfers** API.

3. Feedzai conducts real-time data analysis and scores. The inbound payment is processed by the risk engine platform with rules and/or models. As a result, the risk engine generates a recommendation that the inbound payment is either **genuine** or **fraudulent**.
4. Fraud analysts perform manual investigation on alerts raised by the risk engine. This step only applies to inbound payments subject to longer processing times. For example, for instant inbound transfers, the alert investigation works as a post-mortem analysis because the inbound transfer is processed in real time.
5. The receiver's system makes an automated decision based on Feedzai's recommendation and optionally notifies Feedzai about this decision via **Info** or **Feedback** endpoints.

6. The receiver's account is updated with the funds and, optionally, the receiver's bank informs Feedzai about the payment via the **Settlement** endpoint.
7. A fraud analyst can investigate the event at any point after the real-time event processing is completed. The information is sent back to the risk engine and can be leveraged by rules and models.

Endpoint summaryâ

These endpoints allow users to send information at different stages of a transaction life cycle as follows:

- **Inbound Transfer Initiation:** Allows Feedzai's clients to leverage data science capabilities to score a transaction according to its risk.
- **Info:** Allows Feedzai's clients to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream.
- **Get:** Allows clients to obtain the current transaction status, including all the details of the event, i.e., the information sent through the endpoints during an inbound payment's life cycle.
- **Settlement:** Allows Feedzai's clients to inform Feedzai when a transaction is settled. Note that this endpoint is non-scorable, i.e., it does not flow through rules in order to generate a decision. However, the information is used by Feedzai's risk engine platform to calculate metrics and profiles, which impacts subsequent inbound payments associated with the same customer. It is also shown in the fraud analyst interface (Case Manager) as a new event to support the alert investigation.
- **Feedback:** Allows Feedzai's clients to label a transaction using pre-configured options that classify the inbound payment according to its risk. The labeling of an event is highly recommended since it gives feedback to Feedzai's platform enabling it to learn or consolidate illegal activity patterns. The *x4xx* messages refer to the dispute cases and reverse the action of the previous inbound transfer initiation.

Endpoints and event typesâ

The available endpoints allow you to integrate with the Inbound Payments API at the key stages of the transaction life cycle. This ensures that all the information is used to detect and prevent fraudulent activity efficiently. Since each endpoint is used to accomplish a different goal, the following table summarizes the key information for each of them:

Endpoint/ event type	Event subtype	Mandatory (M), Recommended (R), Optional (O)	Is the event scored?	Recommendation (approve/ decline)	Ability to inform Feedzai about final decision (accepted/ rejected)	Ability to label the transaction
Inbound Transfer Initiation	N/A	M	Yes	Yes	No	No
Info	Additional information	O	No	No	Yes	No
	Pre-decided	R	No	No	Yes	No
	Not-honor	R	No	No	Yes	No
	Out of band (sms/call)	O	No	No	Yes	No
	Dispute	R	No	No	Yes	No
Settlement	N/A	O	No	No	No	No
Feedback	Dispute	-Â M, for Info with subtype dispute - R, all other cases	No	No	No	Yes
	Out of band	Only used in combination with Info - out of band. In this case is mandatory	No	No	No	Yes

This table gives you the following information:

- **Mandatory (M), Recommended (R), Optional (O):** These options relate to the message request. Messages specified as recommended are used by the risk engine platform to improve the system's accuracy to monitor inbound payments.
- **Is the event scored?** An event can be scored by the data science models with a number ranging between 0-1000, where 1000 represents the highest probability of being fraudulent. The scoring is only generated if you have models configured in the risk engine platform.

info

Scorable events: The endpoints and event subtypes marked as scorable are the ones with the `score` field in the API response. However, the actual scoring depends on:

- The platform having a model configured in the runtime workflow.
- The filter conditions defined in the runtime workflow allowing the event to reach the scoring operator.

- **Recommendation (approve/decline):** Generated by the risk engine rules with the possible values "approve" or "decline".

- **Ability to inform Feedzai about final decision (accepted/rejected):** Final decision made by the client, i.e., transaction accepted or declined.
- **Ability to label the transaction:** The `feedback_value` field is populated with the value "true" or "false" representing a fraudulent or genuine transaction, respectively. This is used to improve the accuracy of the data science models, and to calculate metrics such as precision and false positive rate.

Recommendations on cash and ATM depositsâ

To correctly map cash and ATM deposits to inbound payments, and hence score cash and ATM deposits, some fields need to be populated to identify cash or ATM deposit events as follows:

```
    {
      // Required fields
      "event_external_id": "4048611b295e993d1b561d9030c53424bbf2b95e",
      "lifecycle_id": "1df7a7qg78vd7fs",
      "channel_type": "atm", // To indicate it's ATM
      "payment_status": "entered",
      "created_at": 1603207359060,
      "sender_transaction_amount": 1603207359060,
      "sender_transaction_currency": "GBP",
      "sender_transaction_bank_account_number": "5a800244-325a-4086-ab8a-23ecbb927be5",
      "direction": "incoming",
      "event_occurred_at": 1684492270092,

      // Recommended for mapping cash and ATM deposits
      "payment_type": "sepa", // To indicate it's a deposit
      "payment_sub_type": "direct_deposit", // Use the `direct_deposit` value to indicate it's
a direct deposit
    }
```

Read the [Inbound Transfer Initiation [POST]] documentation for more information on other fields you can use in the payload.

Reference Data

Was this page helpful?Â

You can use the Reference Data API to send key business information to Feedzai, such as customer and account data. The reference data is then stored in the database, i.e. Reference Data Storage, and can be used to enrich events in real time, i.e. populate information from the reference data onto the transaction. This allows you to reduce the additional information that you would otherwise send along with each event.

As a best practice, you should send all entity data via Reference Data API. The reference data operations section explains which operations you should execute for data storage and which entities you should send to the Reference Data Storage for further event enrichment.

As a result, other Feedzai APIs can fetch this reference data from the Reference Data Storage each time the Pulse runtime workflow processes an event. Take into consideration that this is only possible if the sent keys have a match in the reference data storage.

Data storageâ

The following image shows the path of the reference data from the customer to Feedzai's Reference Data Storage:

The information sent via Reference Data API is organized into different entities (e.g. Account entity, Customer entity). Continue reading to learn how to store or update data in the Reference Data Storage to further enrich real-time events.

Reference data operationsâ

The following operations can be used with Reference Data API to get, store, update (in bulk or selectively), and delete data:

Method	HTTP request (example)	Purpose
`GET`	`GET <>/api/referenceData/payees/{key}`	Populates entity-related information in Case Manager.
`PUT`	`PUT <>/api/referenceData/customers/{key}`	Updates the full list of fields from an entity in the Reference Data Storage. This means that all entity fields are stored or updated in the Reference Data Storage. Ensure you send all entity-related fields when providing updates about an entity to avoid potential issues with the `PUT` method treating missing fields as removals.
`PATCH`	`PATCH <>/api/referenceData/cards/{key}`	Updates a field, or selection of fields from an entity in the Reference Data Storage. In contrast to `PUT`, which updates an entity in the Reference Data Storage in bulk, `PATCH` allows you to update only the fields you need.
`DELETE`	`DELETE <>/api/referenceData/credentials/{key}`	Deletes an entity from the Reference Data Storage.

To store, update or delete data from the Reference Data Storage, refer to the following table:

Endpoint	Key	Purpose
New Payees	`customer_id`	Enriches the Payee's entity data in the Reference Data Storage. As such, all updates to a payee must be done through Reference Data's Payees endpoint, through: - **`PUT <> /api/referenceData/payees/{key}`**, for an integral field update - **`PATCH <> /api/referenceData/new_payee/{key}`**, for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - **`DELETE <> /api/referenceData/payees/{key}`**.
Credentials	`customer_id`	Enriches the Credential's entity data in the Reference Data Storage. As such, all updates to those credentials must be done through Reference Data's Credentials endpoint, through: - **`PUT <> /api/referenceData/credentials/{key}`**, for an integral field update. - **`Patch <> /api/referenceData/credentials/{key}`**, for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - **`DELETE <>/api/referenceData/credentials/{key}`**.
Cards	`card_id`	The Cards endpoint is necessary both to populate the fields in Case Manager's **Customer** page and to enrich the Card's entity data in the Reference Data Storage. As such, all updates to a card must be done through Reference Data's

Endpoint	Key	Purpose
		Cards endpoint, through: - `PUT <> /api/referenceData/cards/{key}` , for an integral field update. - `Patch <> /api/referenceData/cards/{key}` , for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - `DELETE <>/api/referenceData/cards/{key}` .
Devices	`device_id`	Enriches the Device's entity data in the Reference Data Storage. As such, all updates to a device must be done through Reference Data's Devices endpoint, through: - `PUT <> /api/referenceData/devices/{key}` , for an integral field update. - `Patch <> /api/referenceData/devices/{key}` , for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - `DELETE <>/api/referenceData/devices/{key}` .
Customers	`sender_id` and `receiver_id`	The Customers endpoint is necessary both to populate the fields in Case Manager's **Customer** page and to enrich the Customer's entity data in the Reference Data Storage. As such, all updates to a customer must be done through Reference Data's Customers endpoint, through: - `PUT <> /api/referenceData/customers/{key}` , for an integral field update. - `Patch <> /api/referenceData/customers/{key}` , for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - `DELETE <>/api/referenceData/customers/{key}` .
Aliases	`customer_id`	Enriches the Alias's entity data in the Reference Data Storage. As such, all updates to an alias must be done through Reference Data's Aliases endpoint, through: - `PUT <> /api/referenceData/aliases/{key}` , for an integral field update. - `Patch <> /api/referenceData/aliases/{key}` , for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - `DELETE <>/api/referenceData/aliases/{key}` .
Merchants	`acceptor_id`	Enriches the Merchant's entity data in the Reference Data Storage. As such, all updates to a merchant must be done through Reference Data's Merchants endpoint, through: - `PUT <> /api/referenceData/merchants/{key}` , for an integral field update. - `Patch <> /api/referenceData/merchants/{key}` , for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - `DELETE <>/api/referenceData/merchants/{key}` .
Accounts	`account_iban` for cards `sender_account_iban` **and** `receiver_account_iban` for inbound and outbound transfers	Enriches the Account's entity data in the Reference Data Storage. As such, all updates to an account must be done through Reference Data's Accounts endpoint, through: - `PUT <> /api/referenceData/accounts/{key}` , for an integral field update. - `Patch <> /api/referenceData/accounts/{key}` , for updates to a selection of fields. To remove this entity from the Reference Data Storage you must delete it through: - `DELETE <>/api/referenceData/accounts/{key}` .

Real-life scenarioâ

Imagine that Feedzai receives a card authorization request, where the final output (i.e. payload) includes new information about `card_category` . This information will be present at the root of the payload.

However, the customer array (i.e. `customer_cards`) will not reflect the same changes. That is because this array is part of the customer information that is sent by the client to Feedzai through Reference Data API's Customer entity. To update arrays such as `customer_cards` or `customer_accounts` , you must update the Reference Data API's Customer entity through either `PUT` or `PATCH` . Refer to the [Reference Data Operations] section to understand the difference between these methods.

Digital Activity's use caseâ

You can trigger updates to the Reference Data Storage via the Digital Activity API. These updates apply exclusively to fields starting with the `new_*` prefix as described in the following examples:

- The customer wants to update the status of his card. He opens the app and momentarily disables the card. The bank calls the Update Card endpoint in Digital Activity and sets `new_card_status` = `blocked` . If approved, this will update the `card_status` for the Card entity in the Reference Data Storage.
- The customer changes his phone number on the bank's app. The bank sends a Profile Update request to Feedzai with `new_customer_phone_number` = `1-970-664-2654` . If approved, this will update the `customer_phone_number` = `1-970-664-2654` for the Customer entity in the Reference Data Storage.

Notice that updates from the Digital Activity API occur under the following conditions:

- `POST` events whose system decision is `approve` .
- `PUT` events regardless of the system decision.

caution

To amend information, you need to resend the entire entity data (i.e. populate all the `new_*` fields). If you want to update pieces of information instead of updating the entire entity, use the Reference Data API with the PATCH method. Refer to the Reference Data Operations section to understand the difference between these methods.

Real-time event enrichmentâ

caution

Real-time event enrichment can only occur if the relevant data (e.g. card, account, device) already exists in the Reference Data Storage. Learn how to store data in the Reference Data Storage using Reference Data's API endpoints.

When processing an event, if the Reference Data Storage contains more data about the entity whose event relates to, this extra data enriches the event. Take into consideration that this is only possible if the keys sent in the event (e.g. card_id, customer_id) have a match in the reference data storage.

The following image shows the field enrichment process for a payment initiation request.

The following scenarios might occur during the enrichment:

- **Scenario 1:** The field sent via the Transaction API (i.e. original field) is not populated, and the information is not in the Reference Data Storage. As a result, the field remains unpopulated.

- **Scenario 2:** The original field is not populated, but the corresponding field in the Reference Data Storage contains data. As a result, the field from the event is enriched with reference data. This is important for scoring purposes, as well as to help analysts throughout their investigation (i.e. the additional data populates Case Manager's fields).
- **Scenario 3:** The original field is populated with a different value than the value that is in the Reference Data Storage. As a result, the value sent via **the Transaction API prevails over the reference data**. To update the Reference Data Storage with the value sent via the Transaction API, you must use the Reference Data API to update the corresponding entity, namely through the `PATCH` method (see the difference between the `PATCH` and `PUT` methods).

Entitiesâ

To enrich the event with reference data as described in Scenario 2, the ID sent in the event must be associated with the entity that you want to enrich. For instance, to enrich the real-time event with card data, its `card_id` must match the `card_id` of Reference Data Storage.

The following sections show which entities can be used to enrich each event type with reference data.

caution

Note that for real-time event enrichment to take place, the relevant data must already exist in the Reference Data Storage. Learn how to store data in the Reference Data Storage using Reference Data's API endpoints.

Digital activity eventsâ

- **Payee data**: `new_payees`
  - **Key**: `customer_id`
- **Credential data**: `credentials`
  - **Key**: `customer_id`
- **Device data**: `devices`
  - **Key**: `device_id`
- **Alias data**: `aliases`
  - **Key**: `customer_id`
- **Profile data**: `customers`
  - **Key**: `customer_id`
- **Card data**: `cards`
  - **Key**: `card_id`
- **Account data**: `accounts`
  - **Key**: `account_iban`

Card eventsâ

- **Card data**: `cards`
  - **Key**: `card_id`
- **Customer data**: `customers`
  - **Key**: `customer_id`
- **Merchant data**: `merchants`
  - **Key**: `acceptor_id`
- **Account data**: `accounts`
  - **Key**: `account_iban`

Inbound payment eventsâ

- **Customer data**: `customers`
  - **Key**: `sender_id` and `receiver_id`
- **Account data**: `accounts`
  - **Key**: `sender_account_iban` and `receiver_account_iban`
- **Device data**: `devices`
  - **Key**: `device_id`

Outbound transfer eventsâ

- **Customer data**: `customers`.
  - **Key**: `sender_id` and `receiver_id`
- **Account data**: `accounts`.
  - **Key**: `sender_account_iban` and `receiver_account_iban`.
- **Device data**: `devices`
  - **Key**: `device_id`

snippet-reference-data-enrichment-scenarios

- **Scenario 1:** The field sent via the Transaction API (i.e. original field) is not populated, and the information is not in the Reference Data Storage. As a result, the field remains unpopulated.
- **Scenario 2:** The original field is not populated, and the information is in the Reference Data Storage. As a result, the field is enriched with reference data.
- **Scenario 3:** The original field is populated with a different value than the value that is in the Reference Data Storage. As a result, the value sent via **the Transaction API prevails over the reference data**.

snippet-reference-data-entity-ids

- **Card data:** `card_id`
- **Account data:** `account_iban`
- **Merchant data:** `acceptor_id`
- **Customer data:** `customer_id`

Automated Clearing House

Was this page helpful?Â

Learn about the Automated Clearing House (ACH) end-to-end life cycle to smoothly integrate with the transfers API.

Transaction initiationâ

The scenarios described in this section explain the transfer life cycle for both single counterparty and multiple counterparties. This page covers the end-to-end default scenario in which only the Transfer Initiation endpoint is called to process transfers.

info

Endpoints Integration: Check the [Transaction Life Cycle] page to learn more about the life cycles for transfer updates, settlements, and feedback actions, corresponding to requests to the Info, Settlement and Feedback endpoints, respectively.

Single counterpartyâ

The life cycle that involves a single counterparty applies to large financial institutions that communicate directly with a clearing mechanism. The following diagram depicts a typical flow:

The steps depicted in the previous image are the following:

1. The sender, also referred to as originator, initiates the transfer using any available channel - for example, mobile, internet banking, phone, bank branch, contact center, email, file transfer system.
2. The sender bank or the payment service provider (PSP) - a direct participant in this scenario - receives the transfer instructions and debits the sender's account. The sender bank sends the transfer instructions

corresponding to several senders to the clearing mechanism via batch (i.e. a bulk message with multiple transfers instructions).

3. The clearing mechanism processes the transfers instructions, and reconciles them. In some cases, it also confirms the payments orders or secures the transfers instructions prior to settlement, including the netting of instructions and the establishment of the final positions for settlement. After the netting of the instructions or at the end of the day, the clearing mechanism processes the settlement of funds between two or more parties.
4. The receiver bank - a direct participant in this scenario - receives the instructions to credit the receiver's account.
5. The receiver's account is updated with the funds.

info

Integration with the transfers API: The transfers API was designed to allow banks that operate at different points of the transfers life cycle to integrate with Feedzai. This flexible integration capability is shown in the previous figure with the API label.

Multiple counterpartiesâ

The scenario of multiple counterparties occurs when the sender's bank is an institution that does not communicate directly with the ACH mechanism. In this case, a third-party is required to establish the communication between the direct participant and a clearing mechanism:

The steps depicted in the previous image are the following:

1. The sender, also referred to as originator, initiates the transfer using any available channel - for example, mobile, internet banking, phone, bank branch, contact center, email, file transfer system.
2. The sender bank or the PSP - an **indirect** participant in this scenario - receives the transfer instructions and debits the sender's account. This institution sends the transfer instructions to the corresponding financial institution that has the direct participance in this scenario.
3. The sender bank - a **direct** participant in this scenario - receives the transfer instructions. The direct sender bank sends the transfer instructions corresponding to several senders to the clearing mechanism via batch (i.e. bulk messages with multiple transfers instructions).
4. The clearing mechanism processes the transfers instructions, and reconciles them. In some cases, it also confirms the payments orders or secures the transfers instructions prior to settlement, including the netting of instructions and the establishment of the final positions for settlement. After the netting of the instructions or at the end of the day, the clearing mechanism processes the settlement of funds between two or more parties.
5. The receiver bank - a **direct** participant in this scenario - receives the instructions and sends it to the corresponding indirect participant.
6. The receiver bank - an **indirect** participant in this scenario - receives the transfer instructions and credits the receiver's account.
7. The receiver's account is updated with the funds.

info

Integration with the transfers API: The transfers API was designed to allow banks that operate at different points of the transfers life cycle to integrate with Feedzai. This flexible integration capability is shown in the previous figure with the API label.

Learn moreâ

Check the Correspondent Banking page to learn how to integrate with other transfer methods.

Correspondent Banking

Learn about the Correspondent Banking end-to-end life cycle to smoothly integrate with the transfers API.

Transaction initiationâ

The scenarios described in this section explain the transfer life cycle for both single counterparty and multiple counterparties. This page covers the end-to-end default scenario in which only the Transfer Initiation endpoint is called to process transfers.

info

Endpoints Integration: Check the [Transaction Life Cycle] page to learn more about the life cycles for transfer updates, settlements, and feedback actions, corresponding to requests to the Info, Settlement and Feedback endpoints, respectively.

Single counterpartyâ

The life cycle that involves a single counterparty applies to large financial institutions. The following diagram depicts a typical flow:

The steps depicted in the previous image are the following:

1. The sender, also referred to as originator, initiates the transfer using any available channel - for example, mobile, internet banking, phone, bank branch, contact center, email, file transfer system.
2. The sender bank receives the transfer instructions and debits the sender's account. The sender bank sends the transfer instructions to the receiver bank. A combination of message MT103 and MT202 (or MT202 COV) are used in this process. The MT103 is the transfer message and is in the critical. The MT202 (or MT202 COV) is the

correspondent message to settle the funds between the banks, is sent afterwards, and does not include information about the originating financial institution.

3. The receiver bank receives the message from the sender bank, acknowledges and deducts a transaction fee. It also receives the instructions to credit the receiver's account.
4. The receiver's account is updated with the funds.

info

Integration with Feedzai API: The transfers API was designed to allow banks that operate at different points of the transfers life cycle to integrate with Feedzai. This flexible integration capability is shown in the previous figure with the API label.

Multiple counterpartiesâ

The scenario of multiple counterparties occurs when the sender and the receiver's bank are small-sized institutions. In this case, a third-party is required to establish the main communication (e.g. overseas connection) between the correspondents.

The steps depicted in the previous image are the following:

1. The sender, also referred to as originator, initiates the transfer using any available channel - for example, mobile, internet banking, phone, bank branch, contact center, email, file transfer system.
2. The sender bank receives the transfer instructions and debits the sender's account. The sender bank sends the transfer instructions to the sender correspondent bank. A combination of message MT103 and MT202 (or MT202 COV) are used in this process. The former is in the critical path whereas the latter is sent afterwards and does not include information about the originating financial institution.
3. The sender correspondent bank receives the message from the sender bank, acknowledges, deducts a fee from this transaction and forwards the transfer messages to the receiver correspondent bank.
4. The receiver correspondent bank receives the message from the sender correspondent bank, acknowledges, deducts a fee from this transaction and forwards the transfer messages to the receiver bank.
5. The receiver bank receives the transfer messages and credits the receiver's account.
6. The receiver's account is updated with the funds.

info

Integration with Feedzai API: The transfers API was designed to allow banks that operate at different points of the transfers life cycle to integrate with Feedzai. This flexible integration capability is shown in the previous figure with the API label.

Learn moreâ

Check the Real-time Gross Transfers page to learn how to integrate with other transfer methods.

- About TFB
- Event Life Cycles
- Outbound Transfers
- Feedback Endpoint

On this page

Feedback Endpoint

Was this page helpful?Â

The Feedback endpoint allows you to send label information about the transfer. The feedback might come from the following sources:

- **Analyst:** Using the analyst interface to mark alerts as being legitimate or not.
- **Feedzai customers:** This scenario occurs after the info - out of band and info - dispute use cases to communicate the final decision.

The following diagram shows the end-to-end process for feedback as a result of the out of band use case:

Steps 1-7 are the same as the in the Info Out of band scenario, and then:

- **Step 8:** The account holder confirms or denies the transfer authenticity.
- **Step 9:** The sender bank sends an API request via the **Feedback** endpoint to inform Feedzai about the final label.

The following diagram shows the end-to-end process for feedback as a result of the dispute use case:

Steps 1-6 are the same as the in the Info Dispute scenario, and then:

- **Step 7:** The account holder confirms or denies the transfer authenticity.
- **Step 8:** The sender bank sends an API request via the **Feedback** endpoint to inform Feedzai about the final label.

Learn moreâ

The communication between the sender's bank, processing model, and the receiver's bank have extra details that are not covered in the life cycles included in each endpoint page. Check the Processing Model Details page to understand how Feedzai covers each processing model: Automated Clearing House (ACH), Correspondent Banking, Real-Time Gross Transfers, and Peer-to-Peer.

-
- [About TFB]
- [Event Life Cycles]
- [Outbound Transfers]
- Info Endpoint

On this page

Info Endpoint

Was this page helpful?Â

The Info endpoint enables you to update and add new information to an existing transfer, as well as to notify Feedzai about transactions processed upstream. The following section explain the life cycle for each info subtype.

info

When you use the Info endpoint to perform an update, you only need to provide the new information (i.e. the original information does not need to be sent).

Additional informationâ

Additional Information is used to update information sent in a previous request. The following diagram shows the end-to-end process:

The steps depicted in the previous image are the following:

1. The sender initiates a transfer directly with the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint.
3. The sender bank sends the transfer instructions to the receiver bank.
4. The receiver's account is updated with the funds.
5. The sender bank sends an API request via the **Info** endpoint with subtype `additional_info` to update or add new information.

reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities (e.g. Card, Customer, or Account) have a match in the Reference Data Storage.

info

Note that any new data from transfer info events (e.g. new `sender_card_category`) will take precedence over what's in the Reference Data Storage. Read the Reference Data Life Cycle page to understand the logic behind reference data's entity-based enrichment.

Pre-decidedâ

Pre-decided is used with transfers that have been either rejected or accepted before arriving at Feedzai (i.e. does not have the Feedzai recommendation or score). Typical transfers use cases include:

- Rejection at the network level before it arrives at the receiver.
- Rejection at the sender side, for example, for lack of account balance or blocked account.
- Internal transfer that is automatically accepted.
- Initiated and immediately canceled by the sender.

The following diagram shows the end-to-end process:

The steps depicted in the previous image are the following:

1. The sender initiates a transfer directly with the sender bank. If the transfer is accepted by the sender bank, then the following steps 2 and 3 take place.
2. The sender bank sends the transfer instructions to the receiver bank.
3. The receiver's account is updated with the funds.
4. The sender bank sends an API request via the **Info** endpoint with subtype `pre_decided` to inform Feedzai about the existing transfer with the final decision if the transactions were accepted or rejected.

Not-honorâ

Not-honor is used when you take a different decision than the one recommended by Feedzai:

- Feedzai generates an 'approve' recommendation, and you decide to reject the transfer.
- Feedzai generates a 'decline' recommendation, and you decide to accept the transfer.

The following diagram shows the end-to-end process for the two examples described above:

The steps depicted in the previous image for the **first scenario** are the following:

1. The sender initiates a transfer directly with the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint. Feedzai generates a recommendation to accept the transfer.
3. The sender bank rejects the transaction and sends the information to the sender.
4. The sender bank sends an API request via the **Info** endpoint with subtype `not_honor` to inform Feedzai about the transfer's final decision.

In the second scenario, the transfer is accepted by the sender bank and follows the normal path towards updating the receiver account. At the end, sender bank sends an API request via the **Info** endpoint with subtype `not_honor` to inform Feedzai about the transfer's acceptance.

info

If the transfer is not originally sent to Feedzai, then pre-decided subtype needs to be used instead of the not-honor to communicate the transfer information to Feedzai.

Out of bandâ

You can define policies for advising downstream systems to contact an account holder (e.g. via sms or call) to validate a transfer. These policies can be defined, for example, based on the response provided by the Feedzai's model score or rules, another output parameter (e.g. an action triggered by a rule), or any external reason. Out of band allows you to notify Feedzai that the account holder has been contacted. If the response provided by the account holder leads to a decision about whether or not the transfer is legitimate, then this information is sent via the Feedback use case.

Consider the following example of a typical scenario: An account holder can be contacted and not answer synchronously. In this situation, this information is collected via info, subtype `out_of_band`. Once the bank has the account holder's confirmation, the information is sent via the Feedback endpoint.

The following diagram shows the end-to-end process:

The steps depicted in the previous image for the API integration at the **sender bank (top figure)** are the following:

1. The sender sends the transfer instructions to the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint. Feedzai generates a recommendation to decline the transfer.
3. The sender bank contacts the account holder to verify the transfer authenticity.
4. The account holder checks the information provided by the bank.

    info

    Transfer's confirmation: If the account holder confirms the transfer, the information is sent via the Feedback endpoint. Based on the answer, the transfer may or may not follow the normal path towards updating the receiver account.

5. The sender bank sends an API request via the **Info** endpoint with subtype `out_of_band` to inform Feedzai that the account holder has been contacted.

As for the API integration at the receiver bank, the steps follow the same logic as the sender bank integration; however, the receiver bank contacts the receiver to verify the transfer authenticity. After that, the receiver bank sends the API call to inform Feedzai that the account holder has been contacted.

Now consider a second scenario in which the integration is done via the sender bank and the transfer is rejected by the receiver bank:

In this situation, the logic of steps is the same as the previous scenario and you can use the **Info** endpoint with subtype `out_of_band` to inform that the account holder has been contacted.

Disputeâ

An account holder can contact a bank to file a complaint for different reasons (e.g. an unauthorized payment). In this case, the bank and the account holder may initiate a dispute (e.g. reversal, recall, and refund). The Info endpoint with subtype `dispute` can be used to inform Feedzai about the details of this process. If the process results in a refund, you can use the Feedback endpoint to inform Feedzai about refund-related reasons.

The following diagram shows the end-to-end process:

The steps depicted in the previous image are the following:

1. The sender sends the transfer instructions to the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint.
3. The sender bank sends the transfer instructions to the receiver bank.
4. The receiver's account is updated with the funds.
5. The sender contacts the sender bank to initiate a dispute.
6. The sender bank sends an API request via the **Info** endpoint with subtype `dispute` to inform Feedzai about this process.

info

Transfer's confirmation: Once the dispute is resolved, the final information needs to be sent via the Feedback endpoint.

Learn moreâ

Check the Settlement Endpoint page to continue learning about the endpoints' life cycles.

-
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- Event Life Cycles
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- Peer-to-Peer

On this page

Peer-to-Peer

Was this page helpful?Â

Learn about the Peer-to-Peer end-to-end life cycle to smoothly integrate with the transfers API.

Transaction initiationâ

The scenarios described in this section explain the transfer life cycle for direct peers. This page covers the end-to-end default scenario in which only the Transfer Initiation endpoint is called to process transfers.

info

Endpoints Integration: Check the Transaction Life Cycle page to learn more about the life cycles for transfer updates, settlements, and feedback actions, corresponding to requests to the Info, Settlement and Feedback endpoints, respectively.

Direct peersâ

The following diagram depicts typical use cases associated with this transfer model:

The steps depicted in the previous image for the top-up use case are the following:

1. Before the transfer, the sender does a top-up using any available channel - for example, mobile, internet banking - to ensure enough funds to perform the transfer.
2. The request is sent to the top-up bank.
3. The top-up bank transfers the funds to the sender.
4. The sender initiates the transfer using any available channel - for example, mobile, internet banking.
5. The receiver peer (e.g. mobile device) receives the funds.

6. The receiver is notified and may transfer the funds to a bank account.

The steps depicted in the previous image for the card/account charge use case are the following:

1. The sender initiates the transfer using any available channel - for example, mobile, internet banking.
2. After transfer initiation, the sender peer sends the transfer instruction to the sender bank to charge the sender's debit/credit card or bank account.
3. The receiver peer receives the transfer instructions.
4. The receiver bank receives the payout or settlement via card or bank account.
5. The receiver peer notifies the receiver.

info

Integration with the transfers API: The transfers API was designed to allow banks that operate at different points of the transfers life cycle to integrate with Feedzai. This flexible integration capability is shown in the previous figure with the API label.

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- Processing Model Details

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About Processing Model Details

Was this page helpful?Â

Learn how transfers are processed and understand how to integrate with the Feedzai's transfers API at the different points of the transfer life cycle. The pages in this chapter provide you a functional description for a successful integration with a focus on key models: Automated Clearing House (ACH), Correspondent Banking, Real-Time Gross Transfers and Peer-to-Peer.

Processing modelsâ

The transfers API was designed to allow integration at any point of a transfer life cycle and account for any processing model (i.e. model agnostic). Take the following models as examples:

ACHâ

ACH is an electronic clearing system in which payment orders are exchanged among financial institutions, primarily via telecommunication networks, and handled by a data processing center.

The Clearing House is a central location or central processing mechanism through which financial institutions agree to exchange payment instructions or other financial obligations (e.g. securities). The institutions settle for items exchanged at a designated time based on the rules and procedures of the clearing house. In some cases, the clearing house may assume significant counterparty, financial or risk management responsibilities for the clearing system.

The clearing is the process of transmitting, reconciling and, in some cases, confirming payment orders or security transfer instructions prior to settlement, possibly including the netting of instructions and the establishment of final positions for settlement.

This system is designed to work with payment batches, allowing you to process a large number of credit transfers at once. Examples of typical transfers include:

- Payroll transfers
- Social security benefits
- Tax refunds

Learn how you can integrate with Feedzai to protect transfers in the ACH model.

Correspondent Bankingâ

Correspondent Banking is an arrangement under which one bank (correspondent) holds deposits owned by other banks (respondents) and provides payment and other services to those respondent banks. Such arrangements may also be known as agency relationships in some domestic contexts. The key characteristic of this model is to ensure efficient reachability between correspondent banks, independently on the currency.

In international banking, balances held for a foreign respondent bank may be used to settle foreign exchange transactions. Reciprocal correspondent banking relationships may involve the use of so-called *nostro* and *vostro* accounts to settle foreign exchange transactions.

This model is typically used in:

- Overseas transfers.
- When the financial institutions involved in the transfer have different currencies.

Learn how you can integrate with Feedzai to protect transfers in the Correspondent Banking model.

Real-time Gross Transfersâ

Real-Time Gross Transfers is a model for real time settlement of funds, individually on an order by order basis (i.e. without netting). The "real-time" component means that the instructions are processed at the time they are received rather than at some later time. The "gross transfers" component means that the settlement funds corresponding to the transfers instruction occurs individually (on an instruction by instruction basis).

This model is typically used in:

- Settlement between financial institutions on the retail infrastructures in single transfers or batches (e.g. batch of transfers and clearing of card transactions of a correspondent financial institution). Support real-time transactions in two use cases:
  - Above a certain threshold, e.g. 100.000â‚¬ is mandatory in some countries to transfer via RTGSs for two main reasons:
    - Decrease systemic risk.
    - Provide visibility to the supervisor high volume transactions, e.g. AML activities.
  - Offer a real-time transfer option for consumers who have this need.

Examples of systems that operate under this model include Fedwire and TARGET 2.

info

Real-Time Gross Tansfers model: The transactions processed by this model are known as urgent transfers. This model is not used to process all transactions due to systemic reasons and the high costs associated with a very resilient and available system.

Learn how you can integrate with Feedzai to protect transfers in the Real-time Gross Transfers model.

Peer-to-Peerâ

Peer-to-Peer payments (P2P) is an online technology that allows customers to transfer funds from their online application connect to Bank Account, Wallet or credit card to another individualâs via the Internet or a mobile phone.

The following use cases are examples of Peer-to-Peer payments:

- Users establish secure accounts with a trusted third-party vendor, designating their bank account or credit card information to be used to transfer and accept funds. Using the third partyâs website or mobile application, individuals can complete the process of sending or receiving funds. Users are generally identified by their phone number or email address and can send funds to anyone who is a member of the network. Paypal is an example of this approach.

- Users access an online interface or mobile application (developed by their bank or financial institution) to choose the amount of funds to be transferred. The recipient is designated by their email address or phone number. Once the transfer has been initiated by the sender, the recipient then receives a notification to use the online interface to input his or her bank account information and routing number to accept the transfer of funds. In this method, recipients do not need to have an account with the financial institution of the sender in order to receive a money transfer.

Transfers timelineâ

There are two scenarios in transfers regarding timeline:

- **Real time:** Transfers that are processed in real time undergo an API behavior similar to card transactions. The Transfer Initiation endpoint should be used for any update (e.g. any failure is communicated via the pre-decided type of the Info endpoint).
- **Non-real time:** Transfers that take longer to process (i.e. in the order of hours). The Transfer Initiation endpoint is used to start the process.

Learn moreâ

Learn more about the transfer life cycle for the following models:

- ACH
- Correspondent Banking
- Real-time Gross Transfers
- Peer-to-Peer

Real-time Gross Transfers

Learn about the Real-time Gross Transfers end-to-end life cycle to smoothly integrate with the transfers API.

Transaction initiationâ

The scenarios described in this section explain the transfer life cycle for the **single counterparty**. This page covers the end-to-end default scenario in which only the Transfer Initiation endpoint is called to process transfers.

info

Endpoints Integration: Check the Transaction Life Cycle page to learn more about the life cycles for transfer updates, settlements, and feedback actions, corresponding to requests to the Info, Settlement and Feedback endpoints, respectively.

Single counterpartyâ

The life cycle involving a single counterparty applies to large financial institutions that communicate directly with a clearing/settlement mechanism. The following diagram depicts a typical flow:

The steps depicted in the previous image are the following:

1. The sender, also referred to as originator, initiates the transfer using any available channel - for example, mobile, internet banking, phone, bank branch, contact center, email, file transfer system.
2. The sender bank or the payment service provider (PSP) - a direct participant in this scenario - receives the transfer instructions and debits the sender's account. The sender bank sends the transfer instruction to a settlement platform.

3. The settlement platform processes the transfer instruction and sends it to the receiver bank.
4. The receiver bank - a direct participant in this scenario - receives the instructions to credit the receiver's account.
5. The receiver's account is updated with the funds.

info

Integration with the transfers API: The transfers API was designed to allow banks that operate at different points of the transfers life cycle to integrate with Feedzai. This flexible integration capability is shown in the previous figure with the API label.

Multiple counterpartiesâ

The scenario of multiple counterparties occurs when the sender and the receiver's bank are small-sized institutions. In this case, a third-party is required to establish the main communication (e.g. overseas connection) between the indirect participants.

info

Multiple counterparties: See the Multiple counterparties section under the Automated Clearing House page to learn about the role of indirect participants in the transfers life cycle.

Learn moreâ

Check the Peer-to-Peer page to learn how to integrate with other transfer methods.

-
- About TFB
- Event Life Cycles
- Outbound Transfers
- Settlement Endpoint

On this page

Settlement Endpoint

Was this page helpful?Â

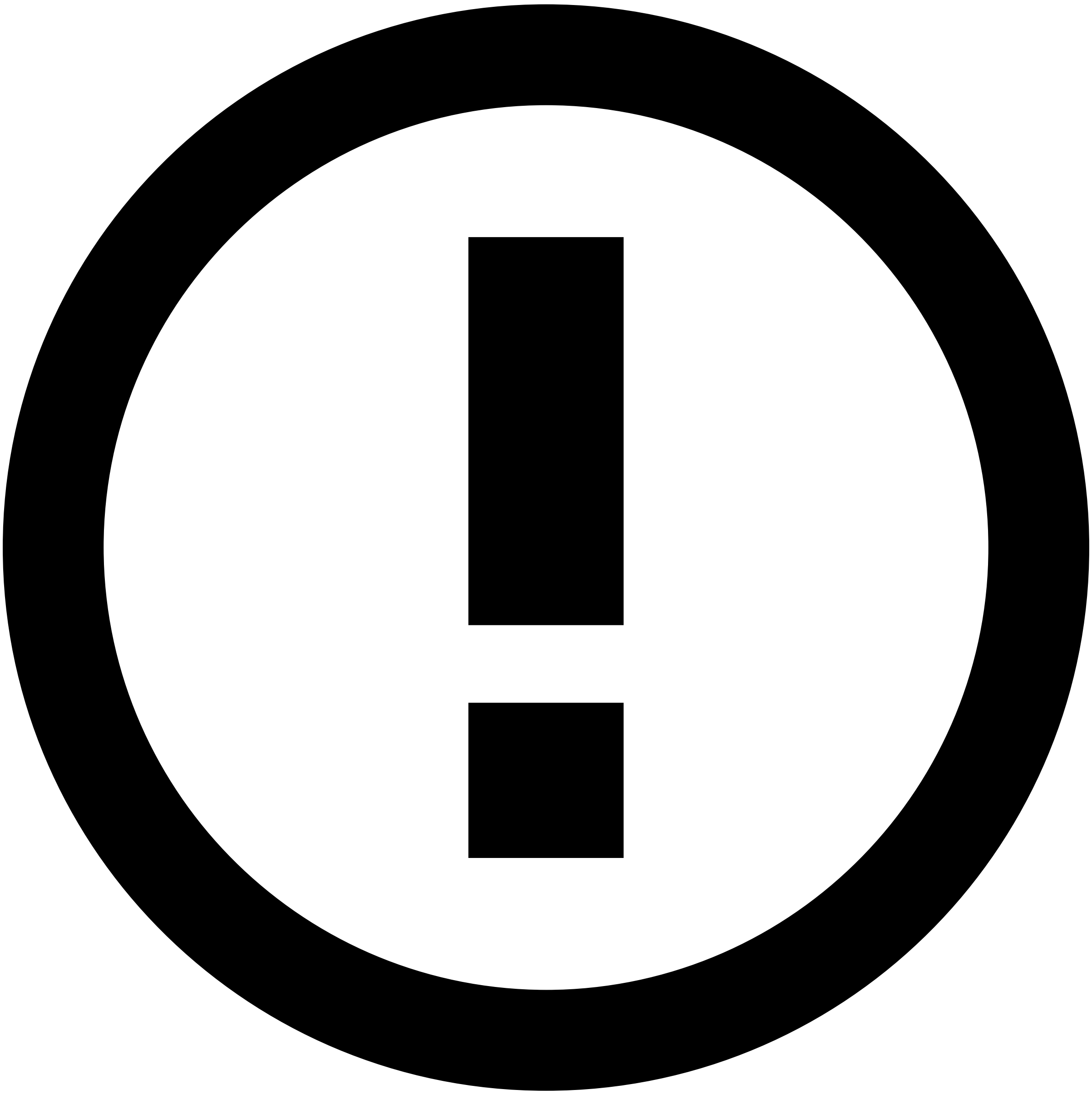
The Settlement endpoint allows you to inform Feedzai when a transaction is settled. Note that this endpoint is non-scorable, i.e. it does not flow through rules in order to generate a decision. However, the information is used by the Feedzai's risk engine platform to calculate metrics and profiles, and thus, impacts subsequent transfers associated with the same customer. It is also shown in the fraud analyst interface (Case Manager) as a new event to support the alert investigation.

Since the settlement of a transfer is strongly dependent on the processing model, this section presents the end-to-end life cycle for the most common processing models.

The following diagram shows the end-to-end process for the ACH model:

The steps depicted in the previous image for the integration at the **sender bank (top figure)** are the following:

1. The sender initiates a transfer directly with the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint.
3. The sender bank sends the transfer instructions corresponding to several senders to the clearing mechanism (e.g. ACH) via batch.
4. The clearing mechanism processes the transfers instructions, and reconciles them.
5. The ACH **accepts/acknowledges** to initiate the settlement.
6. The sender bank sends an API request via the **Settlement** endpoint to inform Feedzai about this process.



reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities (e.g. Card, Customer, or Account) have a match in the Reference Data Storage.

info

Note that any new data from transfer settlement events (e.g. new `sender_card_category`) will take precedence over what's in the Reference Data Storage. Read the [Reference Data Life Cycle] page to understand the logic behind reference data's entity-based enrichment.

1. The ACH sends the transfer instructions to the receiver bank.
2. The RGTS sends the settlement message to the receiver bank.

The steps for the API integration at the receiver bank follow the same logic as the sender bank end; however, in this case, the receiver bank sends the API calls with the transfer initiation and settlement.

The following diagram shows the end-to-end process for the Correspondent Banking and Peer-to-Peer models:

The steps depicted in the previous image for the integration at the **sender bank (top figure)** are the following:

1. The sender initiates a transfer directly with the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint.

3. The sender bank receives the transfer instructions and debits the sender's account. The sender bank sends the transfer instructions to the receiver bank.
4. The receiver bank receives the message from the sender bank, **accepts/acknowledges** and deducts a transaction fee. It also receives the instructions to credit the receiver's account.
5. Based on the **accept/acknowledge** message, the sender bank sends the settlement instructions to the receiver bank.
6. The receiver bank receives the settlement instructions.
7. The sender bank sends an API request via the **Settlement** endpoint to inform Feedzai about this process.
8. The receiver's account is updated with the funds.

For the integration with the receiver bank (example on the bottom figure), the transfer needs to be sent to Feedzai for scoring (see step 4) before the settlement take place (see step 5).

The following diagram shows the end-to-end process for the Real-time Gross Transfer model:

The steps depicted in the previous image for the integration at the **sender bank (top figure)** are the following:

1. The sender initiates a transfer directly with the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint.
3. The sender bank receives the transfer instructions and debits the sender's account. The sender bank sends the transfer instructions to the RGTS (i.e. settlement platform).
4. The RGTS **accepts/acknowledges** to initiate the settlement.
5. The sender bank sends an API request via the **Settlement** endpoint to inform Feedzai about this process.
6. The RGTS sends the transfer instructions to the receiver bank.
7. The RGTS sends the settlement message to the receiver bank.

The steps for the API integration at the receiver bank follow the same logic as the sender bank end; however, in this case, the receiver bank sends the API calls with the transfer initiation and settlement.

Learn moreâ

Check the Feedback Endpoint page to continue learning about the endpoints' life cycles.

Outbound Transfers

Was this page helpful?Â

Learn about the transaction life cycle and understand how the transfers API is used in the different stages of a transaction, e.g. transfer initiation, disputes, and settlements.

Transaction life cycleâ

The following diagram shows how the transfers API endpoints come into play in the course of a typical transaction life cycle:

tip

Step nuances: The steps illustrated in the diagram above and the endpoint evocations represent a scenario in which the API is used to score individual transactions. A different use case might undergo different steps. For example, in AML transaction monitoring, the identification of suspicious activity is performed by conducting a behavioral analysis of the customer activity. Since the transactions have been settled already, the first API call might be done directly via the settlement endpoint, replacing the transfer initiation request.

The diagram above represents an integration between the sender's bank and Feedzai. For inbound transfers at the receiver's end, see Inbound Payments' Life Cycle. The following steps describe the integration at the sender's bank-end as depicted in the diagram. Refer to the Transfer Initiation page to request a decision for an incoming transfer.

1. **Sender initiates transfer** : The sender, also referred to as **originator**, initiates the transfer using any available channel - for example, mobile, internet banking, phone, bank branch, contact center, email, file transfer system.

2. **Sender's bank invokes the Transfers API** : The sender's bank or the payment service provider (PSP) receives the transfer instructions and debits the sender's account, also referred to as **originator**. The sender's bank invokes the Transfer Initiation endpoint with the payload related to the given transfer. In some situations, the Info endpoint is also used at this point, for example, to communicate pre-decided transfers. Transfers sent via the Info endpoint are not scored.

   info

   For transactions that occur between customers from the same bank, two events should be sent to Feedzai, one via **Inbound Payments API**, and another one via **Outbound Transfers** API.

3. **Feedzai conducts real-time data analysis and scores** : The transfer is processed by the risk engine platform with rules and/or models. As a result, the risk engine generates a recommendation that the transfer is either **legitimate** or **suspicious**. The recommendation is configured at the risk engine level and may vary according to

the use case. It may also have intrinsic logic to respond to the type of transfers. For example, instant transfers may only result in recommendations that indicate legitimate or suspicious activity. On the other hand, transfers that are subject to longer processing times - credit transfers and direct debit - may generate a Pending for Review recommendation to indicate that the transaction is being investigated by a fraud analyst.

4. **Analysts perform alert manual revision** : Fraud analysts perform manual investigation on alerts raised by the risk engine. This step only applies to credit transfers and direct debit since these are bank payments subject to longer processing times. For instant transfers, the alert investigation is a post-mortem-like analysis because the payment is processed in real time.

info

Customer notification: When an analyst decides that a transfer is fraudulent or legitimate, Feedzai sends the information to the customer through a webhook notification. The client receives a `lifecyle_id` that they can use by invoking the Transfers Get endpoint to obtain more information about a specific event. Customers can benefit from this notification service by implementing mechanisms that act upon the analyst's decision. This notification is **particularly important for non-instant transfers**, where analysts can manually review the payment before deciding whether to include it or not in the batch of transfers. For example, the bank can leverage this information when creating the file that is sent to the clearing mechanism via batch (i.e. a bulk message with multiple transfer instructions). The bank can either include the transfer only when the analyst confirms the event as not fraud, or include all the transfers by default and then exclude the ones marked as fraudulent. Read more about webhooks.

5. **Sender's bank makes an automated decision based on Feedzai's recommendation** : The sender's system makes an automated decision based on Feedzai's recommendation and optionally notifies Feedzai about this decision via Info or Feedback endpoints. The bank sends the transfer instructions to the receiver's bank following the processing model that applies to the given transfer.

6. **Transfer is processed by the given payment model** : The transfer is processed by the processing model. Common models are **Automated Clearing House** (ACH), **Correspondent Banking**, and **Real-time Gross Transfer** (RTGT). Peer-to-peer, a model that is being rapidly adopted, is also covered by the Transfers Connector integration; however, the life cycle has to be adjusted to this processing model. Check Peer-to-Peer for more information.

tip

Processing model: This communication between the sender's bank, processing model, and receiver's bank contains extra details that are not included in the diagram for simplicity. Check the Processing Model Details to learn more about these details.

7. **Receiver's bank receives the transfer instructions** : The receiver's bank receives the instructions to credit the receiver's account.
8. **Receiver's account is updated with the funds** : The receiver's account is updated with the funds and the sender's bank informs Feedzai about the transfer settlement.
9. **Analysts perform alert manual revision** : At any point after the real-time event processing, a fraud analyst can investigate the event. The information is sent back to the risk engine and can be leveraged by rules and models.

Endpoint summaryâ

These endpoints allow users to send information at different stages of a transaction life cycle as follows:

- **Transfer Initiation:** Allows Feedzai's customers to leverage data science capabilities to score a transaction according to its risk.
- **Info:** Allows Feedzai's customers to update and add new information to an existing transaction, as well as to notify Feedzai about transactions processed upstream.
- **Get:** Allows the client to obtain the current payment status, including all the details of the event, i.e. the information sent through the endpoints during a transfer's life cycle.
- **Settlement:** Allows Feedzai's customers to inform Feedzai when a transaction is settled. Note that this endpoint is non-scorable, i.e. it does not flow through rules in order to generate a decision. However, the information is used by the Feedzai's risk engine platform to calculate metrics and profiles, and thus, impacts subsequent transfers associated with the same customer. It is also shown in the fraud analyst interface (Case Manager) as a new event to support the alert investigation.
- **Feedback:** Allows Feedzai's customers to label a transaction using pre-configured options that classify the transfer according to its risk. The labeling of an event is highly recommended since it gives feedback to Feedzai's platform enabling it to learn or consolidate illegal activity patterns. The x4xx messages refer to the dispute cases and reverse the action of the previous transfer initiation.

Integrating with transfer typesâ

The Feedzai API endpoints can be invoked at different times in the different transfer types (credit transfer, instant credit transfer or direct debit). Also, not all transfer types are supported by each processing model.

The following table shows the supported transfer types for each processing model:

Processing model	Instant Transfer	Direct Debit	Direct Deposit	Credit Transfer (cut-off time)	Internal Transfer	Bill Payment	Other transfers types
SEPA	Yes	Yes	N/A	Yes (1 business day)	Yes	N/A	N/A
BACS	N/A	Yes	N/A	Yes (3 working days)	Yes	N/A	N/A
CHAPS	N/A	N/A	N/A	Yes	Yes	N/A	N/A
WIRE	N/A	N/A	N/A	Yes (Only available on business days)	Yes	N/A	N/A
SWIFT	N/A	N/A	N/A	Yes	Yes	N/A	N/A
ACH	N/A	Yes	N/A	Yes (After a certain hour, a transfer wonât be processed until the next business day)	Yes	N/A	Direct deposits*
Faster Payments	Yes	N/A	N/A	Yes	Yes	N/A	Forward-dated Payments**, Standing Orders**
NPP	Yes	N/A	N/A	Yes (depends on the connection hub)	Yes	N/A	N/A
BPAY	N/A	N/A	N/A	N/A	N/A	Yes	N/A
PIX	Yes	N/A	N/A	Yes (depends on the connection hub)	Yes	N/A	N/A
BOLETO	N/A	N/A	N/A	N/A	N/A	Yes	E-Commerce
Zelle	Yes	N/A	N/A	Yes (depends on the connection hub)	Yes	N/A	N/A
EMT	Yes	N/A	N/A	Yes (depends on the connection hub)	Yes	N/A	N/A
RTP TCH	Yes	N/A	N/A	Yes (depends on the connection hub)	Yes	N/A	N/A

*Direct deposits are considered as normal credit transfers.

**Forward-dated Payments and Standing Orders are scheduled in the sender bank. After they are initiated, they are handled in the same way as credit transfers.

info

Batch files: One characteristic of the ACH model is to process transactions in batches. Note that, in this case, batch files need to be parsed and converted into individual transactions before sending them to Feedzai.

Endpoints and event typesâ

The available endpoints allow you to integrate with the Transfers Connector API at the key stages of the transfer life cycle. This ensures that all the information is used to detect and prevent suspicious activity efficiently. Since each endpoint is used to accomplish a different goal, the following table summarizes the key information for each of them:

Endpoint/ event type	Event subtype			Recommendation (approve/decline)	Ability to inform Feedzai about final	

		Mandatory (M), Recommended (R), Optional (O)	Is the event scored?		decision (accepted/ rejected)	Ability to label the transfer
Transfer Initiation	NA	M	Yes	Yes	No	No
Info	Additional information	O	No	No	Yes	No
	Pre-decided	R	No	No	Yes	No
	Not-honor	R	No	No	Yes	No
	Out of band (sms/call)	O	No	No	Yes	No
	Dispute	R	No	No	Yes	No
Settlement	NA	M	No	No	No	No
Feedback	Dispute	-Â M, for Info with subtype dispute - R, all other cases	No	No	No	Yes
	Out of band	Only used in combination with Info - out of band. In this case is mandatory	No	No	No	Yes

This table gives you the following information:

- **Mandatory (M), Recommended (R), Optional (O):** These options relate to the message request. Messages specified as recommended are used by the risk engine platform to improve the system's accuracy to monitor transfers.
- **Is the event scored?** An event can be scored by the data science models with a number ranging between 0-1000, where 1000 represents the highest probability of being illegitimate. The scoring is only generated if you have models configured in the risk engine platform.

info

Events being scored: The endpoints and event subtypes marked as scorable are the ones that have the `score` field in the API response. However, the actual event scoring depends on whether:

- The platform has a model configured in the runtime workflow.
- The filter conditions defined in the runtime workflow allow the event to reach the scoring operator.

- **Recommendation (approve/decline):** Generated by the risk engine rules with the possible values 'approve' or 'decline'.
- **Ability to inform Feedzai about final decision (accepted/rejected):** Final decision made by your system, i.e. transaction accepted or declined.
- **Ability to label the transfer:** The feedback_value field is populated with the value 'true' or 'false' representing an illegitimate or a legitimate transfer, respectively. This is used to improve the accuracy of the data science models, and to calculate metrics such as precision and false positive rate.

Learn moreâ

Learn how each endpoint is used along the transaction life cycle:

- Transfer Initiation
- Info
- Settlement
- Feedback

Additionally, as described above, the communication between the sender's bank, processing model, and the receiver's bank have extra details that are not covered in the overall life cycle diagram. Check the Processing Model Details page to understand how Feedzai covers each processing model: Automated Clearing House (ACH), Correspondent Banking, Real-Time Gross Transfers, and Peer-to-Peer.

- About TFB
- Event Life Cycles
- Outbound Transfers
- Transfer Initiation Endpoint

On this page

Transfer Initiation Endpoint

Was this page helpful?Â

The Transfer Initiation endpoint allows you to leverage Feedzai's data science capabilities to generate a recommendation (e.g. approve/decline) according to the transfer's risk. The recommendation is configured at the risk engine level and may vary with the use case.

This page shows the end-to-end process. Note that the examples described in this page apply to a simplified processing model. To learn more about processing models, check the following pages which cover the transfer initiation use case with more details on the end-to-end life cycle:

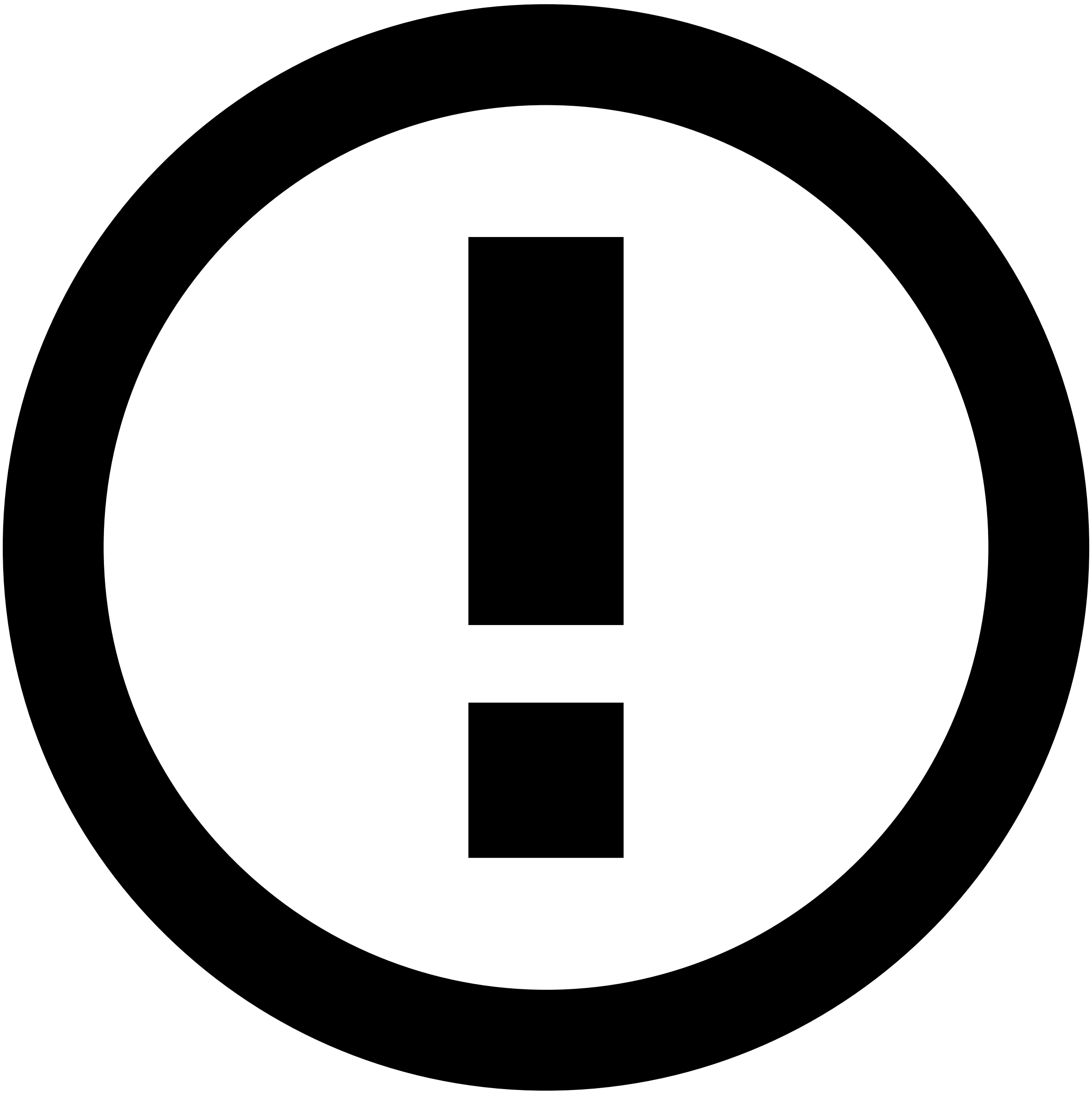
- Automated Clearing House

- [Corresponding Model]
- [Real-time Gross Transfers]
- [Peer-to-Peer]

Funds transfer initiationâ

The following image illustrates the integration between the sender bank and Feedzai.

1. The sender (i.e. originator) initiates a transfer directly with the sender bank.
2. The sender bank sends an API request via the **Transfer Initiation** endpoint.



reference data enrichment

The event may become enriched with data fetched from the Reference Data Storage if any of the event's entities (e.g. Card, Customer, or Account) have a match in the Reference Data Storage.

info

Note that any new data from transfer initiation events (e.g. new `sender_card_category`) will take precedence over what's in the Reference Data Storage. Read the Reference Data Life Cycle page to understand the logic behind reference data's entity-based enrichment.

1. Feedzai sends the recommendation, i.e marks the event as legitimate or fraudulent. If the event is marked as legitimate, the instructions go to the Clearing House.
2. After the transfer is approved by Feedzai and accepted by the sender bank, the latter sends the transfer instructions to the receiver bank.
3. The receiver's account is updated with the funds.

Funds transfer confirmationâ

The following image illustrates the integration between the receiver bank and Feedzai.

1. The sender (i.e. originator) initiates a transfer directly with the sender bank.
2. The receiver (i.e. beneficiary) bank receives the instructions to credit the receiver's account.

3. The receiver bank sends an API request via the **Transfer Initiation** endpoint.
4. Feedzai sends the recommendation, i.e marks the event as legitimate or fraudulent. If the event is marked as legitimate, the bank can forward the transfer instructions to the processing entity (e.g Clearing House).
5. After the transfer is approved by Feedzai and accepted by the receiver bank, the receiver's account is updated with the funds.

Note that, if you don't require a recommendation but still need the incoming transfers to contribute to metrics, you can invoke Feedzai's **Info** endpoint as follows:

1. The sender (i.e. originator) initiates a transfer directly with the sender bank.
2. The receiver (i.e. beneficiary) bank receives the instructions to credit the receiver's account.
3. The receiver's bank invokes Feedzai's **Info** endpoint.
4. Feedzai processes the transaction, which contributes to segment-of-one-profiles. If the event is marked as legitimate, the bank can forward the transfer instructions to the processing entity (e.g the Clearing House).
5. The receiver's account is updated with the funds.

tip

At any point after the real-time event processing, an analyst can investigate the event. The information is sent back to the risk engine and can be leveraged by rules and models.

Learn moreâ

Check the Info Endpoint page to continue learning about the endpoints' life cycles.

-

- Operational Work
- Fraud Prevention Manager
- Manage Workload and Rules
- Operationalize Fraud Teams
- Configure Automation Rules

On this page

Configure Automation Rules

Was this page helpful?Â

Automation rules are used to automatically handle the distribution of workload in Case Manager. This page provides you with examples of automation rules that you can use in Case Manager.

tip

Out-of-the-box settings: Note that Case Manager offers default configurations for automation rules. Check them here.

Create automation rules in Case Managerâ

To create automation rules in Case Manager, execute the following steps:

1. In the general menu, hover over the **Settings** icon.
2. Select the **Automation Rules** option.

3. Use the button to open the rule creation panel.
4. Start by adding the rules's basic information:
    - **Name**: Rule's name. For example, "Set new alerts as unreviewed".
    - **Description**: Explanation of the rule's purpose. For example, "Set all incoming alerts as unreviewed".
    - **Channel**: Defines that only events from the selected channel are considered in the rule's conditions.
    - **Type**: Defines if the rule should be triggered based on **Events** or **Cases**.
5. Specify the rule's conditions and outputs according to your needs. In this example, the conditions would be the following:

6. Select **Enabled** to immediately activate the rule after it's created.
7. **Create** the rule.

Enable webhooksâ

Webhooks can be enabled using Automation Rules via the **Call Webhook** action as the automation's output. With this capability, Case Manager is able to send HTTP requests to external endpoints.

Consider the following example for a card webhook automation rule:

1. **Input**: Set up an automation rule to trigger when an event arrives.
2. **Conditions**: When a payment is scored by Feedzai, each rule triggered by Pulse returns an action, which is a string specified in the rule definition within Pulse (e.g. **[Velocity-Cards-1]**). You can use the **Triggered Actions** condition to verify which rules have triggered for this event. Additionally, you can include another condition to check that the System Decision for this event is "decline." You can add as many conditions as necessary.
3. **Outputs**: The **Call Webhook** automation action allows you to configure the URL where the webhook will be sent. In the dropdown menu, select the option corresponding to the channel you're creating the automation for. For this example, the **Cards Webhook Transformation** would be selected to ensure the Cards template is used in the payload of the webhook.

Check some example payloads from each available channel for more information on the possible fields and their corresponding values.

Webhook retry policy and authentication mechanism

The following default values are applied to new webhook connections:

- **Retry policy**:
  - `max_attempts` : `8` . This refers to the maximum number of attempts that failed requests will be retried.
  - `max_delay_millis` : `45000` (45 seconds). This refers to the maximum delay between retries.
  - `request_timeout_seconds` : `10` . This refers to the maximum time for a webhook request attempt. When the attempt takes longer than the time defined on this property, there is a timeout.
- **Authentication mechanism**: Via Bearer authentication token.

These defaults and other configurations can be adjusted on a need basis. Contact Feedzai for further configurations.

HTTP requestsâ

The webhook request has the following attributes:

- **Method**: POST

- **Headers**:
    - Send an Authorization header if authentication is configured.
    - Send an X-Request-Id header to uniquely identify the requests.

Payloadâ

The payload of the HTTP request includes the details of the event in JSON format.

Consider the following payload examples for each available channel:

- Cards
- Outbound Transfers
- Ad-hoc Customer Investigations
- Default


```json
{
    "lifecycle_id": "0d1c1529-a34e-3542-917a-07ca56815037",
    "event_external_id": "event_external_id",
    "event_timestamp": 1705498953008,
    "action_timestamp": 1705502592467,
    "risk_assessment": {
        "alert": true,
        "score": 0,
        "decision": "approve",
        "sca_required": true
    },
    "operational_process": {
        "operational_status": "closed",
        "status": "genuine",
        "reasons": {
            "operational_status": [],
            "status": []
            }
        },
    "analyst_assessment": {
        "status_cards": "fraud_cards",
        "reasons": {
            "status_cards": [
                "Confirmed Fraud",
                "Authorization Dispute"
            ]
        }
    }
}
```

info

The payload includes the `lifecycle_id`, a unique identifier for the payment, which you can use to retrieve the remaining fields from your database. You can also call the GET endpoint from Feedzai, but notice that this action may increase the volume of events and impact performance.

For details on the different field types, refer to the Cards API documentation.

Create automatic casesâ

This section explains how to configure cases using an example that implements the following logic:

- Creates a case automatically when an alert arrives to Case Manager.
- Adds to the case all the subsequent events/alerts that have the same card PAN as the one that opened the case.
- Associates the case to a queue.

To achieve this, you need to create two automation rules:

Case automatic openedâ

This rule creates a case when a new alert arrives at the workflow. If there is already a case with an alert from the same card PAN, then this rule adds the alert to the existing case.

- **Rule name:** Case Automatic Opened
- **Description field:** Open case automatically when an alert arrives at Case Manager and associate the case to a queue.

The following image show you how to configure the remaining fields:

Link non-alerted transactionsâ

The automation rule described in the Case Automatic Opened section creates cases when new alerts arrive at Case Manager, and links alerts with the same card PAN to existing cases. However, if you want to link events to an existing case based on a different rule condition, you need to:

1. Create a new automation rule.
2. Define the new condition.
3. Select the 'Link to existing Case' output.

For example, imagine that you also want to link future events (that are not alerts) with the same card PAN to existing cases. To achieve this, you can create a new automation rule as described below:

- **Rule name:** Link non-alerted transactions
- **Description field:** Link non-alerted transactions to an existing automatic case that contains at least one alert.

Next stepsâ

In the next step of this user journey you will learn how to configure SLAs to automate the organization of events into queues based on a time period.

- Operational Work
- Fraud Prevention Manager
- Manage Workload and Rules
- Operationalize Fraud Teams
- Configure Queues

On this page

Configure Queues

Was this page helpful?Â

Start configuring your standard setup with queues that reflect different priorities to organize events/alerts, cases, and to assign team members. Check the Access Workload page to see how fraud analysts use queues to quickly access their work.

tip

Out-of-the-box settings: Note that Case Manager offers default configurations for queues. Check them here.

Create queues in Case Managerâ

To create queues in Case Manager, execute the following steps:

1. In the general menu, hover over the **Settings** icon.
2. Select the **Queues** option.

3. Use the button to create a queue.

Queue examplesâ

The following table provides you with examples of queues that you can use in Case Manager.

Queue Name	Description	Type
Event - High Priority	Event without analysis for more than 2 hours	Event
Automatically Opened Case	Automatically opened cases with alerted transactions	Case
Manual Case - Approved	Manual case for suspicious approved transactions	Case
Manual Case - Declined	Manual case for suspicious declined transactions	Case

tip

Manual cases can be used to manually organize cases in queues.

The field options available in Case Manager to configure queues are:

- **Name:** Queue name.
- **Description:** Brief description of a queue's purpose (e.g. "All alerts in this queue failed to be reviewed in the 24-hour window and their SLA status is 'Breached'").
- **Type:** Lets you make the queue specific for either **Events** or **Cases**.
- **Users:** Lets you make the queue visible only for certain users. For the standard setup, keep the default option.

Next stepsâ

Now that you know how to configure queues, learn how to use them with automation rules.

Configure SLAs

Service Level Agreements (SLAs) allow you to automate the organization of events into queues based on a time period. This page helps you create SLAs in Case Manager.

tip

Out-of-the-box settings: Note that Case Manager offers default configurations for SLAs. Check them here.

Create SLAs in Case Managerâ

To create SLAs in Case Manager, execute the following steps:

1. In the general menu, hover over the **Settings** icon.
2. Select the **SLAs** option.

3. Use the button to open the SLA creation panel.

The next section shows you an example of an SLA that you can add to Case Manager.

Prioritize events without analysisâ

This SLA adds events to a queue when they are not analyzed within a certain period.

- **SLA name:** Event - High Priority
- **Description field:** Events without analysis for more than 2 hours are sent to a queue with higher priority.

The following image shows you how to configure the remaining fields:

Learn moreâ

The configuration of SLAs is the last step of the user journey to operationalize fraud teams. However, you may need to manually add events/alerts or cases to queues, or to manually assign fraud analysts to events/alerts or cases. You can learn more about this in Organize Workload Manually.

On this page

Out-of-the-Box Settings

Was this page helpful?Â

The out-of-the-box settings include default configurations for queues, automation rules, and SLAs. These settings are defined to:

- Offer an automated transition between the various Case Manager's statuses.
- Automatically organize alerts in queues with different priorities to help the analyst investigation.

Queuesâ

The out-of-the-box queues are configured according to the channel (i.e. transfers, cards and digital activity):

Breach queues are used in conjunction with SLAs. These settings allow analysts to have a central place to look into alerts at risk of breaching.

Card alert, transfer alert and digital activity alert queues contain alerts that are investigated as a post-mortem analysis, i.e. these alerts are associated with events that have been already declined or accepted by the financial institution, but their analysis is important to:

- Provide feedback to the system on whether the event was correctly accepted or rejected.
- Prevent future card payments when the analyst identifies that a card is being used for fraudulent purposes.

SLAsâ

The out-of-the-box SLAs are the following:

These SLAs account for:

- Transfers that are at risk of breaching.
- Alerts (card, transfer and digital activity events) that have been assigned to an analyst but the review process is taking longer than expected, thus preventing the investigation from being forgotten.

Automation rulesâ

The out-of-the-box automation rules are the following:

These automation rules are used in conjunction with SLAs and queues, ensuring the automated flow of the Case Manager statuses. They are configured for event arrival, event assignee changes, and event status changes.

Learn Moreâ

If you want to build on the top of the out-of-the-box settings by creating queues, SLAS, and automation rules, check the About Operationalizing Fraud Teams page.

-

- [Operational Work]
- [Fraud Prevention Manager]
- [Manage Workload and Rules]
- Operationalize Fraud Teams

On this page

Operationalize Fraud Teams

Was this page helpful?Â

Learn how to automate the workload management in Case Manager, helping fraud teams to review alerts efficiently.

Operational strategyâ

Different events might need to be reviewed with different priorities and might require analysts with a specific skill set depending on the complexity of the situation. The high flux of events constantly reaching Case Manager make it virtually impossible to manually distribute events by priority and fraud analyst in an prompt and efficient way. Therefore, Case Manager has various tools that allow you to automate the way events are organized to build useful workloads.

The strategy to automate workload management may vary with the business specificity. However, you can find a list of recommendations below:

- Organize queues with different priorities to assign to the team members.
- Add events to queues automatically.
- Inside queues, assign events to team members automatically.
- Configure Service Level Agreements (SLAs) to add events to queues based on a time period.

These recommendations can be implemented in Case Manager using a combination of queues, automation rules, and SLAs.

Queues, automation rules and SLAsâ

To configure Case Manager autonomously, it is important to understand the following concepts:

Queues: Bundles of cases and events, which may or may not include alerts, grouped according to a business context to make workload access and distribution simpler (learn more about events, alerts, and cases). As an example, you can create a queue to group all the alerts that added a card ID to a cards PosNeg list, in Pulse. in Pulse. Fraud analysts can then use queues in the alerts and cases pages as filters to quickly find their workload.

Events and cases are associated to queues either manually or through automation processes. The automation processes are configured using a combination of queues and automation rules, and queues and SLAs. The following figure shows how queues can be used:

Automation Rules: Perform automatic tasks using a when-if-then structure, i.e. when something occurs, if certain conditions apply to that event, then an action is taken. You can use automation rules to add events to a queue automatically. For example, when an event arrives at Case Manager, if a transaction is declined and the card is on the PosNeg listed cards with the value declined (i.e. the card is blacklisted), then add the event to a specific queue.

SLAs (Service Level Agreements): Work similarly to automation rules, but follow an if-then structure. The SLAs are generally used to create automation that requires a condition with a time period. For example, if an event has not been reviewed for over two hours, then add the event to a queue.

User journeyâ

The following user journey shows the different steps to fully configure Case Manager with the operational strategy described above. Each step is further described in separate pages using practical examples to ensure that, after completing this journey, you have a standard setup in place to help the daily work of fraud teams.

Next stepsâ

If you don't already have the necessary queues in place to configure automation rules and SLAs, start with the queues Queues step to configure part of your standard setup using practical examples. These queues will be used in the subsequent steps of the user journey.

If you already have queues in place that you want to use to configure automation rules and SLAs, proceed to the next steps:

- Automation Rules: Automate the organization of events in Case Manager, including:
- Add events to a queue - e.g. add an alert to a queue every time that the transaction is declined and card ID is added to a PosNeg list.
- Assign users to queues - e.g. assign an event that has been declined to a certain user.
- Create cases - e.g. create a case every time that an alert with different card ID arrives at Case Manager.
- SLAs: Automate the organization of events into queues based on a time period (e.g. add to queue all the events that are without analysis for over two hours).

Analyze the AutoML Output

Was this page helpful?Â

The AutoML output analysis allows gives you an insight on the end results, i.e. if they have fulfilled your business needs or if you should continue with data science iterations. This page walks you through an example of an AutoML run output analysis.

You can find the output of an AutoML run in the **Data Science** tab on the following panels:

- Data sources: Includes the 'Feature Generation Results' data source.
- Plans: Includes the plan that has the definition of the new features.

- Model Projects: Includes the recommended model.

How to analyze an AutoML runâ

An AutoML run generates various resources including data sources, plans and model projects. However, if you want to quickly assess the quality of the results, focus on the model project output. More specifically, use the following available options:

- **ROC curve:** Check how to analyze the performance of the recommended model in the Model evaluations section.
- **Feature Importance:** Identify the subset of features that contribute the most to the recommended model in the Model evaluations section.
- **Hyperparameter results:** See the combination of different parameters used to train various models in the Hyperparameter tunings section.

AutoML outputâ

This section provides a description for the output of each resource generated by AutoML. Note that, as mentioned in the How to analyze an AutoML run section, the most relevant results that define the quality of the AutoML run are described in the Model projects section.

Data sourcesâ

When executing an AutoML run, Pulse processes the selected input data source and creates new features. Along with this process, Pulse generates and makes the following data sources available:

- **Feature Generation Results:** Contains the data enriched with the new features.
- **Train dataset / Validation dataset / Test dataset:** Individual data sources that contain data corresponding to training, validation and test time ranges, as defined during the AutoML setup.

You can use the generated data sources to quickly explore the results:

1. Open the **Data Science** tab and select a data source (e.g. 'Feature Generation Results').
2. Select the button to open the **Data Source Analysis** menu:
3. Select the **Field available for breakdown**, i.e. a field corresponding to the fraud label.
4. Select the **Analyse** button to compute the statistics.

tip

Breakdowns: The breakdown option allows you to select a field to later use as a filter for your analysis. Consider that you select the 'fraud label' field and compute statistics for the 'fraud' and 'not fraud' breakdown values. Once the analysis job is computed, you can open the results of each feature and view the data associated to the 'fraud' or the 'not fraud' labels individually.

5. Once the statistics are computed, you can use the **View Statistics** button to view the results for each feature. You can see the data distribution for numerical and categorical values. You can use this functionality to perform quick validations - for example, to confirm that the metrics related to sum of amounts are not negative.

Plansâ

During an AutoML run, Pulse generates the following plans:

- **Datasource Split: Non-deployable plan** that defines which part of the input data source is used for training, validation and, optionally, testing the data science models. You can open the plan to see the time ranges that were selected in the setup of the AutoML run.
- **Feature Generation: Deployable plan** that contains all the fields that are generated during the AutoML run. The operators inside this plan are described in the Feature Generation plan section.
- **Feature Generation Reduced: Deployable plan** that contains only the relevant features selected by the AutoML during the feature selection step.

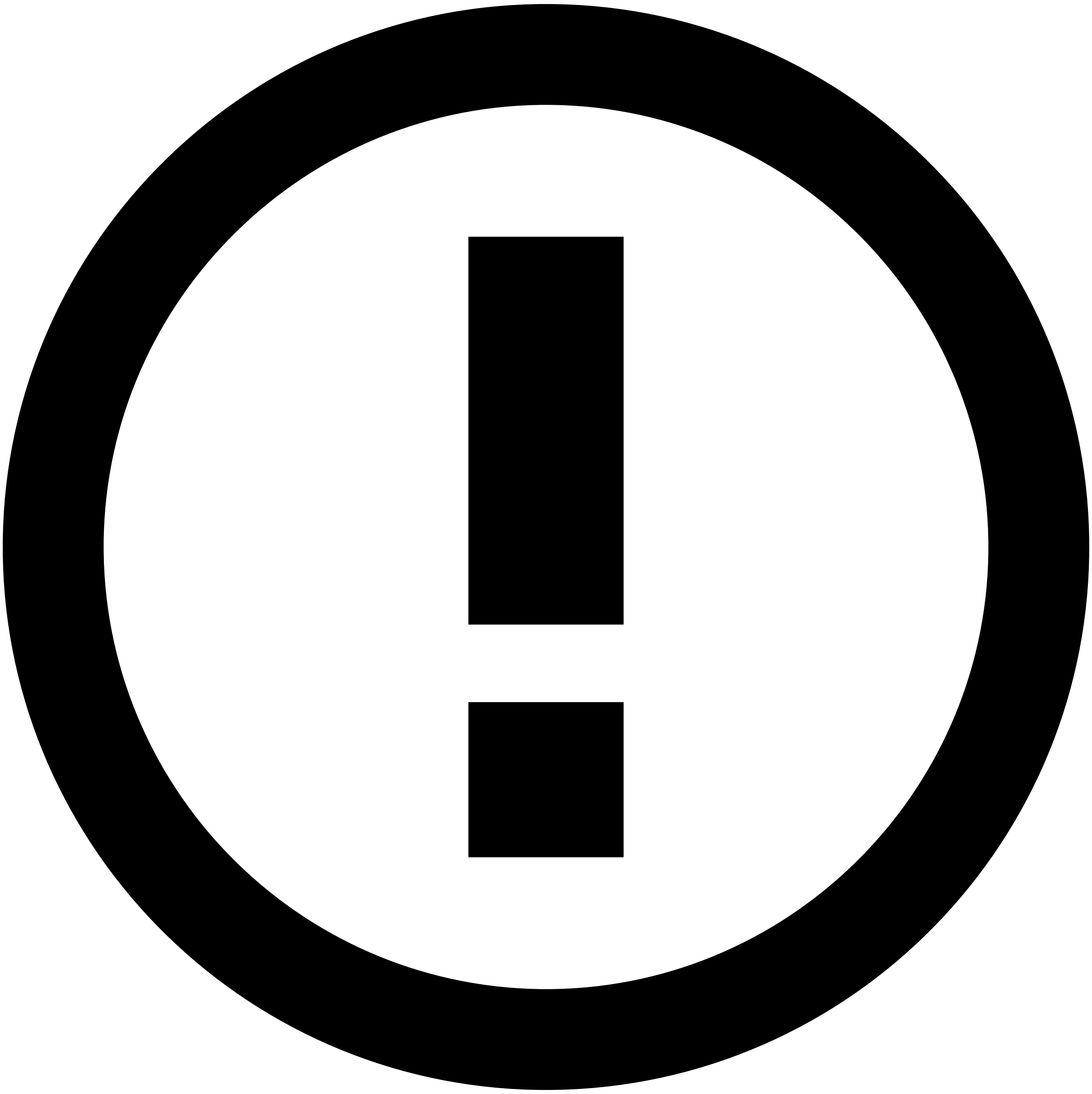
info

Ready for deployment: Both feature generation and reduced plans can be directly integrated in the runtime workflow:

- Use the feature generation plan if you want to get the most of all the features generated by AutoML.
- Use the feature generation reduced plan if you want to integrate it with the model recommended by AutoML.

Feature Generation planâ

The following image shows an example of a feature generation plan created by an AutoML run:



info

Operators and their features definition: The number of operators - and hence number and quality of the features - depends significantly on how the fields are tagged in the semantic file. Tagging the input data fields incorrectly may not produce meaningful features (or features at all). Note that the quality of models depends on the features, and consequently on model results, including ROC curve and metrics, depend heavily on the tagging quality.

The map and metric operators in this plan are explained as follows:

- **Feature 1:** Contains basic data transformations and data normalizations such as:
    - Extracting the month day from a given field, e.g. the month of the account open date.
    - Calculating the time difference between the `event_occurred_at` and the `account_open_date` fields.

- **Profiles:** Contains the field definition that represents the behavior of a given entity, such as card or merchant, in a specific type frame (i.e. also known as segments-of-one profiles). Examples include:
    - Average amount spent by a given card in the last month.
    - Number of transactions in the last month.
- **Feature 2:** Contains calculations performed over the fields defined in the profiles operator, such as:
    - Ratio between two fields, e.g. a certain count per hour divided by the count per day.

Remember that this output is used as an example. The number and type of operators depend on semantic tags and on the input data source, and therefore, your plan may have different operators.

Model projectsâ

For each run, AutoML stores all the information related to models in a new model project.

info

Ready for deployment: AutoML recommends the best model, which can be directly integrated in the runtime workflow.

The following image shows an example of a model project for an AutoML run:

You can quickly assess the performance of the trained models with:

- [Model Evaluations]: Check the ROC curve and Feature Importance results.
- [Hyperparameter Tunings]: Check the results of testing different sets of parameters.

Model evaluationsâ

Inside the model project, open the model evaluation generated by AutoML to check the ROC curves, metrics and Feature Importance:

1. Navigate to the **Data Science** tab and open the model project generated by AutoML.
2. Under the **Model Evaluations** table, select the one that you want to analyze.
3. Check the following subsections to learn how to analyze ROC curves, metrics and Feature Importance.

ROC curveâ

The ROC curve shows the recall (percentage of events that are fraud in the real world, and that were classified as fraud by the risk engine) for a given false-positive rate (FRP - percentage of events that are not fraud in the real world, and that were classified as fraud by the risk engine). An example of a good metric indicator is a recall above 70% at an FPR of 1%. You can use the range options to easily check this value by narrowing down the axis range:

Metricsâ

You can also see a list of metrics associated with the recommended model. You can use the **Threshold** bar to visualize the metrics at a certain value, e.g. at a FPR of 1%:

Feature Importanceâ

To access the Feature Importance results, you may have to compute it first. Inside the model evaluation, use the button to run the job. Once the job is finished, a new tab is displayed in the **Model Evaluation** panel with a list of features that impact the model the most:

Hyperparameter tuningsâ

An AutoML run trains several models using a hyperparameter optimization method:

1. Navigate to the **Data Science** tab and open the model project generated by AutoML.
2. Select the tuning result under the **Hyperparameter Tunings** table.
3. Select the **Exploration Mode** tab. The following figure shows an example of the hyperparameter tuning result at a 1.0% FPR:

Next stepâ

Read the Integrate Resources in Runtime to start scoring transactions in real time.

-
- Operational Work
- Data Scientist & Fraud Strategist
- Use Models
- AutoML
- AutoML Stages

On this page

AutoML Stages

Was this page helpful?Â

You can use AutoML to create features and train models in an automated way. AutoML evaluates the models and recommends the most performant one. To accomplish this, the risk engine platform (Pulse) performs a number of operations (i.e. jobs) described in the following sections.

AutoML processâ

AutoML relies on a smart engine that generates features automatically based on prior data science knowledge. In addition to generating features, the AutoML also performs the following tasks:

1. Uses the most important features to train multiple models.
2. Recommends the most suitable model for the use case. You can then integrate this model into the Pulse workflow to score transactions in real time.

To give you a more granular view about what happens, the following figure summarizes the key stages of AutoML:

To run AutoML, you need to provide:

- A data source with historical data.
- A semantic file that maps the input data fields to semantic tags.

danger

Environment to run AutoML: Since the AutoML computation is quite resource-intensive for the system, Feedzai recommends running AutoML in a staging environment dedicated to the data science work (i.e. typically a Data Science environment). This is important to ensure that the production environment is not compromised by this functionality. Once you achieve results that meet your business needs, you can [migrate](#) the AutoML model and plan to the production environment and [integrate these new resources into the runtime workflow](#).

Semantic fileâ

To give meaning to input data fields in an AutoML run, you need to have a semantic file that maps the data fields of the input data source to semantic tags. The semantic tags translate the meaning of the fields, thus providing the only portion of human knowledge that is required to power AutoML.

Download the standard semantic files that apply to your use case.

info

Importance of the semantic tags: The number and quality of the features generated by AutoML depend significantly on how the fields are tagged in the semantic file. Tagging the input data fields incorrectly may not produce meaningful features (or features at all). Note that the quality of models depends on features, and consequently on model results (e.g. the ROC curve and metrics depend heavily on the tagging quality).

The following example is an excerpt of a semantic file:

```
[
  {
```

```
    "@type": "simple",
    "tags": [
      "amount"
    ],
    "fieldDesc": "amount"
  },
  {
    "@type": "simple",
    "tags": [
      "grouping-entity",
      "entity"
    ],
    "fieldDesc": "acceptor_id"
  },
  {
    "@type": "simple",
    "tags": [
      "entity",
      "country",
      "location"
    ],
    "fieldDesc": "acceptor_country_code"
  }
  ...
```

The `fieldDesc` corresponds to the tagged field, and the `tag` corresponds to the semantic meaning given to that field.

Each stage of the AutoML process is further explained in the following sections.

Feature engineering

Feature engineering is the process of applying domain knowledge to derive new features from existing data. These features in Pulse are represented by maps and metric operators in a plan. During the AutoML computation, Pulse generates a plan with these operators.

AutoML has a list of data field transformations defined according to Feedzai's experience in the fraud-fighting domain. Those transformations are based on the semantics of the data fields, combining fields with specific meaning into features. For example, if a data source has two location data fields, it is useful to calculate the distance between the two. This means that the AutoML has a transformation that computes the distance between two location fields.

To allow AutoML to apply those transformations, when defining an AutoML run you need to provide a semantic file that maps the input data fields to semantic tags. For example, the `customer_name` field can be tagged with the 'name' tag. This way, AutoML knows the meaning of this input field, and it can compute the number of words in that field, which can be useful to fight fraud.

Without tagging the input data fields in a semantic file, AutoML cannot generate features. The quality of the models generated by AutoML relies heavily on the quality of tagging.

In summary, AutoML follows these steps to create features:

1. Looks at the semantic file and maps the data fields to their meaning.

2. Based on the data fields' meaning and the list of transformations, AutoML creates a plan containing map and metric operators (see Features Plan in the AutoML figure).
3. Executes the plan on the input data source, and saves the result in a features data source that contains the data enriched with the new data fields.

Dataset splitâ

When setting up an AutoML run, you must define which subsets of the input data source will be used for training and validation. Optionally, you can also choose to have a test dataset to simulate a blind evaluation of the best model found.

AutoML creates the training, validation and the optional test data sources based on the defined time ranges (see Split Plan in the AutoML figure).

info

Planning training, validation and test data sources: To avoid under- or overfitting, plan the time ranges carefully. Ensure that the resulting datasets contain enough data for their purpose. A typical dataset split is 50% for training, 30% for validation and 20% for evaluation, as long as the data source size is sufficient for that. The time windows need to be in the past and follow the order training-validation-evaluation.

If you don't define a test dataset, AutoML uses the validation dataset for the final evaluation of the best model found. Since the validation dataset was used for finding the best model, it is not an optimal solution to use it to test the data. Therefore, you should use a test dataset.

Feature selectionâ

In this step, AutoML evaluates which features will better contribute to the model performance. Pulse uses Gradient Boosted Trees to evaluate the features over the training dataset, and ranks the features accordingly.

Then Pulse reduces the features to a subset that has the most impact in performance, keeping only a subset of the features for the model training step. The features that AutoML keeps contribute with more than 90% of the importance rating.

Model training and evaluationâ

After selecting the most important features, AutoML creates a hyperparameter tuning job to train and evaluate models. Hyperparameter tuning is the process of fine-tuning parameters to obtain the optimal set of parameters for a specific use case. The risk engine generates the performance statistics for a number of different algorithms, testing multiple parameter combinations. This is done based on a grid search that iterates over multiple parameters, a method also known as parameter sweep. Hyperparameter tuning jobs distribute the model training and evaluation in the Spark cluster.

Finally, AutoML compares the models based on the metric that you selected to be optimized, and recommends the best model.

Next stepâ

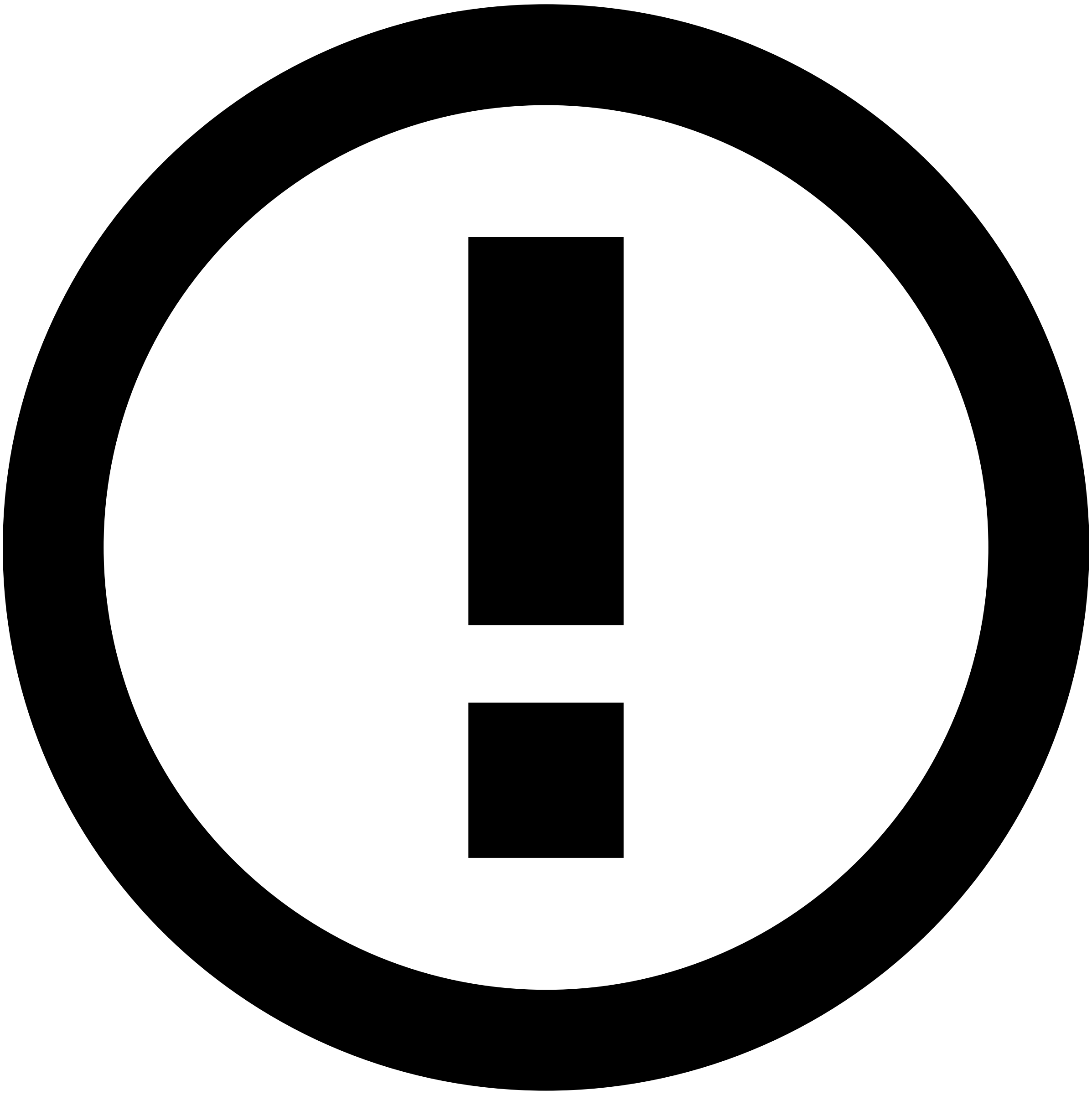
Now that you know how AutoML works, learn how to load and prepare the data that will be then used to run AutoML.

Complete the Schema

This page helps you complete the schema of the data source that will be used to set up AutoML. This step is important to prevent schema conflicts during the integration of the AutoML output (i.e. plan and model) with the existing resources in the runtime workflow.

danger

Environment to run AutoML: Since the AutoML computation is quite resource-intensive for the system, Feedzai recommends running AutoML in a staging environment dedicated to the data science work (i.e. typically a Data Science environment). This is important to ensure that the production environment is not compromised by this functionality. Once you achieve results that meet your business needs, you can migrate the AutoML model and plan to the production environment and integrate these new resources into the runtime workflow.



info

The applicability of this step depends on your product setup. If you already have a data source loaded in your Pulse application that has a schema compatible with the one used by the rules, you can skip this step entirely. In that case, check the [Create an AutoML Run] page to complete the next user journey step.

Whyâ

The goal of running AutoML is to generate a model to integrate into the runtime workflow. To perform this integration, you need to add the AutoML plan and model inline with other operators that are already in your workflow (i.e. the standard plan and rule blocks).

To prevent schema conflicts in the workflow integration, it is important to set up an AutoML run with a data source that has the same schema as the one used by the rules. You can prepare the data source with such schema using plans in the **Pulse Data Science** tab. The general idea is to run the plans with the data source to mimic the changes that a schema undergoes in the runtime workflow. This is explained in more detail in the following section.

Howâ

To make the schema of your data source identical to the rules' schema, it is important to understand the different schemas that flow through each operator in the workflow. The following image shows the operators that are configured in the standard workflow and highlights the schemas along the workflow path:

tip

Main goal: You need to prepare the schema of your data source so that it contains the fields that are in the 'Workflow Schema' plus any additional fields. By doing so, you ensure that the AutoML plan output keeps the fields that are required for your rules in addition to creating new fields that are relevant for the model.

To prepare your data source:

1. Identify the current schema of your data source. It should match one of the schemas highlighted in the figure.
2. Use Pulse plans to mimic the transformations that a schema undergoes to reach the 'Workflow Schema'. For example, if your data source has a schema identical to the 'Input Schema', you need to transform it into the 'Extended Schema' and then into the 'Workflow Schema'. The next section shows an example of a data source's schema transformation from 'Extended Schema' to 'Workflow Schema'.

Modify the schema with Pulse plansâ

You can add fields to the schema of your data source with Pulse plans.

As an example, consider that you have the data source with a schema identical to the 'Extended Schema'. To extend this schema and make it identical to the 'Workflow Schema', perform the following steps:

1. Open the **Data Science** tab.
2. Open the standard plan that is responsible for adding the extra fields in the runtime workflow (i.e. 'Plan v_N').
3. Select the button to open the **Configure Plan Execution** menu and:
4. Select the data source that has the initial schema.
5. Select to execute the job.
6. Select the button to open the **Configure Plan Snapshot** menu and:
7. Select the **Export** box.
8. Choose the new schema name.
9. Select the button to create a new data source with the target schema. The new data source is stored under the **Data Sources** table in the **Data Science** tab.

Next stepâ

Now that you have completed the schema, read the [Create an AutoML Run] page to generate features and models automatically.

Create an AutoML Run

Was this page helpful?Â

This page explains how to create an AutoML run to generate features and models automatically.

danger

Environment to run AutoML: Since the AutoML computation is quite resource-intensive for the system, Feedzai recommends running AutoML in a staging environment dedicated to the data science work (i.e. typically a Data Science environment). This is important to ensure that the production environment is not compromised by this functionality. Once you achieve results that meet your business needs, you can migrate the AutoML model and plan to the production environment and integrate these new resources into the runtime workflow.

tip

Data Sampling: For large datasets, the computation of AutoML may stop the system. To prevent this, before creating an AutoML run, use a sampling strategy to remove part of the original dataset, while keeping a fair data representation.

Prerequisites:â

Before running AutoML on a data source, you need to ensure that:

- The data values are valid. Check the complete list of data conversions in the [Prepare the data values - AutoML prerequisites] section.
- [Download] the semantic file that maps the data values to semantic tags that are relevant to your use case.

Create an AutoML runâ

To create an AutoML run that finds features and train models, perform the following actions:

1. In the **Data Science** tab, expand the AutoML section using the button.
2. Select to start a new AutoML run.
3. Configure the data source and basic settings:
    1. Select the **Data Source** you will use to create features and train models.

info

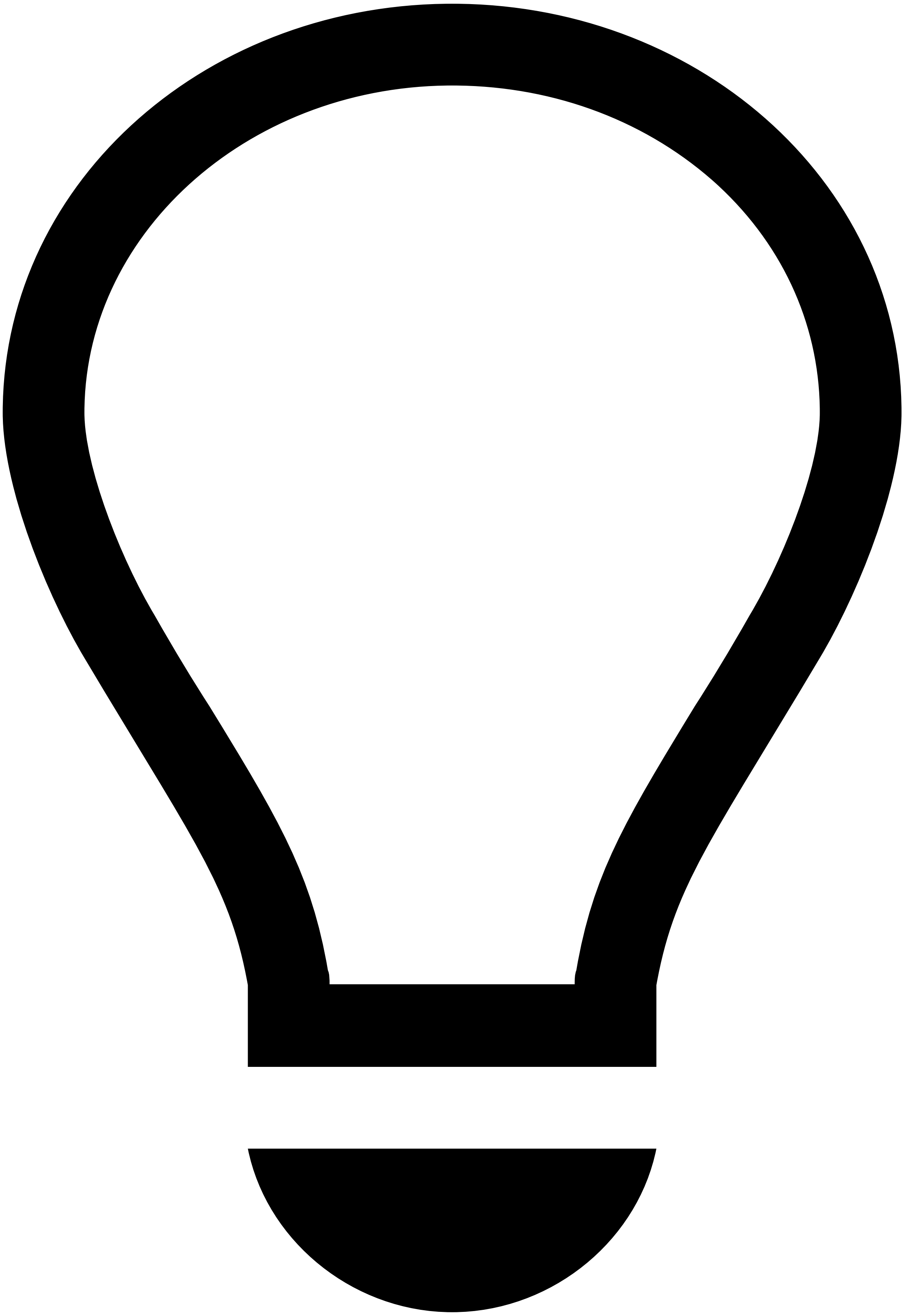
Data Source: If you want to integrate the recommended model that is generated by AutoML with the standard rules in the runtime workflow, make sure you select a data source that has the same schema instead of the one that is used by rules. Check the [Complete the Schema] page for more information about this step.

2. Upload the **Semantic File** that contains the AutoML tags for the data fields. Check the [AutoML Stages] page to learn more about the semantic file. The tags link the fields to their meaning. This file is provided by the standard Transaction Fraud for Banking package.
3. Select the **Timestamp Field** from the list of available fields (e.g. `event_occurred_at`). Pulse uses the timestamp field to aggregate the data by time windows for the calculation of metrics.
4. Select a categorical data field for the **Target Variable**. This is the variable that the models will predict, and is typically a field named as `fraud`.
5. Select the **Target Value**, which is the value of the **Target Variable** that you want to detect. For example, in the fraud detection scenario, the **Target Variable** would be the `fraud` field, and the possible **Target Values** would be 'True' and 'False'. To detect fraud, select 'True' for the **Target Value**.
6. Enable an **Additional Cost Field** for model evaluations. Select a field that represents the transaction amount to have metrics such as the Money Recall and Money False Positive Rate (FPR).
7. Select the option **Next** to proceed to the next stage.

4. Split the data source into smaller datasets for training, validation and evaluation. To do so, define the following:
   1. **Time range for training**: The training dataset will be used for training models during the hyperparameter tuning phase of the AutoML run.
   2. **Time range for validation**: The validation dataset will be used to evaluate the models during the hyperparameter tuning phase to find the best model.
   3. Use the **Enable test dataset** switch to define the time range for the dataset that will be used for the final evaluation of the best model found. If you don't define a test dataset, Pulse will use the validation dataset for this purpose.

info

Time ranges: In time ranges, the beginning timestamp is inclusive, while the ending timestamp is exclusive.



tip

Planning training, validation and test data sources: To avoid under- or overfitting, plan the time ranges carefully. Ensure that the resulting datasets contain enough data for their purpose. A typical dataset split is 50% for training, 30% for validation and 20% for evaluation, as long as the data source size is sufficient for that. The time windows need to be in the past and follow the order training-validation-evaluation.

If you don't define a test dataset, AutoML uses the validation dataset for the final evaluation of the best model found. Since the validation dataset was used for finding the best model, it is not an optimal solution to use it to test the data. Therefore, you should use a test dataset.

1. Select **Metric to optimize**, and a fixed metric value that you want to achieve. For example, you may want to optimize your Recall as your metric while ensuring an FPR around 1% as your fixed metric value.
2. You can configure **Advanced options** as follows:

    1. Set the **Number of Models** you want to train in this AutoML run.

        info

        Computational weight: This value heavily influences the runtime of the AutoML job. Feedzai recommends that you test the runtime with a few models first.

    2. Select the **Path to generated data source** if you want to save the data in a specific path.

    3. Select the **Window size** for metrics. If the window size of a metric is 1 hour, it means that the metric calculated for a certain point in time is based on the data of the last hour before that given time. Each defined window size will generate another metric.

        - Metrics with **Real-time Updates** will be updated in every event.
        - Metrics with **Scheduled Updates** will be updated at the start of each day.
3. Select **Create** to start the AutoML job.

Follow up the AutoML runâ

After setting up the AutoML process, you can follow up on its status at the top of the **Data Science** tab. When the AutoML job is finished, you can view the results in the same place:

- The **Best Model** trained in the last AutoML run.
- The **Features Plan** that created a data source with features based on Feedzai's experience in fraud detection.
- The **Details** about the job that executed the last AutoML run. If the AutoML run fails, you can start investigating here.

Next stepsâ

Once the AutoML job has finished, analyze the AutoML Output. One of the key outputs of an AutoML run is the recommended model. The next page helps you analyze the performance of the model and provides details about additional resources produced by the AutoML such as plans and data sources.

On this page

Download Semantic File

Was this page helpful?Â

TFB provides two semantic files, one for cards and another one for transfers (i.e. addresses both instant and non-instant transfers). Select the following buttons to download the files, and then run AutoML.

Semantic file - Cards

Semantic file - Transfers

With the semantic file for cards, you can run AutoML and generate approximately 800 new features relevant to the card (issuing) use case. The semantic file for transfers allows you to generate about 490 new features that apply to instant and non-instant transfers.

If you already have your data pre-loaded with the data type matching the tags' meaning in these semantic files, you can go directly to Create an AutoML Run. If you are new to AutoML, the next step of the user journey helps you to get started.

Next stepâ

Start by loading and preparing the data which will be used to run AutoML.

On this page

Integrate Resources in Runtime

Was this page helpful?Â

This page helps you integrate the plan with relevant features and the recommended model generated by the AutoML into the runtime workflow.

An AutoML run generates a features plan with a plan execution and a recommended model. If the model fits your business requirements, you can already deploy these resources in the runtime workflow.

Integrate resources into workflowâ

To integrate the plan and recommended model into the workflow, perform the following steps:

1. Use the **Workflow** tab to open the Cards or Transfers Workflow, depending on your use case.
2. Remove the connection between the plan and the first rules Service.
3. Drag and add a new **Plan Service** between the existing plan and the rules Service.
4. Choose the **Edit Service** button to configure the plan.
5. Select the 'Feature Generation' plan or the 'Feature Generation Reduced' plan, depending on whether you want to include all the AutoML generated features or only the relevant ones, respectively.
6. Select the most recent **Plan Execution**.
7. **Save** the service.
8. Connect the plan to the adjacent services.
9. Remove the connection between the 'Self Service' rules and 'Decision' rule services.
10. Drag and add a new **Scoring service** between the services mentioned above.
11. Choose the **Edit Service** button to configure the scoring.
12. Select the **Target Outcome** 'fraud'.
13. Select the model project generated by AutoML.
14. Select the model name recommended by AutoML.
15. **Save** the service. The following image shows the final diagram architecture:
16. **Save** to apply the changes in the workflow.
17. Publish your application using the button.
18. Apply your changes to complete the process using the button.

info

Feedzai recommends that you use the **Rolling Publish** option. This option publishes application changes on one Pulse replica at a time, thus reducing the downtime, as there may be other online replicas still able to handle events.

-
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On this page

AutoML

The Automated Machine Learning (AutoML) guide helps data scientists use the risk engine platform's built-in AutoML tool. With this tool, you can automatically generate features for machine learning models and obtain a trained model with a set of optimal parameters.

This guide covers the following end-to-end AutoML process:

- Prepare the data that is used to generate features and train models.
- Create an AutoML run.
- Analyze the results.

- Integrate the new features and trained model with the existing resources in the runtime workflow.

Why use AutoML?â

Finding the right features and models for a dataset can take several weeks because it requires a number of repetitive and time-consuming tasks: creating features, training models, tuning parameters, and testing changes. AutoML is a functionality available in the risk engine platform that automates these tasks, thus helping data scientists to quickly bootstrap new projects. More specifically, AutoML runs the following tasks behind the scenes:

1. Generates hundreds of features.
2. Selects the features that contribute the most to the model performance.
3. Trains multiple models and optimizes model parameters (i.e. hyperparameter optimization).
4. Recommends the best model.

As a result, after running AutoML, the risk engine platform generates the following output:

- A plan with meaningful features that are derived from a dataset (referred to as features plan in this guide).
- A model project with the trained model, which shows the best results among all the model tests performed by the AutoML run (referred to as recommended model in this guide).

The features plan and the recommended model are readily available to be integrated into the Pulse workflow to process transactions in real time.

How does it map to each use case?â

TFB packages several use cases, namely Transaction Fraud for Cards (issuing), Transaction Fraud for Transfers, and Transaction Fraud for Instant Transfers into a single risk management solution. Learn more about these use cases in the About the Solution page.

The solution provides two semantic files, one for cards and another one for transfers (i.e. addresses both instant and non-instant transfers). You can download the files, and then run AutoML to generate the set of features and the model for each use case.

User journeyâ

The following user journey describes the steps to prepare your data, run AutoML, analyze the results and integrate the AutoML output (i.e. features plan and recommended model) in the runtime workflow:

Check the AutoML Stages page to understand what are the operations executed by AutoML behind the scenes.

To run AutoML end to end, perform the following steps:

1. Load your historical data: Prepare your input data source, which will be used to derive features and train models. In this step, you will need to define the type of each field in your data source's schema. This is important because AutoML relies on a semantic file that maps the input data fields to semantic tags. Therefore, each field type in the input data source needs to match the type that is defined in the semantic file. Check the AutoML Stages page to learn more about the role of the semantic file in the AutoML process.
2. Complete the schema: Before running AutoML, ensure that the data source's schema is identical to the one used by the standard rules. This is important when integrating the recommended model with the standard rules in the runtime workflow to avoid conflicts due to different schemas.
3. Create an AutoML run: Generate a features plan and a recommended model with AutoML.
4. Analyze the AutoML output: Analyze results to understand whether the output (i.e. features plan and recommended model) fits your business needs.
5. Integrate resources in runtime: Integrate the features plan and recommended model in runtime workflow and publish your work to start scoring events.

-

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- Load your Data Source

On this page

Load your Data Source

The first step of the AutoML journey is to import historical data into Pulse. You will use this data to run AutoML to generate features and train models.

danger

Environment to run AutoML: Since the AutoML computation is quite resource-intensive for the system, Feedzai recommends running AutoML in a staging environment dedicated to the data science work (i.e. typically a Data Science environment). This is important to ensure that the production environment is not compromised by this functionality. Once you achieve results that meet your business needs, you can migrate the AutoML model and plan to the production environment and integrate these new resources into the runtime workflow.

info

The applicability of this user journey step depends on your product setup. If you already have a historical data source loaded in your Pulse application, you can skip this step entirely. In that case, check the Complete the Schema page to ensure that the schema of your data source is compliant with the workflow schema.

Import dataâ

1. In the **Data Science** tab, under the Data Sources table, select the button to add a new data source.
2. Select the path from which you want to import the data and select **Next** to go to the next stage.

info

File storage: For cloud projects, this step may require further assistance from your project team to access and store the data source in the right location, depending on the system architecture and permissions. For on-premise projects, the data source storage is highly dependent on each customer's architecture. Note that if you have the file in a local folder, for example, */tmp*, the path is *file://tmp/*.

3. See how your data will be imported by previewing the sample provided by the wizard. If your data is not displayed correctly with the values separated by columns, try different delimiter options. This adjustment is common because there are different methods to separate values in data source files, e.g. using ',' or '|'. Adjust **Quotes**, **Line Separator** and **Missing Values** fields to define how Pulse should interpret and import the data from your file. Select **Next** to go to the data source characterization stage.
4. In this stage, Pulse automatically defines the schema and infers a data type. However, you may have to change the type and configuration based on your schema definition. This is a key step because the data type needs to match the meaning of the tags in the [semantic file]() used to run AutoML. For example, all the `longitude` and `latitude` fields need to be configured as numerical of double type since this is the expected type of 'latitude' and 'longitude' semantic tags.

tip

Categorical values: Pay special attention to the categorical fields, as all the possible values must be defined, and Pulse infers from a subset of rows only.

info

Immutability: Data sources are immutable. This means that after saving your changes you won't be able to edit this data source. However, you can use the **Duplicate and Change** Pulse option to [generate a new data source based on the original]() and quickly make new changes in the schema.

You can configure the schema using the following methods.

- Manually:
  - Select the **Type** field.
  - Under **Data Configuration**, you can further specify the type, for example, define a numerical type as 'integer', 'long', 'float', or 'double', or define a categorical type as 'string', 'integer', 'long' or 'boolean'.
- Based on an existing schema:
  - Use the button and select the target schema.

5. Once you have defined the full schema, select the **Create Data Source** button.

Duplicate and change a data sourceâ□□

Since data schemas in data sources are immutable, if you need to make changes to a data source, you must create a new one. However, you can use an existing data source as your starting point by selecting the **Duplicate** option in the **Data Science** tab. This will open a configuration panel for you to edit the schema (see the Import data section).

Prepare the data valuesâ□□

To be able to run AutoML against a data source, first you need to prepare the data.

- Date and datetime fields, which are not supposed to be in the YYYYMMDD or YYMM format as described in the API specification pages, need to be converted to Unix timestamp format.
- Values associated with amount fields need to be normalized into a single currency.
- The data cannot contain NULL values. Instead, you need to add the default values:
  - Strings should be coalesced to an empty string, for example 'EMPTY'. However, it cannot be ('').
  - Categoricals should be coalesced to an existing value or to a new categorical value representing the NULL, typically a value called 'other'.
  - Numericals should be coalesced to a meaningful number, for example -999, -1 or 0, depending on their valid range.
  - Dates and datetimes should be coalesced to a default value, for example -2208988800, which corresponds to 01-01-1900 in the Unix timestamp format.
- The input data source cannot use empty string ('') as NULL.

Next stepsâ□□

To complete the preparation steps for AutoML, check the Complete the Schema page. The page will help you extend your data source so that it has the same fields as the ones used by the Pulse rules. This will eliminate schema conflicts when integrating the recommended model with the standard rules in the runtime workflow.

- References
- Data Exporting Tool

On this page

Data Exporting Tool

Was this page helpful?Â

The **Data Exporting Tool** is a data delivery system that provides a reliable and consistent stream of data. It allows you to ingest data into your internal systems and analytics tools without requiring it to be manipulated/transformed in advance.

The exported data is presented in pre-defined, stable schemas, ensuring consistency and simplifying it for downstream integrations to consume.

How It Worksâ

Relevant data from Pulse and Case Manager is automatically exported and made available in a secure, data-extraction S3 bucket managed by Feedzai. The exported data is stored as CSV files according to a time-partitioned export structure.

For more information about the available data, check the [entities model](#).

The data flow is completely automated and scheduled to **generate exports every hour with a two-hour delay**.

note

Other timeframes may be configured depending on business needs upon activation. Reach out to your Feedzai team for support.

Export Structureâ

Data exported and made available as CSV files in the data-extraction S3 bucket is **organized by entity in time-partitioned folders** under the `data-platform/` folder.

Entity Foldersâ

Each entity has a dedicated folder under `data-platform/data-exports/[domain]/`. This ensures datasets are isolated and easy to navigate.

The domain is the name of the domain that the data is exported for. It can be tfb, rmpsp or other.

```
data-platform/
    data-exports/
        [domain]/
            [entity]/...
```

For more information about the available entities, check the [entities model](#).

Time-Partitioned Foldersâ

Within each entity folder, data is partitioned hierarchically by year, month, day and hour. This allows you to quickly find and retrieve data exported for specific time ranges.

```
data-platform/
    data-exports/
        [domain]/
            [entity]/
                _year=[yyyy]/
                    _month=[mm]/
                        _day=[dd]/
                            _hour=[hh]/
```

CSV Filesâ

At the lowest level of each time-partitioned folder (i.e., `_hour/`), you'll find one or more `.gz` files containing the exported records for that time window in CSV format.

info

The **complete dataset** for a given entity at a specific date and time is the **combination of all CSV files within that partition**.

This structure is optimized for parallel processing, ensuring faster data ingestion and analysis. It also keeps the files lightweight and easier to transfer.

```
data-platform/
    data-exports/
        [domain]/
            [entity]/
                _year=[yyyy]/
                    _month=[mm]/
                        _day=[dd]/
                            _hour=[hh]/
                                *.gz    # One or more zipped CSV files
```

File Structureâ

Each CSV file is structured according to the following properties:

- **Header**: `name` for each field.
- **Complex fields**: Stored as JSON blobs with type `array`, `map` or `row`.
  The types for each field inside the JSON blob can be found in the data entity models and need to be accounted for during deserialization.
- **Delimiter**: `,`
- **Escape character**: `"`

Practical Exampleâ

Here's the folder structure for the `eventstorage_cards_feedback` entity on 2025/08/22 at 13:00.

```
data-platform/
    data-exports/
        tfb/
            eventstorage_cards_feedback/
                _year=2025/
                    _month=08/
                        _day=22/
                            _hour=13/
                                2b16d9fd57144b50a061425f776764c2.gz
                                faff739cc02d476e82dc9463d4a20610.gz
```

After unzipping each of the following files:

- `2b16d9fd57144b50a061425f776764c2.gz`:

```
eventid,timestamp,feedbacklabel
af872fe8-2fdf-43a1-97b2-8b74f8867b13,1757477792,fraud
```

- `faff739cc02d476e82dc9463d4a20610.gz` :

```
eventid,timestamp,feedbacklabel
826636c0-556f-4dc5-bcfd-f4cf959864bc,1757471765,fraud
0e58187c-735b-436b-a143-d2beea33b05a,1757436330,genuine
```

Both files form the complete dataset for that time window:

eventid	timestamp	feedbacklabel
af872fe8-2fdf-43a1-97b2-8b74f8867b13	1757477792	fraud
826636c0-556f-4dc5-bcfd-f4cf959864bc	1757471765	fraud
0e58187c-735b-436b-a143-d2beea33b05a	1757436330	genuine

Integration Considerationsâ

When integrating with the exported data, keep the following in mind:

Consideration	Details
Data guarantees	Data is delivered at least once. Duplicates may occur in the exported data.
Null handling	Null values are represented as empty fields in CSV files. In JSON outputs (complex fields), they may appear explicitly as `null` .
Encoding	All CSV files are encoded in UTF-8.
Missing partitions	A missing file for a given partition means either an export failure or that there was no data for that entity during that hour. If thereâs data in the downstream partitions, it indicates the system is working as expected and there was simply no data available to export for the missing partition.
Data retention	Data is available for 60 days after it was exported.
Schema evolution	Schemas are additive only: new columns may be added but never removed. CSV headers indicate the available columns.
Timezone	All time partitions (`_year` , `_month` , `_day` , `_hour`) are in UTC.

Entities Modelâ

Event Storageâ

The following are the different entities exported for event storage:

- Event Storage - Cards
- Event Storage - Cards Feedback
- Event Storage - Transfers
- Event Storage - Transfers Feedback
- Event Storage - Digital Activity
- Event Storage - Digital Activity Feedback
- Event Storage - Ad-hoc Customer Investigations
- Event Storage - Ad-hoc Customer Investigations Feedback

Case Managerâ

The following are the different entities exported for Case Manager:

- Case Manager Case
- Case Manager Event - Cards
- Case Manager Event - Digital Activity
- Case Manager Event - Transfers
- Case Manager Event - Ad-hoc Customer Investigations

Pulse Audit Eventsâ

The following are the different entities exported for Pulse Audit Events:

- Pulse Audit - Users
- Pulse Audit - Self Service Rules
- Pulse Audit - Rules
- Pulse Audit - Lists
- Pulse Audit - List Items

-
- Operational Work
- Data Scientist & Fraud Strategist
- Work with Rules
- Create and Manage Rules
- Change the Plan

On this page

Change the Plan

The plan is where you apply transactions and all the data enrichment. You have two types of plans:

- **Deployable plans**: can be used in production.
- **Non-deployable plans**: allow for more flexibility on data exploration, but are not deployable in production.

info

If you need to change the out-of-the-box metrics / fields, create a copy of the original plan first, and make your changes there.

There are five blocks in the out-of-the-box Cards plan, and they are processed sequentially.

- **List Enrichment**: picks up values in the list, if the entity exists in that list.
  - For example, if `card_id` exists in PosNeg Listed Cards and has a `decline` value, then `posneg_card_value` is `decline`.
- **Derived Fields / Metrics Based Fields**: enriches the data using fields/metrics created previously.
  - This is where you can use user-defined functions (UDFs), implement conditional logic with if/else statements, update fields, and delete fields as needed.
- **Metrics**: creates metrics based on a window.

Create a new metricâ

In the side panel, go to **Data Science**.

1. In the **Plans** widget, open the plan where you want to create a new metric.

   danger

   Never change out-of-the-box metrics.

2. Select the **Pencil** icon from the **Metrics** operator.
3. Select **Add new Metric** and fill in the information as explained in the following sections.

Group Byâ

- **Field(s)**: this is where you select the fields by which you want to group events for your metric.

For example, if you want each card to have a different value, you can group by `card_id`.

Windowâ

- **Expression**: determines how long the window is (e.g. 1 minute, 1 hour, 1 day, 1 week). Note that a metric with a window of less than or equal to 24 hours can be set as real-time metric (i.e. "Update on every event" in the **Edit Update Period** section). A metric with a window greater than 24 hours must be set as scheduled metrics (i.e. "Start of day").

- **Delay**: allows you to delay the metric for a certain time to exclude recent transactions to be included in the metric.

- **Is tumbling**: if enabled, refreshes metrics when they hit the window duration.

- **Filter**: ensures that only Authorization or pre-decided Info will enter this metric, as mentioned in [Understand the Workflow](). If you need additional filters, feel free to add them here.

  - **Cards**

```
( event_type == 'authorization' or (event_type == 'info' and info_type ==
'pre_decided') ) and (card_id ?? '' != '')
```

If you group by any entity other than `card_id`, replace `(card_id ?? '' != '')` with `(YOUR_ENTITY ?? '' != '')`.

  - **Transfers**

```
(event_type == 'transfer_initiation' or (event_type == 'info' and info_type == 'pre_decid
ed'))
```

Edit Update Periodâ

- **Update Period**: if it's a metric with a window of less than or equal to 24 hours, select **Update on every event**, otherwise select **Start of day**.

Aggregationâ

- **Aggregation type**: select how you would like to aggregate the metric.

For example, if you want to use `amount`, use `cardholder_bill_amount` for Cards.

Create snapshotsâ

After creating new metrics, you need to create a new snapshot to [deploy this plan] to production.

Tick the Schema's **Export** box and name it.

This schema will contain the newly-created metrics, and will be used in your rules project.

New metric - practical exampleâ

Consider the scenario where you need to create a rule that declines transactions from any card that has spent over $1,000.00 on casino transactions over the last 3 hours. You can also assume casino transactions are identified as MCC = 7800. Note that, if you need to use a field that is not in Feedzai's schema, e.g. you have a boolean flag that already identifies gambling transactions in your system, then you need to contact Feedzai to add this custom field into Feedzai's schema.

This scenario involves creating a metric that will aggregate the amount for gambling transactions. This means you'll have to create a metric in "Cards Plan v4", the plan used in the out-of-the-box workflow:

1. In the side panel, go to **Data Science** and then, in the **Plans** widget, open "Cards Plan v4".
2. Using the **Pencil** open the **Metrics** operator.
3. Select **Add new Metric**.

Group Byâ

- **Field(s)**: `card_id`

Windowâ

1. **Expression**: 3 hours
2. **Filter**: (event_type == 'authorization' or (event_type == 'info' and info_type == 'pre_decided')) and (card_id ?? '' != '') and (mcc ?? '' == '7800')

What does ?? mean?

- ?? means coalesce. If card_id is null, it will fill with empty string.
- card_id ?? '' != '' simply means it must not be null or empty string.

Edit Update Periodâ

1. Since the window is less than or equal to 24 hours, it will be a real-time metric.

Aggregationâ

1. **Function**: sum
2. **Field**: cardholder_bill_amount

Name & Descriptionâ

info

If you don't fill in the name, Pulse will automatically generate a metric name for you. Note that metric names should clearly indicate the grouping entities, aggregation, window, and possibly filter. It is also recommended that you define a prefix to indicate they are custom metrics (e.g. `c_`). Also, as a rule of thumb, name your metrics in lower case, and separate each word using underscores.

1. **Name**: `c_cardid_sum_amt_3h_mcc7800`

In this scenario, the name generated by Pulse is `cardid_sum_cardholderbillamount_3h`, but you can easily edit it directly in the **Name** field.

Go to the next building block, Change Rules Projects, to learn how your rules project picks up these metrics.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/work-with-rules-and-lists/create-rules/list-management-service

On this page

Configure List Management Service

Was this page helpful?Â

The list management service allows you to configure the automatic list transitions that an entity (e.g. card ID) undergoes when triggered by different rule actions.

As described in the Create and Manage Rules page, rule actions have standard values to identify decline, approve and review lists in a PosNeg List. The list management service is responsible for:

1. Checking when a rule triggers one of the above actions.
2. Adding the entity to the PosNeg List and tagging it accordingly as **decline**, **approve** or **review**.

You can extend and adapt the list management service to fulfill your use cases.

For your convenience, the following user journey focuses on the Cards use case.

tip

Advanced functionality: Configurations in the list management service should be performed by advanced users to ensure that the transitions between lists work as expected. Note that, if the transitions are not well configured, the rules that depend on these transitions will not yield the targeted results.

List management serviceâ

The list management service allows you to configure the automatic list transitions that an entity (e.g. card ID) undergoes when triggered by different rule actions.

Rule actions have standard values to identify decline and approve lists in a PosNeg List. The list management service is responsible for:

1. Checking when a rule triggers one of those actions.
2. Adding the entity to the PosNeg List and tagging it as decline or approve.

You can extend and adapt the list management service to fulfill your use cases.

To open the list management service:

1. Select the **Workflows** tab.
2. Open the desired workflow.
3. In the Diagram area, select the pencil to edit the *List Management Service*.

tip

Advanced functionality: Configurations in the list management service should be performed by advanced users to ensure that the transitions between lists work as expected. Note that, if the transitions are not well configured, the rules that depend on these transitions will not yield the targeted results.

How the list management service worksâ

The list management service is configured out of the box to classify card IDs as:

- Listed to be declined
- Listed to be approved
- Listed to be reviewed

Configure a list to work with List Management Serviceâ

List Management Service requires the following structure for lists:

1. The type of each value must be set to "complex".

2. The list of complex fields must include a field named "value" and of type string.

info

If the `rules.X.listFieldName` config is overridden, the list must contain a complex field with the same name instead of the default one ("value").

Decline listâ

The entries in the **Configuration** table with the *rules.0* prefix are associated with the same action (i.e. to list a card ID as declined). In this case, this configuration is set up to tag a card as **decline**:

Key	Value	Mandatory	Explanation
rules.0.actions	Auto Decline Card	Yes	The rule action that triggers this service to list a card ID to be declined.
rules.0.entity	card_id	Yes	The entity that is added to the list (i.e. the key of the key-value pair in the list configuration options).
rules.0.listid	ffce7e5ce8f4013a	Yes	The list ID that is used to store the card ID (i.e. PosNeg Listed Cards).
rules.0.appid	c145d72d00d46bab	No	The app ID where the list exists. If not defined, the default is the app ID where the service exists.
rules. 0.listFieldName	value	No	The name of the list's field that is used in the list ID to store the card ID.
rules.0.entries. 0.value	decline	Yes	The value used to list a card ID to be declined (i.e. the value of the key-value pair in the list configuration options).

Check the Card listed to be declined section to learn how PosNeg Listed Cards stores items tagged as **decline** in Pulse.

Approve listâ

The **Configuration** table entries with the prefix *rules.1* are used to list card IDs to be approved:

Key	Value	Mandatory	Explanation
rules.1.actions	Auto Approve Card	Yes	The rule action that triggers this service to list a Card ID to be approved.
rules.1.entity	card_id	Yes	The entity that is added to the list (i.e. the key of the key-value pair in the list configuration options).
rules.1.listid	ffce7e5ce8f4013a	Yes	The list ID that is used to store the card ID (i.e. PosNeg Listed Cards).
rules.1.appid	c145d72d00d46bab	No	The app ID where the list exists. If not defined, the default is the app ID where the service exists.
rules. 1.listFieldName	value	No	The name of the list's field that is used in the list ID to store the card ID.
rules.1.entries. 0.value	approve	Yes	The value used to list a card ID to be approved (i.e. the value of the key-value pair in the list configuration options).
rules.1.entries.0.ttl	7200000	No	The duration that a card remains in the PosNeg Listed Cards list as approved in miliseconds.

Check the Card listed to be approved section to learn how PosNeg Listed Cards stores items tagged as **approve** for two hours in Pulse.

Review listâ

The **Configuration** table entries with the prefix *rules.2* are used to tag cards listed to be reviewed:

Key	Value	Mandatory	Explanation
rules.2.actions	Limit/Review Card	Yes	The rule action that triggers this service to list a card ID to be reviewed.
rules.2.entity	card_id	Yes	The entity that is added to the list (i.e. the key of the key-value pair in the list configuration options).
rules.2.listid	ffce7e5ce8f4013a	Yes	The list ID that is used to store the card ID (i.e. PosNeg Listed Cards).
rules.2.appid	c145d72d00d46bab	No	The app ID where the list exists. If not defined, the default is the app ID where the service exists.
	value	No	The name of the list's field that is used in the list ID to store the card ID.

Key	Value	Mandatory	Explanation
rules. 2.listFieldName			
rules.2.unless	review, decline, approve	No	The validation to add a card ID to the configured list only if the item is not already listed with the value `review` , `decline` or `approve` .
rules.2.entries. 0.value	review	Yes	The value used to list a card ID to be reviewed (i.e. the value of the key-value pair in the list configuration options).
rules.2.entries. 0.ttl	432000000	No	The duration that a card remains in the PosNeg Listed Cards list as listed to be reviewed in miliseconds.
rules.2.entries. 1.value	decline	Yes	The value used to list a card ID to be declined after expiring the duration in which the card remained as listed to be reviewed (i.e. 432000000 miliseconds).

Check the Review list section to learn how PosNeg Listed Cards stores items tagged as **review** for five hours, and after that, how the card ID is tagged as **decline** until the end of time.

info

Behind the scenes: When the `<prefix>.unless` property is not associated to an action in the configuration table, the list management service replaces a list item as follows:

1. Checks if an item exists in the list.
2. If the item exists, it deletes the item.
3. Adds the item to the list with the correct status value.

For example, if a rule triggers the action Auto Approve Card multiple times, the **from** - **to** time that the card remains on the list will keep updating, and thus the card may effectively remain on a list for more than two hours.

Next stepsâ

Now that you have defined all the aspects related to your rule system, read on to learn about Publish Rules to Workflow.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/work-with-rules-and-lists/create-rules/lists

Create Lists

Was this page helpful?Â

A list is a resource that allows you to store items (e.g. card IDs) which can be then referenced in rules.

You can create lists in Pulse with different types of information. Use lists when writing rule conditions to confirm if the value of a given item is stored in a certain list. For example, you can create rules that decline or accept transactions performed by transfers, credit cards or digital events whose IDs are listed to be declined or approved.

In your out-of-the-box settings, approve, decline and review lists are not configured as three different lists in Pulse. Instead, all the entities are added to a single list called *PosNeg Listed entity-name*. In general, PosNeg lists are typically configured to identify items as "approve", "decline" or "review" using a combination key-value pair and activation timeframe.

Default listsâ

Transaction Fraud for Banking includes default lists to speed up your work. Check the Lists page under the References to learn more about out-of-the-box lists.

As an example, the following sections focus on the Cards use case.

PosNeg lists in Pulseâ

In Pulse, instead of using separate lists to classify a card ID listed to be declined, approved, or reviewed, all this logic is hosted in a single list named *PosNeg Listed Cards*. The tags used to classify cards are the following:

- **approve**: Card listed to be approved.
- **decline**: Card listed to be declined.
- **review**: Card listed to be reviewed.

To add new list items, in the sidebar, go to **Lists** .

The following image illustrates the main panel of a PosNeg Listed Cards in Pulse.

List transitionsâ

The following diagram illustrates the transition between lists that a card can undergo when triggering different rules.

The *Unlisted* label in this diagram represents a card that is not included in a *PosNeg Listed Card*.

info

Standard transitions: The image shows an example of the standard transitions; however, you can configure new transitions to fulfill your business needs (e.g. transition from unlisted to listed to be declined).

Decline listâ

The following image illustrates an example of a card listed to be **decline**:

info

Timeframe: As a standard configuration, a card is permanently stored as listed to be declined. Therefore, the **Active from** option only specifies the initial date.

However, in the case of a fraud feedback event (e.g. cardholder response via API or fraud analyst feedback via Case Manager) identifying the transaction as *Not Fraud*, the card is automatically tagged as `approve` (i.e. listed to be approved), thus overriding the `decline` value.

Approve listâ

The following image illustrates an example of a card listed to be **approve**:

info

Timeframe: As a standard configuration, a card is listed to be approved within 2 hours (the time is configurable). After that, the card becomes automatically an inactive item (but still on the list). The **Active from** and **To** options are used to define this time interval.

Review listâ

When a card is listed as **review**, Pulse creates two entries for the same card ID in the *PosNeg Listed Cards*:

- **Listed to be reviewed within 5 days**: This item is created with the tag **review** to keep the card as listed to be reviewed for 5 days. The **Active from** and **To** options are used to define this time interval. After the review period the card is removed from the review list. Note that after 5 days the card may still exist on the list, e.g. if within the review period the card event triggered an Entity Limit rule which listed it to be declined.
- **Transition card to `declined`**: This item is created because the card is automatically tagged as `decline` after 5 days. Therefore, this item only specifies the initial date, which is set to start immediately after the **To** date of the aforementioned pair.

Request a list review / review lists (four-eyes check)â

Ensuring every change is reviewed by a second pair of eyes helps prevent unwanted or unplanned changes from being published.

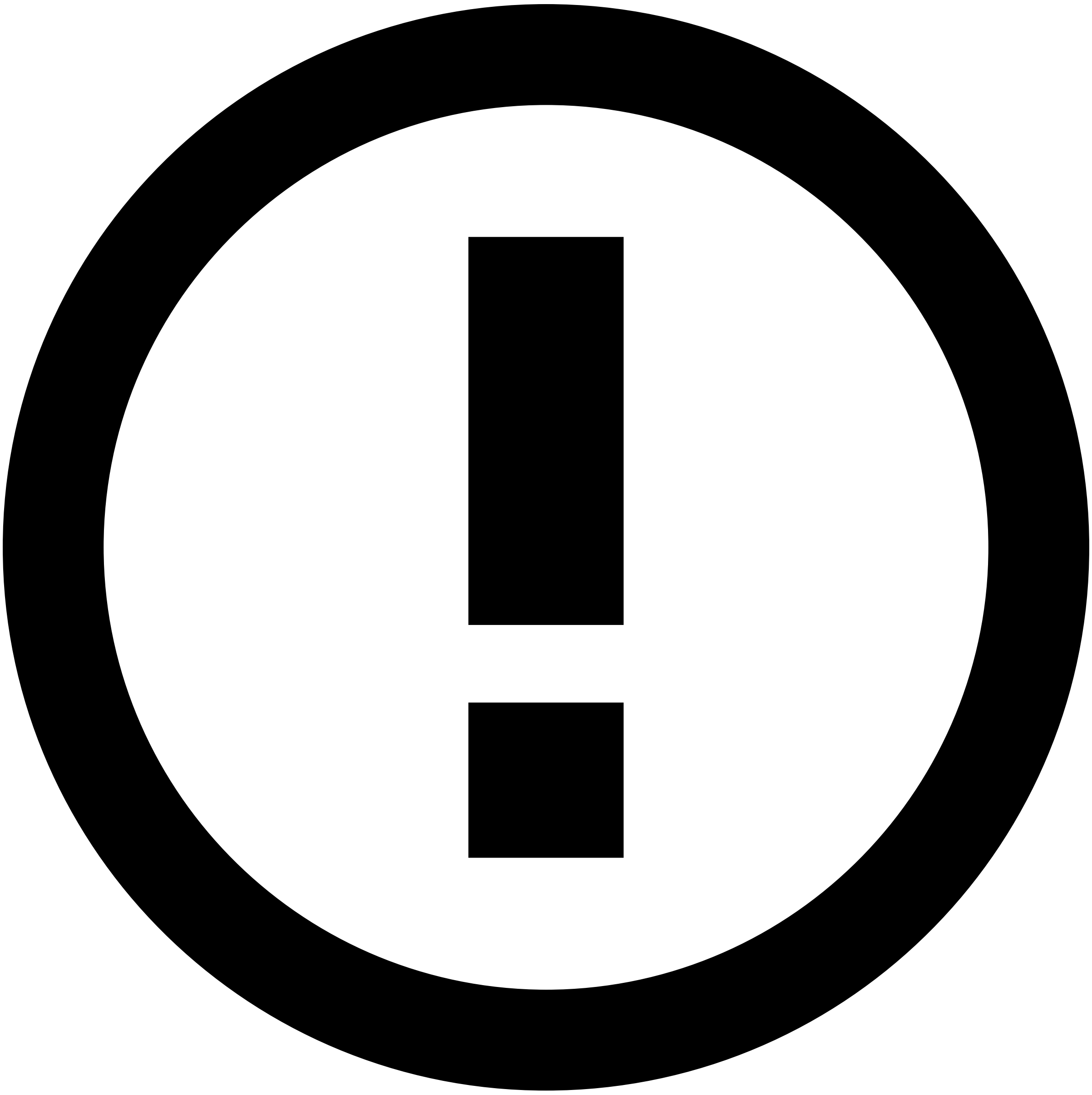
info

The four-eyes check for lists capability is disabled by default. Reach out to your Customer Success Manager to enable it.

tip

By default, requesting reviews is an optional action. To make reviews mandatory, reach our to your Customer Success Manager. With mandatory reviews, changes to lists will not be published right away, since you cannot approve your own requests. Lists will be queued for review and only published after the reviewer approves them.

Request a reviewâ



info

Typically, fraud strategists are entitled to request reviews for lists.

To request a review of a change made to a list, perform the following steps:

1. In the sidebar, select the **Lists** icon.
2. Select the list you want to change.
3. Make the desired changes to the list. For example, adding a new item, editing or deleting an existing item, or importing a batch of items from a CSV file.

caution

You won't be able to make changes to list items that already have pending changes. Attempting to do so will show an error message indicating the ID of the pending review.

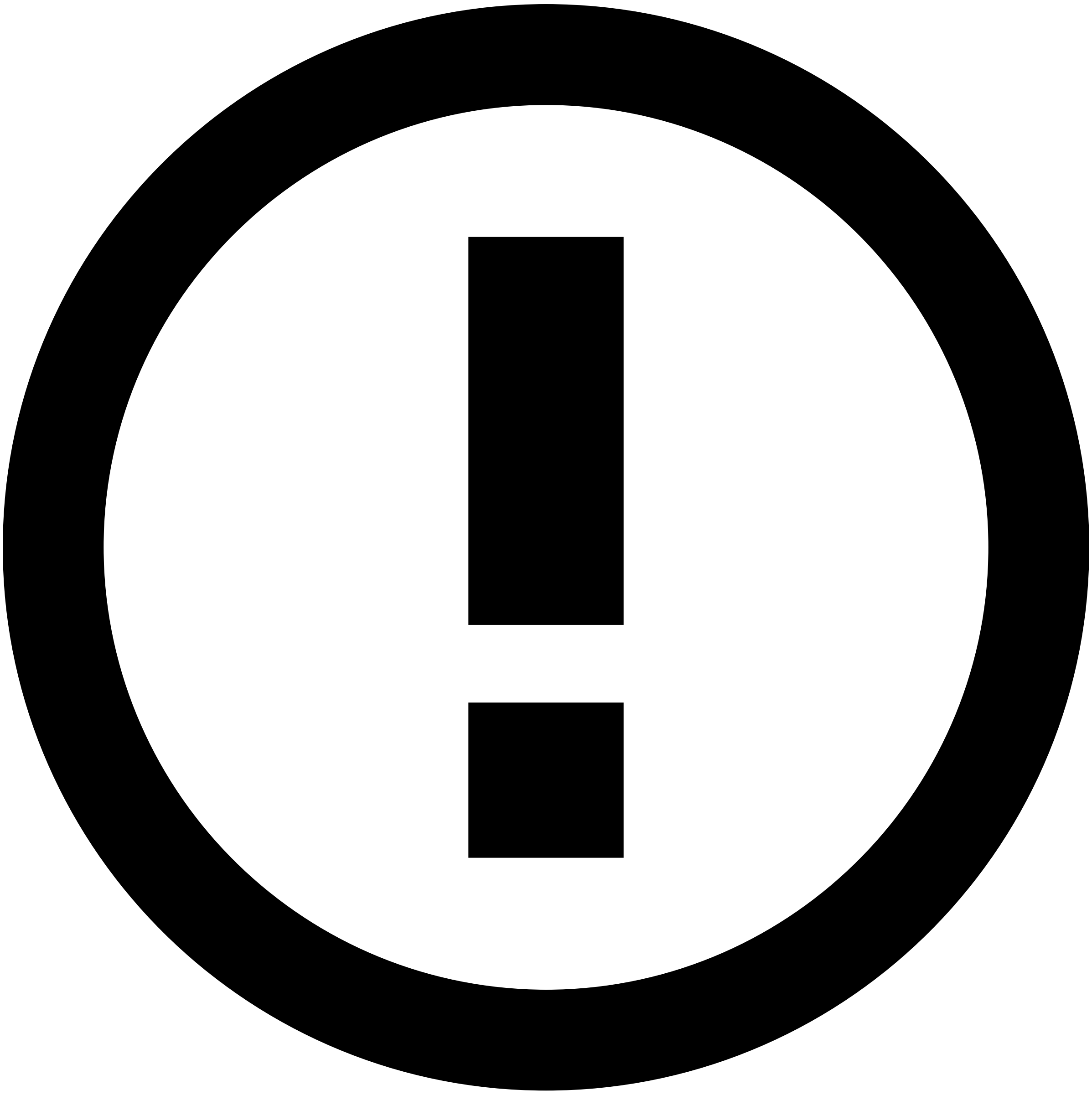
4. Optionally, fill the **Notify users** field to indicate specific users you wish to inform of your review request.

info

To notify users via email, the notification settings must be configured. Contact your Customer Success Manager if you want to enable notifications.

5. Select the **Submit for review** button to add the changes to the review queue. A floating success message will be displayed confirming that the changes have been submitted for review.

Review changes to listsâ



info

Typically, fraud strategist managers are entitled to review list changes.

To review lists, perform the following steps:

1. In the sidebar, select the icon. Lists with pending changes will be marked with an **Unreviewed** label.
2. Open the list with pending changes. Each item with changes will be marked with a label indicating the type of change made (e.g. "Edited", "Added", "Deleted", etc).
3. Select the **Review changes (N)** button. The number in parenthesis is the count of unreviewed changes.
4. Review the overall changes to the list. Change details will at first be collapsed, as indicated by the **Plus** â symbol.

5. Expand the details of each change using the **Plus** â button. This will display a side-by-side comparison of the old and new data associated with the change.

    1. Created items will have no content on the left side.
    2. Deleted items will have no content on the right side.

    The image below shows an example with two changes â one adding a new item, and the other editing an existing item:

    caution

    Changes that consist of batch imports of list items will have no preview (as those can typically include thousands of items). In such cases, the only details that are shown are the number of imported items (in the "Name" column), the import mode (append without overwriting existing items, append with overwriting existing items, and complete rewrite), the change author, and the change timestamp.

6. Select "accept" (âï¸ icon) or "decline" (â icon). For either action, a confirmation dialog is shown. It's possible to disable the display of this dialog in the future, but it's strongly recommended to keep it. A floating success message will be displayed, confirming that the changes have been reviewed and (if approved) will be applied to the list.

Next stepsâ

Now that you know how lists work, learn about the List Management Service.

Create and Manage Rules

Was this page helpful?Â

Creating and managing rules is the first step of your user journey to help you set up your tools to detect and mitigate fraudulent transactions. In this page, you will learn how to map your custom business logic to the basic rule configurations in Pulse to get the best out of the risk engine platform.

High-level building blocks for rule creationâ

To successfully create rules, you need to consider the following high-level building blocks:

- Understand the workflow and snapshots
- Change the plan (i.e. create new metrics)
- Create rules projects
- Create rules
- Migrate Pulse resources between environments
- Publish rules to production

Create Rules

Rules allow you to implement your business logic by defining actions that are taken when a transaction matches a criterion that you define. To help you get started, this page gives you an example of how a typical rule specified in your business language can be translated to Pulse using a Cards' default velocity rule that is included in the Fraud rules project.

Rule configurations in Pulseâ

To define a rule in Pulse, you have the following main configurations available:

- **Variables**: Allows you to declare variables that are further used in the rule's condition including thresholds or transformation/conversion of the input value (e.g. variable that converts an amount from cents to US dollar). Variables are useful to avoid having large condition statements, which are error-prone and difficult to maintain.
- **Conditions**: The condition expression must return a boolean value that dictates whether the event triggers a rule.
- **Explanations**: Allows you to understand why a rule has triggered in a human-readable way. If an event triggers, the explanation is displayed in Case Manager to provide additional insights to analysts.
- **Actions**: Operations that are performed when the **rule is triggered**. The standard values that help you meet your custom business logic are:
  - "**Alert**": Sends the transaction to the Case Manager's alerts page for manual revision by analysts.
  - "**Limit/Review**": Flags entities such as cards, accounts, devices, among others, to be added to the PosNeg list with the **review** value. See the List Management Service section for more information about automatic list transitions.
  - "**Auto Decline**": Flags entities such as cards, accounts, devices, among others, to be added to the PosNeg list with the **decline** value. See the List Management Service section for more information about automatic list transitions.
  - "**Auto Approve**": Flags entities such as cards, accounts, devices, among others, to be added to the PosNeg list with the **approve** value. See the List Management Service section for more information about automatic list transitions.
- "**Mark As**": The rule decision on the outcome of the transaction. The standard values that help you meet your custom business logic are:
  - "**approve**": The decision of this rule is to approve the event.
  - "**decline**": The decision of this rule is to decline the event.
  - "**--**": This rule does not decide the event's outcome.
  - "**Review**": This rule does not decide the event's outcome. However, the 'Review' label recommends that an analyst should further investigate the event in Case Manager.
- **Control Flow**: Allows you to choose what happens after your rule triggers.
  - "**Continue**": Continues to the next rule.
  - "**Stop Workflow**": Stops the workflow immediately after the your rule is triggered.
  - "**Next Workflow Element**": Skips all rules within the same rules project and moves on to the next element / block.

Custom business logicâ

The combination of rule actions and decisions described in the previous section allows you to easily define custom business logic. For example, imagine that you want to define the transaction responses:

- **Soft decline**: This use case involves rules that decline the transaction and that limit card transactions and spending amount for a certain period of time (i.e. through the "Limit/Review Card" action). Furthermore, the transaction should be displayed in Case Manager's alert page, for analyst review.
- **Hard decline**: This use case involves rules that decline the transaction and that send the card to an auto-decline list (i.e. through the "Auto Decline Card" action).
- **Soft approve**: This use case involves rules that do not decide the outcome of the transaction but limit the card transactions and spending amount for a period of time (i.e. through the "Limit/Review Card" action). Furthermore, the transaction should be displayed in Case Manager's alert page, for analyst review.
- **Hard approve**: This use case involves rules that approve the transaction without further actions.
- **Review**: This use case involves rules that do not decide the card or transfer event's outcome.
- **Challenge**: This use case involves rules that do not decide the digital event's outcome.

The following image shows how these responses are achieved by combining the standard actions and decisions.

How to write a ruleâ

Consider the following Cards-Velocity-4 rule example:

To implement this rule in Pulse, consider the following main configurations:

1. **Template**: leave it as `none`.
2. **Variables**: If you are creating a variable for a threshold, the threshold can be hardcoded initially. However, you should change it to looking up the value in a list.

   Example:

   ```
   COUNT_LIMIT = cards__rules_thresholds["max_card_appr_count_3min"].threshold;
   ```

tip

Always end each variable with a semicolon, including the final one (as shown in the example above).

3. Add **Conditions**, such as:

   Example:

```
 // Lists
card_id_not_blacklisted_nor_whitelisted
and
// Filters
(card_id ?? '' != '')
and
```

```
// Limits
APPR_AUTH_cardId_count_3m >= COUNT_LIMIT
```

info

As opposed to variables, you don't need to add a semicolon when adding conditions.

4. **Explanation**

Typically starts with the rule name.

Example:

```
[Cards-Velocity-4] - Decline - Card transactions ({event.appr_auth_cardid_count_3m}) in the last 3 minutes is over the limit ({localvar.COUNT_LIMIT}).
```

5. **Actions**: `Alert` `[Cards-Velocity-4]`

   As a best practice, it is common to add an action with the name of the rule (e.g. Cards-Velocity-4) to easily identify which rules are triggering for debugging purposes. Note that, if you write a new rule, you also need to make this label available in **Actions** by editing the rules project configurations in the Pulse **Data Science** tab.

6. **Mark As**: `decline`

info

To put the rule in shadow (NO-OP) mode, select None. The rule will still trigger, but no decision will be made on this rule.

7. **Score contribution**: This function is currently deprecated, please leave it as `none`.

8. **Control flow**: after this rule is triggered, you can choose what happens next.

    - Continue: continue to the next rule
    - Stop workflow: stop the workflow immediately after this rule is triggered.
    - Next workflow element: it will skip all rules within the same project and move on to the next element/block.

How to write a rule with Digital Trust's enriched fieldsâ

Consider the following configuration when creating a rule that depends on Digital Trust's enriched fields.

info

The examples used in this section refer to the Digital Activity - DNI9 rule configuration.

1. **Variables** contain user-defined functions (UDFs), such as `existsAlertOfType` (see example below). The Digital Trust Enricher includes out-of-the-box UDFs that you can use in new rules.

    1. For custom UDFs, create a new rule on the Digital Trust's console. Make sure that the Digital Trust's rule is also raising a risk indicator. This is important if you want the rule to trigger an alert in TFB as well (see **step 4.i**).

Example:

```
is_RAT =
    (
        Revelock.existsAlertOfType(event.revelock_alerts ,'RAT') ?? false
        or Revelock.existsAlertOfType(event.revelock_alerts ,'RAT TeamViewer') ?? false
    )
```

```
    and (event.revelock_score ?? 0 >= 90 and event.revelock_score ?? 0 <= 99)
    ;
```

tip

Always end each variable with a semicolon, including the final one (as shown in the example above).

2. Add **Conditions**, such as:

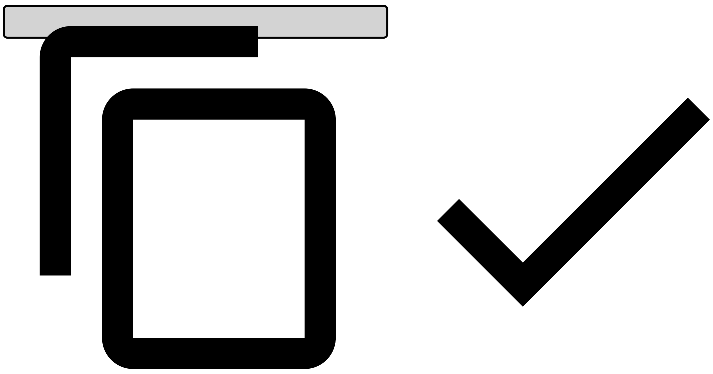
```
is_RAT
```

info

As opposed to variables, you don't need to add a semicolon when adding conditions.

3. **Explanation**: Typically starts with the rule name, as follows:

```
[DA-DNI9] - Medium Risk - Device ({event.device_id}) is suspected to be running a RAT
```



4. **Actions**: As a best practice, it is common to add a rules project action with the name of the rule (e.g. `[DA-DNI9]`) to easily identify which rules are being triggered for clarity purposes. Note that, if you write a new rule, you also need to make this label available in **Actions** by editing the rules project configurations in the Pulse **Data Science** tab.

    1. If you're using a custom Digital Trust action, which is defined when creating the rule in Digital Trust's console - as recommended in **step 1.i** -, add the name of that action here too. This means that, if the rule that you created in the Digital Trust console triggers that action, the rule that you are creating in Pulse will trigger that action too.

5. **Mark As**: `medium`

info

To put the rule in shadow (NO-OP) mode, select None. The rule will still trigger, but no decision will be made on this rule.

6. **Score contribution**: This function is currently deprecated, please leave it as `none`.

Rules that include Digital Trust's enriched fields are typically used in the Digital Activity and Transfers use cases and will have an impact in Case Manager's **Event** widgets, namely in the score:

- Digital Activity:
  - Session
  - Session Alerts
- Outbound Transfers:
  - Session
  - Session Alerts

How to write a rule using TrustScoreâ

Consider the following configuration when creating a rule that scores a card event with TrustScore.

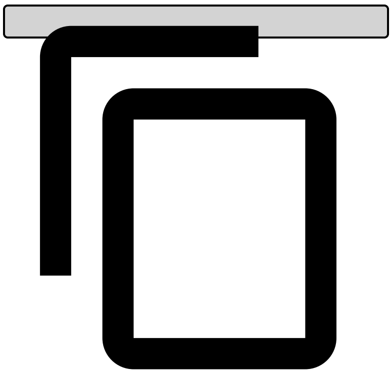
info

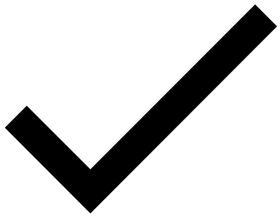
The examples used in this section refer to the Cards-DR1 rule configuration.

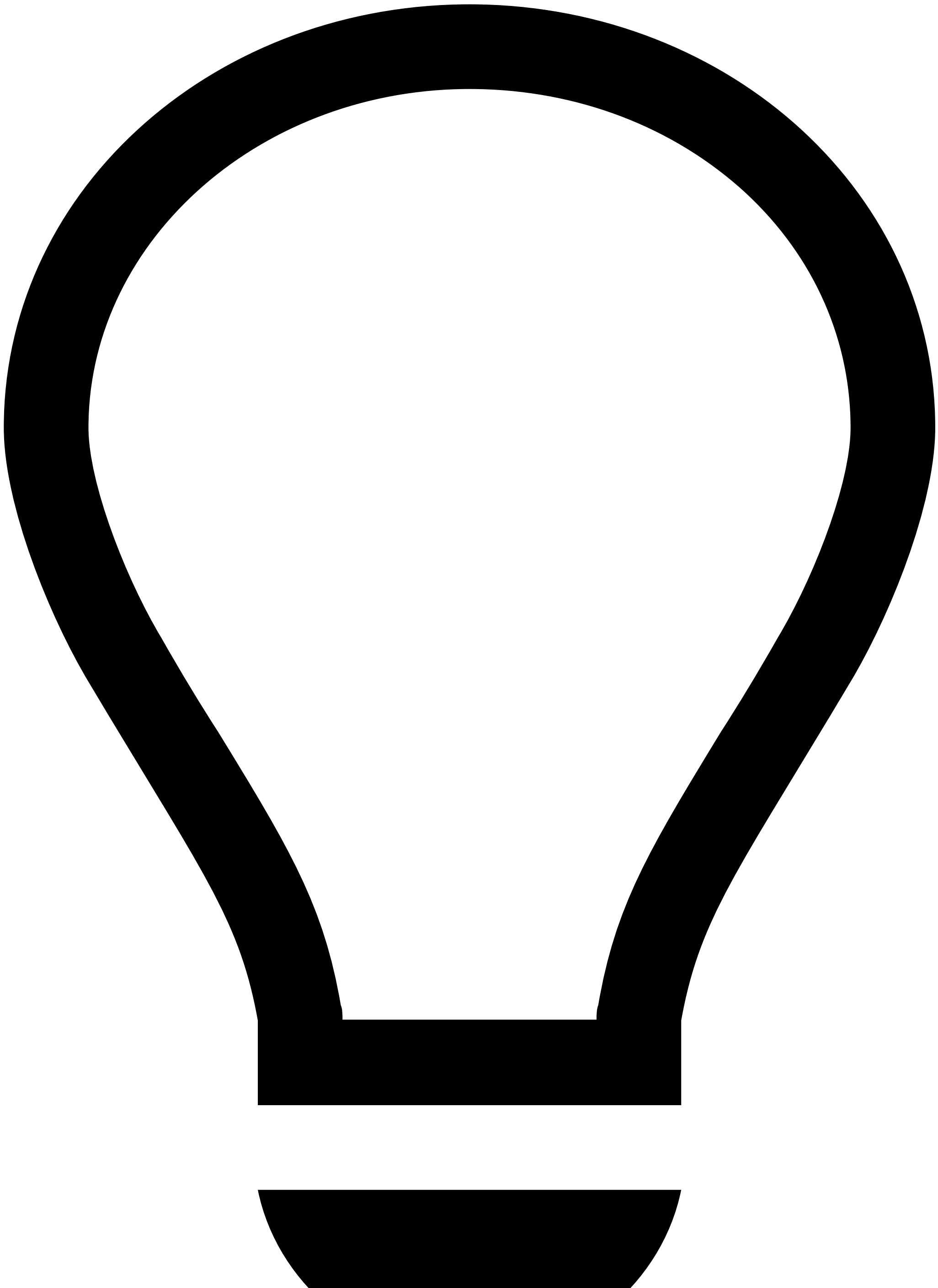
1. **Template**: Leave it as `none`.
2. **Variables**: For threshold variables, you can begin by hardcoding the value. Later, you should modify this to look up the value from a list.

```
## Example
HIGH_SCORE_THRESHOLD = cards__rules_thresholds["model_high_score"].threshold;

score = fraud_score * 1000;
```

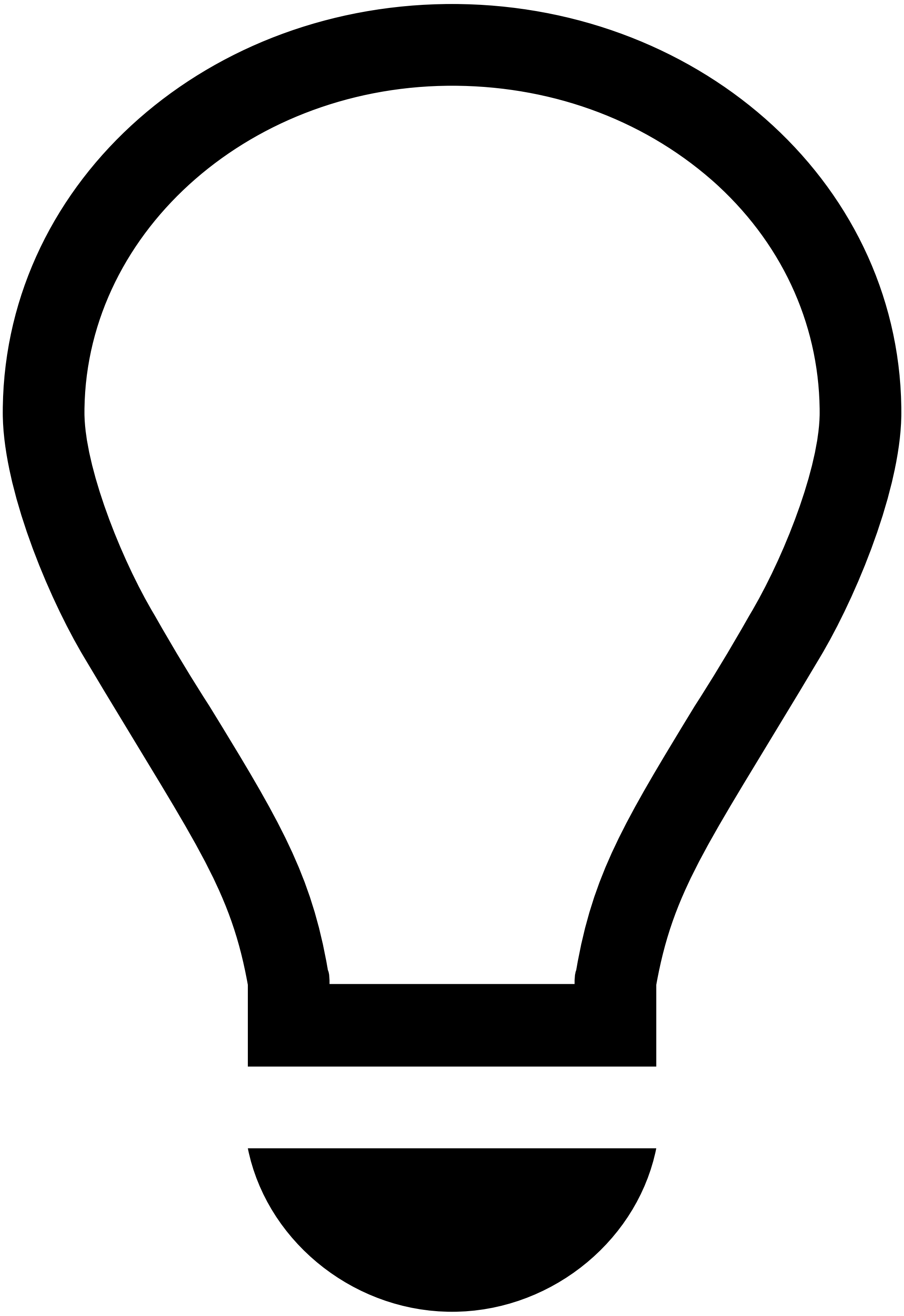






tip

Always end each variable with a semicolon, including the final one (as shown in the example above).



TrustScore - `fraud_score`

By adding `fraud_score` to your variables, TrustScore will start scoring events based on its [intelligent scoring system].

3. **Conditions**

```
## Example
(score ?? 0) >= HIGH_SCORE_THRESHOLD
```

info

As opposed to variables, you don't need to add a semicolon when adding conditions.

4. **Explanation**: Typically starts with the rule name.

```
## Example
[Cards-DR1] - Decline - Model score ({localvar.score}) was above the threshold
({localvar.HIGH_SCORE_THRESHOLD}).
```

5. **Actions**: As a best practice, it is common to add an action with the name of the rule (e.g. `[Cards-DR1]`) to easily identify which rules are triggering for debugging purposes. Note that, if you write a new rule, you also need to make this label available in **Actions** by editing the rules project configurations in the Pulse **Data Science** tab.
6. **Mark As**: `decline`

info

To put the rule in shadow (NO-OP) mode, select None. The rule will still trigger, but no decision will be made on this rule.

7. **Score contribution**: This function is currently deprecated, leave it as `none`.

Create snapshotsâ

After finishing creating / changing rules, you need to create a rule snapshot. A rule snapshot is the collection of rule objects to be deployed in a workflow for production. You can select which ones to enable or disable.

1. In the **Data Science** tab, open the rules project that you have created.
2. In the **Snapshots** widget, create a new snapshot using the **New Snapshot** button.
3. Select the rules that you want to include in the snapshot.

Request a rule snapshot review / review a rule snapshot (four-eyes check)â

Ensuring every change is reviewed by a second pair of eyes helps prevent unwanted or unplanned changes in production. As such, the Four-Eyes Check feature allows you to request reviews from your colleagues.

info

The four-eyes check capability is disabled by default. Reach out to your Customer Success Manager to enable it.

tip

By default, requesting reviews is an optional action. To make reviews mandatory, reach our to your Customer Success Manager. With mandatory reviews, you won't be able to approve your own rule snapshots. This also means that if rule snapshots don't go through the four-eyes check process, they won't be saved into the workflow.

Request a reviewâ

info

Typically, fraud strategists are entitled to request reviews for rule snapshots.

To request a review of a snapshot, execute the following steps:

1. In the **Data Science** tab, go to the rules project that contains the rule snapshot. Under **Snapshots**, select the snapshot that you created.
2. Use the **Request Review** button to submit the snapshot for review.
3. In the **Submit Snapshot for Review** window, fill in the following fields:
    1. **Change Description**: Describe the changes you made to rules, data fields and lists. Include the name of the snapshot that contains the rule's previous state for reference, if there is such a snapshot. This will let the reviewer know the name of the snapshot they should compare yours to. Usually it is the snapshot that was last published live.

2. **Notify User(s)**: Type the name of the Pulse users that you want to invite to review the snapshot. In this field, you can select only users that have an email address configured in Pulse. If you don't find a user, contact your Pulse administrator.

4. Use the **Submit** button to send the snapshot for review and request the selected users to review the changes included in the snapshot.

5. As a result, the users you selected will receive an email with the following information:

    1. Review submitted on your behalf.
    2. The name of the rules project.
    3. The name of the snapshot.
    4. A link to the snapshot.

6. You can follow up on the status of the review next to the snapshot's name:

7. You will receive an email notification when the snapshot is accepted or rejected.

Review rulesâ

info

Typically, fraud strategist managers are entitled to review rule snapshots.

If you are asked to review a rule snapshot, perform the following steps:

1. Open the link from your email, or go to Pulse in the **Data Science** tab and look for the snapshot submitted for review.

2. If there is a previous snapshot that you want to compare it with, use the **Compare with another Snapshot** button to see the differences.

3. When you have decided whether the snapshot should be published or not, select the **Review Snapshot** button.

4. Add your feedback in the **Explanation** field, and select the **Accept** or **Reject** button according to your decision about the snapshot.

5. The user who requested this review will be notified of your decision via email.

New rule - practical exampleâ

tip

The following scenario assumes the new metric example described in Change Plan.

Assumptionsâ

- Casino transactions are identified as MCC = 7800.
- A new metric called `c_cardid_sum_amt_3h_mcc7800` has been created.
- A new rules project has been created with the **Action** code `[Cards-Gambling-1]`.

1. Open the new rules project.
2. Select **New Rule**.
    1. **Name**: Cards-Gambling-1.
    2. **Comment**: Declines transactions from a card that has spent over $1,000.00 on casino transactions over the past 3 hours.
    3. **Template**: None.
    4. **Variables**:
        1. `SPENT_LIMIT = cards__rules_thresholds["gamble-1_spent_limit"].threshold;`
            - `cards__rules_thresholdsCards` (i.e. "Cards - Rules Thresholds") is a list that contains all the thresholds.
            - Add `gamble-1_spent_limit` as a new item in Cards - Rules Thresholds (learn how to add list items in [Create List Items]).
            - `Gamble-1_spent_limit"` is an element from the "Cards - Rules_Thresholds" list.
            - `.threshold` is the threshold stored in `gamble-1_spent_limit`.
        2. `SPENT_LIMITED_FORMATTED = SPENT_LIMIT/100.0;`
            - This variable will be used in the **Explanation** section. Since the amount is stored in cents, you need to convert it to dollars for better interpretation in Case Manager.
    5. **Condition**: `event.c_cardid_sum_amt_3h_mcc7800 >= SPENT_LIMIT and event.mcc == '7800'`.
    6. **Explanation**: [Cards-Gambling-1] - Decline - Gambling transactions (mcc=7800) over the past 3 hours are above the limit ($SPENT_LIMITED_FORMATTED).
    7. **Actions**: `Alert`, `[Cards-Gambling-1]`, `NO-OP` - you can choose to remove `Alert` in the shadow (NO-OP) mode.
    8. **Mark As**: `decline` (or leave it as `None` for shadow (NO-OP) mode).
    9. **Score contribution**: `None`
    10. **Control Flow**: Continue

Evaluate the right rules projectsâ

You can use conditions to control the rules that should be triggered for cards that are unlisted, listed to be declined, approved, or reviewed. Feedzai recommends processing the following rules projects when a card, transfer or digital event is in a certain list:

When card, transfer, or digital event is:	PosNeg rules	Fraud rules	Entity Limit rules
listed to be declined	yes	--	--
listed to be approved	yes	--	--
listed to be reviewed	yes	yes	yes
unlisted	yes	yes	--

Next stepsâ

Learn how cards, transfers and digital events are listed to be approved, declined, or reviewed in Pulse using *PosNeg lists*.

Alternatively, jump to Migrate Pulse Resources, in case you're ready to publish your rules to production, but need to import your changes to the **production environment** first.

Create Rules Projects

Was this page helpful?Â

A rules project is an ordered collection of rules that share similar traits. Grouping your rules in different rules projects allows you to:

- Keep your work organized.
- Set general configurations (e.g. a schema) that apply to all the rules that share similarities.
- Order rules projects in a workflow so that real-time events are first processed by rules with higher priority and then by rules with lower priority. Find more information about this on the Rules precedence section.

Create rules projectsâ

Since you are not allowed to change existing out-of-the-box rules, you need to create a new rules project first.

1. In the side panel, go to **Data Science**.
2. Scroll through the page until you get to the **Rules Projects** widget.
3. Select **New Rules Project**.

This is how you should set up a new rules project:

- **Name**: New rule project name.
- **Type**: Advanced.
- **Schema**: If you changed the out-of-the-box plan, select the schema you created in the previous step. Otherwise, select **Cards Workflow Schema v4**.
- **Confirm** this action.
    - If the schema has changed after you created the rules project, you need to update the schema, even if the schema names are the same.
- **Target Variable**: `decision`.
    - Do NOT select `fraud` as the target variable. It should only be `decision` in almost all cases.
- **State**: Leave it unchecked.
- **Actions**:
    - Add the default action codes `Alert`, `Limit/Review Card`, `Auto Decline Card`, `Auto Approve Card`, `NO-OP`.
    - You also need to create action codes for any new rules. This is because these codes will be displayed in Case Manager when rules are triggered.
        - Naming convention for cards: [Cards-RULE_NAME].
    - Any custom action code needed for Case Manager or other downstream consumption. Contact the Feedzai project team for this customization.
- **Multi-Tenancy**: Global.

To view other rules projects' configurations, or to change your own rules projects' configurations, select the **Configure** button.

Standard rules projectsâ

Transaction Fraud for Banking includes standard rules projects to help you get started:

(*) Fraud Rules are subdivided into several logics, i.e. velocity rules, threshold rules, behavior rules, and card-not-present rules (only applicable to the Cards rules projects).

Conceptually, rules projects are divided into the following main groups according to the event type that arrives to the workflow:

- Feedback events.
- Scorable events.
- All events, i.e. besides the aforementioned events, it includes Cards Clearings and Transfer Settlements.

info

A *Sanctions Rules Project* may also be included. Note that it is only relevant and active if you have Feedzai's Watchlist Transaction Screening solution enabled. It does not impact Transaction Fraud for Banking's performance or results.

Feedback eventsâ

Feedback events are evaluated by the following rules project:

- **Feedback Rules**: Contains rules that classify entities into decline or approve lists according to the analyst's feedback on the event. Examples of rules include:
  - **Feedback fraud**: Card, transfer or digital event is listed to be declined if the customer does not acknowledge the event or the analyst marks it as "Fraud".
  - **Feedback not fraud**: Card, transfer or digital event is listed to be approved if the customer acknowledges the event or the analyst marks it as "Not Fraud".

Scorable events

Scorable events are the events that return a decision outcome. For instance, Cards Authorization, Transfer Initiation, and Digital Activity endpoints return a decision outcome. Scorable events are evaluated by the following rules projects:

- **PosNeg Rules**: Adds entities to a PosNeg list to be approved, declined or reviewed.
- **Fraud Rules**: Contains rules that look for standard fraud patterns.
- **Entity Limit Rules**: Contains rules that allow you to limit the amount and count of transactions for cards and transfers.
- **Decision Rules**: Contains rules that have triggered based on the score produced by a data science model and on rules triggered in the previous blocks.
- **Device Intelligence**: Contains rules to monitor the relevant information about the device that is being used to perform a digital event.

All events

The following rules project is evaluated for all event types:

- **Self Service Rules**: Contains rules that are written in Case Manager by Fraud Managers and are processed in Pulse. Fraud Managers can use this functionality to write rules directly in production to quickly act on an emerging fraud pattern.

Rules precedence

An event can trigger several rules in multiple rules projects. It is important to understand what happens when an event triggers several rules that generate different outcomes. This precedence level in Pulse is explained below for rules that generate decisions and alerts.

Precedence in decisions

The following image shows the precedence level for rules that trigger an approve or decline decision. The label *triggered* in this example means that at least one rule in the rules project has triggered.

Rules precedence regarding decisions works as follows:

- **Scenario 1**: If no rule is triggered, the final outcome is **approve**.
- **Scenario 2**: When one of the rules with the decision decline is triggered, the final outcome is **decline**.
- **Scenarios 3 and 4**: When multiple rules with different decisions are triggered, the final outcome is the decision of the rule that has triggered first.

Precedence in alerts

The following image shows the precedence level for rules projects that trigger an alert:

Rules precedence regarding alerts works as follows:

- **Scenario 1**: If no rule is triggered, the transaction is not displayed in Case Manager's alert page. However, fraud analysts can still access the transaction in Case Manager's search page.
- **Scenario 2**: When one of the rules with the action alert is triggered, the transaction is displayed in Case Manager's alert page. You have access to the reason explaining why the rule was triggered.
- **Scenario 3**: When multiple rules with alert action are triggered, the transaction is displayed in Case Manager's alert page. You have access to the reason explaining why each rule was triggered.

Rules order

The order in which rules are processed can be defined inside each rules project. This allows you to define the order according to the priority of the outcome, i.e. rules with higher priority (e.g. hard decline) can be evaluated before rules with lower priority (e.g. hard approve).

Nextâ□□

Go to the next building block in the rule creation process:

- Create rules

Understand the Workflow

For clarity, this page uses the example of Cards API's workflow to explain the fundamentals of how it works.

tip

Make sure you read Pulse's core concepts to get a better picture of how Pulse works.

Note that all Authorization, Info, Clearing, and Feedback events pass through this workflow. However, not all events are needed for scoring. Therefore, this workflow contains two filters that discard events that aren't needed. Filters are displayed as blue-dashed lines between the blocks.

1. When any event first arrives, Feedzai applies the first filter, discarding all Clearing events.
2. The events will go through the Plan, which is where data is enriched / processed (i.e. where metrics / fields are created to be later used by rules).

3. Any feedback event resulting from fraud analysts' "fraud" or "not fraud" decisions, will pass through Feedback Rules.
    1. These events' entities (cards) will trigger the List Management Service, which will list them to be approved or declined.
4. The second filter is applied in PosNeg Rules to discard all entities listed to be declined.
5. Filtered events (i.e. Authorization events) will go through the remaining rules projects (and model).

Nextâ

Go to the next building block in the rule creation process:

- Change the Plan

Publish Rules to Runtime

warning

If you are working on a **pre-prod environment**, make sure you [migrate resources between environments]() first, i.e. export resources from pre-prod and import them into the production environment.

A workflow consists of processing steps (workflow services) and links between them. A workflow service is a Pulse component that maps to Pulse resources such as plans and rules, and performs their functionality in real time.

These services are configured using static versions of the resources developed in the **Data Science** tab (i.e. snapshots). This allows you to continue evolving resources in the **Data Science** tab while the workflow processes transactions with your current snapshots. Once the work in the data science tab is fully developed, you need to create new snapshots and update the workflow.

To get the best out of rules you need to understand the mapping between resources and the runtime workflow. Learn how to activate your resources in the Pulse runtime workflow to process transactions in real time.

Workflow mappingâ

The following diagram illustrates how Pulse concepts translate from resources to workflow services in runtime:

Standard Workflowâ

Transaction Fraud for Banking (TFB) includes a standard workflow for each API.

Workflow settingsâ

Use the **Workflow** tab to open each API's workflow.

- **Schema**: Workflow's input schema, which contains the API's available fields.
- **Outcomes**: Use this option to configure internal fields that will be used to store the outcome of rules and models. For instance:
  - `decision` : Standard field that is updated when a rule is triggered with the value defined by the **Mark as** option (see the example provided in the Rules page).
- **Outcome Combinator**: This field allows you to combine different scores in case you have more than one element in the workflow contributing to the score (i.e. you can have both models and rules contributing to the score).

info

Outcome fields: The outcome fields are combined with the input schema fields in a new schema named *Extended API Schema version*.

Workflow diagramâ

For your convenience, the following diagram belongs to Transfers API's workflow, which includes the majority of services among the remaining APIs.

- **Input Adapter**: Receives transactions provided by your system using the schema available in the TFB API. The output of the service is an *Extended API Schema version* with the internal fields defined in the **Outcomes** option.
- **Plans**: Computes the fields defined in the plan. Typical examples include profiles such as count of the number of transactions in the last 3 minutes. The input and output schemas are configured when you create a plan execution in the **Data Science** tab, and typically have the following names:
    - **Input Schema**: *Extended API Schema version*
    - **Output Schema**: *Workflow Schema version*
- **Rules**: Applies rules selected by rules projects' snapshots. The input schema of the rules service is the output schema of the last plan that precedes rule services (e.g. *Workflow Schema version*).
- **Custom Workflow Elements**: The standard workflow provides the following services:
    - **Expired Metrics Service**: Service related to metrics such as "how many transactions in the last x days". Due to limitations of real-time metric windows, each of these metrics is composed of a scheduled metric and a shorter real-time metric. They overlap to ensure events are not undercounted. If the job to calculate the schedules metric fails, this service marks the scheduled metric as expired. In this case, Strong Customer Authentication is enforced as it's unclear whether transactions will be undercounted. This ensures the system is compliant with PSD2 requirements.
    - **Workflow Monitoring Service**: Collects metrics regarding how many events passed through the workflow. This service creates a different bucket for each combination of values of all the configured event fields, outcomes and triggered actions.
    - **List Management Service**: Service that checks rule actions generated on each rules service and performs list item transitions as defined in the List Management Service.
    - **Alert Storage Service**: Sends a transaction to Case Manager when a rule is triggered with the `alert` Action.
    - **Outcome Priority Enforcer**: Custom service that looks at the outcome value of each triggered rule (i.e. high, low, and medium) to enforce the final decision outcome. This outcome is written to the `enforced_risk_level field`. Learn more about rule outcomes.
    - **Revelock (i.e. Digital Trust) Enricher Service**: Service that collects data related to device, network and behavior. It returns and processes this raw data to give data scientists indicators of fraud patterns such as account takeover and its threats (phishing, malware, suspicious network activity, among others). Data scientists can create rules, and build models to detect and prevent these fraud scenarios. Note that the **Revelock (i.e. Digital Trust) Enricher** service receives only `revelock` events, while the **Disabled Revelock (i.e. Digital Trust) Enricher** service receives only `non-revelock` events.
    - **Sanctions Enricher Service**: Service that handles the integration with third-party sanctions screening providers and collects screening results. This is only relevant if you have Feedzai's Watchlist Transaction Screening solution enabled. Otherwise, it is disabled, even if visible in the interface.
    - **Data Migration Service**: Service that stores received events into accessible storage in different environments. This service's purpose is to help data scientists conduct research on runtime events. It also converts received events into the Parquet format, thus organizing the information more efficiently.

Publish changes in workflow servicesâ

1. In the **Workflow** tab, edit each operator in the **diagram** of the *Cards Workflow* using the button:
    1. Select the resource and snapshots' latest version. For instance, if in the previous steps you created snapshots for plans and / or rules, use those snapshots in the workflow respective services (plan / rules project, respectively).
    2. **Save** your changes.
    3. **Save** and exit the workflow.
2. Publish the workflow using the **Publish (Live)** button.
3. Apply your changes to complete the process using the **Rolling Publish** button.
    1. Never skip recovery.
    2. Typically, changing schemas, rules, or plans does not require restarting the application process. If in doubt, reach out to Feedzai for help.

info

Feedzai recommends that you use the **Rolling Publish** option. This option publishes application changes on one Pulse replica at a time, thus reducing the downtime, as there may be other online replicas still able to handle events.

If during the creation of your rule, you used the shadow (NO-OP) mode, go back to editing your rule and take the following steps:

1. Remove `NO-OP` from **Actions** and add `Alert` (if needed).
2. **Mark As** `decline` (or `approve/review`), according to the definition of your rule.
3. **Save** the rule.
4. Create a new rule snapshot.
5. Repeat the steps to publish these changes in the workflow services.

Next stepsâ

Now that you know how to apply your rules in runtime to process transactions in real time, learn about testing your work via API.

- Operational Work
- Data Scientist & Fraud Strategist
- Work with Rules
- Send Test Transactions via API
- Check the Response via CM

On this page

Check the Response via Case Manager

Consider the case in which you triggered the Cards-Velocity-4 rule with five transactions in less than three minutes using the Card ID *fdz_testing_rules_1*. This generated a response with the alert action which displays the event in the Case Manager's alert page.

To check the result of this action, open Case Manager and execute the following actions:

1. Select the **Alerts** page.
2. Open the transaction that you have defined in the payload and check the following information:

- **System Decision**: Corresponds to the `decision` field.
- **Operational Status**: Corresponds to the analyst's decision (i.e. 'Fraud' or 'Not Fraud').
- **Rule Explanations**: Reasons why the rule has triggered. Note that this information is defined in the **Explanation** field in Pulse's rules configuration.
- Other values defined in the request payload:
- **Amount**: Transaction amount expressed using the same currency as specified in currency, in cents.
- **Currency**: The amount's currency.
- **Card ID**: Persistent unique card ID used to test the rule.

Next stepsâ

Check the response via Pulse.

-

- [Operational Work]
- [Data Scientist & Fraud Strategist]
- [Work with Rules]
- [Send Test Transactions via API]
- Check the Response via Pulse

On this page

Check the Response via Pulse

Consider the case in which you triggered the Cards-Velocity-4 rule with five transactions in less than three minutes using the Card ID *fdz_testing_rules_1*. This generated a [response] with an action code to list this card ID to be reviewed.

You can open the Pulse application to which you have submitted the request and check if the card was listed as expected:

1. Go to the **Lists** tab.
2. Open the *PosNeg Listed Cards*.
3. Confirm if two Card ID *fdz_testing_rules_1* items are enabled:

One item should contain the card ID tagged as **review** for a period of five days. While the other item should contain the card ID tagged as **decline**, with a date starting from the last review date. See the Lists page for more information about PosNeg lists.

Next stepsâ

Check the response via Case Manager.

- Operational Work
- Data Scientist & Fraud Strategist
- Work with Rules
- Send Test Transactions via API

On this page

Send Test Transactions Via API

Was this page helpful?Â

Learn how to send test transactions via API to test your rules with a default Cards velocity rule from Fraud rules project.

Sending requestsâ

tip

Tooling: Postman is an open source tool recommended to send test transactions via API.

Testing a rule via API using Postman assumes a JWKS and subsequent JWT have been generated, and requires the following:

1. Understand which environment and endpoint you want to submit the request to.
2. Create a payload including the fields that you want to test and the mandatory fields required by the endpoint.

3. Confirm that the sequence of test transactions that you would expect to trigger a rule is successful.

Choose the endpointâ

To test a rule, you can send a POST request to the [Authorization endpoint](). Each endpoint has recommended and mandatory fields. Make sure you use a payload with all the necessary fields. You can check the API Reference section to learn more about API fields.

Create a payloadâ

As an example, consider that you want to trigger the rule Cards-Velocity-4: *Number of transactions performed by card in the last 3 minutes exceeds threshold*, with a threshold equal to five transactions. Check the [Rules]() page for more information about rule Cards-Velocity-4.

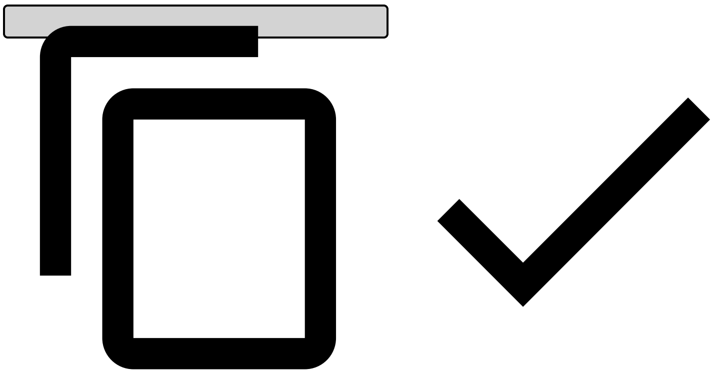
You can trigger this rule by sending five POST requests within three minutes using the following payload:

```
{
    "event_occurred_at": {{nowTimestamp}},
    "lifecycle_id": "{{nowTimestamp}}",
    "card_pan": "fdz_testing_rules_1",
    "card_id": "fdz_testing_rules_1",
    "amount": 1,
    "cardholder_bill_amount": 10,
    "mti": "mti",
    "ttc": "ttc",
    "currency_code": "usd",
    "pos_card_read_method": "pos_card_read_method",
    "mcc": "MCC",
    "acceptor_id": "12",
    "action_code": "01",
    "account_iban": "ex1",
    "customer_id": "fdz_testing_rules_1",
    "card_presence_flag": false,
    "acceptor_country_code": "US",
    "acceptor_zip": "test1234",
    "card_bin": "test1234"
}
```

Remember that the `event_occurred_at` field is the current timestamp. You can set an [environment variable]() to generate this timestamp using the following pre-request script:

Pre-request script to generate current timestamp automatically

```
// The Date.now() method returns the number of milliseconds elapsed since January 1, 1970
00:00:00 UTC.
pm.environment.set("nowTimestamp", Date.now());
```



tip

Lifecycle id: The `lifecycle_id` field should be different for each request that you submit. If you have an environment variable for `event_occurred_at`, you can use this variable in the `lifecycle_id` field to avoid manual change each time you submit a new request.

info

First Request: The first POST request after publishing your application might take longer (approximately one minute) than subsequent requests.

Check the API responseâ

Response - rules not triggeredâ

The first four POST requests with the payload provided in the previous section should not trigger the rule Cards-Velocity-4 because the number of test transactions is below the threshold. The API response should have the `decision` field as `approve`:

```
{
    "lifecycle_id": "12012018194035",
    "decision": "approve",
    "score": 0,
    "action_codes": [],
```

```
    "secondary_action_codes": [],
    "alert": false,
    "status": "ok"
}
```


Response - rules triggeredâ

When fifth request with the same card ID is submitted within the threshold (i.e. 3 minutes), the API response should have the `decision` field as `decline`:


```
{
    "lifecycle_id": "12012018194148",
    "decision": "decline",
    "score": 0,
    "action_codes": [
        "Limit/Review Card",
        "[Cards-Velocity-4]"
    ],
    "secondary_action_codes": [],
    "alert": true,
    "status": "ok"
}
```


- `action_codes` : Populated with the actions of the first triggered rule, other than the alert action. In this example, the actions include:
    - List the card to be reviewed.
    - The name of the rule that was triggered for debugging purposes.
- `secondary_action_codes` : Populated with actions of every other triggered rule (i.e. excluding the first triggered rule), other than the alert action. In this example, only the first rule was triggered.

tip

Secondary action codes and decision: The decision is set only by the first rule that is triggered as explained in the Rules Precedence section of the Rules Projects page.

- `alert`: In this example, the alert is displayed as `true`, and therefore, the event is sent to Case Manager.

Next stepsâ

Now that you have triggered rule *Cards-Velocity-4* as a test, learn how to confirm:

- The response via Pulse: Check if the list transition was successful.
- The response via Case Manager: Check if the event is displayed in Case Manager.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/cards/additional-resources/asynchronous-operation

-

- [API Resources]
- [Additional Resources]
- Cards
- Asynchronous Operation

Asynchronous Operation

Was this page helpful?Â

The Cards Connector supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the asynchronous processing of the request. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using configuration.

When invoking an endpoint using the asynchronous operation mode, request validations such as authentication, authorization and payload specific validations (e.g. schema validations and timestamp validations) are executed immediately. Any invalid request will fail immediately as it would in the synchronous operation mode.

Once validations are successfully passed, the request is sent to a message broker (such as RabbitMQ) where it waits to be processed by the system at a controlled rate. Once the request is successfully persisted in the message broker, the asynchronous request will terminate with an HTTP 202 Accepted response, denoting that the request will be eventually processed. Since the request was not fully processed yet, any response fields that result from processing the request (most notably the score, decision and triggered actions on the Authorization endpoint will not be present in this response.

Requests are consumed from the message broker at a configurable rate and execute the same operations as they would if they were synchronous. Once the request is completed, a response will be produced and stored in a message broker. This response is the same as the one that would have been produced in synchronous operation mode. You can use the Responses Get endpoint to retrieve the responses from the message broker.

The responses obtained from the Responses Get endpoint can be correlated to the original request using the `event_external_id` response field, which will have the same value that was sent in the homonym request field.

ATM Withdrawals

Was this page helpful?Â

To correctly map ATM withdrawals to card authorizations, and hence score ATM withdrawals, some fields need to be populated to identify ATM withdrawal events as follows:

```
{
   // Required fields
   "mti": "0110",
   "card_pan": "8955060600182663", // Which card was used
   "ttc": "001", // Type of transaction
   "currency_code": "EUR",
   "amount": 20,
   "cardholder_bill_amount": 20,
   "lifecycle_id": "1df7a7qg78vd7fs",
```

```
        "pos_card_read_method": "01",
        "event_occurred_at": 1684492270092,
        "card_bin": "123456",
        "account_iban": "482924bf58091603ed69c5bd1221ad67481bfca8",
        "customer_id": "87b2ee22cb0004722a7265fdbbf7be98c681a3bb",
    }
```


Read the [Authorization [POST]] documentation for more information on other fields you can use in the payload.

Authorization Response Scenarios

Was this page helpful?Â

This subsection presents several scenarios that might occur in the Authorization response, depending on the rules that are triggered in the Pulse Workflow. Check the Authorization Endpoint for more information about detailed request and response examples.

The following fields are always present in the Authorization response with their respective default values for the case that they are not triggered:

- `"alert": false` This field is changed to "true" when any rule with the action "alert" is triggered. See examples of possible scenarios below.

- `"action_codes": []` This field is populated with the actions of the first triggered rule, other than the alert action. Note that the alert action is represented by the `alert` field.
- `"secondary_action_codes":[]` This field is populated with actions of every other triggered rule (i.e. excluding the first triggered rule), other than the alert action.

When rules are triggered, they match the following scenarios:

Scenario 1â

- **The first triggered rule:**
  - Action field - Alert
- **Excerpt from the Authorization response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = []
}
```

Scenario 2â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Authorization response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = []
}
```

Scenario 3â

- **The first triggered rule:**
  - Action field - Alert
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Authorization response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Scenario 4â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Authorization response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Scenario 5â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3

- **The third triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 4
- **Excerpt from the Authorization response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3", "Action 4"]
}
```

Scenario 6â

- **The first triggered rule:**
    - This rule doesn't have actions.
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Authorization response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Generated Fields

The Cards API generates some fields which will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Authorization	authorization

Endpoint	event_type
Info	info
Clearing	clearing
Feedback	feedback

event_received_atâ

The `event_received_at` represents the time the request reached the Cards API as a timestamp, in milliseconds.

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/api-reference/digital-activity/additional-resources/asynchronous-operation

- API Resources
- Additional Resources
- Digital Activity
- Asynchronous Operation

Asynchronous Operation

Was this page helpful?Â

The Digital Activity Connector is configured with synchronous operation out of the box. However, it also supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the request's asynchronous processing. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using the configuration.

When invoking an endpoint that is using the asynchronous operation mode, request validations such as authentication (e.g. schema validations and timestamp validations) are executed immediately. Any invalid request will fail immediately as it would in the synchronous operation mode.

Authentication Response Scenarios

This section presents several scenarios that might occur in the Digital Activity's Authentication response, depending on the rules that are triggered in the Pulse workflow. Check the Authentication endpoint for more information about detailed request and response examples.

When no rules are triggered, the following fields and default values are shown in the Authentication endpoint's response:

- `"alert": false` This field is changed to 'true' when any rule with the action 'alert' is triggered. See examples of possible scenarios below.

- `"action_codes": []` This field is populated with the actions of the first triggered rule, other than the 'alert' action. Note that the 'alert' action is represented by the `alert` field.
- `"secondary_action_codes":[]` This field is populated with all the non-'alert' actions of every triggered rule, except the first triggered rule.

When rules are triggered, they match the following scenarios:

Scenario 1â

- **The first triggered rule:**
  - Action field - Alert
- **Excerpt from the Digital Activity's Authentication response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = []
}
```

Scenario 2â

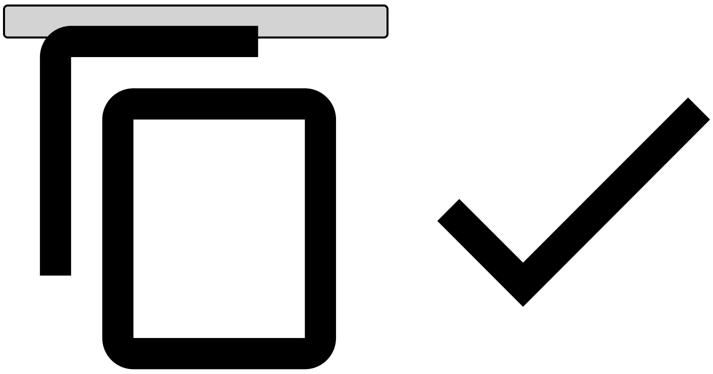
- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = []
}
```

Scenario 3â

- **The first triggered rule:**
  - Action field - Alert
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

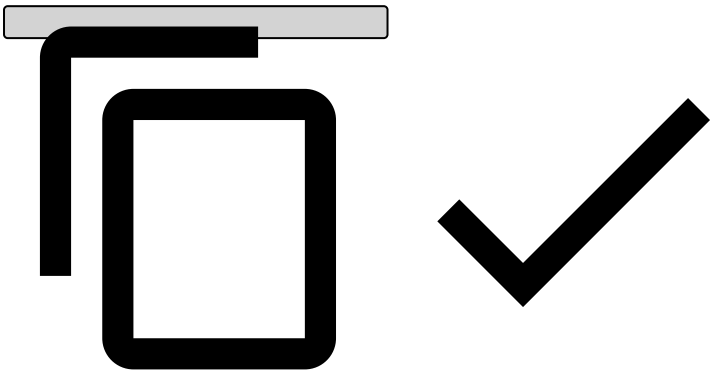
```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```



Scenario 4â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```



Scenario 5â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3

- **The third triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 4
- **Excerpt from the Digital Activity's Authentication response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3", "Action 4"]
}
```

Scenario 6â

- **The first triggered rule:**
    - This rule doesn't have actions.
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Digital Activity's Authentication response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

-
- [API Resources]
- [Additional Resources]
- Digital Activity
- BACEN Fraud Specific Fields

BACEN Fraud Specific Fields

Was this page helpful?Â

The Brazilian Central Bank (BACEN) allows banks to consult fraud records from their customers through DiretÃ³rio de Identificadores de Contas Transacionais (DICT), which provides them with the fraudster history.

DICT is a database that stores the information of recipient users and their transactional accounts. It is the component that enables payment initiation in a practical and fraud risk mitigating manner.

Digital Activity includes the following DICT fields so that events can be enriched by customers and help prevent fraud. These fields are available in all Digital Activity endpoints (all `optional`).

Field	Description
customer_alias_settle_3d	Number of aliases' settlements over the last 3 days

Field	Description
customer_alias_settle_30d	Number of aliases' settlements over the last 30 days
customer_alias_settle_6m	Number of aliases' settlements over the last 6 months
customer_settle_3d	Number of customers' settlements over the last 3 days
customer_settle_30d	Number of customers' settlements over the last 30 days
customer_settle_6m	Number of customers' settlements over the last 6 months
customer_account_settle_3d	Number of accounts' settlements over the last 3 days
customer_account_settle_30d	Number of accounts' settlements over the last 30 days
customer_account_settle_6m	Number of accounts' settlements over the last 6 months
customer_alias_rep_fraud_3d	Number of aliases' reported frauds over the last 3 days
customer_alias_rep_fraud_30d	Number of aliases' reported frauds over the last 30 days
customer_alias_rep_fraud_6m	Number of aliases' reported frauds over the last 6 months
customer_rep_fraud_3d	Number of customers' reported frauds over the last 3 days
customer_rep_fraud_30d	Number of customers' reported frauds over the last 30 days
customer_rep_fraud_6m	Number of customers' reported frauds over the last 6 months
customer_account_rep_fraud_3d	Number of accounts' reported frauds over the last 3 days
customer_account_rep_fraud_30d	Number of accounts' reported frauds over the last 30 days
customer_account_rep_fraud_6m	Number of accounts' reported frauds over the last 6 months
customer_alias_fraud_3d	Number of aliases' confirmed frauds over the last 3 days
customer_alias_fraud_30d	Number of aliases' confirmed frauds over the last 30 days
customer_alias_fraud_6m	Number of aliases' confirmed frauds over the last 6 months
customer_fraud_3d	Number of customers' confirmed frauds over the last 3 days
customer_fraud_30d	Number of customers' confirmed frauds over the last 30 days
customer_fraud_6m	Number of customers' confirmed frauds over the last 6 months
customer_account_fraud_3d	Number of accounts' confirmed frauds over the last 3 days
customer_account_fraud_30d	Number of accounts' confirmed frauds over the last 30 days
customer_account_fraud_6m	Number of accounts' confirmed frauds over the last 6 months
customer_alias_rejected_3d	Number of aliases' rejected transactions over the last 3 days
customer_alias_rejected_30d	Number of aliases' rejected transactions over the last 30 days
customer_alias_rejected_6m	Number of aliases' rejected transactions over the last 6 months
customer_rejected_3d	Number of customers' rejected transactions over the last 3 days
customer_rejected_30d	Number of customers' rejected transactions over the last 30 days
customer_rejected_6m	Number of customers' rejected transactions over the last 6 months
customer_account_rejected_3d	Number of accounts' rejected transactions over the last 3 days
customer_account_rejected_30d	Number of accounts' rejected transactions over the last 30 days
customer_account_rejected_6m	Number of accounts' rejected transactions over the last 6 months
customer_alias_dict_reg_date	Date of registry on DICT

Generated Fields

The Digital Activity API generates fields that will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Alias	alias

Endpoint	event_type
Account Update	account_update
Authentication	authentication
Card Issuing	card_issuing
Card Update	card_update
Credentials Update	credentials_update
Device	device
New Payee	new_payee
Profile Update	profile_update
Info	info

event_received_atâ

The `event_received_at` represents the time the request reached the Digital Activity API as a timestamp, in milliseconds.

- API Resources
- Additional Resources
- Digital Activity
- Responses and Errors

On this page

Responses and Errors

Was this page helpful?Â

When a client makes a request to an HTTP server and the server successfully receives their request, the server notifies the client if the request was successfully processed or not. Read this page to understand what each response and error means.

Successful responseâ

Successful calls to Feedzai's API result in a response with HTTP 200 code. Check each Digital Activity endpoint to see the structure of a successful response.

Warning responseâ

A warning response is a subtype of a successful response which is sent with the HTTP 200 code. The transaction is also processed. The response returns:

- `status` field with the response 'warning'.
- `warnings` field with the reasons for this outcome.

Error responseâ

When a call to Feedzai's API results in an error, the response object returned has always the same fields. In particular, the `status` field is shared with the successful response objects of all endpoints. The **Error Response** object is defined as follows:

Name	Field Type	Description
`code`	String	An indicative code for the error. See possible values below.
`errors`	Array of strings	A list of free text explanations for the error.
`status`	String	Always with value `error`.

Generic error codes - possible valuesâ

Error code	Description	HTTP Code
`authenticationError`	The request could not be authenticated.	`401`
`authorizationError`	The authenticated user does not have permission to access the endpoint.	`403`
`parseError`	The request could not be parsed.	`400`
`validationError`	The request failed to pass payload validations.	`400`
`invalidTimestampError`	The timestamp provided in the request is invalid.	`400`
`nonexistentEndpoint`	The endpoint being called does not exist.	`404`
`unsupportedMethod`	The HTTP method to call this endpoint is not supported.	`405`
`internalError`	Feedzai's system produced an internal error.	`500`
`timeoutError`	Feedzai's system failed to produce a timely response.	`504`

Learn moreâ

Check each Digital Activity endpoint for specific error codes.

Asynchronous Operation

Was this page helpful?Â

The Transfers Connector supports asynchronous operation. This page covers the foundational aspects to help you with the integration process.

Non-get endpoints support an `async` query parameter which, if `true`, will trigger the asynchronous processing of the request. If the `async` query parameter is set to false, the system will assure the synchronous operation. The default value for the `async` query parameter can be defined per endpoint using configuration.

When invoking an endpoint using the asynchronous operation mode, request validations such as authentication, authorization and payload specific validations (e.g. schema validations and timestamp validations) are executed immediately. Any invalid request will fail immediately as it would in the synchronous operation mode.

Once validations are successfully passed, the request is sent to a message broker (such as RabbitMQ) where it waits to be processed by the system at a controlled rate. Once the request is successfully persisted in the message broker, the asynchronous request will terminate with an HTTP 202 Accepted response, denoting that the request will be eventually processed. Since the request was not fully processed yet, any response fields that result from processing the request (most notably the score, decision and triggered actions on the Transfer Initiation endpoint will not be present in this response.

Requests are consumed from the message broker at a configurable rate and execute the same operations as they would if they were synchronous. Once the request is completed, a response will be produced and stored in a message broker. This response is the same as the one that would have been produced in synchronous operation mode. You can use the Responses Get endpoint to retrieve the responses from the message broker.

The responses obtained from the Responses Get endpoint can be correlated to the original request using the `event_external_id` response field, which will have the same value that was sent in the homonym request field.

Generated Fields

Was this page helpful?Â

The Transfers API generates some fields which will be sent to the Pulse Workflow alongside the ones received in the API.

event_typeâ

The `event_type` will be set according to the endpoint used to receive the request:

Endpoint	event_type
Initiation	transfer_initiation

Endpoint	event_type
Info	info
Settlement	settlement
Feedback	feedback

event_received_atâ

The `event_received_at` represents the time the request reached the Transfers API as a timestamp, in milliseconds.

-
- [API Resources]
- [Additional Resources]
- Outbound Transfers
- Transfer Initiation Response Scenarios

On this page

Transfer Initiation Response Scenarios

Was this page helpful?Â

This section presents several scenarios that might occur in the Transfer Initiation response, depending on the rules that are triggered in the Pulse workflow. Check the Transfer Initiation Endpoint for more information about detailed request and response examples.

When no rules are triggered, the following fields and default values are shown in the Transfer Initiation response:

- `"alert": false` This field is changed to 'true' when any rule with the action 'alert' is triggered. See examples of possible scenarios below.

- `"action_codes": []` This field is populated with the actions of the first triggered rule, other than the 'alert' action. Note that the 'alert' action is represented by the `alert` field.
- `"secondary_action_codes":[]` This field is populated with all the non-'alert' actions of every triggered rule, except the first triggered rule.

When rules are triggered, they match the following scenarios:

Scenario 1â

- **The first triggered rule:**
  - Action field - Alert
- **Excerpt from the Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = []
}
```

Scenario 2â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = []
}
```

Scenario 3â

- **The first triggered rule:**
  - Action field - Alert
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Scenario 4â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Scenario 5â

- **The first triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3
- **The second triggered rule:**
  - Action field - Alert + Action 1 + Action 2 + Action 3

- **The third triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 4
- **Excerpt from the Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = ["Action 1", "Action 2", "Action 3"],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3", "Action 4"]
}
```

Scenario 6â

- **The first triggered rule:**
    - This rule doesn't have actions.
- **The second triggered rule:**
    - Action field - Alert + Action 1 + Action 2 + Action 3
- **Excerpt from the Transfer Initiation response:**

```
 {
"alert": true,
"action_codes" = [],
"secondary_action_codes" = ["Action 1", "Action 2", "Action 3"]
}
```

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/life-cycles/cards/cards-alert-life-cycles/alert

- About TFB
- Event Life Cycles
- Cards
- Alert Life Cycle

On this page

Alert Life Cycle

Was this page helpful?Â

Learn about the alert life cycle with a focus on the process within Case Manager. This page helps fraud analysts and fraud managers to understand how alerts associated with each endpoint (i.e. authorization, info, clearing and fraud feedback) are displayed in Case Manager. Check the Card Transaction Life Cycle page to know more about the endpoints and their role within the transaction use case.

Alertsâ

An alert is a special type of event that is created when an event triggers a risk engine rule with the `alert` action, flagging a higher fraud risk. As a consequence, Case Manager displays the alert in the Alerts page. This page is where fraud analysts label alerting transactions as 'Fraud' or 'Not Fraud'.

Life cycleâ

This section describes the alert life cycle for each endpoint, starting with the API call, going through Pulse and finally being sent to Case Manager.

Authorization endpointâ

The following image shows the life cycle of an event corresponding to an Authorization request:

This life cycle is characterized by the following flow:

1. A customer sends an Authorization message via the Transaction Fraud for Banking API.
2. The risk engine platform processes the transaction and sets the `alert` field as 'true' or 'false'.
3. The event/alert is sent to Case Manager and is displayed according to the following rules:

- If the `alert` field is 'false', then the event is displayed in the **Search by Event** page.
- If the `alert` field is 'true', then both the **Search by Event** and the **Alerts** pages show the event. The image also shows a typical journey of an alert in Case Manager:
  (1) The alert goes into a queue.
  (2) The alert is analyzed by an agent, typically a fraud analyst.
  (3) The fraud analyst classifies the alert as 'Fraud' or 'Not Fraud'.
  (4) The event status field is updated with the decision.

Info endpointâ

The following image represents the alert life cycle for the situation in which you call the Info endpoint in addition to the Authorization request.

When information is sent via the Info endpoint, this data is updated in the alert detail's page of the corresponding authorization. Therefore, Case Manager does not generate a new entry in the list of alerts for this Info request. This is represented in the figure by the dashed line.

Clearing endpointâ

The following figure represents the life cycle for a Clearing request:

This type of request generates a new entry in the Case Manager's alerts list. This is represented in the figure by the new 'Case Manager Event = B'.

info

Populate only clearing fields: The information sent in the Clearing request is used to populate only the fields in the page details of the clearing event (i.e. is not used to populate fields in the authorization event).

Transactions may have multiple Clearing requests as shown in the following figure:

Feedback endpointâ

There are two scenarios associated with the Feedback request:

- **Via Case Manager:** Occurs when a fraud analyst reviews the transaction in Case Manager. This behavior is represented in the authorization diagram by the "Agent analyzes the event" step.
- **Via API:** Occurs when the cardholder disputes the transaction either with the issuer or the acquirer. Check the Feedback page to learn more about this scenario.

Via APIâ

The following image shows the diagram of the alert life cycle for the Feedback request via API:

In the previous figure, the Feedback request is represented by a dashed line because the information is updated in the same event entry generated by the Authorization request ("Case Manager Event = A").

A Feedback request can be followed by multiple Clearings, followed by new Feedback requests. The following figure illustrates two Clearings and two Feedbacks requests, however, these two endpoints can be called multiple times.

In the previous figure, the second Feedback request is represented by a dashed line for the same reason as explained in the case of a simple Feedback. Additionally, this figure shows that the information sent via the second Feedback request (e.g. event status) is propagated to all the existing clearings associated with the same life cycle ID.

Learn moreâ

Now that you understand the alert life cycle, you can check the following Case Manager guide's sections:

- Investigate Alerts: Learn how to manually review suspicious transactions to be able to identify and prevent fraudulent behaviors.
- Manage Workload and Rules Guide: Learn how to access analytics dashboards and set up Case Manager to keep the workload of fraud teams organized, preventing fraudulent behavior efficiently.

Alert Life Cycle

Was this page helpful?Â

Learn about the alert life cycle with a focus on the process within Case Manager. This page helps fraud analysts and fraud managers to understand how alerts associated with each endpoint (i.e. Transfer Initiation, Info, Settlement and Fraud Feedback) are displayed in Case Manager. Check the Life Cycle by Processing Model page to know more about the endpoints and their role within the transaction use case.

Alertsâ

An alert is a special type of event that is created when an event triggers a risk engine rule with the `alert` action, flagging a higher fraud risk. As a consequence, Case Manager displays the alert in the Alerts page. This page is where fraud analysts label alerting transactions as 'Fraud' or 'Not Fraud'.

Life cycleâ

This section describes the alert life cycle for each endpoint, starting with the API call, going through Pulse and finally being sent to Case Manager.

Transfer initiation endpointâ

The following image shows the life cycle of an event corresponding to a transfer initiation request:

This life cycle is characterized by the following flow:

1. A customer sends a transfer initiation request via the Transaction Fraud for Banking API.
2. Feedzai's risk engine processes the transaction and sets the `alert` field as 'true' or 'false'.
3. The event/alert is sent to Case Manager and is displayed according to the following rules:

- If the `alert` field is 'false', then the event is displayed in the **Search by Event** page.
- If the `alert` field is 'true', then both the **Search by Event** and the **Alerts** pages show the event. The image also shows a typical journey of an alert in Case Manager:
  (1) The alert goes into a queue.
  (2) The alert is analyzed by an agent, typically a fraud analyst.
  (3) The fraud analyst classifies the alert as 'Fraud' or 'Not Fraud'.
  (4) The event status field is updated with the decision.

Info endpointâ

The following image represents the alert life cycle for the situation in which you call the Info endpoint in addition to the transfer initiation request.

When information is sent via the Info endpoint, this data is updated in the alert detail's page of the corresponding transfer initiation. Therefore, Case Manager does not generate a new entry in the list of alerts for this Info request. This is represented in the figure by the dashed line.

Settlement endpointâ

The following figure represents the life cycle for a settlement request:

This type of request generates an update to the original transfer initiation event in the Case Manager's alerts list. This is represented in the figure by the new 'Case Manager Event = B'.

info

Populate only settlement fields: The information sent in the settlement request is used to populate fields in the transfer initiation event.

Feedback endpointâ

There are two scenarios associated with the Feedback request:

- **Via Case Manager:** Occurs when a fraud analyst reviews the transaction in Case Manager. This behavior is represented in the transfer initiation diagram by the "Agent analyzes the event" step.
- **Via API:** Occurs when the cardholder disputes the transaction either with the issuer or the acquirer. Check the Feedback section to learn more about this scenario.

Via APIâ

The following image shows the diagram of the alert life cycle for the feedback request via API:

In the previous figure, the feedback request is represented by a dashed line because the information is updated in the same event entry generated by the transfer initiation request ("Case Manager Event = A").

A feedback request can be followed by new feedback requests. The following figure illustrates two Feedbacks requests, however, this endpoint can be called multiple times.

In the previous figure, the second feedback request is represented by a dashed line for the same reason as explained in the case of a simple feedback. Additionally, this figure shows that the information sent via the second feedback request (e.g. event status) is propagated to all the existing settlements associated with the same life cycle ID.

Learn moreâ

Now that you understand the alert life cycle, you can check the following Case Manager user guides:

- Investigate Alerts Guide: Learn how to manually review suspicious transactions to be able to identify and prevent fraudulent behaviors.
- Manage Workload and Rules Guide: Learn how to access analytics dashboards and set up Case Manager to keep the workload of fraud teams organized, preventing fraudulent behavior efficiently.

Case Manager Case

Was this page helpful?Â

The **Case Manager - Case** entity stores information from Case Manager cases.

Name	Type	Description	Example
`timestamp`	timestamp with time zone	The original timestamp of the object	2025-09-30 10:25:59.100 +0100
`identifier`	string	The original ID of the object	605e7f99-820b-45b2-ab5e-a4420dae675a
`channel`	string	The channel to which this case is valid for.	cards
`case_type`	string	The type of the case	AML, Fraud

Name	Type	Description	Example
`tenant`	string	The tenant of the case	feedzai
`notification_timestamp`	timestamp with time zone	The timestamp of when this notification took place	2025-09-30 10:25:59.100 +0100
`notification_type`	string	The notification type. Is one of the following: • CASE_CREATE: The case is created • CASE_UPDATE: The case was updated • CASE_EVENT: The case was linked/unlinked with events • ADD_NOTE: A note was added to the case • DELETE_NOTE: A note was removed from the case • CASE_QUEUE: The case was added/removed to/from a queue • CASE_STATE: The case state was changed • CASE_USER: The case was assigned/ unassigned to/from a user	
`user_id`	string	The user id of the user that performed the action	
`user_name`	string	The user name of the user that performed the action	
`queue_id`	string	Id of the queue. This is only filled when on case queue.	
`queue_name`	string	Name of the queue. This is only filled when on case queue.	
`state_id`	string	State identifier: created, Update or State.	
`name`	string	The case name.	
`description`	string	The case description.	
`source_type_id`	string	The source type id	
`source_type_name`	string	The source type name	
`operation`	string	Contains the operation type (ADD/REMOVE). This is only filled when the action is of type CASE_EVENT	
`event_ids`	array<string>	- The list of event ids that were affected by the adding or removing to or from the event. This is only filled when the action is of type CASE_EVENT	
`note_info`	row		
â□□â□□â□□ `note`	string	The note of the case.	
â□□â□□â□□ `note_id`	string	The case note identifier.	
â□□â□□â□□ `note_type_id`	string	Case Note Type identifier	
â□□â□□â□□ `created_at`	timestamp with time zone	When this note was created	2025-09-30 10:25:59.100 +0100
â□□â□□â□□ `created_by`	string	The user that created the note	
â□□â□□â□□ `deleted_at`	timestamp with time zone	When this note was deleted	2025-09-30 10:25:59.100 +0100
â□□â□□â□□ `deleted_by`	string	The user that deleted the note	
`custom_fields`	json	All the custom fields that the case might have.	{"custom_field_1": "value1", "custom_field_2": "value2"}

Name	Type	Description	Example
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/references/data-exporting-tool/case-manager/cm-event

-

- References
- Data Exporting Tool
- Case Manager
- Case Manager Event - Base Schema

Case Manager Event - Base Schema

Was this page helpful?Â

The **Case Manager Event - Base Schema** stores the basic information of Case Manager events.

All Case Manager Event schemas add to this base schema and will be available when extracting information for each specific channel (i.e. Cards, Transfers, Digital Activity or Payments).

Name	Type	Description	Example
`timestamp`	timestamp with time zone	The original timestamp of the object	2025-09-30 10:25:59.100 +0100
`identifier`	string	The original ID of the object	605e7f99-820b-45b2-ab5e-a4420dae675a

Name	Type	Description	Example
`notification_timestamp`	timestamp with time zone	The timestamp of when this notification took place	2025-09-30 10:25:59.100 +0100
`notification_type`	string	The notification type. Is one of the following: • EVENT_ARRIVAL: Indicates this is an event arrival • EVENT_STATUS: Indicates this is a status change on the event • EVENT_USER: Indicates a user was assigned/unassigned to/from the event • EVENT_QUEUE: Indicates the event was added/removed to/from a queue • EVENT_NOTE: Indicates a note was added/removed to/from an event • EVENT_RELATED_TRANSACTIONS: Indicates that related transactions were added/removed to/from the event	
`event_version_id`	string	The unique ID for a specific version of an event, used to track updates or state changes.	4c168e62-48af-4cf3-8948-7e93b32a3f9c
`event_version_timestamp`	bigint	The timestamp of the specific event version.	1588700503112
`event_tenant`	string	The tenant of the event.	feedzai
`is_alert`	boolean	Indicates whether this event is an alert or not	
`access_group_id`	string	The ID for the group of users that can view the event, used to control data visibility.	europe_analysts
`user_id`	string	The Id of the user that made the change. This is only available for event arrival, event queue, event status and event user notification types.	
`user_name`	string	The Id of the user that made the change. This is only available for event note, event status and event user.	
`queue_id`	double	The id of the queue the event is in This is only available for event arrival, event queue, event status and event user.	
`queue_name`	string	The name of the queue the event was added to. This only exists for event queue.	
`state`	string	The event status. This only exists for event status.	
`state_machine_id`	string	The id of the state machine that changed. This only exists on event status.	
`note`	string	The note added to this event. This only exists on event note and event status.	
`reasons`	array<row>	The list of the status and their corresponding value. This only exists on event status.	
`â□□â□□â□□ id`	string	The ID of the status reason provided for the status change	975c5ab6-bd2f-4567-8ae3-6026742f4036
`â□□â□□â□□ status_id`	string	The status of this status change	fraud
`â□□â□□â□□ name`	string	The human-readable name of the status reason provided for the status change	Confirmed Fraud

Name	Type	Description	Example
`state_machines`	map<string, string>	The map of state machines and their corresponding value. This only exists on event arrival, queue, status and user.	
`related_transactions`	json	The list of related transactions for the event. This only exists on EVENT_RELATED_TRANSACTIONS type.	{"rule_001":["{"identifier":"txn_003", "channel":"payments", "timestamp":"2025-10-29 17:12:07.844000 UTC"}"],"rule_002":["{"identifier":"txn_004", "channel":"payments", "timestamp":"2025-10-29 15:27:07.844000 UTC"}"]}
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Case Manager Event - Ad-hoc Customer Investigations

Was this page helpful?Â

The **Case Manager Event - Ad-hoc Customer Investigations** entity stores all events from the Ad-hoc Customer Investigations channel.

It comprises the [Case Manager Event - Base Schema] and the following fields:

Name	Type	Description	Example
`event_payload`	row	The event payload for the adhoc customer investigations event.	
â□□â□□â□□ `lifecycle_id`	string		

Name	Type	Description	Example
â□□â□□â□□ `customer_id`	string		
â□□â□□â□□ `customer_city`	string		
â□□â□□â□□ `customer_country_code`	string		
â□□â□□â□□ `customer_email`	string		
â□□â□□â□□ `customer_name`	string		
â□□â□□â□□ `customer_phone`	string		
â□□â□□â□□ `customer_region`	string		
â□□â□□â□□ `customer_street_addr`	string		
â□□â□□â□□ `customer_zip`	string		
â□□â□□â□□ `customer_dob`	string		
â□□â□□â□□ `customer_occupation`	string		
â□□â□□â□□ `customer_segment`	string		
â□□â□□â□□ `is_client`	boolean		
â□□â□□â□□ `outcome_decision`	string		
â□□â□□â□□ `origin`	string		
â□□â□□â□□ `reason_code`	string		
â□□â□□â□□ `description`	string		
â□□â□□â□□ `event_external_id`	string		
â□□â□□â□□ `event_occurred_at`	double		
â□□â□□â□□ `event_received_at`	double		
â□□â□□â□□ `event_type`	string		
â□□â□□â□□ `feedback_label`	string		
â□□â□□â□□ `feedback_type`	string		
â□□â□□â□□ `feedback_value`	string		
â□□â□□â□□ `info_type`	string		
â□□â□□â□□ `tenant`	string		
â□□â□□â□□ `source`	string		
â□□â□□â□□ `req_9483_review_trigger_type`	string		
â□□â□□â□□ `req_9483_screening_tags`	string		
â□□â□□â□□ `req_9483_sanctions_outcome_decision`	string		
â□□â□□â□□ `score`	double		
â□□â□□â□□ `req_9483_session_id`	string		
â□□â□□â□□ `req_9483_system_risk_rating`	string		
â□□â□□â□□ `req_9483_system_risk_rating_color`	string		
â□□â□□â□□ `req_9483_system_score`	double		
â□□â□□â□□ `req_9483_total_number_of_hits`	double		
â□□â□□â□□ `req_9483_max_hit_score`	double		

Name	Type	Description	Example
â□□â□□â□□ `req_9483_sanctions_alert`	boolean		
â□□â□□â□□ `req_9483_merchant_country_code`	string		
â□□â□□â□□ `req_9483_merchant_id`	string		
â□□â□□â□□ `req_9483_merchant_mcc`	string		
â□□â□□â□□ `req_9483_merchant_name`	string		
â□□â□□â□□ `req_9483_merchant_url`	string		
â□□â□□â□□ `req_9483_amount`	double		
â□□â□□â□□ `req_9483_beneficiary_name`	string		
â□□â□□â□□ `req_9483_currency_code`	string		
â□□â□□â□□ `req_9483_device_id`	string		
â□□â□□â□□ `req_9483_alert_reasons`	string		
â□□â□□â□□ `req_9483_aml_version`	string		
â□□â□□â□□ `req_9483_parent_merchant_id`	string		
â□□â□□â□□ `prodserv_148_originator_name`	string		
â□□â□□â□□ `req_9483_alert_close_at`	double		
â□□â□□â□□ `req_9483_onboarding_origin`	string		
â□□â□□â□□ `req_9483_assessment_type`	string		

Case Manager Event - Cards

The **Case Manager Event - Cards** entity stores all events from the Cards channel.

It comprises the Case Manager Event - Base Schema and the following fields:

Name	Type	Description	Example
`event_payload`	json	The event payload for the cards event.	{ "merchant_city": "New York", "merchant_country_code": "US", "merchant_legal_name": "Merchant Legal Name", "merchant_id": "1234567890", "merchant_mastercard_risk": 1 }
âââ `merchant_city`	string		

Name	Type	Description	Example
â□□â□□â□□ `merchant_country_code`	string		
â□□â□□â□□ `merchant_legal_name`	string		
â□□â□□â□□ `merchant_id`	string		
â□□â□□â□□ `merchant_mastercard_risk`	integer		
â□□â□□â□□ `merchant_name`	string		
â□□â□□â□□ `merchant_open_date`	bigint		
â□□â□□â□□ `merchant_risk`	integer		
â□□â□□â□□ `merchant_risky_flag`	boolean		
â□□â□□â□□ `merchant_street_addr`	string		
â□□â□□â□□ `merchant_visa_risk`	integer		
â□□â□□â□□ `merchant_zip`	string		
â□□â□□â□□ `account_available_credit`	bigint		
â□□â□□â□□ `account_balance`	bigint		
â□□â□□â□□ `account_bic`	string		
â□□â□□â□□ `account_credit_limit`	bigint		
â□□â□□â□□ `account_iban`	string		
â□□â□□â□□ `account_id`	string		
â□□â□□â□□ `account_last_activity`	bigint		
â□□â□□â□□ `account_number_of_cards`	integer		
â□□â□□â□□ `account_open_date`	bigint		
â□□â□□â□□ `account_status`	string		
â□□â□□â□□ `acquirer_id`	string		
â□□â□□â□□ `action_code`	string		
â□□â□□â□□ `amount`	bigint		
â□□â□□â□□ `atc1`	string		

Name	Type	Description	Example
â□□â□□â□□ `atc2`	string		
â□□â□□â□□ `card_bin`	string		
â□□â□□â□□ `card_brand`	string		
â□□â□□â□□ `card_category`	string		
â□□â□□â□□ `card_credit_limit`	bigint		
â□□â□□â□□ `card_effective_date`	string		
â□□â□□â□□ `card_expiration_date`	string		
â□□â□□â□□ `card_id`	string		
â□□â□□â□□ `card_last4`	string		
â□□â□□â□□ `card_pan`	string		
â□□â□□â□□ `card_presence_flag`	boolean		
â□□â□□â□□ `card_status`	string		
â□□â□□â□□ `card_type`	string		
â□□â□□â□□ `cardholder_bill_amount`	bigint		
â□□â□□â□□ `currency_code`	string		
â□□â□□â□□ `customer_city`	string		
â□□â□□â□□ `customer_country_code`	string		
â□□â□□â□□ `customer_email`	string		
â□□â□□â□□ `customer_id`	string		
â□□â□□â□□ `customer_name`	string		
â□□â□□â□□ `customer_phone`	string		
â□□â□□â□□ `customer_region`	string		
â□□â□□â□□ `customer_street_addr`	string		
â□□â□□â□□ `customer_zip`	string		
â□□â□□â□□ `device_id`	string		
â□□â□□â□□ `event_occurred_at`	bigint		

Name	Type	Description	Example
â□□â□□â□□ `last_sca_at`	bigint		
â□□â□□â□□ `lifecycle_id`	string		
â□□â□□â□□ `mcc`	string		
â□□â□□â□□ `pos_card_capture_cap`	string		
â□□â□□â□□ `pos_cardholder_ver_cap`	string		
â□□â□□â□□ `pos_read_cap`	string		
â□□â□□â□□ `secure_3d_val_code`	string		
â□□â□□â□□ `session_id`	string		
â□□â□□â□□ `sic_code`	string		
â□□â□□â□□ `terminal_city`	string		
â□□â□□â□□ `terminal_country_code`	string		
â□□â□□â□□ `terminal_id`	string		
â□□â□□â□□ `terminal_latitude`	double		
â□□â□□â□□ `terminal_longitude`	double		
â□□â□□â□□ `terminal_poc_flag`	string		
â□□â□□â□□ `terminal_region`	string		
â□□â□□â□□ `terminal_street_addr`	string		
â□□â□□â□□ `terminal_zip`	string		
â□□â□□â□□ `ttc`	string		
â□□â□□â□□ `capture_date`	string		
â□□â□□â□□ `transaction_status`	string		
â□□â□□â□□ `dispute_end_date`	string		
â□□â□□â□□ `dispute_feedback_notes`	string		
â□□â□□â□□ `dispute_refunded`	string		
â□□â□□â□□ `feedback_type`	string		

Name	Type	Description	Example
â□□â□□â□□ `oob_end_date`	bigint		
â□□â□□â□□ `oob_feedback_notes`	string		
â□□â□□â□□ `oob_response_type`	string		
â□□â□□â□□ `dispute_notes`	string		
â□□â□□â□□ `dispute_reason`	string		
â□□â□□â□□ `dispute_start_date`	bigint		
â□□â□□â□□ `dispute_submitted_by`	string		
â□□â□□â□□ `info_type`	string		
â□□â□□â□□ `oob_comm_type`	string		
â□□â□□â□□ `oob_notes`	string		
â□□â□□â□□ `oob_start_date`	bigint		
â□□â□□â□□ `action_alert`	boolean		
â□□â□□â□□ `action_auto_approve_card`	boolean		
â□□â□□â□□ `action_auto_decline_card`	boolean		
â□□â□□â□□ `action_limit_review_card`	boolean		
â□□â□□â□□ `channelId`	string		
â□□â□□â□□ `curated_explanations`	json		
â□□â□□â□□ `device_ip`	string		
â□□â□□â□□ `event_type`	string		
â□□â□□â□□ `fraud_score`	double		
â□□â□□â□□ `outcome_decision`	string		
â□□â□□â□□ `sca_required`	boolean		
â□□â□□â□□ `tenant`	string	The tenant of the event.	feedzai
â□□â□□â□□ `account_currency`	string		
â□□â□□â□□ `account_type`	string		
	double		

Name	Type	Description	Example
â□□â□□â□□ `cardholder_bill_conv_rate`			
â□□â□□â□□ `customer_dob`	string		
â□□â□□â□□ `customer_occupation`	string		
â□□â□□â□□ `fields_customer_segment`	string		
â□□â□□â□□ `event_external_id`	string		
â□□â□□â□□ `event_subtype`	string		
â□□â□□â□□ `payment_sub_type`	string		
â□□â□□â□□ `payment_type`	string		
â□□â□□â□□ `terminal_ip`	string		
â□□â□□â□□ `transaction_notes`	string		
â□□â□□â□□ `sanctions_alert`	boolean		
â□□â□□â□□ `sanctions_outcome_decision`	string		
â□□â□□â□□ `card_format`	string		
â□□â□□â□□ `mc_service_indicator`	string		
â□□â□□â□□ `mc_service_indicator_value`	string		
â□□â□□â□□ `mc_rule_score`	string		
â□□â□□â□□ `mc_rule_score_reason1`	string		
â□□â□□â□□ `mc_rule_score_reason2`	string		
â□□â□□â□□ `mc_final_autho_indicator`	string		
â□□â□□â□□ `mc_trace_id`	string		
â□□â□□â□□ `mc_transaction_type_identifier`	string		
â□□â□□â□□ `mc_authentication_flag`	string		
	string		

Name	Type	Description	Example
â□□â□□â□□ `mc_card_ver_result`			
â□□â□□â□□ `mc_terminal_ver_result`	string		
â□□â□□â□□ `mc_service_data`	string		
â□□â□□â□□ `mc_trx_initiate_indicator`	string		

Case Manager Event - Digital Activity

Was this page helpful?Â

The **Case Manager Event - Digital Activity** entity stores all events from the Digital Activity channel.

It comprises the Case Manager Event - Base Schema and the following fields:

Name	Type	Description	Example
`event_payload`	json	The event payload for the digital activity event.	{ "fraud_score": 0.5, "outcome_decision": "approved", "account_last_successful_login_at": 1719859200, "account_id": "1234567890", "account_login_first_seen": 1719859200}
âââ `fraud_score`	double		

Name	Type	Description	Example
â□□â□□â□□ `outcome_decision`	string		
â□□â□□â□□ `acc_last_successful_login_at`	bigint		
â□□â□□â□□ `account_id`	string		
â□□â□□â□□ `account_login_first_seen`	bigint		
â□□â□□â□□ `amount`	bigint		
â□□â□□â□□ `auth_level`	string		
â□□â□□â□□ `channel_type`	string		
â□□â□□â□□ `currency_code`	string		
â□□â□□â□□ `customer_dob`	string		
â□□â□□â□□ `customer_email`	string		
â□□â□□â□□ `customer_id`	string		
â□□â□□â□□ `customer_name`	string		
â□□â□□â□□ `customer_phone_number`	string		
â□□â□□â□□ `device_app_version`	string		
â□□â□□â□□ `device_brand`	string		
â□□â□□â□□ `device_browser`	string		
â□□â□□â□□ `device_browser_language`	string		
â□□â□□â□□ `device_browser_version`	string		
â□□â□□â□□ `device_city`	string		
â□□â□□â□□ `device_country_code`	string		
â□□â□□â□□ `device_email_addr`	string		
â□□â□□â□□ `device_first_seen`	bigint		
â□□â□□â□□ `device_id`	string		
â□□â□□â□□ `device_ip_address`	string		

Name	Type	Description	Example
â□□â□□â□□ `device_ip_city`	string		
â□□â□□â□□ `device_ip_country_code`	string		
â□□â□□â□□ `device_ip_first_seen`	bigint		
â□□â□□â□□ `device_ip_latitude`	bigint		
â□□â□□â□□ `device_ip_longitude`	bigint		
â□□â□□â□□ `device_ip_proxy`	boolean		
â□□â□□â□□ `device_ip_proxy_type`	string		
â□□â□□â□□ `device_ip_region`	string		
â□□â□□â□□ `device_isp`	string		
â□□â□□â□□ `device_isp_country_code`	string		
â□□â□□â□□ `device_isp_region`	string		
â□□â□□â□□ `device_language`	string		
â□□â□□â□□ `device_latitude`	bigint		
â□□â□□â□□ `device_longitude`	bigint		
â□□â□□â□□ `device_model`	string		
â□□â□□â□□ `device_os`	string		
â□□â□□â□□ `device_os_language`	string		
â□□â□□â□□ `device_os_version`	string		
â□□â□□â□□ `device_phone_number`	string		
â□□â□□â□□ `device_region`	string		
â□□â□□â□□ `device_registration_date`	bigint		
â□□â□□â□□ `device_risk_rating`	string		
â□□â□□â□□ `device_risk_reason`	string		
â□□â□□â□□ `device_risk_score`	bigint		

Name	Type	Description	Example
â□□â□□â□□ `device_screen_height`	bigint		
â□□â□□â□□ `device_screen_width`	bigint		
â□□â□□â□□ `device_timezone`	string		
â□□â□□â□□ `device_type`	string		
â□□â□□â□□ `event_external_id`	string		
â□□â□□â□□ `event_occurred_at`	bigint		
â□□â□□â□□ `event_status`	string		
â□□â□□â□□ `event_status_fail_reason`	string		
â□□â□□â□□ `event_sub_type`	string		
â□□â□□â□□ `session_id`	string		
â□□â□□â□□ `device_ip_latitude_double`	double		
â□□â□□â□□ `device_ip_longitude_double`	double		
â□□â□□â□□ `device_latitude_double`	double		
â□□â□□â□□ `device_longitude_double`	double		
â□□â□□â□□ `status_digital_activity_id`	string		
â□□â□□â□□ `event_status_digital_activity_id`	string		
â□□â□□â□□ `breach_status_id`	string		
â□□â□□â□□ `event_breach_status_id`	string		
â□□â□□â□□ `customer_address`	string		
â□□â□□â□□ `revelock_score`	integer		
â□□â□□â□□ `device_user_agent`	string		
â□□â□□â□□ `session_register_date`	bigint		
	json		

Name	Type	Description	Example
â□□â□□â□□ `enricher_revelock`			
â□□â□□â□□ `customer_segment`	string		

Source: https://cam.feedzai.com/tfrb/latest/docs/common/tfrb/references/data-exporting-tool/case-manager/cm-event-transfers

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Case Manager Event - Transfers

Was this page helpful?Â

The **Case Manager Event - Transfers** entity stores all events from the Transfers channel.

It comprises the Case Manager Event - Base Schema and the following fields:

Name	Type	Description	Example
`event_payload`	json	The event payload for the transfer event.	{ "fraud_score": 0.5, "outcome_decision": "approved", "sca_required": true, "dispute_end_date": 1719859200, "dispute_feedback_notes": "The account holder opened a dispute about a recent transaction.", "dispute_refunded": "partial_refund", "oob_feedback_notes": "The account holder opened a dispute about a recent transaction." }

Name	Type	Description	Example
â□□â□□â□□ `fraud_score`	double		
â□□â□□â□□ `outcome_decision`	string		
â□□â□□â□□ `sca_required`	boolean		
â□□â□□â□□ `dispute_end_date`	double		
â□□â□□â□□ `dispute_feedback_notes`	string		
â□□â□□â□□ `dispute_refunded`	string		
â□□â□□â□□ `oob_feedback_notes`	string		
â□□â□□â□□ `feedback_type`	string		
â□□â□□â□□ `oob_end_date`	double		
â□□â□□â□□ `oob_response_type`	string		
â□□â□□â□□ `dispute_notes`	string		
â□□â□□â□□ `dispute_reason`	string		
â□□â□□â□□ `dispute_start_date`	double		
â□□â□□â□□ `dispute_submitted_by`	string		
â□□â□□â□□ `oob_comm_type`	string		
â□□â□□â□□ `oob_notes`	string		
â□□â□□â□□ `oob_start_date`	double		
â□□â□□â□□ `bulk_id`	string		
â□□â□□â□□ `channel_type`	string		
â□□â□□â□□ `created_at`	double		
â□□â□□â□□ `device_first_seen`	double		
â□□â□□â□□ `device_id`	string		
â□□â□□â□□ `device_imei_number`	string		
â□□â□□â□□ `device_ip_address`	string		
	string		

Name	Type	Description	Example
â□□â□□â□□ `device_ip_address_location`			
â□□â□□â□□ `device_language`	string		
â□□â□□â□□ `device_latitude`	string		
â□□â□□â□□ `device_longitude`	string		
â□□â□□â□□ `device_operating_system`	string		
â□□â□□â□□ `device_type`	string		
â□□â□□â□□ `device_user_agent`	string		
â□□â□□â□□ `direction`	string		
â□□â□□â□□ `event_external_id`	string		
â□□â□□â□□ `event_occurred_at`	double		
â□□â□□â□□ `execution_at`	double		
â□□â□□â□□ `frequency`	string		
â□□â□□â□□ `geographic_scope`	string		
â□□â□□â□□ `last_sca_at`	double		
â□□â□□â□□ `malware_flag`	boolean		
â□□â□□â□□ `number_of_payments`	double		
â□□â□□â□□ `pass_reset`	boolean		
â□□â□□â□□ `payment_status`	string		
â□□â□□â□□ `payment_sub_type`	string		
â□□â□□â□□ `payment_type`	string		
â□□â□□â□□ `product_code`	string		
â□□â□□â□□ `proxy_flag`	boolean		
â□□â□□â□□ `rcv_acc_balance`	double		
â□□â□□â□□ `rcv_acc_bic`	string		

Name	Type	Description	Example
â□□â□□â□□ `rcv_acc_credit_limit`	double		
â□□â□□â□□ `rcv_acc_currency`	string		
â□□â□□â□□ `rcv_acc_debt_balance`	double		
â□□â□□â□□ `rcv_acc_iban`	string		
â□□â□□â□□ `receiver_account_id`	string		
â□□â□□â□□ `rcv_acc_last_activity`	double		
â□□â□□â□□ `rcv_acc_number_of_cards`	integer		
â□□â□□â□□ `rcv_acc_open_date`	double		
â□□â□□â□□ `rcv_acc_status`	string		
â□□â□□â□□ `rcv_acc_type`	string		
â□□â□□â□□ `rcv_alias`	string		
â□□â□□â□□ `rcv_alias_last_update_time`	string		
â□□â□□â□□ `rcv_alias_registration_time`	string		
â□□â□□â□□ `rcv_alias_status`	string		
â□□â□□â□□ `rcv_alias_type`	string		
â□□â□□â□□ `rcv_bank_address`	string		
â□□â□□â□□ `receiver_bank_branch_id`	string		
â□□â□□â□□ `rcv_bank_city`	string		
â□□â□□â□□ `rcv_bank_name`	string		
â□□â□□â□□ `rcv_citizen_id_number`	string		
â□□â□□â□□ `rcv_city`	string		
â□□â□□â□□ `rcv_country_code`	string		
â□□â□□â□□ `rcv_currency`	string		
â□□â□□â□□ `rcv_dob`	string		

Name	Type	Description	Example
â□□â□□â□□ `rcv_email`	string		
â□□â□□â□□ `rcv_id`	string		
â□□â□□â□□ `rcv_kyc_lvl`	string		
â□□â□□â□□ `rcv_name`	string		
â□□â□□â□□ `rcv_number_of_accounts`	double		
â□□â□□â□□ `rcv_number_of_cards`	double		
â□□â□□â□□ `rcv_occupation`	string		
â□□â□□â□□ `rcv_passport_number`	string		
â□□â□□â□□ `rcv_phone`	string		
â□□â□□â□□ `rcv_region`	string		
â□□â□□â□□ `rcv_registration_date`	double		
â□□â□□â□□ `rcv_risk_rating`	string		
â□□â□□â□□ `rcv_segment`	string		
â□□â□□â□□ `rcv_ssn`	string		
â□□â□□â□□ `rcv_street_addr`	string		
â□□â□□â□□ `rcv_tax_number`	string		
â□□â□□â□□ `rcv_total_balance`	double		
â□□â□□â□□ `rcv_total_credit_limit`	double		
â□□â□□â□□ `rcv_transact_bank_acc_number`	string		
â□□â□□â□□ `recurring_payment`	boolean		
â□□â□□â□□ `reference_number`	string		
â□□â□□â□□ `requested_execution_at`	double		
â□□â□□â□□ `sca_risk_score`	double		
â□□â□□â□□ `sender_account_balance`	double		
â□□â□□â□□ `sender_account_bic`	string		

Name	Type	Description	Example
â□□â□□â□□ `sender_account_credit_limit`	double		
â□□â□□â□□ `sender_account_currency`	string		
â□□â□□â□□ `sender_account_debt_balance`	double		
â□□â□□â□□ `sender_account_iban`	string		
â□□â□□â□□ `sender_account_id`	string		
â□□â□□â□□ `sender_account_last_activity`	double		
â□□â□□â□□ `sender_account_number_of_cards`	double		
â□□â□□â□□ `sender_account_open_date`	double		
â□□â□□â□□ `sender_account_status`	string		
â□□â□□â□□ `sender_account_type`	string		
â□□â□□â□□ `sender_alias`	string		
â□□â□□â□□ `sender_alias_last_update_time`	string		
â□□â□□â□□ `sender_alias_registration_time`	string		
â□□â□□â□□ `sender_alias_status`	string		
â□□â□□â□□ `sender_alias_type`	string		
â□□â□□â□□ `sender_bank_address`	string		
â□□â□□â□□ `sender_bank_branch_id`	string		
â□□â□□â□□ `sender_bank_city`	string		
â□□â□□â□□ `sender_bank_name`	string		
â□□â□□â□□ `sender_citizen_id_number`	string		
â□□â□□â□□ `sender_city`	string		

Name	Type	Description	Example
â□□â□□â□□ `sender_country_code`	string		
â□□â□□â□□ `sender_currency`	string		
â□□â□□â□□ `sender_dob`	string		
â□□â□□â□□ `sender_email`	string		
â□□â□□â□□ `sender_email_change_flag`	boolean		
â□□â□□â□□ `sender_id`	string		
â□□â□□â□□ `sender_initiated_at`	double		
â□□â□□â□□ `sender_kyc_lvl`	string		
â□□â□□â□□ `sender_name`	string		
â□□â□□â□□ `sender_number_of_accounts`	double		
â□□â□□â□□ `sender_number_of_cards`	double		
â□□â□□â□□ `sender_occupation`	string		
â□□â□□â□□ `sender_passport_number`	string		
â□□â□□â□□ `sender_phone`	string		
â□□â□□â□□ `sender_phone_change_flag`	boolean		
â□□â□□â□□ `sender_region`	string		
â□□â□□â□□ `sender_registration_date`	double		
â□□â□□â□□ `sender_risk_rating`	string		
â□□â□□â□□ `sender_segment`	string		
â□□â□□â□□ `sender_ssn`	string		
â□□â□□â□□ `sender_street_addr`	string		
â□□â□□â□□ `sender_tax_number`	string		
	double		

Name	Type	Description	Example
â□□â□□â□□ `sender_total_balance`			
â□□â□□â□□ `sender_total_credit_limit`	double		
â□□â□□â□□ `sender_transact_amount`	double		
â□□â□□â□□ `sender_transact_bank_accnumber`	string		
â□□â□□â□□ `sender_transact_currency`	string		
â□□â□□â□□ `sender_urgency`	boolean		
â□□â□□â□□ `session_id`	string		
â□□â□□â□□ `user_email_address`	string		
â□□â□□â□□ `user_phone_number`	string		
â□□â□□â□□ `user_phone_number_country_code`	string		
â□□â□□â□□ `rcv_zip`	string		
â□□��â□□â□□ `sender_zip`	string		
â□□â□□â□□ `customer_id`	string		
â□□â□□â□□ `customer_trx_bank_acc_number`	string		
â□□â□□â□□ `customer_transaction_amount`	double		
â□□â□□â□□ `revelock_score`	double		
â□□â□□â□□ `decision_id`	string		
â□□â□□â□□ `event_decision_id`	string		
â□□â□□â□□ `fields_customer_segment`	string		
â□□â□□â□□ `account_open_date`	double		
â□□â□□â□□ `receiver_transaction_currency`	string		
â□□â□□â□□ `sender_bank_bic`	string		

Name	Type	Description	Example
â□□â□□â□□ receiver_bank_bic	string		
â□□â□□â□□ sender_corr_bank_bic	string		
â□□â□□â□□ receiver_corr_bank_bic	string		
â□□â□□â□□ sender_bank_country	string		
â□□â□□â□□ receiver_bank_country	string		
â□□â□□â□□ sender_corr_bank_ctr	string		
â□□â□□â□□ receiver_corr_bank_ctr	string		
â□□â□□â□□ payment_reference	string		
â□□â□□â□□ receiver_corr_bank_name	string		
â□□â□□â□□ receiver_corr_bank_address	string		
â□□â□□â□□ transaction_notes	string		
â□□â□□â□□ tenant	string		
â□□â□□â□□ sender_correspondent_bank_name	string		
â□□â□□â□□ sender_corr_bank_addr	string		
â□□â□□â□□ ttc	string		
â□□â□□â□□ rcv_bank_frgn_exchange_rate	double		
â□□â□□â□□ sndr_bank_frgn_exchange_rate	double		
â□□â□□â□□ counterparty_country_code	string		
â□□â□□â□□ customer_account_iban	string		
â□□â□□â□□ account_balance	double		
â□□â□□â□□ account_credit_limit	double		
â□□â□□â□□ card_credit_limit	double		
	double		

Name	Type	Description	Example
â□□â□□â□□ `receiver_transaction_amount`			
â□□â□□â□□ `receiver_trx_amount_usd`	double		
â□□â□□â□□ `aml_version`	string		
â□□â□□â□□ `customer_email`	string		
â□□â□□â□□ `customer_name`	string		
â□□â□□â□□ `customer_phone`	string		
â□□â□□â□□ `transaction_goal`	string		
â□□â□□â□□ `transaction_method`	string		
â□□â□□â□□ `sanctions_outcome_decision`	string		
â□□â□□â□□ `device_browser`	string		
â□□â□□â□□ `device_browser_version`	string		
â□□â□□â□□ `device_ip_country_code`	string		
â□□â□□â□□ `device_ip_region`	string		
â□□â□□â□□ `device_isp`	string		
â□□â□□â□□ `device_os_version`	string		
â□□â□□â□□ `session_register_date`	double		
â□□â□□â□□ `enricher_revelock`	string		
â□□â□□â□□ `sanctions_alert`	boolean		
â□□â□□â□□ `ultimate_debtor_name`	string		
â□□â□□â□□ `ultimate_creditor_name`	string		
â□□â□□â□□ `screenable_transaction_notes`	string		
â□□â□□â□□ `originator_screening_failed`	boolean		
	boolean		

Name	Type	Description	Example
â□□â□□â□□ `beneficiary_screening_failed`			
â□□â□□â□□ `trx_notes_screening_failed`	boolean		
â□□â□□â□□ `ult_creditor_screening_failed`	boolean		
â□□â□□â□□ `ult_debtor_screening_failed`	boolean		
â□□â□□â□□ `customer_screening_failed`	boolean		
â□□â□□â□□ `cust_entity_screening_skipped`	boolean		
â□□â□□â□□ `prt_dom_screening_skipped`	boolean		
â□□â□□â□□ `trx_notes_dom_screening_skip`	boolean		
â□□â□□â□□ `trx_notes_screening_skipped`	boolean		
â□□â□□â□□ `ult_creditor_screening_skipped`	boolean		
â□□â□□â□□ `ult_debtor_screening_skipped`	boolean		
â□□â□□â□□ `beneficiary_search_id`	double		
â□□â□□â□□ `originator_search_id`	double		
â□□â□□â□□ `transaction_notes_search_id`	double		
â□□â□□â□□ `ultimate_creditor_search_id`	double		
â□□â□□â□□ `ultimate_debtor_search_id`	double		
â□□â□□â□□ `total_results_hashcode`	string		
â□□â□□â□□ `originator_results_hashcode`	string		
	string		

Name	Type	Description	Example
â□□â□□â□□ `beneficiary_results_hashcode`			
â□□â□□â□□ `trx_notes_results_hashcode`	string		
â□□â□□â□□ `ult_creditor_results_hashcode`	string		
â□□â□□â□□ `ult_debtor_results_hashcode`	string		
â□□â□□â□□ `transfer_user_id`	string		

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- References
- Data Exporting Tool
- Event Storage
- Event Storage - Ad-hoc Customer Investigations

Event Storage - Ad-hoc Customer Investigations

Was this page helpful?Â

The **Event Storage - Ad-hoc Customer Investigations** entity stores ad-hoc customer investigation related events.

A new record becomes available for this entity for each ad-hoc customer investigation event.

Name	Type	Description	Example
`fdz_partition_index`	integer	Partition index.	79
`fdz_tenant_key`	string	Tenant key from Feedzai perspective.	sample_value
`fdz_timestamp`	bigint	Unix timestamp.	1756463007000

Name	Type	Description	Example
`fdz_uuid`	string	UUID from Feedzai perspective.	d392378a-f4b2-49b5-8814-e3ea65543420
`rules_explanations`	string	Explanation of rules triggered in text format.	Rule X triggered because X amount surpasses the Y limit.
`rules_ids`	string	Identifier of the rules.	
`action_names`	string		
`decision`	string	Output decision from the system.	approve
`is_alert`	string	Identifies whether the event is an alert.	True
`lifecycle_id`	string		
`event_external_id`	string		
`reason_code`	string		
`origin`	string		
`description`	string		
`customer_id`	string		
`customer_name`	string		
`customer_email`	string		
`customer_phone`	string		
`customer_phone_val_code`	string		
`customer_citizen_id_number`	string		
`customer_passport_number`	string		
`customer_driver_license`	string		
`customer_ssn`	string		
`customer_tax_number`	string		
`customer_registration_date`	bigint		
`customer_street_addr`	string		
`customer_zip`	string		
`customer_zip_area_code`	string		
`customer_city`	string		
`customer_region`	string		
`customer_country_code`	string		
`customer_dob`	string		
`customer_number_of_accounts`	integer		
`customer_total_balance`	bigint		
`customer_total_credit_limit`	bigint		
`customer_kyc_lvl`	string		
`customer_occupation`	string		
`customer_active`	string		
`customer_segment`	string		

Name	Type	Description	Example
`customer_number_of_cards`	integer		
`customer_occupation_status`	string		
`customer_risk_rating`	string		
`customer_currency`	string		
`customer_accounts`	array<row>		
â□□â□□â□□ `account_id`	string		
â□□â□□â□□ `account_iban`	string		
â□□â□□â□□ `account_bic`	string		
â□□â□□â□□ `account_open_date`	bigint		
â□□â□□â□□ `account_balance`	bigint		
â□□â□□â□□ `account_credit_limit`	bigint		
â□□â□□â□□ `account_otb_limit`	bigint		
â□□â□□â□□ `account_available_credit`	bigint		
â□□â□□â□□ `account_debt_balance`	bigint		
â□□â□□â□□ `account_cash_advance_balance`	bigint		
â□□â□□â□□ `account_status`	string		
â□□â□□â□□ `account_last_activity`	bigint		
â□□â□□â□□ `account_number_of_cards`	integer		
â□□â□□â□□ `atc1`	string		
â□□â□□â□□ `atc2`	string		
â□□â□□â□□ `account_last_payment_date`	bigint		
â□□â□□â□□ `account_last_payment_amount`	bigint		
â□□â□□â□□ `account_type`	string		
â□□â□□â□□ `account_currency`	string		
â□□â□□â□□ `account_pw_change_flag`	boolean		
â□□â□□â□□ `account_rewards_used_count`	integer		
â□□â□□â□□ `account_debt_payment_method`	string		
`customer_cards`	array<row>		
â□□â□□â□□ `card_pan`	string		
â□□â□□â□□ `card_bin`	string		
â□□â□□â□□ `card_effective_date`	string		

Name	Type	Description	Example
â□□â□□â□□ `card_expiration_date`	string		
â□□â□□â□□ `card_type`	string		
â□□â□□â□□ `card_brand`	string		
â□□â□□â□□ `card_last4`	string		
â□□â□□â□□ `card_id`	string		
â□□â□□â□□ `card_category`	string		
â□□â□□â□□ `card_program`	string		
â□□â□□â□□ `card_country_code`	string		
â□□â□□â□□ `card_format`	string		
â□□â□□â□□ `card_status`	string		
â□□â□□â□□ `card_available_balance`	bigint		
â□□â□□â□□ `card_transaction_amount_limit`	bigint		
â□□â□□â□□ `cardholder_name`	string		
â□□â□□â□□ `card_issued_date`	bigint		
`customer_address_validated`	boolean		
`customer_email_validated`	boolean		
`customer_contactoptout`	boolean		
`customer_pwd`	string		
`customer_gender`	string		
`customer_mothersname`	string		
`customer_type`	string		
`customer_username`	string		
`customer_branch_id`	string		
`customer_preferred_language`	string		
`customer_reported_income`	bigint		
`customer_alias_status`	string		
`customer_alias_last_update_time`	bigint		
`customer_alias_registration_time`	bigint		
`customer_alias`	string		
`customer_alias_type`	string		
`customer_citizenship`	string		
`customer_credit_score`	integer		
`customer_mailing_city`	string		
`customer_mailing_country_code`	string		
`customer_mailing_street_addr`	string		
`customer_mailing_zip`	string		

Name	Type	Description	Example
`is_client`	boolean		
`customer_products_portfolio`	array<string>		
`event_occurred_at`	bigint		
`event_received_at`	bigint		
`event_type`	string		
`info_type`	string		
`feedback_value`	boolean		
`feedback_label`	string		
`feedback_type`	string		
`source`	string		
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Event Storage - Ad-hoc Customer Investigations Feedback

Was this page helpful?Â

The **Event Storage Feedback Schema** stores the basic information of any feedback event.

Name	Type	Description	Example
`fdz_uuid`	string	Identifier of each event in UUID	061ad3f3-9991-49fe-850f-e49d6d981028
`timestamp`	double	Unix Timestamp in milliseconds	1760000000.0
	string		

Name	Type	Description	Example
`statusreasons`		A JSON Array with the status reasons. This only exists if lifecycle_id is null.	('id': '1234-12345-12345', 'name': 'At your discretion', 'statusid': 'real_event')
`lifecycle_id`	string	The lifecycle ID of the event. This only exists when statusreasons is null.	
`realoutcome`	boolean	Indicates whether this was the real outcome or not.	
`feedbacklabel`	string	Label of the Feedback event	fraud
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Event Storage - Cards

Was this page helpful?Â

The **Event Storage - Cards** entity stores card related events, such as authorizations, information and clearings.

A new record becomes available for this entity for every card transaction processed by the risk engine.

Name	Type	Description	Example
`account_associated_cards`	array<string>	List of the PANs (hashed or tokenized) associated with the account.	["PAN1", "PAN2", ..., "PAN N"]
`account_limits`	array<row>	The account limits.	

Name	Type	Description	Example
â□□â□□â□□ `event_type`	string	The type of events associated with this limit.	
â□□â□□â□□ `day_amount`	bigint	Total amount on the account day period	123
â□□â□□â□□ `daily_amount`	bigint	Total daily amount	100
â□□â□□â□□ `night_amount`	bigint	Total amount on the account night period	123
â□□â□□â□□ `single_day_amount`	bigint	Single transaction amount limit on the day period.	333
â□□â□□â□□ `single_night_amount`	bigint	Single transaction amount limit on the night period.	222
â□□â□□â□□ `start_day_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
â□□â□□â□□ `start_night_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
`amount_fees`	array<row>	Fees associated with this transaction. This is a constructed data element comprising up to six sets of 36-digit values. |ISO 8583, DE 46: Amount fees	
â□□â□□â□□ `fee_type`	string	Fee type associated to the transaction	
â□□â□□â□□ `fee_amount`	bigint	Fee amount (in cents)	
â□□â□□â□□ `fee_conv_rate`	double	Fee conversion rate	
â□□â□□â□□ `fee_reconciliation_amount`	bigint	The original amount of the fee expressed in the reconciliation currency (in cents)	
`customer_accounts`	array<row>	Set of tuples with information on all customer accounts.	
â□□â□□â□□ `account_id`	string	A bank's internal account number. This field is optional and cannot be used as a replacement for `account_iban` . | Feedzai	
â□□â□□â□□ `account_iban`	string	International Bank Account Number (IBAN) in ISO 13616:2020 format or any bank account unique identifier. This field should be sent as a token or hash. Example: 482924bf58091603ed69c5bd1221ad67481bfca8	
	string		

Name	Type	Description	Example
â□□â□□â□□ `account_bic`		Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code. Example: DABAIE2D	
â□□â□□â□□ `account_open_date`	bigint	Opening date of the account (in milliseconds since the Unix epoch).	
â□□â□□â□□ `account_balance`	bigint	Current account balance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_credit_limit`	bigint	Account credit limit (in cents or equivalent). Not relevant for debit. For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_otb_limit`	bigint	Account open-to-buy limit: the available amount between the account balance and unused credit limit (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_available_credit`	bigint	Available credit within the credit limit provided by the bank (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_debt_balance`	bigint	Accountâ□□s debt balance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_cash_advance_balance`	bigint	Debt amount in an account due to cash advance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_status`	string	The status of the account. Possible values: `open`, `blocked`.	
â□□â□□â□□ `account_last_activity`	bigint	Identifies the timestamp of the last activity registered at this account (in milliseconds since the Unix epoch).	
â□□â□□â□□ `account_number_of_cards`	integer	The number of cards associated with this account.	
â□□â□□â□□ `atc1`	string	Account type code 1. Optional code that identifies the type of account to be updated for debits and inquiries ("from" account). Used in conjunction with transaction type code (ttc). For example, 10 = Savings account - default. | ISO 8583	
â□□â□□â□□ `atc2`	string	Account type code 2. Optional code that identifies the type of account to be updated for credits ("to" account). Used in conjunction with transaction type code (ttc). For example, 50 = Investment account - default. | ISO 8583	
â□□â□□â□□ `account_l`	bigint		

Name	Type	Description	Example
`ast_payment_date`		Date when the customer pays off the balance on the account (in milliseconds since the Unix epoch).	
â□□â□□â□□ `account_last_payment_amount`	bigint	Amount of the last payment made on the account (in cents).	
â□□â□□â□□ `account_type`	string	Type of account (e.g. credit, debit, investment).	
â□□â□□â□□ `account_currency`	string	Currency of the account (ISO 4217 code).	
â□□â□□â□□ `account_pw_change_flag`	boolean	For deployments that do not include Digital Activity, you can use this field to detect if the customer changed the password on the account. This refers to whether the password is changed over the last day or within a single session. | Feedzai	
â□□â□□â□□ `account_rewards_used_count`	integer	The number of times that the customer used the rewards.	
â□□â□□â□□ `account_debt_payment_method`	string	Payment method of outstanding balance on the account. Example: Visa	
`customer_cards`	array<row>	Set of tuples with information on customer cards.	
â□□â□□â□□ `card_pan`	string	'The primary account number (PAN), or simply a card number, is the card identifier found in payment cards, such as credit cards and debit cards. This field can be sent in the multiple formats (from most to least secure): - token - hash - encrypted - clear text. For PCI compliance, follow the security guidelines to protect the data: 1) When data is sent as clear text or encrypted, then it''s mandatory to use Feedzai Tokenizer. 2) When data is sent as hash, then Feedzai Tokenizer is recommended (if not used, it will be disabled). 3) When data is sent as token, then Feedzai Tokenizer is not necessary and will be disabled. | ISO 8583, DE 2: Primary Account Number'	
â□□â□□â□□ `card_bin`	string	Card's bank identification number. A card BIN comprises the first 4 to 8 digits of a card PAN. | Feedzai Example: 123456	
â□□â□□â□□ `card_effective_date`	string	Year and month in which the card becomes effective (YYYYMM). Example: 211901	
â□□â□□â□□ `card_expi`	string		

Name	Type	Description	Example
`ration_date`		Specifies the expiration date of the card (YYMM). | ISO 8583, DE 9: Conversion rate reconciliation	
â□□â□□â□□ `card_type`	string	Specifies whether the card is debit, credit.	
â□□â□□â□□ `card_brand`	string	Card brand. Example: Visa	
â□□â□□â□□ `card_last4`	string	Last four digits of the card PAN.	
â□□â□□â□□ `card_id`	string	Persistent unique card ID (even after operations that might lead to an ID change, such as card re-issuing). | Feedzai	
â□□â□□â□□ `card_category`	string	Specifies whether the card is 'business' or 'consumer'. | Feedzai Example: consumer	
â□□â□□â□□ `card_program`	string	Specifies whether the card is black, platinum, etc.	
â□□â□□â□□ `card_country_code`	string	Represents the country (in accordance with ISO 3166, Alpha-2) where the card was issued. | Feedzai	
â□□â□□â□□ `card_format`	string	The card format. The possible values are: virtual, physical, and other.	
â□□â□□â□□ `card_status`	string	The status of the card (e.g. active, blocked, expired).	
â□□â□□â□□ `card_available_balance`	bigint	The available balance on the card (in cents or equivalent).	
â□□â□□â□□ `card_transaction_amount_limit`	bigint	Maximum transaction amount allowed on the card (in cents or equivalent).	
â□□â□□â□□ `cardholder_name`	string	The full name of the cardholder.	
â□□â□□â□□ `card_issued_date`	bigint	Date when the card was issued (in milliseconds since the Unix epoch). | Feedzai	
â□□â□□â□□ `card_first_activation_date`	bigint	Date of the first activation of this card (in milliseconds since the Unix epoch). | Feedzai	
`original_amount_data`	row		
â□□â□□â□□ `currency`	string		EUR

Name	Type	Description	Example
		The amount of data elements from the original transaction.|ISO 8583 , DE 30: Amounts original (Field 1)	
`â□□â□□â□□ currency_unit`	integer	The reconciliation amount of data elements from the original transaction. | ISO 8583, DE 30: Amounts original (Field 2)	2
`â□□â□□â□□ amount`	bigint	Original receiver amount	314
`original_rec_amount_data`	row	The amount of data elements from the original transaction. | ISO 8583	
`â□□â□□â□□ currency`	string	The amount of data elements from the original transaction.|ISO 8583 , DE 30: Amounts original (Field 1)	EUR
`â□□â□□â□□ currency_unit`	integer	The reconciliation amount of data elements from the original transaction. | ISO 8583, DE 30: Amounts original (Field 2)	2
`â□□â□□â□□ amount`	bigint	Original receiver amount	314
`rules_explanations`	string	Explanation of rules triggered in text format.	Rule X triggered because X amount surpasse the Y limit.
`rules_ids`	string	Identifier of the rules.	
`action_names`	string		
`decision`	string	Output decision from the system.	approve
`fdz_partition_index`	integer	Partition index.	79
`fdz_tenant_key`	string	Tenant key from Feedzai perspective.	sample_value
`fdz_timestamp`	bigint	Unix timestamp.	1756463007000
`fdz_uuid`	string	UUID from Feedzai perspective.	d392378a-f4b2-49b5-8814-e3ea65543420
`is_alert`	string	Identifies whether the event is an alert.	True
`acceptor_add_contact_info`	string	'Card acceptor additional contact information.| ISO 8583, DE 43: Card acceptor name/location (Field 10)'	3248371648
`acceptor_certificate_number`	string	'Value assigned to a card acceptor certificate issued by the acquirer''s certificate authority.|ISO 8583, DE 34: Electronic commerce data (Field 7)'	680a503b53268a615b7b2d8b81a61e2c5e38c68a
`acceptor_city`	string	'Card acceptor city.|ISO 8583, DE 43: Card acceptor name/location (Field 4)'	Andersonbury
`acceptor_country_code`	string	'Card acceptor country code (in accordance with ISO 3166, Alpha-2).|ISO 8583, DE 43: Card acceptor name/location (Field 7)'	US
`acceptor_cust_service_phone`	string	'Card acceptor customer service phone number.| ISO 8583, DE 43: Card acceptor name/location (Field 9)'	42340072319124
`acceptor_dba_name`	string	'"Merchant Does Business As" name.|Feedzai'	Hunt, Harris and Ruiz
	string		

Name	Type	Description	Example
`acceptor_email`		'Card acceptor email address.|ISO 8583, DE 43: Card acceptor name/location (Field 12)'	
`acceptor_id`	string	'Code used to uniquely identify the card acceptor originating a transaction; this code is also referred to as Merchant Identifier, or ''MID''. | ISO 8583, DE 42: Card acceptor identification code'	c7a2c7c5850fc9625a015ea0c2ea1d61635763d6
`acceptor_latitude`	double	Latitude of the acceptor. This location derives from GPS coordinates. | Feedzai	
`acceptor_longitude`	double	Longitude of the acceptor. This location derives from GPS coordinates. | Feedzai	
`acceptor_mastercard_risk`	integer	Risk level of the acceptor determined by Mastercard. Ranges from 0 to 999.|Feedzai	
`acceptor_name`	string	'Name of the card acceptor as known to the cardholder.|ISO 8583, DE 43: Card acceptor name/location (Field 2)'	Hunt, Harris and Ruiz
`acceptor_open_date`	integer	Represents the date from which the acceptor is active (in milliseconds since the Unix epoch).| Feedzai	
`acceptor_phone`	string	'Card acceptor phone number.|ISO 8583, DE 43: Card acceptor name/location (Field 8)'	3240396761
`acceptor_phone_val_code`	string	Acceptor phone validation code indicates whether the acceptor phone number is valid.| Feedzai	A
`acceptor_region`	string	'Card acceptor region code.|ISO 8583, DE 43: Card acceptor name/location (Field 5)'	CA
`acceptor_risk`	integer	Represents the risk level of the acceptor determined by the issuing bank. Ranges from 0 to 5.|Feedzai	
`acceptor_risky_flag`	boolean	Indicator on whether the acceptor is considered to be risky or not.|Feedzai	
`acceptor_street_addr`	string	'Card acceptor street address.|ISO 8583, DE 43: Card acceptor name/location (Field 3)'	6316 Christopher Ramp Suite 987
`acceptor_tax_number`	string	Merchant's tax number.|Feedzai	91-83714
`acceptor_url`	string	'Card acceptor internet URL.|ISO 8583, DE 43: Card acceptor name/location (Field 11)'	huntharrisruiz.com
`acceptor_visa_risk`	integer	Risk level of the acceptor determined by Visa. Ranges from 0 to 99.|Feedzai	
`acceptor_zip`	string	'This data element is used to convey the ZIP code or postal code of the merchant''s or card acceptor''s registered location. | ISO 8583, DE 43: Card acceptor name/location (Field 6)'	64465
`account_available_credit`	integer	Available credit within the credit limit provided by the bank (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`account_balance`	integer	Current account balance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	

Name	Type	Description	Example
`account_bic`	string	Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code.|Feedzai	DABAIE2D
`account_cash_advance_balance`	integer	Debt amount in an account due to cash advance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`account_credit_limit`	integer	Account credit limit (in cents or equivalent). Not relevant for debit. For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`account_currency`	string	Currency associated with the account. Refers to accounts' balances and credit limit.|Feedzai	USD
`account_debt_balance`	integer	Accountâ□□s debt balance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`account_debt_payment_method`	string	Payment method of outstanding balance on the account.|Feedzai	Visa
`account_digital_signature`	string	'A digital signature created by the private part of the private/public key pair supplied by a card issuer to a cardholder and linked to the cardholder''s account on which the card is issued.|ISO 8583, DE 34: Electronic commerce data (Field 2)'	b1367b34bfe36c56fa1e1e53a9326105ba59be1d
`account_iban`	string	International Bank Account Number (IBAN) in ISO 13616:2020 format or any bank account unique identifier. This field should be sent as a token or hash.|Feedzai	482924bf58091603ed69c5bd1221ad67481bfca8
`account_id`	string	A bank's internal account number. This field is optional and cannot be used as a replacement for `account_iban` . | Feedzai	c0e68e4e-215f-47cb-b3cd-0d67a9ad881f
`account_last_activity`	integer	Identifies the timestamp of the last activity registered at the account (in milliseconds since the Unix epoch).|Feedzai	
`account_last_payment_amount`	integer	Amount of the last payment made on the account (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Feedzai	
`account_last_payment_date`	integer	Date when the customer pays off the balance on the account (in milliseconds since the Unix epoch).|Feedzai	
`account_number_of_cards`	integer	Number of cards associated with the account.| Feedzai	
`account_open_date`	integer	Opening date of the account (in milliseconds since the Unix epoch).|Feedzai	
`account_otb_limit`	integer	Account open-to buy-limit is the available amount between the account balance and unused credit limit (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
	boolean		

Name	Type	Description	Example
`account_pw_change_flag`		For deployments that do not include Digital Activity, you can use this field to detect if the customer changed the password on the account. This refers to if the password is changed over the last day or within a single session. | Feedzai	
`account_rewards_used_count`	integer	The number of times that the customer used the rewards.|Feedzai	
`account_status`	string	'Status of the account. Possible values: `open` , `blocked` , `closed` , `waiting_for_kyc` .| Feedzai'	open
`account_type`	string	Type of the account. For example, savings or main account.|Feedzai	savings
`acquirer_id`	string	'Identifies the acquiring institution.|ISO 8583, DE 32: Acquiring institution identification code'	18493716481
`acquirer_number`	string	'Identifies the acquirer''s merchant.|ISO 8583, DE 31: Acquirer reference number (Field 2)'	789342
`acquirer_seq_number`	string	'Data supplied by the acquirer.|ISO 8583, DE 31: Acquirer reference number (Field 4)'	18472819401
`action_code`	string	'The result code returned to indicate whether the transaction is approved or declined.|ISO 8583, DE 39: Action code / Result code'	0000
`amount`	integer	'Transaction amount in the original currency as specified by the `currency_code` field, in cents or equivalent (e.g. 1000 to represent $10.00). | ISO 8583, DE 4: Amount transaction (Field 3)'	
`application_transaction_counter`	integer	The 4-character numeric (binary) Application Transaction Counter contains the counter value maintained by the chip card. The chip card increments this value for each transaction (including failed transactions)	
`approval_code`	string	'Code used when the transaction is approved.| ISO 8583, DE 38: Approval Code'	019201
`atc1`	string	'Account type code 1. Optional code that identifies the type of account to be updated for debits and inquiries ("from" account). Used in conjunction with transaction type code (ttc). For example, 10 = Savings account - default. | ISO 8583, DE 3: Processing Code (Field 2)'	10
`atc2`	string	'Account type code 2. Optional code that identifies the type of account to be updated for credits ("to" account). Used in conjunction with transaction type code (ttc). For example, 50 = Investment account - default. | ISO 8583, DE 3: Processing Code (Field 3)'	50
`auth_agent_id`	string	'The authorizing agent institution identification code.|ISO 8583, DE 58: Authorizing agent institution identification code.'	17281947362
`auth_type`	string	Authentication option used by the consumer during the payment process.|Feedzai	fingerprint
`avs_code`	string		A

Name	Type	Description	Example
		'Defines the result of the address verification process.|ISO 8583, DE 49: Verification Data (Field 2 - Response)'	
`business_app_transfer_purpose`	string	Contains information about the payment's purpose	AA
`capture_date`	string	'Month and day of the capture (MMDD).|ISO 8583, DE 17: Date of capture'	0820
`card_available_balance`	integer	The card available balance (in cents or equivalent). For example, for debit cards, this could be the total account amount, which can include an overdraft facility. For credit cards, this is often limited to the card credit limit amount or remaining unspent credit amount. For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`card_bin`	string	Card's bank identification number. A card BIN comprises the first 4 to 8 digits of a card pan. | Feedzai	123456
`card_brand`	string	Card Brand.|Feedzai	Visa
`card_category`	string	Specifies whether the card is 'business' or 'consumer'.|Feedzai	consumer
`card_country_code`	string	Represents the country (in accordance with ISO 3166, Alpha-2) where the card was issued. | Feedzai	US
`card_credit_limit`	integer	Card credit limit (in cents or equivalent, e.g.), as set by the institution according to the cardholder profile. For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`card_credit_score`	integer	Credit Bureau (or similar institution) credit score of the cardholder. Useful for client onboarding, and as context on an individual in Case Manager. | Feedzai	
`card_effective_date`	string	'Identifies the year and month in which the card becomes effective (YYYYMM).|ISO 8583, DE 13: Date effective'	211901
`card_expected_payments_frequency`	string	'The frequency defined with the bank to repay the credit card debt. Possible values: `monthly` , `weekly` , `bi_weekly` .'	monthly
`card_expiration_date`	string	'Specifies the expiration date of the card (YYMM). | ISO 8583, DE 9: Conversion rate reconciliation'	2405
`card_first_activation_date`	integer	Date of the first activation of this card (in milliseconds since the Unix epoch).|Feedzai	
`card_format`	string	'The card format. The possible values are: virtual, physical and other'	physical
`card_id`	string	Persistent unique card ID (even after operations that might lead to an ID change, such as card re-issuing). | Feedzai	eb7a757cc59081a84fe7f827b649e9c6c7b12338
	integer		

Name	Type	Description	Example
`card_issued_date`		Date when the card was issued (in milliseconds since the Unix epoch).|Feedzai	
`card_issuer`	string	Identifies the issuing institution.|Feedzai	Issuer Bank
`card_last4`	string	Card's last 4 digits.|Feedzai	1234
`card_number_of_cardholders`	integer	Number of cardholders associated with the card. For example, in the case of joint accounts, it is possible to have multiple cardholders associated with a card. | Feedzai	
`card_pan`	string	'The primary account number (PAN), or simply a card number, is the card identifier found in payment cards, such as credit cards and debit cards. This field can be sent in the multiple formats (from most to least secure): - token - hash - encrypted - clear text. For PCI compliance, follow the security guidelines to protect the data: 1) When data is sent as clear text or encrypted, then it''s mandatory to use Feedzai Tokenizer. 2) When data is sent as hash, then Feedzai Tokenizer is recommended (if not used, it will be disabled). 3) When data is sent as token, then Feedzai Tokenizer is not necessary and will be disabled. | ISO 8583, DE 2: Primary Account Number'	8955060600182663
`card_presence_flag`	boolean	'Flag indicating whether the card was present during the transaction.|ISO 8583, DE 61: Point of Service (POS) Data (Field 5)'	
`card_program`	string	Specifies whether the card is black, platinum, gold , silver, etc.|Feedzai	silver
`card_seq_number`	string	'Used to distinguish between separate cards with the same primary account number.|ISO 8583, DE 23: Card sequence number'	001
`card_status`	string	Status of the card (active, blocked, under validation, etc.).|Feedzai	active
`card_transaction_amount_limit`	integer	Card's amount limit that can be spent per transaction (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Feedzai	
`card_type`	string	Specifies whether the card is debit, credit, pre-paid, etc. Useful for tuning risk strategy thresholds based on the type of card. | Feedzai	credit
`card_ver_data`	string	'Number that is only printed on the card which is not included in any other technology e.g. magnetic stripe or ICC.|ISO 8583, DE 49: Verification Data (Field 2)'	0129
`cardholder_add_id`	string	'Identification number on the additional identification document.|ISO 8583, DE 49: Verification Data (Field 7)'	bb066a51bcfa18b42ac7c646ce012fce5b394c88
`cardholder_add_id_type`	string	'Represents additional identification offered by the cardholder.|ISO 8583, DE 49: Verification Data (Field 6)'	passport
`cardholder_authentication_verification_value`	string	'CAVV Results Code. Contains a one-character code that indicates the following: - The classification of the transaction, i.e. either an authentication transaction, where the Issuer	2

Name	Type	Description	Example
		ACS has created the CAVV, or an `attempts` transaction, where the Visa Attempts Server has created the CAVV - For an authentication transaction, where the Issuer ACS has created the CAVV - For an `attempts` transaction, where the Visa Attempts Server has created the CAVV'	
`cardholder_bill_amount`	integer	'Amount billed to the cardholder in the issuing bank''s currency, exclusive of cardholder billing fees, as specified by the `cardholder_bill_currency_code` field, in cents or equivalent (e.g. 1000 to represent $10.00). It should be consistent across all transactions as this value is used to calculate profiles in a unified currency. | ISO 8583, DE 6: Amount, cardholder billing'	
`cardholder_bill_conv_rate`	double	'Factor used in the conversion from transaction to cardholder billing amount.|ISO 8583, DE 10: Conversion rate, cardholder billing'	
`cardholder_bill_currency_code`	string	'Code defining the currency of `cardholder_bill_amount`, usually where the account is based (in accordance with ISO 4217, Alpha-3). | ISO 8583, DE 51: Currency code, cardholder billing'	USD
`cardholder_bill_street_addr`	string	'Cardholder billing street address.|ISO 8583, DE 49: Verification Data (Field 3)'	707 Valerie Corners
`cardholder_bill_zip`	string	'ZIP of the cardholder billing address.|ISO 8583, DE 49: Verification Data (Field 4)'	70783
`cardholder_certificate_number`	string	'Value assigned to a cardholder certificate issued by the card issuer''s certificate authority.| ISO 8583, DE 34: Electronic commerce data (Field 6)'	13e3acec30b1dd45689afbd914d2074af222195b
`cardholder_compress_bill_addr`	string	'Represents only the numeric and postcode elements of the cardholder billing address.|ISO 8583, DE 49: Verification Data (Field 5)'	707 70783
`cardholder_name`	string	Cardholder name.|Feedzai	John Doe
`cardholder_presence_indicator`	string	'Indicator informing about the presence of the cardholder.|ISO 8583, DE 61: Point of Service (POS) Data (Field 4)'	1
`cardholder_verification_rule`	string	'A Cardholder Verification Rule (CVR) consists of 2 bytes: the first indicates the type of Cardholder Verification Method (CVM) to be used, while the second specifies in which condition this CVM will be applied'	
`company_id`	string	Company ID.|Feedzai	c3ae6b7f-1605-4c38-a7dc-e67819559a82
`currency_code`	string	'This code specifies the original currency in which transactions are conducted in, and consequently defines the currency for the `amount` field (in accordance with ISO 4217, Alpha-3). | ISO 8583, DE 4: Amount transaction (Field 1)'	EUR
`currency_unit`	integer	'Indicates the number of places that the decimal point of the `amount` field is moved to the left, starting from the rightmost numeric digit. For	

Name	Type	Description	Example
		example, if the currency is US dollar and the `amount` is 1000, a `currency_unit: 2` represents $10.00.|ISO 8583, DE 4: Amount transaction (Field 2)'	
`customer_citizen_id_number`	string	Identification number defined in the citizen identity document associated with the customer's home country.|Feedzai	42611351
`customer_city`	string	Customer city.|Feedzai	South Leahfort
`customer_country_code`	string	Customer country code (in accordance with ISO 3166, Alpha-2).|Feedzai	US
`customer_country_code_change_flag`	boolean	Whether the customer is perfoming the first payment after changing country. This is a reference data field. | Feedzai	
`customer_currency`	string	Currency associated with the customer. Refers to customers' total balance and credit limit.|Feedzai	USD
`customer_dob`	string	Customer date of birth (YYYYMMDD).|Feedzai	20871211
`customer_driver_license`	string	Customer driver's license.|Feedzai	F4540525
`customer_email`	string	Customer email.|Feedzai	
`customer_email_change_flag`	boolean	For deployments that do not include Digital Activity, you can use this field to detect if the customer is perfoming the first payment after changing email. This refers to if the email is changed over the last day or within a single session. This is a reference data field. | Feedzai	
`customer_id`	string	'A unique transaction identifier that should remain consistent across multiple events (auth, info, clearing) associated with the same transaction. | ISO 8583, DE 21: Transaction life cycle identification data (Field 2)'	87b2ee22cb0004722a7265fdbbf7be98c681a3bb
`customer_kyc_lvl`	string	Know Your Customer (KYC) level. This is a reference data field. | Feedzai	A
`customer_name`	string	Customer name.|Feedzai	John Doe
`customer_number_of_accounts`	integer	Number of accounts associated with the customer.|Feedzai	
`customer_number_of_cards`	integer	Number of cards associated with the customer.|Feedzai	
`customer_occupation`	string	Customer's main professional activity.|Feedzai	actor
`customer_occupation_status`	string	Customer's current occupation status.|Feedzai	employee
`customer_passport_number`	string	Customer passport number.|Feedzai	640635130
`customer_phone`	string	Customer phone number.|Feedzai	217823554

Name	Type	Description	Example
`customer_phone_change_flag`	boolean	For deployments that do not include Digital Activity, you can use this field to detect if the customer is perfoming the first payment after changing phone number. This refers to if the phone number is changed over the last day or within a single session. This is a reference data field. | Feedzai	
`customer_phone_val_code`	string	Indicates whether the customer phone number is valid.|Feedzai	partially valid
`customer_region`	string	Customer region code.|Feedzai	WA
`customer_registration_date`	integer	Customer registration date (in milliseconds since the Unix epoch).|Feedzai	
`customer_risk_rating`	string	Customer's risk rating.|Feedzai	A
`customer_segment`	string	Customer's segment within the banks' segmentation.|Feedzai	retail
`customer_ssn`	string	Customer social security number.|Feedzai	204541535
`customer_street_addr`	string	Customer street address.|Feedzai	2087 Laura Courts Suite 109
`customer_tax_number`	string	Customer tax number.|Feedzai	951755612
`customer_total_balance`	integer	Sum of the balances on all customer's accounts (in cents or equivalent). This is a reference data field and can be used to calculate overall balance (debit and credit) to compare against an individual transaction. For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`customer_total_credit_limit`	integer	Sum of the credit limit on all customer's accounts (in cents or equivalent). This is a reference data field. For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Feedzai	
`customer_zip`	string	The customer's registered ZIP code, i.e. the postal code from the customer's residential address registered with the bank. This is a reference data field. | Feedzai	62986
`cvv2_presence_indicator`	string	Code provided by the merchant to indicate whether the CVV2 value is on the card	0
`cvv_val_code`	string	CVV validation code.|Feedzai	1432
`destination_id`	string	'Identification code of the destination which is maintained throughout the life of the transaction.|ISO 8583, DE 93: Transaction destination identification code'	18391847582
`device_fingerprint`	string	Information collected about the software and hardware of the device used to perform the transaction. Could be encrypted information used for device enrichment.|Feedzai	0400YbjucgPeDpqN9ZBLUITS+9VZ4/+YT+qYutCN/R8/6uxqb6W7w==
`device_id`	string	Unique identification number assigned to a device and should be consistent across all transactions. This could come from a computer's	JRO03C

Name	Type	Description	Example
		CPU serial number, MAC address or browser fingerprint. Using the IP address is not recommended. If you use Digital Trust, then this field is optional. | Feedzai	
`device_ip`	string	The IP of the device used to perform the transaction.|Feedzai	13.8.170.179
`device_ip_country_code`	string	The country code derived from the device's IP (in accordance with ISO 3166, Alpha-2). | Feedzai	US
`device_ip_latitude`	double	Device IP latitude. This location derives from the web connection. | Feedzai	
`device_ip_longitude`	double	Device IP longitude. This location derives from the web connection. | Feedzai	
`device_language`	string	Device language code.|Feedzai	US
`digital_wallet_token`	string	Contains the token information of tokenized transactions	
`digital_wallet_token_assurance_level`	string	Contains the token assurance level value information of tokenized transactions	00
`digital_wallet_token_requester`	string	Contains the token requester ID information of tokenized transactions	40010030273
`digital_wallet_token_type`	string	Contains the token type information of tokenized transactions	Q
`digital_wallet_tokenized_transaction`	boolean	Identifies transactions that are tokenized (PANs replaced with a VISA/Mastercard provided cloud-based token)	
`dispute_end_date`	integer	Timestamp corresponding to the moment when the dispute was closed (in milliseconds since the Unix epoch).|Labeling	
`dispute_feedback_notes`	string	Notes related to the dispute feedback event.| Labeling	The client has called to open a dispute on a transaction that was not made by him.
`dispute_notes`	string	Contextual information left by the analyst at the time of the dispute opening.|Dispute	
`dispute_reason`	string	'Why the dispute was opened. Possible reasons: - `duplicate` - Cardholder claims multiple charges for the same product or service; - `fraudulent` - Cardholder claims an unauthorized payment. This can happen if the card was lost or stolen and used to make a fraudulent purchase. It can also happen if the cardholder does not recognize the payment as it appears on the billing statement sent by the card issuer; - `subscription_canceled` - Cardholder claims continuous charges after the subscription being canceled; - `product_unacceptable` - Cardholder claims that a received product or service is defective, damaged, or not as described; - `product_not_received` - Cardholder claims an unreceived purchased of products or services; - `unrecognized` - Cardholder claims an unrecognized payment	

Name	Type	Description	Example
		appearing on the card statement; - `credit_not_processed` - Cardholder claims that the purchased product was returned or the transaction was otherwise canceled, but the merchant has not yet provided a refund or credit; - `general` - Uncategorized dispute.|Dispute'	
`dispute_refunded`	string	'Whether the funds were refunded to the cardholder. Possible values: `no_refund` , `partial_refund` , `total_refund` .|Labeling'	partial_refund
`dispute_start_date`	integer	Timestamp that corresponds to the time when the dispute was opened (in milliseconds since the Unix epoch).|Dispute	
`dispute_submitted_by`	string	'Who submitted the dispute. Possible values: `issuer` or `acquirer` .|Dispute'	
`event_external_id`	string	Unique identifier, on the customer's platform, of this event.|Feedzai	4048611b295e993d1b561d9030c53424bbf2b95e
`event_occurred_at`	integer	Timestamp corresponding to the moment when the current event occurred (i.e. is not the original event in the case of Info and Clearing events, with the exception of the Pre-decided Info events; in milliseconds since the Unix epoch).| Feedzai	
`feedback_type`	string	'It identifies the different types of feedback. Supported values: - `out_of_band` for conclusions of out-of-band communication - `chargeback` for closing disputes. If `feedback_type` equals `out_of_band` , the required fields are `oob_response_type` and `oob_end_date` . If `feedback_type` equals `chargeback` , the required fields are `dispute_refunded` and `dispute_end_date` . | Labeling'	out_of_band
`feedback_value`	boolean	This field only applies to the Feedback endpoint and refers to value that should be associated with the `feedback_label` . For example, if the label is `fraud` , the value `true` labels the transaction as fraudulent and the value `false` labels as non-fraudulent. | Labeling	
`file_transfer_ack_code`	string	'Indicates whether the sender requires an acknowledgement from the receiver. When this sub-element indicates â□□acknowledgement required,â□□the sender shall not transmit any other notification or instruction message before receiving a notification acknowledgement or instruction acknowledgement message.|ISO 8583, DE 68: Batch/file transfer message control (Field 1)'	7
`file_transfer_seq_number`	string	'Provides the unique sequence number assigned to the message within the batch.|ISO 8583, DE 68: Batch/file transfer message control (Field 2)'	00000001
`first_recurring_payment`	boolean	Whether the payment is the first occurrence of a series of recurring payments set up by the customer with the same amount and with the same payee. | Feedzai	
	string		19382917391

Name	Type	Description	Example
`forwarding_institution_id`		'Code identifying the forwarding institution. Not used when the acquirer and the card issuer can communicate directly.|ISO 8583, DE 33: Forwarding institution identification code'	
`function_code`	string	'Purpose of a message within its message class. It adds granularity about the message type, e.g. type of authorization or verification transaction, type of financial presentment. |ISO 8583, DE 24: Function code'	841
`info_type`	string	'It indicates the sub-type of the info event and helps to determine why the event is being sent. Supported values: - `dispute` - `not_honor` - `additional_info` - `pre_decided`. In case of `dispute`, `not_honor` or `additional_info`, this field updates an existing transactional event. When itâs `pre_decided`, this field creates a new transactional event that has already been decided by the source system, i.e. the bank. | Feedzai'	dispute
`julian_processing_date`	string	'Date that the acquirer processed the original transaction, e.g. first presentment.|ISO 8583, DE 31: Acquirer reference number (Field 3)'	8241
`last_sca_at`	integer	Timestamp corresponding to the last time that the consumer performed Strong Customer Authentication (SCA).|Feedzai	
`lifecycle_id`	string	'Unique transaction identifier.|ISO 8583, DE 21: Transaction life cycle identification data (Field 2)'	1df7a7qg78vd7fs
`lifecycle_indicator`	string	'Indicates the point in the transaction life cycle at which the life cycle trace identifier is assigned.|ISO 8583, DE 21: Transaction life cycle identification data (Field 1)'	1
`lifecycle_seq_number`	string	'Used with the life cycle trace identifier to uniquely identify when multiple financial presentments are generated from a single authorization transaction.|ISO 8583, DE 21: Transaction life cycle identification data (Field 3)'	01
`lifecycle_token`	string	'Represents a code calculated using an algorithm against key transaction data elements that are common to both authorization messages and financial presentment messages.|ISO 8583, DE 21: Transaction cycle identification data (Field 4)'	2718
`local_datetime`	string	'Identifies the local year, month, day, and time in which the transaction takes place at the location of the card acceptor in Authorization and Financial Presentment messages (YYYYMMDDhhmmss).|ISO 8583, DE 12: Date and time local transaction'	20181201012233
`login_session_id`	string	Deprecated - Use 'session_id' instead.	fb2e77d
`luhn_check_digit`	integer	'Check sum on the previous 22 digits of acquirer reference number calculated in accordance with ISO 7812-1.|ISO 8583, DE 31: Acquirer reference number (Field 5)'	
`malware_flag`	boolean		

Name	Type	Description	Example
		Indicates whether the device is infected with malware.|Feedzai	
`mc_fraud_score`	integer	Master card fraud score	
`mcc`	string	'The merchant category code (MCC) classifies the type of business being done by the card acceptor for this transaction (in accordance with ISO 18245). | ISO 8583, DE 26: Merchant category code (MCC)'	4722
`merchant_initiated`	boolean	Whether the transaction was initiated by a merchant / acceptor, e.g. a recurring payment. | Feedzai	
`mti`	string	'The Message Type Indicator (MTI) is a four-digit numeric field which indicates the overall function of the message. A message type indicator includes the ISO 8583 version, the Message Class, the Message Function, and the Message Origin. For example, 0110 indicates an authorization. | ISO 8583 '	0110
`oob_comm_type`	string	'The type of communication made with the client. Possible values: `sms` , `call` , `email` .|Out of Band'	
`oob_end_date`	integer	Timestamp corresponding to the moment when the client responded to the out-of-band communication (in milliseconds since the Unix epoch).|Labeling	
`oob_feedback_notes`	string	Notes related to the out-of-band feedback event.|Labeling	The client has confirmed that he didn't make the transaction.
`oob_notes`	string	Notes related to the out-of-band communication.|Out of Band	
`oob_response_type`	string	'The type of out-of-band response made by the client. For example, possible values are: `call` , `sms` , `email` , `branch` , or any other value deemed necessary by the client. Note that this field is required if the `feedback_type` field has the value `out_of_band` . | Labeling'	call
`oob_start_date`	integer	Timestamp that corresponds to the moment when the client was contacted (in milliseconds since the Unix epoch).|Out of Band	
`originator_id`	string	'Identification code of the originator which is maintained throughout the life of the transaction.|ISO 8583, DE 94: Transaction originator identification code'	18493049285
`pos_acceptor_disp_data_len`	integer	'Represents the number of characters of data from a card issuer that can be displayed on the POS for the cardholder.|ISO 8583, DE 27: Point of service capability (Field 7)'	
`pos_acceptor_rec_data_len`	integer	'Represents the number of characters of data from a card issuer that can be printed on a POS receipt for the card acceptor.|ISO 8583, DE 27: Point of service capability (Field 5)'	
`pos_approval_code_len`	integer	'Represents the maximum length of the approval code which the acquirer may accommodate. The	

Name	Type	Description	Example
		card issuer or agent shall limit the approval code to this length.|ISO 8583, DE 27: Point of service capability (Field 3)'	
`pos_card_capture_cap`	string	'Indicates if the card can be captured.|ISO 8583, DE 27: Point of service capability (Field 10)'	N
`pos_card_read_method`	string	'Series of code values indicating the card reading capabilities employed during the transaction, such as magnetic swipe, EMV, or manual entry. Each issuer can define their own code values to specify the POS entry mode for Card Present transactions. For example: 01 - manual entry; 02 - magnetic stripe; 91 - contactless card; 95 - smart card. | ISO 8583, DE 22: Point-of-Service data code (Field 7)'	01
`pos_cardholder_disp_data_len`	integer	'Identifies the card issuer data to be displayed on the POS for the cardholder.|ISO 8583, DE 27: Point of service capability (Field 6)'	
`pos_cardholder_rec_data_len`	integer	'Represents the number of characters of data from a card issuer that can be printed on a POS receipt for the cardholder.|ISO 8583, DE 27: Point of service capability (Field 4)'	
`pos_cardholder_ver_cap`	string	'Series of code values that indicate the POS capabilities available to identify the cardholder.| ISO 8583, DE 27: Point of service capability (Field 2)'	8080
`pos_cardholder_ver_method`	string	'Series of code values that indicate the cardholder verification method (CVM) used by the POS to identify the cardholder. Each issuer can define their codes. For example: 0 - Fail CVM processing; 1 - Plaintext PIN verification performed by ICC (Integrated Circuit Card); 2 - Enciphered PIN verified on-line; 3 - Plaintext PIN AND Signature-Paper; 4 - Enciphered PIN by ICC; 5 - Enciphered PIN by ICC AND Signature-Paper; 30 - Signature-Paper; 31 - No CVM required | ISO 8583, DE 22: Point-of-Service data code (Field 8)'	31
`pos_environment`	string	'Series of code values that indicate if the card acceptor was present. Typically, this field takes the values CP (for Card Present) or CNP (for Card Not Present). If more granularity is required, use `pos_card_read_method` to specify the POS entry mode for Card Present. If not present, you can use this field to also indicate the type of environment in which the transaction took place. Card Not Present transactions could be, for example, MOTO (from mail order/Telephone order) or WEB. This field may be relevant as a machine learning model feature. | ISO 8583, DE 22: Point-of-Service data code (Field 4)'	CP
`pos_icc_data_len`	integer	'Number of characters of data from a card issuer that can be returned to the ICC at the POS.|ISO 8583, DE 27: Point of service capability (Field 8)'	
`pos_pin_len_cap`	integer	'Number of PIN digits that the point of service device can accept.|ISO 8583, DE 27: Point of service capability (Field 11)'	

Name	Type	Description	Example
`pos_read_cap`	string	'Series of code values that indicate the card reading capabilities of the POS.|ISO 8583, DE 27: Point of service capability (Field 1)'	8080
`pos_security_characteristics`	string	'Series of codes that indicate the security characteristics applicable to a transaction. This field may be relevant as a machine learning model feature. | ISO 8583, DE 22: Point-of-Service data code (Field 2)'	0001
`pos_track1_data`	string	'Information encoded on the track 1 of the magnetic stripe 2 including field separators and excluding the start sentinel, end sentinel and longitudinal redundancy check ("LRC") characters.|ISO 8583, DE 45: Track 1 data'	6239184a00000120768=08021010912346567?4
`pos_track2_data`	string	'Information encoded on track 2 of the magnetic stripe as specified in ISO 7813, excluding beginning and ending sentinels and longitudinal redundancy check characters as defined therein.|ISO 8583, DE 35: Track 2 data'	675918450000120768=08051010912346567?4
`pos_track3_data`	string	'Information encoded on track 3 of the magnetic stripe as specified in ISO 4909, including field separators, but excluding beginning and ending sentinels and longitudinal redundancy check characters as defined therein.|ISO 8583, DE 36: Track 3 data'	6333000012345679=080510109123450789O1?7
`pos_track3_rw`	string	'Indicates if the POS can rewrite Track 3.|ISO 8583, DE 27: Point of service capability (Field 9)'	Y
`proxy_flag`	boolean	Whether the device IP is a proxy.|Feedzai	
`reason_code`	string	'Provides the receiver of a request, advice or notification message with the reason, or purpose of that message. For original authorizations and financial presentments, it identifies why the type of message was sent, e.g. why an advice versus a request. For other messages it identifies states why this action was taken.|ISO 8583, DE 25: Message Reason Code'	8002
`receiving_institution_id`	string	'Code identifying the receiving institution. Not used when the acquirer and the card issuer can communicate directly.|ISO 8583, DE 100: Receiving institution identification code'	18493059382
`reconciliation_amount`	integer	'Reconciliation amount in the currency specified by the `reconciliation_currency_code` field, in cents or equivalent (e.g. 1000 to represent $10.00). | ISO 8583, DE 5: Amount Reconciliation (Field 3)'	
`reconciliation_conv_rate`	double	'Factor used in the conversion from transaction to cardholder billing amount.|ISO 8583, DE 9: Conversion rate reconciliation'	
`reconciliation_currency_code`	string	'Code defining the currency of the `reconciliation_amount` (in accordance with ISO 4217, Alpha-3). | ISO 8583, DE 5: Amount Reconciliation (Field 1)'	EUR
`reconciliation_currency_unit`	integer	'Indicates the number of places that the decimal point of the `reconciliation_amount` field is moved to the left, starting from the rightmost numeric digit. For example, if the reconciliation	

Name	Type	Description	Example
		currency is US dollar and the `reconciliation_amount` is 1000, a `reconciliation_currency_unit: 2` represents $10.00.|ISO 8583, DE 5: Amount Reconciliation (Field 2)'	
`reconciliation_date`	string	'The year, month and day for which financial totals are reconciled between the acquirer and the card issuer (YYYYMMDD).|ISO 8583, DE 28: Date reconciliation'	20180829
`reconciliation_indicator`	string	'A value used to allow reconciliation of time periods within a reconciliation date. The value is subject to bilateral agreement.|ISO 8583, DE 29: Reconciliation indicator'	223
`recurring_payment`	boolean	Whether the payment is part of a series of recurring payments set up by the customer with the same amount and with the same payee. | Feedzai	
`reference_fraud_rate`	double	Fraud rate for card payments. Calculated as the total value of unauthorized or fraudulent remote transactions, divided by the total value of all remote card transactions, in the previous calendar quarter (i.e. three-month window).| Feedzai	
`retrieval_reference_number`	string	'Reference supplied by the system retaining the original source information and used to assist in locating that information or a copy thereof.|ISO 8583, DE 37: Retrieval reference number'	129829381948
`return_amount`	integer	'New actual amount (in cents) necessary to perform a partial or full reversal of a financial transaction or a partial completion amount.|ISO 8583, DE 95: Replacement Amounts (Field 1).'	
`secure_3d_indicator`	string	'Identifies the authentication method used by the Issuerâ□□s ACS to authenticate the cardholder (e.g. risk-based authentication, OTP, etc.) '	2
`secure_3d_protocol_version`	string	Authentication data. Includes the 3-D Secure Protocol version number	2.x.x
`secure_3d_val_code`	string	3D secure validation code.|Feedzai	558251
`secure_3d_val_status`	string	3D secure validation status. Possible values are `failed`, `passed`, and `disabled`.| Feedzai (PSD2)	passed
`service_code`	string	'Identifies the geographic/service availability (in accordance with ISO 7813).|ISO 8583, DE 40: Service code'	001
`session_id`	string	Id of the session that the user logged into. This should be consistent across all events within the same banking session, including payment initiations as it will be used to associate events to a session.|Feedzai	9be65da3-ec5d-45c0-93c0-03afc5f7633b
`sic`	string	Standard industrial classification merchant category code.|Feedzai	2000
`stan`	string	'System trace audit number. This field is assigned by a transaction originator to assist in uniquely identifying a transaction. The trace	103405661402

Name	Type	Description	Example
		number remains unchanged for all messages within a two-message exchange, e.g. request/ repeat and response.|ISO 8583, DE 11: Systems trace audit number (STAN)'	
`tenant_id`	string	''	
`terminal_city`	string	Terminal city.|Feedzai	Yvetteville
`terminal_country_code`	string	Terminal country code (in accordance with ISO 3166, Alpha-2).|Feedzai	US
`terminal_id`	string	'Applicable to Card Present transactions. This field represents the card acceptor terminal identification number.|ISO 8583, DE 41: Card acceptor terminal identification'	44b77c377266be3ca36ee2b5bcedde09222d346a
`terminal_ip`	string	IP of the terminal address.|Feedzai	13.59.108.243
`terminal_ip_city`	string	Estimated city of the terminal based on IP.|Feedzai	Yvetteville
`terminal_ip_country_code`	string	Estimated country code (in accordance with ISO 3166, Alpha-2) of the terminal based on IP.|Feedzai	US
`terminal_ip_latitude`	double	Estimated latitude of the terminal based on IP. This location derives from the web connection. | Feedzai	
`terminal_ip_longitude`	double	Estimated longitude of the terminal based on IP. This location derives from the web connection. | Feedzai	
`terminal_ip_region`	string	Estimated region of the terminal based on IP.|Feedzai	CA
`terminal_ip_zip`	string	Estimated ZIP of the terminal based on IP.|Feedzai	73412
`terminal_latitude`	double	Latitude coordinate of the terminal. This location derives from GPS coordinates. | Feedzai	
`terminal_longitude`	double	Longitude coordinate of the terminal. This location derives from GPS coordinates. | Feedzai	
`terminal_poc_flag`	boolean	Whether the terminal was previously a point of compromise (POC).|Feedzai	
`terminal_region`	string	Terminal region code.|Feedzai	CA
`terminal_street_addr`	string	Street address where the terminal is located.|Feedzai	66192 Stephen Bridge Suite 829
`terminal_timezone`	string	Time zone in which the terminal is located expressed as +/- hh:mm format (using UTC as reference).|Feedzai	+01:00
`terminal_zip`	string	Terminal ZIP.|Feedzai	73412
`transaction_category_code`	string	'Data object defined by MasterCard which indicates the type of transaction being performed, and which may be used in card risk management.|ISO 8583, DE 55: TransactionCategoryCode_9F53 (EMV Tag)'	A
	integer	'Counter maintained by the terminal that is incremented by one for each transaction.|ISO	

Name	Type	Description	Example
`transaction_seq_counter`		8583, DE 55: TransactionSequenceCounter_9F41 (EMV Tag)'	
`transaction_status`	string	'Indicates the different status of the transaction. Updates the same life cycle ID with an additional event entry. Possible values: - `accepted` - `rejected` - `cancelled` - `reversed` - `refunded` . | Feedzai'	accepted
`transaction_status_reason`	string	'Explains the transaction status. Possible values are: - `account_closed` - The bank account has been closed; - `account_frozen` - The bank account has been frozen; - `account_restricted` - The bank account has restrictions on either the type, or the number, of payments allowed. This normally indicates that the bank account is a savings or other non-checking account; - `account_ownership_changed` - The bank account is no longer valid because its branch has changed ownership; - `could_not_process` - The bank could not process this payment; - `debit_not_authorized` - Debit transactions are not permitted on the account; - `rejected` - The bank has rejected this payment; - `insufficient_funds` - The user account has insufficient funds to cover the payout; - `invalid_account_number` - The account number is invalid; - `incorrect_account_holder_name` - The account holder name on file is incorrect; - `invalid_currency` - The bank was unable to process this payout because of its currency. This is probably because the bank account cannot accept payments in that currency; - `no_account` - No bank account could be located with these details; - `unsupported_card` - The bank no longer supports payouts to this card; - `accepted` - The bank processed the payment; - `feedzai_unavailable` - Feedzai was unavailable to score the transaction on the Authorization endpoint and it had to be sent to the Info endpoint, as a pre-decided transaction.|Feedzai'	account_closed
`transmission_datetime`	string	'Specifies the date and time when the message initiator sends the message. This field is expressed in the Coordinated Universal Time (UTC) (in accordance with ISO 8601), in the format MMDDhhmmss.|ISO 8583, DE 7: Date and time transmission'	1201112233
`transstain`	string	'A hash value calculated by applying a secure hash algorithm to the XID and card secret (a secret defined value known only to the cardholder and issuer of the cardholder certificate).|ISO 8583, DE 34: Electronic commerce data (Field 9)'	e528318831716fa8a878e4f3c0e1fe5eeecbc292
`trusted_acceptor`	boolean	Whether the acceptor is included in the cardholder's list of trusted acceptors. | Feedzai	
`trx_to_rec_conv_date`	string	'Month and day the conversion rate is effective to convert the transaction amount from the original to reconciliation currency (MMDD).|ISO 8583, DE 9: Conversion rate reconciliation'	0829

Name	Type	Description	Example
`ttc`	string	'The Transaction Type Code (TTC) specifies the type of transaction. For example, 01 means Cash(ATM). | ISO 8583, DE 3: Processing Code (Field 1)'	01
`tvr`	string	'Terminal verification result.|ISO 8583, DE 55: TerminalVerificationResults_95 (EMV Tag)'	8080
`user_format_id`	string	'Identifier applied by the acquirer based on bilateral agreements.|ISO 8583, DE 31: Acquirer reference number (Field 1)'	1
`visa_fraud_score`	integer	Visa fraud score	
`xid`	string	'Value assigned to a transaction as a unique transaction identifier.|ISO 8583, DE 34: Electronic commerce data (Field 8)'	dca8c172f2d02402be16ab906b8db1024824b271
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Event Storage - Cards Feedback

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The **Event Storage Feedback Schema** stores the basic information of any feedback event.

Name	Type	Description	Example
`fdz_uuid`	string	Identifier of each event in UUID	061ad3f3-9991-49fe-850f-e49d6d981028
`timestamp`	double	Unix Timestamp in milliseconds	1760000000.0
`statusreasons`	string	A JSON Array with the status reasons. This only exists if lifecycle_id is null.	('id': '1234-12345-12345', 'name': 'At your discretion', 'statusid': 'real_event')
`lifecycle_id`	string	The lifecycle ID of the event. This only exists when statusreasons is null.	

Name	Type	Description	Example
`realoutcome`	boolean	Indicates whether this was the real outcome or not.	
`feedbacklabel`	string	Label of the Feedback event	fraud
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

-
- [References]
- [Data Exporting Tool]
- Event Storage
- Event Storage - Digital Activity

Event Storage - Digital Activity

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The **Event Storage - Digital Activity** entity stores digital activity related events, such as authentication and alias.

A new record becomes available for this entity for every digital activity event processed by the risk engine.

Name	Type	Description	Example
`account_active`	string	Whether the account is active.	True
`account_atc1`	string		atc1

Name	Type	Description	Example
		Describes the account type affected for debits and inquiries and the â□□fromâ□□account for transfers.	
`account_atc2`	string	Describes the account type affected for credits and the â□□toâ□□account for transfers.	Atc2
`account_cards`	array<string>	The cards associated with the account. Each value in the array corresponds to a persistent unique card ID (even after operations that might lead to an ID change such as card re-issuing).	['3135166684','3866457590']
`account_currency`	string	Currency associated with the account. Refers to accounts' balances and credit limit.	
`account_customers`	array<string>	The customers associated with the account. Each value in the array corresponds to a customer ID generated by the issuing bank.	['6618859991','7271539065']
`card_credit_limit`	bigint	Card credit limit (in cents, e.g.). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`card_credit_score`	integer	Credit Bureau (or similar institution) credit score of the card.	40
`card_expected_payments_frequency`	string	New payment method of outstanding balance on the account.	card
`card_first_activation_date`	bigint	Date of the first activation of this card (in milliseconds since the Unix epoch).	30
`card_format`	string	The card format. The possible values are: virtual, physical and other.	physical
`card_issued_date`	bigint	Date when the card was issued (in milliseconds since the Unix epoch).	
`card_last4`	string	Card's last 4 digits.	1234
`card_number_of_cardholders`	integer	Number of cardholders associated with the card. For example, in the case of joint accounts it is possible to have multiple cardholders associated with a card.	
`card_seq_number`	string	Used to distinguish between separate cards with the same primary account number. | ISO 8583, DE 23: Card sequence number	1
`card_transaction_amount_limit`	bigint	Card's amount limit that can be spent per transaction (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	
`card_type`	string	Whether the card is debit, credit, pre-paid, etc.	credit
`card_ver_data`	string	Number that is only printed on the card which is not included in any other technology (e.g. magnetic stripe or ICC). | ISO 8583, DE 49: Verification Data (Field 2)	129
`customer_accounts`	array<row>	Set of tuples with information on all customer accounts.	
`â□□â□□â□□ account_id`	string	A bank's internal account number. This field is optional and cannot be used as a replacement for `account_iban` . | Feedzai	

Name	Type	Description	Example
â□□â□□â□□ `account_iban`	string	International Bank Account Number (IBAN) in ISO 13616:2020 format or any bank account unique identifier. This field should be sent as a token or hash. Example: 482924bf58091603ed69c5bd1221ad67481bfca8	
â□□â□□â□□ `account_bic`	string	Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code. Example: DABAIE2D	
â□□â□□â□□ `account_open_date`	bigint	Opening date of the account (in milliseconds since the Unix epoch).	
â□□â□□â□□ `account_balance`	bigint	Current account balance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_credit_limit`	bigint	Account credit limit (in cents or equivalent). Not relevant for debit. For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_otb_limit`	bigint	Account open-to-buy limit: the available amount between the account balance and unused credit limit (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_available_credit`	bigint	Available credit within the credit limit provided by the bank (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_debt_balance`	bigint	Accountâ□□s debt balance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_cash_advance_balance`	bigint	Debt amount in an account due to cash advance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency).	
â□□â□□â□□ `account_status`	string	The status of the account. Possible values: `open`, `blocked`.	
â□□â□□â□□ `account_last_activity`	bigint	Identifies the timestamp of the last activity registered at this account (in milliseconds since the Unix epoch).	
â□□â□□â□□ `account_number_of_cards`	integer	The number of cards associated with this account.	
â□□â□□â□□ `atc1`	string	Account type code 1. Optional code that identifies the type of account to be updated for debits and inquiries ("from" account). Used in conjunction with transaction type code (ttc). For example, 10 = Savings account - default. | ISO 8583	
	string	Account type code 2. Optional code that identifies the type of account to be updated for	

Name	Type	Description	Example
â□□â□□â□□ `atc2`		credits ("to" account). Used in conjunction with transaction type code (ttc). For example, 50 = Investment account - default. | ISO 8583	
â□□â□□â□□ `account_last_payment_date`	bigint	Date when the customer pays off the balance on the account (in milliseconds since the Unix epoch).	
â□□â□□â□□ `account_last_payment_amount`	bigint	Amount of the last payment made on the account (in cents).	
â□□â□□â□□ `account_type`	string	Type of account (e.g. credit, debit, investment).	
â□□â□□â□□ `account_currency`	string	Currency of the account (ISO 4217 code).	
â□□â□□â□□ `account_pw_change_flag`	boolean	For deployments that do not include Digital Activity, you can use this field to detect if the customer changed the password on the account. This refers to whether the password is changed over the last day or within a single session. | Feedzai	
â□□â□□â□□ `account_rewards_used_count`	integer	The number of times that the customer used the rewards.	
â□□â□□â□□ `account_debt_payment_method`	string	Payment method of outstanding balance on the account. Example: Visa	
`customer_cards`	array<row>	Set of tuples with information on customer cards.	
â□□â□□â□□ `card_pan`	string	'The primary account number (PAN), or simply a card number, is the card identifier found in payment cards, such as credit cards and debit cards. This field can be sent in the multiple formats (from most to least secure): - token - hash - encrypted - clear text. For PCI compliance, follow the security guidelines to protect the data: 1) When data is sent as clear text or encrypted, then it''s mandatory to use Feedzai Tokenizer. 2) When data is sent as hash, then Feedzai Tokenizer is recommended (if not used, it will be disabled). 3) When data is sent as token, then Feedzai Tokenizer is not necessary and will be disabled. | ISO 8583, DE 2: Primary Account Number'	
â□□â□□â□□ `card_bin`	string	Card's bank identification number. A card BIN comprises the first 4 to 8 digits of a card PAN. | Feedzai Example: 123456	
â□□â□□â□□ `card_effe`	string	Year and month in which the card becomes effective (YYYYMM). Example: 211901	

Name	Type	Description	Example
ctive_date			
â□□â□□â□□ card_expiration_date	string	Specifies the expiration date of the card (YYMM). | ISO 8583, DE 9: Conversion rate reconciliation	
â□□â□□â□□ card_type	string	Specifies whether the card is debit, credit.	
â□□â□□â□□ card_brand	string	Card brand. Example: Visa	
â□□â□□â□□ card_last4	string	Last four digits of the card PAN.	
â□□â□□â□□ card_id	string	Persistent unique card ID (even after operations that might lead to an ID change, such as card re-issuing). | Feedzai	
â□□â□□â□□ card_category	string	Specifies whether the card is 'business' or 'consumer'. | Feedzai Example: consumer	
â□□â□□â□□ card_program	string	Specifies whether the card is black, platinum, etc.	
â□□â□□â□□ card_country_code	string	Represents the country (in accordance with ISO 3166, Alpha-2) where the card was issued. | Feedzai	
â□□â□□â□□ card_format	string	The card format. The possible values are: virtual, physical, and other.	
â□□â□□â□□ card_status	string	The status of the card (e.g. active, blocked, expired).	
â□□â□□â□□ card_available_balance	bigint	The available balance on the card (in cents or equivalent).	
â□□â□□â□□ card_transaction_amount_limit	bigint	Maximum transaction amount allowed on the card (in cents or equivalent).	
â□□â□□â□□ cardholder_name	string	The full name of the cardholder.	
â□□â□□â□□ card_issued_date	bigint	Date when the card was issued (in milliseconds since the Unix epoch). | Feedzai	
â□□â□□â□□ card_first_activation_date	bigint	Date of the first activation of this card (in milliseconds since the Unix epoch). | Feedzai	
	string	New whether the account is active.	"true"

Name	Type	Description	Example
`new_account_active`			
`new_account_atc1`	string	New account type affected for debits and inquiries and the â□□fromâ□□account for transfers.	atc1
`new_account_atc2`	string	New account type affected for credits and the â□□toâ□□account for transfers.	atc2
`new_account_available_credit`	bigint	New available credit within the credit limit provided by the bank (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`new_account_balance`	bigint	New current account balance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`new_account_bic`	string	New Bank Identifier Code (BIC) - in SWIFT format or any other bank unique identifier code - of the account linked to this alias.	NWBK
`new_account_cards`	array<string>	New set of cards associated with the account. Each value in the array corresponds to a persistent unique card ID (even after operations that might lead to an ID change such as card re-issuing).	['3135166684', '3866457590']
`new_account_cash_advance_balance`	bigint	New debt amount in an account due to cash advance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`new_account_credit_limit`	bigint	The new account credit limit (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	
`new_account_currency`	string	Currency associated with the new account. Refers to accounts' balances and credit limit. | Feedzai	USD
`new_account_customers`	array<string>	New set of customers associated with the account. Each value in the array corresponds to a customer ID generated by the issuing bank.	['6618859991', '7271539065']
`new_account_debt_balance`	bigint	New accountâ□□s debt balance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`new_account_debt_payment_method`	string	New payment method of outstanding balance on the account.	card
`new_account_last_activity`	bigint	New timestamp of the last activity registered at the account (in milliseconds since the Unix epoch).	30
`new_account_last_payment_amount`	bigint	New amount of the last payment made on the account (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`new_account_last_payment_date`	bigint	New date when the customer pays off the balance on the account (in milliseconds since the Unix epoch).	30
	integer	New number of cards associated with the account.	40

Name	Type	Description	Example
`new_account_number_of_cards`			
`new_account_open_date`	bigint	New opening date of the account (in milliseconds since the Unix epoch).	1625148450432
`new_account_otb_limit`	bigint	New account open to buy limit is the available amount between the account balance and unused credit limit (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`new_account_pw_change_flag`	boolean	New whether the customer changed the password on the account.	True
`new_account_rewards_used_count`	integer	New number of times that the customer used the rewards.	40
`new_account_status`	string	New status of the account. Possible values: open, blocked, closed, waiting_for_kyc.	open
`new_account_status_last_update_at`	bigint	Timestamp of last update made to the customer's new account status (in milliseconds since the Unix epoch).	
`new_account_type`	string	Type of the account. For example, savings or main account. | Feedzai	savings
`new_account_limits`	array<row>		
â□□â□□â□□ `event_type`	string	The type of events associated with this limit.	
â□□â□□â□□ `day_amount`	bigint	Total amount on the account day period	123
â□□â□□â□□ `daily_amount`	bigint	Total daily amount	100
â□□â□□â□□ `night_amount`	bigint	Total amount on the account night period	123
â□□â□□â□□ `single_day_amount`	bigint	Single transaction amount limit on the day period.	333
â□□â□□â□□ `single_night_amount`	bigint	Single transaction amount limit on the night period.	222
â□□â□□â□□ `start_day_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
â□□â□□â□□ `start_night_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
	bigint		

Name	Type	Description	Example
`new_card_available_balance`		The new card available balance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	
`new_card_bin`	string	Card's bank identification number.	123456
`new_card_brand`	string	Card brand.	Visa
`new_card_category`	string	Specifies whether the card is physical, virtual or digital.	physical
`new_card_country_code`	string	Represents the country (in accordance with ISO 3166, Alpha-2) where the card was issued.	US
`new_card_credit_limit`	bigint	The new card credit limit (in cents).	
`new_card_credit_score`	integer	The new card credit score.	
`new_card_effective_date`	string	Year and month in which the card becomes effective (CCYYMM).	201701
`new_card_expected_payments_frequency`	string	New payment method of outstanding balance on the account.	card
`new_card_expiration_date`	string	New specification of the expiration date of the card.	2001
`new_card_first_activation_date`	bigint	New date of the first activation of this card (in milliseconds since the Unix epoch).	30
`new_card_format`	string	The card format. The possible values are: virtual, physical and other.	physical
`new_card_issued_date`	bigint	Date when the card was issued (in milliseconds since the Unix epoch).	
`new_card_issuer`	string	Identifies the issuing institution.	
`new_card_last4`	string	Card's last 4 digits.	
`new_card_number_of_cardholders`	integer	Number of cardholders associated with the card. For example, in the case of joint accounts, it is possible to have multiple cardholders associated with a card.	
`new_card_program`	string	New specification whether the card is black, platinum, gold, silver, etc.	silver
`new_card_seq_number`	string	New number used to distinguish between separate cards with the same primary account number.	
`new_card_status`	string	New status of the card (active, blocked, under validation, etc.).	
`new_card_transaction_amount_limit`	bigint	New card's amount limit that can be spent per transaction (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency and a currency unit of 2).	20
`new_card_type`	string	New specification whether the card is debit, credit, pre-paid, etc.	credit
	string		129

Name	Type	Description	Example
`new_card_ver_data`		New number that is only printed on the card which is not included in any other technology e.g. magnetic stripe or ICC.	
`new_card_pan`	string	The primary account number (PAN), or simply a card number, is the card identifier found in payment cards, such as credit cards and debit cards. This field can be sent in the multiple formats (from most to least secure): - token - hash - encrypted - clear text. For PCI compliance, follow the security guidelines to protect the data: 1) When data is sent as clear text or encrypted, then it's mandatory to use Feedzai Tokenizer. 2) When data is sent as hash, then Feedzai Tokenizer is recommended (if not used, it will be disabled). 3) When data is sent as token, then Feedzai Tokenizer is not necessary and will be disabled. | ISO 8583, DE 2: Primary Account Number	8955060600182663
`rules_explanations`	string	Explanation of rules triggered in text format.	Rule X triggered because X amount surpasse the Y limit.
`rules_ids`	string	Identifier of the rules.	
`action_names`	string		
`decision`	string	Output decision from the system.	approve
`fdz_partition_index`	integer	Partition index.	79
`fdz_tenant_key`	string	Tenant key from Feedzai perspective.	sample_value
`fdz_timestamp`	bigint	Unix timestamp.	1756463007000
`fdz_uuid`	string	UUID from Feedzai perspective.	d392378a-f4b2-49b5-8814-e3ea65543420
`is_alert`	string	Identifies whether the event is an alert.	True
`account_age`	integer	The account age (i.e. the number of days since the opening of the bank account)	
`account_id`	string	The account Id used to log in	2802717d-82dc-4060-9e97-0157989ec55d
`account_last_login_attempt_at`	integer	Timestamp of last login attempt (in milliseconds since the Unix epoch)	
`account_last_successful_login_at`	integer	Timestamp of last successful login (in milliseconds since the Unix epoch)	
`account_login_first_seen`	integer	Timestamp of the account's first login (in milliseconds since the Unix epoch)	
`account_open_date`	integer	Account opening's timestamp (in milliseconds since the Unix epoch)	
`account_status_last_update_at`	integer	Timestamp of last update made to the customer account status (in milliseconds since the Unix epoch)	
`alias_bank_account_bic`	string	Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code linked to this alias	NWBK
	string	Number of the account linked to this alias	0000 0700 9312 3456 78

Name	Type	Description	Example
`alias_bank_account_number`			
`alias_bank_account_sort_code`	string	Sort code of the account linked to this alias	09-01-31
`alias_last_added_at`	integer	Timestamp of the last time an alias was added to this account number (in milliseconds since the Unix epoch)	
`alias_registration_date`	integer	Alias registration timestamp (in milliseconds since the Unix epoch)	
`alias_type`	string	'Alias type. Possible values are: phone, email, business_number, customer_id, qr_code, random_code, other'	email
`alias_value`	string	Alias value	
`auth_level`	string	'Level of authentication: 1fa, mfa'	mfa
`auth_password_pasted_flag`	boolean	Validates if the password was pasted by the user during the authentication	
`auth_type`	string	'Method used to authenticate. Possible values are: password, biometric, pin, other'	password
`auth_username_pasted_flag`	boolean	Validates if the username was pasted by the user during the authentication	
`card_available_balance`	integer	The card available balance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).	
`card_bin`	string	Card's bank identification number.	123456
`card_brand`	string	Card Brand.	Visa
`card_country_code`	string	Represents the country (in accordance with ISO 3166, Alpha-2) where the card was issued.	US
`card_id`	string	Persistent unique card ID (even after operations that might lead to an ID change such as card re-issuing).|Feedzai	eb7a757cc59081a84fe7f827b649e9c6c7b12338
`card_issuer`	string	Identifies the issuing institution.	Issuer Bank
`card_status`	string	Status of the card (active, blocked, under validation, etc.)	active
`cardholder_name`	string	Cardholder name.	John Doe
`challenge_requested`	boolean	Whether a second update (challenge) was performed for the current event	
`challenge_status`	string	'Challenge status. Possible values are: success, fail, pending, cancelled, unknown'	pending
`challenge_status_fail_reason`	string	'Reason why the challenge failed. Possible values are: cancelled, timeout, error, unknown'	cancelled
`challenge_type`	string	The method used by the customer to complete the challenge (e.g. PIN, 2FA, 3DS, ONE_TIME_CODE)	PIN

Name	Type	Description	Example
`channel_type`	string	'Bank channel the customer is accessing. Examples: web/online, mobile, branch, telephone and other'	web
`credentials_attempts_remaining`	integer	The remaining number of credential retrieval attempts	
`credentials_exp_date`	integer	Credentials' expiration timestamp (in milliseconds since the Unix epoch)	
`credentials_last_update_at`	integer	Timestamp of credentials' last changes (in milliseconds since the Unix epoch)	
`customer_account_fraud_30d`	integer	Number of account's confirmed frauds on the last 30 days	
`customer_account_fraud_3d`	integer	Number of account's confirmed frauds on the last 3 days	
`customer_account_fraud_6m`	integer	Number of account's confirmed frauds on the last 6 months	
`customer_account_rejected_30d`	integer	Number of account's rejected transactions on the last 30 days	
`customer_account_rejected_3d`	integer	Number of account's rejected transactions on the last 3 days	
`customer_account_rejected_6m`	integer	Number of account's rejected transactions on the last 6 months	
`customer_account_rep_fraud_30d`	integer	Number of account's reported frauds on the last 30 days	
`customer_account_rep_fraud_3d`	integer	Number of account's reported frauds on the last 3 days	
`customer_account_rep_fraud_6m`	integer	Number of account's reported frauds on the last 6 months	
`customer_account_settle_30d`	integer	Number of account's settlements on the last 30 days	
`customer_account_settle_3d`	integer	Number of account's settlements on the last 3 days	
`customer_account_settle_6m`	integer	Number of account's settlements on the last 6 months	
`customer_active`	boolean	Whether the customer is active.	
`customer_activity_status`	string	'Customer activity status. Possible values are: employed, unemployed, other'	employed

Name	Type	Description	Example
`customer_addr_last_update_at`	integer	Timestamp of last update made to the customer address (in milliseconds since the Unix epoch)	
`customer_address`	string	Customer address	757 Corner Street
`customer_address_validated`	boolean	Whether the customer address is validated.	
`customer_alias`	string	Customer's alias value. This field only applies when the payment is made with an alias, such as using email or phone number as the key to transfer the funds. An alias is mostly used in instant payments.	
`customer_alias_dict_reg_date`	integer	Date of registry - milliseconds - on DICT (DiretÃ³rio de Identificadores de Contas Transacionais)	
`customer_alias_fraud_30d`	integer	Number of alias' confirmed frauds on the last 30 days	
`customer_alias_fraud_3d`	integer	Number of alias' confirmed frauds on the last 3 days	
`customer_alias_fraud_6m`	integer	Number of alias' confirmed frauds on the last 6 months	
`customer_alias_last_update_time`	integer	Identifies the timestamp of the date the customer's alias is updated (in milliseconds since the Unix epoch).	
`customer_alias_registration_time`	integer	Identifies the timestamp of the date the customer's alias is registered (in milliseconds since the Unix epoch).	
`customer_alias_rejected_30d`	integer	Number of alias' rejected transactions on the last 30 days	
`customer_alias_rejected_3d`	integer	Number of alias' rejected transactions on the last 3 days	
`customer_alias_rejected_6m`	integer	Number of alias' rejected transactions on the last 6 months	
`customer_alias_rep_fraud_30d`	integer	Number of alias' reported frauds on the last 30 days	
`customer_alias_rep_fraud_3d`	integer	Number of alias' reported frauds on the last 3 days	
`customer_alias_rep_fraud_6m`	integer	Number of alias' reported frauds on the last 6 months	
`customer_alias_settle_30d`	integer	Number of alias' settlements on the last 30 days	
`customer_alias_settle_3d`	integer	Number of alias' settlements on the last 3 days	

Name	Type	Description	Example
`customer_alias_settle_6m`	integer	Number of alias' settlements on the last 6 months	
`customer_alias_status`	string	The status of the customer's alias. The possible values are `active` and `deleted` .	active
`customer_alias_type`	string	Customer's alias type. This field only applies when the payment is made with an alias.	email
`customer_branch_id`	string	Customer branch id	960e6560-450f-401e-805a-b565becb4a45
`customer_citizen_id_number`	string	Identification number defined in the citizen identity document associated with the customer's home country.	
`customer_city`	string	Customer city	Rome
`customer_contactoptout`	boolean	Whether the customer opted out of receiving communications.	
`customer_country_code`	string	Customer country code (in accordance with ISO 3166, Alpha-2)	US
`customer_currency`	string	Currency associated with the customer. Refers to customers' total balance and credit limit.	USD
`customer_dob`	string	Customer date of birth (YYYYMMDD)	19730312
`customer_driver_license`	string	Customer driver's license.	F4540525
`customer_email`	string	Customer email	
`customer_email_last_update_at`	integer	Timestamp of last update made to the customer email address (in milliseconds since the Unix epoch)	
`customer_email_validated`	boolean	Whether the customer email is validated.	
`customer_fraud_30d`	integer	Number of customer's confirmed frauds on the last 30 days	
`customer_fraud_3d`	integer	Number of customer's confirmed frauds on the last 3 days	
`customer_fraud_6m`	integer	Number of customer's confirmed frauds on the last 6 months	
`customer_gender`	string	Customer gender.	female
`customer_id`	string	Customer's unique identifier	453b9489-6c57-4b87-b1e5-8468faf07b26
`customer_kyc_lvl`	string	Know your customer (KYC) level.	A
`customer_mothersname`	string	Customer mother's name.	Jane Doe
`customer_name`	string	Customer name	Keneth Boyle
`customer_number_of_accounts`	integer	Number of accounts associated with the customer.	
	integer	Number of cards associated with the customer.	

Name	Type	Description	Example
`customer_number_of_cards`			
`customer_occupation`	string	Customer occupation	actor
`customer_passport_number`	string	Customer passport number.	640635130
`customer_phone_last_update_at`	integer	Timestamp of last update made to the customer phone number (in milliseconds since the Unix epoch)	
`customer_phone_number`	string	Customer phone number	3240396761
`customer_phone_val_code`	string	Indicates whether the customer phone number is valid.	partially valid
`customer_preferred_language`	string	Customer preferred language	en-US
`customer_pwd`	string	Customer's person with disability status.	visual impairment
`customer_region`	string	Customer region	CA
`customer_registration_date`	integer	Customer registration date (in milliseconds since the Unix epoch).	
`customer_rejected_30d`	integer	Number of customer's rejected transactions on the last 30 days	
`customer_rejected_3d`	integer	Number of customer's rejected transactions on the last 3 days	
`customer_rejected_6m`	integer	Number of customer's rejected transactions on the last 6 months	
`customer_rep_fraud_30d`	integer	Number of customer's reported frauds on the last 30 days	
`customer_rep_fraud_3d`	integer	Number of customer's reported frauds on the last 3 days	
`customer_rep_fraud_6m`	integer	Number of customer's reported frauds on the last 6 months	
`customer_reported_income`	integer	Amount of annual income reported by the customer.	5000
`customer_risk_rating`	string	Customer's risk rating.	A
`customer_segment`	string	Specifies the segment of the payment channel (e.g. retail_banking, business_banking, commercial_banking)	retail_banking
`customer_settle_30d`	integer	Number of customer's settlements on the last 30 days	
`customer_settle_3d`	integer	Number of customer's settlements on the last 3 days	
`customer_settle_6m`	integer	Number of customer's settlements on the last 6 months	
`customer_ssn`	string	Customer social security number.	204541535
	string	Customer tax number.	951755612

Name	Type	Description	Example
`customer_tax_number`			
`customer_total_balance`	integer	Sum of the balances on all customer's accounts (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).	
`customer_total_credit_limit`	integer	Sum of the credit limit on all customer's accounts (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).	
`customer_type`	string	Customer type.	individual
`customer_username`	string	Customer username.	johnDoe
`customer_zip_area_code`	string	Customer ZIP area code	77334
`customer_zip_code`	string	Customer ZIP code	77334-3501
`daily_amount_limit`	integer	Maximum daily amount that can be spent for this account	
`device_account_id`	string	Unique device identification of a device/account combination	a85044ba0c1d45dbb64fedde94c16923
`device_android_id`	string	The device Android ID	723dc71eec6ccb31
`device_app_version`	string	App version	1.0
`device_bot`	boolean	Indicates the presence of BOT-like behavior during the use of the digital channel	
`device_brand`	string	The user's visible brand name, and whose product/hardware will be associated with, if any	Samsung
`device_browser`	string	The browser used by the device	Chrome
`device_browser_language`	string	The device's browser language code	en-US
`device_browser_no_tracking`	boolean	Whether the device browser has no tracking enabled	
`device_browser_private_mode`	boolean	Whether the device browser has private mode enabled	
`device_browser_version`	string	The browser version this device is using	89.0.4389.72
`device_cbf`	string	The device CBF	723dc71eec6ccb31
`device_city`	string	Device city	London
`device_cookies_enabled`	boolean	Whether cookies are enabled on this device's browser	
`device_country_code`	string	Device country code (in accordance with ISO 3166, Alpha-2)	US
	string	Email associated with the device.	

Name	Type	Description	Example
`device_email_addr`			
`device_emulator`	boolean	Indicates that the digital channel is being used via an emulated device	
`device_fingerprint`	string	Information collected about the software and hardware of the device used to perform the transaction. Could be encrypted information used for device enrichment.|Feedzai	0400YbjucgPeDpqN9ZBLUITS+9VZ4/+YT+qYutCN/R8/6uxqb6W7w==
`device_first_seen`	integer	Timestamp of the first time a device Id was seen (in milliseconds since the Unix epoch)	
`device_flash_enabled`	boolean	Whether Flash is enabled on this device	
`device_flash_version`	string	The Flash code this device is using	32.0.0.465
`device_id`	string	Device from which the event was initiated	6a340affb04e4c3a97754828d6c6b550
`device_imei`	string	IMEI, required if the request originated from a mobile device	49015420323751
`device_ip_address`	string	Device IP address	13.59.108.243
`device_ip_address_location`	string	Address location of the device.|Device information	Manchester
`device_ip_botnet_intensity`	integer	A measurement of the frequency of suspected botnet activity associated with the IP address. This is a ranking between 0-10 where 0 indicates minimal botnet activity and 10 is a very high frequency of botnet activity.	
`device_ip_city`	string	The device IP's city	London
`device_ip_country_code`	string	The device IP's country	US
`device_ip_first_seen`	integer	Timestamp of the first time a device IP was seen (in milliseconds since the Unix epoch)	
`device_ip_latitude`	double	Latitude corresponding to the location where the device IP is at.	25.582727
`device_ip_longitude`	double	Longitude corresponding to the location where the device IP is at	-17.295952
`device_ip_proxy`	boolean	Whether the device IP corresponds to a proxy	
`device_ip_proxy_type`	string	Type of proxy used (e.g. Anonymous, Hidden)	Hidden
`device_ip_region`	string	The device IP's region	CA
`device_is_new`	boolean	Whether the device has never been seen before.	
`device_isp`	string	The Internet Service Provider used by the device	Verizon
`device_isp_country_code`	string	The country code of the Internet Service Provider used by the device (in accordance with ISO 3166, Alpha-2)	US

Name	Type	Description	Example
`device_isp_region`	string	'The region of the Internet Service Provider used by the device '	CA
`device_java_script_enabled`	boolean	Whether JavaScript is enabled on this device	
`device_language`	string	The device language settings	en-US
`device_last_update_at`	integer	Timestamp of device's last registration, change or deletion in this account (in milliseconds since the Unix epoch)	
`device_latitude`	double	Latitude corresponding to the location where the device is at	12.343423
`device_longitude`	double	Longitude corresponding to the location where the device is at	29.234232
`device_mac_address`	string	The device Mac Address	20:f4:78:dd:33:9b
`device_malware`	boolean	Indicates the presence of malware in the customer's device during the use of the digital channel	
`device_mobile_system_root_detection_disabled`	boolean	Whether the user has attempted to hide detection of rooting or jailbreaking.	
`device_mobile_system_simulator`	boolean	Whether the transaction was initiated from a simulator.	
`device_model`	string	The end user's device model name (empty for desktop)	iPhone6
`device_network_type`	string	The device network type	STD
`device_os`	string	Device operating system	WINDOWS NT 6.3
`device_os_language`	string	The device's operating system language code	en-us
`device_os_version`	string	The device's operating system version	10.10.3
`device_phishing`	boolean	Whether a phishing flag was detected	
`device_phone_number`	string	Device phone number	4378630800
`device_rat`	boolean	Indicates the presence of remote access to the customer's device during the use of the digital channel	
`device_region`	string	Device region	CA
`device_registered`	boolean	Whether the device has been registered	false
`device_registration_date`	integer	Device registration's timestamp (in milliseconds since the Unix epoch)	
	string		low

Name	Type	Description	Example
`device_risk_rating`		Risk mapped according to the enricher policy score (low, medium, high)	
`device_risk_reason`	string	Risk mapped according to the enricher policy score	Device_Negative_History
`device_risk_score`	integer	Risk mapped according to the enricher policy score (0-999)	
`device_rooted`	boolean	Indicates that the device in use has been jailbroken/rooted	
`device_screen_height`	integer	The device screen's total height in pixels	
`device_screen_width`	integer	The device screen's total width in pixels	
`device_serial_id`	string	The device serial ID	12345678
`device_session_duration`	integer	The device session duration in milliseconds	
`device_timezone`	string	Device timezone	+01:00
`device_tor`	boolean	Whether the request is coming from a TOR-enabled network	
`device_type`	string	Type of device from which the event was initiated	web browser
`device_user_agent`	string	User Agent string from browser. Required if the request originated from a web browser	Mozilla/5.0 (Linux; Android 8.0.0; SM-G960F Build/R16NW) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/62.0.3202.84 Mobile Safari/537.36
`device_user_name`	string	Device hostname	Daisey's phone
`event_external_id`	string	Event's unique identifier on the customer's platform	4048611b295e993d1b561d9030c53424bbf2b95e
`event_occurred_at`	integer	Timestamp of the current event occurred in the client's system (in milliseconds since the Unix epoch)	
`event_status`	string	'Status of the update attempt. The possible values are: success, fail'	success
`event_status_fail_reason`	string	'Reason why the update fails. The possible values are: invalid_credentials, device_verification_failed, captcha_failed, challenge_failed, other'	invalid_credentials
`event_sub_type`	string	'Event sub-type. Possible values: new_device, change_device, delete_device'	new_device
`fake_location`	string	Includes information when the geolocation is fake	Geolocation is fake
`fuzzy_device_id`	string	'Device''s unique fingerprint. It identifies the device even if cookies/persistent objects have been deleted or if a user has invoked â□□Private Browsing Modeâ□□'	19473c81b27f47fb90a077b6da3587b7
`lifecycle_id`	string	Unique identifier update	ce8e8f0e-a0da-4d26-a0a0-6e71b33a13c1
`max_daily_amount_limit`	integer	The upper limit for the maximum daily amount that can be spent for this account	

Name	Type	Description	Example
`network_bssid`	string	'Basic Server Set Identifier '	router
`network_ssid`	string	Service Set Identifier - the wifi network name	network_name
`new_alias_bank_account_bic`	string	New Bank Identifier Code (BIC) - in SWIFT format or any other bank unique identifier code - of the account linked to this alias	NWBK
`new_alias_bank_account_number`	string	New number of the account linked to this alias	0000 0700 9312 3456 78
`new_alias_bank_account_sort_code`	string	New sort code of the account linked to this alias	53-44-31
`new_alias_last_added_at`	integer	Timestamp of the last time an alias was added to this account number (in milliseconds since the Unix epoch)	
`new_alias_registration_date`	integer	New alias registration timestamp (in milliseconds since the Unix epoch)	
`new_alias_type`	string	'New alias type. Possible values are: phone, email, business_number, customer_id, qr_code, random_code, other'	email
`new_alias_value`	string	New alias value	
`new_credentials_attempts_remaining`	integer	The new remaining number of credential retrieval attempts	
`new_credentials_exp_date`	integer	New credentials' expiration timestamp (in milliseconds since the Unix epoch)	
`new_credentials_last_update_at`	integer	Timestamp of the last changes made to new credentials (in milliseconds since the Unix epoch)	
`new_customer_addr_last_update_at`	integer	Timestamp of last update made to the new customer's address (in milliseconds since the Unix epoch)	
`new_customer_address`	string	New address associated with this account	
`new_customer_email_addr`	string	New email address associated with this account	
`new_customer_email_last_update_at`	integer	Timestamp of last update made to the new customer's email address (in milliseconds since the Unix epoch)	
`new_customer_phone_last_update_at`	integer	Timestamp of last update made to the new customer's phone number (in milliseconds since the Unix epoch)	
`new_customer_phone_number`	string	New phone number associated with this account	1-970-664-2654
`new_customer_username`	string	New username associated with this account	amos.luettgen
`new_daily_amount_limit`	integer	New maximum daily amount that can be spent for this account	

Name	Type	Description	Example
`new_device_account_id`	string	Unique device identification of a device/account combination	a85044ba0c1d45dbb64fedde94c16923
`new_device_android_id`	string	The device Android ID	723dc71eec6ccb31
`new_device_app_version`	string	App version	1.0
`new_device_bot`	boolean	Indicates the presence of BOT-like behavior during the use of the digital channel	
`new_device_brand`	string	The user's visible brand name, and whose product/hardware will be associated with, if any	Samsung
`new_device_browser`	string	Browser of the new device associated with this account	Chrome
`new_device_browser_language`	string	Browser language of the new device associated with this account	en-US
`new_device_browser_no_tracking`	boolean	Whether the device browser has no tracking enabled	
`new_device_browser_private_mode`	boolean	Whether the device browser has private mode enabled	
`new_device_browser_version`	string	Browser version of the new device associated with this account	89.0.4389.72
`new_device_cbf`	string	The device CBF	723dc71eec6ccb31
`new_device_city`	string	City of the new device associated with this account	Arronbury
`new_device_cookies_enabled`	boolean	Whether cookies are enabled on this device's browser	
`new_device_country_code`	string	Country of the new device associated with this account	US
`new_device_email`	string	Email address of the new device associated with this account	
`new_device_emulator`	boolean	Indicates that the digital channel is being used via an emulated device	
`new_device_first_seen`	integer	Timestamp of the first time a device Id was seen (in milliseconds since the Unix epoch)	
`new_device_flash_enabled`	boolean	Whether Flash is enabled on this device	
`new_device_flash_version`	string	The Flash code this device is using	32.0.0.465
`new_device_id`	string	Id of the new device that is being associated with this account	6a340affb04e4c3a97754828d6c6b550
`new_device_imei`	string	IMEI, required if the request originated from a mobile device	49015420323751
`new_device_ip_address`	string	Device IP address	13.59.108.243

Name	Type	Description	Example
`new_device_ip_city`	string	The device IP's city	London
`new_device_ip_country_code`	string	The device IP's country	US
`new_device_ip_first_seen`	integer	Timestamp of the first time a device IP was seen (in milliseconds since the Unix epoch)	
`new_device_ip_latitude`	double	Latitude corresponding to the location where the device IP is at.	25.582727
`new_device_ip_longitude`	double	Longitude corresponding to the location where the device IP is at	-17.295952
`new_device_ip_proxy`	boolean	Whether the device IP corresponds to a proxy	
`new_device_ip_proxy_type`	string	Type of proxy used (e.g. Anonymous, Hidden)	Hidden
`new_device_ip_region`	string	The device IP's region	CA
`new_device_isp`	string	The Internet Service Provider used by the device	Verizon
`new_device_isp_country_code`	string	The country code of the Internet Service Provider used by the device (in accordance with ISO 3166, Alpha-2)	US
`new_device_isp_region`	string	'The region of the Internet Service Provider used by the device '	CA
`new_device_java_script_enabled`	boolean	Whether JavaScript is enabled on this device	
`new_device_language`	string	Language of the new device associated with this account	en-US
`new_device_latitude`	double	Latitude corresponding to the location where the device is at	12.343423
`new_device_longitude`	double	Longitude corresponding to the location where the device is at	29.234232
`new_device_mac_address`	string	The device Mac Address	20:f4:78:dd:33:9b
`new_device_malware`	boolean	Indicates the presence of malware in the customer's device during the use of the digital channel	
`new_device_model`	string	Model of the new device associated with this account	iPhone6
`new_device_network_type`	string	The device network type	STD
`new_device_os`	string	Device operating system	WINDOWS NT 6.3
`new_device_os_language`	string	The device's operating system language code	en-us
`new_device_os_version`	string	The device's operating system version	10.10.3
	boolean	Whether a phishing flag was detected	

Name	Type	Description	Example
`new_device_phishing`			
`new_device_phone_number`	string	Phone number of the new device associated with this account	893-631-5041
`new_device_rat`	boolean	Indicates the presence of remote access to the customer's device during the use of the digital channel	
`new_device_region`	string	Device region	CA
`new_device_registered`	boolean	Whether the device has been registered	false
`new_device_registration_date`	integer	Device registration's timestamp (in milliseconds since the Unix epoch)	
`new_device_risk_rating`	string	Risk mapped according to the enricher policy score (low, medium, high)	low
`new_device_risk_reason`	string	Risk mapped according to the enricher policy score	Device_Negative_History
`new_device_risk_score`	integer	Risk mapped according to the enricher policy score (0-999)	
`new_device_rooted`	boolean	Indicates that the device in use has been jailbroken/rooted	
`new_device_screen_height`	integer	The device screen's total height in pixels	
`new_device_screen_width`	integer	The device screen's total width in pixels	
`new_device_serial_id`	string	The device serial ID	12345678
`new_device_session_duration`	integer	The device session duration in milliseconds	
`new_device_timezone`	string	Timezone of the new device associated with this account	+01:00
`new_device_tor`	boolean	Whether the request is coming from a TOR-enabled network	
`new_device_type`	string	Type of device from which the event was initiated	web browser
`new_device_user_agent`	string	User Agent string from browser. Required if the request originated from a web browser	Mozilla/5.0 (Linux; Android 8.0.0; SM-G960F Build/R16NW) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/62.0.3202.84 Mobile Safari/537.36
`new_device_user_name`	string	Device hostname	Daisey's phone
`new_fuzzy_device_id`	string	'Device''s unique fingerprint. It identifies the device even if cookies/persistent objects have been deleted or if a user has invoked â□□Private Browsing Modeâ□□'	19473c81b27f47fb90a077b6da3587b7
`new_payee_account_type`	string	Account type of the new payee (business, personal, savings)	
`new_payee_alias_type`	string		email

Name	Type	Description	Example
		'New payee alias type. Possible values: phone, email, business_number, customer_id, qr_code, random_code, other'	
`new_payee_alias_value`	string	New payee alias	
`new_payee_bank_account_bic`	string	New payee's Bank Identifier Code (BIC) - in SWIFT format or any other bank unique identifier code - added to this account	NWBK
`new_payee_bank_account_number`	string	Bank account number of the new payee	0000 0700 9312 3456 78
`new_payee_bank_account_sort_code`	string	Sort code of the new payee that is being added to this account	09-01-31
`new_payee_last_added_at`	integer	Timestamp of last time a new payee was added to the account (in milliseconds since the Unix epoch)	
`new_payee_name`	string	First name of the new payee	Bea Smith
`previous_auth_result`	string	'Result of previous authentication attempt. Possible values are: success, fail'	success
`session_id`	string	The session Id used to log in. This should be consistent across all events within the same banking session, including payment initiations, as it will be used to associate events with a session	b6afe37d-2b05-4104-98be-1cbf25327e95
`session_register_date`	integer	Session first seen timestamp, (in milliseconds since the Unix epoch)	
`tenant_id`	string	''	
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

-
- References
- Data Exporting Tool
- Event Storage
- Event Storage - Digital Activity Feedback

Event Storage - Digital Activity Feedback

Was this page helpful?Â

The **Event Storage Feedback Schema** stores the basic information of any feedback event.

Name	Type	Description	Example
`fdz_uuid`	string	Identifier of each event in UUID	061ad3f3-9991-49fe-850f-e49d6d981028
`timestamp`	double	Unix Timestamp in milliseconds	1760000000.0
`statusreasons`	string	A JSON Array with the status reasons. This only exists if lifecycle_id is null.	('id': '1234-12345-12345', 'name': 'At your discretion', 'statusid': 'real_event')
`lifecycle_id`	string	The lifecycle ID of the event. This only exists when statusreasons is null.	

Name	Type	Description	Example
`realoutcome`	boolean	Indicates whether this was the real outcome or not.	
`feedbacklabel`	string	Label of the Feedback event	fraud
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Event Storage - Transfers

Was this page helpful?Â

The **Event Storage - Transfers** entity stores transfer related events, such as transfer initiations, information and settlements.

A new record becomes available for this entity for every transfer processed by the risk engine.

Name	Type	Description	Example
`receiver_account_limits`	array<row>	The receiver account limits.	
	string		

Name	Type	Description	Example
â□□â□□â□□ `event_type`		The type of events associated with this limit.	
â□□â□□â□□ `day_amount`	bigint	Total amount on the account day period	123
â□□â□□â□□ `daily_amount`	bigint	Total daily amount	100
â□□â□□â□□ `night_amount`	bigint	Total amount on the account night period	123
â□□â□□â□□ `single_day_amount`	bigint	Single transaction amount limit on the day period.	333
â□□â□□â□□ `single_night_amount`	bigint	Single transaction amount limit on the night period.	222
â□□â□□â□□ `start_day_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
â□□â□□â□□ `start_night_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
`sender_account_limits`	array<row>	The sender account limits.	
â□□â□□â□□ `event_type`	string	The type of events associated with this limit.	
â□□â□□â□□ `day_amount`	bigint	Total amount on the account day period	123
â□□â□□â□□ `daily_amount`	bigint	Total daily amount	100
â□□â□□â□□ `night_amount`	bigint	Total amount on the account night period	123
â□□â□□â□□ `single_day_amount`	bigint	Single transaction amount limit on the day period.	333
â□□â□□â□□ `single_night_amount`	bigint	Single transaction amount limit on the night period.	222
â□□â□□â□□ `start_day_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
â□□â□□â□□ `start_night_hour`	bigint	Used to compare the event date and the period transaction limit specified for the client.	
`external_fraud_service`	row	External fraud service score.	
â□□â□□â□□ `external_score`	string		

Name	Type	Description	Example
â□□â□□â□□ external_score_reason	string		
â□□â□□â□□ external_score_recommendation	string		
â□□â□□â□□ sender_fraud_date	bigint		
â□□â□□â□□ sender_fraud_involvement	string		
â□□â□□â□□ sender_fraud_reason_code	string		
â□□â□□â□□ sender_fraud_reason_desc	string		
â□□â□□â□□ sender_fraud_source	string		
â□□â□□â□□ sender_fraud_channel	string		
â□□â□□â□□ sender_fraud_product	string		
â□□â□□â□□ sender_fraud_flag	boolean		
â□□â□□â□□ receiver_fraud_date	bigint		
â□□â□□â□□ receiver_fraud_involvement	string		
â□□â□□â□□ receiver_fraud_reason_code	string		
â□□â□□â□□ receiver_fraud_reason_desc	string		
	string		

Name	Type	Description	Example
â□□â□□â□□ `receiver_fraud_source`			
â□□â□□â□□ `receiver_fraud_channel`	string		
â□□â□□â□□ `receiver_fraud_product`	string		
â□□â□□â□□ `receiver_fraud_flag`	boolean		
`rules_explanations`	string	Explanation of rules triggered in text format.	Rule X triggered because X amount surpasse the Y limit.
`rules_ids`	string	Identifier of the rules.	
`uuid`	string	UUID.	2bf75ff1-0b6f-48fd-bdf2-cf458243eaf2
`direction`	string	Payment direction flow. Possible values are incoming (or outgoing in legacy systems)	incoming
`decision`	string	Output decision from the system.	approve
`action_names`	string		
`fdz_partition_index`	integer	Partition index.	79
`fdz_tenant_key`	string	Tenant key from Feedzai perspective.	sample_value
`fdz_timestamp`	bigint	Unix timestamp.	1756463007000
`fdz_uuid`	string	UUID from Feedzai perspective.	d392378a-f4b2-49b5-8814-e3ea65543420
`is_alert`	string	Identifies whether the event is an alert.	True
`approval_current_number`	integer	Number of current approvals.|Approval information	
`approval_final_approval`	boolean	Identifies whether the final approval has been reached so that the transfer can continue according with its life cycle.|Approval information	
`approval_primary_admin`	boolean	Identifies whether the transaction contains the approval of the primary administrator or not.|Approval information	
`approval_signatures_applied`	integer	Number of physical signatures applied to this transaction.|Approval information	
`approval_status`	string	Approval status of the transaction. The values of this field depend on the `approval_type` field.|Approval information	pending
`approval_submitted_at`	integer	Date at which the approval was submitted (in milliseconds since the Unix epoch).|Approval information	

Name	Type	Description	Example
`approval_total_approves`	integer	Total number of necessary approvals for the payment to be completed.|Approval information	
`approval_type`	string	Each institution might have a different combination of approvals and an associated nomenclature. This field enables the institutions to explicitly qualify the group of approvals present on this transaction.|Approval information	type 1
`approver_customer_id`	string	Unique customer ID.|Approver information	29298w9askajakjsk
`approver_login_id`	string	Customer internet banking ID.|Approver information	2l209wms209029w
`approver_type`	string	Each institution might have a different combination of approvals and an associated nomenclature. This field enables the institutions to explicitly qualify the group of approvals present on this transaction. This field is related to `approval_type` and `approval_status` fields.|Approver information	type 1
`auth_type`	string	Authentication option used by the consumer during the payment process.| Transfer information	fingerprint
`bulk_id`	string	Unique reference ID that refers to the bulk transaction. This can be used by banks for internal control.|Transfer information	b398a9e1-3900-4753-ad45-8310ea583724
`challenge_risk_flag`	boolean	Identifies whether the transfer has an authentication challenge or not.| Authentication information	
`challenge_risk_score`	integer	Quantifies the risk associated with automatic challenges sent to the clients to allow authentication.|Authentication information	
`channel_type`	string	Channel through which the transaction occurred. For example, possible values are `branch` , `telephone` , `atm` , `online` / `web` , `mobile` and `other` . | Transfer information	branch
`created_at`	integer	Timestamp corresponding to the moment when the payment is created (in milliseconds since the Unix epoch). Note that the payment can be created, saved and then sent later. | Transfer information	
`cutoff_at`	integer	Specified cut-off date and time for the payment consent (in milliseconds since the Unix epoch).|Transfer information	
`dd_mandate_name`	string	The name of the direct debit mandate initiator.|Transfer information	Mandate name
`dd_mandate_number`	string	The direct debit mandate identification number.|Transfer information	Mandate number 123
	string	The device Android ID	723dc71eec6ccb31

Name	Type	Description	Example
`device_android_id`			
`device_app_version`	string	App version of the device. This field is only requested if the transfer is originated from a mobile app.|Device information	2.10.1
`device_browser`	string	The browser used by the device	Chrome
`device_browser_language`	string	The device browser language	es-ES
`device_browser_no_tracking`	boolean	Whether the device browser has no tracking enabled	
`device_browser_private_mode`	boolean	Whether the device browser has private mode enabled	
`device_browser_version`	string	The browser version this device is using	89.0.4389.72
`device_cbf`	string	The device CBF	723dc71eec6ccb31
`device_cookies`	boolean	Whether the cookies are enabled on the web browser. Sending this information is optional because it's not always available to the Feedzai's client.|Device information	
`device_fingerprint`	string	Information collected about the software and hardware of the device used to perform the transaction. Could be encrypted information used for device enrichment.|Feedzai	0400YbjucgPeDpqN9ZBLUITS+9VZ4/+YT+qYutCN/R8/6uxqb6W7w==
`device_first_seen`	integer	Timestamp corresponding to the first time the device is used for the transaction (in milliseconds since the Unix epoch).|Device information	
`device_id`	string	Unique identifier of the device.|Device information	d5e8c80119a87f4c
`device_imei_number`	string	International Mobile Equipment Identity (IMEI). This field is only requested if the transfer is originated from a mobile device.|Device information	447292738273482
`device_ip_address`	string	IP address of the device used to conduct the transaction. | Device information	219.6.112.233
`device_ip_address_location`	string	Address location derived from the device's IP address. | Device information	Manchester
`device_ip_country_code`	string	The country code derived from the device's IP (in accordance with ISO 3166, Alpha-2). | Transfer information	US
`device_ip_region`	string	The device IP's region	CA
`device_is_new`	boolean	Whether the device has never been seen before.	
`device_isp`	string	The Internet Service Provider used by the device	Verizon

Name	Type	Description	Example
`device_language`	string	Device language code.|Transfer information	US
`device_latitude`	string	Geographic latitude of the device where the transfer originated. | Device information	53.48
`device_longitude`	string	Geographic longitude of the device where the transfer originated. | Device information	-2.24
`device_mac_address`	string	The device Mac Address	20:f4:78:dd:33:9b
`device_model`	string	The end user's device model name (empty for desktop)	iPhone6
`device_network_type`	string	The device network type	STD
`device_operating_system`	string	'Device''s operating system. Examples include: `MacOS` , `Windows 10` , `Linux` .|Device information'	MacOS
`device_os_version`	string	The device's operating system version	10.10.3
`device_registration_date`	integer	Device registration's timestamp (in milliseconds since the Unix epoch)	
`device_screen_height`	integer	The device screen's total height in pixels	
`device_screen_width`	integer	The device screen's total width in pixels	
`device_serial_id`	string	The device serial ID	12345678
`device_session_duration`	integer	The device session duration in milliseconds	
`device_tor_mode`	boolean	Whether the request is coming from a TOR enabled network. Sending this information is optional because it's not always available to the Feedzai's client.| Device information	
`device_type`	string	'Examples include: `Android` , `IOS` , `Windows Mobile` .|Device information'	android
`device_user_agent`	string	The user agent string from the browser used to initiate the transfer. This field is only requested if the transfer is originated from a web browser.|Device information	Mozilla/5.0 (Linux; Android 8.0.0; SM-G960F Build/R16NW) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/62.0.3202.84 Mobile Safari/537.36
`dispute_end_date`	integer	Timestamp corresponding to the moment when the dispute was closed (in milliseconds since the Unix epoch).| Transfer information	
`dispute_feedback_notes`	string	Contextual information associated with the dispute communication.|Transfer information	The account holder opened a dispute about a recent transaction.
`dispute_notes`	string		The device information seems suspicious.

Name	Type	Description	Example
		Contextual information left by the analyst at the time of the dispute opening.|Transfer information	
`dispute_reason`	string	Why the dispute was opened.|Transfer information	The payroll shows an unauthorized transaction.
`dispute_refunded`	string	'Whether the funds were refunded to the cardholder. Possible values: `no_refund`, `partial_refund`, `total_refund`.|Transfer information'	partial_refund
`dispute_start_date`	integer	Timestamp that corresponds to the time when the dispute was opened (in milliseconds since the Unix epoch).| Transfer information	
`dispute_submitted_by`	string	Who submitted the dispute.|Transfer information	Joe Dow
`end_to_end_identification`	string	Unique reference to unambiguously refer to the payment transaction across systems. This can be used by banks for internal control.|Transfer information	a2d311ac-c9c3-4ea7-9537-1db8dea0534c
`event_external_id`	string	Unique identifier on the client's side, typically used for the client to cross-reference it back into their own systems. For example, you can copy this field from Case Manager and look for it in another CRM used by your organization. | Transfer information	4048611b295e993d1b561d9030c53424bbf2b95e
`event_occurred_at`	integer	Timestamp corresponding to the moment when the current event occurred (in milliseconds since the Unix epoch).	
`execution_at`	integer	Timestamp corresponding to the moment when the payment is settled (in milliseconds since the Unix epoch). This can be in the past or in the future. | Transfer information	
`fake_location`	string	Includes information when the geolocation is fake	Geolocation is fake
`feedback_type`	string	'Identifies the different types of feedback. Possible values: `out_of_band` for conclusions of out-of-band communication and `chargeback` for closing disputes. Note that, in TFB, for the feedback endpoint, if `feedback_type`, if equals `out_of_band`, the required fields are `oob_response_type`, `oob_end_date, and ` oob_feedback_notes`.|Transfer information'	out_of_band
`feedback_value`	boolean	This field only applies to the Feedback endpoint. The value that should be associated with the `feedback_label`. E.g. if the label is `fraud`, the value `true` labels the transaction as fraudulent and the value `false` labels as non-fraudulent. | Transfer information	

Name	Type	Description	Example
`first_recurring_payment`	boolean	Whether the payment is the first occurrence of a series of recurring payments with the same amount and with the same payee.|Transfer information	
`frequency`	string	Recurrent scheduled transfer. Expected values include `weekly`, `each 15 days`, `monthly`, `quarter`, `bi-annual`, `annual` and `every two years`.|Transfer information	weekly
`geographic_scope`	string	'Possible values are: `domestic` - Between the same country and the same currency. `one_leg_out` - Between the same country and different currency. `cross_border` - Between different countries and different currencies.|Transfer information'	domestic
`info_type`	string	'Indicates the type of the info event. Possible values: `dispute`, `out_of_band`, `pre_decided`, `not_honor` and `additional_info`. Note that, in TFB, for the info endpoint, if `info_type` equals `out_of_band`, the required fields are `oob_comm_type` and `oob_start_date`.|Transfer information'	out_of_band
`instant_transfer_processing_method`	string	Method used to process the instant transfer between the sender and the receiver as defined by their profiles both in the instant transfer and in the initiation payment methods (e.g. App vs Home banking). Possible values are `account_to_account`, `account_to_card`, `card_to_account` and `card_to_card`.|Transfer information	account_to_account
`instant_transfer_type`	string	Type of instant transfer. Possible values are `purchase`, `purchase_with_change`, `withdrawal`.|Transfer information	purchase
`is_domestic`	boolean	Indicates if the current transaction is domestic.	
`last_sca_at`	integer	Timestamp, in milliseconds since the Unix epoch, corresponding to the last time that the customer performed Strong Customer Authentication (SCA) when SCA is enabled (mandatory within the European Union; optional outside the EU). | Transfer information	
`lifecycle_id`	string	Identifier of the transfer sent to Feedzai. Each new Info or Settlement request share the same life cycle ID as the original transfer.|Transfer information	1df7a7qg78vd7fs
`login_session_id`	string	Deprecated - Use 'session_id' instead.	fb2e77d
`malware_flag`	boolean		

Name	Type	Description	Example
		Indicates whether the device is infected with malware.|Transfer information	
`max_number_of_authentications`	integer	Number of max attempts authentications allowed for this client.|Authentication information	
`network_bssid`	string	'Basic Server Set Identifier '	router
`network_ssid`	string	Service Set Identifier - the wifi network name	network_name
`number_of_authentications`	integer	Number of attempts to authenticate.|Authentication information	
`number_of_payments`	integer	Specifies the number of payments from the same order.|Transfer information	
`oob_comm_type`	string	'The type of communication made with the client. Possible values: `sms` , `call` , `email` .|Transfer information'	sms
`oob_end_date`	integer	Timestamp corresponding to the moment when the client responded to the out-of-band communication (in milliseconds since the Unix epoch).|Transfer information	
`oob_feedback_notes`	string	Contextual information associated with the out-of-band communication.|Transfer information	The account holder denied the authenticity of the transaction.
`oob_notes`	string	Notes related to the out of band communication.|Transfer information	The customer has been already called.
`oob_response_type`	string	'The type of out-of-band response made by the client. Possible values: `call` , `sms` , `email` , `branch` . **Note that this field is required if the `feedback_type` field has the value `out_of_band`** |Transfer information'	call
`oob_start_date`	integer	Timestamp that corresponds to the moment when the client was contacted (in milliseconds since the Unix epoch).|Transfer information	
`pass_reset`	boolean	If a password reset has been recently initiated by the sender.|Authentication information	
`payment_reference`	string	Unique reference to unambiguously refer to the payment transaction. This can be used by banks for internal control.|Transfer information	bb3fd51c-049e-4bfb-8a22-36e6143da4a8
`payment_status`	string	'Life cycle status of the transfer. The following stage status may occur with the associated values: 1st stage status - `entered` , `approval pending` , `bank received` . 2nd stage status - `processing` , `accepted` , `rejected` , `return` , `recall` , `request for recall` , `waiting for recall` , `recall accepted` , `recall rejected` .|Transfer information'	entered

Name	Type	Description	Example
`payment_sub_type`	string	Specific service inside each payment type. Possible values are `instant_transfer`, `direct_debit`, `direct_deposit`, `credit_transfer`, `internal_transfer` and `bill_payment`.|Transfer information	instant_transfer
`payment_type`	string	Payment type. Possible values are `sepa`, `bacs`, `chaps`, `wire`, `swift`, `ach`, `fps`, `npp`, `bpay`, `pix`, `boleto`, `zelle`, `emt`, `rtp_tch`, `ted`, `tef`, `doc`, `internal`, and `other`.|Transfer information	sepa
`product_code`	string	The code of the product offered by the bank that was used to initiate the transfer.|Transfer information	68002
`proxy_flag`	boolean	Whether the device IP is a proxy.|Transfer information	
`purchased_amount`	integer	Purchase amount in cents (e.g. 1000 represents $10.00).|Transfer information	
`receiver_account_atc1`	string	Describes the account type affected for debits and inquiries and the â□□fromâ□□ account for transfers.|Receiver's account	atc1
`receiver_account_atc2`	string	Describes the account type affected for credits and the â□□toâ□□account for transfers.|Receiver's account	atc2
`receiver_account_available_credit`	integer	Available credit within the credit limit provided by the bank (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Receiver's account	
`receiver_account_balance`	integer	Current account balance of the person receiving the request (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Receiver's account	
`receiver_account_bic`	string	Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code.|Receiver's account	PTCDG
`receiver_account_cash_advance_balance`	integer	Debt amount in an account due to cash advance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Receiver's account	
`receiver_account_credit_limit`	integer	Account credit limit (in cents or equivalent). Not relevant for debit. For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Sender's account	
`receiver_account_currency`	string	Currency associated to the receiver's account (in accordance with ISO 4217, Alpha-3). | Receiver's account	GBP
	integer	Accountâ□□s debt balance (in cents or equivalent). For example, 1000 to	

Name	Type	Description	Example
`receiver_account_debt_balance`		represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Receiver's account	
`receiver_account_debt_payment_method`	string	Payment method of outstanding balance on the account.|Receiver's account	card
`receiver_account_fraud_30d`	integer	Number of account's confirmed frauds on the last 30 days	
`receiver_account_fraud_3d`	integer	Number of account's confirmed frauds on the last 3 days	
`receiver_account_fraud_6m`	integer	Number of account's confirmed fraudson the last 6 months	
`receiver_account_iban`	string	International Bank Account Number (IBAN) in ISO 13616:2020 format or any bank account unique identifier. This field should be sent as a token or hash.|Receiver's account	PT50 1234 5678 90
`receiver_account_id`	string	Banking App account id. This field should be sent as a token or hash.|Receiver's account	26574591-19c5-4ad6-9d22-e966cbcf2df7
`receiver_account_last_activity`	integer	Identifies the timestamp of the last activity registered at the account (in milliseconds since the Unix epoch).|Receiver's account	
`receiver_account_last_payment_amount`	integer	Amount of the last payment made on the account (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Receiver's account	
`receiver_account_last_payment_date`	integer	Date when the receiver pays off the balance on the account (in milliseconds since the Unix epoch).|Receiver's account	
`receiver_account_number_of_cards`	integer	Number of cards associated with the account.|Receiver's account	
`receiver_account_open_date`	integer	Opening date of the account (in milliseconds since the Unix epoch).|Receiver's account	
`receiver_account_otb_limit`	integer	Account open to buy limit is the available amount between the account balance and unused credit limit (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Receiver's account	
`receiver_account_pw_change_flag`	boolean	Whether the receiver changed the password on the account.|Receiver's account	
`receiver_account_rejected_30d`	integer	Number of account's rejected transactions on the last 30 days	
	integer		

Name	Type	Description	Example
`receiver_account_rejected_3d`		Number of account's rejected transactions on the last 3 days	
`receiver_account_rejected_6m`	integer	Number of account's rejected transactions on the last 6 months	
`receiver_account_rep_fraud_30d`	integer	Number of account's reported frauds on the last 30 days	
`receiver_account_rep_fraud_3d`	integer	Number of account's reported frauds on the last 3 days	
`receiver_account_rep_fraud_6m`	integer	Number of account's reported fraudson the last 6 months	
`receiver_account_rewards_used_count`	integer	The number of times that the receiver used the rewards.|Receiver's account	
`receiver_account_settle_30d`	integer	Number of account's settlements on the last 30 days	
`receiver_account_settle_3d`	integer	Number of account's settlements on the last 3 days	
`receiver_account_settle_6m`	integer	Number of account's settlementson the last 6 months	
`receiver_account_status`	string	'Status of the account. Possible values: `open`, `blocked`, `closed`, `waiting_for_kyc`.|Receiver''s account'	open
`receiver_account_type`	string	Type of the account. For example, savings or main account.|Receiver's account	credit
`receiver_alias`	string	Receiver's alias value. This field only applies when the payment is made with an alias, such as using email or phone number as the key to transfer the funds. An alias is mostly used in instant payments, such as PayPal or Venmo. | Receiver general information	
`receiver_alias_dict_reg_date`	integer	Date of registry on DICT (DiretÃ³rio de Identificadores de Contas Transacionais)	
`receiver_alias_fraud_30d`	integer	Number of alias' confirmed frauds on the last 30 days	
`receiver_alias_fraud_3d`	integer	Number of alias' confirmed frauds on the last 3 days	
`receiver_alias_fraud_6m`	integer	Number of alias' confirmed fraudson the last 6 months	
`receiver_alias_last_update_time`	integer	Identifies the timestamp of the date the receiver's alias is updated (in milliseconds since the Unix epoch).| Receiver general information	
	integer		

Name	Type	Description	Example
`receiver_alias_portab_date`		Start date of the portability or ownership claim process	
`receiver_alias_registration_time`	integer	Identifies the timestamp of the date the receiver's alias is registered (in milliseconds since the Unix epoch).|Receiver general information	
`receiver_alias_rejected_30d`	integer	Number of alias' rejected transactions on the last 30 days	
`receiver_alias_rejected_3d`	integer	Number of alias' rejected transactions on the last 3 days	
`receiver_alias_rejected_6m`	integer	Number of alias' rejected transactionson the last 6 months	
`receiver_alias_rep_fraud_30d`	integer	Number of alias' reported frauds on the last 30 days	
`receiver_alias_rep_fraud_3d`	integer	Number of alias' reported frauds on the last 3 days	
`receiver_alias_rep_fraud_6m`	integer	Number of alias' reported fraudson the last 6 months	
`receiver_alias_settle_30d`	integer	Number of alias' settlements on the last 30 days	
`receiver_alias_settle_3d`	integer	Number of alias' settlements on the last 3 days	
`receiver_alias_settle_6m`	integer	Number of alias' settlements on the last 6 months	
`receiver_alias_status`	string	The status of the receiver's alias. The possible values are `active` and `deleted` .|Receiver general information	active
`receiver_alias_type`	string	Receiver's alias type. This field only applies when the payment is made with an alias.|Receiver general information	email
`receiver_bank_address`	string	Address of the receiver's bank.|Receiver's bank	Newton Avenue 77
`receiver_bank_bic`	string	Bank Identifier Code (BIC) in SWIFT format. | Receiver's bank	HBUKGB4BXXX
`receiver_bank_branch_id`	string	Account bank branch id.|Receiver's bank	12-20-99
`receiver_bank_city`	string	City in which the receiver's bank is located.|Receiver's bank	Manchester
`receiver_bank_country_code`	string	The country code of where the receiver's bank is based (in accordance with ISO 3166, Alpha-2). | Receiver's bank	US
`receiver_bank_currency`	string	Currency of the receiver's bank (in accordance with ISO 4217, Alpha-3).|Receiver's bank	GBP
`receiver_bank_fee_amount`	integer	Fee that is applied to the receiver's bank included in the `sender_transactio`	

Name	Type	Description	Example
		`n_amount` field, in cents (e.g. 1000 to represent $10.00).|Receiver's bank	
`receiver_bank_fee_currency`	string	Defines the currency of the `receiver_bank_fee_amount` field (in accordance with ISO 4217, Alpha-3).|Receiver's bank	GBP
`receiver_bank_fee_type`	string	'Defines the method that is used to handle the payment charges or payment fees. Expected values include: - `OUR` : The sender of the payment will bear all of the payment transaction fees. The recipient of the payment will not pay any payment fees. - `SHA` (SHAred): The sender of the payment will pay all fees charged by the sending bank. The recipient of the payment will pay all fees charged by the receiving bank. - `BEN` (BENeficiary): The recipient of the payment will incur all of the payment transaction fees. The sender of the payment will not pay any payment fees. NOTE: Fees can happen at any stage of the transaction, it will depend on the Bilateral or multilateral agreements.|Receiver''s bank'	SHA
`receiver_bank_foreign_exchange_rate`	double	Foreign exchange rate, defined by the sender and receiver currencies.|Receiver's bank	
`receiver_bank_name`	string	Name of the receiver's bank.|Receiver's bank	Receiver Bank
`receiver_bank_postal_code`	string	Postal code of the receiver's bank.|Receiver's bank	WC7N 7EW
`receiver_bank_routing_number`	string	Account bank routing number.|Receiver's bank	201-192-111
`receiver_bank_sort_code`	string	Account bank sort code.|Receiver's bank	09-01-31
`receiver_citizen_id_number`	string	Identification number defined in the citizen identity document associated with the receiver's home country.|Receiver general information	12345678901234
`receiver_city`	string	Receiver's city.|Receiver general information	Manchester
`receiver_company_id`	string	Receiver company ID.|Receiver general information	852814c8-a24f-4269-a566-a1d10364b3cb
`receiver_correspondent_bank_address`	string	Bank address.|Receiver correspondent's bank	River Street 54
`receiver_correspondent_bank_bic`	string	Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code.|Receiver correspondent's bank	0ed7c577-b503-448a-95c8-b92342e0051e
`receiver_correspondent_bank_city`	string	City in which the bank is located.|Receiver correspondent's bank	London

Name	Type	Description	Example
`receiver_correspondent_bank_country_code`	string	The country code of the correspondent's bank (in accordance with ISO 3166, Alpha-2). | Receiver correspondent's bank	US
`receiver_correspondent_bank_currency`	string	Currency used by the Bank (in accordance with ISO 4217, Alpha-3).| Receiver correspondent's bank	GBP
`receiver_correspondent_bank_fee_amount`	integer	Fee that is applied to the sender's bank included in the `sender_transaction_amount` field, in cents (e.g. 1000 to represent $10.00).|Receiver correspondent's bank	
`receiver_correspondent_bank_fee_currency`	string	Defines the currency of the `fee_amount` field (in accordance with ISO 4217, Alpha-3).|Receiver correspondent's bank	GBP
`receiver_correspondent_bank_fee_type`	string	'Defines the method that is used to handle the payment charges or payment fees. Expected values include: - `OUR` : The sender of the payment will bear all of the payment transaction fees. The recipient of the payment will not pay any payment fees. - `SHA` (SHAred): The sender of the payment will pay all fees charged by the sending bank. The recipient of the payment will pay all fees charged by the receiving bank. - `BEN` (BENeficiary): The recipient of the payment will incur all of the payment transaction fees. The sender of the payment will not pay any payment fees. NOTE: Fees can happen at any stage of the transaction, it will depend on the Bilateral or multilateral agreements.| Receiver correspondent''s bank'	SHA
`receiver_correspondent_bank_foreign_exchange_rate`	double	Foreign exchange rate defined by the sender and receiver currencies.| Receiver correspondent's bank	
`receiver_correspondent_bank_name`	string	Bank name.|Receiver correspondent's bank	Receiver's Correspondent Bank SA
`receiver_correspondent_bank_postal_code`	string	Postal code of the bank.|Receiver correspondent's bank	AWB3 7Y3
`receiver_country_code`	string	The country code of the receiver / beneficiary (in accordance with ISO 3166, Alpha-2). | Receiver general information	US
`receiver_currency`	string	Currency associated to the receiver. Refers to receivers' total balance and credit limit.|Receiver general information	GBP
`receiver_dob`	string	Receiver's date of birth (YYYYMMDD).| Receiver general information	19900101
	string		1234567

Name	Type	Description	Example
`receiver_driver_license`		Receiver driver's license.|Receiver general information	
`receiver_email`	string	Receiver's main email contact. | Receiver general information	
`receiver_fraud_30d`	integer	Number of customer's confirmed frauds on the last 30 days	
`receiver_fraud_3d`	integer	Number of customer's confirmed frauds on the last 3 days	
`receiver_fraud_6m`	integer	Number of customer's confirmed fraudson the last 6 months	
`receiver_id`	string	Receiver ID generated by the receiver's bank.|Receiver general information	ADG13421
`receiver_kyc_lvl`	string	Know your receiver (KYC) level.| Receiver general information	A
`receiver_name`	string	The name of the individual receiving the payment. | Receiver general information	John Smith
`receiver_number_of_accounts`	integer	Number of accounts associated with the receiver.|Receiver general information	
`receiver_number_of_cards`	integer	Number of cards associated with the receiver.|Receiver general information	
`receiver_occupation`	string	Receiver's main professional activity.| Receiver general information	engineer
`receiver_occupation_status`	string	Receiver's current occupation status.| Receiver general information	employee
`receiver_passport_number`	string	Receiver passport number.|Receiver general information	PT-019796bb-783c-4020-8110-780e2900515b
`receiver_phone`	string	Receiver's main telephone contact. | Receiver general information	001122334455
`receiver_phone_val_code`	string	Indicates whether the receiver phone number is valid.|Receiver general information	001122
`receiver_region`	string	Receiver's region.|Receiver general information	Noth West
`receiver_registration_date`	integer	Receiver registration date (in milliseconds since the Unix epoch).| Receiver general information	
`receiver_rejected_30d`	integer	Number of customer's rejected transactions on the last 30 days	
`receiver_rejected_3d`	integer	Number of customer's rejected transactions on the last 3 days	
`receiver_rejected_6m`	integer	Number of customer's rejected transactions on the last 6 months	
`receiver_rep_fraud_30d`	integer	Number of customer's reported frauds on the last 30 days	
`receiver_rep_fraud_3d`	integer	Number of customer's reported frauds on the last 3 days	
`receiver_rep_fraud_6m`	integer	Number of customer's reported fraudson the last 6 months	

Name	Type	Description	Example
`receiver_risk_rating`	string	Receiver's risk rating.|Receiver general information	B
`receiver_segment`	string	Receiver's segment within the banks' segmentation.|Receiver general information	retail
`receiver_settle_30d`	integer	Number of customer's settlements on the last 30 days	
`receiver_settle_3d`	integer	Number of customer's settlements on the last 3 days	
`receiver_settle_6m`	integer	Number of customer's settlements on the last 6 months	
`receiver_settlement_at`	integer	States the settlement date for the transfer (in milliseconds since the Unix epoch).|Receiver general information	
`receiver_ssn`	string	Receiver social security number.| Receiver general information	123456789
`receiver_street_addr`	string	The address of the individual receiving the payment. | Receiver general information	Lincoln Avenue 77
`receiver_tax_number`	string	Receiver tax number.|Receiver general information	987654321
`receiver_total_balance`	integer	Sum of the balances on all receiver's accounts (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Receiver general information	
`receiver_total_credit_limit`	integer	Sum of the credit limit on all receiver's accounts (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Receiver general information	
`receiver_transaction_amount`	integer	Transaction amount received by the receiver.|Receiver transaction information	
`receiver_transaction_bank_account_number`	string	The bank account number of the receiver.|Receiver transaction information	9a08f316-df1f-4096-9a2d-a904a82bb65e
`receiver_transaction_currency`	string	Defines the currency of the `receiver_transaction_amount` field (in accordance with ISO 4217, Alpha-3).|Receiver transaction information	GBP
`receiver_transaction_fee_amount`	integer	Fee that is applied to the sender's bank included in the `sender_transaction_amount` field, in cents (e.g. 1000 to represent $10.00).|Receiver transaction information	
`receiver_transaction_fee_currency`	string	Defines the currency of the `receiver_bank_fee_currency` field (in accordance with ISO 4217, Alpha-3).|Receiver transaction information	GBP

Name	Type	Description	Example
`receiver_transaction_fee_type`	string	'Defines the method that is used to handle the payment charges or payment fees. Expected values include: - `OUR` : The sender of the payment will bear all of the payment transaction fees. The recipient of the payment will not pay any payment fees. - `SHA` (SHAred): The sender of the payment will pay all fees charged by the sending bank. The recipient of the payment will pay all fees charged by the receiving bank. - `BEN` (BENeficiary): The recipient of the payment will incur all of the payment transaction fees. The sender of the payment will not pay any payment fees. NOTE: Fees can happen at any stage of the transaction, it will depend on the Bilateral or multilateral agreements.| Receiver transaction information'	SHA
`receiver_transaction_foreign_exchange_rate`	double	Foreign exchange rate, defined by the sender and receiver currencies.| Receiver transaction information	
`receiver_transaction_notes`	string	Notes added by the receiver.|Receiver transaction information	The receiver notes.
`receiver_transaction_original_amount`	integer	Transaction amount on the receiver's end. This amount is in the receiver's account currency in cents (e.g. 1000 represents $10.00), and includes fees.| Receiver transaction information	
`receiver_transaction_original_currency`	string	Defines the currency of the `receiver_transaction_original_amount` field (in accordance with ISO 4217, Alpha-3).|Receiver transaction information	GBP
`receiver_zip`	string	Receiver's postal code.|Receiver general information	WC7N 7EW
`recurring_payment`	boolean	Whether the payment is part of a series of recurring payments with the same amount and with the same payee.| Transfer information	
`recurring_payment_day`	integer	'Defines the concrete date of the recurring payment. For example: true the 15th day of each month, or on the on the 1st day of each month (i.e. value 15 or 1 in this example).|Transfer information'	
`recurring_payment_hour`	integer	Defines the hour of the expected recurring day to initiate a transfer.| Transfer information	
`reference_fraud_rate`	double	Fraud rate for transfers. Calculated as the total value of unauthorized or fraudulent remote transfers, divided by the total value of all remote transfers, in the previous calendar quarter (i.e. three-month window).|Transfer information	

Name	Type	Description	Example
`reference_number`	string	Numeric representation of payment ID.|Transfer information	Boleto ID
`reference_number_entry_mode`	string	The entry mode of the payment ID. Possible values are `manual` , `barcode` and `qr_code` .|Transfer information	manual
`requested_execution_at`	integer	Date at which the initiating party requests the settlement agent to process the payment. The sender in this field will be able to specify the expected due date (in milliseconds since the Unix epoch).|Transfer information	
`sca_risk_flag`	boolean	Identifies whether the transfer has strong customer authentication (SCA) or not.|Authentication information	
`sca_risk_score`	integer	Quantifies the risk associated with automatic strong customer authentication (SCA) sent to the clients to allow authentication.|Authentication information	
`segment_channel_type`	string	Specifies the transaction payment channel. Supported values are `retail_banking` , `business_banking` , `commercial_banking` , `treasury` , and `cash_management` . | Transfer information	retail_banking
`sender_account_atc1`	string	Describes the account type affected for debits and inquiries and the â□□fromâ□□ account for transfers.|Sender's account	atc1
`sender_account_atc2`	string	Describes the account type affected for credits and the â□□toâ□□account for transfers.|Sender's account	atc2
`sender_account_available_credit`	integer	Available credit within the credit limit provided by the bank (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Sender's account	
`sender_account_balance`	integer	Current account balance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Sender's account	
`sender_account_bic`	string	Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code.|Sender's account	NWBK
`sender_account_cash_advance_balance`	integer	Debt amount in an account due to cash advance (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Sender's account	
`sender_account_credit_limit`	integer	Account credit limit (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Sender's account	

Name	Type	Description	Example
`sender_account_currency`	string	Currency associated to the sender's account (in accordance with ISO 4217, Alpha-3). | Sender's account	GBP
`sender_account_debt_balance`	integer	Accountâ□□s debt balance (in cents or equivalent). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2). | Sender's account	
`sender_account_debt_payment_method`	string	Payment method of outstanding balance on the account.|Sender's account	card
`sender_account_iban`	string	International Bank Account Number (IBAN) in ISO 13616:2020 format or any bank account unique identifier. This field should be sent as a token or hash.| Sender's account	GB98 MIDL 0700 9312 3456 78
`sender_account_id`	string	Banking App account id. This field should be sent as a token or hash.| Sender's account	6edd8107-f24f-4bdd-ba86-4baf21f27c1b
`sender_account_last_activity`	integer	Identifies the timestamp of the last activity registered at the account (in milliseconds since the Unix epoch).| Sender's account	
`sender_account_last_payment_amount`	integer	Amount of the last payment made on the account (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Sender's account	
`sender_account_last_payment_date`	integer	Date when the sender pays off the balance on the account (in milliseconds since the Unix epoch).|Sender's account	
`sender_account_number_of_cards`	integer	Number of cards associated with the account.|Sender's account	
`sender_account_open_date`	integer	Opening date of the account (in milliseconds since the Unix epoch).| Sender's account	
`sender_account_otb_limit`	integer	Account open to buy limit is the available amount between the account balance and unused credit limit (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).| Sender's account	
`sender_account_pw_change_flag`	boolean	For deployments that do not include Digital Activity, you can use this field to detect if the customer changed the password on the account. This refers to if the password is changed over the last day or within a single session. | Sender's account	
`sender_account_rewards_used_count`	integer	The number of times that the sender used the rewards.|Sender's account	

Name	Type	Description	Example
`sender_account_status`	string	'Status of the account. Possible values: `open` , `blocked` , `closed` , `waiting_for_kyc` .|Sender''s account'	open
`sender_account_type`	string	Type of the account. For example, savings or main account.|Sender's account	credit
`sender_alias`	string	Sender's alias value. This field only applies when the payment is made with an alias, such as using email or phone number as the key to transfer the funds. An alias is mostly used in instant payments, such as PayPal or Venmo. | Sender general information	
`sender_alias_last_update_time`	integer	Identifies the timestamp of the date the sender's alias is updated (in milliseconds since the Unix epoch).| Sender general information	
`sender_alias_registration_time`	integer	Identifies the timestamp of the date the sender's alias is registered (in milliseconds since the Unix epoch).| Sender general information	
`sender_alias_status`	string	The status of the sender's alias. The possible values are `active` and `deleted` .|Sender general information	active
`sender_alias_type`	string	Sender's alias type. This field only applies when the payment is made with an alias.|Sender general information	email
`sender_bank_address`	string	Sender's bank address.|Sender's bank	Diagon Alley 33
`sender_bank_bic`	string	Bank Identifier Code (BIC) in SWIFT format. | Sender's bank	HBUKGB4BXXX
`sender_bank_branch_id`	string	Account bank branch id.|Sender's bank	77-71-13
`sender_bank_city`	string	Sender's bank city.|Sender's bank	London
`sender_bank_country_code`	string	The country code of where the sender's bank is based (in accordance with ISO 3166, Alpha-2). | Sender's bank	UK
`sender_bank_currency`	string	Sender's bank currency (in accordance with ISO 4217, Alpha-3).|Sender's bank	GBP
`sender_bank_fee_amount`	integer	Fee that is applied to the sender's bank included in the `sender_transaction_amount` field, in cents (e.g. 1000 to represent $10.00).|Sender's bank	
`sender_bank_fee_currency`	string	Defines the currency of the `sender_bank_fee_amount` field (in accordance with ISO 4217, Alpha-3).|Sender's bank	GBP
`sender_bank_fee_type`	string	'Defines the method that is used to handle the payment charges or payment fees. Expected values include: - `OUR` : The sender of the payment will bear all of the payment transaction fees. The recipient of the payment will not pay any	SHA

Name	Type	Description	Example
		payment fees. - `SHA` (SHAred): The sender of the payment will pay all fees charged by the sending bank. The recipient of the payment will pay all fees charged by the receiving bank. - `BEN` (BENeficiary): The recipient of the payment will incur all of the payment transaction fees. The sender of the payment will not pay any payment fees. NOTE: Fees can happen at any stage of the transaction, it will depend on the Bilateral or multilateral agreements.| Sender''s bank'	
`sender_bank_foreign_exchange_rate`	double	Foreign exchange rate, defined by the sender and receiving currency.|Sender's bank	
`sender_bank_name`	string	Sender's bank name.|Sender's bank	Gringotts
`sender_bank_postal_code`	string	Sender's bank postal code.|Sender's bank	WC2N 5DU
`sender_bank_routing_number`	string	Account bank routing number.|Sender's bank	192-211-329
`sender_bank_sort_code`	string	Account bank sort code.|Sender's bank	09-01-31
`sender_citizen_id_number`	string	Identification number defined in the citizen identity document associated with the sender's home country.|Sender general information	12345678901234
`sender_city`	string	Sender's city.|Sender general information	Edinburgh
`sender_company_id`	string	Sender company ID.|Sender general information	1af402d5-ea3b-4f81-acae-80934a325daf
`sender_correspondent_bank_address`	string	Bank address.|Sender correspondent's bank	River Street 55
`sender_correspondent_bank_bic`	string	Bank Identifier Code (BIC) in SWIFT format or any other bank unique identifier code.|Sender correspondent's bank	0ed7c577-b503-448a-95c8-b92342e0051e
`sender_correspondent_bank_city`	string	City in which the bank is located.|Sender correspondent's bank	London
`sender_correspondent_bank_country_code`	string	The country code of the correspondent's bank (in accordance with ISO 3166, Alpha-2). | Sender correspondent's bank	US
`sender_correspondent_bank_currency`	string	Currency used by the Bank (in accordance with ISO 4217, Alpha-3).| Sender correspondent's bank	GBP
`sender_correspondent_bank_fee_amount`	integer	Fee that is applied to the sender's bank included in the `sender_transaction_amount` field, in cents (e.g. 1000 to represent $10.00).|Sender correspondent's bank	

Name	Type	Description	Example
`sender_correspondent_bank_fee_currency`	string	Defines the currency of the `fee_amount` field (in accordance with ISO 4217, Alpha-3).|Sender correspondent's bank	GBP
`sender_correspondent_bank_fee_type`	string	'Defines the method that is used to handle the payment charges or payment fees. Expected values include: - `OUR`: The sender of the payment will bear all of the payment transaction fees. The recipient of the payment will not pay any payment fees. - `SHA` (SHAred): The sender of the payment will pay all fees charged by the sending bank. The recipient of the payment will pay all fees charged by the receiving bank. - `BEN` (BENeficiary): The recipient of the payment will incur all of the payment transaction fees. The sender of the payment will not pay any payment fees. NOTE: Fees can happen at any stage of the transaction, it will depend on the Bilateral or multilateral agreements.|Sender correspondent''s bank'	SHA
`sender_correspondent_bank_foreign_exchange_rate`	double	Foreign exchange rate defined by the sender and receiver currencies.|Sender correspondent's bank	
`sender_correspondent_bank_name`	string	Bank name.|Sender correspondent's bank	Correspondent Bank SA
`sender_correspondent_bank_postal_code`	string	Postal code of the bank.|Sender correspondent's bank	YEB3 UD3
`sender_country_code`	string	The country code of the sender / originator (in accordance with ISO 3166, Alpha-2). | Sender general information	US
`sender_country_code_change_flag`	boolean	Whether the sender is perfoming the first payment after changing country in their address. | Transfer information	
`sender_currency`	string	Currency associated to the sender. Refers to senders' total balance and credit limit.|Sender general information	GBP
`sender_dob`	string	Sender's date of birth (YYYYMMDD).|Sender general information	19900101
`sender_driver_license`	string	Sender driver's license.|Sender general information	1234567
`sender_email`	string	Sender's main email contact. | Sender general information	
`sender_email_change_flag`	boolean	Whether the sender is perfoming the first payment after changing email.|Transfer information	
`sender_id`	string	Sender ID generated by the sender's bank.|Transfer information	MSG12345
`sender_initiated_at`	integer	Time at which the sender makes the transfer (in milliseconds since the Unix epoch). This happens after the payment	

Name	Type	Description	Example
		creation. Both events can happen in sync or independently.|Transfer information	
`sender_kyc_lvl`	string	Know your sender (KYC) level.|Sender general information	A
`sender_name`	string	The name of the individual sending the payment. | Sender general information	Joe Dow
`sender_number_of_accounts`	integer	Number of accounts associated with the sender.|Sender general information	
`sender_number_of_cards`	integer	Number of cards associated with the sender.|Sender general information	
`sender_occupation`	string	Sender's main professional activity.|Sender general information	actor
`sender_occupation_status`	string	Sender's current occupation status.|Sender general information	employee
`sender_passport_number`	string	Sender passport number.|Sender general information	PT-019796bb-783c-4020-8110-780e2900515b
`sender_phone`	string	Sender phone number.|Sender general information	001122334455
`sender_phone_change_flag`	boolean	Whether the sender is perfoming the first payment after changing phone number.|Transfer information	
`sender_phone_val_code`	string	Indicates whether the sender phone number is valid.|Sender general information	001122
`sender_region`	string	Sender's region.|Sender general information	Edinburgh
`sender_registration_date`	integer	Sender registration date (in milliseconds since the Unix epoch).|Sender general information	
`sender_risk_rating`	string	Sender's risk rating.|Sender general information	B
`sender_segment`	string	Sender's segment within the banks' segmentation.|Transfer information	retail
`sender_ssn`	string	Sender social security number.|Sender general information	123456789
`sender_street_addr`	string	The address of the individual sending the payment. | Sender general information	94 Morningside Rd, Edinburgh, EH10 4BY
`sender_tax_number`	string	Sender tax number.|Sender general information	987654321
`sender_total_balance`	integer	Sum of the balances on all sender's accounts (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Sender general information	
`sender_total_credit_limit`	integer	Sum of the credit limit on all sender's accounts (in cents). For example, 1000 to represent $10.00 (assuming the US dollar currency, and a currency unit of 2).|Sender general information	

Name	Type	Description	Example
`sender_transaction_amount`	integer	Amount of the transaction performed by the sender. This amount is in the sender's account currency in cents or equivalent (e.g. 1000 represents $10.00), and includes fees. It should be consistent across all transactions as this value is used to calculate profiles in a unified currency. | Sender transaction information	
`sender_transaction_bank_account_number`	string	Sender's bank account number.|Sender transaction information	5a800244-325a-4086-ab8a-23ecbb927be5
`sender_transaction_category_purpose`	string	Category of the purpose of the transaction (ISO 20022).|Sender transaction information	cash_management
`sender_transaction_currency`	string	This field defines the currency of the `sender_transaction_amount` field (in accordance with ISO 4217, Alpha-3) and should be consistent across all transactions. This should also be the bank's official currency. | Sender transaction information	GBP
`sender_transaction_fee_amount`	integer	Fee that is applied to the sender included in the `sender_transaction_amount` field, in cents (e.g. 1000 to represent $10.00).|Sender transaction information	
`sender_transaction_fee_currency`	string	Defines the currency of the `sender_transaction_fee_amount` field (in accordance with ISO 4217, Alpha-3).| Sender transaction information	GBP
`sender_transaction_fee_type`	string	'Defines the method that is used to handle the payment charges or payment fees. Expected values include: - `OUR` : The sender of the payment will bear all of the payment transaction fees. The recipient of the payment will not pay any payment fees. - `SHA` (SHAred): The sender of the payment will pay all fees charged by the sending bank. The recipient of the payment will pay all fees charged by the receiving bank. - `BEN` (BENeficiary): The recipient of the payment will incur all of the payment transaction fees. The sender of the payment will not pay any payment fees. NOTE: Fees can happen at any stage of the transaction, it will depend on the Bilateral or multilateral agreements.| Sender transaction information'	SHA
`sender_transaction_foreign_exchange_rate`	double	Foreign exchange rate, defined by the sender and receiving currencies.|Sender transaction information	
`sender_transaction_notes`	string	Notes added by the sender, when performing the transfer, including note to self. | Sender transaction information	my last rent installment
	integer		

Name	Type	Description	Example
`sender_transaction_original_amount`		Amount of the transaction performed by the sender, in the original currency, in cents or equivalent (e.g. 1000 to represent $10.00). This amount includes the fees. | Sender transaction information	
`sender_transaction_original_currency`	string	Defines the currency of the `sender_transaction_original_amount` field (in accordance with ISO 4217, Alpha-3).|Sender transaction information	GBP
`sender_transaction_purpose`	string	Purpose of the transaction (ISO 20022).|Sender transaction information	cash
`sender_urgency`	boolean	Identifies whether a transfer should be processed as urgent or not.|Transfer information	
`sender_zip`	string	Sender's postal code.|Sender general information	EH10 4BY
`session_id`	string	Id of the session that the user logged into. This should be consistent across all events within the same banking session, including payment initiations as it will be used to associate events to a session.|Authentication information	9be65da3-ec5d-45c0-93c0-03afc5f7633b
`session_register_date`	integer	Session first seen timestamp, (in milliseconds since the Unix epoch)	
`text_for_receiver`	string	Free text message from the sender to the receiver. If the sender does not provide a message, this field is populated with the value `NOTPROVIDED` .|Receiver transaction information	Transfer associated with the last rent installment. Please let me know once you receive the payment.
`transaction_goal`	string	Goal of the transfer. Each institution might have specific goals for certain types of transactions. These goals can be, for instance, the payment of taxes, bills or deposit.|Transfer information	
`transaction_method`	string	Method of the transfer. Each institution might have specific payment methods for certain types of transactions. These methods can be, for instance, balance in the account or credit card.|Transfer information	
`trusted_receiver`	boolean	Whether the receiver is included in the customer's list of trusted receivers.|Transfer information	
`ultimate_creditor_country_code`	string	Ultimate Creditor's country code (in accordance with ISO 3166, Alpha-2).|Sender general information	US
`ultimate_creditor_dob`	string	Ultimate creditor's date of birth (YYYYMMDD).|Sender general information	19900101
`ultimate_creditor_name`	string	The transfer ultimate creditor name	John Doe
	string		US

Name	Type	Description	Example
`ultimate_debtor_country_code`		Ultimate Debtor's country code (in accordance with ISO 3166, Alpha-2).| Sender general information	
`ultimate_debtor_dob`	string	Ultimate debtor's date of birth (YYYYMMDD).|Sender general information	19900101
`ultimate_debtor_name`	string	The transfer ultimate debtor name	John Doe
`user_email_address`	string	Email associated with the bank's user. | Authentication information	
`user_id`	string	The ID of the user that is working on a banking session on behalf of a main account, that could be joint or business and can be different from the customer ID.	3b74891a-04f2-43a0-9519-8e0c2d841986
`user_name`	string	The name of the user that is working on a banking session on behalf of a main account, that could be joint or business and can be different from the customer name.	John Doe
`user_phone_number`	string	Phone number associated with the bank user. | Authentication information	447292738273482
`user_phone_number_country_code`	string	Country code associated with the bank user's phone number (in accordance with ISO 3166, Alpha-2). | Authentication information	US
`withdrawal_amount`	integer	Withdrawal amount in cents (e.g. 1000 represents $10.00).|Transfer information	
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

- References
- Data Exporting Tool
- Event Storage
- Event Storage - Transfers Feedback

Event Storage - Transfers Feedback

Was this page helpful?Â

The **Event Storage Feedback Schema** stores the basic information of any feedback event.

Name	Type	Description	Example
`fdz_uuid`	string	Identifier of each event in UUID	061ad3f3-9991-49fe-850f-e49d6d981028
`timestamp`	double	Unix Timestamp in milliseconds	1760000000.0
`statusreasons`	string	A JSON Array with the status reasons. This only exists if lifecycle_id is null.	('id': '1234-12345-12345', 'name': 'At your discretion', 'statusid': 'real_event')
`lifecycle_id`	string	The lifecycle ID of the event. This only exists when statusreasons is null.	

Name	Type	Description	Example
`realoutcome`	boolean	Indicates whether this was the real outcome or not.	
`feedbacklabel`	string	Label of the Feedback event	fraud
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Pulse Audit - Base Schema

Was this page helpful?Â

The **Pulse Audit - Base Schema** stores the basic information of Pulse Audit events.

All Pulse Audit schemas add to this base schema and will be available when extracting information for each specific audit type (i.e. Users, SSRs, Rules, Lists or List Items).

Name	Type	Description	Example
`entity_type`	string	The type of entity that was audited.	rules, lists, users, etc.
`action_timestamp`	timestamp with time zone	The timestamp at which the action took place.	2025-09-30 10:25:59.100 +0100

Name	Type	Description	Example
`action_type`	string	The type of action that was performed.	login, logout, create, update, delete, etc.
`action_user`	string	The user that performed the action.	john.doe
`_year`	integer	Year of the export	2011
`_month`	integer	Month of the export	10
`_day`	integer	Day of the export	15
`_hour`	integer	Hour of the export	13

Pulse Audit - List Items

Was this page helpful?Â

The **Pulse Audit - List Items** entity stores all audit events for Lists Items. For every action in Pulse related to List Items, this entity captures the audit trail.

It comprises the [Pulse Audit - Base Schema] and the following fields:

Name	Type	Description	Example
`audit_payload`	row	The audit payload associated with the list items action.	
âââ `app_name`	string		
âââ `list_name`	string		

Name	Type	Description	Example
â□□â□□â□□ `context`	string		

Pulse Audit - Lists

Was this page helpful?Â

The **Pulse Audit - Lists** entity stores all audit events for Lists. For every action in Pulse related to Lists, this entity captures the audit trail.

It comprises the [Pulse Audit - Base Schema] and the following fields:

Name	Type	Description	Example
`audit_payload`	row	The audit payload associated with the lists action.	
â□□â□□â□□ `labels`	string		
â□□â□□â□□ `context`	string		

Pulse Audit - Rules

Was this page helpful?Â

The **Pulse Audit - Rules** entity stores all audit events for Rules. For every action in Pulse related to Rules, this entity captures the audit trail.

It comprises the Pulse Audit - Base Schema and the following fields:

Name	Type	Description	Example
`audit_payload`	row	The audit payload associated with the rules action.	
âââ `labels`	string		
âââ `context`	string		

Pulse Audit - Self Service Rules

Was this page helpful?Â

The **Pulse Audit - Self Service Rules** entity stores all audit events for Self Service Rules (SSRs). For every action in Pulse related to SSRs, this entity captures the audit trail.

It comprises the [Pulse Audit - Base Schema] and the following fields:

Name	Type	Description	Example
`audit_payload`	row	The audit payload associated with the SSR action.	
â â â `app_name`	string		
â â â `workflow_name`	string		

Name	Type	Description	Example
â□□â□□â□□ `rules_project_name`	string		
â□□â□□â□□ `rule_name`	string		
â□□â□□â□□ `context`	string		

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- References
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Pulse Audit - Users

Was this page helpful?Â

The **Pulse Audit - Users** entity stores all audit events for Users. For every action in Pulse related to Users, this entity captures the audit trail.

It comprises the Pulse Audit - Base Schema and the following fields:

Name	Type	Description	Example
`audit_payload`	row	The audit payload associated with the users action.	
â□□â□□â□□ `labels`	string		
â□□â□□â□□ `context`	string		